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SUPERSEDING
MIL-HDBK-516C
12 December 2014

DEPARTMENT OF WAR HANDBOOK

AIRWORTHINESS CERTIFICATION CRITERIA

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of War (DoW).
2. Beyond the technical updates from MIL-HDBK-516C to MIL-HDBK-516D, the revision effort had the following objectives:
    a. Standardize the style and format of content such that airworthiness (AW) criteria are concise, correct, consistent, unambiguous, singular, and verifiable in the standards and methods of compliance (MoCs).
    b. Update the standards to address only design attributes of the air system. Any necessary processes should be evaluated as part of MoCs to support assessments of the design attributes.
    c. Ensure criteria are written in support of the definition of AW, which is defined as the property of an air system configuration to safely attain, sustain, and complete flight in accordance with (IAW) approved usage limits. Criteria do not address civilian harm or the safety of ground crew unless it impacts the AW of the air system. Criteria do not address the as-built, as-maintained, or as-operated air system.
        a. Identify a specific location within a reference (e.g., paragraph, table, or figure) in the content of a standard or MoC to align criteria directly to pertinent documentation. References cited will generally utilize the document number and paragraph title to provide content location but will not include the associated verbiage. There are unique cases where the entire reference applies.
    d. Add narratives to provide clarity on the section or subsection criteria.
    e. Delineate MoCs within each criterion by two distinct phases of the program (flight test and operational test or operations) and by verification methodology (examination, analysis, demonstration, and test).
    f. Criteria across sections were reviewed and consolidated, where appropriate.
    g. Write criteria to be applicable to all DoW Components as much as possible (e.g., MIL-HDBK-516D Section 19, Materials, is applicable to all services instead of being limited to Navy and Marine Corps as it was in MIL-HDBK-516C).
    h. Ensure compatibility with DoW digital engineering/digital transformation processes, methodologies, tools, and models.
3. MIL-HDBK-516D consolidates criteria that address similar attributes and separates subsection content into a unique section as compared to MIL-HDBK-516C. Significant changes at the section level include:
    a. Section 10, Diagnostics Systems, criteria were combined into Section 11, Avionics.
    b. Section 14, System Safety, criteria were combined into Section 4, Systems Engineering.
    c. Section 18, Passenger Safety, criteria were combined into Sections 9, Crew Systems.
    d. Section 16 contains all standards associated with maintenance publications.
    e. Aerial refueling (AR) criteria in MIL-HDBK-516C, subsection 8.7, were separated from Section 8, Subsystems and moved to the renamed Section 10, Aerial Refueling.
4. It is recognized that, as part of this release, not all sections meet all objectives of the MIL-HDBK-516D effort. Due to the large effort necessary to focus on design attributes, Section

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15 was not updated to meet the MIL-HDBK-516D objectives. The Section 15 narrative provides clarity on specific format, structure, and content changes of the section. Section 15 standard and method of compliance language were modified to include multi-core processing. Section 8.7, Mechanisms, was converted to the new formatting, with no update to the technical content. Sections 15 and 8.7 will be updated in a future revision to focus on design attributes and align with MIL-HDBK-516D objectives.

5. Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, Bldg 28, 2145 Monahan Way, Wright-Patterson AFB OH 45433-7017 or emailed to Engineering.Standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database https://assist.dla.mil.

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1. SCOPE

1.1 Scope.

Prior to flight, an air system AW approval issued by the DoW Component AW authority is required IAW DoDI 5030.61. The AW authority initiates the AW approval process by conducting comprehensive AW assessments. These assessments capture the description of the air system's configuration, operating limitations, and any other information necessary for safe operation. An AW approval affirms an appropriate level of safety-of-flight (SOF) is provided, the tenets of the AW process are met, and residual risks are accepted by the appropriate authority. Integrating AW early in the systems engineering process will inform the design, development, and verification of a new or modified air system, enabling AW assessment and approval in support of program milestones.

MIL-HDBK-516 establishes the AW criteria, defined as both the standard and its associated MoCs, to be used in the AW assessment of the design of all fixed-wing and rotary-wing air systems, whether manned or unmanned. This handbook defines a minimum set of criteria that only relate to AW and are verifiable by reproducible methods, independent of design approach. The DoW Component AW authorities use this foundational document to define an air system's AW certification basis. The criteria focus on the ability of the air system to safely attain, sustain, and complete flight IAW approved usage limits, including the required interactions by crewmembers to operate the air system. The sections herein represent a minimal set of AW evaluation criteria; MIL-HDBK-516 is not a design guide and is not intended to establish a design specification. Rather, it is intended to provide a focused look at the minimal standards required for AW and is expected to be used in conjunction with other specifications, standards, and engineering to fully implement Operational Safety, Suitability, and Effectiveness (OSS&E). This handbook is for guidance only and cannot be cited as a requirement.

1.2 Applicability.

This handbook may be applied at any point throughout the life cycle of an air system when a design-based AW assessment is necessary. Early understanding of a design's applicable criteria improves a program's ability to identify hazards and assess risk. An AW assessment does not include an air system's ability to meet mission-specific requirements, life cycle costs, or survive threat environments.

Each criterion is evaluated based upon the definition of AW as the property of an air system to attain, sustain, and complete flight IAW approved usage limits. Attaining flight begins with the intent for flight IAW DoDI 6055.07 or the potential for flight (e.g., high speed taxi, tethered rotorcraft). Ground activities such as pre-flight maintenance, hot refueling, stores loading and unloading, towing, and start-up will not be covered in an AW assessment unless it impacts the ability of the air system to attain, sustain, or complete flight.

To ensure consistent and accurate AW assessments, the AW certification basis fundamentally relies on complete and accurate documentation of both applicability and system-specific, measurable criteria. Each MIL-HDBK-516 criterion consists of a standard and at least one MoC. The standards are specific metrics or principles that, when met, enable an air system to safely attain, sustain, and complete flight IAW approved usage limits; standards are based on aviation design principles and cover all aspects needed for the safe design of an air system. Failure to

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show compliance to a standard indicates a potential hazard to the air system, crewmembers, and passengers and may require risk acceptance by the appropriate authority. The MoCs are a logical or systematic means to demonstrate or verify compliance with AW standards. Executing MoCs will generate artifacts and objective evidence used in an AW authority's determination of AW. Within each criterion, MoCs are organized by two distinct phases of the program (flight test and operational test or operations) and by verification methodology (examination, analysis, demonstration, and test). Numbering of MoCs is continuous across phases by verification methodology (e.g., if there are two analyses needed prior to flight test and one needed prior to operations, the numbering schema is Analysis (1) - (3)). Test MoCs are categorized as lab, ground, or flight test. Multiple test categories may be selected for a single Test MoC in cases where more than one test category is acceptable to show compliance with the standard. Any MoC identified as requiring compliance data before flight test should also be reassessed for operational test or operations. Additionally, compliance evidence may require an update if it is incomplete at the time of a first flight authorization or relies on evidence from flight testing to show compliance to enter operational test or operations.

The criteria herein are designed for implementation by engineers in industry or government with demonstrable experience and technical expertise in the relevant areas. These engineers are responsible for understanding and correctly applying the criteria, including tailoring, application, and evaluation of data for a particular air system's compliance through execution of applicable MoCs.

For additional information or clarification call the appropriate service point of contact provided in Appendix A.

1.3 Airworthiness for the air system life cycle.

The AW for the air system life cycle encompasses the as-designed, as-built, as-maintained, and as-operated air system. This revision of MIL-HDBK-516 only addresses the as-designed phase, establishing the certified type design of the air system. It defines the AW criteria and supporting rationale necessary to evaluate whether the air system's design is capable of safe operation within its intended flight envelope, configuration, and mission profile (MP).

The remaining life cycle phases (as-built, as-maintained, and as-operated) are governed by service-specific policies, processes, and authorities not detailed in MIL-HDBK-516. These phases fall under the purview of continuing and continued AW management.

In the as-built phase, AW is supported by stable manufacturing processes, configuration management (CM) (including configuration audits), and effective quality management systems. These processes and systems form the basis for an airworthy, as-built air system and ensure each product conforms to its approved design and is in a condition for safe operation. Consideration for producibility during the design phase establishes requisite production methods and manufacturing process controls to support conformity of the as-built system to the approved design. This precludes manufacturing and quality issues during production that could result in configuration changes, necessitating an additional AW assessment. Guidance on proven manufacturing management practices and quality management systems to promote delivery of affordable and capable air systems is contained in MIL-HDBK-896, SAE International (SAE) AS6500, SAE AS9100, and SAE AS9145.

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The as-maintained and as-operated phases of AW ensure an air system remains safe and compliant for its intended usage throughout the service life. This is accomplished through establishment and execution of continuing AW and continued AW.

Continuing AW focuses on operational sustainment and the ongoing, day-to-day activities required to ensure each individual serialized air system is maintained in a condition for safe operation. It establishes the processes to ensure that, at any time in its life, an air system complies with the technical conditions fixed to its AW approval and is in a condition for safe operation. This includes scheduled and unscheduled maintenance, configuration tracking, inspections, and integrity programs.

Continued AW establishes the design integrity and safety of the air system type throughout its service life. Focusing on design sustainment, it includes the ongoing actions required to maintain the integrity of the certified type design. This includes engineering support activities such as design updates, safety directives, changes to TOs, and life-extension assessments. Continued AW activities verify technical data for compliance with MIL-HDBK-516 criteria.

The design-based AW assessment is fixed to a defined operational usage, flight profile, configuration, and intended operating environment for the air system. Any changes to usage, configuration, or operating environment necessitate an AW assessment IAW service-specific guidance to ensure the certified type design remains current and relevant. Over the course of its service life, air systems typically undergo multiple modifications that may result in different permutations fielded simultaneously. The AW authorities are responsible for the continued assessment of AW for modifications, maintaining configuration control of approved type designs, and ensuring applicable and current technical data support the safe operation of the air system.

1.4 Creating the airworthiness certification basis.

Programs utilize MIL-HDBK-516 by determining the applicable criteria (standards and MoCs) appropriate for each unique design, considering system configuration, flight profile, usage, and operating environment. Identifying these criteria establishes the AW certification basis for a unique air system configuration. The AW certification basis should be fully documented and maintained under configuration control.

Criteria may require tailoring of standards and MoCs based on the specific air system or program requirements. This tailoring may also include developing additional standards and MoCs (e.g., air systems utilizing emerging technology not currently covered may require unique standards or MoCs to establish AW). The act of tailoring is used to ensure that the appropriate design level for a specific operation is established.

All criteria (standards and MoCs) may be tailored considering the system type, capabilities, and intended use; service-specific procedures define the tailoring approval process. Frequent tailoring of criteria and any new criteria developed should be considered for future inclusion in this handbook. Tailoring rules are as follows:

a. Tailoring includes rationale for change.
b. Specific revisions of any government or commercial standards referenced (e.g., MIL-STD-810) may be added to standards and MoCs.

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c. An equivalent level of safety is maintained. Alterations to the baseline level of safety are approved IAW service-specific procedures. Tailoring rationale should indicate whether the equivalent level of safety is maintained or the baseline level of safety is altered.
d. Standards and MoCs, in their entirety or in part, that are not applicable may be removed or modified.
e. Development of new criteria, as appropriate, incorporates both a standard and any associated MoCs. Justification for new criterion should indicate the capabilities or technology not fully addressed by the existing handbook criterion.

1.5 Configuration control of airworthiness products.

Configuration management utilizing an artifact repository is essential for maintaining the authoritative source of truth for all AW products. This controlled repository contains a comprehensive record of the air system design and the specific AW artifacts used to show compliance with the AW standards. The record includes the AW plan, AW certification basis, the AW assessment report with associated compliance artifacts, any identified risks associated with non-compliance, documented acceptance of these risks, and the AW approval. More details on these products are addressed in service-specific guidance. Programs may use an integrated digital environment that includes AW products and artifacts but should ensure a static record is stored in the controlled repository for historical purposes.

1.6 Application and validation of analysis tools.

Modeling and simulation tools are frequently used to generate data in support of showing compliance with AW standards. The AW assessment requires supporting evidence clarifying the configuration, intended usage, fidelity level, and alignment to the air system for the tool. The documentation includes accuracy requirements for production representation from the perspective of the intended usage and identifies requisite training for modeling and simulation users.

Modeling and simulation data generated to directly show compliance to AW criteria requires production representative tools with evidence of verification, validation, and accreditation (VV&A). MIL-STD-3022 provides guidance and templates for developing reports to show this evidence. The reports include the intended use, modeling and simulation requirements and acceptability criteria, assumptions, capabilities, limitations, risks, impacts, and verification and validation (V&V) or accreditation agents. Validation of modeling and simulation tools involves a closed-loop process that uses evidence from comparison between or application of actual test data to the tool.

1.7 Considerations for commercial derivative aircraft.

Air systems that are commercial derivative aircraft (CDA) use a Federal Aviation Administration (FAA) certification as the basis for DoW Component AW approval. Title 14 of the Code of Federal Regulations (CFR) governs all civil aviation activities, including FAA type certification. The FAA Order 8110.101 explains the FAA type certification procedures for CDAs. The DoW Component AW approval addresses any gaps or differences beyond the FAA type certification in configuration, flight profile, intended use, and operating environment. The gaps or differences from FAA certification should be addressed by establishing a MIL-HDBK-516 AW certification basis to ensure a comprehensive AW assessment for the air system. Criteria in sections 4.1, 4.2,

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and 4.3 include MoCs to assess understanding of the gaps or differences, definition of the certification seam between FAA and DoW Component, and verification of comprehensive air system AW.

1.8 Considerations for artificial intelligence usage and airworthiness.

1.8.1 Introduction.

The use of artificial intelligence (AI) in military air systems can offer many benefits. However, when AI has the potential to impact the integrity of safety-critical function (SCF) operation, its incorporation also presents unique challenges. One of these challenges is assessment of the AI's impact to AW. Unlike traditional systems (systems which do not implement AI) that exhibit predictable, deterministic behavior, many systems which implement AI can exhibit emergent, non-linear behavior, to include adapting their functionality based on input and model training data. As a result, many systems which implement AI are not fully explainable or understandable and cannot be assessed with the traditional development and verification standards. The purpose of Section 1.8.1 is to help make programs aware of the challenges associated with the assessment of systems implementing AI and additional considerations that may be needed as part of AW planning. As part of this update, it was determined that the development of AI AW standards was nascent and therefore were not included.

1.8.2 Definition of artificial intelligence.

When used within MIL-HDBK-516 and associated AW publications, the definition for AI is as stated in the Section 15 narrative. The term "AI" is an umbrella term for a type of technology for which there are varying types and implementations. Examples include, but are not limited to, machine learning models, neural networks, large language models, and Bayesian networks.

1.8.3 Artificial intelligence and challenges to traditional methodologies for airworthiness.

In traditional methodologies, the systems engineering process is utilized to define functionality, define requirements, create a design, and implement the design. This allows the integrator and developer to have full control and insight into what is in the design along with the ability to fully explain the design choices and decisions. Unlike these traditional methodologies, however, AI introduces characteristics that challenge this approach:

a. Explainability: Many AI algorithms, especially those involving deep learning, function as "black boxes", making it difficult to trace decision-making processes or to verify correctness.
b. Emergent behaviors: Machine learning systems can evolve based on new data, raising concerns about how these changes impact functionality and integration into air systems.
c. Uncertainty: Unlike traditional deterministic systems, systems implementing AI can produce probabilistic outputs, leading to variability in behavior under similar conditions.
d. Complex interactions: Integration and use of AI by air systems, operators, maintainers, and environments can create unexpected or unplanned outcomes arising from complex interactions that are difficult to anticipate.

These characteristics may necessitate additional AW considerations to establish the baseline level of safety for air systems which utilize AI.

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1.8.4 Usage and integration of artificial intelligence in air systems.

The integration of AI into air systems is not solely a technological challenge focusing on the development, training, and verification of AI itself. This challenge extends to how the AI is utilized and integrated with the air system. The criticality of the air system functionality that the AI element supports is crucial in understanding the integrity expectations that apply due to the inherent risk of certain functionality. For example, utilizing AI elements to support flight-critical functionality will require a higher level of integrity due to the AW risk it could introduce compared to mission-critical functionality. Additionally, the interaction and interfacing of AI in the air system will be thoroughly evaluated as part of the AI's Operational Design Domain. There are architectural considerations to apply when implementing any functionality within an air system, which is equally important when utilizing AI. Utilizing these architectural considerations, AI can be integrated into air systems in various ways ranging from fully autonomous operation (with or without human override), to aiding human decision making. Both the usage and integration of AI into air systems need to be determined and evaluated to support the AW of the air system.

1.8.5 Airworthiness assessments involving artificial intelligence elements.

When AI elements are implemented as part of an air system, additional standards and MoCs beyond what is defined in this handbook may need to be applied to ensure the AW baseline level of safety has been properly established. When considering the full impact of AI design on the AW of the air system, it is important to realize that these considerations apply not only to computer systems and software (CS&S), but also to other functional areas impacted by the design and data management. If a program determines that an AI element does not impact the AW of the air system, an analysis should be provided showing that the integration of the AI within the air system does not have the ability, under all expected operating conditions, to adversely impact the integrity of SCF operation. For those air systems where an AI element either supports the integrity of SCF operation or has the ability to adversely impact SCF operation, the program should consult with the DoW Component AW authority to understand additional AW considerations which may be required.

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2. APPLICABLE PUBLICATIONS

2.1 General.

The publications listed below are not necessarily all of the publications referenced herein, but are those needed to understand the information provided by this handbook. For military references, version identifiers are not included per MIL-STD-967, Government specifications, standards, and handbooks paragraph. To help readers find content, paragraph, table, and figure titles are included (instead of numbers) since the associated numbers may change when a document is revised. For commercial references, versions are permitted.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

INTERNATIONAL STANDARDIZATION AGREEMENT

AIR FORCE INTEROPERABILITY COUNCIL

AIR-STD-1052	Minimal Protection for Aircrew Exposed to Altitude Above 50,000 Feet
INFO-PUB-4048	Ejection Acceleration Limits

ALLIED ORDINANCE PUBLICATION (AOP)

AOP-52	Guidance on Software Safety Design and Assessment of Munition-Related Computing Systems

ALLIED TACTICAL PUBLICATION (ATP)

ATP-3.3.4.2	Air-to-Air Refuelling
ATP-3.3.4.5	Air-to-Air Refuelling Equipment: Boom-Receptacle System and Interface Requirements
ATP-3.3.4.6	Air-to-Air (Aerial) Refuelling Equipment: Probe-Drogue Interface Characteristics

NATO STANDARDIZATION OFFICE

STANAG-4671	Unmanned Aircraft Systems Airworthiness Requirements (USAR) – AEP-4671 Edition B

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DEPARTMENT OF DEFENSE SPECIFICATIONS

JSSG-2001	Air Vehicle
JSSG-2006	Aircraft Structures
JSSG-2007	Engines, Aircraft, Turbine
JSSG-2009	Air Vehicle Subsystems
JSSG-2010	Crew Systems
MIL-W-5013	Wheel and Brake Assemblies, Aircraft General Specification For
MIL-DTL-5041	Tires, Ribbed Tread, Pneumatic, Aircraft
MIL-DTL-5096	Manuals, Technical - Inspection and Maintenance Requirements; Inspection Work Cards; and Checklists; Preparation of
MIL-E-7016	Electric Load and Power Source Capacity, Aircraft, Analysis of
MIL-DTL-7700	Detail Specification Manuals, Technical – Flight Manual Program Publications
MIL-DTL-7788	Panels, Information, Integrally Illuminated
MIL-B-8584	Brake Systems, Wheel, Aircraft, Design of
MIL-DTL-8815	Filter and Filter Elements, Fluid Pressure, Hydraulic Line, 15 Micron and 5 Micron, Type II, General Specification for
MIL-M-8856	Missiles, Guided, Structural Integrity General Specification for
MIL-A-8863	Airplane Strength and Rigidity Ground Loads for Navy Acquired Airplanes
MIL-A-8870	Airplane Strength and Rigidity Vibration, Flutter, and Divergence
MIL-E-18927	Environmental Control Systems, Aircraft, General Requirements for
MIL-A-19736	Air Refueling Systems, General Specification for
MIL-DTL-32742	Handling Qualities for Military Rotorcraft
MIL-DTL-38449	Signal and Signal/Intercommunication Amplifiers, Universal, Aerial Refueling
MIL-C-83124	Cartridge Actuated Devices/Propellant Actuated Devices General Design Specification for

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MIL-C-83125	Cartridges for Cartridge Actuated/Propellant Actuated Devices, General Design Specification for
MIL-P-83126	Propulsion Systems, Aircrew Escape, Design Specification for
MIL-DTL-85025	NATOPS Program Technical Publications and Products; Style, Format and Common Technical Content
MIL-PRF-85676	Lighting, Emergency Egress, Subassembly
MIL-L-85762	Lighting, Aircraft, Interior, Night Vision Imaging System (NVIS) Compatible

(Copies of these documents are available online at https://quicksearch.dla.mil).

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-411	Aircrew Station Alerting Systems
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-1289	Airborne Stores, Ground Fit and Compatibility Requirements
MIL-STD-1290	Light Fixed and Rotary-Wing Aircraft Crash Resistance
MIL-STD-1366	Transportability Criteria
MIL-STD-1425	Safety Design Requirements for Military Lasers and Associated Support Equipment
MIL-STD-1472	Human Engineering
MIL-STD-1530	Aircraft Structural Integrity Program
MIL-STD-1553	Digital Time Division Command/Response Multiplex Data Bus
MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems

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MIL-STD-1587	Material and Process Requirements for Aerospace Weapons Systems
MIL-STD-1760	Aircraft/Store Electrical Interconnection System
MIL-STD-1787	Aircraft Display Symbology
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MIL-STD-1791-1	Criteria for Nonstandard Airdrop Equipment and Payloads
MIL-STD-1796	Avionics Integrity Program (AVIP)
MIL-STD-1797	Flying Qualities of Piloted Aircraft
MIL-STD-1798	Mechanical Equipment and Subsystems Integrity Program
MIL-STD-3009	Lighting, Aircraft, Night Vision Imaging System (NVIS) Compatible
MIL-STD-3013	Glossary of Definitions, Ground Rules, and Mission Profiles to Define Air Vehicle Performance Capability
MIL-STD-3022	Documentation of Verification, Validation, and Accreditation (VV&A) for Models and Simulations
MIL-STD-3024	Propulsion System Integrity Program (PSIP)
MIL-STD-3050	Aircrew Breathing Systems (ACBS)
MIL-STD-3063	Rotorcraft Structural Integrity Program (RSIP)
MIL-STD-7080	Selection and Installation of Aircraft Electric Equipment
MIL-STD-8591	Airborne Stores, Suspension Equipment and Aircraft-Store Interface (Carriage Phase)
MIL-STD-38784	General Style and Format Requirements for Technical Manuals
MIL-STD-40051	Preparation of Digital Technical Information for Technical Manuals (TMs)

(Copies of these documents are available online at https://quicksearch.dla.mil).

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-235-1	Military Operational Electromagnetic Environment Profiles
MIL-HDBK-240-2	Electromagnetic Environmental Effects to Ordnance Guide Part 2 Hazards of Electromagnetic Radiation to Ordnance Testing

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MIL-HDBK-240-3	Electromagnetic Environmental Effects to Ordnance Guide Part 3 Electrostatic Discharge to Ordnance
MIL-HDBK-240-4	Electromagnetic Environmental Effects to Ordnance Guide Part 4 External Radio Frequency Electromagnetic Environments
MIL-HDBK-244	Guide to Aircraft/Stores Compatibility
MIL-HDBK-310	Global Climatic Data for Developing Military Products
MIL-HDBK-704-1	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics (Part 1 of 8 Parts)
MIL-HDBK-1763	Aircraft/Stores Compatibility: Systems Engineering Data Requirements and Test Procedures
MIL-HDBK-1823	Nondestructive Evaluation System Reliability Assessment

(Copies of these publications are available online at https://quicksearch.dla.mil).

2.2.2 Other government publications.

The following other government publications form a part of this document to the extent specified herein.

AERONAUTICAL DESIGN STANDARD

ADS-33E-PRF	Handling Qualities Requirements for Military Rotorcraft

(Copies of this standard are available at https://www.avmc.army.mil/Directorates/STI/TechDataMgmt/).

AIRWORTHINESS ADVISORY AND STRUCTURES BULLETINS

EN-SB-12-002	Methodology to Establish Bird Strike Design Criteria
EZ-SB-22-01	Joint USAF/USN Methodology to Establish the Aircraft Allowable Load Limit Related to Static Strength Capability

(Copies of these publications are available at https://www.daytonaero.com/).

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CODE OF FEDERAL REGULATIONS (CFR)

Title 14, Part 23	Airworthiness Standards: Normal Category Airplanes
Title 14, Part 25	Airworthiness Standards: Transport Category Airplanes
Title 14, Part 27	Airworthiness Standards: Normal Category Rotorcraft
Title 14, Part 29	Airworthiness Standards: Transport Category Rotorcraft
Title 14, Part 33	Airworthiness Standards: Aircraft Engines
Title 14, Part 35	Airworthiness Standards: Propellers

(Copies of this publication are available online at https://www.ecfr.gov).

DEPARTMENT OF THE AIR FORCE PUBLICATIONS

AIR FORCE TECHNICAL ORDER (TO)

TO 00-5-3	Methods and Procedures AF Technical Order Life Cycle Management
TO 1-1B-50	Aircraft Weight and Balance
TO 42B2-1-12 / NAVAIR 01-1A-17	Technical Manual Organizational, Intermediate and Depot Maintenance Aviation Hydraulics Manual

(Copies of these TO's are available online at https://www.tinker.af.mil/Resources/Technical-Orders/ or by requesting them from: Commander, Naval Air Systems Command, 47123 Buse Rd, Bldg 2272, Patuxent River, MD 20670-1547).

DEPARTMENT OF THE ARMY PUBLICATIONS

ADS-44-HDBK	Armament Airworthiness Qualification for U.S. Army Aircraft
DA PAM 25-40	Army Publishing Program Procedures

(Copies of these publications are available at https://armypubs.army.mil/ or https://www.avmc.army.mil/).

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DEPARTMENT OF THE NAVY PUBLICATIONS

NAVY MANUALS

JSSSEH	Joint Software Systems Safety Engineering Handbook

(Copies of these manuals are available at https://www.cto.mil/sea/sse/).

NAVAIR MANUALS

NAVAIR 00-25-604	Naval Air Systems Command Fleet Support/Integrated Program Team Acquisition and Sustainment of NAVAIR Technical Manuals

(Copies of this document can be requested from NATEC, ATTN: FOIA Program, PO Box 357031, BLDG 90, NAS North Island, San Diego, CA 92135-07031).

NAVAL ORDNANCE SAFETY AND SECURITY ACTIVITY (NOSSA)

JOTP-062	Personnel-borne ElectroStatic Discharge (PESD) and Helicopter-borne ElectroStatic Discharge (HESD) Requirements for Ordnance

(Copies of this document are available online at https://quicksearch.dla.mil).

NAVAL SEA SYSTEM COMMAND (NAVSEA)

NAVSEAINST 8020.5	Qualification and Final (TYPE) Qualification Procedures for Navy Explosives
NAVSEA TM-S9310-AQ-SAF-010	Batteries, Navy Lithium Safety Program Responsibilities and Resources
NAVSEA OP-3565 Vol 1	Electromagnetic Radiation Hazards (Hazards to Personnel and Fuel)

(Copies of these publications are available at https://discover.dtic.mil/).

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DEPARTMENT OF WAR PUBLICATIONS

DD FORM 365-1	Chart A - Basic Weight Checklist Record
DD FORM 365-2	Form B - Aircraft Weighing Record
DD FORM 365-4	Weight And Balance Clearance Form F

(Copies of these publications are available at https://www.esd.whs.mil).

FEDERAL AVIATION ADMINISTRATION

FAA TSO-C77b	Gas Turbine Auxiliary Power Units

(Copies of these publications are available at https://www.faa.gov/aircraft/air_cert/design_approvals/engine_prop/apu_approvals/apu_auxpoweru).

NATIONAL TECHNICAL INFORMATION SERVICE

PB94-195047	Documentation for Immediately Dangerous to Life or Health Concentrations (IDLHs)

(Copies of these publications are available at https://ntrl.ntis.gov/NTRL/dashboard/searchResults/titleDetail/PB94195047.xhtml).

2.3 Non-government publications.

The following publications form a part of this document to the extent specified herein.

AEROSPACE INDUSTRY ASSOCIATION

NAS412	Foreign Object Damage (FOD) Prevention Guidance Document

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(Copies of this publication is available at https://www.aia-aerospace.org/).

AIRLINES FOR AMERICA PUBLICATIONS (A4A)

A4A MSG-3	Operator/Manufacture Scheduled Maintenance Development Volume 1 & 2, Fixed Wing Aircraft and Rotorcraft

(Copies of this publication is available at https://publications.airlines.org/).

AMERICAN NATIONAL STANDARDS

ANSI Z136.1	American National Standard for Safe Use of Lasers
ANSI Z136.4	Recommended Practice for Laser Safety Measurements for Classification and Hazard Evaluation

(Copies of these publications are available at https://webstore.ansi.org/).

ARINC INDUSTRY ACTIVITIES

ARINC 429	Digital Information Transfer System (DITS)

(Copies of this publication is available at https://aviation-ia.sae-itc.com/).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F330-21	Standard Test Method for Bird Impact Testing of Aerospace Transparent Enclosures

(Copies of these publications are available at https://www.astm.org/).

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COMPOSITE MATERIALS HANDBOOK

CMH-17	Composite Materials Handbook

(Copies of this publication are available at https://www.cmh17.org/).

INSTITUTE OF ELELCTRICAL AND ELECTRONIC ENGINEERS (IEEE)

IEEE C95.1™-2019	IEEE Standard for Safety Levels with Respect to Human Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz
IEEE C95.1-2345™-2014	IEEE Standard for Military Workplaces--Force Health Protection Regarding Personnel Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz

(Copies of this publication are available at https://www.ieee.org/).

METALLIC MATERIAL PROPERTIES DEVELOPMENT STANDARDIZATION

MMPDS	Metallic Materials Properties Development and Standardization Handbook

(Copies of this publication are available at https://www.mmpds.org/).

NATIONAL LIBRARY OF MEDICINE

Bass, C.R., Rafaels, K.A. and Salzar, R.S., 2008. Pulmonary injury risk assessment for short-duration blasts. Journal of Trauma and Acute Care Surgery, 65(3), pp.604-615.

Rowson, Steven and Duma, Stefan, "Brain Injury Prediction: Assessing the Combined Probability of Concussion Using Linear and Rotational Head Acceleration," Annals of Biomedical Engineering, Vol. 41, No.5, May 2013.

(Copies of these publications are available at https://pubmed.ncbi.nlm.nih.gov/).

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RADIO TECHNICAL COMMISSION FOR AERONAUTICS (RTCA)

RTCA DO-160	Environmental Conditions and Test Procedures for Airborne Equipment

(Copies of this publication are available at https://www.rtca.org/).

SAE INTERNATIONAL

SAE AIR1419	Inlet Total-Pressure Distortion Considerations for Gas-Turbine Engines
SAE AIR1703	In-Flight Thrust Determination
SAE AIR5686	A Methodology for Assessing Inlet Swirl Distortion
SAE AIR5687	Inlet / Engine Compatibility – From Model to Full Scale Development
SAE AIR5866	An Assessment of Planar Waves
SAE AIR5867	Assessment of the Inlet/Engine Total Temperature Distortion Problem
SAE AIR6345	Processing and Validating CFD Data for Estimating Inlet Dynamic Total-Pressure Distortion
SAE ARP1185C	Flexure Testing of Hydraulic Tubing Joints and Fittings
SAE ARP1420	Gas Turbine Engine Inlet Flow Distortion Guidelines
SAE ARP1493C	Wheel and Hydraulically Actuated Brake Design and Test Requirements for Military Aircraft
SAE ARP4392	Lighting, Aircraft Exterior, Night Vision Imaging System (NVIS) Compatible
SAE ARP4761A	Guidelines for Conducting the Safety Assessment Process on Civil Aircraft, Systems, and Equipment
SAE ARP5412	Aircraft Lightning Environment and Related Test Waveforms
SAE ARP5415	User's Manual for Certification of Aircraft Electrical/Electronic Systems for the Indirect Effects of Lightning
SAE ARP5416	Aircraft Lighting Test Methods

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SAE ARP5672	Aircraft Precipitation Static Certification
SAE ARP5825	Design Requirements and Test Procedures for Dual Mode Exterior Lights
SAE ARP6420	Guidelines for Characterization of Gas Turbine Engine Total-Pressure, Planar-Wave, and Total-Temperature Inlet-Flow Distortion
SAE ARP94910	Aerospace – Vehicle Management Systems – Flight Control Design, Installation and Test of Military Unmanned Aircraft, Specification Guide For
SAE AS603B	Impulse Testing of Hydraulic Hose, Tubing, and Fitting Assemblies
SAE AS4059G	Contamination Classification for Hydraulic Fluids
SAE AS5440B	Hydraulic Systems, Military Aircraft, Design and Installation, Requirements For
SAE AS5586A	General Requirements for Hydraulic System Reservoirs
SAE AS6053A	Tests, Impact, Shock Absorber Landing Gear, Aircraft
SAE AS6870	Nondestructive Inspection Program Requirements for Aerospace Systems
SAE AS8049D	Performance Standard for Seats in Civil Rotorcraft, Transport Aircraft, and General Aviation Aircraft
SAE AS8584B	Brake Systems, Wheel, Military Aircraft
SAE AS8775	General Specification for Military Aircraft and Missile Hydraulic System Components
SAE AS9100	Quality Systems – Aerospace – Model for Quality Assurance in Design, Development, Production, Installation and Servicing
SAE AS50881	Wiring, Aerospace Vehicle
SAE AS94900A	Vehicle Management Systems – Flight Control Function, Design, Installation and Test of Piloted Military Aircraft, General Specification For
SAE EIA-649-1	Configuration Management Requirements for Defense Contracts
SAE JA1011	Evaluation Criteria for Reliability-Centered Maintenance (RCM) Processes

(Copies of these publications are available at https://www.sae.org/).

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3. DEFINITIONS AND ACRONYMS

3.1 Definitions.

All definitions, unless otherwise referenced, are to be considered within the context of this document.

3.1.1 Airframe.

The structure of an air vehicle apart from accessories and power plant. The principal parts of the airframe include the fuselage, wings, empennage, landing gear, and nacelles or pods.

3.1.2 Air system.

An air vehicle plus the training and support systems for the air vehicle (e.g., communications, control, ground/surface/control station, launch and recovery, and support elements), and any weapons to be employed on the air vehicle.

3.1.3 Air vehicle.

An air vehicle includes the installed equipment (hardware, software, and firmware) for airframe, propulsion, onboard vehicle and applications software, communications/identification, navigation/guidance, central computer, fire control, data display and controls, survivability, reconnaissance, automatic flight control, central integrated checkout, antisubmarine warfare, armament, weapons delivery, auxiliary equipment, and all other installed equipment.

3.1.4 Airworthiness.

The property of an air system configuration to safely attain, sustain, and complete flight IAW approved usage limits.

3.1.5 Airworthiness approval.

Documents issued by an empowered AW authority and may take a number of different forms (e.g., AW release, military-type certificate, flight clearance), depending on specific AW authority policy. An AW approval affirms that the appropriate tenets of the AW process are met and that the air vehicle or air system was assessed against the required AW standards and any residual risk to crewmembers, ground crew, passengers, or to third parties has been accepted by the appropriate authority.

3.1.6 Airworthiness assessment.

A technical evaluation of data against specific AW criteria and determination of residual risk. An AW assessment is a critical step in the AW approval process but itself does not necessarily result in the issuance of an AW approval.

3.1.7 Airworthiness authority.

An individual who has the legal mandate to develop and enforce pertinent rules, regulations, and policy governing AW.

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3.1.8 Airworthiness certification basis.

A document comprised of a system description, the set of approved AW certification criteria that applies to a specific air system design. It is typically derived from MIL-HDBK-516, including approved change notices (CNs).

3.1.9 Airworthy.

Conforming to a type design that has a valid AW approval that is in a condition of safe operation.

3.1.10 Allocated baseline.

The approved requirements for a product, subsystem, or component, describing the functional, performance, interoperability, and interface requirements, that are allocated from higher-level requirements, and the verifications required to demonstrate achievement of those requirements, as established at a specific point in time and documented in the allocated configuration documentation. The allocated baseline for each lower-level system element (hardware and software) is usually established and put under configuration control at the system element Preliminary Design Review.

3.1.11 Analysis.

A verification method used to verify requirements by determining quantitative properties and performance conducting modeling, simulation, or calculations and interpreting the results for components or systems interacting with their intended environment. Analysis includes specific methods, procedures, formulas, or mathematical modeling based on physical laws, for any technical calculation to be performed. Analysis techniques include interpretation or interpolation/extrapolation of analytical or empirical data collected via test under defined conditions. Sensitivity, similarity, and failure effects analyses are forms of this method.

3.1.12 Catastrophic severity category.

A hazard where the severity results in a mishap that could result in one or more of the following: death, permanent total disability, irreversible significant environmental impact, or monetary loss equal to or exceeding $10M.

3.1.13 Commercial derivative aircraft.

An aircraft procured as a commercial off-the-shelf (COTS) non-development item, and whose serial number is listed on the FAA Type Certificate Data Sheet.

3.1.14 Common mode failure.

A type of failure that occurs when multiple systems or components fail in the same way at the same time. This can be caused by a single external event or internal design defect.

3.1.15 Computer software configuration item.

A logically grouped collection of software components having a definitive interface and functional purpose that are developed and integrated together into a configuration item (CI).

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3.1.16 Concept of operations/Operational mode summary/Mission profile.

A component-approved document that is derived from and consistent with the validated/approved capability requirements document. The CONOPS/OMS/MP describes the operational tasks, events, durations, frequencies, and environments in which the materiel solution is expected to perform each mission and each phase of the mission.

3.1.17 Configuration baseline.

Configuration of a product, at a specific point in time, which serves as a basis for defining change, for conducting verifications, and for other management activities.

3.1.18 Configuration control.

The systematic proposal, justification, evaluation, coordination, disposition of proposed changes or requested variances, and the implementation of all approved changes or variances in the configuration of a CI after establishment of the configuration baseline.

3.1.19 Configuration item.

A product or an aggregation of products (hardware, software, firmware) and associated documentation that accomplishes an end-use function, is under separate configuration control and is designated for CM control. It is the primary identifier for referencing a product or allocated components that satisfies an end use function. An item is designated as a CI for purposes of additional CM focus due to its complexity, logistic support requirements, acquisition strategy, or because it is intended to undergo configuration status accounting (CSA) or verification and audit separately from other items. Configuration items are end items or major components of end items, which typically have performance requirements allocated to them and documented in their own specification.

3.1.20 Configuration status accounting.

The CM function that formalizes the recording and reporting of the established product configuration information (including historical information), the status of proposed changes, and the implementation of approved changes and changes occurring to the product units due to operation and maintenance. Configuration status accounting implementation includes assurance that the information is current, accurate, and retrievable.

3.1.21 Continued airworthiness.

All the actions taken for the upkeep of a certified type design and associated approved data of air systems, engines, propellers, and other airworthy systems, so they can maintain a condition in which they can be operated safely for their intended purpose.

3.1.22 Continuing airworthiness.

The process by which an air system is kept in a condition where it remains airworthy throughout its life (i.e., technically fit for flight).

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3.1.23 Control station.

A facility or device, consisting of hardware, software, and firmware from which one or more unmanned air vehicles (UAVs) are controlled, monitored, or controlled and monitored for any phase of flight operations. The control station may be stationary or mobile (e.g., as carried by another ground vehicle, air vehicle, naval vessel, or ground crew).

3.1.24 Copilot.

The member of the flight crew whose duties include assisting or relieving the pilot in operating the flight controls of an air vehicle.

3.1.25 Crewmember.

Military personnel on competent flight orders or civilian personnel whose duties require frequent and regular participation in aerial flights to perform inflight functions. Crewmembers are also referred to as aircrew.

3.1.26 Crew station.

The physical environment in which a crewmember performs mission duties.

3.1.27 Criteria.

Within MIL-HDBK-516D, a criterion refers to both the standard and the MoC. MIL-HDBK-516D is a collection of criteria for an air system configuration to safely attain, sustain, and complete flight IAW approved usage limits.

3.1.28 Critical safety item.

A part, assembly, installation equipment, launch equipment, recovery equipment, or support equipment for an air vehicle or aviation weapon system for which such contains a characteristic any failure, malfunction, or absence of which could cause a catastrophic or critical failure resulting in the loss of, or serious damage to, the air vehicle or weapon system; an unacceptable risk of personal injury or loss of life; or an uncommanded engine shutdown that jeopardizes safety.

3.1.29 Critical severity category.

A hazard where the severity results in a mishap that could result in one or more of the following: permanent partial disability, injuries or occupational illness that may result in hospitalization of at least three personnel, reversible significant environmental impact, or monetary loss equal to or exceeding $1M but less than $10M.

3.1.30 Demonstration.

A verification method, without the use of precision measurement equipment, to verify hardware or software requirements by exercising or operating the system or components under specified controlled conditions on the air system (on-ground or in-flight) or in an equivalent environment. Functional success is determined on a qualitative or pass-fail basis and generally denotes the actual operation, adjustment, or reconfiguration of items to provide evidence that the designed functions were accomplished under specific scenarios.

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3.1.31 Design envelope.

Envelope that includes ground and flight modes, profiles, and operations of an air vehicle.

3.1.32 Digital engineering.

An integrated digital approach that uses authoritative sources of systems' data and models as a continuum across disciplines to support lifecycle activities from concept through disposal.

3.1.33 Electrical wiring interconnect system.

Any wire, electrical cabling, or fiber optic cabling, or a combination of these items, including terminations, installed in any area of the air vehicle for the purpose of transmitting electrical or optical energy, signals or data between two or more electrical or optical end points.

3.1.34 Enhanced vision flight system.

A system which combines flight information, symbology, navigation guidance, and a real-time image of the external scene and presents them to the pilot on a Head-Up Display (HUD) or an equivalent presentation.

3.1.35 Enhanced vision system.

A system where imaging sensors produce a real-time image of the outside scene.

3.1.36 Examination (previously inspection).

A verification method used to verify by direct inspection that the physical features of an actual component, system, or function conform to specified requirements as identified by applicable drawings, specifications, and supportive data through non-destructive means. Examination includes visual inspection, simple physical manipulation, gauging, and measurements without the use of precision measurement equipment, including review of authenticated documentation.

3.1.37 Failure.

The inability of a system or system component to perform a required function within specified limits.

3.1.38 Failure modes, effects, and criticality analysis.

A procedure for identifying potential failure modes in a system and classifying them according to their severity. A FMECA is usually carried out progressively in two parts. The first part identifies failure modes and their effects (also known as failure modes and effects analysis (FMEA)). The second part ranks the failure modes according to the combination of their severity and the probability of occurrence (criticality analysis).

3.1.39 Fault.

A manifestation of an error in an item.

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3.1.40 Fault tolerance.

The ability of a system to provide an acceptable level of operational performance and safety in the event of one or more failures.

3.1.41 Flight crew.

Crewmember on board a manned air vehicle or at a control station for Unmanned Aircraft System (UAS) responsible for performing specific aeronautical tasks in support of the assigned mission.

3.1.42 Flight-critical.

A term applied to any condition, event, operation, process, or item whose proper recognition, control, performance, or tolerance is necessary to attain, sustain, or complete controlled flight of an air vehicle during flight operations.

3.1.43 Flight termination system.

A system which provides the ability to end flight in a controlled manner. The FTS includes all systems, subsystems and components that control and execute the termination procedure. The function may be commanded from a control station or by the air vehicle itself.

3.1.44 Functional baseline.

Describes the system's performance (functional, interoperability, and interface characteristics) and the verification required to demonstrate the achievement of those specified characteristics. It is directly traceable to the operational requirements contained in the Initial Capabilities Document or equivalent document.

3.1.45 Hazard.

A real or potential condition that could lead to an unplanned event or series of events (e.g., mishap) resulting in death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.

3.1.46 Integrity.

The essential characteristics of a system, subsystem, or equipment that allow specific performance, reliability, safety, and supportability to be achieved under specified operational and environmental conditions, and over a specific service life.

3.1.47 Interface.

The performance, functional, and physical attributes required to exist at a common boundary.

3.1.48 Interlock.

System design mechanization to enable or disable systems, functions, subsystems, or modes at given times and conditions.

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3.1.49 Iron bird.

A functional simulator containing hardware-in-the-loop (HITL), used for integrated testing of new air system designs (e.g., hydraulic systems and flight controls).

3.1.50 Level of safety.

In the context of AW, the degree of probability and consequence (risk) of harm to persons or property due to the utilization of an air system IAW its approved usage limits.

3.1.51 Life-limited parts.

Life-limited parts are any part for which a mandatory replacement interval is specified in the approved military design series configuration, through integrity programs, and within approved maintenance publications. The replacement interval-can be defined in accumulated cycles, hours, calendar time, or other defined limits.

3.1.52 Maintenance publications.

Maintenance manuals, ground crew checklists, weapons loading checklists, inspection manuals, and all other forms of printed or digital maintenance related documentation.

3.1.53 Method of compliance.

A logical or systematic means to demonstrate or verify compliance with AW standards. MoCs will generate artifacts and objective evidence used in an AW authority's determination of AW.

3.1.54 Mishap.

An event or series of events resulting in unintentional death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.

3.1.55 Mission.

The profiles flown within the operating environment, including duration and severity factor, particularly as they relate to air system cyclic, stress, or fatigue limits and service life. Mission also referred to as mission usage.

3.1.56 Mission-critical.

A term applied to any condition, event, operation, process or item, the failure of which may result in the inability to achieve successful mission completion or to maintain combat capability.

3.1.57 Modeling and simulation.

The use of models and simulations, either statically or overtime, to develop data as a basis for making managerial or technical decisions. This includes but is not limited to emulators, prototypes, simulators, and stimulators.

3.1.58 Modification.

Changes to hardware, software, firmware, technical data, or mission usage by removing or adding a capability or function, enhancing technical performance or suitability, to include form,

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fit, function, and interfaces replacement of an approved configuration.

3.1.59 Operating envelope.

The operating limitations for a specific configuration defined by the AW authority.

3.1.60 Operating environment.

The surroundings or conditions in which an air system operates including, but not limited to temperatures, loads, ambient environmental conditions, moisture and fluid exposures, electromagnetic spectrum, radiation, maintenance, and ground handling.

3.1.61 Operational state I.

The normal state of vehicle control function (VCF) operation providing sufficient performance, safety and reliability to accomplish the mission tasks and system requirements of the air vehicle within the required flight envelope and airspace. This state satisfies Level 1 flying qualities within the Region of Satisfactory Handling (ROSH) and no worse than Level 2 flying qualities within the Region of Tolerable Handling (ROTH).

3.1.62 Operational state II.

The state of less than normal VCF operational performance, safety, or reliability with no loss of flight-critical functions. Any degradation in mission effectiveness and restrictions of flight envelopes will not prevent the air vehicle from completing a modified mission or from making a degraded landing at the destination of original intent. This state results in minimal increase in pilot workload and may result in a limited selection of VCF modes. This state satisfies Level 1 flying qualities within the ROSH and no worse than Level 2 flying qualities within the ROTH.

3.1.63 Operational state III.

The state of degraded VCF performance, safety, or reliability which permits the air vehicle to safely abort the current mission task; maneuver capability is limited to cruise, navigation, transition to appropriate operating areas, and an emergency landing. This state results in objectionable pilot workload such that phases of the intended mission involving precision tracking or maneuvering cannot be completed satisfactorily . This state satisfies no worse than Level 2 flying qualities for flight tasks required for landing (e.g., cruise, navigation, transit to appropriate operating areas, landing, waveoff, go-around, and bolter).

3.1.64 Operational state IV.

The state of degraded VCF operation at which continued safe flight is not possible; however, sufficient control remains to allow engine restart attempt(s), controlled descent and immediate emergency landing (or safe termination for UAS).

3.1.65 Operational state V.

The immediate loss of VCF state which allows for no control of the air vehicle impact site or of heading prior to impact. The VCF capability is limited to maneuvers required to reach a flight condition at which crewmember evacuation may be safely accomplished.

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3.1.66 Passenger.

Any occupant on board an air vehicle who does not have active crewmember duties and is not essential for accomplishing mission tasks.

3.1.67 Performance.

A quantitative or qualitative measure characterizing a physical or functional attribute relating to the execution of an operation or function. Performance attributes include quantity (how many or how much), quality (how well), coverage (how much area, how far), timeliness (how responsive, how frequent), and readiness (availability, mission/operational readiness). Performance is an attribute for all systems, people, products, and processes including those for development, production, verification, deployment, operations, support, training, and disposal. Supportability parameters, manufacturing process variability, and reliability are all performance measures.

3.1.68 Photometric.

Relating to the measurement of visible light in units that are weighted according to the sensitivity of the human eye or test instrumentation.

3.1.69 Pilot.

The member of the flight crew who is responsible for operating the flight controls of an air vehicle when under way, including the landing and takeoff of an air vehicle.

3.1.70 Probability.

An expression of the likelihood of occurrence of a mishap.

3.1.71 Probability of loss of control.

An expression of the likelihood of loss of control during flight operations, based upon the levels of redundancy of the air system architecture, robustness of the failure management design, and the failure probabilities of those items contributing to control of the air vehicle, where loss of control has the potential to result in a catastrophic or critical hazard.

3.1.72 Product baseline.

Describes the detailed design at a specific point in time, for production, fielding/deployment, operations, and support. The product baseline prescribes all necessary physical (form, fit, or function) characteristics and selected functional characteristics designated for production acceptance testing and production test requirements. The initial product baseline includes "build-to" specifications for hardware (product, process, material specifications, engineering drawings, 3D Computer Aided Design (CAD) models, and other related data) and software (software module design - "code-to" specifications). The as-delivered and subsequent product baselines add product operational information needed to operate and maintain the product.

3.1.73 Qualified.

The resulting state or condition from a formal verification process being applied to hardware and software systems to verify that requirements have successfully been met.

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3.1.74 Redundancy.

Utilization of two or more components, subsystems, or channels so that the functions that they support are capable of being sustained in the event of a failure.

3.1.75 Resonance assessment profile testing.

Testing conducted to measure modal response characteristics of a system or component.

3.1.76 Risk.

A combination of the severity of the mishap and the probability that the mishap will occur.

3.1.77 Safety-critical.

A term applied to a condition, event, operation, process, or item whose mishap severity consequence is either Catastrophic or Critical.

3.1.78 Safety-critical function.

A function whose loss, degradation, or erroneous operation can potentially result in a mishap of either Catastrophic or Critical severity.

3.1.79 Safety-critical function thread.

The collection of equipment within a system and the associated data parameters, interfacing, and interaction of that equipment that supports the operation of a given SCF.

3.1.80 Safety-critical function thread analysis.

An analysis of the functional integration of a SCF thread and the V&V activities applied to the thread. A SCFTA is intended to ensure that the complete SCF thread has been identified, analyzed, and by collectively analyzing all levels of the design, complete end-to-end V&V of the thread has been performed.

3.1.81 Safety-critical item.

A hardware or software item that has been determined through analysis to potentially contribute to a hazard with Catastrophic or Critical mishap potential, or that may be implemented to mitigate a hazard with Catastrophic or Critical mishap potential.

3.1.82 Service life.

The maximum permissible period of air vehicle operational service expressed in units of actual flight hours, equivalent flight hours, and time in service. Service life limits may be defined for the air system, individual major structural components, or air system and individual major structural components.

3.1.83 Severity.

The magnitude of potential consequences of a mishap to include death, injury, occupational illness, damage to or loss of equipment or property, damage to the environment, or monetary loss.

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3.1.84 Single point failure.

A failure of a single component which causes a loss or unacceptable degradation of a function.

3.1.85 Situational awareness.

The ability to identify, process, and comprehend critical, perceived elements of information in one's environment in order to make decisions about a future state or needed actions. Any information that is presented to the crewmember can affect situational awareness. If a system or information is presented “for situational awareness”, then performance is demonstrated to prove that the information or system contributes (increases situational awareness), does not increase workload, or present potentially hazardous and misleading information.

3.1.86 Standard.

Specific metrics that, when met, enable an air system can safely attain, sustain, and complete flight IAW its approved usage limits. They are based on aviation design principles and cover all aspects in the design of safe air systems.

3.1.87 System.

The organization of hardware, software, material, facilities, personnel, data, and services needed to perform a designated function within a stated environment with specified results.

3.1.88 Test.

A verification method used to exercise or operate the system, or components, to verify requirements or validate design and functional operation. Testing provides quantitative data under specified conditions and predetermined procedures to evaluate the functional characteristics against expected results. Software verification using the test method employs structured verification using predetermined initial conditions; scripted sequence of stimuli, and re-defined quantitative results to verify that software performs as required.

Tests are categorized as lab, ground, or flight test. Lab tests are air vehicle, or air vehicle component, tests in a ground-based facility with a physical simulation or representation of the operating environment. Ground tests are conducted on the air vehicle with no intent or potential for flight. Flight tests are on the air vehicle in conditions when there is intent for flight, there is sufficient kinetic energy for flight, or the air vehicle is in flight.

3.1.89 Type Design.

The engineering definition of a particular product, which consists of: the drawings and specifications, and a listing of those drawings and specifications, necessary to define the configuration and the design features of the air system shown to comply with the AW criteria applicable to the air system. Information on dimensions, materials, materiel properties, and processes necessary to define the structural strength of the product. Any AW limitations required for safe operation and maintenance.

3.1.90 Unmanned aircraft system.

A system comprised of individual elements consisting of the UAV, the control station and/or control device, and any other support elements necessary to enable operation including, but not

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limited to data links, communications systems/links, and UAV-unique launch and recovery equipment. There may be multiple UAVs, control stations or control devices, and support elements within an UAS. The control station or control device may be located on the ground (stationary or mobile), on a ship, submarine, or air vehicle.

3.1.91 Unmanned air vehicle.

A remotely operated, semi-autonomous, or autonomous air vehicle and its onboard operating system. This does not include air vehicles designed for one-time use as a weapon (e.g., cruise missile).

3.1.92 User interface.

A physical, graphical, or perceptual interaction between the crewmember and the system.

3.1.93 Vehicle control functions.

Vehicle control functions include an integrated combination of functions which are critical to SOF and which permit an air vehicle to be controlled in a manner that allows its specified missions to be accomplished satisfactorily. Vehicle control functions include all components and functions used to sense vehicle position, velocity, speed, inertial attitudes, rates and accelerations, heading and altitude, and to generate and transmit appropriate commands to force and moment producers in response to flight path commands, whether internally generated or received from an on- or offboard flight crew. These commands, often generated by guidance and navigation elements of the VCFs, normally result in control of air vehicle altitude, airspeed, heading, attitude, aerodynamic or geometric configuration, ride quality, and structural modes.

3.1.94 Verification.

Confirmation through the provision of evidence provided by examination, analysis, demonstration, and test that specified requirements have been fulfilled.

3.2 Abbreviations and acronyms.

AA	Airworthiness Advisory
AC	Advisory Circular
ACAS X	Next Generation Airborne Collision Avoidance System
ACBS	Aircrew Breathing System
ADS	Aeronautical Design Standard
ADS-B	Automatic Dependent Surveillance-Broadcast
ADS-C	Automatic Dependent Surveillance-Contract
AFE	Aircrew Flight Equipment
AFHA	Aircraft Functional Hazard Assessment
AFMAN	Air Force Manual
AFN	Air Facilities Notification
AGL	Above Ground Level
AI	Artificial Intelligence
AIMS	Air Traffic Control Radar Beacon System, Identification Friend or Foe, Mark XII/Mark XIIA, Systems

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AIP	Aerodynamic Interface Plane
AIR	Aerospace Information Report
AMAD	Airframe Mounted Accessory Drive
AMP	Asymmetric Multiprocessing
AMT	Accelerated Mission Test
ANSI	American National Standards Institute
AOA	Angle of Attack
AOP	Allied Ordnance Publication
AOS	Angle of Sideslip
APS	Auxiliary Power System
APU	Auxiliary Power Unit
AR	Aerial Refueling
ARINC	Aeronautical Radio, Incorporated
ARP	Aerospace Recommended Practice
AS	Aeronautical Standard
ASE	Aeroservoelastic
ASIP	Aircraft Structural Integrity Program
ASME	American Society of Mechanical Engineers
ASMET	Accelerated Simulated Mission Endurance Test
ASTM	American Society of Testing and Materials
ATC	Air Traffic Control
ATD	Anthropometric Test Device
ATP	Allied Tactical Publication
AUR	All-Up-Round
AW	Airworthiness
BER	Bit Error Rate
BIT	Built-In-Test
BLOS	Beyond Line of Sight
BMP	Bound Multiprocessing
BTU	British Thermal Unit
C2	Command and Control
CAD	Computer Aided Design
CAD/PAD	Cartridge Actuated Devices/Pyrotechnic Actuated Devices
CAS	Control Augmentation System
CBIT	Continuous Built-In-Test
CBRN	Chemical, Biological, Radiological, and Nuclear
CC	Critical Characteristic
CDA	Commercial Derivative Aircraft
CFD	Computational Fluid Dynamics
CFR	Code of Federal Regulations
CG	Center of Gravity
CI	Configuration Item
CM	Configuration Management
CMF	Common Mode Failure

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CONOPS	Concept of Operations
COTS	Commercial Off-the-Shelf
CNS/ATM	Communications, Navigation and Surveillance/Air Traffic Management
CPCP	Corrosion Prevention and Control Plan
CPDLC	Controller-Pilot Data Link Communications
CS&S	Computer Systems and Software
CSA	Configuration Status Accounting
CSCI	Computer Software Configuration Item
CSI	Critical Safety Item
CSIL	Computer System Integrity Level
CW	Continuous Wave
DAA	Detect and Avoid
DCMA	Defense Contract Management Agency
DD	Department of Defense Form
DEW	Directed Energy Weapon
DIPES	Deck Interface Pilot Effort Scale
DLL	Design Limit Load
DoD	Department of Defense
DoDD	Department of Defense Document
DoDI	Department of Defense Instruction
DoW	Department of War
DOD	Domestic Object Damage
DOF	Degrees of Freedom
DR	Dynamic Response
DSL	Design Service Life
E3	Electromagnetic Environmental Effects
ECS	Environmental Control System
EFH	Engine Flight Hours
ELA	Electrical Load Analysis
EMAD	Engine Mounted Accessory Drive
EMC	Electromagnetic Compatibility
EMCON	Emission Control
EME	External Electromagnetic Environment
EMI	Electromagnetic Interference
ENSS	Element Not Supporting Safety-Critical Function(s)
EPS	Electrical Power System
EPU	Emergency Power Unit
ESD	Electrostatic Discharge
ESS	Element Supporting Safety-Critical Function(s)
E-TCAS	Enhanced-Traffic Collision Avoidance System
EVFS	Enhanced Vision Flight System
EVS	Enhanced Vision System
EW	Electronic Warfare

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EWIS	Electrical Wiring Interconnect System
FAA	Federal Aviation Administration
FCG	Fatigue Crack Growth
FCS	Flight Control System
FEA	Finite Element Analysis
FFT	Fast Fourier Transform
FHA	Functional Hazard Assessment
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes, Effects, and Criticality Analysis
FMET	Failure Modes and Effects Testing
FOD	Foreign Object Damage
FOR	Field of Regard
FOV	Field of View
FPS	Fire Protection System
FQT	Formal Qualification Test
FRIES	Fast Rope Insertion and Extraction Systems
FSDT	Full-Scale Durability Test
FTA	Fault Tree Analysis
FTS	Flight Termination System
g-LOC	G-Induced Loss of Consciousness
GOTS	Government Off-the-Shelf
GVE	Good Visual Environment
GVT	Ground Vibration Test
HAT	Height Above Target
HCF	High Cycle Fatigue
HDBK	Handbook
HEDAD-O	Human Engineering Design Approach-Operator
HERF	Hazards of Electromagnetic Radiation to Fuel
HERO	Hazards of Electromagnetic Radiation to Ordinance
HERP	Hazards of Electromagnetic Radiation to Personnel
HESS	Hardware Element Supporting Safety-Critical Function(s)
HF	High Frequency
HFDL	High Frequency Data Link
HITL	Hardware-in-the-Loop
HMI	Hazardously Misleading Information
HPT	High Pressure Turbine
HQR	Handling Qualities Rating
HUD	Head-Up Display
HWCI	Hardware Configuration Item
I/O	Input/Output
IAW	In Accordance With
IBIT	Initiated Built-In-Test
ICAO	International Civil Aviation Organization
ICD	Interface Control Document

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IDS	Independent Disconnect System
IEEE	Institute of Electrical and Electronics Engineers
IGE	In Ground Effect
IMC	Instrument Meteorological Conditions
IMV	Instrumented Measurement Vehicle
JSSG	Joint Service Specification Guide
JSSSEH	Joint Software Systems Safety Engineering Handbook
KEAS	Knots Equivalent Airspeed
Kf	Fatigue Stress Concentration Factor
KGS	Knots Ground Speed
Kt	Stress Concentration Factor
LCF	Low Cycle Fatigue
LCO	Limit Cycle Oscillations
LEP	Laser Eye Protection
LF	Load Factor
LIAeq	A-weighted sound pressure level of the impulse(s) averaged over a specific time period
LLP	Life-Limited Parts
LOS	Line of Sight
LPV/LP	Localizer Performance with Vertical Guidance/Localizer Performance
LRU	Line Replaceable Unit
LSRA	Laser Safety Review Authority
MCP	Multicore Processor
MDRC	Multi-Axis Dynamic Response Criteria
MECSIP	Mechanical Equipment and Subsystems Integrity Program
MIL	Military
MILACAS-FR	Military Airborne Collision Avoidance System-Formation Rendezvous
MILACAS-XR	Military Airborne Collision Avoidance System-Extended Range
MIL-STD	Military Standard
MMPDS	Metallic Materials Properties Development and Standardization
MoC	Method of Compliance
Mode S	Mode Select
MP	Mission Profile
MRT	Modified Rhyme Test
MSL	Mean Sea Level
MTBF	Mean Time Between Failure
MTE	Mission Task Elements
NASA	National Aeronautics Space Administration
NATOPS	Naval Aviation Training and Operating Procedures Standardization
NAVAIR	Naval Air Systems Command
NDI	Non-Destructive Inspection

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NDIP	Non-Destructive Inspection Plan
Nr	Rotor Speed
NRI	NVIS Radiance Intensity
NVD	Night Vision Device
NVIS	Night Vision Imaging System
Nz	Normal Force in Z (down) Direction
OEI	One Engine Inoperative
OFP	Operational Flight Program
OML	Outer Mold Line
OMS	Operational Mode Summary
OSS&E	Operational Safety, Suitability, and Effectiveness
PA	Public Address
PBG	Pressure Breathing for G
PCP	Propeller Critical Part
PDL	Pilot Director Light
PFD	Primary Flight Display
PFIBIT	Pre-Flight Initiated Built-In-Test
PFR	Primary Flight Reference
Ph	Probability of Hit
Ph/EB	Probability of Hit due to an Early Burst
PICFO	Power-on/off, Initialization, Checkout, and Flight Operations
PIO	Pilot-In-The-Loop Oscillation
PITL	Pilot-In-The-Loop
Pk	Probability of Kill
PLOA	Probability of Loss of Aircraft
PLOC	Probability of Loss of Control
POD/CL	Probability of Detection/Confidence Level
PSIP	Propulsion System Integrity Program
PTO	Power Take-Off
PVI	Pilot Vehicle Interface
q	Notch Sensitivity Factor
QPL	Qualified Products List
RAP	Resonance Assessment Profile
RBD	Reliability Block Diagram
RCM	Reliability-Centered Maintenance
RF	Radio Frequency
RNAV	Area Navigation
RNP	Required Navigation Performance
RORH	Regions of Recoverable Handling
ROSH	Regions of Satisfactory Handling
ROTH	Regions of Tolerable Handling
RPM	Revolutions per Minute
RSIP	Rotorcraft Structural Integrity Program
RTCA	Radio Technical Committee for Aeronautics

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RTO	Refused Takeoff
RVSM	Reduced Vertical Separation Minimum
SAE	SAE International
SATCOM	Satellite Communication
SATVOICE	Satellite Voice
SCF	Safety-Critical Function
SCFTA	Safety-Critical Function Thread Analysis
SCI	Safety-Critical Item
SDP	Software Development Plan
SEA	Safe Escape Analysis
SEE	Single Event Effect
SEF	Stability Enhancement Function
SENSS	Software Element Not Supporting Safety-Critical Function(s)
SESS	Software Element Supporting Safety-Critical Function(s)
SFE	Service Flight Envelope
SHA	System Hazard Analysis
SIL	System Integration Laboratory
SME	Subject Matter Expert
SMI	Structural Mode Interaction
SMS	Stores Management System
SOF	Safety-of-Flight
SPA	System Processing Architecture
SRVL	Shipboard Rolling Vertical Landing
SSA	System Safety Assessment
SSE	Safety Supporting Element
SSHA	Subsystem Hazard Analysis
SSHE	Safety Supporting Hardware Element
SSPP	System Safety Program Plan
SSSE	Safety Supporting Software Element
STD	Standard
STIPA	Speech Transmission Index for Public Address
STO	Short Takeoff
STOVL	Short Takeoff Vertical Landing
SVT	Software Verification Testing
SW	Software
TAC	Translational Acceleration Command
TAWS	Terrain Awareness and Warning System
TCAS	Traffic Collision Avoidance System
TEMP	Test and Evaluation Master Plan
TLV	Threshold Limit Values
TMDE	Test Measurement and Diagnostics Equipment
TO	Technical Order
TRC	Translational Rate Command
TSO	Technical Standard Order

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UAS	Unmanned Aircraft System
UAV	Unmanned Air Vehicle
USAF	United States Air Force
USG	United States Government
USN	United States Navy
V&V	Verification and Validation
VCF	Vehicle Control Function
VDD	Version Description Document
VDL	VHF Data Link
VHF	Very High Frequency
VL	Vertical Landing
Vmax	Maximum Velocity
VM	Virtual Machine
VMC	Visual Meteorological Conditions
VNAV	Vertical Navigation
Vs	Velocity for Stall
V/STOL	Vertical/Short Takeoff and Landing
VTO	Vertical Takeoff
VTOL	Vertical Takeoff and Landing
VV&A	Verification, Validation, and Accreditation
WCA	Warning, Caution, and Advisory
WCET	Worst-Case Execution Time
WILI	Weapon Inhibit, Limit, and Interrupt
WOD	Wind Over Deck

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4. SYSTEMS ENGINEERING

Section 4 defines a methodical and disciplined approach that leverages processes to support a consistent and repeatable means to deliver a capability into an air system and sustain it in service. This approach plays a critical role in achieving and maintaining an air system's AW by considering all system interactions and potential safety risks. Systems engineering, CM, and system safety practices are applied throughout an air system's life cycle, from initial design and development through maintenance and modifications, to ensure continued AW and continued operational safety.

Execution of the systems engineering, CM, and system safety processes generates products unique to each air system and its modifications. These products, including architecture, requirements, configuration baselines, verification activities, risks, and problem reporting and tracking, define the operational usage, configuration, and operating environment for an air system and form the basis of an AW assessment and subsequent AW approval. Incorrect or insufficient systems engineering or CM execution can result in propagation of errors and the introduction of hazards to any attribute of the air system and its modification.

This section establishes AW criteria applicable to all air systems and their modifications. Criteria herein include establishing the design boundaries for the intended operational usage, configuration, and operating environment and addressing the total air system integration to safeguard against gaps in AW assessment coverage.

The AW assessment requires selection of the correct CIs as part of establishing the allocated baseline to ensure the intended usage of the air system can be accomplished. Consult SAE EIA-649-1, Configuration Item Selection section for guidance on CI selection. For the purposes of Section 4, CI encompasses all types of CIs, including CSCIs.

4.1 Operational usage.

Systems engineering translates operational needs into materiel solutions. The component-approved CONOPS/OMS/MP describes the operational tasks, events, durations, frequencies, and environments in which the materiel solution is expected to perform each mission and each phase of the mission. Criteria herein refer to the CONOPS/OMS/MP as the document and operational usage as content within that document. The systems engineering process translates the CONOPS/OMS/MP into technical requirements and decomposes these top-level requirements to produce a complete set of system requirements, functional and physical. As part of requirements verification, 4.1.1 addresses identification of planned technical reviews and audits; 4.2.1 addresses closure of these events.

This section addresses design and SCF establishment with traceability throughout the system architecture hierarchy and alignment to the operational usage and limits. Assessment of SCFs depends on verification of the functions both under nominal and off-nominal conditions, with injection of failure conditions to assess loss, degraded, and erroneous operation, as well as during transition to defined accommodation mechanism(s) and any subsequent return to the primary modes or system. Criterion 4.1.1 addresses the complete verification of functions and 4.1.3 specifically addresses failure mode testing and fault accommodation, integration issues, including combinations of failures and failures not directly addressed by another criterion in this handbook.

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The operational usage for UAS necessitates unique systems engineering considerations. The planned capability, level of supervision, and residual role of the operator for each autonomy function, the broader C2 and datalink requirements, changes to the criticality of traditional systems (such as existing C2 links), and the intended use of the FTS all contribute to the development of the technical requirements and identification of SCFs for the air system. This includes ensuring functional and physical separation of special military modes from SCFs. Special military modes and their design safety precepts are outlined in the Unmanned Systems Safety Guide for DoD Acquisition, DSP-4, DSP-6, DSP-7, among others. Further discussion on UAS AW requirements is contained in NATO Standard AEP-4671, Special Military Modes of Operation Resulting in Reduction of Level of Safety section. Computer systems and software contribution to air system design are evaluated under Section 15.

Many MoCs provide a list of items to support a show of compliance. Tailor the MoC to align with the specific assessment being conducted and specific program requirements.

4.1.1 Operational usage.

Standard: Design criteria derive from the intended operational usage and comprehensively address AW for the mission usage, full permissible flight envelope, duty cycle, interfaces, induced and natural environment, inspection capability, and maintenance philosophy.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) The CONOPS/OMS/MP establishes the intended operational usage. This includes the intended mission usage, flight envelope, maintenance philosophy, and the tasks, events, durations, frequencies, and environments associated with air system operations.

(2) For UAS, the intended operational usage includes expectations for levels of air system autonomy, C2 requirements, and intended usage of the FTS. It also addresses any special military modes requiring functional and physical separation from SCFs.

(3) Systems engineering planning documentation addresses definition of the functional and allocated baselines for translation into the system architecture hierarchy. It also describes the planned activities, including technical reviews and audits, to show verification to the specification requirements and validation of the operational requirements.

(4) For air systems reliant upon DoW Component AW authority-accepted design approvals (e.g., FAA type design approval for CDA), systems engineering planning and specification documentation accounts for variations and deviations in operational usage. At the discretion of the issuing authority, the variations and deviations do not invalidate the underlying design approval.

(5) The specification requirements translate operational needs into a set of a technical requirements and comprehensively define safety-critical functionality at the system level. Allocation of specification requirements associated with SCFs and sub-functions occurs throughout the system architecture hierarchy and addresses the intended function of the physical architecture, including hardware, software, and crewmember actions. This requirements

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allocation and traceability shows comprehensive materiel solution alignment to the entirety of the intended operational usage.

(6) The TEMP, flight test plans, and flight test procedures include verification activities that align with the intended operational usage and confirm the air system functions as intended. The scope of testing addresses the requirements that need flight test to complete verification. The TEMP identifies the resources to execute a production-relevant test to support validation of operational usage of the air system.

(7) Systems engineering planning documentation describes a closed-loop process for addressing deficiency and problem reports identified during verification activities and ensures traceability for deficiency and problem reports that impact SCFs. Evidence for process adherence demonstrates clear identification, analysis, and resolution of issues with record of planned or completed changes to design, specification requirements, verification activities, and mitigations. Deficiency and problem reports associated with SCFs are corrected or mitigated.

(8) Verification of SCFs indicates readiness to enter flight test. Evidence from completed verification activities supports execution to intended operational usage and positioning for AW approval to begin flight test.

(9) Validation or accreditation and acceptance of models and simulations used to verify SCFs ensures the intended usage aligns with the air system's operational usage. This includes verification of analysis tools (e.g., FEA) that are inputs to the model or simulation. When documenting this validation or accreditation, include the intended use, modeling and simulation requirements and acceptability criteria, assumptions, capabilities, limitations, risks, impacts, and V&V or accreditation agent IAW MIL-STD-3022, V&V Report Template appendix or Accreditation Report Template appendix.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(10) Evidence of a closed-loop process for addressing deficiency and problem reports associated with SCFs shows completion of changes to design, specification requirements, verification activities, and mitigations documented in USG verified technical data. Deficiencies associated with SCFs are corrected or mitigated.

(11) Verification of SCFs, using a production-representative article, shows air system AW for the intended operational usage. Evidence from completed verification activities supports validation of production-representative articles to the intended operational usage and entry into operations.

4.1.2 System architecture and interface management.

Standard: The system architecture hierarchy contains the relationship between the end item product and its components, including identification of interfaces and interactions required for safety-critical functionality. Internal and external interface definition contains the functional and physical characteristics of each interface with traceability to all levels identified in the system architecture hierarchy and aligns with the intended operational usage and allocated design requirements.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) The system architecture provides a hierarchical view of the system design with identification of CIs and interfaces and interactions between components of the design. The hierarchy addresses different levels of assembly with consistent terminology throughout design documentation.

(2) The system architecture identifies SCFs on the hierarchical view. Accompanying documentation to the system architecture captures the flow of each SCF through the physical architecture, including redundancy, and includes comprehensive horizontal and vertical traceability that defines all interfaces and interactions, internal and external, that support safety-critical functionality.

(3) Definition of interfaces internal to the air system captures any interactions between and within CIs associated with or connected to SCF(s). This includes any interaction required to prevent, mitigate, or provide notification to crewmembers of degradation or failure of SCFs. The documented interfaces trace to the system architecture hierarchy and the allocated design requirements.

(4) Definition of interfaces between the air system and external systems and equipment addresses connections to SCFs, including support activities necessary to execute these functions for the intended operational usage. This includes interfaces related to servicing, data transfer, stores integration, and AR that support execution of SCFs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(1) Conduct analysis on configuration and requirement changes to determine impacts on interface definitions and verify that interface documentation is updated to preserve traceability, safety-critical functionality, and alignment with intended operational usage.

4.1.3 Air system fault accommodation.

Standard: An air system-level assessment, with integration of hardware and software, confirms that the fault accommodation features of the system provide a reaction, means for return to intended function, and vehicle recovery following a single failure or combination of failures more likely than an unacceptable rate as determined for the air system or program (e.g., improbable or lower quantitative probabilities in MIL-STD-882, Quantitative Probability Example). Means for return to intended function and vehicle recovery may include crewmember actions with WCAs and procedures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) The system safety planning documentation includes the air system or program definition of the unacceptable rate for the air system or program (e.g., improbable or lower quantitative probabilities in MIL-STD-882, Quantitative Probability Example).

(2) The FMET plan addresses verification of SCFs in addition to assessment of loss, degraded, and erroneous operation. The FMET addresses intended functionality of mitigations such as redundancy, monitoring, WCAs, and diagnostics. The FMET plan identifies, with rationale, the levels of the design intended for testing.

(3) Review of hazard analyses for failure conditions with a catastrophic and critical severity, including review of FTAs, confirms no single failures or combination of failures more likely than the unacceptable rate. Analysis and frequency determination addresses the intended operational usage. Review design changes and mitigations, including mitigations in test planning documentation, that reduce the severity or frequency of any failure conditions to meet this standard.

(4) Hazard tracking system review confirms verification of mitigation effectiveness to support execution to intended operational usage and positioning to begin flight test.

Analysis.

(1) Conduct failure mode and hazard analyses for the air system to determine the probability of failure occurrence. Analysis methodologies include the FMEA, FMECA, SSHA, SSA, and FTA; analyses may be documented in a comprehensive Safety Assessment Report.

(2) The FMEA, FMECA, and other analyses address loss, degradation, and erroneous operation of SCFs and are inputs to FMET test plans and procedures.

(3) Unsatisfactory results and unintended behavior observed during FMET, captured in problem or deficiency reports, undergo analysis and evaluation to identify corrective actions. Corrective actions may include mitigation measures, substantiated by the analysis, to support entry into flight test.

(4) Conduct analysis of FMET results to assess the integration of hardware and software for verification of SCFs as well as the ability to react, means for return to intended function, and recover from a single failure or combination of failures. Means for return to intended function and vehicle recovery may include crewmember actions with WCAs and procedures.

Test.

(1) Lab test: The FMET comprehensively tests SCFs by injecting failure conditions into the system at appropriate levels of the design to confirm the ability to react, means for return to intended function, and recover from a single failure or combination of failures. Means for return to intended function and vehicle recovery may include crewmember actions with WCAs and procedures.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Examination.

(5) Review of hazard analyses for failure conditions, updated based on flight test results, with a catastrophic and critical severity, including review of FTAs, confirms no single failures or combination of failures more likely than the unacceptable rate. Review design changes and mitigations, with evidence in requirements documentation and USG verified technical data, that reduce the severity or frequency of any failure conditions to meet this standard.

(6) Hazard tracking system review confirms completion of mitigation effectiveness verification with evidence in requirements documentation and USG verified technical data.

Analysis.

(5) Conduct analysis of test results to assess the integration of hardware and software for verification of SCFs as well as the ability to react, means for return to intended function, and recover from a single failure or combination of failures. Means for return to intended function and vehicle recovery may include crewmember actions with WCAs and procedures.

(6) Unsatisfactory results and unintended behavior observed during testing, captured in problem or deficiency reports, undergo analysis and evaluation to identify corrective actions. Evidence of corrective action implementation shows completion of changes to design, specification requirements, verification activities, and mitigations documented in USG verified technical data.

(7) A SCF V&V coverage analysis confirms that the collective set of data provides for end-to-end verification for the function thread under nominal and off-nominal conditions for the intended air system operational usage. Analysis identifies gaps in V&V activities (e.g., incomplete end-to-end test coverage, failures in testing) and verifies adequacy of mitigations to ensure integrity of SCF operation.

Test.

(2) Flight or Ground test: For safety-critical functionality that was not evaluated during lab testing, conduct test to collect verification data on safety-critical functionality and effectiveness of mitigations, to include redundancy, monitoring, diagnostics, and notification to crewmembers. Problem or deficiency reports capture any unintended behaviors observed during verification and identify mitigations to address these behaviors.

(3) Flight, Ground, or Lab test: Conduct regression testing to verify completion of mitigations or configuration changes implemented to address deficiency and problem reports identified in FMET and other test activities.

4.1.4 Unmanned aircraft system flight termination system.

Standard: The UAS FTS functions such that air system safety-critical functionality is not adversely impacted by FTS normal operations, functional failure, or physical failure, including installation. This includes assuring unintentional activation occurs with likelihood no greater than 10^{-7} per flight hour.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Allocation of specification requirements associated with the FTS occurs throughout the system architecture hierarchy and addresses the intended functions of the physical architecture, including hardware, software, and crewmember actions. This requirements allocation defines the methods of FTS activation (e.g., fuel starvation, zero-lift and zero-thrust, and explosive charge) and addresses requirements to prevent unintentional activation.

(2) Review of design documentation and specification verification confirms a single crewmember action does not result in activation of the FTS.

Analysis.

(1) Conduct FMECA, FMEA, or other analyses to identify failure conditions associated with physical or functional failures of the FTS and its components, including those that could lead to unintentional activation of the FTS.

(2) Conduct FTA to verify probability for an unintentional activation of the FTS occurs with a likelihood no greater than 10^{-7} per flight hour and that no single point failures in the FTS and connected systems cause unintentional activation of flight termination.

(3) Conduct analysis of SIL test results, including any unintended behavior, and failure mode analyses to confirm the FTS functions as intended and unintentional activation occurs with likelihood no greater than 10^{-7} per flight hour.

(4) Conduct analysis of the FTS, its components, and SIL test results to confirm that air system safety-critical functionality is not adversely impacted by FTS normal operations, functional failure, or physical failure, including installation.

4.2 Configuration management.

The AW assessment centers around a specific air system configuration. Configuration management aids in establishing crucial air system characteristics. Configuration change management processes are adhered to for controlling, documenting, and making changes to the baseline through permanent or temporary modifications. Permanent modifications alter the air system configuration, while temporary modifications, including roll-on/roll-off equipment and test equipment, cause short-term changes to the air system configuration. The CSA offers an accurate and up-to-date information source regarding a product and its product configuration info throughout the product's life cycle. The criteria herein refer to CM terminology defined in GEIA-HB-649, SAE EIA-649, and SAE EIA-649-1.

4.2.1 Configuration baseline.

Standard: The air system configuration is defined, approved, and maintained. Configuration management processes are in place to maintain and control the air system configuration baselines to preclude any unauthorized changes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Functional and allocated baseline establishment identifies verification methodologies for all specification requirements. Verification methodologies and traceability matrices confirm verifiability of all allocated design criteria and ensure required levels of AW for the defined operational flight envelope, environment, usage, and life.

(2) Initial product baseline establishment defines the authorized physical configuration of the air system. It consists of applicable technical data necessary to define the configuration and the design features of the air system shown to comply with the AW criteria applicable to the air system. Applicable technical data includes models, engineering design data, associated lists, specifications, standards, performance requirements, quality assurance provisions, software documentation, and packaging details.

(3) For air systems reliant upon DoW Component AW authority-accepted design approvals (e.g., FAA type design approval for CDA), systems engineering planning and specification documentation accounts for variations and deviations in configuration. At the discretion of the issuing authority, the variations and deviations do not invalidate the underlying design approval.

(4) The completion and status of action items from technical reviews and audits demonstrates the execution of events to support positioning for AW approval to begin flight test.

(5) Tools used for the creation, management, and maintenance of requirements and verification artifacts preclude unauthorized changes to the configuration baselines and include documentation defining the environment and intended use of the tool to ensure alignment with the overall needs of the air system and program.

Analysis.

(1) As part of establishing the allocated baseline, the CIs are selected based on analysis of the air system to ensure the intended usage of the air system can be accomplished. Selection of CIs is conducted based on guidance contained in SAE EIA-649-1, Configuration Item Selection section.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(2) Final product baseline establishment defines the authorized physical configuration of the air system and the basis for review and evaluation of the detailed design for production, fielding and deployment, and operations and support of the equipment or system. It consists of applicable technical data necessary to define the configuration and the design features of the air system shown to comply with the AW criteria applicable to the air system. Applicable technical data includes models, engineering design data, associated lists, specifications, standards, performance requirements, quality assurance provisions, software documentation and packaging details.

(3) Product baseline management and maintenance includes documented process specifications necessary for manufacturing, operating, maintaining, or calibrating items within the design.

(4) The completion and status of action items from technical reviews and audits demonstrates the closure of events and validates that specification and operational requirements have been fulfilled.

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4.2.2 Configuration status accounting.

Standard: The CSA system captures, stores, tracks, and maintains all product configuration information.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) The CSA records and reports accurately reflect the current physical configuration of the air system.

(2) The CSA tool used for capturing, storing, tracking, and maintaining production configuration information preclude unauthorized changes and include documentation defining the environment and intended use of the tool to ensure alignment with the overall needs of the air system and program.

4.2.3 Temporary modification baseline.

Standard: The temporary and test configurations are documented, approved, and managed to preclude any unauthorized changes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Configuration documentation defines the authorized physical configuration of the air system with the temporary modifications completed. It consists of applicable technical data necessary to define the configuration and the design features of the air system shown to comply with the AW criteria applicable to the air system. Applicable technical data includes models, engineering design data, associated lists, and software documentation.

4.3 Operating environment.

Verification confirms air system operation in its intended natural and induced environments. Determine and tailor environmental requirements and the necessary verification for the specific configuration and intended usage, including that documented in the CONOPS/OMS/MP. Environmental assessments occur at the air system level, with consideration for items like water intrusion, solar radiation, and temperatures, including thermal loading from equipment operation. Assessments also occur for the components within the integrated air system: verifying safety-critical components operate throughout environmental conditions and verifying that neither normal operations nor physical failure, including installation, of non-safety-critical components adversely impacts safety-critical functionality or the safety of the crewmembers and passengers. Criteria herein do not address the operating environment assessment of the engine; Section 7 addresses the engine.

Tailor the MoC to align with the specific assessment being conducted and requirements captured in the air system's environment criteria documentation.

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4.3.1 Zone establishment for air system environments.

Standard: Zones established for the air system through analytical and measured characterization ensure total coverage for the intended environment. Zone establishment includes divisions of the air vehicle, internal and external, as well as the integrated air vehicle as a whole. Air system zones divide the system such that all items within a zone are expected to encounter the same natural and induced environment for the service life and CONOPS/OMS/MP.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Environment criteria documentation contains information on the air system, including all air vehicle components and control station components, with the intended natural and induced environmental conditions to understand the intended environment of the air system. Definition of the intended natural and induced environmental conditions addresses flight envelope and design usage for the air system in addition to the climatic information. This definition derives from sources including the CONOPS/OMS/MP, MIL-HDBK-310, MIL-STD-810, and JSSG-2001, Environment paragraph.

(2) Air system specification documentation includes complete operating environment requirements for the flight envelope, design usage, CONOPS/OMS/MP, and intended climatic information that align to the environment criteria documentation. Design documentation shows requirement allocation into the design of the air system with zone identification and addresses environmental concerns at the integrated air vehicle level (e.g., water intrusion, solar radiation). The air system specification derives from sources including MIL-HDBK-310, MIL-STD-810, and JSSG-2001, Environment paragraph.

Analysis.

(1) Conduct analysis to confirm environmental design requirements are consistent with MIL-HDBK-310 climatic information given planned areas of geographic deployment, handling, and logistics requirements as defined in the CONOPS/OMS/MP. The design requirements encompass the 1% frequencies listed in MIL-HDBK-310 for each environment except cold temperature (20% frequency) and rainfall (0.5% frequency).

(2) Conduct analysis of the allocated environmental requirements for the defined air system zones to confirm that all items within the same zone encounter commensurate natural and induced environments for the service life. Zone definition includes impacts from open doors on the ground or in flight, which may result in environmental impacts, such as intrusion of water, sand, and dust.

(3) Conduct analysis of air system design to characterize induced environments that are a combination of natural environmental forcing functions and the physical and chemical characteristics of the materiel itself, such as temperature, pressure, and solar radiation.

(4) Conduct analysis based on transportation and storage requirements to identify the environmental requirements for the defined air system zones, including the integrated air vehicle.

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(5) For UAS control stations, conduct analysis based on transportation and mobility requirements to identify the appropriate vibration spectrum using guidance as specified in NATO Standard AEP-4671, Other Aspects of Transportation and Factors to Consider appendix.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight or Ground test: Conduct test to provide data to define, substantiate, and validate air vehicle zone identification and characterization.

4.3.2 Air system environmental requirements.

Standard: The integrated air system operates as intended in the natural and induced environments it will encounter for its service life. TABLE I lists the environments assessed in this criterion.

TABLE I. *Natural and induced environments for the air system.*

Environment	Natural, Induced, or Both	Notes
Temperature	Natural and Induced	Includes the ambient temperatures for all intended ground and flight envelope altitudes, thermal loading from subsystems and components of the air system, and cooling considerations.
Temperature shock	Induced	
Pressure	Natural and Induced	Static and dynamic pressures for all intended ground and flight envelope altitudes.
Rapid decompression	Induced	
Solar radiation (sunshine)	Natural and Induced	Includes the combination of natural environment forcing functions and the physical and chemical characteristics of the materiel itself.
Wind speed and wind shear	Natural	
Humidity	Natural and Induced	
Rainfall	Natural	Includes identification of any areas for water intrusion, such as open doors and seepage through the air vehicle outer skin.
Blowing snow and snow load	Natural	
Icing conditions and freeze-thaw cycles	Natural	

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TABLE I. *Natural and induced environments for the air system – Continued.*

Environment	Natural, Induced, or Both	Notes
Sand and dust	Natural	
Salt fog and corrosive environments	Natural and Induced	Includes gunfire gases, missile gases, sulfur and nitrogen oxides.
Acceleration (Environment characterization)	Induced	Addresses the steady-state inertia loads induced by platform acceleration, deceleration, and maneuver in the service environment, including crash landing loads (static acceleration test).
Vibration (Environment characterization)	Induced	Includes mechanical and gunfire vibration.
Sonic fatigue (Environment characterization)	Induced	Includes all sources of acoustic loads.
Shock (Environment characterization)	Induced	Addresses the materiel's ability to physically and functionally withstand the shocks encountered in handling, transportation, and service environments including crash situations (transient shock test).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) The TEMP and the Environmental TEMP, as specified in MIL-STD-810, TEMP paragraph, contain the planned verification activities and testing environments to comprehensively address required levels of AW for the defined environment. Verification activities include any subset of service life testing to support entry into flight test and the tailored MIL-STD-810 requirements.

(2) Systems engineering planning and specification documentation account for variations and deviations in operating environment for air systems reliant upon DoW Component AW authority-accepted design approvals (e.g., FAA type design approval for CDA). At the discretion of the issuing authority, the variations and deviations do not invalidate the underlying design approval.

Analysis.

(1) Analysis of available environment verification data for the air system and its components, as collected in this section, confirms air system operation in all intended natural and induced environments. While the analysis addresses any available environment verification data, the

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minimum data set addresses twice the intended duration of flight test and the environment to be encountered during flight test. The analysis addresses any gaps between actual test data and environmental requirements and any configuration changes made after completion of verification activities, providing an equivalent level of safety assessment and showing no increased risk to operations. Examples of environment verification data: test data, drawings, digital twins of the air system, specification sheets, and ICDs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(2) Analysis of environment verification data for the air system and its components, as collected in this section, confirms air system operation in all intended natural and induced environments. The analysis addresses any gaps between actual test data and environmental requirements and any configuration changes made after completion of verification activities, providing an equivalent level of safety assessment and showing no increased risk to operations. Examples of environment verification data: test data, drawings, digital twins of the air system, specification sheets, and ICDs.

Test.

(1) Flight or Ground test: Air system testing verifies operations for natural and induced environmental conditions as defined in environment criteria documentation, the TEMP, and the Environmental TEMP. In addition to the test data showing operation in the natural and induced environment, data collected validates or substantiates modeling and analysis, including life cycle analysis. Testing may be conducted in full-scale climatic laboratories and in naturally occurring environments.

4.3.3 Safety-critical component environment qualification.

Standard: Safety-critical air system components of the air system perform their intended functions in the natural and induced environment for the defined service life. This criterion excludes the engine. The environments assessed in this criterion are listed in TABLE II.

TABLE II. *Natural and induced environments for safety-critical components.*

Environment	**Natural, Induced, or Both**	**Notes**
Temperature	Natural and Induced	Includes the ambient temperatures for all intended ground and flight envelope altitudes, thermal loading from subsystems and components of the air system, and cooling considerations.
Temperature shock	Induced	
Pressure	Natural and Induced	Pressures for all intended ground and flight envelope altitudes. Includes considerations for air vehicle cabin pressurization.

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TABLE II. *Natural and induced environments for safety-critical components – Continued.*

Environment	**Natural, Induced, or Both**	**Notes**
Rapid decompression	Induced	
Solar radiation (sunshine)	Natural and Induced	Includes the combination of natural environment forcing functions, including wind speed, and the physical and chemical characteristics of the materiel itself.
Humidity	Natural	
Rain	Natural	Includes components in areas that would be exposed to water intrusion such as those near open doors. Addresses wind velocity.
Icing or Freezing rain	Natural	Addresses wind velocity.
Immersion	Induced	
Contamination by fluids	Induced	
Fungus	Natural Induced	
Sand and dust	Natural	Includes wind speeds.
Salt fog and corrosive environments	Natural and Induced	Includes gunfire gases, missile gases, sulfur and nitrogen oxides.
Acceleration	Induced	Addresses the steady-state inertia loads induced by platform acceleration, deceleration, and maneuver in the service environment, including crash landing loads (static acceleration test). Obtain acceleration values for testing from the air vehicle structural loads analyses.
Vibration	Induced	Includes mechanical, acoustic, and gunfire vibration and addresses the effect of dynamic pressures. Acquire measured data from the air vehicle for these vibration parameters.
Shock	Induced	Addresses the materiel's ability to physically and functionally withstand the shocks encountered in handling, transportation, and service environments including crash situations (transient shock test). Tailor the shock values to align with air vehicle representative field measurements of shock.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of post-test data for component qualification testing to identify and assess any failures and design changes (proposed or implemented) to ensure final, tested configuration is understood.

(2) Conduct analysis for components qualified by TSO authorization to address differences with air system intended usage, any changes from the original certification, and deviations documented in the TSO authorization to ensure an equivalent level of safety and that the component performs its intended functions. Analysis identifies environment verification, including those altered or impacted by installation or integration (e.g., vibration and temperature), to be addressed through other MoCs of this criterion. Note: a TSO authorization is not an approval to install and use the component in the air system.

(3) Analysis for components that are qualified by similarity address differences in installation, integration, and intended usage from the originally qualified component to ensure an equivalent level of safety and no increased risk to operations through use of qualification by similarity.

(4) Analysis of available environment verification data for the components confirms operation in all intended natural and induced environments. While the analysis addresses any available environment verification data, the minimum data set addresses twice the intended duration of flight test and the environment to be encountered during flight test. The analysis addresses any gaps between actual test data and environmental requirements and any configuration changes made after completion of verification activities, providing an equivalent level of safety assessment and that the component performs its intended functions. Verification data includes test data, specification sheets, and ICDs that show intended function for all natural and induced environments.

Test.

(1) Lab test: Conduct component testing IAW the tailored MIL-STD-810 for a subset of the life cycle of the component and in modes representative of flight (e.g., powered-on testing). At a minimum, the testing encompasses twice the intended duration of flight test and the environment to be encountered during flight test. MIL-STD-810 contains tailoring instructions with general guidelines in Part One and specific guidelines in each method of Part Two.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(5) Analysis of environment verification data for the components confirms operation in all intended natural and induced environments. The analysis addresses any gaps between actual test data and environmental requirements and any configuration changes made after completion of verification activities, providing an equivalent level of safety assessment and that the component performs its intended functions. Verification data includes test data, specification sheets, and ICDs that show intended function for all natural and induced environments.

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Test.

(2) Lab test: Conduct component testing IAW the tailored MIL-STD-810 for the life cycle of the component, including shipping, transportation, storage, logistics supply, mission use, and sortie use. Conduct testing in modes representative of flight (e.g., powered-on testing). Tailoring instructions are contained within MIL-STD-810 with general guidelines in Part One and specific guidelines in each method of Part Two.

4.3.4 Non-safety-critical component environment qualification.

Standard: Components of the air system that do not support a SCF are qualified to the natural and induced environment such that neither normal operations nor physical failure, including installation, of the component adversely impacts safety-critical functionality or the safety of the crewmembers and passengers. This criterion excludes the engine. The minimum environments assessed in this criterion are listed in TABLE III; additional environments may be appropriate for the air system and program being assessed.

TABLE III. *Natural and induced environments for non-safety-critical components.*

Environment	Natural, Induced, or Both	Notes
Temperature	Natural and Induced	Includes the ambient temperatures for all intended ground and flight envelope altitudes, thermal loading from subsystems and components of the air system, and cooling considerations.
Pressure	Natural and Induced	Pressures for all intended ground and flight envelope altitudes. Includes considerations for air vehicle cabin pressurization.
Rapid decompression	Induced	Rapid decompression testing conducted for components in occupied compartments.
Acceleration	Induced	Component retained as installed to prevent injury or damage caused by loose items with exposure to aerodynamic loads, including crash landing loads.
Vibration	Induced	Component retained as installed to prevent injury or damage caused by loose items with exposure to vibration. Includes mechanical, acoustic, and gunfire vibration and addresses the effect of dynamic pressures.
Shock	Induced	Component retained as installed to prevent injury or damage caused by loose items with exposure to the shocks encountered, including catapult launch, arrested landing, and crash landings.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of non-safety-critical components to confirm that neither normal operations nor physical failure, including installation, of the component adversely impacts safety-critical functionality or the safety of crewmembers and passengers.

(2) Conduct analysis of post-test data for component qualification testing to identify and assess any failures and design changes (proposed or implemented) to ensure final, tested configuration is understood.

(3) Conduct analysis for components qualified by TSO authorization to address differences with air system intended usage, any changes from the original certification, and deviations documented in the TSO authorization to ensure an equivalent level of safety. Analysis identifies environment verification, including those altered or impacted by installation or integration (e.g., vibration and temperature), to be addressed through other MoCs of this criterion. Note: a TSO authorization is not an approval to install and use the component in the air system.

(4) Analysis for components that are qualified by similarity address differences in installation, integration, and intended usage from the originally qualified component to ensure an equivalent level of safety.

(5) Analysis of available environment verification data for the components confirms neither normal operations nor physical, including installation, failure of the component introduces or propagates a hazard in all intended natural and induced environments. While the analysis addresses any available environment verification data, the minimum data set addresses twice the intended duration of flight test and the environment to be encountered during flight test. The analysis addresses any gaps between actual test data and environmental requirements and any configuration changes made after completion of verification activities, providing an equivalent level of safety assessment. Verification data includes test data, specification sheets, and ICDs that show operation in the intended natural and induced environments.

Test.

(1) Lab test: Conduct component testing IAW the tailored MIL-STD-810 for a subset of the life cycle of the component and for the minimum environments listed in TABLE III. At a minimum, the testing encompasses the intended duration and environment to be encountered during flight test. Tailoring instructions are contained within MIL-STD-810 with general guidelines in Part One and specific guidelines in each method of Part Two.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(6) Analysis of environment verification data for the components confirms neither normal operations nor physical failure, including installation, of the component introduces or propagates a hazard in all intended natural and induced environments. The analysis addresses any gaps between actual test data and environmental requirements and any configuration changes made

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after completion of verification activities, providing an equivalent level of safety assessment. Verification data includes test data, specification sheets, and ICDs that show operation in the intended natural and induced environments.

Test.

(2) Lab test: Conduct component testing IAW the tailored MIL-STD-810 for the life cycle of the component, including shipping, transportation, storage, logistics supply, mission use, and sortie use and for the minimum environments listed in TABLE III. Tailoring instructions are contained within MIL-STD-810 with general guidelines in Part One and specific guidelines in each method of Part Two.

4.4 System safety.

This section facilitates a review of the air system considering the application of fundamental system safety engineering processes that are directly related to AW and SOF. Criteria within this section represent the minimal set of AW evaluation criteria necessary to establish, verify, and maintain the system safety pedigree of the air system under review.

The AW assessment requires system safety methodology IAW MIL-STD-882, General Requirements section and is bound by MIL-STD-882 definitions for any DoW system. The elements of the system safety process establish the system safety risk baseline of the air system and ensure OSS&E for the intended usage of the air system. The basic building block for this section is the identification of SCFs for the air system.

System safety is an engineering discipline that employs specialized knowledge and skills in applying scientific and engineering principles, criteria, and techniques to identify hazards and then to eliminate those hazards or reduce risks associated with them if they cannot be eliminated. The application of these methodologies and principles within the constraints of operational effectiveness and suitability, time and cost throughout all phases of the system life-cycle, and the products generated through execution of the system safety process are unique to each air system and modification. If system safety is not executed correctly, there is an increased likelihood the air system or modification could introduce a propagation of errors, unrealized hazards, or an increase in the severity or probability of known hazards.

Artifacts applicable for analysis, as identified in SAE ARP4761A and MIL-STD-882 (Notes chapter and the tasks), include a range of documentation and analyses, such as (but not limited to): SSPP, hazard tracking system, preliminary hazard list, SHA, SSHA, Safety Assessment Report, FHA, safety V&V reports, software SHA, zonal safety analysis, Operating and Support Hazard Analysis, FMEA, and FMECA.

Specific remarks for standards are as follows:

a. Safety-critical function identification. Hazards may develop from the failure of a function as well as the failure of related hardware, a software anomaly or failure of the human interfaces to the system. Even if no item has failed mechanically, physically, systemically, or logically, a function can fail resulting in a catastrophic or critical hazard. All SCFs can be either "must-work" or "must-not-work" functions. Preparers should examine functions using SAE ARP4761A, Aircraft Functional Hazard Assessment appendix methodology, and consider the consequence if the function fails to initiate, is degraded, malfunctions, or operates when unintended or out of sequence with other

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functions as well as the interfaces between subsystems and components. The FHA is primarily used to identify and classify the system functions and the safety consequences of a functional failure or malfunction. The FHA may require revision if any anomalies, unexpected behavior, or unpredicted functionality observed during air system verification activities, including testing, result in identification of a previously unidentified SCF. A revision may also be required to ensure all functional failures with catastrophic or critical severity hazards are associated with SCFs, as a part of the overall system safety process.

b. Safety-critical assembly and installation hazard prevention. An installation design for a safety-critical component or item should consider the potential hazards from interactions and interferences with other systems and equipment located in the same zone of the air vehicle. The scope of potential hazards should include those that could occur during installation or removal, operation, and maintenance of equipment within the zone. A guideline for analysis of these hazards is provided in the Zonal Safety Analysis section of the SAE ARP4761A.
c. Fail-safe design concept or single-point of failure. The failure of any single element, component, or connection during any one flight should not result in a catastrophic mishap at a frequency greater than improbable, IAW MIL-STD 882, Probability Levels table definitions. Analysis of the design verifies that the required level of safety is achieved through design alteration, design integrity, redundancy, backup systems or functions, isolation, proven reliability, designed failure effect limits or failure path, factors of safety or failure warning to enable continued safe flight and landing. The FMEA, FMECA, FTA, TOs and engineering data may contribute to finding compliance with this standard.
d. Isolation of hazardous substances, components and operations. Hazards may develop due to proximity of personnel or equipment to energy (e.g., kinetic, potential, heat, electrical, acoustic, chemical) sources or hazardous materials. Isolation of these hazards from personnel and safety-critical equipment is the preferred method of mitigation when these hazards cannot be eliminated. Other corrective measures should be considered when isolation is not possible consistent with the system safety design order of precedence found in MIL-STD 882. For AW consideration, the primary concern is preservation of the system's ability, including required flight crew, to attain, sustain, and complete flight. However, maintenance actions without proper controls for isolation can contribute to system safety issues related to AW. A Zonal Safety Analysis as described in SAE ARP4761A can often be helpful when assessing this criterion.
e. Foreign object damage prevention. Foreign objects are a major cause of air vehicle damage. Damage can result in minor repairs or catastrophic events. Any foreign object that can find its way into an air vehicle can create a FOD related hazard. A FOD tolerant design ensures that the air system and associated support equipment does not allow the introduction of FOD which could adversely affect SOF. This standard will ensure technical drawings and TOs capture FOD tolerant design practices in air vehicles, parts, component assemblies and sub-assemblies. NAS412, Product and Process Design Considerations section, is a reference germane to evaluating this criterion, as is DCMA INST 8210-1, Contractor's Flight and Ground Operations.

4.4.1 Safety-critical function identification.

Standard: All air system functions whose loss, degradation, or erroneous operation can potentially result in a mishap of either catastrophic or critical severity are identified as SCFs.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) The list of air system functions, SCFs list, function descriptions, and associated analyses are documented, maintained, and under configuration control.

Analysis.

(1) A systems engineering functional analysis identifies all air system functions and provides a description of their functionality for the purposes of identifying which of those functions are SCFs.

(2) An air system FHA is performed IAW SAE ARP4761A, Aircraft Functional Hazard Assessment (AFHA) appendix, that identifies functional failure conditions along with the rationale for each severity classification for each air system function. This includes assessing loss of function, degradation of function, malfunction, functioning out of time, and functioning out of sequence. The AFHA severity classification is conducted with or correlated to the categories in the MIL-STD-882, Severity categories table.

(3) The AFHA is performed to identify all SCFs. When utilizing an AFHA for the purposes of SCF identification, the AFHA:

a. Provides a description of each air system-level function.
b. Evaluates functional failure modes for each air system function.
c. Assesses each functional failure condition for its potential, without consideration to probability, to produce a catastrophic or critical severity mishap, or adversely impact the function's ability to provide information or detect and mitigate conditions to prevent the occurrence of a catastrophic or critical severity mishap.
d. Performs a purely functional assessment that does not account for a function's design or implementation.
e. Accounts for the effects of flight phases (e.g., taxi, cruise, and landing), operational conditions (e.g., air vehicle weight and air vehicle altitude), and operating environment for each functional failure condition.
f. Assumes any preconditions under which an air system function is expected to operate exist while the functional failure conditions are assessed (e.g., when assessing functional failure conditions of a fire detection and suppression function, a fire is assumed to exist).

4.4.2 Safety-critical assembly and installation hazard prevention.

Standard: The air system design does not expose safety-critical systems, subsystems and assemblies to errors during installation, assembly, use, or connection.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine design requirements and drawings of safety-critical removable assemblies and components of the air system to confirm the design and incorporation of mitigations prevents damage or loss of a SCF due to errors in installation, assembly, use, or connections.

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(2) Review design requirements, verification artifacts, and drawings of removable components to ensure their access for installation and maintenance prevents loss of function for associated or nearby safety-critical assemblies and components.

(3) Review design documentation to ensure that safety-critical assemblies and components can be assembled or installed without damage or loss of safety-critical functionality.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight, Ground, or Lab test: Conduct test to verify standard maintenance and operations procedures to ensure they have no adverse effect on safety-critical systems.

4.4.3 Fail-safe design concept or single-point of failure.

Standard: The failure of any single element, component, or connection for the system should be assumed and loss of an SCF for the system falls below the Improbable level.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review program documentation to ensure the probability of loss of an SCF for the system falls below the Improbable level for any single design element, component, or connection of a system or subsystem..

Analysis.

(1) Verify completion of risk reduction analyses for the air system to determine the probability of SCF failure occurrence falls below the Improbable level. Analysis methodologies may include the FMEA, FMECA, SSHA, SSA, or FTA.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Hazard tracking system verifies implementation of validated risk mitigation measures.

4.4.4 Isolation of hazardous substances, components, and operations.

Standard: The air system design prevents damage to safety-critical subsystems and components and injury to crewmembers and passengers by ensuring isolation of hazardous substances or exposures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design, crewmember procedures, and maintenance documentation to ensure the air system is designed, built, operated, and maintained with isolation of hazardous substances to

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prevent exposure to crewmembers and passengers and damage to safety-critical subsystems and components. This includes verifying any corresponding corrective measures (e.g., separation, shielding, isolation, monitors, fail-safe, alerting systems).

4.4.5 Foreign object damage prevention.

Standard: The air system and peculiar support equipment will not result in FOD (to include FOD from system or subsystem components) that can be introduced into safety-critical systems and subsystems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation for the air system and peculiar support equipment (e.g., fabrication, manufacturing, assembly and installation drawings); and operating documentation (e.g., flight and maintenance manuals) to ensure design does not allow FOD introduction that could affect safety-critical systems.

(2) Review design documentation of safety-critical systems to verify the systems incorporate FOD free design considerations, IAW NAS412, Product Design Considerations section.

(3) Review design documentation to verify the design allows for FOD inspections of safety-critical systems, subsystems, and openings, compartments or cavities as described in NAS412, Product Design Considerations section.

(4) Review FOD inspection results documentation to ensure that none of the FOD found is a result of system design or improper product design considerations.

4.5 Critical safety items and critical characteristics.

The AW of an air system relies on the consistent and correct identification, documentation, management, and control of CSIs and their product and process CCs. Consult SECNAVINST 4140.2/AFI 20-106 for guidance on management of aviation CSIs. As part of this process, CSIs should be recorded in a service-specific repository and in technical data associated with the air system. Consult SAE AS9017, SAE AS9138, and ASTM E2782 for guidance on managing CSIs and their product and process CCs. Recommended technical data with CSI and CC information include drawings, drawing trees, and specification trees. For CSI and CC markings in drawings, ASME Y14.100-2017, Markings on Drawings section, should be consulted for information. Consult MIL-STD-31000 for guidance on Technical Data Packages.

For CSIs with a National Stock Number, CSIs should be identified with the applicable criticality code in the Federal Logistics Information System IAW DoDM 4140.01, Volume 11, Managing and Controlling CSI section.

4.5.1 Critical safety item and critical characteristic identification.

Standard: Air system, to include propulsion systems, CSIs are identified and documented to enable management of CSIs for continued AW. Associated product CCs (e.g., tolerance, surface finish, materials, and dimensions) and process CCs (e.g., manufacturing, assembly, and inspection processes) are defined in the technical data.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the CSI list (e.g., CSI Management Plan) verifies documentation of CSIs.

(2) Engineering technical data, including models and drawings, identifies the CSIs and their product CCs (e.g., tolerance, surface finish, materials, and dimensions) and process CCs (e.g., manufacturing, assembly, or inspection processes).

Analysis.

(1) Assessment of current safety hazard, failure mode, functional thread and testing documentation (e.g., FMECA, SHA, SSHA, SCFTA, FMET, FTA, CSI determination decision documentation, to include any design-based factors or mitigations considered during the identification process) ensures identification of all CSIs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) The final product baseline and technical data, including models and drawings, identifies the CSIs and their product and process CCs.

4.6 Flight manuals.

Central to achieving and maintaining continued AW for an air system is the reliance on comprehensive flight manuals IAW the certified configuration, operational usage, and operating environment. Flight manuals serve the primary function of describing the equipment, characteristics, operation, and restrictions necessary for crewmembers to operate the air system during all phases of flight. The term “flight manuals” encompasses the full set of procedural and informational documents containing system descriptions, limitations, normal and emergency procedures, warnings, cautions, notes, performance data, weight and balance information, equipment list, associated subsystem supplemental manuals, and supplements.

This section addresses capstone criteria for the formatting, content, and accuracy of flight manuals for the air system. This handbook contains additional criteria to address specific flight manual information and instructions. TABLE IV below lists these criteria.

TABLE IV. *Flight manual related criteria (standard or MoC).*

Criteria	Heading/Criterion Title	Flight Manual Content in Standard	Flight Manual Content in MoC
5.1.1	Flight loads.		X
5.1.2	Ground loads.		X
5.1.3	Miscellaneous loads.		X

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TABLE IV. *Flight manual related criteria (standard or MoC) – Continued.*

Criteria	Heading/Criterion Title	Flight Manual Content in Standard	Flight Manual Content in MoC
5.2.1	Aeroelasticity.		X
5.2.2	Aeroservoelasticity.		X
5.2.3	Vibration and sonic fatigue.		X
5.5.2	Weight and balance documentation.	X	X
6.1.2	Manuals.	X	X
6.2.8.1	Pre-flight procedures.		X
6.2.8.2	Normal procedures and limits.		X
6.2.8.3	Emergency procedures.		X
6.2.10.3	Ground provisions.		X
6.4.1	Performance information.		X
6.4.2	Performance limits.		X
6.4.3	Failures affecting performance.		X
7.1.2	Propulsion technical data.	X	X
7.2.5	Air start.	X	X
7.2.8	Exhaust gas impingement.		X
7.4.12	Propulsion monitoring system.		X
7.5.7	Vibration and balancing.		X
8.1.3.2	Transition of power to backup and emergency systems.		X
8.1.5	Flight crew displays.		X
8.1.6	Technical manuals.	X	X
8.1.10.1	Component qualification.		X
8.2.4	Technical manuals.	X	X
8.2.7.2	Isolation of occupied compartments.		X
8.2.8.3	Ice detection.		X
8.2.11	Smoke and fume removal.		X
8.3.2	Qualification tests.		X

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TABLE IV. *Flight manual related criteria (standard or MoC) – Continued.*

Criteria	Heading/Criterion Title	Flight Manual Content in Standard	Flight Manual Content in MoC
8.3.14	Compatibility with approved fuels.		X
8.3.16	Technical manuals.	X	X
8.3.17	Fuel system center of gravity.		X
8.4.6	Single failures.		X
8.5.1.9	Landing gear hiking and dehiking system.		X
8.5.10.1	Air vehicle arrestment performance (loads).		X
8.5.10.4	Arresting hook position status.		X
8.5.11.1	Drag and spin chutes.		X
8.5.11.2	Drag or spin chute release.		X
8.5.11.6	Aborted landing.		X
8.5.15.4	Flight crew interface.		X
8.5.15.5	Technical manuals.	X	X
8.5.15.6	Qualification testing.		X
8.6.1	Qualification of components.		X
8.6.10.1	Flight crew displays.		X
8.6.10.2	Flight crew manual controls.		X
8.6.11	Technical manuals.	X	X
8.8.22	Health monitoring system.		X
8.9.2	Cargo hook system operation.		X
8.9.4	Release of external cargo.		X
8.10.1	Rescue hoist system operation.		X
8.10.4	Jettison capability for cable.		X
8.11.1	Fast rope insertion and extraction system operation.		X
9.4.3	Flight manuals.	X	X
9.4.5	User interface error handling.		X

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TABLE IV. *Flight manual related criteria (standard or MoC) – Continued.*

Criteria	Heading/Criterion Title	Flight Manual Content in Standard	Flight Manual Content in MoC
9.6.3.9	Emergency egress exits and routes - documentation.	X	X
10.1.2.1	Aerial refueling envelope.		X
10.1.3.6	Service life.	X	X
10.1.6.1	Identification of fuels.		X
10.3	Failure modes.	X	X
10.5	Technical manuals.	X	X
11.5.1	Standard for flight manuals.	X	X
12.1.7	Electrical system limitation.		X
12.1.8	Electrical system procedures.	X	X
12.1.10	Notification of battery discharge.		X
15.1.11	Notification of loss of critical processing.		X
15.2.4	Critical discrepancies.	X	X
16.2.2	Special inspections.		X
17.2.3.5	Armament and stores electrostatic charge control.		X
17.2.3.6	Armament and stores hazards of electromagnetic radiation to ordnance.		X
17.3.3.2	Stores critical function.		X
17.4.1	Crewmember exposure.		X
17.4.2	Laser boresight.		X
17.4.3	Laser fire control.		X
17.4.4	Latching.		X
17.4.5	Display indications.		X
17.4.7	Ground lasing.		X
17.5.1	Armament and store configuration.		X
17.5.3	Store loadouts and limitations.		X

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TABLE IV. *Flight manual related criteria (standard or MoC) – Continued.*

Criteria	Heading/Criterion Title	Flight Manual Content in Standard	Flight Manual Content in MoC
17.5.4	Armament and store operation and delivery procedures.	X	X
20.6	Aerial delivery manuals.	X	X

4.6.1 Flight manual format.

Standard: Flight manuals conform to a format that enables crewmembers to distinguish emergency procedures, limitations, warnings, cautions, and notes from normal procedures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Flight manuals and crewmember checklists are arranged IAW one of the following:

a. MIL-DTL-7700, Flight Manual (-1) arrangement section.
b. MIL-DTL-85025, Arrangement of publication section.

(2) Flight manuals and crewmember checklists contain emergency procedures formatted IAW MIL-STD-38784, Emergency Page Markings paragraph, and one of the following:

a. MIL-DTL-7700, Checklist Emergency Procedures Arrangement section.
b. MIL-DTL-85025, Emergency Procedures Border paragraph, Steps in Emergency Procedures section, and Emergency Procedures subsection of the Pocket Checklist section.

(3) Flight manuals contain limitations IAW one of the following:

a. MIL-DTL-7700, Section V – Operating Limitations section.
b. MIL-DTL-85205, Chapter 4 – Operating Limitations section.

(4) Flight manuals and crewmember checklists contain warnings, cautions, and notes formatted IAW MIL-STD-38784, Warnings, Cautions, and Notes paragraph, and one of the following:

a. MIL-DTL-7700, Warnings, Cautions, and Notes paragraph and Checklist Emergency Procedures Arrangement subsection.
b. MIL-DTL-85205, Warnings, Cautions, and Notes subsections of the Technical Content Requirements for NATOPS Products section and Format Requirements for NATOPS Products section.

(5) For air systems that do not require typical TO formatting, such as CDAs that use FAA-formatted material, USG approval of the technical data format and equivalent level of safety for the formatting standard is documented.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(6) Supplements required for operational test or operations conform to the formatting of the manual it supplements IAW MIL-STD-38784, Supplements section.

4.6.2 Flight manual content.

Standard: Flight manual content derives from system design information and safety analyses resulting in a systems description, complete set of normal and emergency procedures, operating limitations, warnings, cautions, notes, and performance data for crewmembers to operate the air system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Flight manuals contain an air system description aligned to the intended type design and IAW one of the following:

a. MIL-DTL-7700, Section I Description and Operation section.
b. MIL-DTL-85025, Part I – The Aircraft section.

(2) Flight manuals and crewmember checklists contain normal procedures aligned to the intended type design and AW substantiating data, including safety and failure mode analyses. The normal procedure content is IAW one of the following:

a. MIL-DTL-7700, Section II – Normal Procedures sections, Section II – Normal Aircrew Procedures sections, and Abbreviated Checklist Requirements paragraph. The MIL-DTL-7700 provides tailored guidance on the content of the Normal Procedures section based on the specific manual (also referred to as Section II – Normal Aircrew Procedures); ensure review of all applicable section guidance.
b. MIL-DTL-85025, Part III – Normal Procedures section and Normal Procedures paragraph of the Content Requirements for the Pocket Checklist subsection.

(3) Flight manuals and crewmember checklists contain emergency procedures aligned to the intended type design and AW substantiating data, including safety and failure mode analyses. The emergency procedure content is IAW one of the following:

a. MIL-DTL-7700, Section III – Emergency Procedures sections, Section III – Emergency Aircrew Procedures sections, and Abbreviated Checklist Requirements paragraph. The MIL-DTL-7700 provides tailored guidance on the content of the Emergency Procedures section based on the specific manual (also referred to as Section III – Emergency Aircrew Procedures); ensure review of all applicable section guidance.
b. MIL-DTL-85025, Part V – Emergency Procedures section and Emergency Procedures paragraph of the Content Requirements for the Pocket Checklist subsection.

(4) Flight manuals and crewmember checklists contain operating limitations, including system and subsystem envelopes, aligned to the intended type design and AW substantiating data. The operating limitations section is IAW one of the following:

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a. MIL-DTL-7700, Section V – Operating Limitations section
b. MIL-DTL-85205, Chapter 4 – Operating Limitations section.

(5) Flight manuals and crewmember checklists contain warnings, cautions, and notes aligned to the intended type design and AW substantiating data, including safety and failure mode analyses. The warnings, cautions, and notes content is IAW MIL-STD-38784, Warnings, Cautions, and Notes paragraph, and one of the following:

a. MIL-DTL-7700, Warnings, Cautions, and Notes paragraph.
b. MIL-DTL-85205, Warnings, Cautions, and Notes subsection of the Technical Content Requirements for NATOPS Products section.

(6) Flight manuals contain performance data aligned to the intended type design and AW substantiating data. The performance data content is IAW with one of the following:

a. MIL-DTL-7700, Performance Data Appendix/Manual section
b. MIL-DTL-85025, Part XI – Performance Data section.

(7) Ancillary flight test documentation (e.g., flight clearance, test operating limitations, engineering source data, test cards) provides additional information and restrictions associated with the flight test operational usage.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(8) Supplements required for operational test or operations do not contradict the final reproducible copy of the flight manuals.

4.6.3 Flight manual accuracy.

Standard: Execution of documented crewmember procedures in the flight manuals results in the intended function of the air system for its operational usage.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Preparation and review of preliminary flight manuals, modification flight manuals, crewmember checklists, and ancillary flight test documentation (e.g., flight clearance, test operating limitations, engineering source data, test cards) includes affirmation of readiness to begin flight test.

(2) Flight manuals reflect the current configuration, operational usage, and operating environment for the air system.

(3) Normal procedures in flight manuals and crewmember checklists describe the clear and precise step-by-step actions that the crewmembers need to take to result in the intended function of the air system for its operational usage.

(4) Emergency procedures in flight manuals and crewmember checklists describe the clear and precise step-by-step actions that the crewmembers need to take after a non-normal operation or system condition occurs.

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Test.

(1) Lab test: Execution of crewmember procedures, utilizing the flight manuals, crewmember checklists, and ancillary flight test documentation (e.g., flight clearance, test operating limitations, engineering source data, test cards) in a representative environment, results in the intended function of the air system. Crewmember procedures address planned flight test conditions as well as potential emergency situations.

(2) Lab test: Execution of procedural mitigations identified through safety analyses verifies mitigation of hazards.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(5) The flight manuals and crewmember checklists include any procedural mitigations identified by safety analyses and hazards.

(6) The flight manuals and crewmember checklists incorporate any necessary revisions identified during testing and documented in the flight test technical report(s).

Test.

(3) Flight or Ground test: Execution of crewmember procedures, utilizing the preliminary flight manuals, modification flight manuals, crewmember checklists, and ancillary flight test documentation (e.g., flight clearance, operating limitations, engineering source data, test cards), results in the intended function of the air system.

(4) Flight, Ground, or Lab test: Execution of procedural mitigations and procedural revisions identified during testing verifies mitigation of hazards and confirms accuracy of the flight manuals.

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5. STRUCTURES

The air vehicle structure consists of those parts and subassemblies that are required to carry and transfer loads to perform their primary intended functions. The structure of an air vehicle includes the wing, fuselage, empennage, flight control surfaces (e.g., flap, aileron, rudder, elevator, speed brakes, and spoilers), mechanical control systems, leading edges, trailing edges, radomes, inlets, nacelles, engine mounts, stores mounts, structural aspects for air vehicle windshields and canopy systems, landing gear structural parts (e.g., axles, pistons, cylinders, trunnions, drag braces, and connecting pins), arresting gear structural parts (e.g., arresting hooks, arresting hook shanks, pivot fittings, damper fittings, and connecting pins), additional shipboard launch and recovery gear structural parts (e.g., launch bars, bridal attach fittings), in-flight refueling mechanisms, shipboard related apparatus and devices, structural attachment provisions (e.g., seats, payload, air vehicle tie-down and hoist points, medical evacuation equipment, personnel accommodations, and onboard facilities), rotary-wing rotor and drive systems, and other parts as described in the contract specification. The air vehicle provides the necessary strength, rigidity, and aerodynamic shape to enable controlled flight.

Structural integrity is the attribute when a structure is sound and unimpaired while providing the desired level of structural safety, performance, durability, and supportability. The application of either MIL-STD-1530 ASIP or MIL-STD-3063 RSIP for an air vehicle design series is established by the procuring agency.

The ASIP Master Plan and Structural Design Criteria are living documents that define the air vehicle specific requirements necessary to achieve structural integrity. These documents are established early in program development with coordination by the assigned technical authority and updated as required by the procuring agency. At time of AW certification basis review, these artifacts should document the minimum requirements to demonstrate compliance. Structural certification plans document unique design criteria and planned verification methods for modifications impacting structure.

The following criteria have applicability to new air vehicle designs, modifications to the air vehicle, system modifications or software changes that impact the geometry, material, load path, load magnitude, weight, CG, usage spectra, or structural stability of the air vehicle or any part thereof.

5.1 Loads.

5.1.1 Flight loads.

Standard: The air vehicle design flight loads resulting from the maximum, minimum, and most critical combinations of all known externally or internally applied pressures, forces, or moments are defined and commensurate with all points within the structural design envelope accounting for:

a. Flight crew or autonomous input.
b. Emergency maneuvers.
c. Pressurization.
d. Power plant.
e. Configuration.
f. Environmental effects.

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g. Aerial refueling.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the flight design loads; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for flight loads, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all significant flight loads to allow for tracking margins to achieve envelope expansion.

Analysis.

(1) Verify loads analysis models are updated with developmental tests, and predictions demonstrate flight loads are compliant with structural design criteria.

Test.

(1) Lab test: Verify developmental tests substantiate the aerodynamic and mechanical models used for loads analyses predictions.

(2) Ground test: Verify the as-completed full-scale ground tests support the initial flight envelope IAW EZ-SB-22-01. Verify loads calibration test results in loads equations that achieve accuracy requirements and the required loads parameters for flight test.

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(4) Verify strength summary and operating restrictions define the operational envelope and necessary restrictions; reflect all analyses and tests; satisfy the structural design criteria; and are consistent with EZ-SB-22-01.

(5) Verify the flight manual is consistent with the envelope and limitations defined in the strength summary and operating restrictions for all air vehicle configurations; and contains all warnings, cautions, notes, and emergency procedures required to operate within the design loads.

Analysis.

(2) Verify loads analysis models are updated and correlated to flight test, and final analyses predictions demonstrate flight loads are compliant with structural design criteria and are incorporated into the strength summary and operating restrictions.

Test.

(3) Flight test: Verify loads flight test substantiates the models used for loads analyses predictions.

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5.1.2 Ground loads.

Standard: The air vehicle design ground loads resulting from the maximum, minimum, and most critical combinations of all known externally or internally applied pressures, forces, or moments are defined and commensurate with all points within the structural design envelope accounting for:

a. Takeoff, ground and water landing, taxiing, and towing.
b. Emergency conditions.
c. Turning, braking, pivoting, and steering.
d. Shipboard launch and recovery.
e. Jacking, hoisting, arrestment extension and retraction.
f. Loads induced while parked and moored.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the ground design loads; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for ground loads, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all significant ground loads to allow for tracking margins to achieve envelope expansion.

Analysis.

(1) Verify loads analysis models are updated with developmental tests, and predictions demonstrate ground loads are compliant with structural design criteria.

Test.

(1) Ground test: Verify the as-completed full-scale ground tests support initial ground operations and flight envelope IAW with EZ-SB-22-01. Verify loads calibration test results in loads equations that achieve accuracy requirements and the required loads parameters for ground and flight test.

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(4) Verify strength summary and operating restrictions define the operational envelope and necessary restrictions; reflect all analyses and tests; satisfy the structural design criteria; and are consistent with EZ-SB-22-01.

(5) Verify the flight manual is consistent with the envelope and limitations defined in the strength summary and operating restrictions for all air vehicle configurations; and contains all warnings, cautions, notes, and emergency procedures required to operate within the design loads.

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Analysis.

(2) Verify loads analysis models are updated and correlated to ground test, and final analyses predictions demonstrate ground loads are compliant with structural design criteria and are incorporated into the strength summary and operating restrictions.

Test.

(2) Ground test: Verify full-scale drop test results are used to update the design.

(3) Ground test: Verify loads ground test substantiates the models used for loads analyses predictions.

5.1.3 Miscellaneous loads.

Standard: The air vehicle design includes loads derived from events not captured in normal ground and flight operations, including:

a. Crash.
b. Bird strike and foreign object debris.
c. Hail and lightning.
d. Transport of air vehicle by ground, water or air.
e. Maintenance and abuse loads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the miscellaneous design loads; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for miscellaneous loads, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all miscellaneous flight loads to allow for tracking margins to achieve envelope expansion.

Analysis.

(1) Verify loads analysis models are updated with developmental tests, and predictions demonstrate the miscellaneous loads are compliant with structural design criteria.

Test.

(1) Lab test: Verify bird strike, hail strike, and lightning tests demonstrate integrity for flight-critical structure or structure protecting flight-critical systems, and the results are used to update analyses and validate structural capabilities.

Prior to entering operational test or operations, verify the standard by the following method(s):

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Examination.

(4) Verify strength summary and operating restrictions define the operational envelope and necessary restrictions; reflect all analyses and tests; and satisfy the structural design criteria.

(5) Verify the flight manual is consistent with the envelope and limitations defined in the strength summary and operating restrictions for all air vehicle configurations.

5.1.4 Repeated loads.

Standard: The loads spectra used to support the DSL of the air vehicle is defined using all sources of repeated loads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the repeated design loads; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for repeated loads, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all significant repeated loads to allow for tracking margins to achieve envelope expansion.

Analysis.

(1) Verify appropriate loads spectra are developed for all repeated load sources to support initial durability and damage tolerance analyses and flight test plans.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Verify loads spectra in the durability and damage tolerance analyses and loads analyses are updated based on flight and ground load testing to demonstrate repeated loads are compliant with structural design criteria.

Test.

(1) Flight or Ground test: Verify flight and ground load test substantiates the models used for repeated loads predictions.

5.2 Structural dynamics.

5.2.1 Aeroelasticity.

Standard: The air vehicle and its components are designed with margins for airspeed, Mach, and damping to operate without aeroelastic instability throughout the structural design envelope accounting for:

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a. All configurations and operating conditions.
b. Thermal, Mach, and gyroscopic effects.
c. Control surface mass balancing.
d. Freeplay and rigidity of hinged mechanisms.
e. Service life wear of components.
f. Failures or malfunctions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the aeroelastic stability; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for aeroelasticity, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all metrics for dynamic stability to allow for tracking margins to achieve envelope expansion.

Analysis.

(1) Verify flutter and divergence models are correlated with all lab and GVTs, and predictions demonstrate the aeroelastic design envelopes are compliant with structural design criteria.

(2) Verify manufacturing design tolerance for control surface and actuated door linkages:

a. Define the largest sources for freeplay.
b. Ensure limits meet requirements specified in the structural design criteria or provide engineering rationale for deviation.
c. Provide margin to support the freeplay management strategy for the operational fleet DSL.

Test.

(1) Lab test: Verify developmental tests substantiate the aerodynamic, mechanical, and mass models used for the flutter and divergence analyses predictions.

(2) Ground test: Verify freeplay and rigidity test demonstrates control surface and in-flight actuated door freeplay values are within tolerance requirements.

(3) Ground test: Verify component and full-scale GVTs capture the modal parameters used to correlate the mechanical and mass models.

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(4) Verify strength summary and operating restrictions define the operational envelope and necessary restrictions, reflect all analyses and tests, and satisfy the structural design criteria.

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(5) Verify the flight manual is consistent with the envelope and limitations defined in the strength summary and operating restrictions for all air vehicle configurations; and contains all warnings, cautions, notes, and emergency procedures required to maintain aeroelastic stability margins for the DSL.

Analysis.

(3) Verify the flutter and divergence models are correlated to flight test, and the final analyses predictions demonstrate the aeroelastic design envelopes are compliant with structural design criteria and are incorporated into the final strength summary and operating restrictions.

Test.

(4) Flight test: Verify flutter flight test results are used to update and correlate the flutter and divergence analyses, project stability margins for all analytical flutter mechanisms, and substantiate the fleet freeplay management strategy for the DSL.

5.2.2 Aeroservoelasticity.

Standard: The air vehicle flight control system is designed with gain and phase margins to operate without instabilities induced through coupling of aerodynamic, structural, thermal, and inertial forces throughout the structural design envelope and accounting for:

a. All configurations and operating conditions.
b. Any nominal, failed, or degraded flight control mode of operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the ASE stability; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for aeroservoelasticity, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all metrics for ASE stability to allow for tracking margins to achieve envelope expansion.

Analysis.

(1) Verify ASE models are correlated with all lab and ground structural coupling tests and results are compliant with the structural design criteria.

Test.

(1) Lab test: Verify developmental tests substantiate the aerodynamic, mechanical, flight control, and mass models used for the ASE stability analyses predictions.

(2) Ground test: Verify full-scale structural coupling test results are used to update and correlate the ASE analyses.

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Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(4) Verify strength summary and operating restrictions define the operational envelope and necessary restrictions; reflect all analyses and tests; and satisfy the structural design criteria.

(5) Verify the flight manual is consistent with the envelope and limitations defined in the strength summary and operating restrictions for all air vehicle configurations; and contains all warnings, cautions, notes, and emergency procedures required to maintain ASE stability margins for the DSL.

Analysis.

(2) Verify ASE models are updated for all flight control software updates and flight test results, and the final analyses predictions are compliant with the structural design criteria and are incorporated into the final strength summary and operating restrictions.

Test.

(3) Flight test: Verify ASE flight test results are used to update the ASE analyses and demonstrate closed loop stability for all flight control modes of operation.

5.2.3 Vibration and sonic fatigue.

Standard: The air vehicle is designed with margin to withstand the induced vibration and acoustic environments for the DSL accounting for:

a. All vibration and acoustic sources.
b. Configurations.
c. Flight and ground operations.
d. Design usage spectra.
e. Safety factors.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063, respectively, documents the analysis and test verification methods required to certify the design for vibration and sonic fatigue; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for vibration and sonic fatigue, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all metrics for vibration and sonic fatigue to allow for tracking margins to achieve envelope expansion.

Analysis.

(1) Verify the environment criteria documentation defines vibration and acoustic environments throughout the air vehicle; predicted environments consider all sources, ground and flight

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operational conditions, air vehicle configurations, and usage spectra; spectra compression and test exposure durations are traceable to equivalent operational exposure durations; and analytical predictions are updated and correlated with developmental and ground tests.

(2) Verify initial vibration and sonic fatigue analysis models are updated and correlated with developmental tests; and analyses predictions are compliant with structural design criteria.

Test.

(1) Lab test: Verify developmental test results are used to update the environment criteria documentation and substantiate the vibration and sonic fatigue predictions.

(2) Ground test: Verify vibration and acoustic ground tests substantiate the models used to update the environmental predictions.

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(4) Verify strength summary and operating restrictions define the operational envelope and necessary restrictions; reflect all analyses and tests; and satisfy the structural design criteria.

(5) Verify the flight manual is consistent with the envelope and limitations defined in the strength summary and operating restrictions for all air vehicle configurations; and contains all necessary warnings, cautions, notes, emergency procedures, and inspection criteria required to operate in the design vibration and acoustic environments for the DSL.

Analysis.

(3) Verify the environment criteria documentation is updated with ground and flight tests to define the vibration and acoustic environments.

(4) Verify vibration and sonic fatigue analysis models are updated and correlated to ground and flight test; and final analyses predictions are compliant with the structural design criteria and are incorporated into the final strength summary and operating restrictions.

Test.

(3) Flight or Ground test: Verify ground and flight test results are used to update the environment criteria documentation and substantiate the analytical vibration and sonic fatigue lives.

5.3 Strength.

5.3.1 Detrimental deformation.

Standard: The air vehicle is designed with margin to withstand DLLs without yielding, or detrimental deformation accounting for:

a. Material and processes.
b. Joining methods.
c. Safety factors.
d. Fail-safe design.
e. Static strength, rigidity, and stability.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063, respectively, documents the analysis and test verification methods required to certify the design for detrimental deformation; that structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for detrimental deformation, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all internal load paths and demonstrate the strength design envelope.

Analysis.

(1) Verify strength analysis models are updated with ground tests, and predictions demonstrate the design strength is compliant with the structural design criteria.

Test.

(1) Lab test: Verify development tests substantiate:

a. Material allowables.
b. Material processing.
c. Strength capability of joining methods.

(2) Ground test: Verify the as-completed full-scale ground tests support the initial flight envelope IAW EZ-SB-22-01.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Verify strength analysis models are based on flight test correlated loads, static test correlated internal loads, and the current air vehicle configuration, and analysis predictions are compliant with the structural design criteria and incorporated into the final strength summary and operating restrictions.

Test.

(3) Ground test: Verify the full-scale ground test substantiates the models used for strength analysis predictions and demonstrates the air vehicle can withstand DLLs without detrimental deformation.

5.3.2 Failure or collapse.

Standard: The air vehicle is designed with margin to withstand design ultimate loads without rupture or collapsing failures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063, respectively, documents the analysis and test verification methods required to certify the design for failure or collapse; that structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for failure or collapse, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to measure all internal load paths and demonstrate the strength design envelope.

Analysis.

(1) Verify strength analysis models are updated with ground tests, and analyses predictions are compliant with the structural design criteria.

Test.

(1) Lab test: Verify development tests substantiate:

a. Material allowables.
b. Material processing.
c. Strength capability of joining methods.

(2) Ground test: Verify the as-completed full-scale ground tests support the initial flight envelope IAW EZ-SB-22-01.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Verify strength analysis models are based on flight test correlated loads, static test correlated internal loads, and the current air vehicle configuration, and analysis predictions are compliant with the structural design criteria and incorporated into the final strength summary and operating restrictions.

Test.

(3) Ground test: Verify the full-scale ground test substantiates the models used for strength analysis predictions and demonstrates the air vehicle can withstand design ultimate loads without failure or collapse.

5.4 Durability and damage tolerance.

Methodologies for durability and damage tolerance may differ between services. The appropriate fatigue and fracture methodology utilized for design of each structural element or category of structural elements is specified and agreed to in the ASIP Master Plan or RSIP Master Plan.

5.4.1 Durability.

Standard: The air vehicle and its components are designed with durability margins such that it resists cracking, corrosion, thermal degradation, delamination, erosion, wear, and the effects of

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FOD to maintain structural integrity and mechanical operation of the air vehicle throughout the specified service life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the design for durability; that structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for durability, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify durability test plans are scoped to apply a representative service life spectrum and monitor critical control points.

(4) Verify the CPCP documents corrosion prevention systems, corrosion prevention and control measures, procedures, and processes are defined to ensure structural integrity and support the operational environment and maintenance capability throughout the specified service life of the air vehicle.

(5) Verify the Durability and Damage Tolerance Control Plan or Fatigue and Fracture Control Plan defines the methodology used for durability analysis and identifies durability critical parts to ensure structural integrity and supports the operational and maintenance capability required of the air vehicle structure.

(6) Verify the NDIP defines methods, instrumentation, procedures, intervals, and detection capabilities to ensure structural integrity.

Analysis.

(1) Verify durability analyses predictions are compliant with structural design criteria and support flight test.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Verify durability models are updated with repeated loads from flight test; are correlated to the FSDT; and analyses predictions are compliant with the structural design criteria.

Test.

(1) Ground test: Verify the FSDT uses representative loading spectra and demonstrates durability requirements are achieved; test results are used to establish maintenance requirements.

5.4.2 Damage tolerance.

Standard: Air vehicle structure determined to be SOF is designed with damage tolerant margins and assumptions to maintain residual strength throughout the specified service life of the air vehicle.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the design for damage tolerance; that structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for damage tolerance, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify damage tolerance test plans are scoped to apply a representative service life spectrum and monitor critical control points.

(4) Verify the CPCP documents corrosion prevention systems, corrosion prevention and control measures, and procedures; processes are defined to ensure structural integrity and support the operational environment and maintenance capability throughout the specified service life of the air vehicle.

(5) Verify the Durability and Damage Tolerance Control Plan or Fatigue and Fracture Control Plan defines the methodology used for durability analysis and identifies durability critical parts to ensure structural integrity and supports operational and maintenance capability required of the air vehicle structure.

(6) Verify the NDIP defines methods, instrumentation, procedures, intervals, and detection capabilities to ensure structural integrity.

Analysis.

(1) Verify damage tolerance analyses predictions are compliant with structural design criteria and support flight test.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Verify damage tolerance models are updated with repeated loads from flight test; are correlated to the full-scale damage tolerance test; and analyses predictions are compliant with structural design criteria.

Test.

(1) Ground test: Verify the full-scale damage tolerance test uses representative loading spectra; demonstrates durability requirements are achieved; results are used to establish maintenance requirements.

5.4.3 Safe-life fatigue.

Standard: The air vehicle is designed with margin such that it will not initiate fatigue damage and maintains full static strength capability for the DSL.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the design for safe-life; that structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for safe-life, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify test plans are scoped to apply a representative service life spectrum, monitor critical control points, and demonstrate static strength capability.

(4) Verify the CPCP documents corrosion prevention systems, corrosion prevention and control measures, procedures, and processes are defined to ensure structural integrity and support the operational environment and maintenance capability throughout the specified service life of the air vehicle.

(5) Verify the Durability and Damage Tolerance Control Plan or Fatigue and Fracture Control Plan defines the methodology used for safe-life analyses and identifies applicable structural elements to ensure structural integrity and supports the operational and maintenance capability required of the air vehicle structure.

(6) Verify the NDIP defines the methods, instrumentation, procedures, intervals, and detection capabilities to ensure structural integrity is maintained.

Analysis.

(1) Verify safe-life analyses predictions are compliant with structural design criteria and support flight test.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Verify the safe-life models are updated with the repeated loads from flight test; are correlated to the full-scale fatigue test; and analyses predictions are compliant with the structural design criteria.

Test.

(1) Ground test: Verify the full-scale fatigue test uses representative loading spectra and demonstrates safe-life requirements are achieved; test results are used to establish maintenance requirements.

5.4.4 Service life.

Standard: The service lives of the air vehicle and its parts are specified and maintained in terms of years, flight hours, flight cycles, landings, equivalent flight hours or other measures of damage in which the structure is expected to maintain its structural integrity when flown to the design loads spectra.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the DSL; that structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for service life, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

(3) Verify the initial or modified service life is specified to an original design loads spectrum.

(4) Verify the Force Structural Maintenance Plan is established or a documentation plan for inspections and modifications such that methodologies, intervals, location, and cost are detailed and support the continued AW throughout the specified service life.

(5) Verify the air vehicle is designed with an individual aircraft tracking or individual rotary-wing tracking system to record usage parameters that supports updates to the inspection, modification, overhaul, and replacement times based on actual, measured usage of each air vehicle.

Analysis.

(1) Verify the durability, damage tolerance, and safe-life analyses predictions demonstrate residual strength and support flight test.

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(6) Verify the Loads/Environmental Spectra Survey is developed and supports updates to the baseline operational loads spectrum.

(7) Verify the Loads/Environmental Spectra Survey and individual aircraft tracking program support updates to individual air vehicle usage severity used to determine equivalent flight hours or other measures of damage.

Analysis.

(2) Verify durability, damage tolerance, and safe-life models are updated with repeated loads from flight test; are correlated to durability and damage tolerance tests; and analyses predictions demonstrate the service life is compliant with structural design criteria.

5.5 Weight and balance.

5.5.1 Weight and center of gravity.

Standard: Technical mass properties data for the air vehicle is defined and shown to remain within the air vehicle weight and CG limits for all defined missions, loadings, and configurations for the operating envelope.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASIP Master Plan or RSIP Master Plan, as specified in MIL-STD-1530 or MIL-STD-3063 respectively, documents the analysis and test verification methods required to certify the design weight and CG; structural certification tasks are updated for modifications or changes to the air vehicle following initial AW certification basis; and all artifacts are developed IAW applicable sections of MIL-STD-1530 or MIL-STD-3063.

(2) Verify detailed structural design criteria for mass properties, as specified in MIL-STD-1530 or MIL-STD-3063, documents the requirements to achieve structural integrity.

Analysis.

(1) Verify for a new air vehicle, weight and balance data are shown to be within design limits for all defined flight and ground operations.

(2) Verify the air vehicle calculated basic weight and CG reflects the current configuration and is shown to not exceed any defined weight or CG limits.

5.5.2 Weight and balance documentation.

Standard: The Weight and Balance Handbook is accurate and IAW the Joint Service Technical Manual, 1-1B-50, Aircraft Weight and Balance; Sample Basic Weight Checklists (Chart A) and Loading Data (Chart E) are consistent and contain all required checklists and loading data necessary to conduct required weight and balance checks.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify Weight and Balance Handbook reflects current weight and balance information for each specific air vehicle.

(2) Verify the air vehicle is weighed and documented with DD 365-2, Form B – Aircraft Weighing Record.

(3) Verify DD 365-4, Weight and Balance Clearance Form F is used to validate the air vehicle configuration will remain within all weight and CG limits for flight.

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(4) Verify flight manuals, Sample Basic Weight Checklists (DD Form 365-1, Chart A), and Loading Data (Chart E) reflect the current configuration of the air vehicle.

(5) Verify Weight and Balance Handbook reflects current weight and balance information for each specific air vehicle.

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6. FLIGHT TECHNOLOGY

This section establishes flight technology AW criteria applicable to all air systems and their modifications. Flight Technology consists of four subsections representing the flight mechanics functional areas: Flying Qualities, VCF, Internal Aerodynamics, and External Aerodynamics and Performance. Unless explicitly noted, the standards and MoC apply to all air vehicles regardless of the vehicle type.

Flying Qualities (6.1) standards address the flying qualities, handling qualities, and stability and control characteristics of the air vehicle in normal and degraded states, including in various atmospheric conditions, required to ensure AW and operational safety throughout the entire ground and flight envelope.

The VCF standards (6.2) address the safety-critical hardware and software functions that stabilize and control the air vehicle throughout all flight and ground operations in both normal and degraded operational states. The VCF standards address the integrated systems architecture, redundancy management, command signal integrity from flight crew input to control effector response, and demands on hardware items such as the VCF computer, inertial sensors, actuators, and air data sources to ensure AW and operational safety throughout the entire flight envelope.

Internal Aerodynamics (6.3) standards address the compatibility of integrated propulsion system components (airframe, inlet, engine, nozzle). These standards address propulsion-integration effects on the aerodynamic force and moment accounting system as well as propulsion-integration effects on the engine stability margin throughout the entire ground and flight envelope.

External Aerodynamics and Performance (6.4) standards address the air vehicle performance throughout all flight and ground operations in both normal and degraded operational states. These standards include assessment of the vehicle aerodynamic force and moment accounting, takeoff and landing performance, mass properties, and air vehicle limitations throughout the entire ground and flight envelope.

Documents referenced under any standard and MoC provide other important supporting rationale, guidance, lessons learned and information useful in properly understanding, interpreting, and applying the relevant standard and MoC.

6.1 Flying qualities.

Flying qualities are quantitative characteristics and metrics of the complete air vehicle which have been determined to allow the pilot to perform to his/her qualitative satisfaction, via handling qualities assessment, the flying and ground handling tasks required to safely accomplish the mission, with acceptable pilot compensation, while operating in the real-world environment for which it is intended to operate. The flying qualities quantitative Levels in MIL-STD-1797 have been developed over many decades of research, development, flight test, and operational experience, and have been determined to reflect that meeting the quantitative metrics correlate closely with producing the expected qualitative assessments. These flying characteristics and metrics are also applicable for assuring the flight safety of an UAS.

Compliance with the criteria in this section is demonstrated using analysis, simulation, and flight test at different phases of air system design and development. By preliminary design review and continuously evaluated up to critical design review and beyond, the following steps define the

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scope of analysis and simulation compliance activities to ensure SOF throughout the entire flight envelope. Air system requirements, design, and configuration documentation are necessary information sources to define these tasks:

a. Determine operational missions and associated flight phases. For rotary-wing, also determine the required agility and response types.
b. Define air vehicle states, including internal and external stores combinations, loadings, mass properties, air vehicle configurations, and control modes.
c. Identify air vehicle normal and extreme states.
d. Identify air vehicle failure, and special and specific failure states, including loss of control of thrust (e.g., engine control, no thrust, full power, propeller pitch).
e. Define the regions of handling (fixed-wing) or the operational and SFEs (rotary-wing).

For fixed-wing air systems, use the methods defined in the following MIL-STD-1797 paragraphs: Determining Operational Missions, Determining Applicable Flight Phases, and Defining Air Vehicle States paragraphs, including sub-paragraphs and associated figures and tables, as appropriate. For rotary-wing air systems, use the methods defined in MIL-DTL-32742, General Requirements paragraph, including sub-paragraphs and associated figures and tables.

The procuring activity and the contractor work together to define regions of the flight envelope where Level 1 flying/handling qualities (quantitative and qualitative) are desired for meeting mission objectives, which constitutes the ROSH, still referred to as the operational flight envelope in the rotary-wing community. Furthermore, they should define regions where gradual degradation to Level 2 flying/handling qualities will be acceptable for safely meeting mission objectives, which constitutes the ROTH, and similarly defined to be the SFE for rotary-wing. Lastly, for fixed-wing only, they should determine the outer-most regions of the flight envelope which can be attained by the air system where degradation to Level 3 flying/handling qualities is defined, which constitutes RORH, historically, the “Permissible Flight Envelope”.

The flying and handling qualities standards and MoC in this handbook are specified in terms of quantitative and qualitative degrees of suitability and Levels of flying qualities, respectively. The quantitative characteristics are numerical metrics as defined in MIL-STD-1797 for fixed-wing, MIL-DTL-32742 for rotary-wing, MIL-F-83300 for fixed-wing in hover, and other referenced documents which define flying qualities. The qualitative measures are based on pilot assessment following completion of a set of well-defined, operationally representative, handling qualities tasks (MTEs) using the Cooper-Harper rating scale and the PIO Rating Scale. The degrees of suitability are defined as:

a. Satisfactory (Level 1): Flying qualities clearly adequate for the mission flight phase. Desired performance is achievable with no more than minimal pilot compensation.
b. Tolerable (Level 2): Flying qualities adequate to accomplish the mission flight phase. Increase in pilot compensation, degradation in mission effectiveness, or both, exists.
c. Controllable (Level 3): Flying qualities such that the air vehicle/system can be controlled in the context of the mission flight phase, even though pilot compensation is excessive, mission effectiveness is inadequate, or both. The pilot can transition from rapid maneuvering or precision control flight phase tasks to benign maneuvering flight phases that are less demanding of the pilot and air vehicle/system. Level 3 is not necessarily safe.

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d. Recoverable flying qualities such that control can be regained following loss of control due to departure, failures, or atmospheric disturbances.

Air vehicles/systems should be designed to have Level 1 flying qualities (quantitative metrics) in the ROSH (fixed-wing) or operational flight envelope (rotary-wing) to safely accomplish the mission and achieve compliance with this handbook throughout the life-cycle of the air system. A Level 1 flying qualities design has inherent system robustness to failure conditions, degraded environments, new mission operations and air vehicle configurations, and enables graceful degradation in flying qualities to protect the air vehicle from non-airworthy operations.

When showing compliance with referenced MIL-STD-1797 and MIL-DTL-32742 criteria, it is imperative that MoC data artifacts reflect the intent to meet the expected verification methods outlined within the respective document. For pre-flight test AW compliance assessments, the flying qualities engineers will examine artifacts that show compliance to quantitative metrics for all air vehicle states, configurations, and phases of flight for which the air vehicle is to perform its operational missions and tasks. Concurrently, artifacts should also show that test pilots have evaluated the handling qualities (qualitative assessment) for every mission/task at the most critical flight conditions (e.g., mass properties, altitude, airspeed) of the entire flight envelope utilizing the highest fidelity handling qualities simulator(s), including updates from available flight and ground test results. Furthermore, test pilots should assess critical tasks at critical conditions (e.g., takeoff, landing, AR) in the piloted simulator(s) with critical failure modes, with and without various atmospheric disturbances. Assessments and artifacts for critical failure modes and in atmospheric conditions are especially important since most of these will not be tested during flight test; therefore, these artifacts will be used for AW compliance of critical failure modes with and without atmospheric disturbances. At least three test pilots (government and contractor) with different piloting techniques (e.g., high and low gain pilot bandwidth control and input frequency and amplitude) should assess the handling qualities to be no worse than Level 2 up to the critical flight conditions, to ensure handling qualities robustness to various pilot techniques. Level 3 flying/handling qualities in the RORH constitute the outermost regions of the flight envelope, where handling qualities are not necessarily safe, but provide flight envelope margin for the air system to return to operations in the ROTH and ROSH (fixed-wing) and SFE (rotary-wing). Unlike simulation, flight test has some amount of inherent risk, just due to its exploratory nature. Therefore, the procuring activity and contractor will define the selected combinations evaluated by flight test that balances the need for AW data collection against the risks of flight test and the associated programmatic cost and schedule impacts. High-risk, on air vehicle testing, like failure modes, high AOA and departure susceptibility testing, will be carefully approached and conducted to ensure that the results determine safe air vehicle limitations, safe piloting techniques and procedures to maintain control, and repeatable recovery techniques from loss of control. To the greatest extent practical, these should be determined using a high-fidelity handling qualities simulation to the boundaries of the flight envelope (typically defining the RORH).

For UAS, in areas where qualitative assessments are not feasible, current quantitative metrics (e.g., quantitative metrics in MIL-STD-1797 and MIL-DTL-32742) may be tailored, and complementary quantitative metrics and analysis methods may be developed, with the approval of the procuring activity, to assure safe conduct of all required UAS MTEs.

For non-traditional air systems (e.g., tiltrotor and electric vertical take-off and landing air vehicles), that blend wing-borne forward flight and rotor-borne slow speed and hovering flight

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modes, the appropriate fixed-wing or rotary-wing standards should be used for those respective areas of flight and air system behavior.

For legacy air systems, unless a different method is specified or approved by the procuring activity, modifications to legacy air systems will be assessed against their existing and most current baseline flying/handling qualities (quantitative and qualitative), including any waivers, to include subsequent approved changes, that have been accepted by the procuring activity. All changes to flight characteristics that impact flying/handling qualities, either to improve or impair (deviations from the legacy baseline), will be rigorously documented with sufficient artifacts to identify all AW non-compliances to the baseline as well as all risks and hazards that will be mitigated. If the legacy baseline characteristics are not provided or are unknown, the air vehicle will be assessed against the standards and MoC herein.

6.1.1 Modeling, simulation, and analysis.

Standard: The modeling, simulation, analysis tools, and databases for the evaluations of flying qualities characteristics and Levels are of appropriate maturity, fidelity, and accuracy to represent the air vehicle's flight characteristics for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the documentation to assess model requirements and development process, the fidelity and pedigree of the frequency and time domain-based analysis tools, models, simulations, and all the associated databases and components thereof (e.g., aerodynamics, stability and control, ground and ground effects, FCS and flight control laws, sensors, actuators, equations of motion, environmental conditions, terrain and ship models). Documentation is at the simulation level and also at the simulation subsystem level, to include subsystem models generated by the contractor (e.g., aerodynamic database, mass properties, equations of motion, force and moment accounting), and other subsystem models generated by subcontractors (e.g., propulsion system database, flight control actuator models). Documentation exists for all simulation environments used for Section 6 AW data generation, including all software simulations and HITL simulations. Hardware-in-the-loop simulation documentation includes time history, frequency response, and time delay comparisons with all software simulations and desktops simulations. Documentation includes limitations for use and any known deficiencies for the models, simulations, and tools, along with plans to correct them. Review design documentation to verify decomposition of air vehicle design requirements which define the required models, simulations, associated databases, and components and their associated levels of fidelity and accuracy to support the AW assessment of the air vehicle for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Review configuration control documentation to verify all of the above components for currency and traceability during development (e.g., preliminary design review, critical design review, test readiness review, first-flight readiness review, functional configuration audit, physical configuration audit, production), including incorporation of updates as a result of component testing, propulsion system testing, ground testing, and flight testing.

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Analysis.

(1) Conduct a VV&A process, as specified in MIL-STD-3022, to show that the results created from the tools, models, simulations and all the associated databases and components are complete and representative of the flight characteristics for AW and SOF analyses. Verify and validate that modeling, simulation, analysis tools, and databases, which are utilized for evaluating AW criteria across the flight envelope, for all expected CG ranges and mass properties, for all flight phases, tasks and flight control modes, for all configurations and store loadings, and in the expected atmospheric disturbances for which the air vehicle is to perform its mission(s) are of sufficient fidelity and accuracy. The VV&A analysis should include:

a. Model development reference documentation, the identification of the intended use of the model, the adequacy of the achieved model fidelity for the intended use, model limitations, model configuration variations used for different testing activities, and the model V&V approach. For example, landing gear and tire models should include, as a minimum, tire cornering coefficients as a function of slip angle, tire sides force, and braking force as a function of longitudinal slip. This data should be used in conjunction with aerodynamic effects on the air vehicle during ground roll.
b. Model validation to experimental data or model verification to the actual subsystem or component specification requirements. When validating models to experimental data, an independent data set is used instead of a data set used for model development.
c. The identification of the air vehicle subsystem key characteristics including performance (actuation system capacity), mechanical (hysteresis, backlash, break-out forces, threshold), and processing (data rates, latencies).
d. Flexible air vehicle models included in the simulation or used for analysis are validated by ground vibration testing as well as airframe and control system structural coupling ground testing.
e. The basis for demonstrating that the models (e.g., aerodynamic, propulsion, landing gear, and air data system) are representative of air vehicle ground and flight test results and uninstalled or installed propulsion system tests.
f. Air vehicle simulation models used for mission planning or other performance predictions (e.g., fuel flow, climb and descent rates) are validated against the performance presented in a basic aerodynamic data and substantiating performance report.
g. Uncertainty quantification for use in air vehicle flying qualities robustness analyses and simulations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Review of documentation as stated in Examination MoC (1) to verify all documents incorporate findings from testing and analysis results up to the conclusion of developmental test and evaluation phase.

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Analysis.

(2) Conduct a VV&A process as stated in Analysis MoC (1) to verify all models incorporate findings from testing and analysis results up to the conclusion of developmental test and evaluation phase.

6.1.2 Manuals.

Standard: The flight, performance, operations manuals, and any supplements, contain the air vehicle's FCS descriptions, flight characteristics, and operating limits that bound regions of handling qualities and procedures to assure flight safety and recovery from ROTH, RORH, and post-stall for all air vehicle states, flight phases, and environmental conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the manuals to assess the contents for accuracy, completeness, and significance to safety with respect to flying/handling qualities.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Review of the manuals that are updated with ground and flight test results to assess the contents for accuracy, completeness, and significance to safety with respect to flying/handling qualities.

6.1.3 Display characteristics.

Standard: For the MTEs in which the pilot's use of primary and standby flight displays and instruments is critical in performing the task, the characteristics of these displays and instruments do not lead to air vehicle flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the attributes of the primary and standby flight displays and instruments to include display latency and how the time delay is determined across all flight phases, loadings, configurations, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the attributes of the primary and standby flight displays and instruments to include display latency and how the time delay is determined across all flight phases, loadings, configurations, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess the attributes of the primary and standby flight displays and instruments to include display latency and how the time delay is determined across all flight phases and environmental conditions.

6.1.4 Fixed-wing general flying qualities.

6.1.4.1 Control margin.

Standard: For all normal, extreme, and failure states, there is adequate control margin (aerodynamic control power, control surface rate, propulsion control, and hinge moment capability) for restoring moments throughout the RORH.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control margin as specified within MIL-STD-1797, Control Margin paragraph and Control Power for Stall Recovery paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's control margin as specified within MIL-STD-1797, Control Margin paragraph and Control Power for Stall Recovery paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's control margin as specified within MIL-STD-1797, Control Margin paragraph and Control Power for Stall Recovery paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.2 Environmental conditions.

Standard: The air vehicle flying qualities are no worse than Level 2 in uncommon atmospheric disturbances and no worse than Level 3 in temporary extraordinary atmospheric disturbances, and the air vehicle flying qualities are no worse than Level 2 in sustained extraordinary atmospheric disturbances if mission requirements dictate.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis and simulation to assess the air vehicle flying qualities in degraded environmental conditions as specified in MIL-STD-1797, Flying Qualities in Degraded Environmental Conditions paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle handling qualities in degraded environmental conditions as specified within MIL-STD-1797, Flying Qualities in Degraded Environmental Conditions paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle handling qualities in degraded environmental conditions as specified within MIL-STD-1797, Flying Qualities in Degraded Environmental Conditions paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.3 Icing conditions.

Standard: The air vehicle meets the flight in icing conditions standard as specified in 14 CFR 23.2165, and the applicable flight in icing conditions requirements in 14 CFR 25; unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2 with ice accretion on the lifting surfaces (wings, tails) during and after flight through icing conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle flying qualities in icing conditions as specified in 14 CFR 23.2165, and applicable flight in icing conditions requirements in 14 CFR 25 for each combination of regions of handling qualities, air vehicle states, flight phases, and icing conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle handling qualities in icing conditions as specified in 14 CFR 23.2165, and applicable flight in icing conditions requirements in 14 CFR 25 for each combination of regions of handling qualities, air vehicle states, flight phases, and icing conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle handling qualities in icing conditions as specified in 14 CFR 23.2165, and applicable flight in icing conditions requirements in 14 CFR

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25 for selected combinations of regions of handling qualities, air vehicle states, flight phases, and icing conditions.

6.1.4.4 Probability of degraded flying qualities due to failures.

Standard: In no case can a failure state (except a special failure state) degrade flying qualities to worse than Level 3.

In calm air, the overall probability of encountering Level 2 flying qualities due to one or more failures is less than 10^{-5} per flight hour within the ROSH, and the overall probability of encountering Level 3 flying qualities due to one or more failures is less than 10^{-6} per flight hour within the ROSH and less than 10^{-3} per flight hour within the ROTH.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct FMECA to assess the overall probability of encountering Level 2 flying/handling qualities due to one or more failures within the ROSH, and the overall probability of encountering Level 3 flying/handling qualities due to one or more failures within the ROSH and within the ROTH.

Test.

(1) Lab test: Conduct FMET to assess the overall probability of encountering Level 2 flying/handling qualities due to one or more failures within the ROSH, and the overall probability of encountering Level 3 flying/handling qualities due to one or more failures within the ROSH and within the ROTH.

(2) Lab test: Conduct piloted simulation and evaluation, including open-loop flying qualities maneuvers and closed-loop handling qualities tasks, to assess flying/handling qualities for the failure states for each combination of regions of handling qualities, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test, including open-loop flying qualities maneuvers and closed-loop handling qualities tasks, to assess flying qualities for the failure states for selected combinations of regions of handling qualities, flight phases, and environmental conditions.

6.1.4.5 Flying qualities levels for failure states.

Standard: The air vehicle meets the standards as specified in MIL-STD-1797, Allowable Levels for Specific Air Vehicle Failure States paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct FMECA to assess the air vehicle flying qualities in failure states as specified in MIL-STD-1797, Allowable Levels for Specific Air Vehicle Failure States paragraph for each combination of regions of handling qualities, specific failure states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct FMET to assess the air vehicle flying qualities in failure states as specified in MIL-STD-1797, Allowable Levels for Specific Air Vehicle Failure States paragraph for each combination of regions of handling qualities, specific failure states, flight phases, and environmental conditions.

(2) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle flying qualities in failure states as specified in MIL-STD-1797, Allowable Levels for Specific Air Vehicle Failure States paragraph for each combination of regions of handling qualities, specific failure states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to assess the air vehicle flying qualities in failure states as specified in MIL-STD-1797, Allowable Levels for Specific Air Vehicle Failure States paragraph for selected combinations of regions of handling qualities, specific failure states, flight phases, and environmental conditions.

6.1.4.6 Failures outside the regions of tolerable handling.

Standard: The air vehicle is capable of returning to the ROSH and ROTH after encountering any failure outside of the ROTH but within the RORH.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct FMECA to assess the air vehicle flying qualities after encountering any failure outside of the ROTH but within the RORH.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that air vehicle can be flown to particular flight conditions outside of the ROTH and return to the ROSH or ROTH under the failure conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess that air vehicle can be flown to selected flight conditions outside of the ROTH and return to the ROSH or ROTH under the failure conditions.

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6.1.4.7 Approach to dangerous conditions.

Standard: When approaching dangerous flight conditions, it is possible by clearly discernible means for the pilot to recognize the impending dangers and take preventive action.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to identify potentially dangerous flight conditions (e.g., stall, spin, minimum control speeds) and onset conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess timely, clear, and unambiguous warnings and indications of approach to dangerous flight conditions for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess timeliness, clear and unambiguous warnings and indications of approach to dangerous flight conditions for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.8 Devices for warning of or preventing dangerous conditions.

Standard: Special devices incorporated to eliminate dangerous flight conditions perform their functions whenever needed and do not limit flight within the ROSH. Normal or inadvertent operation of such devices does not create a hazard to the air vehicle. There is no nuisance operation of these devices. Functional failure of these devices are indicated to the pilot.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the operations of special devices incorporated to eliminate dangerous flight conditions are not hazardous to the air vehicle for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the operations of special devices incorporated to eliminate dangerous flight conditions are not hazardous to the air vehicle for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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6.1.4.9 Buffet.

Standard: The effect of buffet does not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess normal acceleration excursions about trim at the pilot station as specified in MIL-STD-1797, Buffet paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle handling qualities in buffet conditions as specified in MIL-STD-1797, Buffet paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle handling qualities in buffet conditions for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.10 Store separation.

Standard: The intentional release or ejection of any stores does not result in dangerous flight characteristics or flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess store separation characteristics as specified in MIL-STD-1797, Release of Stores paragraph for all store loadout configurations, to include retained stores due to failures (e.g., missile hang fire, hung stores), for each combination of store release conditions, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct wind tunnel and piloted simulations and evaluations to assess the aerodynamic changes and handling qualities during a store separation event for all store loadout configurations for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Analysis.

(2) Conduct analysis of flight test data to assess the impact on flying qualities due to worse case lateral asymmetry, to include effects due to hung stores or missile hang fires.

Test.

(2) Flight test: Conduct flight test to assess the aerodynamic changes and handling qualities during a store separation event for all store loadout configurations for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions. Flight test data shows that worst-case lateral asymmetry, to include effects due to hung stores or missile hang fires, do not degrade flying qualities, induce departure from sustained flight, or prevent recovery of the air vehicle.

6.1.4.11 Flying qualities due to operation of movable parts.

Standard: Operation of movable parts (e.g., weapons bay doors, cargo doors, armament pods, refueling devices) does not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle flying qualities due to operation of movable parts as specified in MIL-STD-1797, Effects of Armament Delivery and Special Equipment paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct wind tunnel and piloted simulations and evaluations to assess the aerodynamic changes and handling qualities during the activation of movable parts for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental /conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the aerodynamic changes and handling qualities during the activation of movable parts for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.12 Flying qualities with failures.

Standard: The air vehicle meets the standards as specified in MIL-STD-1797, Failures paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct FMECA and simulation to assess the air vehicle flying qualities due to failure as specified in MIL-STD-1797, Failures paragraph for each combination of regions of handling qualities, air vehicle state, and environmental conditions.

Test.

(1) Lab test: Conduct FMET and piloted simulations and evaluation to assess the air vehicle flying/handling qualities due to failures for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle flying/handling qualities due to failures for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.13 Transients following failures.

Standard: For at least two seconds following a failure, the air vehicle meets the standards as specified in MIL-STD-1797, Transients Following Failures paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle flying qualities due to transients following failure as specified in MIL-STD-1797, Transients Following Failures paragraph for each combination of regions of handling qualities, air vehicle state, and environmental conditions.

Test.

(1) Lab test: Conduct FMET and piloted simulations and evaluations to assess the air vehicle flying/handling qualities due to transients following failures for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle flying/handling qualities due to transients following failures for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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6.1.4.14 Trim changes due to failure.

Standard: The air vehicle meets the standards as specified in MIL-STD-1797, Trim Changes due to Failures paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle flying qualities due to trim changes due to failure as specified within MIL-STD-1797, Trim Changes due to Failures paragraph for each combination of regions of handling qualities, air vehicle state, and environmental conditions.

Test.

(1) Lab test: Conduct FMET and piloted simulations and evaluation to assess the air vehicle flying/handling qualities due to trim changes due to failures for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle flying/handling qualities due to trim changes due to failures for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.15 Pilot-in-the-loop oscillations.

Standard: In any axis or in a combination of axes, there is no tendency for sustained or uncontrollable unintentional oscillations, in which the air vehicle attitude, angular rate, acceleration, or other quantity derived from these states, is approximately 180 degrees out of phase with the pilot's or control system's commands.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the frequency response as specified in MIL-STD-1797, Pilot-in-the-loop Oscillations paragraph, Pitch PIOs paragraph, and Roll PIO paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation and perform operationally representative, high gain and bandwidth, closed-loop handling qualities tasks to assess PIO tendency as specified in MIL-STD-1797, Pilot-in-the-loop Oscillations paragraph, Pitch PIOs paragraph, and Roll PIO paragraph for each combination of regions of handling qualities, air vehicle states,

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flight phases, and environmental conditions. This evaluation requires at least three developmental test pilots with various piloting techniques (e.g., high and low gain pilot bandwidth control and input frequency and amplitude).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess PIO tendency as specified in MIL-STD-1797, Pilot-in-the-loop Oscillations paragraph, Pitch PIOs paragraph, and Roll PIO paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions. This evaluation requires at least three developmental test pilots with various piloting techniques (e.g., high and low gain pilot bandwidth control and input frequency and amplitude).

6.1.4.16 Residual oscillations.

Standard: Air vehicle flying qualities due to experiencing any sustained residual oscillations in calm air are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess any sustained residual oscillations in normal acceleration about trim at the pilot station in calm air as specified in MIL-STD-1797, Residual Oscillations paragraph for each combination of regions of handling qualities, air vehicle states, and flight phases.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's residual oscillation tendencies and characteristics in calm air as specified in MIL-STD-1797, Residual Oscillations paragraph for each combination of regions of handling qualities, air vehicle states, and flight phases.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's residual oscillation tendencies and characteristics in calm air as specified in MIL-STD-1797, Residual Oscillations paragraph for selected combinations of regions of handling qualities, air vehicle states, and flight phases.

6.1.4.17 Ride qualities.

Standard: The air vehicle meets the standards as specified in MIL-STD-1797, Ride Qualities paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis and simulation to assess the maximum ride discomfort index as specified in MIL-STD-1797, Ride Qualities paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's ride qualities as specified in MIL-STD-1797, Ride Qualities paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's ride qualities as specified in MIL-STD-1797, Ride Qualities paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.18 Pilot control inceptor characteristics.

Standard: Pilot control inceptor characteristics do not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the cockpit controller characteristics as specified in MIL-STD-1797, Cockpit Controller Characteristics paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the cockpit controller characteristics as specified in MIL-STD-1797, Cockpit Controller Characteristics paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the cockpit controller characteristics as specified in MIL-STD-1797, Cockpit Controller Characteristics paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.19 Pitch inceptor characteristics.

Standard: Pitch inceptor characteristics do not result in flying qualities worse than Level 2.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control forces and displacements as specified in MIL-STD-1797, Longitudinal Control Forces and Displacements paragraph, as well as its eight subparagraphs, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle flying/handling qualities for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle flying/handling qualities for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.20 Roll inceptor characteristics.

Standard: Roll inceptor characteristics do not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control forces and displacements as specified in MIL-STD-1797, Lateral-Directional Control with Speed Changes paragraph, Lateral-Directional Control with Speed Changes-Asymmetric Loads paragraph, and Lateral-Directional Control Forces and Displacements paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle flying/handling qualities for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle flying/handling qualities for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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6.1.4.21 Yaw inceptor characteristics.

Standard: Yaw inceptor characteristics do not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control forces and displacements as specified in MIL-STD-1797, Lateral-Directional Control with Speed Changes paragraph, Lateral-Directional Control with Speed Changes-Asymmetric Loads paragraph, and Lateral-Directional Control Forces and Displacements paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle flying/handling qualities for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle flying/handling qualities for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.22 Axial inceptor characteristics.

Standard: Axial inceptor characteristics do not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the axial inceptor control forces and displacements for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle flying/handling qualities for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess the air vehicle flying/handling qualities for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.23 Alternate control modes.

Standard: The air vehicle's motion transients and the control forces required to maintain attitude following a control mode change are no greater than the limits, including the time component, as specified in MIL-STD-1797, Transfer to Alternate Control Modes paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the motion transients resulting from intentional mode changes and the changes in control forces required to maintain attitude and sideslip following an intentional mode change as specified in MIL-STD-1797, Transfer to Alternate Control Modes paragraph for each combination of regions of handling qualities, flight phases, and environmental conditions with the air vehicle in the normal states.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the motion transients resulting from intentional mode changes and the changes in control forces required to maintain attitude and sideslip following an intentional mode change as specified in MIL-STD-1797, Transfer to Alternate Control Modes paragraph for each combination of regions of handling qualities, flight phases, and environmental conditions with the air vehicle in the normal states.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the motion transients resulting from intentional mode changes and the changes in control forces required to maintain attitude and sideslip following an intentional mode change as specified in MIL-STD-1797, Transfer to Alternate Control Modes paragraph for selected combinations of regions of handling qualities, flight phases, and environmental conditions with the air vehicle in the normal states.

6.1.4.24 Augmentation systems.

Standard: Operation of stability augmentation and CASs and devices, including in the presence of saturation or rate limiting, do not result in flying qualities worse than Level 2 in up to uncommon atmospheric disturbances. Control rates are able to retain stabilization and control in extraordinary atmospheric disturbances.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis and simulation to assess the augmentation system's robustness as specified in MIL-STD-1797, Rate of Control Surface Displacement paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the augmentation system's robustness as specified in MIL-STD-1797, Rate of Control Surface Displacement paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the augmentation system's robustness as specified in MIL-STD-1797, Rate of Control Surface Displacement paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.25 Trim system characteristics.

Standard: The air vehicle's trim system characteristics do not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the trim system characteristics as specified in MIL-STD-1797, Trim System paragraph, Trim System Irreversibility paragraph, Rate of Trim Operation paragraph, Stalling of Trim Systems paragraph, Trim for Asymmetric Thrust paragraph, and Automatic Trim System paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the trim system characteristics as specified in MIL-STD-1797, Trim System paragraph, Trim System Irreversibility paragraph, Rate of Trim Operation paragraph, Stalling of Trim Systems paragraph, Trim for Asymmetric Thrust paragraph, and Automatic Trim System paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the trim system characteristics as specified in MIL-STD-1797, Trim System paragraph, Trim System Irreversibility paragraph, Rate of Trim Operation paragraph, Stalling of Trim Systems paragraph, Trim for Asymmetric Thrust

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paragraph, and Automatic Trim System paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.26 Secondary control devices and configuration changes.

Standard: The transients and steady-state trim changes for normal operation of secondary control devices do not result in air vehicle worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the characteristics of secondary control devices and impact on flying qualities due to in-flight configuration changes as specified in MIL-STD-1797, Transient Response Limits to Configuration or Control Mode Change table and Control Force Limits for Configuration or Control Mode Change table, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the characteristics of secondary control devices and impact on flying/handling qualities due to in-flight configuration changes as specified in MIL-STD-1797, Transient Response Limits to Configuration or Control Mode Change table and Control Force Limits for Configuration or Control Mode Change table, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the characteristics of secondary control devices and impact on flying/handling qualities due to in-flight configuration changes as specified within MIL-STD-1797, Transient Response Limits to Configuration or Control Mode Change table and Control Force Limits for Configuration or Control Mode Change table, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.4.27 Pitch trim changes.

Standard: The pitch controller force required to maintain the trim conditions for at least 5 seconds following the initiation of a configuration change does not exceed the Level 2 limits as specified in MIL-STD-1797, Pitch Trim Changes paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the pitch control forces to maintain the trim condition after a configuration change as specified in MIL-STD-1797, Pitch Trim Changes

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paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the pitch control forces to maintain the trim condition after a configuration change as specified in MIL-STD-1797, Pitch Trim Changes paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight to assess the pitch control forces to maintain the trim condition after a configuration change as specified in MIL-STD-1797, Pitch Trim Changes paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5 Fixed-wing longitudinal flying qualities.

6.1.5.1 Long-term longitudinal response.

Standard: Long-term longitudinal time responses with a period of 15 seconds or longer has a positive damping ratio, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2. There are no unstable aperiodic modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's long-term longitudinal dynamics as specified in MIL-STD-1797, Phugoid Dynamics paragraph and Long-Term Longitudinal Response paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's long-term longitudinal dynamics as specified in MIL-STD-1797, Phugoid Dynamics paragraph and Long-Term Longitudinal Response paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's long-term longitudinal dynamics as specified in MIL-STD-1797, Phugoid Dynamics paragraph and Long-Term Longitudinal Response paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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6.1.5.2 Short-term longitudinal response.

Standard: Short-term longitudinal responses are no worse than Level 2 as specified in MIL-STD-1797, Short-Period Dynamics paragraph and Short-Term Pitch Response to Pitch Controller paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's short-term longitudinal responses as specified in MIL-STD-1797, Short-Period Dynamics paragraph and Short-Term Pitch Response to Pitch Controller paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's short-term longitudinal responses as specified in MIL-STD-1797, Short-Period Dynamics paragraph and Short-Term Pitch Response to Pitch Controller paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's short-term longitudinal responses as specified in MIL-STD-1797, Short-Period Dynamics paragraph and Short-Term Pitch Response to Pitch Controller paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.3 Effective pitch time delay.

Standard: The air vehicle's effective pitch time delay does not exceed 0.17 seconds, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's effective pitch time delay as specified in MIL-STD-1797, Short-Term Pitch Response to Pitch Controller paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's effective pitch time delay as specified in MIL-STD-1797, Short-Term Pitch Response to Pitch Controller

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paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's effective pitch time delay as specified in MIL-STD-1797, Short-Term Pitch Response to Pitch Controller paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.4 Steady-state flight path response.

Standard: In the landing approach flight phase, the air vehicle flight path angle versus true airspeed slope is no greater than 0.15 degrees per knot, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's steady-state flight path and airspeed response as specified in MIL-STD-1797, Steady-State Flight-Path Response to Pitch Controller paragraph for each combination of air vehicle states and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's steady-state flight path and airspeed response as specified in MIL-STD-1797, Steady-State Flight-Path Response to Pitch Controller paragraph for each combination of air vehicle states and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's steady-state flight path and airspeed response as specified in MIL-STD-1797, Steady-State Flight-Path Response to Pitch Controller paragraph for selected combinations of air vehicle states and environmental conditions.

6.1.5.5 Airspeed response due to pitch attitude changes.

Standard: The air vehicle's airspeed does not diverge when the pitch attitude is disturbed from trim, unless the time for the airspeed perturbation to double amplitude following a disturbance from trim is greater than six seconds, and further modified as specified in MIL-STD-1797, Relaxation in Transonic Flight paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's airspeed response due to pitch attitude changes as specified in MIL-STD-1797, Direction of Speed Change paragraph and Relaxation in Transonic Flight paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's airspeed response due to pitch attitude changes as specified in MIL-STD-1797, Direction of Speed Change paragraph and Relaxation in Transonic Flight paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's airspeed response due to pitch attitude changes as specified in MIL-STD-1797, Direction of Speed Change paragraph and Relaxation in Transonic Flight paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.6 Pitch attitude bandwidth.

Standard: The air vehicle bandwidth and phase delay of the pitch attitude response to pitch control input is no worse than Level 2 as specified in MIL-STD-1797, Pitch Attitude Bandwidth paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's pitch attitude bandwidth and phase delay as specified in MIL-STD-1797, Pitch Attitude Bandwidth paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's pitch attitude bandwidth and phase delay as specified in MIL-STD-1797, Pitch Attitude Bandwidth paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's pitch attitude bandwidth and phase delay as specified in MIL-STD-1797, Pitch Attitude Bandwidth paragraph for selected

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combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.7 Normal acceleration at the pilot station.

Standard: The dynamic normal acceleration response at the pilot station to the pitch controller does not result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the center of rotation in pitch is appropriately situated in relation to the pilot station as specified in MIL-STD-1797, Normal Acceleration at the Pilot Station paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the dynamic normal acceleration response at the pilot station as specified in MIL-STD-1797, Normal Acceleration at the Pilot Station paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the dynamic normal acceleration response at the pilot station as specified in MIL-STD-1797, Normal Acceleration at the Pilot Station paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.8 Pitch control power throughout the range of attainable angle of attack.

Standard: The air vehicle's control power, control surface rate, and hinge moment capability is sufficient for recovery throughout the range of attainable AOA to prevent the occurrence of a deep stall. For all normal states, the air vehicle exhibits no deep stall trim point within the CG limits and has no objectionable nose-up or nose-down recovery characteristics, such as positive or negative AOA hang-up.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess aerodynamic control power, control surface rate, and hinge moment capability as specified within MIL-STD-1797, Longitudinal Control Margin paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that aerodynamic control power, control surface rate, and hinge moment capability as specified within MIL-STD-1797, Longitudinal Control Margin paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess that aerodynamic control power, control surface rate, and hinge moment capability as specified within MIL-STD-1797, Longitudinal Control Margin paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.9 Longitudinal control power throughout speed range.

Standard: The air vehicle longitudinal control power does not limit the ability to attain and hold any speed from Vs to Vmax in steady 1g flight at any altitude within the ROTH.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess control power required to achieve pitch trim from Vs to Vmax in steady 1g flight in the ROTH as specified in MIL-STD-1797, Longitudinal Control Power in Unaccelerated Flight paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess control power required to achieve pitch trim from Vs to Vmax in steady 1g flight within the ROTH as specified in MIL-STD-1797, Longitudinal Control Power in Unaccelerated Flight paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess control power required to achieve pitch trim from Vs to Vmax in steady 1g flight within the ROTH as specified in MIL-STD-1797, Longitudinal Control Power in Unaccelerated Flight paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.10 Pitch control power throughout normal acceleration range.

Standard: The air vehicle, by the use of pitch control alone, is capable of achieving no worse than Level 2 Load Factor Range in a maneuvering flight within the ROSH as specified in MIL-

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STD-1797, Longitudinal Control Power in Maneuvering Flight paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess pitch control required for a maneuvering flight within the ROSH as specified in MIL-STD-1797, Longitudinal Control Power in Maneuvering Flight paragraph for each combination of air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess pitch control required for a maneuvering flight within the ROSH as specified in MIL-STD-1797, Longitudinal Control Power in Maneuvering Flight paragraph for each combination of air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess pitch control required for a maneuvering flight within the ROSH as specified in MIL-STD-1797, Longitudinal Control Power in Maneuvering Flight paragraph for selected combinations of air vehicle states, flight phases, and environmental conditions.

6.1.5.11 Abrupt pitch onset.

Standard: Following an abrupt maximum pitch control step input to elevated LFs, the air vehicle LF onset does not result in pilot g-LOC, air vehicle departure, or exceedance of LF limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the flying qualities and the susceptibility to pilot g-LOC, departure from controlled flight, or structural exceedances following an abrupt maximum pitch control step input to elevated LFs for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the flying/handling qualities and the susceptibility to pilot g-LOC, departure from controlled flight, or structural exceedances following an abrupt maximum pitch control step input to elevated LFs for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess the flying/handling qualities and the susceptibility to pilot g-LOC, departure from controlled flight, or structural exceedances following an abrupt maximum pitch control step input to elevated LFs for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.12 Pitch control for takeoff.

Standard: The air vehicle has sufficient pitch control to prevent overrotation (tail strike and stall) with handling qualities no worse than Level 2 during all types of takeoff as specified in MIL-STD-1797, Longitudinal Control Power for Takeoff paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the pitch control required during a takeoff as specified in MIL-STD-1797, Longitudinal Control Power for Takeoff paragraph for each combination of air vehicle states, runway conditions, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the pitch control required during a takeoff as specified in MIL-STD-1797, Longitudinal Control Power for Takeoff paragraph for each combination of air vehicle states, runway conditions, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the pitch control required during a takeoff as specified in MIL-STD-1797, Longitudinal Control Power for Takeoff paragraph for selected combinations of air vehicle states, runway conditions, and environmental conditions.

6.1.5.13 Pitch control for assisted takeoff.

Standard: The air vehicle has sufficient control power to maintain control (pitch attitude, AOA, bank angle, heading, pitch rate, and specific excess energy) during launch assisted takeoffs through transition to normal flight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control power, engine response, and margin to stall speed needed to maintain control and complete the flyaway from minimum end airspeed to maximum speed (to include any potential system motion effects, 2-D performance analysis for pitch-rate control, 6-DOF simulations for automatic control and trim settings for controls free

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launches, and release dynamics) for each combination of air vehicle states, control modes, environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the control power, engine response, and margin to stall speed needed to maintain control and complete the flyaway from minimum end airspeed to maximum speed, including any potential system motion effects, automatic control and trim settings for controls free launches, and release dynamics, for each combination of air vehicle states, control modes, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the control power, engine response, and margin to stall speed needed to maintain control and complete the flyaway from minimum end airspeed to maximum speed, including any potential system motion effects, automatic control and trim settings for controls free launches, and release dynamics, for selected combinations of air vehicle states, control modes, and environmental conditions.

6.1.5.14 Pitch control for landing.

Standard: The air vehicle has sufficient pitch control during a landing to maintain pitch attitude and AOA to prevent overrotation (stall and tail strike), to ensure landing dynamics do not exceed landing gear limits, and results in handling qualities no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the pitch control required during a landing, in and out of ground effect, as specified in MIL-STD-1797, Longitudinal Control Power in Landing paragraph for each combination of air vehicle states, runway conditions, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the pitch control required during a landing as specified in MIL-STD-1797, Longitudinal Control Power in Landing paragraph for each combination of air vehicle states, runway conditions, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the pitch control required during a landing as specified in MIL-STD-1797, Longitudinal Control Power in Landing paragraph for selected combinations of air vehicle states, runway conditions, and environmental conditions.

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6.1.5.15 Pitch control for alternate recovery methods.

Standard: The air vehicle has sufficient control power to maintain control (pitch attitude, AOA, bank angle, heading, pitch rate, and specific excess energy) during approach to alternate or unique recovery methods.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the pitch control required to maintain control during an approach to alternate or unique recovery methods, in and out of ground effect, for each combination of regions of handling qualities, air vehicle states, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the pitch control required to maintain control during an approach to alternate or unique recovery methods for each combination of regions of handling qualities, air vehicle states, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the pitch control required to maintain control during an approach to alternate or unique recovery methods for selected combinations of regions of handling qualities, air vehicle states, and environmental conditions.

6.1.5.16 Control force variations during rapid speed changes.

Standard: The air vehicle flying qualities are no worse than Level 2 as it accelerates and decelerates rapidly through the operational speed range to maintain the desired LF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the flying qualities due to pitch control variations during a rapid speed change as specified in MIL-STD-1797, Control Force Variations during Rapid Speed Changes paragraph for each combination of regions of handling qualities, air vehicle states, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the flying/handling qualities due to pitch control variations during a rapid speed change as specified in MIL-STD-1797, Control Force Variations during Rapid Speed Changes paragraph for each combination of regions of handling qualities, air vehicle states, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess the flying/handling qualities due to pitch control variations during a rapid speed change as specified in MIL-STD-1797, Control Force Variations during Rapid Speed Changes paragraph for selected combinations of regions of handling qualities, air vehicle states, and environmental conditions.

6.1.5.17 Longitudinal control forces and displacements in takeoff.

Standard: The air vehicle's longitudinal control force and travel in takeoff meets the standards as specified in MIL-STD-1797, Longitudinal Control Force and Travel in Takeoff paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the controller forces and travel in takeoff as specified in MIL-STD-1797, Longitudinal Control Force and Travel in Takeoff paragraph for each combination of air vehicle states, runway conditions, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the flying/handling qualities during a takeoff as specified in MIL-STD-1797, Longitudinal Control Force and Travel in Takeoff paragraph for each combination of air vehicle states, runway conditions, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the flying/handling qualities during a takeoff as specified in MIL-STD-1797, Longitudinal Control Force and Travel in Takeoff paragraph for selected combinations of air vehicle states, runway conditions, and environmental conditions.

6.1.5.18 Longitudinal control forces in dives and recoveries.

Standard: Longitudinal control forces in dives and recoveries result in air vehicle flying qualities no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the controller force limits in dives and recoveries as specified in MIL-STD-1797, Longitudinal Control Force Limits in Dives within the ROTH paragraph and Longitudinal Control Force Limits in Dives within the RORH paragraph, for each combination of air vehicle states, flight phases, and environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the flying/handling qualities in dives as specified in MIL-STD-1797, Longitudinal Control Force Limits in Dives within the ROTH paragraph and Longitudinal Control Force Limits in Dives within the RORH paragraph, for each combination of air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the flying/handling qualities in dives as specified in MIL-STD-1797, Longitudinal Control Force Limits in Dives within the ROTH paragraph and Longitudinal Control Force Limits in Dives within the RORH paragraph, for selected combinations of air vehicle states, flight phases, and environmental conditions.

6.1.5.19 Flight path response to designated flight path controller.

Standard: The air vehicle flight path response to designated flight path controller is no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's flight path response as specified in MIL-STD-1797, Flight Path Response to Designated Flight Path Controller paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's flight path response as specified in MIL-STD-1797, Flight Path Response to Designated Flight Path Controller paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's flight path response as specified in MIL-STD-1797, Flight Path Response to Designated Flight Path Controller paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.20 Flight path control power.

Standard: The air vehicle's flight path control power is no worse than Level 2 as specified in MIL-STD-1797, Flight Path Control Power paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's flight path control power as specified in MIL-STD-1797, Flight Path Control Power paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's flight path control power as specified in MIL-STD-1797, Flight Path Control Power paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's flight path control power as specified in MIL-STD-1797, Flight Path Control Power paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.5.21 Flight path controller characteristics.

Standard: In all flight conditions the pilot applied force and deflection required to maintain a change in flight path are in the same sense as those required to initiate the change.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control force and deflection characteristics for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the control force and deflection characteristics for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the control force and deflection characteristics for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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6.1.6 Fixed-wing lateral-directional flying qualities.

6.1.6.1 Roll mode time constant.

Standard: The air vehicle's maximum roll mode time constants are no worse than Level 2 as specified in MIL-STD-1797, Roll Mode paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the equivalent roll mode time constant as specified in MIL-STD-1797, Roll Mode paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the equivalent roll mode time constant as specified in MIL-STD-1797, Roll Mode paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the equivalent roll mode time constant as specified in MIL-STD-1797, Roll Mode paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.2 Dutch roll characteristics.

Standard: The air vehicle's Dutch roll characteristics are no worse than Level 2 as specified in MIL-STD-1797, Dutch Roll Frequency and Damping paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's Dutch roll characteristics as specified in MIL-STD-1797, Dutch Roll Frequency and Damping paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's Dutch roll characteristics as specified in MIL-STD-1797, Dutch Roll Frequency and Damping paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's Dutch roll characteristics as specified in MIL-STD-1797, Dutch Roll Frequency and Damping paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.3 Spiral stability.

Standard: From a wings-level attitude following a disturbance in bank angle of up to 20 degrees, the time for the bank angle to double is at least 8 seconds, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's spiral characteristics as specified in MIL-STD-1797, Spiral Stability paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's spiral characteristics as specified in MIL-STD-1797, Spiral Stability paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's spiral characteristics as specified in MIL-STD-1797, Spiral Stability paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.4 Roll-spiral characteristics.

Standard: The air vehicle does not exhibit a coupled roll-spiral mode in Category A Flight Phases, and the air vehicle's roll-spiral damping coefficients for Category B and C Flight Phases are no less than 0.3 rad/s as specified in MIL-STD-1797, Coupled Roll-Spiral Oscillation paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's roll-spiral damping as specified in MIL-STD-1797, Coupled Roll-Spiral Oscillations paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's roll-spiral damping as specified in MIL-STD-1797, Coupled Roll-Spiral Oscillations paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's roll-spiral damping as specified in MIL-STD-1797, Coupled Roll-Spiral Oscillations paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.5 Equivalent roll time delay.

Standard: The air vehicle's equivalent roll time delay does not exceed 0.2 seconds, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's equivalent roll time delay as specified in MIL-STD-1797, Roll Time Delay paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's equivalent roll time delay as specified in MIL-STD-1797, Roll Time Delay paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's equivalent roll time delay as specified in MIL-STD-1797, Roll Time Delay paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.6 Equivalent yaw time delay.

Standard: The air vehicle's equivalent yaw time delay does not exceed 0.2 seconds, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's equivalent yaw time delay as specified in MIL-STD-1797, Yaw Time Delay paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's equivalent yaw time delay as specified in MIL-STD-1797, Yaw Time Delay paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's equivalent yaw time delay as specified in MIL-STD-1797, Yaw Time Delay paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.7 Heading attitude bandwidth for ground operations.

Standard: While on the ground, the air vehicle heading attitude bandwidth, defined as the highest frequency at which the air vehicle has less than 45 degrees of phase lag between heading command input and air vehicle yaw response. The bandwidth is no less than 1 radian per second at 40 kts, varying linearly to 2 radians per second at 120 kts and extrapolated down to administrative taxi speeds and up to the maximum landing touchdown speed, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2. Parameters of interest are phase and gain bandwidth frequency and phase delay.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's heading attitude bandwidth using heading command control effector frequency sweeps at discrete groundspeeds (described in rad/s) for each combination of air vehicle states, runway conditions, and environmental conditions. Heading attitude bandwidth should be analyzed as a function of speed with and without braking.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's heading attitude bandwidth using heading command control effector frequency sweeps at discrete airspeeds for each combination of air vehicle states, runway conditions, tire pressures and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight or Ground test: Conduct ground or flight test to assess the air vehicle's heading attitude bandwidth using heading command control effector frequency sweeps at discrete airspeeds for selected combinations of air vehicle states, runway conditions, tire pressures, and environmental conditions.

6.1.6.8 Roll rate response to small roll commands.

Standard: The air vehicle's roll rate response to small roll inputs is no worse than Level 2 as specified in MIL-STD-1797, Roll Rate Response to Small Roll Commands paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's roll rate response as specified in MIL-STD-1797, Roll Rate Response to Small Roll Commands paragraph and Linearity of Roll Response to Roll Controller paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's roll rate response as specified in MIL-STD-1797, Roll Rate Response to Small Roll Commands paragraph and Linearity of Roll Response to Roll Controller paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's roll rate response as specified in MIL-STD-1797, Roll Rate Response to Small Roll Commands paragraph and Linearity of Roll Response to Roll Controller paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.9 Bank angle response to roll commands.

Standard: The air vehicle's bank angle response to roll inputs is no worse than Level 2 as specified in MIL-STD-1797, Bank Angle Response to Roll Commands paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's bank angle response as specified in MIL-STD-1797, Bank Angle Response to Roll Commands paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's bank angle response as specified in MIL-STD-1797, Bank Angle Response to Roll Commands paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's bank angle response as specified in MIL-STD-1797, Bank Angle Response to Roll Commands paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.10 Roll rate response to large roll commands.

Standard: The air vehicle's roll rate response to large roll inputs is no worse than Level 2 as specified in MIL-STD-1797, Roll Rate Response to Large Roll Commands paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2. Abrupt roll terminations do not cause air vehicle motions which result in roll reversal, roll ratcheting, loss of control, stall, or exceedance of structural limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's roll rate response as specified in MIL-STD-1797, Roll Rate Response to Large Roll Commands paragraph, Linearity of Roll Response to Roll Controller paragraph, and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's roll rate response as specified in MIL-STD-1797, Roll Rate Response to Large Roll Commands paragraph, Linearity of Roll Response to Roll Controller paragraph, and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's roll rate response as specified in MIL-STD-1797, Roll Rate Response to Large Roll Commands paragraph, Linearity of Roll Response to Roll Controller paragraph, and Roll Termination paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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6.1.6.11 Yaw rate response to small roll commands.

Standard: The air vehicle's yaw response to small roll inputs is no worse than Level 2 as specified in MIL-STD-1797, Yaw Response to Small Roll Control Commands paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's yaw response as specified in MIL-STD-1797, Yaw Rate Response to Small Roll Control Commands paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's yaw response as specified in MIL-STD-1797, Yaw Rate Response to Small Roll Control Commands paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's yaw response as specified in MIL-STD-1797, Yaw Rate Response to Small Roll Control Commands paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.12 Yaw rate response to large roll commands.

Standard: The air vehicle's yaw response to large roll inputs is no worse than Level 2 as specified in MIL-STD-1797, Yaw Response to Large Roll Control Commands paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2. Abrupt roll terminations do not cause air vehicle motions that result in roll reversal, roll ratcheting, loss of control, stall, or exceedance of structural limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's yaw rate response as specified in MIL-STD-1797, Yaw Response to Large Roll Control Commands paragraph and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's yaw rate response as specified in MIL-STD-1797, Yaw Response to Large Roll Control Commands paragraph and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's yaw rate response as specified in MIL-STD-1797, Yaw Response to Large Roll Control Commands paragraph and Roll Termination paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.13 Turn coordination.

Standard: The air vehicle's turn coordination characteristics are no worse than Level 2 as specified in MIL-STD-1797, Coordination in Turn Entry and Exit paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's turn coordination characteristics as specified in MIL-STD-1797, Coordination in Turn Entry and Exit paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's turn coordination characteristics as specified in MIL-STD-1797, Coordination in Turn Entry and Exit paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's turn coordination characteristics as specified in MIL-STD-1797, Coordination in Turn Entry and Exit paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.14 Time to bank.

Standard: The air vehicle's time to bank from both wings-level and steady coordinated bank attitudes is no worse than Level 2 as specified in MIL-STD-1797, Roll Control Effectiveness

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paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2. Abrupt roll terminations do not cause air vehicle motions that result in roll reversal, roll ratcheting, loss of control, stall, or exceedance of structural limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's time to bank characteristics as specified in MIL-STD-1797, Roll Control Effectiveness paragraph and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's time to bank characteristics as specified in MIL-STD-1797, Roll Control Effectiveness paragraph and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's time to bank characteristics as specified in MIL-STD-1797, Roll Control Effectiveness paragraph and Roll Termination paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.15 Time to bank for class IV air vehicle.

Standard: For the Class IV air vehicle, the air vehicle's time to bank during 360-degree rolls initiated at 1g is no worse than Level 2 as specified in MIL-STD-1797, Additional Roll Requirements for Class IV Air Vehicles paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2. Abrupt roll terminations do not cause air vehicle motions that result in roll reversal, roll ratcheting, loss of control, stall, or exceedance of structural limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the Class IV air vehicle's time to bank characteristics in the air-to-air combat and ground attack flight phases as specified in MIL-STD-1797, Additional Roll Requirements for Class IV Air Vehicles paragraph and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to verify the Class IV air vehicle's time to bank characteristics in the air-to-air combat and ground attack flight phases as specified in MIL-STD-1797, Additional Roll Requirements for Class IV Air Vehicles paragraph and Roll Termination paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the Class IV air vehicle's time to bank characteristics in the air-to-air combat and ground attack flight phases as specified in MIL-STD-1797, Additional Roll Requirements for Class IV Air Vehicles paragraph and Roll Termination paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.16 Pedal force during wave off (go-around).

Standard: The pedal control force required to maintain straight flight during waveoff (go-around) due to thrust, configuration, and airspeed changes is no worse than Level 2 as specified in MIL-STD-1797, Yaw Control Forces in Waveoff (Go-Around) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's yaw control force during waveoff (go-around) as specified in MIL-STD-1797, Yaw Control Forces in Waveoff (Go-Around) paragraph for each combination of air vehicle states and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the yaw control forces in during waveoff (go-around) as specified in MIL-STD-1797, Yaw Control Forces in Waveoff (Go-Around) paragraph for each combination of air vehicle states and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the yaw control forces in during waveoff (go-around) as specified in MIL-STD-1797, Yaw Control Forces in Waveoff (Go-Around) paragraph for selected combinations of air vehicle states and environmental conditions.

6.1.6.17 Steady sideslips.

Standard: In steady sideslips, the air vehicle has conventional roll and yaw control forces and inceptor deflections, and bank angle response as specified in MIL-STD-1797, Steady Sideslips

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paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the roll and yaw control and air vehicle's bank angle response in steady sideslips as specified in MIL-STD-1797, Steady Sideslips paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the roll and yaw control and air vehicle's bank angle response in steady sideslips as specified in MIL-STD-1797, Steady Sideslips paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the roll and yaw control and air vehicle's bank angle response in steady sideslips as specified in MIL-STD-1797, Steady Sideslips paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.18 Lateral-directional control in crosswinds.

Standard: During takeoff and landing, roll and yaw control forces in up to maximum crosswind are no worse than Level 2 as specified in MIL-STD-1797, Lateral-Directional Control in Crosswinds paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the roll and yaw control forces required for a takeoff or landing in up to maximum crosswinds as specified in MIL-STD-1797, Lateral-Directional Control in Crosswinds paragraph, Final Approach in Crosswinds paragraph, Takeoff Run and Landing Rollout in Crosswinds paragraph, and Taxiing Wind Speed Limits paragraph, for each combination of air vehicle states, runway conditions, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the roll and yaw control forces required for a takeoff or landing in up to maximum crosswinds as specified in MIL-STD-1797, Lateral-Directional Control in Crosswinds paragraph, Final Approach in Crosswinds paragraph, Takeoff Run and Landing Rollout in Crosswinds paragraph, and Taxiing Wind Speed Limits

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paragraph, for each combination of air vehicle states, runway conditions, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the roll and yaw control forces required for a takeoff or landing in up to maximum crosswinds as specified in MIL-STD-1797, Lateral-Directional Control in Crosswinds paragraph, Final Approach in Crosswinds paragraph, Takeoff Run and Landing Rollout in Crosswinds paragraph, and Taxiing Wind Speed Limits paragraph, for selected combinations of air vehicle states, runway conditions, and environmental conditions.

6.1.6.19 Minimum control speed-ground.

Standard: Following sudden asymmetric loss of thrust from any factor while on the ground, the air vehicle is controllable with no worse than Level 2 flying qualities, including in crosswinds up to takeoff and landing maximum limits from the unfavorable direction, as specified in MIL-STD-1797, Lateral-Directional Control with Asymmetric Thrust paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's lateral-directional control for takeoff and landing with asymmetric thrust as specified in MIL-STD-1797, Lateral-Directional Control with Asymmetric Thrust paragraph, Thrust Loss during Takeoff Run paragraph, Thrust Loss after Takeoff paragraph, Thrust Loss during Waveoff/Go-Around paragraph, and Two-Engine Failures in Multi-Engine Air Vehicles paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, runway conditions, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's lateral-directional control for takeoff and landing with asymmetric thrust as specified in MIL-STD-1797, Lateral-Directional Control with Asymmetric Thrust paragraph, Thrust Loss during Takeoff Run paragraph, Thrust Loss after Takeoff paragraph, Thrust Loss during Waveoff/Go-Around paragraph, and Two-Engine Failures in Multi-Engine Air Vehicles paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, runway conditions, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's lateral-directional control for takeoff and landing with asymmetric thrust as specified in MIL-STD-1797, Lateral-Directional Control with Asymmetric Thrust paragraph, Thrust Loss during Takeoff Run paragraph, Thrust Loss after Takeoff paragraph, Thrust Loss during Waveoff/Go-Around paragraph, and Two-Engine

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Failures in Multi-Engine Air Vehicles paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, runway conditions, and environmental conditions.

6.1.6.20 Minimum control speed-air.

Standard: Following sudden asymmetric loss of thrust from any factor while in the air, the air vehicle is controllable with no worse than Level 2 flying qualities as specified in MIL-STD-1797, Lateral-Directional Control with Asymmetric Thrust paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's lateral-directional control with asymmetric thrust as specified in MIL-STD-1797, Yaw Controls Free paragraph and Two-Engine Failures in Multi-Engine Air Vehicles paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's lateral-directional control for takeoff and landing with asymmetric thrust as specified in MIL-STD-1797, Yaw Controls Free paragraph and Two-Engine Failures in Multi-Engine Air Vehicles paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's lateral-directional control for takeoff and landing with asymmetric thrust as specified in MIL-STD-1797, Yaw Controls Free paragraph and Two-Engine Failures in Multi-Engine Air Vehicles paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.6.21 Air vehicle response to direct side force controller/mode.

Standard: The air vehicle's lateral-directional (e.g., lateral nose-pointing, wings-level turn) and lateral-translational response to the direct side force controller/mode is no worse than Level 2 as specified in MIL-STD-1797, Wings-Level Turn paragraph and Lateral Translation paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's lateral-directional and lateral-translational response to the direct side force controller as specified in MIL-STD-1797, Wings-

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Level Turn paragraph and Lateral Translation paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's lateral-directional and lateral-translational response to the direct side force controller as specified in MIL-STD-1797, Wings-Level Turn paragraph and Lateral Translation paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's lateral-directional and lateral-translational response to the direct side force controller as specified in MIL-STD-1797, Wings-Level Turn paragraph and Lateral Translation paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.7 Fixed-wing cross-axis response.

6.1.7.1 Longitudinal control in sideslips.

Standard: The longitudinal control forces in sideslip are no worse than Level 2 as specified in MIL-STD-1797, Longitudinal Control Forces in Sideslips paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the longitudinal control force required to maintain a constant air speed in steady sideslips as specified in MIL-STD-1797, Longitudinal Control Forces in Sideslips paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the longitudinal control force required to maintain a constant air speed in steady sideslips as specified in MIL-STD-1797, Longitudinal Control Forces in Sideslips paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the longitudinal control force required to maintain a constant air speed in steady sideslips as specified in MIL-STD-1797, Longitudinal Control

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Forces in Sideslips paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.7.2 Lateral-directional control in dives and pullouts.

Standard: The roll and yaw control forces to maintain wings-level and zero sideslip, without retrimming, throughout dives and recoveries within the ROTH are no worse than the limits as specified in MIL-STD-1797, Lateral-Directional Control in Dives and Pullouts paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the roll and yaw control forces in dives and pullouts, including in the ROTH, as specified in MIL-STD-1797, Lateral-Directional Control in Dives and Pullouts paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the roll and yaw control forces in dives and pullouts, including in the ROTH, as specified in MIL-STD-1797, Lateral-Directional Control in Dives and Pullouts paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the roll and yaw control forces in dives and pullouts, including in the ROTH, as specified in MIL-STD-1797, Lateral-Directional Control in Dives and Pullouts paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.7.3 Cross-axis coupling in roll.

Standard: The air vehicle response due to cross-axis coupling in roll maneuvers is no worse than Level 2 and does not lead to exceedance of structural limits, roll autorotation, or departure from controlled flight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle response due to cross-axis coupling in roll maneuvers as specified in MIL-STD-1797, Cross-Axis Coupling in Roll Maneuvers paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle response due to cross-axis coupling in roll maneuvers as specified in MIL-STD-1797, Cross-Axis Coupling in Roll Maneuvers paragraph for each combination of regions of handling qualities air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle response due to cross-axis coupling in roll maneuvers as specified in MIL-STD-1797, Cross-Axis Coupling in Roll Maneuvers paragraph for selected combinations of regions of handling qualities air vehicle states, flight phases, and environmental conditions.

6.1.7.4 Pitch and roll controller crosstalk.

Standard: The pitch and roll control force and displacement sensitivities and breakout forces are such that intentional inputs to one control axis do not cause inputs into the other axis that result in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the controller crosstalk characteristics for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the controller crosstalk characteristics for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the controller crosstalk characteristics for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.7.5 Control harmony.

Standard: Simultaneous control forces do not exceed the limits as specified in MIL-STD-1797, Control Harmony paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control forces required to execute maneuvers requiring combined control input as specified in MIL-STD-1797, Control Harmony paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the control forces required to execute maneuvers requiring combined control input as specified in MIL-STD-1797, Control Harmony paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the control forces required to execute maneuvers requiring combined control input as specified in MIL-STD-1797, Control Harmony paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.7.6 Control cross-coupling.

Standard: Controllers (continuous or discrete) that affect the motion of the air vehicle do not create a secondary response which results in flying qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the secondary response due to controller inputs as specified in MIL-STD-1797, Control Cross-Coupling paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the secondary response due to controller inputs as specified in MIL-STD-1797, Control Cross-Coupling paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess the secondary response due to controller inputs as specified in MIL-STD-1797, Control Cross-Coupling paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8 Fixed-wing high angle of attack.

6.1.8.1 Stall and loss of control warning.

Standard: Warning or indication of approach to stall or to departure from controlled flight are clear and unambiguous, and the onset of stall warning occurs within the speed and lift range as specified in MIL-STD-1797, Stall Approach paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the timeliness and appropriateness of warning and indication of approach to stall or loss of air vehicle control as specified in MIL-STD-1797, Warning Cues paragraph, Stall Approach paragraph, Stall Warning paragraph, Uncommanded Oscillations prior to Stall paragraph, Cockpit Controls prior to Stall paragraph, and Departure Warning paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the timeliness and appropriateness of warning and indication of approach to stall or loss of air vehicle control as specified in MIL-STD-1797, Warning Cues paragraph, Stall Approach paragraph, Stall Warning paragraph, Uncommanded Oscillations prior to Stall paragraph, Cockpit Controls prior to Stall paragraph, and Departure Warning paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the timeliness and appropriateness of warning and indication of approach to stall or loss of air vehicle control as specified in MIL-STD-1797, Warning Cues paragraph, Stall Approach paragraph, Stall Warning paragraph, Uncommanded Oscillations prior to Stall paragraph, Cockpit Controls prior to Stall paragraph, and Departure Warning paragraph, for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.2 Stall characteristics.

Standard: The air vehicle's stall characteristics meet the standards as specified in MIL-STD-1797, Stall Characteristics paragraph.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the stall characteristics as specified in MIL-STD-1797, Stall Characteristics paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the stall characteristics as specified in MIL-STD-1797, Stall Characteristics paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the stall characteristics as specified in MIL-STD-1797, Stall Characteristics paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.3 Stall prevention and recovery.

Standard: A stall can be prevented by the use of the pitch control alone at the onset of the stall warning, and the maximum control forces required for stall recovery do not exceed the limits as specified in MIL-STD-1797, Stall Recovery paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the stall prevention and recovery as specified in MIL-STD-1797, Stall Recovery paragraph and Control Power for Stall Recover paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the stall prevention and recovery as specified in MIL-STD-1797, Stall Recovery paragraph and Control Power for Stall Recover paragraph, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the stall prevention and recovery as specified in MIL-STD-1797, Stall Recovery paragraph and Control Power for Stall Recover paragraph, for

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selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.4 Stall recovery with the critical engine inoperative.

Standard: The multi-engine air vehicle can be recovered from stalls with the critical engine inoperative.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's recovery characteristics from stalls with the critical engine inoperative and thrust on the remaining engines at the settings as specified in MIL-STD-1797, One-Engine-Out Stalls for Multi-Engine Air Vehicles paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's recovery characteristics from stalls with the critical engine inoperative and thrust on the remaining engines at the settings as specified in MIL-STD-1797, One-Engine-Out Stalls for Multi-Engine Air Vehicles paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's recovery characteristics from stalls with the critical engine inoperative and thrust on the remaining engines at the settings as specified in MIL-STD-1797, One-Engine-Out Stalls for Multi-Engine Air Vehicles paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental condition.

6.1.8.5 Departure resistance.

Standard: The air vehicle is resistant to departure from controlled flight, post-stall gyrations, and spins. For trainers, the air vehicle is capable of a developed spin and consistent recovery.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's departure resistance as specified in MIL-STD-1797, Departure from Controlled Flight paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's departure resistance as specified in MIL-STD-1797, Departure from Controlled Flight paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's departure resistance as specified in MIL-STD-1797, Departure from Controlled Flight paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.6 Uncommanded motions.

Standard: During maneuvering up to the AOA and AOS limits, the air vehicle does not exhibit uncommanded motions that cannot be arrested promptly by simple application of pilot control.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's uncommanded motions, if encountered, up to the AOA and AOS limits as specified in MIL-STD-1797, Uncommanded Motions paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's uncommanded motions, if encountered, up to the AOA and AOS limits as specified in MIL-STD-1797, Uncommanded Motions paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's uncommanded motions, if encountered, up to the AOA and AOS limits as specified in MIL-STD-1797, Uncommanded Motions paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.7 Departure resistance to a sudden asymmetric loss of thrust.

Standard: Following sudden asymmetric loss of thrust from the most critical factor, a departure from controlled flight can be avoided.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle's departure resistance following a sudden asymmetric loss of thrust as specified in MIL-STD-1797, Departure Avoidance Following Sudden Asymmetric Loss of Thrust paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle's departure resistance following a sudden asymmetric loss of thrust as specified in MIL-STD-1797, Departure Avoidance Following Sudden Asymmetric Loss of Thrust paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle's departure resistance following a sudden asymmetric loss of thrust as specified in MIL-STD-1797, Departure Avoidance Following Sudden Asymmetric Loss of Thrust paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.8 Recovery from post-stall gyrations and spins.

Standard: The proper recovery techniques are easily apparent to the pilot, are simple and easy to apply under the motions encountered, and provide prompt recovery from all post-stall gyrations, incipient spins, and developed spins.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess recovery the techniques for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the recovery techniques for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the recovery techniques for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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6.1.8.9 Turns and altitude loss for spin recovery.

Standard: The number of turns and altitude loss for recovery for all modes of spin are no greater than the limits as specified in MIL-STD-1797, Turns and Altitude Loss for Recovery paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the number of turns and altitude loss for all modes of spin as specified in MIL-STD-1797, Turns and Altitude Loss for Recovery paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the number of turns and altitude loss for all modes of spin as specified in MIL-STD-1797, Turns and Altitude Loss for Recovery paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the number of turns and altitude loss for all modes of spin as specified in MIL-STD-1797, Turns and Altitude Loss for Recovery paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.10 Adverse spin mode.

Standard: Avoidance of a spin reversal or an adverse mode change does not depend upon precise pilot control, timing, or deflection.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess piloting techniques and timing to avoid adverse spin mode change or spin reversal for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess piloting techniques and timing to avoid adverse spin mode change or spin reversal for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess piloting techniques and timing to avoid adverse spin mode change or spin reversal for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.11 Control forces for recovery.

Standard: The control forces for recovery from post-stall gyrations and spins do not exceed the limits as specified in MIL-STD-1797, Control Forces for Recovery paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the control forces for recovery from post-stall gyrations and spins as specified in MIL-STD-1797, Control Forces for Recovery paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the control forces for recovery from post-stall gyrations and spins as specified in MIL-STD-1797, Control Forces for Recovery paragraph for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the control forces for recovery from post-stall gyrations and spins as specified in MIL-STD-1797, Control Forces for Recovery paragraph for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.8.12 Automatic stall and departure prevention and recovery functions.

Standard: Automatic stall, departure, and spin prevention and recovery functions perform as effective as or better than pilot initiated procedures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the effectiveness of the stall, departure, and spin prevention and recovery functions, for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the effectiveness of the stall, departure, and spin prevention and recovery functions for each combination of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the effectiveness of the stall, departure, and spin prevention and recovery functions for selected combinations of regions of handling qualities, air vehicle states, flight phases, and environmental conditions.

6.1.9 Shipboard operations.

Operating air vehicles from aviation ships has many unique characteristics related to air vehicle design, control, and handling qualities that the other flying qualities standards in this section do not adequately address. The most significant difference is that air vehicles will launch/takeoff and recover/land to or from a moving and highly constrained takeoff and landing area, which justifies additional standards to ensure safe air vehicle operations in the shipboard environment. These standards address the unique considerations required for shipboard systems like catapult and arresting equipment for fixed-wing air vehicles, securing and traversing of rotary-wing air vehicles, and visual landing aids for both the pilot and shipboard observers. Operational considerations extend further to special control law modes, ship motion (pitch, roll, heave), WOD, individual ship airwake effects, multi-air vehicle operations, waveoff, bolter, and air vehicle taxi movements on deck. Air vehicle operations around and aboard ships are highly choreographed by standard procedures and further refined by shipboard air traffic and deck handler control. Therefore, special consideration needs to be given to the influence of air vehicle and rotor air wakes and the ingestion of steam or jet exhaust and ship exhaust, from various relative angles, when operating on and around a ship. Additionally, certain failure modes deserve special consideration in the shipboard environment because land-based mitigations may not be viable. For example, if an air vehicle loses an engine (multi-engine only) or control surface during a carrier catapult stroke, it will still be able to attain a safe fly away speed and pitch attitude because it cannot abort the takeoff.

Most air vehicle design attributes like high strength internal structures, robust landing gear, launch bar, arresting hook, and air vehicle tiedown hardpoints that enable shipboard operations do not directly impact flying qualities. However, shipboard requirements often place physical constraints on the design that may cause handling qualities issues. For instance, a smaller than optimal wingspan may drive a high approach AOA, limiting the pilot's FOV, requiring more effective control surfaces and shipboard-specific flight control software for slow speed flight. Similarly, an engine optimized for range and endurance may have a slow engine response to throttle inputs, increasing the difficulty of an arrested landing. The balance between up and away mission performance and low speed shipboard operations is a complex design trade space, which may result in design compromises that are sub-optimal for both flight phases.

Some piloted air vehicles primarily use a precision landing mode in which the FCS targets air vehicle trajectory (e.g., glideslope) during an approach to reduce pilot workload and touchdown dispersion aboard ships. Rotary-wing air vehicles are increasingly employing highly augmented

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flight controls to similarly ease the pilot compensation of landing to ships. These piloted control modes and autonomous landing capabilities use ship relative navigation sources, like Joint Precision Approach and Landing System, as part of the flight control laws. Therefore, variations in ship relative navigation sources will be evaluated to identify total system dependencies and performance capabilities to maintain safe shipboard operations. Similarly, existing unaugmented control modes, backup systems, or reversionary modes will still be certified for shipboard operations and be evaluated during the AW process in case of blue water operators where a shore diversion is not possible. There should be no potential for cognitive error based on proficiency in a primary system and the sudden need to switch to a reversionary mode, or vice versa, during shipboard operations.

Certification for initial shipboard operations is the result of the combination of piloted and offline/desktop simulations, land-based flight test, and analysis against historical standards to show safety. A logical and measured flight test build-up from land-based, shipboard unique handling qualities tasks to shipboard testing, starting with benign conditions and building to more stressing conditions using unique tasks on land before attempting them on a ship. Shipboard capability does not happen by chance but rather is designed into the air vehicle from the outset and the standards herein assure safe operations from aviation ships. In addition to the flying qualities references identified in the 6.1 narrative, more information on shipboard specific terms, design considerations, and test and evaluation methods can be found in MIL-STD-3013, Technical Report NAWCADPAX/TR-2002/71, Carrier Suitability Test Manual (SA FTM-01), NAWCAD Dynamic Interface Test Manual (NTWL/25-13840), and MIL-STD-1797, Carrier Operations and V/STOL Specific Requirements paragraphs.

6.1.9.1 Shipboard ground handling.

Standard: The air vehicle has ground handling characteristics that are no worse than qualitative Level 2 when taxiing aboard ships.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify the pilot and air vehicle are able to taxi in confined spaces on land and aboard ships with various obstacles (e.g., other air vehicles and arresting wire), and in precise control scenarios (e.g., catapult lineup and parking), including direction control power, braking performance, aerodynamic effects, engine power and control, and timing delays from command (e.g., from the air vehicle director) to vehicle response for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: surface conditions (e.g., tire friction, wet, oil), ship motion, and WOD from all relative directions.

Test.

(1) Lab test: Mission representative ground handling tasks include maneuvering in the tight confines of the carrier hangar and flight deck, during flight deck taxiing, during shipboard long taxi, while performing final alignment on the catapult and in tight turns (e.g., catapult spin-out and deck edge parking), during low speed taxi on a pitching deck, when clearing the landing area following shipboard arrested landings, and as specified in SA FTM-01, Flight and Hangar Deck Handling and Wing Fold/Sweep and Canopy Operations paragraphs.

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Conduct piloted simulation and evaluation of mission representative ground handling tasks to verify the pilot and air vehicle are able to taxi in confined spaces aboard ships with various obstacles (e.g., other air vehicles and arresting wire), and in precise control scenarios (e.g., catapult lineup and parking) for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: surface conditions (e.g., tire friction, wet, oil), ship motion, and WOD from all relative directions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based piloted evaluations of mission representative ground handling tasks to verify the pilot and air vehicle are able to taxi in confined spaces aboard ships with various obstacles (e.g., other air vehicles and arresting wire), and in precise control scenarios (e.g., catapult lineup and parking) for all control modes across selected combinations of air vehicle states, control modes, environmental conditions: surface conditions (e.g., tire friction, wet, oil) and winds from all relative directions.

(3) Flight test: Conduct shipboard piloted evaluations of mission representative ground handling tasks to verify the pilot and air vehicle are able to taxi in confined spaces aboard ships with various obstacles (e.g., other air vehicles and arresting wire), and in precise control scenarios (e.g., catapult lineup and parking) for all control modes across selected combinations of air vehicle states, environmental conditions, and ship conditions: surface conditions (e.g., tire friction, wet, oil), ship motion, and WOD from all relative directions.

6.1.9.2 Carrier catapult launch.

Standard: The air vehicle is able to maintain control of flight path and attitude during carrier catapult launch.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of shore-based and shipboard catapult launches to determine the control power, power and engine response, and margin to stall speed needed to maintain control, including 2-D performance analysis for pitch-rate control and sink off bow, 6-DOF simulations for automatic control and trim settings for controls free launches, and gear/catapult release dynamics, for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: catapult type and shipboard location, ship departure patterns, WOD headwind and crosswind conditions from catapult minimum end airspeed to maximum WOD, including carrier airwake and ship motion motion effects, subject to constraints of the catapult's minimum and maximum end speeds.

Test.

(1) Lab test: Conduct piloted simulation and evaluation of land-based and shipboard catapult launches as specified in SA FTM-01, Launching Tests section for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: catapult type and shipboard location, ship departure patterns, WOD headwind and crosswind conditions from

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catapult minimum end airspeed to maximum WOD, including carrier airwake and ship motion motion effects, subject to constraints of the catapult's minimum and maximum end speeds.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based catapult launches as specified in SA FTM-01, Launching Tests section for nominal and critical combinations of air vehicle states, control modes, environmental conditions, and ship conditions: catapult type and shipboard location, ship departure patterns, headwind and crosswind conditions from catapult minimum end airspeed to maximum shore-based WOD equivalent, subject to constraints of the catapult's minimum and maximum end speeds.

(3) Flight test: Conduct shipboard catapult launches as specified in SA FTM-01, Launching Tests section for nominal and critical combinations of air vehicle states, control modes, environmental conditions, and ship conditions: catapult type and shipboard location, ship departure patterns, multiple levels of ship motion, WOD magnitudes and directions from catapult minimum end airspeed to maximum WOD and end airspeed, subject to constraints of the catapult's minimum and maximum end speeds.

6.1.9.3 Carrier approach.

Standard: The air vehicle is no worse than Level 2 handling qualities during carrier approach and landing to carrier arresting gear.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of shore-based and shipboard carrier approach and landing to determine the control power, roll and flight path control power and bandwidth, power and engine response, margin to stall speed, and defined minimum and maximum approach airspeed needed to complete the mission task, including 6-DOF simulations for automatic control modes, for each combination of air vehicle states, control modes, approach patterns, off-nominal approaches, environmental conditions, and ship conditions: ship class and arresting gear layout, WOD magnitude and direction, steady and turbulence carrier airwake and ship motion effects, ship speed and turn rate, subject to the constraints of the arresting gear capabilities and airframe structural limits.

Test.

(1) Lab test: Conduct piloted simulation and evaluation of land-based and shipboard carrier approach and landing as specified in SA FTM-01, Landing Tests section for each combination of air vehicle states, control modes, approach patterns, off-nominal approaches, environmental conditions, and ship conditions: ship class and arresting gear layout, approach speed, WOD magnitude and direction, steady and turbulence carrier airwake and ship motion effects, ship speed and turn rate, subject to the constraints of the arresting gear capabilities and airframe structural limits. Reversionary control modes are included in simulation evaluations to assess

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capabilities of potential backup systems and the potential for cognitive error between the primary system and the reversionary mode.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based arrested landings as specified in SA FTM-01, Landing Tests section for nominal and critical combinations of air vehicle states, control modes, approach patterns, off-nominal approaches, environmental conditions, and ship conditions: arresting gear type and layout, approach speed, headwind and crosswind conditions from minimum to maximum shore-based WOD equivalent, subject to the constraints of the arresting gear capabilities and airframe structural limits. Reversionary control modes will be included in test evaluations to assess capabilities of potential backup systems and the potential for cognitive error between the primary system and the reversionary mode.

(3) Flight test: Conduct shipboard arrested landings for nominal and critical combinations of air vehicle states, control modes, approach patterns, off-nominal approaches, environmental conditions, and ship conditions: arresting gear type and shipboard layout and configuration, multiple levels of ship motion, approach speed, WOD magnitudes and directions from minimum to maximum WOD, subject to the constraints of the arresting gear capabilities and airframe structural limits. Reversionary control modes will be included in test evaluations to assess capabilities of potential backup systems and the potential for cognitive error between the primary system and the reversionary mode.

6.1.9.4 Carrier bolter.

Standard: The air vehicle can maintain control during a carrier bolter, with flying qualities no worse than qualitative Level 2, including maintaining wingtips within the foul lines of the landing area.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of shore-based and shipboard carrier bolter, while maintaining wingtips within the carrier foul lines (e.g., landing area) and control, within adequate tolerances, of pitch attitude, AOA, bank angle, heading, pitch rate, and specific excess energy, to determine needed aerodynamic and ground handling control power and bandwidth, sink off deck performance, and specific excess energy needed to complete the mission task, including 6-DOF simulations for automatic control modes, for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: ship class and landing area dimensions, ship departure patterns, approach speed, flight path angle, WOD magnitude and direction, steady and turbulence carrier airwake and ship motion effects, ship speed and turn rate, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

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Test.

(1) Lab test: Conduct piloted simulation and evaluation of land-based and shipboard carrier bolter, while maintaining wingtips within the carrier foul lines or land-based representations, as specified in SA FTM-01, Landing Tests section, including 6-DOF simulations for automatic control modes, for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: ship class and landing area dimensions, ship departure patterns, approach speed, flight path angle, WOD magnitude and direction, steady and turbulence carrier airwake and ship motion effects, ship speed and turn rate, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based bolter tests as specified in SA FTM-01, Landing Tests section for nominal and critical combinations of air vehicle states, control modes, environmental conditions, and ship conditions: ship class and landing area dimensions, ship departure patterns, approach speed, flight path angle, headwind and crosswind conditions from minimum to maximum shore-based WOD equivalent, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

(3) Flight test: Conduct shipboard bolter tests as specified in SA FTM-01, Landing Tests section, while maintaining wingtips within the carrier foul lines (e.g., landing area), for nominal and critical combinations of air vehicle states, control modes, environmental conditions, and ship conditions: ship class and landing area dimensions, ship departure patterns, approach speed, flight path angle, WOD magnitude and direction, steady and turbulence carrier airwake and ship motion effects, ship speed and turn rate, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

6.1.9.5 Carrier waveoff.

Standard: The air vehicle is no worse than Level 2 handling qualities during waveoff from a carrier approach.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of shore-based and shipboard carrier waveoff, while maintaining heading control over the landing area and the minimum hook to carrier ramp and deck clearance, to determine the control power, roll and flight path control power and bandwidth, power and engine response, margin to stall speed, and defined minimum and maximum approach airspeed and flight path angle needed to complete the mission task, including 6-DOF simulations for automatic control modes, for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: ship class and arresting gear layout, WOD magnitude and direction, steady and turbulence carrier airwake and ship motion effects, ship speed and turn rate, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

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Test.

(1) Lab test: Conduct piloted simulation and evaluation of land-based and shipboard carrier waveoff as specified in SA FTM-01, Landing Tests section for each combination of air vehicle states, control modes, including reversionary modes, environmental conditions, and ship conditions: ship class and arresting gear layout, approach speed, flight path angle, WOD magnitude and direction, steady and turbulence carrier airwake and ship motion effects, ship speed and turn rate, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based carrier approach-style waveoffs as specified in SA FTM-01, Landing Tests section for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, environmental conditions, and ship conditions: arresting gear type and layout, approach speed, flight path angle, headwind and crosswind conditions from minimum to maximum shore-based WOD equivalent, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

(3) Flight test: Conduct shipboard waveoffs as specified in SA FTM-01, Landing Tests section for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, environmental conditions, and ship conditions: arresting gear type and shipboard layout/configuration, approach speed, flight path angle, multiple levels of ship motion, WOD magnitudes and directions from minimum to maximum WOD, subject to the constraints of the arresting gear capabilities and airframe structural limits (for a successful arrestment).

6.1.9.6 Shipboard launch assisted takeoff.

Standard: The air vehicle is no worse than Level 2 handling qualities for shipboard launch assisted takeoff, not including carrier catapult launch.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of shore-based and shipboard launch assisted takeoff and transition to self-powered or airborne flight to determine the control power, power and engine response, margin to stall speed, needed to maintain control and complete the takeoff and transition between phases of flight, including 2-D performance analysis for pitch-rate control and launch trajectory, and 6-DOF simulations for automatic control and trim settings for controls free launches, and launch system release dynamics, for each combination of air vehicle states, control modes, and environmental conditions: launch method(s), ship departure patterns, headwind and crosswind conditions from minimum assisted launch end or release airspeed to maximum winds, including gusts and turbulence, ship motion effects, ship speed and turn rate, subject to constraints of the assisted launch method minimum and maximum end or release speeds.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation of shore-based and shipboard launch assisted takeoff and transition to self-powered or airborne flight, and 6-DOF simulations for automatic control, for each combination of air vehicle states, control modes, including reversionary modes and environmental conditions: launch method(s), ship departure patterns, headwind and crosswind conditions from minimum launch end or release airspeed to maximum winds, including gusts and turbulence, ship motion effects, ship speed and turn rate, subject to constraints of the assisted launch method minimum and maximum end or release speeds.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based launch assisted takeoffs for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, and environmental conditions: launch method(s), ship departure patterns, shore-based WOD equivalent magnitude and direction conditions from minimum launch end or release airspeed to maximum winds, subject to constraints of the launch method minimum and maximum end/release speeds.

(3) Flight test: Conduct shipboard launch assisted takeoffs for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, and environmental conditions: launch method(s), ship departure patterns, headwind and crosswind conditions from minimum launch end/release airspeed to maximum winds, multiple levels of ship motion, subject to constraints of the launch method minimum and maximum end or release speeds.

6.1.9.7 Unassisted shipboard takeoff.

Standard: The air vehicle is no worse than Level 2 handling qualities, or DIPES 3 for rotary-wing air vehicles, for unassisted shipboard takeoff.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of shore-based and shipboard takeoff and departure to determine the control power, power and engine response, margin to control and flight envelope limits needed to maintain control and complete the launch and transition between phases of flight, including 2-D performance analysis and launch trajectories, and 6-DOF simulations for automatic control and trim settings for controls free launches, for each combination of air vehicle states, control modes, and environmental conditions: launch method(s), ship departure patterns, ship motion effects, ship speed and turn rate, headwind and crosswind conditions from minimum to maximum winds, including gusts and turbulence.

Test.

(1) Lab test: Conduct piloted simulation and evaluation of shore-based and shipboard takeoff and departure, and 6-DOF simulations for automatic control, for each combination of air vehicle states, control modes, including reversionary modes, and environmental conditions: launch

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method(s), ship departure patterns, ship motion effects, ship speed and turn rate, headwind and crosswind conditions from minimum to maximum winds, including gusts and turbulence.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based takeoff and departure for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, and environmental conditions: launch method(s), ship departure patterns, and shore-based WOD equivalent magnitude and direction conditions from minimum to maximum winds.

(3) Flight test: Conduct shipboard takeoffs for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, and environmental conditions: launch method(s), ship departure patterns, multiple levels of ship motion, and headwind and crosswind conditions from minimum to maximum winds.

6.1.9.8 Shipboard recovery and landing.

Standard: The air vehicle is no worse than Level 2 handling qualities, or DIPES 3 for rotary-wing air vehicles, for shipboard recovery and landing, not including carrier arresting gear recovery.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of land-based and shipboard recovery/landing to determine the control power, roll, heading and flight path control power and bandwidth, power and engine response, margin to control and flight envelope limits, and minimum and maximum approach speeds needed to complete the mission task, including 6-DOF simulations for automatic control modes, for each combination of air vehicle states, control modes, approach patterns, environmental conditions, and ship conditions: ship class, recovery/landing method(s) and locations, WOD magnitude and direction, steady and turbulence airwake and ship motion effects, and ship speed and turn rate.

Test.

(1) Lab test: Conduct piloted simulation and evaluation of land-based and shipboard recovery/landing for each combination of air vehicle states, control modes, including reversionary modes, approach patterns, environmental conditions, and ship conditions: ship class, recovery/landing method(s) and locations, approach speed, WOD magnitude and direction, steady and turbulence airwake and ship motion effects, and ship speed and turn rate.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based recoveries/landings for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, approach patterns, environmental conditions, and ship conditions: ship class, recovery/landing method(s) and

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locations, approach speed, from minimum to maximum shore-based WOD equivalent magnitude and direction, steady and turbulence airwake and ship motion effects, and ship speed and turn rate.

(3) Flight test: Conduct shipboard recoveries/landings for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, approach patterns, environmental conditions, and ship conditions: ship class, recovery/landing method(s) and locations, approach speed, WOD magnitude and direction, multiple levels of ship motion, and ship speed and turn rate.

6.1.9.9 Aborted recoveries and landings.

Standard: The air vehicle is no worse than Level 2 handling qualities, or DIPES 3 for rotary-wing air vehicles, during shipboard aborted recoveries and landings.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of aborted recoveries/landings, while maintaining minimum air vehicle to ship clearances, to determine the control power, roll, heading and flight path control power and bandwidth, power and engine response, margin to control and flight envelope limits, and minimum and maximum approach speeds needed to complete the mission task, including 6-DOF simulations for automatic control modes, for each combination of air vehicle states, control modes, environmental conditions, and ship conditions: ship class, recovery/landings method(s) and locations, aborted recovery/landings flight paths and patterns, WOD magnitude and direction, steady and turbulence airwake and ship motion effects, and ship speed and turn rate.

Test.

(1) Lab test: Conduct piloted simulation and evaluation of land-based and shipboard aborted recovery/landing for each combination of air vehicle states, control modes, including reversionary modes, environmental conditions, and ship conditions: ship class, recovery/landings method(s) and locations, aborted recovery/landings flight paths and patterns, WOD magnitude and direction, steady and turbulence airwake and ship motion effects, and ship speed and turn rate.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based shipboard approach-style aborted recoveries/landings for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, environmental conditions, and ship conditions: ship class, recovery/landings method(s) and locations, aborted recovery/landings flight paths and patterns, headwind and crosswind conditions from minimum to maximum shore-based WOD equivalent.

(3) Flight test: Conduct shipboard aborted recoveries/landings for nominal and critical combinations of air vehicle states, control modes, including reversionary modes, environmental conditions, and ship conditions: ship class, recovery/landings method(s) and locations, aborted

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recovery/landings flight paths and patterns, multiple levels of ship motion, WOD magnitudes and directions from minimum to maximum WOD.

6.1.9.10 Vision aided shipboard operations.

Standard: The air vehicle is no worse than Level 2 handling qualities, or DIPES 3 for rotary-wing air vehicles, using visual cues for both unassisted and vision aiding devices for shipboard operations, day and night.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of shore-based and shipboard takeoff/launch and landing/recovery operations using visual cues, including air vehicle cues to the ground crew aboard the ship (e.g., Landing Signal Officer), to determine the visual aide configuration(s) (types, locations, and variations) needed to complete the tasks for unassisted and vision aiding devices (e.g., NVDs) for each combination of air vehicle states, control modes, and environmental conditions: ambient lighting conditions from day to night, DVEs, ship class, launch/landing/recovery/landing method, approach and departure paths and patterns, minimum to maximum WOD magnitudes and headings, including gusts and turbulence, ship motion effects, ship speed, and turn rate.

Test.

(1) Lab test: Conduct piloted simulation and evaluation of shore-based and shipboard takeoff/launch and landing/recovery operations using visual cues, both unassisted and with vision aiding devices, and 6-DOF simulations for automatic control, for each combination of air vehicle states, control modes, and environmental conditions: visual aide configuration, ambient lighting conditions from day to night, DVEs, ship class, launch/landing/recovery/landing method, approach and departure paths and patterns, minimum to maximum WOD magnitudes and headings, including gusts and turbulence, ship motion effects, ship speed, and turn rate.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct shore-based takeoff/launch and landing/recovery operations using visual cues, both unassisted and with vision aiding devices for nominal and critical combinations of air vehicle states, control modes, and environmental conditions: visual aide configuration, ambient lighting conditions from day to night, DVEs, ship class, launch/landing/recovery/landing method, approach and departure paths/patterns, shore-based WOD equivalent magnitude and direction conditions from minimum to maximum winds.

(3) Flight test: Conduct shipboard takeoff/launch and landing/recovery operations using visual cues, both unassisted and with vision aiding devices for nominal and critical combinations of air vehicle states, control modes, and environmental conditions: visual aide configuration, ambient lighting conditions from day to night, DVEs, ship class, launch/landing/recovery/landing method, approach and departure paths/patterns, multiple levels of ship motion, ship speed and turn rate, WOD magnitudes and directions from minimum to maximum WOD.

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6.1.10 Vertical/short takeoff and landing air vehicles.

While the flying qualities quantitative levels in MIL-STD-1797 for conventional fixed-wing air vehicles have been developed over many decades of research, development, flight test, and operational experience, MIL-STD-1797 does not have as much support for operational STOVL or V/STOL or VTOL fixed-wing air vehicles. There have been fewer DoW operational STOVL, V/STOL, and VTOL fixed-wing air vehicles compared to conventional air vehicles. Nonetheless, there are important reference documents covering many decades that should be considered: Tactical fighter and attack air vehicles: MIL-STD-1797, MIL-F-83300, AFFDL-TR-70-88, AGARD 577, AFFTC-TR-91-29, NASA TM 112213, NASA TP 3356, NASA TP-2000-209591, and other documents. Transport air vehicles: ASD-TR-78-13, NASA TN D-5594, and other documents.

The term ‘V/STOL’ is used loosely as an overarching term in this subsection; however, there are more precise acronyms used for specific air vehicles. Short takeoff vertical landing generally applies to air vehicles that require STOs and powered lift or slow landings with any appreciable combat loadings, with vertical (hover) operations relegated to lower gross weights after stores have been used or jettisoned. Vertical/short takeoff and landing generally applies to air vehicles with vertical operations as a driving design requirement compared to STOVL air vehicles. Vertical takeoff and landing applies to air vehicles that only takeoff and land vertically.

The operational requirements document, air vehicle specification, and all subsequent flowed down requirements documents will set the required basing (land-based only, including runway types, road widths, runway condition ratings for weather, or land-based plus shipboard, including ship types and flat-deck versus ski-jump equipped), required atmospheric conditions, required visibility conditions, required loadings which will be flight phase-dependent (lower gross weights expected for hover operations), and other CONOPS for weapons jettison or usage in the context of V/STOL mode flight.

Several aspects of V/STOL flying/handling qualities are unique. For jet-borne or hovering air vehicles, with direct lift provided by vectored main engine thrust with possible additional lift from lift fans or other embedded propulsion systems, IGE aerodynamics are generally highly destabilizing. As the air vehicle banks or pitches IGE, the reflected jet exhaust off of the landing surface hits the part of the air vehicle already rotated up, which is destabilizing in roll and pitch. As such, unlike helicopters, V/STOL fixed-wing air vehicles with deflected thrust for hovering flight do not spend any appreciable time in IGE. It is a transitory environment that is meant to be punched through repeatably, consistently, and expeditiously, in the context of either VTOs, creeping VTOs, VLs, or creeping VLs.

Jet induced aerodynamics are also very important on V/STOL air vehicles; both Out-of-Ground Effects and IGE. It is very important to develop a high fidelity 6-DOF simulation of the air vehicle, with not only a detailed aerodynamic database and propulsion system(s) model(s), but also detailed V/STOL jet effects modeled to the best extent practicable. Similarly, just like FCS effectors, V/STOL propulsion system effectors which are required for safe air vehicle control and flying/handling qualities should have failure modes adequately modeled in the 6-DOF simulation to assess the degraded mode flying/handling qualities AW standards.

The semi-jetborne region, the transition region between conventional flight and jet-borne or hovering flight, has some unique characteristics and AW standards. Fundamentally, the control strategy provided to the pilot to seamlessly transition between conventional flight, semi-jetborne

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flight, and hovering flight, is integral to AW. The control strategy should be intuitive, easily trainable, and ideally consistent across the speed regime, to avoid risk of modal confusion, cognitive error, and misapplied inputs, which can have flight safety implications. The air vehicle itself is undergoing large changes in the distribution of lift between aerodynamic force and propulsive force, and the distribution of control power between conventional flight control effectors and propulsion system effectors for moment control. Ideally, these large changes should be transparent to the pilot, who should still be able to control the air vehicle the same way with the inceptors, regardless of the large changes happening to lift and control power as the air vehicle decelerates.

The V/STOL standards herein are focused on takeoff, landing, and approach pattern flight phases and associated air vehicle states. It is understood that some air vehicles use thrust vectoring to augment maneuverability in up and away flight. These standards don't explicitly address thrust vectoring for enhanced up and away maneuverability, though it is acknowledged that controllability and departure resistance and possibly departure recovery would need to be assessed with thrust vectoring active for applicable air vehicles, operational missions, and flight phases. For now, the previous fixed-wing subsections of 6.1 should be used for up and away thrust vectoring AW, with thrust vectoring functionality, performance, and failure modes taken into account.

It is important to note that flying/handling qualities standards are overarching, system level by nature due to assuring air vehicle/system control with subsystem failure modes, even in the presence of various atmospheric disturbances; thus, flying/handling qualities standards historically drive numerous subsystems' architectural designs, including but not limited to flight controls, hydraulics, electrical systems, structures, other subsystems, and propulsion systems and thrust vectoring effectors in the case of V/STOL air vehicles. This can mean that some subsystems which historically might not have been flight-critical, are flight-critical in the V/STOL air vehicles. For example, if they are required to open V/STOL doors or to control propulsion system effectors used for vehicle control.

6.1.10.1 Short takeoff.

Standard: The air vehicle is no worse than Level 2 flying qualities for all STOs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all STOs to evaluate air vehicle capabilities and control margins during STOs, to include:

a. Land-based STO.
b. Land-based formation STOs (< maximum performance for lead air vehicle).
c. Land-based ski-jump STO (flight test purposes).
d. Shipboard flat deck STO.
e. Shipboard ski-jump STO.
f. Auto-STO land-based.
g. Auto-STO shipboard.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Short Takeoff (STO) paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Land-based STO.
b. Land-based formation STOs (< maximum performance for lead air vehicle).
c. Land-based ski-jump STO (flight test purposes).
d. Shipboard flat deck STO.
e. Shipboard ski-jump STO.
f. Auto-STO land-based.
g. Auto-STO shipboard.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Short Takeoff (STO) paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Land-based STO.
b. Land-based formation STOs (< maximum performance for lead air vehicle).
c. Land-based ski-jump STO (flight test purposes).
d. Shipboard flat deck STO.
e. Shipboard ski-jump STO.
f. Auto-STO land-based.
g. Auto-STO shipboard.

6.1.10.2 Vertical takeoff.

Standard: The air vehicle is no worse than Level 2 flying qualities for all VTOs, land and shipboard.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all VTOs to evaluate air vehicle capabilities and control margins during VTOs, to include:

a. Land-based hover pit VTO.
b. Land-based hover pad VTO.
c. Shipboard VTO.
d. Rolling VTO.
e. Creeping VTO.
f. Baseline control system.
g. Precision hover modes.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Takeoff (VTO) paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Land-based hover pit VTO.
b. Land-based hover pad VTO.
c. Shipboard VTO.
d. Rolling VTO.
e. Creeping VTO.
f. Baseline control system.
g. Precision hover modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Takeoff (VTO) paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Land-based hover pit VTO.
b. Land-based hover pad VTO.
c. Shipboard VTO.
d. Rolling VTO.
e. Creeping VTO.
f. Baseline control system.
g. Precision hover modes.

6.1.10.3 Shipboard recovery pattern.

Standard: The air vehicle is no worse than Level 2 while performing the Shipboard Recovery Pattern MTE as specified in MIL-STD-1797, Suggested MTE Handling Qualities Evaluation Tasks for V/STOL Operations table.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of the entire shipboard recovery pattern from fully wingborne flight through semi-jetborne flight and culminating in either deceleration to hover and VL or deceleration to SRVL approach speed and SRVL, to evaluate air vehicle capabilities, to include:

a. Single air vehicle recovery pattern, including all subtasks (break turn, conversion from wingborne to semi-jetborne on downwind, turn to final, final deceleration to hover or SRVL speed, hover translation/VL or SRVL).
b. Recovery pattern while in formation with other air vehicle for some percentage of the pattern.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess handling qualities, control strategy, overall workload, cognitive error potential for the entire shipboard recovery pattern related to all procuring agency requirements, including configurations and loadings, winds, sea states, day and night, VMC and IMC, solo recoveries, and formation recoveries.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities, control strategy, overall workload, cognitive error potential for the entire shipboard recovery pattern related to all procuring agency requirements, for selected combinations of configurations and loadings, winds, sea states, day and night, VMC and IMC, solo recoveries, and formation recoveries.

6.1.10.4 Powered-lift landing.

Standard: The air vehicle is no worse than Level 2 flying qualities for all powered-lift landings (landings with combination of aerodynamic lift and propulsive lift (e.g., slow landings, rolling VLs)).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all powered-lift landings to evaluate air vehicle capabilities and control margins, to include:

a. Slow landings (land-based powered-lift landings).
b. Rolling VLs (land-based slow speed powered-lift landings).
c. Shipboard rolling VLs (shipboard slow speed powered-lift landings).

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Powered-lift Landing paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Slow landings (land-based powered-lift landings).
b. Rolling VLs (land-based slow speed powered-lift landings).
c. Shipboard rolling VLs (shipboard slow speed powered-lift landings).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified within MIL-STD-1797, Powered-lift Landing paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Slow landings (land-based powered-lift landings).

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b. Rolling VLs (land-based slow speed powered-lift landings).
c. Shipboard rolling VLs (shipboard slow speed powered-lift landings).

6.1.10.5 Height captures in hover.

Standard: The air vehicle is no worse than Level 2 flying qualities during height captures and maintenance in hover.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all height captures in hover to evaluate air vehicle capabilities and control margins, to include:

a. Baseline control system.
b. Precision hover modes.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Hover paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Hover paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

6.1.10.6 Vertical landing.

Standard: The air vehicle is no worse than Level 2 flying qualities for VLs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all VLs to evaluate air vehicle capabilities and control margins, to include:

a. Vertical landing to pad.
b. Ship VL.
c. Vertical landing to hover pit.

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d. Creeping VL if applicable.
e. Baseline control system.
f. Precision hover modes.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Landing (VL) paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Vertical landing to pad.
b. Ship VL.
c. Vertical landing to hover pit.
d. Creeping VL if applicable.
e. Baseline control system.
f. Precision hover modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Landing (VL) paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Vertical landing to pad.
b. Ship VL.
c. Vertical landing to hover pit.
d. Creeping VL if applicable.
e. Baseline control system.
f. Precision hover modes.

6.1.10.7 V/STOL ground handling.

Standard: The air vehicle is no worse than Level 2 flying qualities for V/STOL-configured ground handling. For conventional mode operations, see the other relevant standards in 6.1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all V/STOL ground handling to evaluate air vehicle capabilities to include:

a. Vehicle configured following powered-lift landings or VLs.
b. Vehicle configured for planned takeoff.
c. Vehicle configured for powered-lift taxiing.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess ground handling qualities as specified in MIL-STD-1797, Ground Handling paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Vehicle configured following powered-lift landings or VLs.
b. Vehicle configured for planned takeoff.
c. Vehicle configured for powered-lift taxiing.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess ground handling qualities as specified in MIL-STD-1797, Ground Handling paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Vehicle configured following powered-lift landings or VLs.
b. Vehicle configured for planned takeoff.
c. Vehicle configured for powered-lift taxiing.

6.1.10.8 Transition and conversion.

Standard: The air vehicle is no worse than Level 2 flying qualities during transition and conversion between forward flight and V/STOL configurations, with no performance deficit conditions that are not automatically prevented by the control system or clearly alerted to the pilot during the deceleration post-conversion.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all transitions and conversions to evaluate air vehicle capabilities and control margins, to include:

a. Single air vehicle operations.
b. Formation transitions and conversions.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Transition/Conversion paragraph, including performance deficit protection, for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Single air vehicle operations.
b. Formation transitions and conversions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities in MIL-STD-1797, Transition/Conversion paragraph, including performance deficit protection, for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Single air vehicle operations.
b. Formation transitions and conversions.

6.1.10.9 Flying qualities in the transition region.

Standard: For all configurations and conditions between hover and maximum conversion speed, the air vehicle control strategy eliminates the potential for pilot cognitive error, misapplied inputs on inceptors, and modal confusion, as the air vehicle transitions from aerodynamic lift to combinations of aerodynamic and propulsive lift to jetborne lift and vice versa.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all V/STOL mode tasks, including an overall ship pattern and recovery task consisting of wingborne flight breakturn, downwind conversion and transition, turn to final in semi-jetborne flight, followed by deceleration to hover, hover translation, and VL, or followed by deceleration to SRVL speed and SRVL, to evaluate the control strategy, to include:

a. Baseline control system.
b. Any alternate pilot selectable modes such as Delta Flight Path, Hover TAC, and Hover TRC.
c. Analysis emphasis pre-simulation should be on command types, and mapping of commands to inceptors in order to ensure a consistent and intuitive control strategy across the wide airspeed regime between wingborne, semi-jetborne, and jetborne flight.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Flying Qualities in the Transition Region paragraph and Control Strategy paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Any alternate pilot selectable modes such as Delta Flight Path, Hover TAC, and Hover TRC.

Pilot will assess overall workload, pilot compensation, and any modal confusion or cognitive error (e.g., misapplied inceptor inputs for a specific V/STOL task), in the context of stressing shipboard V/STOL takeoff and recovery or comparable land-based V/STOL takeoff and recovery using the entire speed range from wingborne to semi-jetborne to jetborne. Failure modes can and should be inserted into the piloted simulation test matrix in order to assess a pilot's ease in assessing the failure and applying correct inputs to the inceptors to continue the mission task evaluation.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Flying Qualities in the Transition Region paragraph and Control Strategy paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Any alternate pilot selectable modes such as Delta Flight Path, Hover TAC, and Hover TRC.

Pilot will assess overall workload, pilot compensation, and any modal confusion or cognitive error (misapplied inceptor inputs for a specific V/STOL task), in the context of stressing shipboard V/STOL takeoff and recovery or comparable land-based V/STOL takeoff and recovery using the entire speed range from wingborne to semi-jetborne to jetborne. Failure modes can and should be inserted into the piloted simulation test matrix in order to assess a pilot's ease in assessing the failure and applying correct inputs to the inceptors to continue the mission task evaluation. Stressing winds and environmental conditions should be assessed in flight test IAW program requirements and fleet expectations for cleared envelope.

6.1.10.10 Hovering translation.

Standard: The air vehicle is no worse than Level 2 flying qualities during hovering translation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all hover translations to evaluate air vehicle capabilities and control margins, to include:

a. Baseline control system.
b. Precision hover modes.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Hovering Translation paragraph and Translational Rate Command (TRC) in Hover paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Hovering Translation paragraph and Translational Rate Command (TRC) in Hover

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paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

6.1.10.11 Pitch response in hovering flight.

Standard: The air vehicle hover pitch response is no worse than Level 2 as specified in MIL-STD-1797, Pitch Axis Response in Hovering Flight paragraph and Angular (Moment-Generating) Control Power in Hovering Flight paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all hover pitch response to evaluate air vehicle capabilities and control margins, to include:

a. Baseline control system.
b. Precision hover modes.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Pitch Axis Response in Hovering Flight paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. For control systems which automatically control pitch attitude, pilot will assess overall workload, pilot compensation, and any undesirable coupling of pitch response into other axes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Pitch Axis Response in Hovering Flight paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. For control systems which automatically control pitch attitude, pilot will assess overall workload, pilot compensation, and any undesirable coupling of pitch response into other axes.

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6.1.10.12 Roll axis response in hovering flight.

Standard: The air vehicle hover roll response is no worse than Level 2 as specified in MIL-STD-1797, Roll Axis Response in Hovering Flight paragraph and Angular (Moment-Generating) Control Power in Hovering Flight paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all hover roll response to evaluate air vehicle capabilities and control margins, to include:

a. Baseline control system.
b. Precision hover modes.

Test.

(1) Lab test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Roll Axis Response in Hovering Flight paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. For control systems which automatically control pitch attitude, pilot will assess overall workload, pilot compensation, and any undesirable coupling of pitch response into other axes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Roll Axis Response in Hovering Flight paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. For control systems which automatically control pitch attitude, pilot will assess overall workload, pilot compensation, and any undesirable coupling of pitch response into other axes.

6.1.10.13 Yaw axis response in hovering flight.

Standard: The air vehicle hover yaw response is no worse than Level 2 as specified in MIL-STD-1797, Yaw Axis Response in Hovering Flight paragraph and Angular (Moment-Generating) Control Power in Hovering Flight paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis of all hover yaw response to evaluate air vehicle capabilities and control margins, to include:

a. Baseline control system.
b. Precision hover modes.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797. Yaw Axis Response in Hovering Flight paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. For control systems which automatically control pitch attitude, pilot will assess overall workload, pilot compensation, and any undesirable coupling of pitch response into other axes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Yaw Axis Response in Hovering Flight paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. For control systems which automatically control pitch attitude, pilot will assess overall workload, pilot compensation, and any undesirable coupling of pitch response into other axes.

6.1.10.14 Thrust margin in hovering flight.

Standard: The air vehicle has a minimum of 5% thrust margin above the thrust required to maintain an out of ground effect hover at maximum hover weight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of hover thrust margin to evaluate air vehicle capabilities for all required hover loadings and configurations, ensuring an allocation of thrust margin is available for control margin for all three rotational axes in hover, to include:

a. Baseline control system.
b. Precision hover modes.

Hover thrust margin calculation ground rules need to be clearly specified by procuring agency, to include accounting for all aerodynamic and propulsive forces, as well as a detailed weight estimate.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Takeoff (VTO) paragraph, Vertical Landing (VL) paragraph, and specifically, waveoff maneuvers and height stabilize maneuvers during VLs for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling of thrust commands into other axes. Pilot will assess timeliness of h-dot response and height lost during a stabilize or waveoff maneuver from a nominal descent rate VL IAW program requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Takeoff (VTO) paragraph, Vertical Landing (VL) paragraph, and specifically, waveoff maneuvers and height stabilize maneuvers during VLs for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling of thrust commands into other axes. Pilot will assess timeliness of h-dot response and height lost during a stabilize or waveoff maneuver from a nominal descent rate VL IAW program requirements.

6.1.10.15 Vertical axis response in hovering flight.

Standard: The air vehicle is no worse than Level 2 as specified in MIL-STD-1797, Vertical Axis Response in Hovering Flight paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of hover vertical velocity response to evaluate air vehicle capabilities for all required hover loadings and configurations, ensuring an allocation of thrust margin is available for control margin for all 3 rotational axes in hover, to include:

a. Baseline control system.
b. Precision hover modes.

Vertical velocity response analysis IGE should be IAW program requirements, which typically include VLs and VTOs, but typically not height captures IGE.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Takeoff (VTO) paragraph, Vertical Landing (VL) paragraph, and specifically, waveoff maneuvers and height stabilize maneuvers during VLs for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling of thrust commands into other axes. Pilot will assess timeliness of h-dot response and height lost during a stabilize or waveoff maneuver from a nominal descent rate VL IAW program requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Vertical Takeoff (VTO) paragraph, Vertical Landing (VL) paragraph, and specifically, waveoff maneuvers and height stabilize maneuvers during VLs for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling of thrust commands into other axes. Pilot will assess timeliness of h-dot response and height lost during a stabilize or waveoff maneuver from a nominal descent rate VL IAW program requirements.

6.1.10.16 Longitudinal translation response in hovering flight.

Standard: The air vehicle is no worse than Level 2 as specified in MIL-STD-1797, Longitudinal Translation Response in Hovering Flight paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of hover longitudinal velocity response to evaluate air vehicle capabilities for all required hover loadings and configurations, to include:

a. Baseline control system.
b. Precision hover modes.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Longitudinal Translation Response in Hovering Flight paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include baseline control system.

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Pilot will assess overall workload, pilot compensation, and any undesirable coupling into other axes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Longitudinal Translation Response in Hovering Flight paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling into other axes.

6.1.10.17 Lateral translation response in hovering flight.

Standard: The air vehicle is no worse than Level 2 as specified in MIL-STD-1797, Lateral Translation Response in Hovering Flight paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of hover lateral translation velocity response to evaluate air vehicle capabilities for all required hover loadings and configurations, to include:

a. Baseline control system.
b. Precision hover modes.
c. Tasks should include both left and right translations and 45-degree diagonal translations.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Lateral Translation Response in Hovering Flight paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. Tasks should include both left and right translations and 45-degree diagonal translations.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling into other axes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Lateral Translation Response in Hovering Flight paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.
c. Tasks should include both left and right translations and 45-degree diagonal translations.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling into other axes.

6.1.10.18 Cross-axis coupling in hovering flight.

Standard: Any cross-axis coupling between attitude or lateral or longitudinal position and height (vertical) control in hovering flight does not result in air vehicle handling qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all hover tasks to evaluate air vehicle capabilities for all required hover loadings and configurations, to include:

a. Baseline control system.
b. Precision hover modes.

Analysis emphasis pre-simulation should be on command types and mapping of commands to inceptors in order to ensure low pilot compensation and minimization of potential cross-axis coupling via command-type mapping to inceptors and inceptor force and gradient design.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities as specified in MIL-STD-1797, Cross-axis Coupling in Hovering Flight paragraph for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.
b. Precision hover modes.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling into other axes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities as specified in MIL-STD-1797, Cross-axis Coupling in Hovering Flight paragraph for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Baseline control system.

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b. Precision hover modes.

Pilot will assess overall workload, pilot compensation, and any undesirable coupling into other axes.

6.1.10.19 Short takeoff longitudinal acceleration and flight path response.

Standard: The air vehicle's longitudinal acceleration during the STO is no less than 0.065g. The minimum longitudinal acceleration exists with no CG sink for shorebased STO, and with no more than 10 ft CG sink from CG position at deck edge for shipboard STO and no more than 10 ft CG sink from CG position at end of ramp for shipboard ski jump STO.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of all STO tasks to evaluate air vehicle capabilities, including longitudinal acceleration for different STO flight path captures, for all required STO loadings and configurations, to include:

a. Shorebased STO.
b. Shorebased Ski-Jump STO.
c. Shipboard Flat-Deck STO.
d. Shipboard Ski-Jump STO.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess flying/handling qualities for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Shorebased STO.
b. Shorebased Ski-Jump STO.
c. Shipboard Flat-Deck STO.
d. Shipboard Ski-Jump STO.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Shorebased STO.
b. Shorebased Ski-Jump STO.
c. Shipboard Flat-Deck STO.
d. Shipboard Ski-Jump STO.

6.1.10.20 Semi-jetborne roll rate response.

Standard: The air vehicle's semi-jetborne roll response, including the minimum bank angle change in one second, results in flying qualities to no worse than Level 2 and has a roll effective

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time delay no greater than 0.20 second for all air vehicle Classes. For Class IV air vehicles, the roll mode time constant is no greater than 1.0 second.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Assess the MIL-HDBK-516, Fixed-Wing Lateral-Directional Flying Qualities paragraphs and conduct analysis of semi-jetborne roll response to evaluate air vehicle capabilities, for all required semi-jetborne loadings and configurations, to include:

a. Semi-jetborne formation, at discrete airspeeds or with inclusion of deceleration to a specific minimum airspeed.
b. Semi-jetborne landings, including offset landings.
c. STOs with specific stressing roll maneuvers such as formation STOs and STOs with aggressive course acquisition.

Test.

(1) Lab test: Asses the MIL-HDBK-516, Fixed-Wing Lateral-Directional Flying Qualities paragraphs and conduct piloted simulation and evaluation to assess flying/handling qualities for each combination of air vehicle states, control modes, and environmental conditions, to include:

a. Semi-jetborne formation, at discrete airspeeds or with inclusion of deceleration to a specific minimum airspeed.
b. Semi-jetborne landings, including offset landings.
c. STOs with specific stressing roll maneuvers such as formation STOs and STOs with aggressive course acquisition.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities for selected combinations of air vehicle states, control modes, and environmental conditions, to include:

a. Semi-jetborne formation, at discrete airspeeds or with inclusion of deceleration to a specific minimum airspeed.
b. Semi-jetborne landings, including offset landings.
c. STOs with specific stressing roll maneuvers such as formation STOs and STOs with aggressive course acquisition.

6.1.10.21 Semi-jetborne flight path response.

Standard: The air vehicle's semi-jetborne flight path response results in flying qualities no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Assess the MIL-HDBK-516, Fixed-Wing Longitudinal Flying Qualities paragraphs and conduct analysis of semi-jetborne flight path response, to evaluate air vehicle capabilities, for all required semi-jetborne loadings and configurations, to include:

a. Semi-jetborne formation, at discrete airspeeds or with inclusion of deceleration to a specific minimum airspeed.
b. Semi-jetborne landings, including waveoffs.
c. STOs with specific stressing flight path maneuvers such as formation STOs and STOs with aggressive flight path acquisition and altitude corrections.

Test.

(1) Lab test: Assess the MIL-HDBK-516, Fixed-Wing Longitudinal Flying Qualities paragraphs and conduct piloted simulation and evaluation to assess flying/handling qualities for each combination of air vehicles states, control modes, and environmental conditions, to include:

a. Semi-jetborne formation, at discrete airspeeds or with inclusion of deceleration to a specific minimum airspeed.
b. Semi-jetborne landings, including waveoffs.
c. STOs with specific stressing flight path control maneuvers such as formation STOs and STOs with aggressive flight path acquisition and altitude corrections.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess flying/handling qualities for selected combinations of air vehicles states, control modes, and environmental conditions, to include:

a. Semi-jetborne formation, at discrete airspeeds or with inclusion of deceleration to a specific minimum airspeed.
b. Semi-jetborne landings, including waveoffs.
c. STOs with specific stressing flight path maneuvers such as formation STOs and STOs with aggressive flight path acquisition and altitude corrections.

6.1.11 Rotary-wing.

Rotary-wing operations are inherently unique in aviation, and it follows that a dedicated set of flying qualities standards have been defined to ensure safe handling qualities. The most prominent differentiator of rotary-wing air vehicles from traditional fixed-wing air vehicles is the introduction of hover and low airspeed operations as a routine and essential function of the vehicle. This capability has proven the utility of rotary-wing air vehicles in both civil and military applications; however, it exposes the air vehicle to unique hazards such as operating in close proximity to natural and manmade obstacles and landing on unprepared, sloped, or obstructed austere surfaces. Rotary-wing air vehicles also have the tendency to obscure its own local environment from entrainment of dirt, sand, or snow into the rotor wash causing degraded visual conditions. To perform reliably and safely in these difficult conditions demands flight control augmentation that enables the pilot to select a response type that provides the desired level of stabilization. The philosophy of selecting an ideal flight control response type for a given

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mission phase ensures the handling qualities are equally satisfactory across all mission tasks and operating environments. Equally important is the correct categorization of the candidate vehicle in terms of its required agility. Rotary-wing air vehicles that serve an attack role will maneuver precisely and aggressively while utility and cargo rotary-wing air vehicles can accomplish the mission with limited agility.

The standards of the rotary-wing section are based on a combination of quantitative criteria as assessed from time or frequency domain analysis in concert with qualitative pilot feedback on the handling qualities as assessed through the execution of MTEs. In addition, categorizing standards as specific to an axis of control (pitch, roll, yaw, heave), they are further broken down into the two primary operational areas: hover/low airspeed and forward flight. A collection of general standards that apply globally across the flight envelope or specifically to system design fill out the remainder of the rotary-wing standards. Key references that form the foundation for these standards include MIL-DTL-32742, ADS-33E-PRF, Special Report AMR-AF-08-07 (Test Guide for ADS-33E-PRF), and the Background Information and User's Guide for Handling Qualities Requirements for Military Rotorcraft (USAAVSCOM Technical Report 89-A-008).

6.1.11.1 Required agility.

Standard: The air vehicle has the required agility to perform the intended MTEs with assigned handling qualities no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the assigned HQRs for the MTEs appropriate to the air vehicle's required level of agility (Limited, Moderate, Aggressive, Target Acquisition and Tracking) and category (Attack, Scout, Utility, Cargo) as specified in MIL-DTL-32742, Required Agility and Mission Task Elements paragraphs for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the assigned HQRs for the MTEs appropriate to the air vehicle's required level of agility (Limited, Moderate, Aggressive, Target Acquisition and Tracking) and category (Attack, Scout, Utility, Cargo) as specified in MIL-DTL-32742, Required Agility and Mission Task Elements paragraphs for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the assigned HQRs for the MTEs appropriate to the air vehicle's required level of agility (Limited, Moderate, Aggressive, Target Acquisition and Tracking) and category (Attack, Scout, Utility, Cargo) as specified in MIL-DTL-32742, Required Agility and Mission Task Elements paragraphs for selected combinations of air vehicle configuration and loading within the SFE.

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6.1.11.2 Flight control system response types.

Standard: The air vehicle has the necessary response types to achieve no worse than Level 2 handling qualities within the SFE in both GVEs and DVEs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the air vehicle has the necessary FCS response types for both GVEs and DVEs as specified in MIL-DTL-32742, Response-Types paragraph and its subparagraphs.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle has the necessary FCS response types for both GVEs and DVEs as specified in MIL-DTL-32742, Response-Types paragraph and its subparagraphs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle has the necessary FCS response types for both GVEs and DVEs as specified in MIL-DTL-32742, Response-Types paragraph and its subparagraphs.

6.1.11.3 Flight control system mode annunciations.

Standard: The air vehicle provides clear and easily interpretable annunciations to the pilot of the engaged or armed control system response type and the status of hold modes unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess that the air vehicle provides clear and easily interpretable annunciations to the pilot of the engaged or armed control system response type and the status of any hold modes as specified in MIL-DTL-32742, Annunciation of Response Type to the Pilot and Position Hold Annunciation paragraphs.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that the air vehicle provides clear and easily interpretable annunciations to the pilot of the engaged or armed control system response type and the status of any hold modes as specified in MIL-DTL-32742, Annunciation of Response Type to the Pilot and Position Hold Annunciation paragraphs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess that the air vehicle provides clear and easily interpretable annunciations to the pilot of the engaged or armed control system response type and the status of any hold modes as specified in MIL-DTL-32742, Annunciation of Response Type to the Pilot and Position Hold Annunciation paragraphs.

6.1.11.4 Controller characteristics.

Standard: The air vehicle's controller characteristics are no worse than Level 2 flying qualities.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the controller characteristics as specified in MIL-DTL-32742, Centering and Breakout Forces, Force Gradients, Limit Control Forces, Sensitivity and Gradients, Cockpit Control Free Play, Control Harmony, Trimming Characteristics, and Dynamic Coupling paragraphs for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the controller characteristics as specified in MIL-DTL-32742, Centering and Breakout Forces, Force Gradients, Limit Control Forces, Sensitivity and Gradients, Cockpit Control Free Play, Control Harmony, Trimming Characteristics, and Dynamic Coupling paragraphs for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the controller characteristics as specified in MIL-DTL-32742, Centering and Breakout Forces, Force Gradients, Limit Control Forces, Sensitivity and Gradients, Cockpit Control Free Play, Control Harmony, Trimming Characteristics, and Dynamic Coupling paragraphs for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.5 Control power.

Standard: The air vehicle has sufficient control power available in the pitch, roll, and yaw axis to generate an angular response not less than 7 deg/s from trim in both directions within 1.5 seconds. The air vehicle has sufficient control power available in the pitch, roll, and yaw axis to generate a delta angular response not less than 7 deg/sec in the restoring (towards trim) direction during non-trimmed flight within 1.5 seconds.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis to assess that for all air vehicle configurations and loadings within the SFE, the air vehicle can generate a minimum angular pitch, roll, and yaw rate of 7 deg/s from trim in both directions within 1.5 seconds of the control input. Conduct analysis to assess that for all air vehicle configurations, loadings, and mission maneuvers within the SFE, the air vehicle can generate a minimum delta angular pitch, roll, and yaw rate of 7 deg/s from trim in both directions within 1.5 seconds of the control input.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that for all air vehicle configurations and loadings within the SFE, the air vehicle can generate a minimum angular pitch, roll, and yaw rate of 7 deg/s from trim in both directions within 1.5 seconds of the control input. Conduct piloted simulation and evaluation to assess that for all air vehicle configurations, loadings, and mission maneuvers within the SFE, the air vehicle can generate a minimum delta angular pitch, roll, and yaw rate of 7 deg/s from trim in both directions within 1.5 seconds of the control input.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test for selected combinations of air vehicle configurations and loadings within the SFE, the air vehicle can generate a minimum angular pitch, roll, and yaw rate of 7 deg/s from trim in both directions within 1.5 seconds of the control input. Conduct flight test for selected combinations of air vehicle configurations, loadings, and mission maneuvers within the SFE, the air vehicle can generate a minimum delta angular pitch, roll, and yaw rate of 7 deg/s from trim in both directions within 1.5 seconds of the control input.

6.1.11.6 Warnings of reduced control margin.

Standard: The air vehicle provides the pilot with clear and unambiguous warnings and indications of less than 10% control margin remaining. For near-earth operations, the warnings and indications are interpretable by the pilot with eyes out of the cockpit. Special devices used for indication, warning, prevention, and recovery during normal or inadvertent operation do not create a hazard to the rotary-wing air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess that warnings and indications of less than 10% control margin remaining are clear and unambiguous to the pilot and that for near earth operations, the warnings and indications are interpretable by the pilot with their eyes focused outside of the cockpit. Verify that special devices, tactile cues (e.g., soft stops, stick shakers, stick pushers), haptic cues, and annunciations (aural and visual) that indicate, warn, prevent, or assist the pilot in recovering the air vehicle during normal or inadvertent operation do not create a hazard to the air vehicle.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that warnings and indications of less than 10% control margin remaining are clear and unambiguous to the pilot and that for near earth operations, the warnings and indications are interpretable by the pilot with their eyes focused outside of the cockpit. Verify that special devices, tactile cues (e.g., soft stops, stick shakers, stick pushers), haptic cues, and annunciations (aural and visual) that indicate, warn, prevent, or assist the pilot in recovering the air vehicle during normal or inadvertent operation do not create a hazard to the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test, for selected combinations of for all air vehicle configurations, loadings, and mission maneuvers within the SFE to assess that warnings and indications of less than 10% control margin remaining are clear and unambiguous to the pilot and that for near earth operations, the warnings and indications are interpretable by the pilot with their eyes focused outside of the cockpit. Verify that special devices, tactile cues (e.g., soft stops, stick shakers, stick pushers), haptic cues, and annunciations (aural and visual) that indicate, warn, prevent, or assist the pilot in recovering the air vehicle during normal or inadvertent operation do not create a hazard to the air vehicle.

6.1.11.7 Disturbance rejection.

Standard: The air vehicle flying qualities in response to atmospheric disturbances are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle response to disturbance inputs as specified in MIL-DTL-32742, Short-term Pitch (Roll) Response to Disturbance Inputs, Short-term Yaw Response to Disturbance Inputs, Short-term Height Response to Disturbance Inputs, Short-term Position Response to Disturbance Inputs, Short-term Translational Rate Response to Disturbance Inputs paragraph in conjunction with qualitative results from piloted simulation for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle response to disturbance inputs as specified in MIL-DTL-32742, for specified MTEs and other mission representative scenarios for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle response to disturbance inputs as specified in MIL-DTL-32742 for specified MTEs and other mission representative scenarios for

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selected combination of air vehicle configuration and loading with the SFE. A final adjudication of airworthiness with respect to disturbance rejection should consider both quantitative and qualitative criteria with greater credence given to qualitative results.

6.1.11.8 Icing conditions.

Standard: The air vehicle meets the flight in icing conditions standard as specified in 14 CFR 27.1093, 29.1093, 27.1325, 29.1325, 27.1419, and 29.1419 including the supplemental appendix on icing certification and AW criteria for rotary-wing instrument flight, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2 with airframe ice accretion during and after flight through icing conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis and simulation to assess the air vehicle flying qualities in icing conditions as specified in 14 CFR 27.1093, 29.1093, 27.1325, 29.1325, 27.1419, and 29.1419 including the supplemental appendix on icing certification for rotary-wing instrument flight for each air vehicle configuration, loading, and icing conditions within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle handling qualities in icing conditions as specified in 14 CFR 27.1093, 29.1093, 27.1325, 29.1325, 27.1419, and 29.1419 including the supplemental appendix on icing certification for rotary-wing instrument flight for each air vehicle configuration, loading, and icing conditions within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle handling qualities in icing conditions as specified in 14 CFR 27.1093, 29.1093, 27.1325, 29.1325, 27.1419, and 29.1419 including the supplemental appendix on icing certification for rotary-wing instrument flight for selected combinations of air vehicle configuration, loading, and icing conditions within the SFE.

6.1.11.9 Ground operations and air/ground transition.

Standard: The air vehicle ground handling characteristics are no worse than Level 2 handling qualities during takeoff, landing, and taxi maneuvers.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the ground handling qualities during takeoff, landing, and ground maneuvers as specified in ADS-33E-PRF, Rotor Start/Stop, Parked Position Requirement, and Wheeled Rotorcraft Ground Requirements paragraphs for each air vehicle configuration and internal loading.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the ground handling qualities during takeoff, landing, and ground maneuvers as specified in ADS-33E-PRF, Rotor Start/Stop, Parked Position Requirement, and Wheeled Rotorcraft Ground Requirements paragraphs for each air vehicle configuration and internal loading.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the ground handling qualities during takeoff, landing, and ground maneuvers as specified in ADS-33E-PRF, Rotor Start/Stop, Parked Position Requirement, and Wheeled Rotorcraft Ground Requirements paragraphs for selected combinations of air vehicle configuration and internal loading.

6.1.11.10 Flight beyond the service flight envelope.

Standard: Flight beyond the SFE that does not involve structural failure or unrecoverable loss of rotor RPM is recoverable to the SFE without undue pilot skill.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the handling qualities that result from transient excursions from, and subsequent recoveries to, the SFE for each air vehicle configuration and loading. Perform intentional excursions from the SFE (e.g., airspeed, bank angle, Nz, Nr, sideslip, CG) and evaluate the handling qualities of recovery to the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the handling qualities that result from transient excursions from, and subsequent recoveries to, the SFE for each air vehicle configuration and loading. Perform intentional excursions from the SFE (e.g., airspeed, bank angle, Nz, Nr, sideslip, CG) and evaluate the handling qualities of recovery to the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the handling qualities that result from transient excursions from, and subsequent recoveries to, the SFE for selected combinations of air vehicle configuration and loading. Perform intentional excursions from the SFE (e.g., airspeed, bank angle, Nz, Nr, sideslip, CG) and evaluate the handling qualities of recovery to the SFE.

6.1.11.11 Pilot-in-the-loop oscillations.

Standard: In any axis or in a combination of axes, there is no tendency for sustained or uncontrollable unintentional oscillations in which the air vehicle attitude, angular rate,

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acceleration, or other quantity derived from these states is approximately 180 degrees out of phase with the pilot or control system's commands.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the hover and low airspeed bandwidth criterion as specified in MIL-DTL-32742, Short-term Response to Control Inputs (Bandwidth) and Short-term Response to Yaw Control Inputs (Bandwidth) paragraphs for each air vehicle configuration and loading within the SFE. Similarly, conduct analysis to assess the forward flight bandwidth criterion as specified in MIL-DTL-32742, Short-term Response (Bandwidth), Small Amplitude Roll Attitude Response to Control Inputs (Bandwidth) paragraphs. Analyze time domain data of inceptor activity and air vehicle response as well as qualitative pilot comments and PIO ratings (as required) that result from the execution of MTEs for PIO tendency.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the hover and low airspeed bandwidth criterion as specified in MIL-DTL-32742, Short-term Response to Control Inputs (Bandwidth) and Short-term Response to Yaw Control Inputs (Bandwidth) paragraphs for each air vehicle configuration and loading within the SFE. Similarly, conduct analysis to assess the forward flight bandwidth criterion as specified in MIL-DTL-32742, Short-term Response (Bandwidth), Small Amplitude Roll Attitude Response to Control Inputs (Bandwidth) paragraphs. Analyze time domain data of inceptor activity and air vehicle response as well as qualitative pilot comments and PIO ratings (as required) that result from the execution of MTEs for PIO tendency.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the hover and low airspeed bandwidth criterion as specified in MIL-DTL-32742, Short-term Response to Control Inputs (Bandwidth) and Short-term Response to Yaw Control Inputs (Bandwidth) paragraphs for selected combinations of air vehicle configuration and loading within the SFE. Similarly, conduct analysis to assess the forward flight bandwidth criterion as specified in MIL-DTL-32742, Short-term Response (Bandwidth), Small Amplitude Roll Attitude Response to Control Inputs (Bandwidth) paragraphs. Analyze time domain data of inceptor activity and air vehicle response as well as qualitative pilot comments and PIO ratings (as required) that result from the execution of MTEs for PIO tendency.

6.1.11.12 Flight control system failures.

Standard: Specific failures of the FCS to include complete or partial loss of any function of the FCS following a single failure, transient motions due to failures, trim changes following reversion to alternate control modes, and required or recommended configuration changes following failures result in handling qualities no worse than Level 3.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess that failures of the FCS to include complete or partial loss of any function of the FCS following a single failure, transient motions due to failures, trim changes following reversion to alternate control modes, and required or recommended configuration changes following failures result in handling qualities no worse than Level 3 as specified in ADS-33E-PRF, Failures of the Flight Control System paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that failures of the FCS to include complete or partial loss of any function of the FCS following a single failure, transient motions due to failures, trim changes following reversion to alternate control modes, and required or recommended configuration changes following failures result in handling qualities no worse than Level 3 as specified in ADS-33E-PRF, Failures of the Flight Control System paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess that failures of the FCS to include complete or partial loss of any function of the FCS following a single failure, transient motions due to failures, trim changes following reversion to alternate control modes, and required or recommended configuration changes following failures result in handling qualities no worse than Level 3 as specified in ADS-33E-PRF Failures of the Flight Control System paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.13 Failures of a single engine.

Standard: From any condition within the SFE, the air vehicle is capable of sustaining a single engine failure, entering into a power off autorotation, and landing on a level surface from all points outside of the height velocity curve.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess transient air vehicle responses to a sudden engine failure. Evaluate the handling qualities of entering and sustaining a power-off autorotative descent from both forward flight and high altitude hovering flight while maintaining operational limitations. Execute a forced landing to a level surface from outside of the height velocity curve. Perform analysis for each air vehicle configuration and loading.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess transient air vehicle responses to a sudden engine failure. Evaluate the handling qualities of entering and sustaining a power-off autorotative descent from both forward flight and high altitude hovering flight while maintaining operational limitations. Execute a forced landing to a level surface from outside of the height velocity curve. Perform analysis for each air vehicle configuration and loading.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess transient air vehicle responses to a sudden engine failure for selected combinations of air vehicle configuration and loading within the SFE. Evaluate the handling qualities of entering and sustaining a power off autorotative descent from both forward flight and high altitude hovering flight while maintaining operational limitations. Execute a forced landing to a level surface from outside of the height velocity curve.

6.1.11.14 Complete or partial loss of engine power.

Standard: Complete or partial loss of engine power results in handling qualities no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess handling qualities following complete or partial loss of engine power. Restarting a failed engine from level flight with OEI or from an autorotative descent does not cause the FCS to fail or become inoperative.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess handling qualities following complete or partial loss of engine power. Restarting a failed engine from level flight with OEI or from an autorotative descent does not cause the FCS to fail or become inoperative.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess handling qualities following complete or partial loss of engine power for selected combinations of air vehicle configuration and loading within the SFE. Restarting a failed engine from level flight with OEI or from an autorotative descent does not cause the FCS to fail or become inoperative.

6.1.11.15 Complete or partial loss of an electrical subsystem.

Standard: Complete or partial loss of an electrical subsystem results in handling qualities no worse than Level 2.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess handling qualities following complete or partial loss of an electrical subsystem.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess handling qualities following complete or partial loss of an electrical subsystem.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess handling qualities following complete or partial loss of an electrical subsystem for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.16 External load jettison.

Standard: The rotary-wing air vehicle is capable of jettisoning external loads without incurring damage to the airframe or resulting in loss of control.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the air vehicle is capable of jettisoning external loads without loss of control or incurring damage to the air vehicle for each air vehicle configuration and loading within the SFE. External loadings should include the minimum and maximum combinations of weight and sling/pendant length approved for service.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the air vehicle is capable of jettisoning external loads without loss of control or incurring damage to the air vehicle for each air vehicle configuration and loading within the SFE. External loadings should include the minimum and maximum combinations of weight and sling/pendant length approved for service.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the air vehicle is capable of jettisoning external loads without loss of control or incurring damage to the air vehicle for selected combinations of air vehicle configuration and loading within the SFE. External loadings should include the minimum and maximum combinations of weight and sling/pendant length approved for service.

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6.1.11.17 Failure of external load suspension system.

Standard: Any single failure of a suspension system element such as attachment fittings, slings, pendants, apex fittings, and cargo hooks does not result in loss of control of the rotary-wing or cause damage to the airframe.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess that any single failure of a suspension system element such as attachment fittings, slings, pendants, apex fittings, and cargo hooks does not result in loss of control of the air vehicle or cause damage to the airframe as specified in ADS-33E-PRF, Failure of External Load System paragraph for each air vehicle configuration and loading within the external loads SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that any single failure of a suspension system element such as attachment fittings, slings, pendants, apex fittings, and cargo hooks does not result in loss of control of the air vehicle or cause damage to the airframe as specified in ADS-33E-PRF, Failure of External Load System paragraph for each air vehicle configuration and loading within the external loads SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess that any single failure of a suspension system element such as attachment fittings, slings, pendants, apex fittings, and cargo hooks does not result in loss of control of the air vehicle or cause damage to the airframe as specified in ADS-33E-PRF, Failure of External Load System paragraph for selected combinations of air vehicle configuration and loading within the external loads SFE.

6.1.11.18 Rotary-wing failures.

Standard: The rotary-wing air vehicle's handling qualities in the presence of one or more system failures are no worse than Level 2 as specified in MIL-DTL-32742, Rotorcraft Failures paragraph, and that indications of failures are immediate and easily interpretable by the flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess that the rotary-wing air vehicle meets the allowable degradation in handling qualities due to one or more system failures as specified in MIL-DTL-32742, Rotorcraft Failures paragraph and subparagraphs for each air vehicle configuration and loading within the SFE.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that the rotary-wing air vehicle meets the allowable degradation in handling qualities due to one or more system failures as specified in MIL-DTL-32742, Rotorcraft Failures paragraph and subparagraphs for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess that the rotary-wing air vehicle meets the allowable degradation in handling qualities due to one or more system failures as specified in MIL-DTL-32742, Rotorcraft Failures paragraph and subparagraphs for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.19 Rotor revolutions per minute governing.

Standard: While executing MTEs, the rotor RPM stays within the limits established by the SFE.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess that while executing the MTEs designated for the air vehicle required agility and category, the rotor RPM does not decay below the lower limit, nor does it exceed the upper limit prescribed by the SFE as specified in MIL-DTL-32742, Rotor RPM Governing paragraph for each air vehicle configuration and loading.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that while executing the MTEs designated for the air vehicle required agility and category, the rotor RPM does not decay below the lower limit nor does it exceed the upper limit prescribed by the SFE as specified in MIL-DTL-32742, Rotor RPM Governing paragraph for each air vehicle configuration and loading.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test, for selected combinations of air vehicle configuration and loading within the SFE, to assess that while executing the MTEs designated for the air vehicle required agility and category, the rotor RPM does not decay below the lower limit nor does it exceed the upper limit prescribed by the SFE as specified in MIL-DTL-32742, Rotor RPM Governing paragraph for each air vehicle configuration and loading.

6.1.11.20 Short-term response (bandwidth) – pitch and roll.

Standard: The air vehicle's short-term response to pitch and roll control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-

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32742, Short-term Response to Control Inputs (Bandwidth) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the pitch and roll axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response to Control Inputs (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the pitch and roll axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response to Control Inputs (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the pitch and roll axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response to Control Inputs (Bandwidth) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.21 Mid-term response – pitch and roll.

Standard: The air vehicle's mid-term response to pitch and roll control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Mid-Term Response to Control Longitudinal (Lateral) Inputs paragraph and subparagraphs, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the mid-term response to pitch and roll control inputs for both fully attended and divided attention operations as specified in MIL-DTL-32742, Mid-Term Response to Control Longitudinal (Lateral) Inputs paragraph and subparagraphs for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the mid-term response to pitch and roll control inputs for both fully attended and divided attention operations as specified in MIL-DTL-32742, Mid-Term Response to Control Longitudinal (Lateral) Inputs paragraph and subparagraphs for each air vehicle configuration and loading within the SFE.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the mid-term response to pitch and roll control inputs for both fully attended and divided attention operations as specified in MIL-DTL-32742, Mid-Term Response to Control Longitudinal (Lateral) Inputs paragraph and subparagraphs for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.22 Moderate amplitude attitude changes – pitch and roll.

Standard: The air vehicle's moderate amplitude response to pitch and roll control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Moderate Amplitude Pitch (Roll) Attitude Changes (Attitude Quickness) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the moderate amplitude response to pitch and roll control inputs as specified in MIL-DTL-32742, Moderate Amplitude Pitch (Roll) Attitude Changes (Attitude Quickness) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the moderate amplitude response to pitch and roll control inputs as specified in MIL-DTL-32742, Moderate Amplitude Pitch (Roll) Attitude Changes (Attitude Quickness) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the moderate amplitude response to pitch and roll control inputs as specified in MIL-DTL-32742, Moderate Amplitude Pitch (Roll) Attitude Changes (Attitude Quickness) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.23 Large amplitude attitude changes – pitch and roll.

Standard: The air vehicle's large amplitude response to pitch and roll control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Large-Amplitude Pitch (Roll) Attitude Changes paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis to assess the achievable angular rate (rate response types) or attitude change (attitude response types) as specified in MIL-DTL-32742, Large-Amplitude Pitch (Roll) Attitude Changes paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the achievable angular rate (rate response types) or attitude change (attitude response types) as specified in MIL-DTL-32742, Large-Amplitude Pitch (Roll) Attitude Changes paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the achievable angular rate (for Rate Response Types) or Attitude change (for attitude Response types) as specified in MIL-DTL-32742 Large-Amplitude Pitch (Roll) Attitude Changes paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.24 Short-term response (bandwidth) – yaw.

Standard: The air vehicle's short-term response to yaw control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Short-term Response to Yaw Control Inputs (Bandwidth) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the yaw axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response to Yaw Control Inputs (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the yaw axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response to Yaw Control Inputs (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the yaw axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response to Yaw Control Inputs (Bandwidth) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

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6.1.11.25 Yaw rate response to lateral gusts.

Standard: In hover and low airspeed (up to 45 KGS), in the 3 seconds following a step lateral gust input, the ratio of peak yaw rate to gust speed is no greater than 1.0 (deg/s)/(ft/s) unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the yaw rate response to lateral gust inputs as specified in MIL-DTL-32742, Yaw Rate Response to Lateral Gusts paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the yaw rate response to lateral gust inputs as specified in MIL-DTL-32742, Yaw Rate Response to Lateral Gusts paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the yaw rate response to lateral gust inputs as specified in MIL-DTL-32742, Yaw Rate Response to Lateral Gusts paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.26 Mid-term response – yaw.

Standard: The air vehicle's mid-term response to yaw control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Mid-Term Response to Yaw Control Inputs paragraph and subparagraphs, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the mid-term response to yaw control inputs for both fully attended and divided attention operations are as specified in MIL-DTL-32742, Mid-Term Response to Yaw Control Inputs paragraph, as well as its two subparagraphs, for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the mid-term response to yaw control inputs for both fully attended and divided attention operations are as specified in MIL-DTL-32742, Mid-Term Response to Yaw Control Inputs paragraph, as well as its two subparagraphs, for each air vehicle configuration and loading within the SFE.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the mid-term response to yaw control inputs for both fully attended and divided attention operations are as specified in MIL-DTL-32742, Mid-Term Response to Yaw Control Inputs paragraph, as well as its two subparagraphs, for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.27 Moderate amplitude heading changes - yaw.

Standard: The air vehicle's moderate amplitude response to yaw control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Moderate Amplitude Heading Attitude Changes (Attitude Quickness) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess that the ratio of peak yaw rate to change in heading is as specified in MIL-DTL-32742, Moderate Amplitude Heading Attitude Changes (Attitude Quickness) paragraph for each configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess that the ratio of peak yaw rate to change in heading is as specified in MIL-DTL-32742, Moderate Amplitude Heading Attitude Changes (Attitude Quickness) paragraph for each configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess that the ratio of peak yaw rate to change in heading is as specified in MIL-DTL-32742, Moderate Amplitude Heading Attitude Changes (Attitude Quickness) paragraph for selected combinations of configuration and loading within the SFE.

6.1.11.28 Large amplitude heading changes - yaw.

Standard: The air vehicle's large amplitude response to yaw control inputs in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Large-Amplitude Heading Changes paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis to assess the achievable yaw rate in hover as specified in MIL-DTL-32742, Large-Amplitude Heading Changes paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the achievable yaw rate in hover as specified in MIL-DTL-32742, Large-Amplitude Heading Changes paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the achievable yaw rate in hover as specified in MIL-DTL-32742, Large-Amplitude Heading Changes paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.29 Yaw due to collective for aggressive agility.

Standard: The air vehicle's yaw rate response to a step collective input in hover and low airspeed (up to 45 KGS) for aggressive agility is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Yaw due to Collective for Aggressive Agility paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the yaw rate response to an abrupt collective control input as specified in MIL-DTL-32742, Yaw Due to Collective for Aggressive Agility paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the yaw rate response to an abrupt collective control input as specified in MIL-DTL-32742, Yaw Due to Collective for Aggressive Agility paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the yaw rate response to an abrupt collective control input as specified in MIL-DTL-32742, Yaw Due to Collective for Aggressive Agility paragraph for selected combinations of air vehicle configuration and loading within the SFE.

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6.1.11.30 Pitch and roll coupling for aggressive agility.

Standard: The air vehicle's pitch and roll coupling in hover and low airspeed (up to 45 KGS) for aggressive agility is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Aggressive Agility paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the ratio of the peak off-axis attitude response from trim to the desired (on-axis) attitude response from trim following an abrupt lateral or longitudinal cockpit control step as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Aggressive Agility paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the ratio of the peak off-axis attitude response from trim to the desired (on-axis) attitude response from trim following an abrupt lateral or longitudinal cockpit control step as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Aggressive Agility paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the ratio of the peak off-axis attitude response from trim to the desired (on-axis) attitude response from trim following an abrupt lateral or longitudinal cockpit control step as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Aggressive Agility paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.31 Pitch and roll coupling for target acquisition and tracking.

Standard: The air vehicle's pitch and roll coupling in hover and low airspeed (up to 45 KGS) for target acquisition and tracking is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the pitch due to roll and roll due to pitch coupling for target acquisition and tracking as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph for each air vehicle configuration and loading within the SFE. Frequency response data generated for forward flight pitch and roll

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bandwidth can be used to assess compliance provided off-axis control inputs are minimized and uncorrelated with the on-axis sweep. Multi-input FFT analysis methods are required to obtain the correct off-axis frequency responses necessary to evaluate against this standard.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the pitch due to roll and roll due to pitch coupling for target acquisition and tracking as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph for each air vehicle configuration and loading within the SFE. Frequency response data generated for forward flight pitch and roll bandwidth can be used to assess compliance provided off-axis control inputs are minimized and uncorrelated with the on-axis sweep. Multi-input FFT analysis methods are required to obtain the correct off-axis frequency responses necessary to evaluate against this standard.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test and piloted simulation and evaluation using models that are updated with flight test results to assess the pitch due to roll and roll due to pitch coupling for target acquisition and tracking as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph for selected combinations of air vehicle configuration and loading within the SFE. Frequency response data generated for forward flight pitch and roll bandwidth can be used to assess compliance provided off-axis control inputs are minimized and uncorrelated with the on-axis sweep. Multi-input FFT analysis methods are required to obtain the correct off-axis frequency responses necessary to evaluate against this standard.

6.1.11.32 Height response characteristics.

Standard: The air vehicle's height response characteristics in hover and low airspeed (up to 45 KGS) are no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Height Response Characteristics paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the height response characteristics as specified in MIL-DTL-32742, Height Response Characteristics paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the height response characteristics as specified in MIL-DTL-32742, Height Response Characteristics paragraph for each air vehicle configuration and loading within the SFE.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the height response characteristics as specified in MIL-DTL-32742, Height Response Characteristics paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.33 Torque response characteristics.

Standard: The air vehicle's torque response in hover and low airspeed (up to 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Torque Response paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the torque response as specified in MIL-DTL-32742, Torque Response paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the torque response as specified in MIL-DTL-32742, Torque Response paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the torque response as specified in MIL-DTL-32742, Torque Response paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.34 Vertical axis control power.

Standard: From an inertially stable out of ground effect hover with the wind from the most critical direction at a velocity of up to 35 kts, in the most critical air vehicle loading, at the most critical density altitude, the air vehicle achieves a minimum vertical climb rate of 55 ft/min in not more than 1.5 seconds, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the vertical axis control power as specified in as specified in MIL-DTL-32742, Vertical Axis Control Power paragraph for each air vehicle configuration within the

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SFE. Collective inputs need only be in the upward direction as this standard is an assessment of climb rate only.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the vertical axis control power as specified in as specified in MIL-DTL-32742, Vertical Axis Control Power paragraph for each air vehicle configuration within the SFE. Collective inputs need only be in the upward direction as this standard is an assessment of climb rate only.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the vertical axis control power as specified in as specified in MIL-DTL-32742, Vertical Axis Control Power paragraph for selected combinations of air vehicle configuration within the SFE. Collective inputs need only be in the upward direction as this standard is an assessment of climb rate only.

6.1.11.35 Short-term response (bandwidth) – pitch and roll.

Standard: The air vehicle's short-term response to pitch control inputs in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Short-Term Response (Bandwidth) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight pitch axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight pitch axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight pitch axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Short-term Response (Bandwidth) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.36 Mid-term response – pitch and roll.

Standard: The air vehicle's mid-term response to pitch control inputs in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Mid-

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Term Response to Control Inputs paragraph and subparagraphs, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight mid-term response to pitch control inputs for both fully attended and divided attention operations as specified in MIL-DTL-32742, Mid-Term Response to Control Inputs paragraph and subparagraphs for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight mid-term response to pitch control inputs for both fully attended and divided attention operations as specified in MIL-DTL-32742, Mid-Term Response to Control Inputs paragraph and subparagraphs for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight mid-term response to pitch control inputs for both fully attended and divided attention operations as specified in MIL-DTL-32742, Mid-Term Response to Control Inputs paragraph and subparagraphs for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.37 Maneuvering stability.

Standard: In forward flight (greater than 45 KGS), the incremental control force required to produce a change in normal LF and pitch rate is in the same sense (aft force – increase LF, forward force – decrease LF) as required to initiate the change and that there is no tendency for the air vehicle pitch attitude or AOA to diverge aperiodically.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the control feel and stability during constant airspeed maneuvering in forward flight for each air vehicle configuration and loading within the SFE. Obtain stick force per g to confirm desired sensing and examine pitch attitude and AOA response for positive stability.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the control feel and stability during constant airspeed maneuvering in forward flight for each air vehicle configuration and loading within the SFE. Obtain stick force per g to confirm desired sensing and examine pitch attitude and AOA response for positive stability.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the control feel and stability during constant airspeed maneuvering in forward flight for selected combinations of air vehicle configuration and loading within the SFE. Obtain stick force per g to confirm desired sensing and examine pitch attitude and AOA response for positive stability.

6.1.11.38 Control forces in maneuvering flight.

Standard: The air vehicle's control forces during maneuvering forward flight (greater than 45 KGS) have no local negative gradients, the deflection of the longitudinal cockpit controller does not lead the control force at any frequency below 5 rad/s, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2 to include any gradient nonlinearities.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the control forces in maneuvering flight as specified in MIL-DTL-32742 Control Forces in Maneuvering Flight paragraph for each air vehicle configuration and loading within the SFE. Obtain stick force per g to confirm desired sensing and gradient behavior. Obtain a frequency response of the control deflection to control force and verify no unfavorable controller dynamics lead to stick force lightening (bobweight effect) whereby stick position leads the force. An established threshold defining the lower limit of Level 2 flying qualities is not provided in MIL-DTL 32742, therefore compliance with the standard is expected to be qualitative only.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the control forces in maneuvering flight as specified in MIL-DTL-32742, Control Forces in Maneuvering Flight paragraph for each air vehicle configuration and loading within the SFE. Obtain stick force per g to confirm desired sensing and gradient behavior. Obtain a frequency response of the control deflection to control force and verify no unfavorable controller dynamics lead to stick force lightening (bobweight effect) whereby stick position leads the force. An established threshold defining the lower limit of Level 2 flying qualities is not provided in MIL-DTL 32742, therefore compliance with the standard is expected to be qualitative only.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the control forces in maneuvering flight as specified in MIL-DTL-32742, Control Forces in Maneuvering Flight paragraph for selected combinations of air vehicle configuration and loading within the SFE. Obtain stick force per g to confirm desired sensing and gradient behavior. Obtain a frequency response of the control deflection to control force and verify no unfavorable controller dynamics lead to stick force lightening

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(bobweight effect) whereby stick position leads the force. An established threshold defining the lower limit of Level 2 flying qualities is not provided in MIL-DTL 32742, therefore compliance with the standard is expected to be qualitative only.

6.1.11.39 Pitch control authority.

Standard: The air vehicle's pitch control power in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Pitch Control Power paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight pitch control power as specified in MIL-DTL-32742, Pitch Control Power paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight pitch control power as specified in MIL-DTL-32742, Pitch Control Power paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight pitch control power as specified in MIL-DTL-32742, Pitch Control Power paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.40 Flight path response to pitch attitude.

Standard: The air vehicle's forward flight (greater than 45 KGS) vertical rate response does not lag the pitch attitude response by more than 45 degrees at all frequencies below 0.25 rad/s, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight vertical rate response as specified in MIL-DTL-32742, Flight Path Control and Flight Path Response to Pitch Attitude (Frontside) paragraph for each air vehicle configuration and loading within the SFE.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight vertical rate response as specified in MIL-DTL-32742, Flight Path Control and Flight Path Response to Pitch Attitude (frontside) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight vertical rate response as specified in MIL-DTL-32742, Flight Path Control and Flight Path Response to Pitch Attitude (Frontside) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.41 Flight path response to collective controller.

Standard: The air vehicle's flight path response to collective inputs in forward flight (greater than 45 KGS) are no worse than Level 2 flying qualities as specified in MIL-DTL-32742 Flight Path Response to Collective Controller (Backside) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the flight path response to collective inputs in forward flight as specified in MIL-DTL-32742, Flight Path Control and Flight Path Response to Collective Controller (Backside) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the flight path response to collective inputs in forward flight as specified in MIL-DTL-32742 Flight Path Control and Flight Path Response to Collective Controller (Backside) paragraphs for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the flight path response to collective inputs in forward flight as specified in MIL-DTL-32742, Flight Path Control and Flight Path Response to Collective Controller (Backside) paragraphs for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.42 Longitudinal static stability.

Standard: In forward flight (greater than 45 KGS) and without retrimming, push forces on the longitudinal controller increase speed and pull forces decrease airspeed, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight longitudinal static stability as specified in MIL-DTL-32742 Longitudinal Static Stability paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight longitudinal static stability as specified in MIL-DTL-32742, Longitudinal Static Stability paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight longitudinal static stability as specified in MIL-DTL-32742, Longitudinal Static Stability paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.43 Small amplitude roll attitude response to control inputs.

Standard: The air vehicle's small amplitude roll attitude response to control inputs in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Small-Amplitude Roll Attitude Response to Control Inputs (Bandwidth) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight roll axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Small-Amplitude Roll Attitude Response to Control Inputs (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight roll axis bandwidth and phase delay parameters as specified in MIL-DTL-32742 Small-Amplitude Roll Attitude Response to Control Inputs (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight roll axis bandwidth and phase delay parameters as specified in MIL-DTL-32742, Small-Amplitude Roll Attitude Response to

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Control Inputs (Bandwidth) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.44 Moderate amplitude roll attitude changes.

Standard: The air vehicle's attitude quickness in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Moderate Amplitude Attitude Changes (Attitude Quickness) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight attitude quickness as specified in MIL-DTL-32742, Moderate Amplitude Attitude Changes (Attitude Quickness) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight attitude quickness as specified in MIL-DTL-32742, Moderate Amplitude Attitude Changes (Attitude Quickness) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight attitude quickness as specified in MIL-DTL-32742, Moderate Amplitude Attitude Changes (Attitude Quickness) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.45 Large amplitude roll attitude changes.

Standard: The air vehicle's large amplitude roll attitude response in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Large-Amplitude Roll Attitude Changes paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight large amplitude roll response as specified in MIL-DTL-32742, Large-Amplitude Roll Attitude Changes paragraph for each air vehicle configuration and loading within the SFE.

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Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight large amplitude roll response as specified in MIL-DTL-32742, Large-Amplitude Roll Attitude Changes paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the forward flight large amplitude roll response as specified in MIL-DTL-32742, Large-Amplitude Roll Attitude Changes paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.46 Bank angle oscillations.

Standard: The air vehicle's bank angle response in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Bank Angle Oscillations paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess bank angle oscillations in forward flight as specified in MIL-DTL-327142, Roll-Sideslip Coupling and Bank Angle Oscillations paragraphs for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess bank angle oscillations in forward flight as specified in MIL-DTL-327142, Roll-Sideslip Coupling and Bank Angle Oscillations paragraphs for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess bank angle oscillations in forward flight as specified in MIL-DTL-327142, Roll-Sideslip Coupling and Bank Angle Oscillations paragraphs for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.47 Turn coordination.

Standard: The air vehicle's turn coordination in forward flight (greater than 45 KGS) does not result in handling qualities worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis to assess turn coordination performance as specified in MIL-DTL-32742, Roll-Sideslip Coupling and Turn Coordination paragraphs for each air vehicle configuration and loading within the SFE. An established threshold defining the lower limit of Level 2 flying qualities is not provided in MIL-DTL 32742, therefore compliance with the standard is expected to be qualitative only.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess turn coordination performance as specified in MIL-DTL-32742, Roll-Sideslip Coupling and Turn Coordination paragraphs for each air vehicle configuration and loading within the SFE. An established threshold defining the lower limit of Level 2 flying qualities is not provided in MIL-DTL 32742, therefore compliance with the standard is expected to be qualitative only.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess turn coordination performance as specified in MIL-DTL-32742, Roll-Sideslip Coupling and Turn Coordination paragraphs for selected combinations of air vehicle configuration and loading within the SFE. An established threshold defining the lower limit of Level 2 flying qualities is not provided in MIL-DTL 32742, therefore compliance with the standard is expected to be qualitative only.

6.1.11.48 Yaw control bandwidth for target acquisition and tracking.

Standard: The air vehicle's small amplitude yaw response for target acquisition and tracking in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Small-Amplitude Yaw Response for Target Acquisition and Tracking (Bandwidth) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the forward flight yaw axis bandwidth and phase delay parameters for target acquisition and tracking as specified in MIL-DTL-32742, Small-Amplitude Yaw Response for Target Acquisition and Tracking (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the forward flight yaw axis bandwidth and phase delay parameters for target acquisition and tracking as specified in MIL-DTL-32742, Small-Amplitude Yaw Response for Target Acquisition and Tracking (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct flight test to assess the forward flight yaw axis bandwidth and phase delay parameters for target acquisition and tracking as specified in MIL-DTL-32742, Small-Amplitude Yaw Response for Target Acquisition and Tracking (Bandwidth) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.49 Large amplitude heading changes for aggressive agility.

Standard: For air vehicles requiring aggressive agility, in forward flight (greater than 45 KGS), heading changes in one second following an abrupt step displacement of the yaw control are at least 8 degrees or 50% of the operational flight envelope sideslip limit, whichever is smaller.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess large amplitude heading changes in forward flight as specified as specified in MIL-DTL-32742, Large-Amplitude Heading Changes for Aggressive Agility (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE. Apply and hold step input for at least one second while holding other controls fixed. Small control movements, however, are allowed to minimize pitch and roll excursions. Conduct in both directions (left and right).

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess large amplitude heading changes in forward flight as specified in MIL-DTL-32742, Large-Amplitude Heading Changes for Aggressive Agility (Bandwidth) paragraph for each air vehicle configuration and loading within the SFE. Apply and hold step input for at least one second while holding other controls fixed. Small control movements, however, are allowed to minimize pitch and roll excursions. Conduct in both directions (left and right).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess large amplitude heading changes in forward flight as specified as specified in MIL-DTL-32742, Large-Amplitude Heading Changes for Aggressive Agility (Bandwidth) paragraph for selected combinations of air vehicle configuration and loading within the SFE. Apply and hold step input for at least one second while holding other controls fixed, Small control movements, however, are allowed to minimize pitch and roll excursions. Conduct in both directions (left and right).

6.1.11.50 Yaw control with speed change.

Standard: The air vehicle's yaw control forces with speed changes in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Yaw Control with Speed Change paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the yaw control forces with speed changes in forward flight as specified in MIL-DTL-32742, Yaw Control with Speed Change paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the yaw control forces with speed changes in forward flight as specified in MIL-DTL-32742, Yaw Control with Speed Change paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the yaw control forces with speed changes in forward flight as specified in MIL-DTL-32742, Yaw Control with Speed Change paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.51 Lateral-directional oscillations.

Standard: The air vehicle's lateral directional oscillations in forward flight (greater than 45 KGS) are no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Lateral-Directional Oscillations paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the natural frequency and damping ratio of the lateral directional oscillations in forward flight as specified in MIL-DTL-32742, Lateral-Directional Oscillations paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the natural frequency and damping ratio of the lateral directional oscillations in forward flight as specified in MIL-DTL-32742, Lateral-Directional Oscillations paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the natural frequency and damping ratio of the lateral directional oscillations in forward flight as specified in MIL-DTL-32742, Lateral-Directional

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Oscillations paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.52 Spiral stability.

Standard: In forward flight (greater than 45 KGS), the air vehicle's spiral mode stability as indicated by the variation of bank angle with time is linear or the time to double in bank angle is greater than 12 seconds.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the spiral mode stability as indicated by the bank angle time to double as specified in MIL-DTL-32742, Spiral Stability paragraph for each air vehicle configuration and loading within the SFE. A linear variation of bank angle with time following lateral pulse input regardless of the time to double is considered compliant. The initiating pulse input should be just large enough to generate a roll disturbance. Spiral mode stability is either stable (bank angle returns to trim), neutral (bank angle remains at new position), or divergent (bank angle continuously increases).

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the spiral mode stability as indicated by the bank angle time to double as specified in MIL-DTL-32742, Spiral Stability paragraph for each air vehicle configuration and loading within the SFE. A linear variation of bank angle with time following lateral pulse input regardless of the time to double is considered compliant. The initiating pulse input should be just large enough to generate a roll disturbance.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the spiral mode stability as indicated by the bank angle time to double as specified in MIL-DTL-32742, Spiral Stability paragraph for selected combinations of air vehicle configuration and loading within the SFE. A linear variation of bank angle with time following lateral pulse input regardless of the time to double is considered compliant. The initiating pulse input should be just large enough to generate a roll disturbance.

6.1.11.53 Directional stability.

Standard: The air vehicle's directional stability in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Yaw Control in Steady Sideslips (Directional Stability) paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct analysis to assess the directional stability in forward flight as specified in MIL-DTL-32742, Yaw Control in Steady Sideslips (Directional Stability) paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the directional stability in forward flight as specified in MIL-DTL-32742, Yaw Control in Steady Sideslips (Directional Stability) paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the directional stability in forward flight as specified in MIL-DTL-32742, Yaw Control in Steady Sideslips (Directional Stability) paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.54 Bank angle in steady sideslips.

Standard: The air vehicle's bank angle response to sideslip in forward flight (greater than 45 KGS) does not have an opposing increase in bank angle opposite the commanded sideslip unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the bank angle response to sideslip as specified in MIL-DTL-32742, Bank Angle in Steady Sideslips paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the bank angle response to sideslip as specified in MIL-DTL-32742, Bank Angle in Steady Sideslips paragraph for each air vehicle configuration and loading within the SFE.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the bank angle response to sideslip as specified in MIL-DTL-32742, Bank Angle in Steady Sideslips paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.55 Lateral control in steady sideslips.

Standard: The air vehicle's lateral control in forward flight (greater than 45 KGS) steady sideslips is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Lateral

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Control in Steady Sideslips paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the lateral control in steady sideslips as specified in MIL-DTL-32742, Lateral Control in Steady Sideslips paragraph for each air vehicle configuration and loading within the SFE. From a trimmed steady heading sideslip, verify that left sideslip does not require right lateral control displacement and force (and vice-versa) and that the variation of lateral control displacement and force with sideslip is essentially linear.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the lateral control in steady sideslips as specified in MIL-DTL-32742, Lateral Control in Steady Sideslips paragraph for each air vehicle configuration and loading within the SFE. From a trimmed steady heading sideslip, verify that left sideslip does not require right lateral control displacement and force (and vice-versa) and that the variation of lateral control displacement and force with sideslip is essentially linear.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the lateral control in steady sideslips as specified in MIL-DTL-32742, Lateral Control in Steady Sideslips paragraph for selected combinations of air vehicle configuration and loading within the SFE. From a trimmed steady heading sideslip, verify that left sideslip does not require right lateral control displacement and force (and vice-versa) and that the variation of lateral control displacement and force with sideslip is essentially linear.

6.1.11.56 Dihedral effect.

Standard: In forward flight (greater than 45 KGS), the air vehicle's positive effective dihedral does not require more than 90% of the roll control power available and does not exceed 13.5 lbs of roll control force for sideslip angles that may be experienced while performing MTEs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the positive effective dihedral as specified in MIL-DTL-32742, Positive Effective Dihedral Limit paragraph for each air vehicle configuration and loading within the SFE while performing the MTEs as defined in MIL-DTL-32742, Mission Task Elements paragraph. Compliance is determined first by calculating the amount of roll controller deflection required to reach 90% of the maximum roll rate or bank angle specified in MIL-DTL-32742 Large Amplitude Roll Attitude Changes paragraph. Then ensuring that for the sideslips

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experienced while performing the designated MTEs, the roll control deflection does not exceed this computed displacement value.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the positive effective dihedral as specified in MIL-DTL-32742, Positive Effective Dihedral Limit paragraph for each air vehicle configuration and loading within the SFE while performing the MTEs as defined in MIL-DTL-32742, Mission Task Elements paragraph. Compliance is determined first by calculating the amount of roll controller deflection required to reach 90% of the maximum roll rate or bank angle specified in MIL-DTL-32742, Large Amplitude Roll Attitude Changes paragraph. Then ensuring that for the sideslips experienced while performing the designated MTEs, the roll control deflection does not exceed this computed displacement value.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test, for selected combinations of air vehicle configuration and loading within the SFE, to assess the positive effective dihedral as specified in MIL-DTL-32742, Positive Effective Dihedral Limit paragraph for each air vehicle configuration and loading within the SFE while performing the MTEs as defined in MIL-DTL-32742, Mission Task Elements paragraph. Compliance is determined first by calculating the amount of roll controller deflection required to reach 90% of the maximum roll rate or bank angle specified in MIL-DTL-32742, Large Amplitude Roll Attitude Changes paragraph. Then ensuring that for the sideslips experienced while performing the designated MTEs, the roll control deflection does not exceed this computed displacement value.

6.1.11.57 Roll due to pitch coupling for aggressive agility.

Standard: The air vehicle's roll due to pitch coupling for aggressive agility in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Roll due to Pitch Coupling for Aggressive Agility paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the roll due to pitch coupling for aggressive agility in forward flight as specified in MIL-DTL-32742, Roll due to Pitch Coupling for Aggressive Agility paragraph for each air vehicle configuration and loading within the SFE.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the roll due to pitch coupling for aggressive agility in forward flight as specified in MIL-DTL-32742, Roll due to Pitch Coupling for Aggressive Agility paragraph for each air vehicle configuration and loading within the SFE.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the roll due to pitch coupling for aggressive agility in forward flight as specified in MIL-DTL-32742, Roll due to Pitch Coupling for Aggressive Agility paragraph for selected combinations of air vehicle configuration and loading within the SFE.

6.1.11.58 Pitch and roll coupling for target acquisition and tracking.

Standard: The air vehicle's pitch and roll coupling for target acquisition and tracking in forward flight (greater than 45 KGS) is no worse than Level 2 flying qualities as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph, unless pilot evaluation determines the air vehicle handling qualities are no worse than Level 2.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to assess the pitch and roll coupling for target acquisition and tracking in forward flight as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph for each air vehicle configuration and loading within the SFE. Frequency response data generated for forward flight pitch and roll bandwidth can be used to assess compliance provided off-axis control inputs are minimized and uncorrelated with the on-axis sweep. Multi-input FFT analysis methods are required to obtain the correct off-axis frequency responses necessary to evaluate against this standard.

Test.

(1) Lab test: Conduct piloted simulation and evaluation to assess the pitch and roll coupling for target acquisition and tracking in forward flight as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph for each air vehicle configuration and loading within the SFE. Frequency response data generated for forward flight pitch and roll bandwidth can be used to assess compliance provided off-axis control inputs are minimized and uncorrelated with the on-axis sweep. Multi-input FFT analysis methods are required to obtain the correct off-axis frequency responses necessary to evaluate against this standard.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to assess the pitch and roll coupling for target acquisition and tracking in forward flight as specified in MIL-DTL-32742, Pitch due to Roll and Roll due to Pitch Coupling for Target Acquisition and Tracking paragraph for selected combinations of air vehicle configuration and loading within the SFE. Frequency response data generated for forward flight pitch and roll bandwidth can be used to assess compliance provided off-axis

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control inputs are minimized and uncorrelated with the on-axis sweep. Multi-input FFT analysis methods are required to obtain the correct off-axis frequency responses necessary to evaluate against this standard.

6.2 Vehicle control function.

The VCF is the SCF that stabilizes and controls the air vehicle throughout all flight and ground operations in both normal and degraded operational states. The VCF is an integrated system that includes, but it is not limited to, manual control, automatic modes, control laws, computer resources, actuation, air data, inertial sensors, PVI, diagnostics, and pre-flight, in-flight and post-flight BITs. For UAS, VCF includes the ground control station interfaces required to safely control the vehicle throughout all phases of fight and ground operations.

The VCF includes all hardware and software components required to sense air vehicle state, determine input commands from the flight crew or guidance and navigation source, compute the control signals, and provide actuation of the force and moment producers. These components commonly include sensors, pilot inceptors, mechanical linkages, flight control computers, control surfaces actuators, and actuators on thrust vectoring devices. Vehicle control function criticality typically drives the highest level of design integrity, robustness, and redundancy to meet safety requirements. As a result, the prime contractor and procuring activity should identify and agree on the list of all hardware and software components that comprise the VCF and are subject to the criteria of this section. Since the VCF depends on integration with other subsystems (e.g., hydraulics, electrical power, ECS, propulsion, avionics, and SMS), certain attributes of those subsystems will be considered as part of the AW assessment of the integrated VCF.

Designing a safety-critical VCF requires a disciplined, requirements-driven process that ensures traceability from concept through operations and sustainment. A structured approach supports compliance with AW standards while promoting robustness by addressing nominal, off-nominal, and worst-case failure conditions. The system safety process is integrated throughout VCF design and development to ensure a robust airworthy vehicle control system. Vehicle control system domain expertise is required throughout this process to ensure VCF related hazards and mitigations are well understood, accurately represent the VCF design, and support safe control of the air vehicle. VCF development, integration, manufacturing, and maintenance should be executed IAW the SSPP. Any deviations from the plan associated with the VCF should be documented and assessed for hazard impact.

Vehicle control function hardware components require hardware qualification analysis and testing to ensure specified performance over the range of operational air vehicle environments representing the air vehicle life without permanent degradation. These include natural and induced environments based on the air vehicle and the associated installed location, endurance testing, E3, lightning, and any specific environments associated with operational capabilities such as shipboard operations or AR. Environmental qualification testing verifies the VCF components (hardware and software) will function properly during and after exposure to various environmental conditions to meet the operational need. Vehicle control function hardware should meet the standards in 4.3 of this document. Endurance qualification testing, covered in 6.2.7, verifies VCF components meet their expected operational lifespan. This testing requires determination of representative loads and duty cycles for the specific hardware component. Box level E3 and lightning qualification should meet the standards in Section 13 of this document.

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Through all of the hardware qualification, the VCF hardware is verified to be in an operationally representative state with the ability to detect failures or performance deviations while exposed to the specific environments.

In support of a robust, fault tolerant VCF design, comprehensive system-level testing on representative hardware and interfaces is performed to provide requirements coverage and ensure integrated components work together as intended to meet safety, performance, and reliability objectives under realistic and adverse operational scenarios. As part of this testing, system-level FMET provides a key element used as a MoC for multiple standards in 6.2. Vehicle control function and integrated subsystems FMET methodology and procedures are developed based on requirements, architecture, detailed design, and failure analyses, including FMECA. Failure modes and effects testing procedures are written to provide evidence that system response to failures (including fault detection, isolation, annunciation, accommodation, and flight crew response) results in expected transient response within limits established for the vehicle. Failure modes and effects testing procedures cover single and multiple failure scenarios (including dissimilar failures) at worst-case flight conditions and include impacts of failure resets or intermittent failures where applicable.

The test environments should be verified to be appropriate for the level of test being conducted. System-level verification tests and FMET should be conducted with physical hardware and interfaces used as much as possible. Modeled or simulated system components (e.g., sensors, hydraulics, engine) should be validated as representative of actual air vehicle components and include the necessary fidelity to support test requirements. The scope of FMET should be documented and agreed early enough in the process to ensure the test environments, including modeled and simulated components, are developed to support failure insertion and accurately reflect responses to failures.

As part of a final AW assessment, open hardware and software deficiencies for the VCF and integrated subsystems are assessed to ensure they do not result in VCF hazards that cannot be mitigated to acceptable risk levels. Any mitigating limitations or procedural workarounds are documented, agreed to by the procuring activity SMEs, and communicated to the flight crew.

6.2.1 Failure response and probability of loss of control.

Probability loss of control, failure immunity, and fail-safe capability constitute a quantitative flight safety assessment of the integrated VCF that includes all flight-critical functions, components, and integrated subsystems. The intention is to provide architectural resistance and robustness to failures to ensure an airworthy design. The PLOC analysis assesses failures and failure combinations to demonstrate that the VCF architecture provides control of the air vehicle in the presence of failures. The failure immunity analysis is complementary to the PLOC analysis and ensures failures, failure combinations, and their associated transients do not cause damage to the air vehicle, departure from the air vehicle envelope, or result in incorrect failure response. Analysis typically uses a combination of FTA, FMECA, RBD’s, and hazard analysis. To prevent architecture redundancies from being traded based on unproven reliability data, a fail-safe requirement is added for single and dual failures, thus ensuring (at a minimum) analysis of all two-fail combinations. Unique failures may be identified that are to be subject to, or excluded from, the standards of 6.2.1.1 through 6.2.1.3, regardless of probability. These unique failures need to be defined, documented, and concurred with by the procuring activity.

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6.2.1.1 Probability of loss of control.

Standard: The VCF PLOC per flight hour does not exceed 10^{-7}. Vehicle control function PLOC analysis includes material failures in the VCF and integrated subsystems required to support flight-critical VCF functionality during flight and ground operations. All PLOC estimates and corresponding assumptions are substantiated. The VCF PLOC analysis is performed IAW SAE AS94900A (if the air vehicle is a UAV, SAE ARP94910), Flight Safety Assessment section. Tailoring of PLOC, if necessary, should be coordinated early in the AW planning process to ensure any impacts to other sections are properly identified, agreed, and addressed.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Failure modes, effects, and criticality assessment at the VCF system, subsystem, and component level is performed for flight and ground operations to identify failure modes, failure rates, failure detection methods, and consequences. As part of this analysis, safety-critical failure modes are identified.

(2) Fault tree analysis or RBDs representing the end-to-end flight-critical VCF functionality, including redundancy management, are developed and documented to include failure modes, failure rates, and consequences for failure combinations that impact the VCF and result in air vehicle loss of control. Failure combinations include VCF subsystems and components as well as integrated subsystems required to support flight-critical VCF functionality.

(3) Failure probabilities are substantiated with appropriate data and analysis. Substantiating reliability data may include field data from similar products and configurations, analysis of individual components with specified reliability in a representative environment (e.g., thermal, vibratory, thermal and mechanical shock, endurance cycles), or reliability testing on a statistically significant sample of components.

(4) Effects of imperfect fault coverage are accounted for in the analysis IAW SAE AS94900A (if the air vehicle is a UAV, SAE ARP94910), Effects of Imperfect Fault Detection section. Any methods used to increase fault detection coverage (e.g., use of dissimilar sources to generate analytical signals for signal selection logic) or mitigate failure consequences (e.g., design robustness to worst case errors) should be considered as part of this analysis.

6.2.1.2 Failure immunity.

Standard: Within the RORH, no single failure or failure combination in the VCF or integrated subsystems, that is more probable than PLOC divided by 10, results in any of the following effects before flight crew or safety device can take effective corrective action:

a. Flutter, divergence, or other aero-elastic instabilities within the RORH of the air vehicle, or a structural damping coefficient for any critical flutter mode below the fail-safe stability limit of MIL-A-8870.
b. Uncontrollable motions of the air vehicle or maneuvers which exceed airframe limit loads.
c. Exceedance of the RORH or inability to return to the ROTH.
d. Vehicle control function failures that could cause loss of total thrust.

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e. Erroneous and misleading air vehicle airspeed, altitude, attitude, AOA, or AOS information displayed to the flight crew that could result in incorrect flight crew inputs to the VCF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The FMECA's, FTA's or RBD's, and safety analysis are reviewed to determine the set of failures that will be analyzed for air vehicle transient response and post failure performance assessment to demonstrate the absence of behaviors a-e.

(2) Analysis of air vehicle transient response and post failure performance demonstrates the absence of behaviors a-f for failures across the envelope.

(3) Analysis of air vehicle transient response and post failure performance used to determine critical failures and corresponding flight conditions (e.g., air vehicle configuration, flight envelope, mode transitions) to be included in FMET.

Test.

(1) Lab test: Failure modes and effects testing test cases developed representing critical failures and corresponding conditions.

(2) Lab test: Failure modes and effects testing test results, with pilot comments where required, confirm absence of behaviors a-e.

6.2.1.3 Fail-safe capability.

Standard: For failures within the VCF and integrated subsystems affecting the flight control function, the VCF demonstrates fail-safe capability (no worse than Operational State III) upon any single failure more probable than PLOC divided by 100 or any combination of two failures (similar or dissimilar) more probable than PLOC divided by ten.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The FMECA's, FTA's, RBD's, and safety analysis are reviewed to determine the set of single failures and failure combinations that will be analyzed for air vehicle transient response and post failure performance assessment to demonstrate continued Operational State III performance or better. For two fail combinations, analysis includes an assessment of the system effect for failure combinations more probable than PLOC divided by 100.

(2) Analysis of air vehicle transient response and post failure performance demonstrates continued Operational State III performance or better across the envelope.

(3) Analysis of air vehicle transient response and post failure performance used to determine critical failures and corresponding flight conditions (e.g., air vehicle configuration, flight envelope, mode transitions) to be included in FMET.

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Test.

(1) Lab test: Failure modes and effects testing test cases developed representing critical failures and corresponding conditions.

(2) Lab test: Failure modes and effects testing test results, with pilot comments where required, confirm continued Operational State III performance or better.

6.2.2 Vehicle control function functional requirements.

Vehicle control function functional requirements and interfaces form the basis of the VCF V&V test scope. Through the requirements development process, SCFs are identified with traceability to lower tier derived requirements. As requirements are derived into lower tiers, analysis is provided to support their derivation. The scope of the analysis is dependent on the complexity of the derivation and can range from simple allocations to more in-depth performance assessments. As part of the process, VCF driven requirements and interfaces are established and flowed to subsystems required to support flight-critical VCF functionality.

6.2.2.1 Functional criteria.

Standard: Functional requirements and interfaces are defined with safety-critical requirements and interfaces identified. Bi-directional traceability demonstrates linkage between lower-level requirements and top-level SCFs. Requirements are defined for redundancy management and integration for all vehicle control aspects.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Functional block diagrams document VCF internal and external (hydraulics, electrical, ECS, propulsion, avionics) interfaces.

(2) Interface control documents have been finalized and are under configuration control.

(3) All SCFs and interfaces are documented.

(4) Functional and interface requirements are documented in a system requirements specification.

(5) Vehicle control function hardware CI requirements are documented.

(6) Vehicle control function software configuration requirements are documented.

Analysis.

(1) Vehicle control function requirements tool demonstrates bi-directional requirements traceability down to individual VCF hardware or software components.

(2) Modeling and simulation are used to verify functionality and interfaces.

6.2.2.2 Subsystem compatibility.

Standard: Vehicle control function driven requirements and interfaces are established and flowed to the integrated subsystems (e.g., electrical power, hydraulics, avionics, propulsion,

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ECS). These requirements capture worst-case operational scenarios, for both nominal (un-failed) and failure conditions, to ensure VCF performance is maintained. Vehicle control function functional performance is demonstrated in all worst-case nominal and degraded conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Vehicle control function requirements and interface documents with all integrated subsystems capture VCF driving requirements and interfaces and are configuration controlled.

Analysis.

(1) Analysis to support VCF driving requirements and interfaces for integrated subsystems is performed. Analysis shows that margins are maintained during worst-case operational scenarios (e.g., hydraulic demand, electrical power, cooling) including, but not limited to the following considerations:

a. Hydraulic capacity is assessed in worst-case nominal (un-failed) and hydraulic degraded conditions and requirements and interfaces coordinated with the hydraulics domain. Hydraulic source switching transients are analyzed to assess hydraulic switching timing.
b. Uninterruptible electrical power is assessed to maintain all VCF critical systems in all worst-case nominal and electrical power degraded conditions, including air vehicle start-up, and requirements and interfaces coordinated with the electrical power domain. Vehicle control function electrical loads required on emergency power are analyzed to ensure electrical capacity is available to support emergency scenarios.
c. Cooling is assessed to enable continuous VCF component operation under worst-case nominal conditions. Upon total loss of cooling, critical VCF components operate without degradation for the period of time it takes to land the air vehicle under worst-case operating conditions.

Test.

(1) Ground or Lab test: Hardware-in-the-loop test (e.g., HITL labs, Iron Bird, air vehicle ground test) verifies VCF functionality and interfaces with all integrated subsystems for nominal (un-failed), worst-case performance.

(2) Lab test: Failure modes and effects testing verifies VCF functional performance in failed and degraded conditions to include backup configurations and in emergency conditions.

(3) Lab test: Following total loss of cooling, thermal testing demonstrates critical VCF components operate without degradation for the period of time it takes to land the air vehicle under worst-case operating conditions.

6.2.3 Vehicle control function architecture.

6.2.3.1 Functional architecture.

Standard: High-level VCF architecture, including integration with supporting subsystems, provides the redundancy and functional separation to meet PLOC, failure immunity, fail-safe, and VCF safety-critical requirements established for the air vehicle. Separation and isolation exist between critical and non-critical functions to prevent propagation of failures.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Vehicle control function functional block diagram defines functional flow with underlying redundancy identified.

(2) Vehicle control function internal and external interfaces are defined.

(3) Vehicle control function component functional requirements are defined to include redundancy and functional separation.

Analysis.

(1) Safety-critical function thread analysis, or equivalent, identifies SCFs and functional flow.

(2) Software architecture identifies separation between critical and non-critical functions.

6.2.3.2 Physical architecture.

Standard: The physical VCF architecture and design, including integration with supporting subsystems, provides the physical redundancy and separation to prevent CMFs or propagation of a failure of one portion of the system to another. The physical architecture meets the PLOC, failure immunity, fail-safe, and VCF safety-critical requirements established for the air vehicle. Separation and isolation between critical and non-critical systems prevent propagation of failures from non-critical systems to the VCF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Vehicle control function design and installation drawings demonstrate physical redundancy and separation of component and signal routing to avoid failure propagation.

(2) Vehicle control function design and installation drawings demonstrate redundant VCF wiring is routed with isolation between redundant components and signals IAW SAE AS94900A (if the air vehicle is a UAV, SAE ARP94910), Isolation and Protection of Redundant Electrical Circuits section.

(3) Interface control documents and installation drawings verify physical separation between critical and non-critical systems to protect VCF hardware.

6.2.4 Vehicle control function integrity.

The system safety process is integrated throughout the VCF design and development to ensure a robust airworthy vehicle control system. Vehicle control system domain expertise is required throughout this process to ensure VCF related hazards and mitigations are well understood, accurately represent the VCF design, and support safe control of the air vehicle. Vehicle control function development, integration, manufacturing, and maintenance are executed IAW the SSPP. Any deviations associated with the VCF are documented and assessed for hazard impact.

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6.2.4.1 Vehicle control function integrity.

Standard: Design protections to prevent catastrophic and critical hazards during nominal operation, as well as in the presence of single failures and failure combinations, are incorporated in the VCF and verified to support the failure response and PLOC requirements in 6.2.1.1 – 6.2.1.3.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The VCF subsystem and SHA are reviewed for completeness IAW with the SSPP for the following:

a. Hazards identified and assessed for completeness.
b. Causal factors established and reviewed for completeness.
c. Mitigating controls are established and reviewed for their effectiveness in hazard mitigation.
d. Requirements for the mitigating controls are in place and allocated appropriately to the hardware, software, or system level, with traceability to verification methods.

(2) For mitigating controls verified by analysis, the analysis demonstrates successful hazard closure evidence.

(3) Open hardware and software deficiencies for the VCF and integrated subsystems are analyzed to ensure they do not result in VCF hazards that cannot be mitigated to risk levels specified by the procuring activity. Any mitigating limitations or procedural workarounds (e.g., envelope limits, procedural limits, warnings, cautions, notes) have been documented, agreed to by the procuring activity SMEs, and communicated to the flight crew.

Test.

(1) Lab test: For mitigating controls verified by test, test results demonstrate successful hazard closure evidence.

6.2.5 Redundancy management.

The VCF redundancy management ensures that redundant data elements are received, analyzed, and source selected to provide valid and coherent data for the control function. As part of this, the VCF redundancy management function provides failure detection, isolation, and accommodation for all signal and component faults that impact the control function. The failure detection scheme may utilize a combination of in-line, cross-channel, or dissimilar sourced failure detection through a combination of BIT fault detection and redundancy management algorithmic voters. Once the failure is detected, it is then isolated from the source selection voter and the voter is reconfigured as appropriate to provide valid data to the control function. The failure detection and isolation design should consider both air vehicle transients as well as susceptibility to nuisance failures. The VCF redundancy management is typically broken into categories covering input signal management and output signal management. Within the context of this document, VCF redundancy management also includes data integrity assurance and coherence, which includes communication of information between VCF components as well as components interfacing with the VCF.

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6.2.5.1 Failure detection and fault isolation.

Standard: Failure detection and fault isolation schemes are established and verified to support the failure response and PLOC requirements. All VCF failure modes are identified and the corresponding detection scheme provides 100% coverage for all safety-critical failures. Threshold and persistence values are set to ensure failures do not result in exceeding air vehicle transient limits for single failures or failure combinations when inserted during credible, worst-case flight conditions. Detected failures are isolated to the failed signal or component to support failure accommodation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the VCF redundancy management design documentation to verify failure detection schemes are in place for all safety-critical signals and functions.

(2) Review failure detection schemes to verify the combination of in-line, cross-channel, or dissimilar sourced failure detection provides failure coverage. For designs that utilize a “last good value”, ensure that the design has a credible last value in all scenarios (e.g., when a dynamic signal is failed on power-up).

Analysis.

(1) Air vehicle transient analysis utilizing finalized threshold and persistence values demonstrates transients are within air vehicle established limits for all safety-critical single failures and combinations for all air vehicle configurations across the envelope, including during the catapult shot, takeoff rotation, and landing gear contact (bolter and touch and go).

(2) Conduct failure detection analysis, including BIT coverage, to demonstrate 100% coverage for all safety-critical failures. Verify failures, which cannot be detected by BIT and are not easily inspected prior to flight, do not result in instability, LCO, or worse than Operational State II.

(3) Verify failed signals and functions are isolated to prevent degraded VCF performance upon the detection of the failure.

Test.

(1) Lab test: Vehicle control function requirements-based testing (e.g., FQT) verifies failure detection and fault isolation schemes are properly implemented to meet system requirements.

(2) Lab test: Failure modes and effects testing demonstrates a safe reaction to (within air vehicle transient limits), and recovery from, all safety-critical single failures and combinations identified from the air vehicle transient analysis. Ensure failures are inserted throughout the envelope, including during the catapult shot, takeoff rotation, and landing gear contact (bolter and touch and go).

6.2.5.2 Failure accommodation.

Standard: The system accommodates identified failures automatically or through flight crew action in time to meet PLOC, failure immunity, fail-safe, and transient requirements. The system may utilize multiple different accommodation approaches (e.g., reconfiguration through

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switching to an alternate redundant source, integrating multiple sources, estimating signals, or using a fixed value).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the VCF redundancy management design documentation to verify failure accommodation schemes are in place for all safety-critical signals and functions.

Analysis.

(1) Air vehicle transient analysis utilizing finalized threshold and persistence values demonstrates transients are within air vehicle established limits for all safety-critical single failures and combinations for all air vehicle configurations across the envelope, including during the catapult shot, takeoff rotation, and landing gear contact (bolter and touch and go).

Test.

(1) Lab test: Vehicle control function requirements-based testing (e.g., FQT) verifies failure accommodation schemes are properly implemented to meet system requirements.

(2) Lab test: Failure modes and effects testing demonstrates a safe reaction to (within air vehicle transient limits), and recovery from, all safety-critical single failures and combinations identified from the air vehicle transient analysis. Ensure failures are inserted throughout the envelope, including during the catapult shot, takeoff rotation, and landing gear contact (bolter and touch and go).

6.2.5.3 In-line fault coverage.

Standard: Where fault detection relies on in-line coverage for isolation, these in-line monitors detect safety-critical faults to support failure response and PLOC requirements. Fault coverage is accounted for in PLOC, failure immunity, and fail-safe assessments and coverage numbers are justified.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Built-in-test analysis of VCF subsystems and components identifies failures that require in-line fault isolation and provides evidence of the in-line fault coverage to support the PLOC, failure immunity, and fail-safe analyses. Analysis of in-line fault coverage is assessed as specified in SAE AS94900A and SAE ARP94910, Effects of Imperfect Fault Detection paragraph.

6.2.5.4 Nuisance failures.

Standard: Failure detection performance is evaluated with credible worst-case component tolerances to ensure failure monitor threshold and persistence values do not result in nuisance failures at a rate that degrades failure response and PLOC compliance.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Vehicle control function component tolerance analysis demonstrates redundancy management and BIT monitor threshold and persistence values are set to provide robustness to nuisance failures while exposed to the component operating environment as well as component variations over life.

Test.

(1) Lab test: Failures during formal hardware and system-level testing are analyzed to determine root cause. Where nuisance faults are the root cause, monitor thresholds and persistence values are updated as required to bound the perceived failure rates (the rate at which a component is predicted to be marked failed by the failure detection logic) to values that do not affect compliance to the failure response and PLOC standards.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Failures during flight test are analyzed to determine root cause. Where nuisance faults are the root cause, monitor thresholds and persistence values are updated as required to bound the perceived failure rates (the rate at which a component is predicted to be marked failed by the failure detection logic) to values that do not affect compliance to the failure response and PLOC standards.

6.2.5.5 Resettable faults.

Standard: The VCF fault reset design identifies which signals/component failures are resettable and which failures are latched. Resettable failures are identified as automatic or flight crew initiated.

Signal/component integrity and validity is confirmed before re-introduction into the control function. The re-engagement of a previously failed signal/component via fault reset does not result in exceeding air vehicle transient limits. In the case of multiple failures of a redundant signal/component, ensure that non-simultaneous failures are not capable of voting out/failing a valid signal/component. If a maximum number of resets exist to prevent an air vehicle hazard, rationale is documented and communicated for operational awareness and execution.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation for fault reset approach. Identify what faults are resettable.

(2) Ensure lower-level design requirements capture the reset functionality.

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Analysis.

(1) Through air vehicle transient analysis, determine any envelope limitations required for fault resets to maintain air vehicle transient limits (e.g., airspeed, altitude, bank angle).

Test.

(1) Lab test: Vehicle control function reset functionality is verified by software and system-level test, including FMET.

6.2.6 Built-in-test.

Vehicle control function BIT provides an assessment of the integrity and health status for all VCF hardware and corresponding functionality. Built-in-test is typically divided into three separate functions that work collectively to provide failure detection and reporting in support of AW:

a. Start-up BIT provides an integrity check for any electronic processing function at power-on prior to BIT testing at higher subsystem and system levels.
b. Pre-flight initiated built-in-test is typically flight crew initiated and allows intrusive tests on the ground that would be hazardous to perform in-flight. This includes testing all isolation and reconfiguration functions.
c. Continuous built-in-test passively monitors VCF functionality both on the ground and in-flight and reports failures to redundancy management to enable safe system response. Continuous built-in-test is sometimes referred to as Periodic BIT.

Since PLOC analysis assumes a healthy VCF prior to each flight, the combination of Start-Up BIT, PFIBIT, and CBIT work together on the ground to identify any failures in the system, including latent failures that impact failure accommodation (e.g., failure of a switching valve solenoid). Clear GO/NO-GO notification should be indicated to the flight crew based on these PFBIT results.

6.2.6.1 Start-up built-in-test.

Standard: For VCF designs utilizing complex electronic hardware, Start-up BIT ensures the integrity of the processing function prior to BIT testing at higher subsystem and system levels. Where Start-Up BIT fails, the failure is detected, isolated, and annunciated to the flight crew. For in-air restarts/power-ups, the restart does not impact VCF functionality. If a fast/warm start-up is required to address a common mode all-channel reset in-flight, identify which Start-up BIT tests are run and the timing is sufficient to maintain air vehicle control within Operational State III.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to verify all complex electronic hardware has coverage of critical processing functionality during box-level power-up and prior to integration with the VCF.

(2) Review design documentation to verify VCF system initialization/synchronization ensures the internal health of the hardware prior to integrating into the VCF, including cold start (on-ground) and any warm starts during operation.

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Analysis.

(1) For an all-channel in-air reset, determine the reset time allocation required to maintain air vehicle transients within limits and verify the time allocation is flowed to the box-level requirements.

Test.

(1) Lab test: For complex electronic hardware, verify box-level FQT demonstrates Start-up BIT runs within its time allocation and provides coverage to determine box-level health prior to being utilized by the VCF.

(2) Lab test: For an all-channel in-air reset, hardware in the loop test demonstrates air vehicle transient response is within limits and maintains no worse than Operational State III.

6.2.6.2 Pre-flight initiated built-in-test.

Standard: Pre-flight initiated built-in-test (both automatic and flight crew initiated) tests VCF safety-critical functionality and component-level health. Pre-flight initiated built-in-test verifies no latent faults are present prior to flight, including all failure isolation and accommodation functions. Pre-flight initiated built-in-test detects 100% of safety-critical faults and at least 98% of all subsystem faults, with less than 1% false indications. Interlocks are provided to prevent in-flight engagement and to terminate PFIBIT when the conditions for engagement no longer exist. Pre-flight initiated built-in-test pass/fail status is annunciated to the flight crew with indication of what was tested and not tested and any failures present.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to verify PFIBIT engagement logic has interlocks and design protections against engagement in-flight.

(2) Review design documentation to verify the PFIBIT annunciates as failed in response to detected failures or if PFIBIT is suspended prior to completion.

(3) Review design documentation to verify the pass/fail status is annunciated to the flight crew with indication of what was tested and not tested and any failures present.

Analysis.

(1) Built-In-Test coverage analysis demonstrates PFIBIT provides coverage for all VCF safety-critical functionality and component-level health.

(2) Probabilistic analysis of component tolerances and IBIT parameters shows less than one false indication of fault during IBIT for every 100X IBIT is run on hardware.

Test.

(1) Lab test: Formal qualification test, system-level testing, and FMET verify PFIBIT functionality, including engagement and disengagement requirements and clear indication of the health of the VCF to the flight crew prior to flight.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Available evidence from the flight test program indicates the operational fleet will see less than one false indication of fault from IBIT for every 100 sorties.

6.2.6.3 Continuous built-in-test.

Standard: Continuous built-in-test does not adversely impact VCF real-time processing and provides fault coverage and health status to the VCF redundancy management function in support of failure detection and fault isolation requirements, as specified in 6.2.5.1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Built-In-Test coverage analysis demonstrates CBIT, in coordination with redundancy management, provides real-time coverage for all VCF safety-critical functionality and component-level health.

Test.

(1) Lab test: Formal qualification test, system-level testing, and FMET verifies CBIT functionality.

6.2.7 Hardware endurance.

Standard: Vehicle control function components are designed and tested according to the expected operational requirements. The operational requirements are based on MPs that determine the duty cycles expected for the life of the air vehicle. Vehicle control function components meet or exceed the expected life duty cycles without degrading below minimum specified performance.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review expected operational MPs.

Analysis.

(1) Determine duty cycles based on the operational MPs.

(2) Document duty cycles expected in the life of the air vehicle.

(3) Analyze VCF components for durability for the life of the air vehicle.

(4) Perform damage tolerance analysis on all single point failures. If damage tolerance analysis is not practical due to material properties, high durability design approach (4 lives) may be substituted.

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Test.

(1) Lab test: At least 25% of single life duty cycles are successfully completed prior to first flight.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Lab test: For designs supported by damage tolerance analysis, design is verified by successful completion of a single life endurance test. Otherwise, the design is verified by successful test completion of 2X the endurance life limit. Successful completion entails no degradation of performance or wear exceeding wear tolerances.

6.2.8 Flight crew procedures.

6.2.8.1 Pre-flight procedures.

Standard: Flight crew pre-flight procedures are in place with clear GO/NOGO criteria for the VCF. Pre-flight procedures may include both automatic PFBIT as well as flight crew-initiated procedures, such as walk-arounds. Pre-flight procedures ensure PFBIT is performed prior to each flight and after any power cycle for which flight will follow. The flight crew has clear guidance on handling failures associated with PFBIT. Any workarounds for pre-flight failures and corresponding mitigations are documented and approved.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manual (or equivalent) to verify clear guidance is in place for flight crew pre-flight procedures.

a. Pre-flight built-in-test is performed prior to each flight and after any power cycle for which flight will follow.
b. The flight crew and test team have clear guidance on handling failures associated with PFBIT, including any specifics on failed pre-flight IBIT re-run criteria.
c. Any workarounds for pre-flight failures and corresponding mitigations are documented and approved.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Review the flight manual to verify clear guidance is in place for flight crew pre-flight procedures.

a. Ensure PFBIT is part of the standard flight crew pre-flight procedure and checklist. Procedure and checklist are clear that pre-flight IBIT is performed prior to each flight and after any power cycle for which flight will follow.
b. The flight crew has clear guidance on handling failures associated with PFBIT, including any specifics on failed pre-flight IBIT re-run criteria.

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c. Any workarounds for pre-flight failures and corresponding mitigations are documented and approved.

6.2.8.2 Normal procedures and limits.

Standard: Flight crew normal procedures and limits are reviewed to ensure they provide the flight crew with clear direction on using the VCF as intended by the design without introducing new hazards. Air vehicle limits are supported by the VCF design and analysis.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manual to verify normal procedures and limits are clear and represent the state of the VCF development going into flight test.

a. Air vehicle limits are supported by the VCF design and analysis.
b. Key deficiencies are documented and presented to the flight crew and test team to characterize user impacts.
c. Any workarounds for normal procedures and corresponding mitigations are documented and approved.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Review the flight manual to verify normal procedures and limits are clear and represent the state of the VCF design.

a. Design changes and inputs from flight test are incorporated in the flight manual.
b. Air vehicle limits are supported by the VCF design, analysis, and flight test.
c. Key deficiencies are documented in the flight manual to characterize flight crew impacts.

6.2.8.3 Emergency procedures.

Standard: For the VCF and applicable integrated subsystems, flight crew emergency procedures provide the flight crew with clear direction to mitigate hazards for identified failure modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manual to verify emergency procedures and corresponding limits are clear and represent the state of the VCF development going into flight test. Emergency procedure specific air vehicle limits are supported by the VCF design and analysis.

Test.

(1) Lab test: Where possible, PITL FMET procedures are performed with the emergency procedures to demonstrate clarity and effectiveness. Any redlines from testing are reviewed and incorporated into the procedures.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Review the flight manual to verify emergency procedures and corresponding limits are clear and represent the state of the VCF design.

a. Emergency procedure updates from flight test are incorporated in the flight manual.
b. Emergency procedure specific air vehicle limits are supported by the VCF design, analysis, and flight test.

6.2.9 Vehicle control function actuators.

6.2.9.1 Actuator redundancy management.

Standard: Actuator redundancy management supports the VCF function throughout the RORH flight envelope.

Actuators meet functional and PLOC requirements for the appropriate operational states. The probability of VCF actuators catastrophic failure due to propagated, common mode, or a single point failure is less probable than PLOC divided by 100.

Switching between redundant hydraulic or electrical systems does not cause VCF actuator transient resulting in departure from controlled flight.

If there is more than one actuator or more than one control valve for any flight control surface, there is no force fight that would degrade air vehicle flying qualities or damage structure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review actuators design, drawings and architecture to identify single point failures and engineering data verifying less probable than PLOC divided by 100 for all single point failures.

(2) Provide FMECA for the actuation system.

(3) Provide actuator reliability data in support of PLOC computation.

Test.

(1) Lab test: Verify all actuator FMET test cases have been successfully completed.

(2) Lab test: Provide handling qualities simulation data verifying no worse than Operational State II after any single actuator failure.

(3) Lab test: Review test data showing fault detection, isolation, and reconfiguration time and transients do not result in below Operational State II.

(4) Lab test: Verify actuator failures are clearly displayed and annunciated to the flight crew.

6.2.9.2 Actuator failure accommodation.

Standard: Actuator failure detection and isolation supports the VCF function throughout the RORH flight envelope or permissible envelope.

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A combination of a single hydraulic, electrical, or electronic system failure and a single actuator failure does not reduce VCF performance below Operational State III.

Vehicle control function actuators failure detection, isolation, and reconfiguration time is less than or equal to 100 milliseconds.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review the documents that list all monitored, detected, and annunciated actuator faults to verify fault coverage.

Test.

(1) Lab test: Verify all actuator FMET test cases have been successfully completed.

(2) Lab test: Verify detection, isolation, and reconfiguration time associated with actuator failure accommodation meets the threshold in the standard.

6.2.9.3 Hydraulic contamination.

Standard: Actuators provide protection from hydraulic contamination due to the largest particle which are able to block or wedge an actuation control spool in the smallest passage.

Actuators control valve components are able to chip shear the largest particle that could pass through the hydraulic system filtration. Hydraulic contamination does not cause cavitation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review actuator drawings to verify locations for possible chip blockage impacting functionality.

Analysis.

(1) Determine the biggest chip size that may pass through the hydraulic system filters.

(2) Provide analysis verifying chip shear force within the actuator control valve to maintain full functionality.

(3) Vehicle control function and hydraulics ICD addresses any effects of cavitation to the VCF.

6.2.9.4 Bottoming and snubbing.

Standard: Actuators design incorporates protection against bottoming, such as the use of snubbing. In the case of control surface mis-rigging, the resulting bottoming loads are less than the actuator limit load.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Provide design analysis to verify actuator rams, gearboxes, or equivalent will not bottom during normal and failure conditions.

(2) Determine the range of mis-rigging that will result in bottoming.

(3) Provide analysis of bottoming load as a result of mis-rigging, including effects of any snubbing features used in the design.

Test.

(1) Lab test: Conduct closed loop actuator testing to verify actuator design eliminates bottoming under normal and failure conditions.

(2) Lab test: Conduct bottoming cycles SOF qualification testing to 25% of the estimated number of bottoming cycles for full life.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Conduct bottoming cycles qualification testing based on the estimated number of bottoming cycles.

6.2.9.5 Actuator structural integrity.

Standard: Actuators meet vibration and thermal environments in their installed location. Actuators meet all stiffness requirements under normal and failed conditions. Actuators can withstand limit loads of 1.15X the maximum possible aerodynamics loads without permanent deformation. Actuators can withstand ultimate loads of 1.5X the aerodynamics limit loads without structural failure. Aerodynamic aiding and opposing loads do not cause actuator buffeting. For dual tandem actuators, rip-stop design approach is implemented.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Design analysis verifies actuator stiffness. For hydro-mechanical actuators, stiffness analysis will include both high and low fluid temperatures.

(2) Analysis verifies actuator capability to withstand aerodynamic limit and ultimate loads.

(3) Provide actuator performance data under maximum aiding and opposing aerodynamic loads to verify absence of buffeting.

Test.

(1) Lab test: Verify actuator meets vibration and thermal environments as specified in the environmental conditions when installed on the air vehicle.

(2) Lab test: Conduct actuator aerodynamics limit loads test.

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6.2.9.6 Actuator pressure vessel integrity.

Standard: Actuators can withstand impulse pressures of 1.5X the normal pressure and burst pressure of 2.5X the normal pressure. Actuator pressure backlash and hysteresis do not exceed maximum of 1-degree deflection measured at the surface. Actuators internal and external leakage meet the allocation as specified by hydraulic system ICD. Actuator seals are designed to leak-before-burst. Actuators expected life operational cycles have been determined using operational usage profile.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify leak-before-burst design approach.

Analysis.

(1) Provide actuator backlash and hysteresis analysis to determine maximum surface deflection.

(2) Provide analysis for actuator internal and external leakage.

Test.

(1) Lab test: Conduct actuator impulse pressure testing. Impulse pressure testing is to be done at both fully retracted and fully extended linear actuator position.

(2) Lab test: Conduct actuator burst pressure testing.

(3) Lab test: Conduct actuator duty cycle testing to verify a minimum of 25% cycles of full life profile.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Lab test: Conduct actuator duty cycle testing to verify full life profile.

6.2.9.7 Actuator performance.

Standard: Actuator loop closure meets 6 dB gain and 45-degree phase stability margins.

Actuators response and time delay prevent PIOs, preclude structural coupling, aero-elastic coupling, and flutter.

Worst-case actuator surface rate, hinge moment, and free-play do not degrade VCF performance below Operational State II. Actuator force stall or blowback does not reduce VCF performance below Operational State III under adverse flight and environmental conditions.

Actuator performance on emergency systems such as an emergency hydraulic system or battery backup, with degraded hinge moment capability does not degrade VCF below Operational State III. Actuator total loss of hydraulic or electrical power do not cause flutter or LCO within the RORH.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review actuator free play allocation documented in the approved structures ICD.

Analysis.

(1) Verify actuator performance requirements and design support required VCF operational states in both normal operation and failure states.

(2) Conduct actuator closed loop analysis to verify stability margins.

(3) Conduct stiffness analysis to verify compliance to allocated flutter and ASE requirements.

Test.

(1) Lab test: Conduct actuator performance test for both high and low fluid temperatures to determine force output, rate, and frequency response meet requirements.

(2) Lab test: Conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to demonstrate actuator performance for normal operation and failure states.

(3) Lab test: Conduct system-level FMET with HITL to verify actuator performance.

(4) Lab test: Review test data validating hinge moments and performance under normal and failed conditions

6.2.9.8 Mechanical stops.

Standard: Control surface installation is designed to limit and constrain motion to within the actuator range of travel. Control surface mechanical stop is capable of withstanding maximum loads including the effects of mis-rigging.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review actuator installation drawings.

(2) Review control surface mechanical stop design.

Analysis.

(1) Analyze mechanical stop design for maximum force and rate impact.

6.2.9.9 Electrical regeneration.

Standard: For all anticipated conditions, the electrical energy regenerated by actuators (e.g., by aerodynamic loading of the control surfaces or actuator reversal) is either stored or dissipated. Regenerated power characteristics are compatible with the energy storage systems or energy dissipation devices. There are no transients or reduction in performance when using stored power for actuation. A single failure in the storage system does not affect actuator performance. Subsystem thermal analysis accounts for heat produced by the actuators in both normal and failed conditions.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Actuator component-level test plans show that mission regeneration requirements are incorporated.

Analysis.

(1) Actuator regeneration supports mission requirements. Actuator duty cycle is developed by the analysis of all mission spectrums to generate worst-case loading and heating characteristics. Any usage assumptions and limitations resulting from the analysis are assessed and documented.

Test.

(1) Lab test: Supplier lab testing verifies regenerative power storage and dissipation meets mission requirements for all normal and required failure modes.

6.2.10 Vehicle control function air data safety.

Air data systems are designed and installed on-air vehicles to measure and compute relevant properties of the air relative to the air vehicle. Flight condition information such as AOA, AOS, altitude, airspeed, Mach number, and vertical speed are either directly measured or derived from other parameters such as static and total pressure and outside air temperature. This information may be corrected by an onboard computer prior to use in control law logic and are provided to the flight crew via flight instruments for situational awareness and possible manual adjustment.

Proper understanding of the accuracy, tolerance, and operating range of the air data sensors, sensor placement on the air vehicle, handling of redundant parameters, and failure modes of the hardware are critical to integrate the air data system into the VCF. Proper functioning of the air data system is critical when flying by the instrument panel is required or to maintain safe vertical separation.

6.2.10.1 Accuracy and tolerances.

Standard: Air data accuracy and tolerances are defined for altitude, AOA, AOS, and airspeed or Mach, including Mach compressibility, which are either directly measured or derived from measurements such as dynamic pressure, static pressure, and total air temperature.

Air data sources are qualified to the boundaries of the RORH accounting for airspeed, altitude, AOA, and AOS limits of the air vehicle. The VCF allows for 10% margin for adverse effects in the air data system without loss of control.

The air data system provides the VCF with airspeed ±5 KCAS or ±0.02 Mach, altitude ±5 ft per 1000 (min ±20 ft), AOA ±1 degree, and AOS ±1 degree in the ROSH.

Accuracy analysis considers scaling, nonlinearity, higher order dynamics, resolution, lag, latency, manufacturing tolerances, aging, wear, maintenance, and any other parameters such as voting thresholds that affect overall accuracy.

When assessing air data system errors and failures, account for:

a. Noise on air data signals.

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b. Calibration table errors in slope and bias.
c. Blockages in pneumatic lines or total and static pressure ports.
d. Total system performance tolerance.
e. Intermittent signal failures (especially failures of duration shorter than the persistence counter).
f. Lags: pneumatic, sensor, computational, electrical.
g. Leaks.
h. Contour changes near the sensors due to maintenance or repair.
i. Installation irregularities.
j. Minimum speed for air data validity.
k. Other areas identified by the vendor that may affect accuracy or tolerance.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation for placement of air data sensors to minimize measurement errors and account for what errors remain.

Analysis.

(1) Conduct air data sensitivity study to determine influence of variations in air data parameters on the VCF.

Test.

(1) Lab test: Conduct lab test to verify measurement accuracy and tolerance of sensors.

(2) Lab test: Conduct wind tunnel test to establish desired location of sensors on the air vehicle and initial estimates for source error correction.

(3) Lab test: Conduct PITL simulation to verify the air vehicle does not lose control when modeling effects of errors in the air data system.

(4) Ground test: Conduct ground test to verify installation of air data system and pressure check of the system for leaks and rated pressures.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct high speed taxi runs to validate air data predictions.

(6) Flight test: Conduct flight testing with boom or other device throughout the ROSH to gather truth data for validating accuracy and performance of the air data system. Other techniques such as pacer air vehicle, tower flyby, and trailing cone may be used to collect additional verification data for the air data system.

6.2.10.2 Integration.

Standard: The VCF air data architecture provides control scheduling and valid inputs for all tasks throughout the RORH and provides processing, memory, and data communications capability to meet the requirements for the VCF integrated functions (e.g., navigation, engines,

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displays, and SMS), both internal and external to the VCF. In order to provide valid inputs, the air data function supports:

a. Fault detection of the system pre-flight, in-flight, and post-flight.
b. Redundancy and fault tolerance: fail-operational/fail-safe capability.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documents to verify redundancy management is implemented within requirements.

Analysis.

(1) Conduct analysis of air data processing and transmission needs of the integrated system to confirm the VCF architecture meets the needs of functions that rely on air data.

(2) Complete fault tree and FMECA to identify failure modes, failure rates, and consequences.

Test.

(1) Lab test: Conduct FMET with VCF HITL to verify fault detection, system redundancy, and fault tolerance.

(2) Lab test: Verify that outputs of the air data system for the nominal case as well as in the presence of air data failures are expected values and are transmitted for use throughout the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight test operations to verify that the air data system operates in-flight as designed.

6.2.10.3 Ground provisions.

Standard: The air data ground provisions:

a. Allow ground checkout prior to flight for functionality.
b. Protect sensors from the elements.
c. Allow for maintenance fault isolation to the individual sensor-level.
d. Include specific ground handling procedures to prevent damage to sensors or surrounding structure when performing maintenance or using heaters.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation of ground provisions in the flight manual and checkout in the pre-flight checklist.

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(2) Review maintenance procedures for protection, fault diagnostics, and handling of air data components.

Demonstration.

(1) Show that protective devices for the air data sensors can be installed and removed without causing damage to the sensors or surrounding structure and without affecting sensor calibration.

Test.

(1) Lab test: Conduct FMET including failure detection and fault handling of the air data system when on the ground.

6.2.10.4 Ice prevention.

Standard: Air data ice prevention and removal is provided. This protection is provided wherever the probes or sensors are located. Design does not allow the entrapment of moisture that can result in the formation of ice. The health of air data ice prevention is monitored and failures are identified and presented to flight crew. Redundancy in ice prevention, including power sources, is provided to meet air data system requirements when integrated with the VCF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation of design specifications for air data sources to ensure the design prevents ice accumulation and facilitates removal of ice.

Analysis.

(1) Verify that the ice prevention and removal method will function normally with primary or backup power sources.

Test.

(1) Lab test: Conduct qualification tests on all air data components for exposure to icing conditions.

(2) Lab test: Verify that the ice prevention and removal method removes ice that has accumulated and prevents buildup of ice when active.

(3) Lab test: Conduct FMET of air data icing protection and verify failures due to exposure to icing conditions are presented to the flight crew.

(4) Lab test: Conduct component-level test to confirm ice prevention and removal capability functions when powered.

6.2.10.5 In-flight provisions.

Standard: The air data in-flight provisions:

a. Monitor air data health and integrity and provide system status to the flight crew.
b. Mitigate or accommodate physical loss of air data sensors.
c. Mitigate or accommodate functional loss of air data sensors.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation and identify whether the design will mitigate or accommodate both physical and functional loss of air data sensors.

Analysis.

(1) Complete fault tree and FMECA to identify failure modes, failure rates, and consequences.

Test.

(1) Lab test: Conduct FMET to verify proper alerting to the flight crew, and the intended design methods to accommodate either physical or functional loss of air data sensors.

6.2.11 Vehicle control function control law safety.

The VCF control law determines the required actuator commands to meet the flight crew input based on the flight condition input from air vehicle sensors. The control law should be robust to varying inputs especially due to nonlinearities and time delay in the system, and uncertainty in inputs to the control law.

The standards for control law apply for the primary air vehicle and when the VCF is integrated with payload and ground functions. Additional considerations for directional control on the ground are found in 8.5.8.5.

Automatic modes include any mode or system where the flight crew is not providing direct input to the VCF including, but not limited to basic autopilots, advanced autopilots, pilot relief modes, automated takeoff and landing, approach power compensation, collision avoidance, ground or target following, and autonomous command generation. For commands generated outside of the VCF, consider the standards of 6.2.13.5.

Vehicle control function control law assessment is primarily based on offline simulation and PITL simulation with validation of airframe aerodynamic and subsystem models occurring in flight test.

6.2.11.1 Nominal performance and robustness.

Standard: The VCF control law provides longitudinal, lateral, directional, lift, drag, performance limiting, and variable geometry control (if applicable) with VCF performance no worse than Operational State II.

The VCF provides phase and gain margins as specified in SAE AS94900A and SAE ARP94910, Stability section, which are a function of frequency and flight envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation for appropriate and effective control in all required axes.

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Analysis.

(1) Verify gain and phase margin with quantitative analysis across the flight envelope and allowed mass properties. Parameter variations will be made such that mode and gain changes are captured.

(2) Regions of flight envelope or mass properties that do not meet the gain and phase standard will be further evaluated in PITL simulation with non-linear air vehicle and aerodynamic models by varying gain, control power, time delay, or other parameters to demonstrate equivalent robustness.

Test.

(1) Lab test: Conduct PITL simulation to verify VCF performance is no worse than Operational State II.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to verify VCF provides longitudinal, lateral, directional, lift, drag, performance limiting, and variable geometry control (if applicable) with VCF performance no worse than Operational State II.

6.2.11.2 Nonlinearities.

Standard: Nonlinearities within the integrated system (e.g., mechanical, electrical, hydraulic, digital, and analog interfaces) as well as computational path nonlinearities do not degrade VCF below Operational State III or result in PIO.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to define sources of nonlinearities in the integrated system, including but not limited to: rate limiting, overshoot, and saturation.

(2) Review simulations model documentation to verify nonlinearities are incorporated.

Test.

(1) Lab test: Conduct piloted simulation to verify nonlinearities do not degrade VCF below Operational State III or cause PIO. Nonlinearities are intentionally exercised as part of this evaluation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight: Conduct flight test to verify nonlinearities within the integrated system (e.g., mechanical, electrical, hydraulic, digital, and analog interfaces) as well as computational path nonlinearities do not degrade VCF below Operational State III or result in PIO.

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6.2.11.3 Transients.

Standard: For manned air vehicles: Air vehicle response due to gain transition does not exceed 0.05g normal or lateral acceleration (at the air vehicle CG), 3 degrees per second roll rate, 5 degrees of sideslip and does not last more than 2 seconds. The flight crew control forces do not exceed 10 lbs in pitch, 5-lb roll, and 5 lbs in yaw from trim.

For UAVs: Air vehicle response due to gain transitions are contained to preclude loss of control, damage to equipment, excitement of structural modes, and interruption to SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to define gain transitions.

Analysis.

(1) Conduct simulation analysis to verify air vehicle response due to gain transitions are within the standard.

(2) Conduct simulation analysis to verify gain transitions create flight crew control forces within the standard.

Test.

(1) Lab test: Conduct PITL simulation to verify air vehicle response due to gain transitions and flight crew control forces are within the standard.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to verify response to transient meets the standard.

6.2.11.4 Uncertainties.

Standard: Anticipated uncertainty in the following do not degrade the VCF below Operational State III:

a. Aerodynamic parameters, including control surface effectiveness and stability derivatives.
b. Air data parameters.
c. Mass properties, including misfires and hung stores.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to identify sources and extent of uncertainties in aerodynamic parameters, air data parameters, and mass properties.

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(2) Conduct sensitivity analysis of the identified aerodynamic parameters, air data parameters, and mass properties uncertainties, including offline simulation.

Test.

(1) Lab test: Based upon sensitivity analysis results, conduct PITL simulation using identified sources of critical aerodynamic parameter, air data parameter, and mass properties variation to verify anticipated uncertainty does not degrade the VCF below Operational State III.

6.2.11.5 Time delays.

Standard: The total delay from the instant a controller input is provided to the time the air vehicle begins to perceptibly respond to the input is such that the VCF performance does not degrade below Operational State II, result in PIO, or result in LCO. Time delay assessment includes worst-case latency for each component in the control loop, including processor timing variations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to determine the allowed time delay and allocate time delay requirements to each element of the control loop.

(2) The time delays in all feedback signals and feedforward elements of the VCF are properly modeled. If data links are used, they are modelled with representative latency and dropout characteristics.

(3) Conduct analysis to determine time delay of the integrated VCF and computational paths between inceptors and effectors, to include aerodynamic delay and aero-elastic influences.

Test.

(1) Lab test: Conduct PITL simulation with expected time delay to verify VCF performance does not degrade below Operational State II, result in PIO, or result in LCO.

6.2.11.6 Functional modes and limiters.

Standard: Vehicle control function limiters and filters incorporated to protect the air vehicle or flight crew (e.g., structural, AOA, sideslip, flight crew command) function as expected throughout the entire flight and ground envelopes without causing loss of the control function, departure from controlled flight, or loss of air vehicle and flight crew. No VCF or integrating control function creates conditions that defeat the limiters. If flight crew override of structural limiters is provided for use in emergency conditions, the achieved flight condition will not result in loss of control of the air vehicle due to structural deformation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation for appropriate and effective use of limiters to protect the vehicle or flight crew.

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(2) Review design documentation for appropriate and effective use of filters to protect the vehicle or flight crew.

Analysis.

(1) Conduct simulation analysis to verify limiters and filters function as expected throughout the flight envelope without indications of loss of control or air vehicle. Verify limiters are not defeated and do not reduce capability in regions where the limiting may be active but not necessary to protect the air vehicle.

Test.

(1) Lab test: Conduct HITL, and PITL for manned air vehicles, simulation to verify limiters, and flight crew override function as expected throughout the flight envelope without indications of loss control of the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to verify limiters function as expected throughout the flight envelope.

6.2.11.7 Automatic modes and functions.

Standard: The VCF uses integrity monitoring and system interlocks to permit engagement of automatic modes or functions. The VCF system interlocks prevent inadvertent engagement of the automatic mode or function.

The automatic mode will disengage and return control to the flight crew upon relevant failures, loss of integrity, when interlocks are not met, or flight crew override.

Feedback is provided to the flight crew of a change in control between manual and automatic modes. Indications are provided that the automatic mode or function is active.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review requirements and design documentation of automatic modes and functions to verify disengagement methods, integrity monitoring, safety interlocks, and flight crew notification.

Analysis.

(1) Complete fault tree and FMECA to identify failure modes, failure rates, and consequences of automatic modes and functions, to include inadvertent activation.

(2) Conduct analysis to determine necessary system interlocks and methods for integrity monitoring of automatic modes and functions.

Test.

(1) Lab test: Conduct offline testing to verify operation of integrity monitoring, system interlocks, engagement and disengagement, and indications.

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(2) Lab test: Conduct PITL simulation to validate normal mode engagements, disengagements, and modal indications.

(3) Lab test: Conduct FMET to verify VCF behavior following automatic mode or function failure.

6.2.12 Vehicle control function pilot vehicle interface.

The PVI includes devices that generate flight crew commands to the VCF and provides the flight crew with situational awareness of the state of the VCF. Devices that generate flight crew commands may also be known as inceptors and would commonly include devices such as stick, yoke, throttle lever, cyclic, collective, pedals, and wheel. Consideration is also required for mechanical transmission devices that are implemented between the inceptor and actuator. For the purpose of this section, devices such as computer mice do not need to be considered.

6.2.12.1 Flight crew commands.

Standard: Flight crew commands that the VCF transmits, generates, or conveys, including manual overrides and engineering test commands, are implemented such that the air vehicle response is predictable and controllable under normal conditions and failure conditions that are more probable than PLOC divided by 100.

All portions of the VCF used for flight crew command signal generation and transmission maintain integrity of the signal. Effects of lost or corrupt messages do not induce a loss of control.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the requirements and design features of flight crew input devices to meet the standard of predictable and controllable air vehicle response and signal integrity. The type of signal transmission (e.g., mechanical, electrical, or optical) will have particular requirements.

Analysis.

(1) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences.

Test.

(1) Lab test: Conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to examine predictable and controllable air vehicle response to flight crew commands under normal and failure conditions.

(2) Ground test: Conduct on-air vehicle ground test to verify signal transmission devices operate as designed.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight operations evaluating flight crew commands for acceptability across the envelope during normal operation.

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6.2.12.2 Functional characteristics.

Standard: The inceptor device functional characteristics of friction levels, breakout forces, dead zones, hysteresis, and backlash do not induce a failure, loss of control, or departure. Non-linear characteristics are properly accounted for in the design, math models, simulations, or emulations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the design of inceptor devices to identify elements that provide for friction, breakout forces, dead zones, hysteresis, and backlash.

(2) Review of the design of inceptor devices to identify failure modes of the elements that provide for friction, breakout forces, dead zones, hysteresis, and backlash.

(3) Review of design of inceptor devices to identify non-linear characteristics.

Analysis.

(1) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences.

(2) Verify model, simulation, and emulation design account for non-linear characteristics.

Test.

(1) Lab test: Conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to examine functional characteristics under normal and failure conditions, including expected variations in friction levels, breakout forces, dead zones, hysteresis, and backlash.

(2) Ground test: Conduct on-air vehicle ground test to verify functional characteristics are as designed.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight operations evaluating functional characteristics for acceptability across the ROTH during normal operation.

6.2.12.3 Jammed inceptor functionality.

Standard: Provisions are incorporated to mitigate or accommodate inceptor jams to prevent loss of control or departure. In the case of connected inceptors, a jam in one does not impede operation of the other.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the design of inceptors and mechanical transmission devices to identify elements that may cause jamming.

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(2) Review of the inceptor and mechanical transmission device design to identify elements intended to mitigate or accommodate the impact of a jammed inceptor.

Analysis.

(1) Verify FMECA and SSHA are completed to identify jam causes, failure rates, and consequences.

(2) Perform structural analysis to evaluate effectiveness of any mechanical mitigation.

Test.

(1) Lab test: For mechanical mitigations, verify that the mitigation performs as required using a representative inceptor assembly.

(2) Lab test: For inceptors that operate in a jam mode (e.g., active inceptors), conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to assess air vehicle response does not induce loss of control or departure.

6.2.12.4 Inceptor mechanical integrity.

Standard: Each inceptor functions to the maximum displacement and maximum force that can be imparted by the flight crew without obstruction. Each mechanical device that transmits flight crew commands, from inceptor to actuator, functions to the maximum displacement and maximum force that can be imparted by the flight crew without obstruction.

Units, components, and parts which transmit command mechanically:

a. Have strength such that the limit load is 1.15X the design load and the ultimate load is 1.5X the design load.
b. Maintain strength for the full-life duty cycle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the requirements and design of inceptors to ensure displacement, flight crew forces, and strength requirements are established.

Analysis.

(1) Conduct material analysis to determine material ultimate load.

(2) Conduct structural analysis to determine unit, component, or part strength.

(3) Conduct analysis to determine the duty cycle of each inceptor.

Test.

(1) Lab test: Conduct tests to verify structural characteristics of components and assemblies.

(2) Lab test: Conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to verify inceptors can achieve maximum displacement and maximum force without obstruction.

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6.2.12.5 Mechanical controls and artificial feel devices.

Standard: The ratio changers, as implemented in mechanical systems, provide feedback to the flight crew and in the case of failure, remain engaged in a position that does not induce departure, loss of control, or PIO.

Any artificial feel device with adjustable features does not induce departure, loss of control, or PIO.

Any loss of artificial feel function does not result in less than Operational State III.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the requirements and design of ratio changers and artificial feel devices provide the necessary characteristics.

Analysis.

(1) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences.

Test.

(1) Lab test: Conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to examine ratio changers and artificial feel devices under normal and failure conditions.

(2) Ground test: Conduct on-air vehicle ground test to verify functional characteristics of the artificial feel devices are as designed.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight test operations evaluating functional characteristics of the artificial feel devices for acceptability across the ROTH during normal operation.

6.2.12.6 Warning, caution, and advisory functions.

Standard: The WCA system provides flight crew with information in sufficient time to provide situational awareness and maintain the operational state commensurate with the severity of the failure.

The VCF displays, panels, annunciators, switches, and indicators provide positive unambiguous state and status information and problem recognition to enable flight crew corrective action.

The WCA functions of the VCF are able to identify the most probable cause of multiple failures occurring simultaneously.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review of the requirements and design of WCA system for required placement, information, and redundancy.

Analysis.

(1) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences with associated displayed and annunciated information.

Test.

(1) Lab test: Conduct FMET to verify the WCA system has been properly implemented.

(2) Lab test: Conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to demonstrate flight crew response to failure conditions maintain operational state commensurate with the failure.

6.2.13 Vehicle control function integrated systems.

The nature of the VCF requires integration with other elements of the air vehicle to execute its function. Primary integrations for airframe or structure, avionics, and propulsion system will be considered in how they will impact the VCF.

6.2.13.1 Vehicle control function mechanical clearances.

Standard: For mechanical elements of the VCF, physical clearances are designed to protect against jamming or binding due to probable combinations of temperature effects, air loads, structural deflections, vibration, buildup of manufacturing tolerances, wear, sag, or installation. In locally congested areas for both static elements and elements with relative motion, minimal clearances are IAW SAE AS94900A, System Separation, Protection and Clearance section (or, if the air vehicle is a UAV, SAE ARP94910, Invulnerability to Natural Environments section).

The vehicle control system is able to obtain the full range of required control surface positions under the most adverse flight, manufacturing, rigging or installation bias, environmental, and load conditions without mechanical interference from the primary structure or surrounding devices.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of design and drawings for mechanical elements of the VCF to ensure clearances are IAW SAE AS94900A, System Separation, Protection and Clearance section (or, if the air vehicle is a UAV, SAE ARP94910, Invulnerability to Natural Environments section) with worst-case tolerances.

(2) Inspection of installed vehicle control system elements to ensure designed clearance is achieved.

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Analysis.

(1) Evaluate clearance requirements of the mechanical components under variations of temperature, air loads, structural deflections, vibration, buildup of manufacturing tolerances, wear, installation, and flight loads.

(2) Verify free movement of the mechanical components under variations of manufacturing tolerances, rigging or installation bias, environmental conditions, and aerodynamic loading.

Test.

(1) Ground test: Conduct ground test to verify nominal clearance.

6.2.13.2 Foreign object damage.

Standard: For mechanical elements of the VCF, physical clearances are designed to protect against jamming due to foreign objects under probable combinations of temperature effects, air loads, structural deflections, vibration, buildup of manufacturing tolerances, wear, sag, or installation. For FOD sensitive areas, protection schemes (e.g., cover, seals, armor plate) are in place to prevent FOD from entering the minimum mechanical clearance aperture resulting in binding or damage to elements. Deformation damage to surfaces or associated structure from foreign objects (e.g., objects from tires, birds) does not result in mechanical interference over the range of the air vehicle operating envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review FOD Control Plan or equivalent to define FOD sensitive areas and mitigations.

(2) Review of requirements, design, and drawings to evaluate FOD protection and mitigation elements meet the intended purpose.

(3) Inspection of installed air vehicle control system elements to ensure designed clearance is achieved. For mechanical elements with relative motion, measurements are taken throughout the full range of all elements within the local area.

Analysis.

(1) Evaluate the deformation from FOD of flight control surfaces and air vehicle skin near regions of mechanical VCF to verify the deformation does not result in mechanical interference.

6.2.13.3 Intermittent devices.

Standard: Normal and failed conditions of control devices that operate intermittently (e.g., trailing edge flaps, leading edge flaps, speed brakes, spoilers, and wing geometry mechanisms), including latent failures, do not cause loss of control or cause PIOs. Failures of these devices more probable than PLOC divided by 10 do not degrade the VCF below Operational State IV.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review of the requirements and design to identify the intermittent devices and their safety criticality.

Analysis.

(1) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences.

Test.

(1) Lab test: Conduct FMET in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to examine intermittent devices under normal and failure conditions, including latent failures.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight test operations evaluating intermittent devices for acceptability across the envelope during normal operation.

6.2.13.4 Structural mode interaction.

Standard: Vehicle control function sensors and bio-dynamic feedback do not drive closed loop SMI instability, including vibration from configuration loading and gun fire. As necessary, the VCF contains features (e.g., SMI filters) to minimize these interactions to at least 6 dB of gain margin at frequencies less than or equal to the first aeroelastic mode and 8 dB of gain margin at frequencies greater than the first aeroelastic mode.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the requirements and design of VCF features that minimize SMI for effectivity and adequacy.

(2) Review of placement of sensors to be in appropriate places with respect to air vehicle structural dynamics.

Analysis.

(1) Informed by ground test, analysis of SMI is conducted with variations in flight condition, CG, mass distribution (including fuel), and external store configuration to determine gain margin across the envelope for each control loop.

(2) Conduct pilot mode analysis to identify sources of bio-dynamic feedback.

Test.

(1) Ground test: Conduct ground test (structural coupling test) to determine zero speed margins. Ground testing is conducted with the air vehicle on a soft suspension system to eliminate any constraints imposed by the landing gear.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Review flight test reports for VCF performance with respect to SMI.

6.2.13.5 Integration with avionics systems.

Standard: The avionics system elements integrated with the VCF meet the criticality levels required by the VCF functions they support or the VCF accommodates for the differences in criticality levels.

Single failures in the avionics system do not degrade the VCF below Operational State II.

Avionics system failures more probable than PLOC divided by ten do not the degrade VCF below Operational State IV.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the interface and criticality requirements between VCF and avionics.

(2) Review of the VCF requirements and design for mitigation of hazards due to integration of avionics systems.

Analysis.

(1) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences. Special consideration is taken where integration has several fault layers, including single, dual or a combination of failures. The integration and fault accommodation may be order sensitive and is considered in the analysis.

Test.

(1) Lab test: Conduct FMET in a HITL/Iron Bird, SIL or high fidelity integration lab with critical hardware and software in the loop to verify interfaces and integration with avionics functions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight test operations evaluating integration with avionics across the envelope and integrated modes during normal operation.

6.2.13.6 Integration with propulsion systems.

Standard: Where the propulsion system is used for control in the pitch, roll, yaw, or heave axis:

a. Single failures in the propulsion system and nominal operation under expected worst-case engine performance do not degrade the VCF below Operational State II.

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b. Propulsion system failures more probable than PLOC divided by ten do not the degrade VCF below Operational State IV.

Failures resulting in asymmetric thrust do not degrade the VCF below Operational State III.

Propulsion system health monitoring is provided to the VCF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the interface and performance requirements between VCF and propulsion.

Analysis.

(1) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences of propulsion system failures.

(2) Identify worst-case engine performance under expected environmental conditions with consideration to end of life performance.

(3) Analyze VCF impact due to identified worst-case engine performance.

Test.

(1) Lab test: Conduct FMET to examine the propulsion system integration.

(2) Lab test: Conduct test in a flight crew evaluation facility, a handling qualities simulation facility, or SIL to verify the VCF performance does not degrade below Operational State II for either worst-case engine performance or single failures of the propulsion system (or below Operational State III for failures resulting in asymmetric thrust).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight operations evaluating integration with propulsion across the envelope during normal operation.

Test.

(3) Flight test: Conduct flight test operations evaluating the performance of the propulsion system across the envelope during normal operation.

6.2.14 Vehicle control function engage, disengage, and interlocks safety.

Engaging or disengaging VCF modes (or modes the VCF manages) may have associated mode transition characteristics. The VCF engineer needs to understand these characteristics and determine if adjustments to the system need to be made to maintain acceptable behavior and stay within limits.

Certain modes should be prevented from engaging during portions of the flight. The VCF engineer is responsible for identifying those modes and implementing necessary protections.

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Modes are checked for compatibility with system state, flight condition, and air vehicle configuration to determine if the mode is allowed and, if not, take action to prevent or remove the incompatibility. If multiple modes cannot be active simultaneously and a mode is requested and allowed, the VCF disengages the incompatible mode prior to engaging the requested mode.

The flight crew is notified regarding mode engagement and disengagement, whether manually selected or automatic.

6.2.14.1 Interlocks.

Standard: The VCF includes redundant interlocks to prevent engagement of VCF modes resulting in a hazardous condition including, but not limited to, test specific functions in a non-test environment and in-flight engagement of on-ground system checks or diagnostics.

For electronically linked dual stations, the VCF provides interlocks to react to inputs from one crew station at a time.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to identify interlocks for normal and failed system states.

(2) Review technical documentation regarding mechanization of interlocks, including levels of redundancy and independence.

Analysis.

(1) Conduct analysis to identify where interlocks are necessary for hazard mitigation.

Test.

(1) Lab test: Conduct lab test of system interlocks to verify proper functionality in normal and failed system states. Confirm that devices or programs which are only meant for use on the ground cannot be engaged in-flight.

(2) Lab test: Conduct FMET to verify attempts to access non-operative devices or programs are not successful.

6.2.14.2 Incompatible modes.

Standard: The VCF prevents engaging a mode, or disengages an active mode, that cannot be supported due to system degradations, current flight condition, or air vehicle configuration.

If a mode change is requested and is supported but incompatible with an existing mode, the VCF disengages the incompatible mode prior to engaging the requested mode.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review technical documentation to verify available VCF modes based on air vehicle health, flight condition, and configuration are incorporated into the VCF architecture.

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(2) Review technical documentation to verify compatibilities of modes for mode engagement and disengagement are identified and incorporated into the VCF architecture.

Analysis.

(1) Complete FMECA and SSHA to identify failure cases and associated loss of available modes.

Test.

(1) Lab test: Conduct lab testing to verify mode selection that is incompatible with an existing mode is not engaged before disengagement of the existing mode.

(2) Lab test: Conduct integrated system testing to verify correct response to modes that cannot be engaged or need to be immediately disengaged due to flight condition or air vehicle configuration.

(3) Lab test: Conduct FMET testing to verify correct response to modes that cannot be engaged or need to be immediately disengaged due to failures.

6.2.14.3 Mode transition transients.

Standard: Engaging, disengaging, and transitioning between modes maintains response characteristics consistent with the operational state of the air vehicle.

The transients associated with the engage, disengage, and mode transitions are supported by the air vehicle dynamics without exceeding any air vehicle limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Complete FMECA and SSHA to identify failure cases and associated mode engagement or disengagement.

Test.

(1) Lab test: Conduct 6-DOF simulation to verify transient behavior associated with mode engagement, disengagement, or transition.

(2) Lab test: Conduct PITL simulation to verify operational state following engagement, disengagement, and transition between modes in normal conditions and due to failures.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight test operations evaluating mode transition transient behavior.

6.2.14.4 Mode transition characteristics.

Standard: Vehicle control function provides protection from rapid mode transitions that can adversely affect controllability of the air vehicle.

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Vehicle control function logic that retains, adjusts, or initializes values following a mode transition or sequences of mode transitions do not lead to loss of control.

The VCF provides notification to the flight crew regarding mode engagement or disengagement.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review technical documentation regarding methods of mode engagement and disengagement.

(2) Review technical documentation regarding VCF logic following mode transition.

Analysis.

(1) Complete FMECA and SSHA to identify failure cases and associated mode engagement or disengagement.

Test.

(1) Lab test: Conduct 6-DOF simulation to verify correct response to rapid mode transition requests to include oscillating between modes.

(2) Lab test: Conduct 6-DOF simulation to verify VCF logic during mode transitions, to include sequential mode transitions or oscillating between modes, does not cause loss of control.

(3) Lab test: Conduct PITL simulation to verify flight crew notification for mode engagement or disengagement.

(4) Lab test: Conduct PITL simulation to verify operational state following engagement, disengagement, and transition between modes in normal conditions and due to failures.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight test operations evaluating mode transition transient behavior.

6.2.15 Vehicle control function UAS command and control communications safety.

6.2.15.1 Unmanned air system command and control integration.

Standard: Command and control processing resources support the functional requirements as allocated to computer hardware and software. The UAS architecture provides C2 2-way data link between the control station and the UAV which allows the flight crew to C2 the UAV movement (e.g., flight path, altitude, airspeed, heading, and ground path). Transition between ground control elements, such as a Launch Recovery Element or Mission Control Element, does not cause loss of the C2 function.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Review C2 system detailed specifications (hardware and software requirements) of expected functionality, performance, and design requirements for effective testing.

Demonstration.

(1) Ground test demonstrates C2 functionality including transitions between control stations.

Test.

(1) Lab test: Conduct C2 component testing to verify hardware and software requirements are met.

(2) Lab test: Conduct UAS SIL test to verify communication interface paths support sending and receiving C2 data to control the UAV.

(3) Lab test: Conduct UAS SIL test to verify transitions between control stations are robust and do not result in loss of the command-and-control function.

(4) Lab test: Conduct UAS SIL test to verify processing resources function without degradation of the C2 function under worst-case conditions for communications bus loading/traffic. This includes the entire communication path between the control station and the VCF.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Flight operations demonstrate C2 functionality, including transitions between control stations, with mission representative communications bus loading/traffic.

6.2.15.2 Command and control situational awareness and health and status.

Standard: The UAS C2 system provides situational awareness and health and status of the UAV to the flight crew. Specifically, the C2 system processes and displays real-time flight-critical parameters (e.g., altitude, airspeed, heading, control surface positions, flight mode), and health and status of the VCF and integrated subsystems (e.g., VCF computers, air data, inertial measurement units, navigation, actuators, propulsion, electrical power, hydraulics).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review software requirement specifications to ensure the requirements for situational awareness flight parameters and health and status are clearly defined.

Test.

(1) Lab test: Conduct SIL testing to verify control station displays function properly and refresh per the design requirements.

a. Verify that the subsystems health and status updates get downlinked to the control station and status is refreshed per the design requirements.
b. Any faults are clearly annunciated or displayed to the flight crew.

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(2) Ground test: Conduct ground testing (air vehicle and control station) to validate SIL testing and verify control station displays function properly and refresh per the design requirements:

a. Verify that the subsystems health and status updates get downlinked to the control station and status is refreshed per the design requirements.
b. Any faults are clearly annunciated or displayed to the flight crew.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight operations to verify display functionality and health and status data is provided to the control station for the various subsystems at the designed refresh rates.

6.2.15.3 Command and control redundancy and safety.

Standard: The UAS provides redundant, fault tolerant, and reliable C2 data links between the UAV and the control station with separation and isolation among internal and external VCF interfaces to meet fail-operational capability of the C2 function for any single failure unless the failure is less probable than PLOC. The predicted performance of the C2 data links meets latency and reliability requirements with confidence levels as specified by the procuring activity.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review C2 integrated system design drawings to ensure physical separation and isolation among internal and external VCF interfaces and to identify single point failures.

(2) Review documentation of design drawings and specification to verify link redundancy and link reconfiguration requirements.

Analysis.

(1) Failure modes, effects, and criticality assessment results for the C2 integrated system show single point failures are less probable than PLOC.

(2) Conduct C2 system reliability analysis to verify system meets predicted reliability requirements.

(3) Conduct hazard analyses to verify the reliability, functionality, and latency of the C2 system is accounted for properly in the hazard analyses.

Test.

(1) Lab test: Conduct C2 system component hardware qualification testing.

(2) Lab test: Conduct UAS SIL test to verify operation of redundant data links meet functional requirements (including latency requirements).

(3) Lab test: Conduct UAS SIL FMET to verify fail-op, fault detection, isolation, and reconfiguration for critical C2 failures.

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(4) Ground test: Conduct ground testing to verify C2 link reliability and latency requirements are met.

6.2.15.4 Command and control integrity data check and protection.

Standard: The UAS C2 integrated system provides communications data transmission integrity checks (e.g., checksums, parity checks, cyclic redundancy checks, range verification checks, or integrity data monitors) to detect any data transmission errors (received, processed, and sent) and provides anti-aliasing filters and debounce protection. The UAV rejects erroneous or invalid input commands and provides clear visual feedback to the flight crew. The probability of data corruption resulting in loss of air vehicle is less than PLOC divided by 100.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review UAS C2 system and software requirements specification for data integrity requirements (data validation and error detection algorithms/monitors).

(2) Review C2 system design documentation to verify the implementation of anti-alias filters and debounce signal protection.

Analysis.

(1) Perform analysis to determine probability of data corruption resulting in loss of air vehicle.

Test.

(1) Lab test: Conduct SIL testing to validate data integrity error detecting algorithm and integrity data monitor works per design.

(2) Lab test: Conduct SIL testing to ensure data consistency (accurate, up-to-date, and identical data) for across the integrated C2 system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) On-air vehicle ground demonstration ensures data consistency across the integrated C2 system.

(2) Flight demonstration ensures data consistency across the integrated C2 system.

6.2.15.5 Loss of command and control and critical failures.

Standard: The air vehicle VCF provides robust contingency logic for lost communications, loss of command, and failures within the VCF and integrated subsystems. In case of total loss of communication or loss of command with the control station, the air vehicle is capable of autonomous guidance, navigation, and control for a safe landing. Total loss of communication or loss of command does not result in instability, LCO, exceeding transient limits, or VCF degradation below Operational State III. Contingency state is downlinked from the UAV and displayed to the flight crew. Changes in contingency state are clearly annunciated.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review contingency logic software requirements to verify lost communication, loss of command, and failures within the VCF and integrated subsystems are complete.

(2) Ensure software requirements define contingency hierarchy, such that the most critical contingency takes precedent over other contingencies, and account for all operating modes (e.g., takeoff, landing, flight path overrides, autonomous navigation, waveoff).

Analysis.

(1) Verify FMECA and SSHA identify failures and consequences for the integrated system.

(2) Analysis verifies contingency logic is implemented properly to handle loss of command, total loss of communication, and critical failures in all operating modes.

Demonstration.

(1) For every contingency type, conduct lab test to verify the contingency logic is initiated, the UAV takes appropriate action, follows appropriate route, and either lands safely or terminates as designed.

Test.

(1) Lab test: Conduct UAS SIL and FMET to simulate in-flight-critical failures, loss of command, and total loss of communication. Ensure contingency logic is properly initiated and does not result in instability, LCO, exceeding transient limits, or VCF degradation below Operational State III and that contingency state is properly displayed and annunciated.

(2) Lab test: Conduct UAS SIL and FMET to demonstrate contingency hierarchy implementation performs as designed and is robust for all operating modes (e.g., takeoff, landing, flight path overrides, autonomous navigation, waveoff).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Simulate a total loss of C2 communication with the UAV to verify that contingency logic is initiated and the UAV follows the loss of communication contingency route and lands safely.

6.2.15.6 Remotely piloted operation.

Standard: For remotely piloted UAVs, sensors that provide feedback to the remote flight crew are fully operational under all natural and induced environmental conditions. The environmental effects on visual cues do not degrade flight crew capability to recognize hazards to SOF. Provisions are incorporated to protect onboard devices (e.g., sensors, cameras) against FOD, obscuration of the sensor transparency, and laser threats. Latencies between flight crew inputs and visual response do not induce instability or PIO and are sufficient to accomplish flight-critical tasks (e.g., takeoff, landing, aerial refueling). No single point failure within the remote-control function of the UAS causes loss of SOF information. In case of total loss of the remote-

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control capability, the UAV has contingency logic to switch to autonomous mode for control and landing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review UAS system design, drawings, and architecture to identify single point failures in the control station, communications path, and air vehicle that impact the remote control function.

(2) Review UAS system design and drawings to verify provisions are incorporated to protect onboard devices (e.g., sensors, cameras).

Analysis.

(1) Verify FMECA and SHA are completed to identify failures, failure rates, and consequences of the remote-control function.

(2) Conduct PITL simulation analysis to determine latency requirements and verify worst-case latencies do not adversely impact C2 of the UAV.

Test.

(1) Lab test: Conduct UAS component hardware functional and environmental qualification testing.

(2) Lab test: Conduct UAS SIL tests to verify UAS performance under required natural and induced environmental conditions. Tests are conducted with all applicable data links and include V&V of the latency requirements.

(3) Lab test: Conduct FMET and UAS SIL testing to verify fault identification and contingency logic.

(4) Ground test: Conduct ground test, with the UAV in motion (e.g., low speed taxi), to verify system functionality and that system latencies are within requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Flight testing in operationally representative environments demonstrates UAS remote control functionality and performance.

6.2.16 Vehicle control function hydraulic power source safety.

6.2.16.1 Hydraulic system integrity.

Standard: Hydraulic system failures more probable than PLOC divided by ten do not degrade the VCF below Operational State IV. No single failure within the hydraulic system degrades VCF below Operational State II. Hydraulic system health monitoring is provided to the VCF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Review air vehicle hydraulic system functional architecture schematic to identify the VCF interface design and failures points.

(2) Review physical VCF and hydraulic system installation drawings.

(3) Verify FMECA and SSHA are completed to identify failures, failure rates, and consequences of the hydraulic system.

(4) Identify all hydraulic system fault monitoring and detection schemes.

Test.

(1) Lab test: Conduct hydraulic system performance and integrity test on a flight representative hydraulic system.

(2) Lab test: Conduct FMET with flight representative hydraulic system and VCF HITL to verify fault detection, system redundancy, and fault tolerance.

6.2.16.2 Hydraulic system dynamics.

Standard: Under maximum representative hydraulic load demand, VCF system performance meets actuator loaded rate and stability requirements for normal, backup, and emergency hydraulic operation. Emergency and backup hydraulic power systems do not cause loss of VCF or any instability due to flow rates, lower pressure, or reduced flutter margin.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Validate model, simulation, and emulation design are representative of the hydraulic system design.

Test.

(1) Lab test: Conduct VCF and hydraulic system integrated loaded rate performance test on a flight representative hydraulic system.

(2) Lab test: Conduct VCF and hydraulic system integrated performance test to validate 6 dB and 45-degree stability margins.

6.2.16.3 Transients, fluctuations, and transitions.

Standard: Hydraulic power transients due to switching sources, accumulators, valves, relays, controllers, and any other devices do not cause any hydraulic power fluctuations that result in loss of VCF.

Vehicle control function does not degrade below Operational State III as a result of transition to backup or emergency hydraulic operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Test.

(1) Lab test: Conduct VCF and hydraulic system integrated loaded rate performance test on a flight representative hydraulic system.

(2) Lab test: Conduct VCF and hydraulic system integrated performance test to evaluate actuator stability.

(3) Lab test: Conduct PITL simulation to verify hydraulic transitions will not cause the VCF to degrade below Operational State III.

6.2.17 Vehicle control function electrical power systems safety.

The EPS provides the means to supply the power necessary for VCF operation. Power coming to the VCF via the EPS should be safe, independent and redundant. Vehicle control function power should be transient-free and reliable. To prevent loss of VCF in the event of a failure of primary power generation equipment, a backup source is used capable of operating flight-critical elements of the VCF. When operating, power generation sources do not result in the loss of VCF. There is more than one generation source and these sources can operate independently. Any power transients do not degrade proper operation of the VCF. If a battery is used for a backup source, they are sized for capacity, or if used to suppress transients, they satisfy that purpose. These batteries will be exclusive to the VCF. Electrical buses for VCF operation are separated to preclude failures from propagating. Any dedicated VCF bus is intended only for flight-critical devices.

6.2.17.1 Backup.

Standard: Following a total loss of engines or generators, electrical power is provided from a source independent from the primary power source with capacity to maintain Operational State IV from anywhere inside the RORH. Critical VCF displays are maintained via the backup power source.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis of the backup power design architecture determines what functions are shed and what functions are retained with backup power.

(2) Complete an ELA and verify proper coverage of electrical loads with backup electrical power.

Test.

(1) Lab test: Conduct FMET of the EPS to verify proper transfer of required functions to the backup power source(s) and that backup power can support those required functions.

(2) Lab test: Conduct VCF FMET to verify proper response to EPS failures.

(3) Lab test: Conduct simulation to verify capacity of the power source meets time to land requirements.

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6.2.17.2 Independent sources.

Standard: The electrical power generators used to power the VCF components: (e.g., equipment, wiring, controls and associated grounding schemes or backup to the VCF) do not induce any conditions that result in loss of the VCF.

Each component within the VCF has multiple, independent power sources.

No single failure of a power source degrades VCF below Operational State II.

No single VCF failure due to wiring or grounding architecture can cause loss of power to the VCF.

Electrical system failures more probable than PLOC divided by ten do not degrade the VCF below Operational State IV.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review EPS architecture for independent sources of power to VCF components.

Analysis.

(1) Electrical power system FMECA is used to verify failure modes, failure rates, and impact to the VCF.

Test.

(1) Lab test: Conduct EPS FMET to verify functionality of independent power sources.

(2) Lab test: Conduct VCF FMET to verify proper response to EPS failures.

6.2.17.3 Transients.

Standard: Electrical power transients do not result in degradation of VCF below Operational State II.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify electrical power quality requirements are captured for VCF components.

Test.

(1) Lab test: Conduct VCF component level power quality interface test IAW MIL-STD-704, General Requirements and Detailed Requirements sections, or as specified by the procuring activity.

(2) Lab test: Complete a power transient evaluation during EPS integrated test.

(3) Ground test: Complete a power transient evaluation for startup and switching.

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6.2.17.4 Battery.

Standard: Where batteries are employed as backup power source to VCF, they provide sufficient capacity for VCF not to degrade below Operational State IV. If batteries are used for transient suppression, they provide suppression capability for VCF not to degrade below Operational State II. Battery health is monitored and status provided to the flight crew. Batteries that are exclusive to VCF operation are not to be shared with other subsystems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review VCF and EPS ICD for use of batteries.

(2) Review EPS architecture with respect to battery integration. Identify functions which are retained and those which are shed when batteries are used as backup power.

Analysis.

(1) Complete an ELA and verify proper coverage of electrical loads when using batteries for backup power.

Test.

(1) Lab test: Conduct FMET of the EPS to verify redundancy management including adequacy of battery usage coverage.

(2) Lab test: Conduct HITL simulation to verify no worse than VCF Operational State IV while on battery power only.

(3) Lab test: Conduct HITL simulation to verify battery suppression capability meets no worse than VCF Operational State II.

6.2.17.5 Bus separation.

Standard: Vehicle control function electrical buses are physically separated such that failure on one bus will not propagate to another.

If a dedicated VCF electrical bus is provided, devices not classified as flight-critical are not permitted use of the VCF bus.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Electrical power system architecture verifies functional separation, and dedicated VCF buses.

(2) Electrical power system installation drawings verifies physical bus separation.

6.2.18 Vehicle control function computer processing.

For air vehicles requiring electronic hardware, software, and firmware to provide stability and control as well as outer loop control functions, the VCF is typically a safety-critical, software-intensive system. In such designs, software functionality is required to provide time critical,

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frame coherent data in a deterministic manner to support the control function. While the standards for AW of VCF CS&S is managed by a variety of standards and MoC in Section 15 of this document, it is the responsibility of the VCF system design to place key requirements on the CS&S to ensure its support of overall VCF performance.

The VCF requires precise scheduling and execution of tasks to enable time critical, frame coherent operation in a deterministic manner in support of the control system. Requirements should be flowed to the software to ensure its architecture and design provide a mechanism for scheduling tasks at required (possibly multiple) rates which support acceptable VCF performance in terms of control loop closure timing, signal latency, redundancy management synchronization, and communication response.

Tight time synchronization of software execution among redundant processing channels of the VCF provides a significant opportunity for a software error, such as divide by zero, to occur simultaneously in all channels. Requirements, including recovery time limitations, will be flowed to the software to ensure that a multi-channel common mode software error recovery mechanism is provided in support of VCF fault tolerance and PLOC requirements.

As part of this effort, an analysis is performed to identify the criticality levels of the hardware, software, and firmware components in the VCF design. The analysis evaluates the severity of loss, erroneous, or otherwise degraded operation of the VCF functionality.

6.2.18.1 Design assurance.

Standard: The VCF functionality has been fully allocated to hardware, software, and firmware components within the architecture, and each component is assigned and developed to a criticality level as determined by an analysis of the severity of loss, erroneous, or otherwise degraded operation of the VCF functionality. Lesser criticality functions do not impact the performance and execution of higher criticality functions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review VCF hardware, software, and firmware development plans and test plans to ensure processes are aligned with CS&S expectations for the required criticality levels.

Analysis.

(1) Conduct analysis of the severity of loss, erroneous, or otherwise degraded operation of the VCF functionality and the support provided by the hardware, software, and firmware components to verify that VCF component criticality levels are correctly assigned. Conduct analysis of the VCF hardware and software architectures and high-level software and firmware designs to verify that the VCF design supports the required criticality levels.

(2) For designs with mixed criticality levels within a common processor, ensure lesser criticality functions do not impact the performance and execution of higher criticality functions through hardware and architectural protections (e.g., brick-wall partitioning).

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6.2.18.2 Vehicle control function timing requirements.

Standard: Software and firmware execution timing (e.g., frame rates, control loop closure timing, signal latency, redundancy management synchronization, data transfers, and communication responses) supports VCF performance, redundancy management, and stability robustness requirements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review VCF software and firmware specifications to ensure execution timing requirements are captured.

Analysis.

(1) Conduct analysis in the discrete time domain for VCF control loops (e.g., inner loops, outer loops, and servo loops) to verify software and firmware execution timing supports VCF performance and the VCF control law safety standards herein.

(2) Conduct analysis to verify software and firmware execution timing supports VCF redundancy management.

6.2.18.3 Vehicle control function timing architecture.

Standard: The software architecture, along with its underlying real-time operating system and task scheduler, provides time deterministic scheduling with frame coherent data of all tasks required to implement the VCF. The software architecture identifies any critical within-frame timing for all internal data transfers to execute the control function. For each partition or minor frame, margin for unused, available processor throughput is reserved to accommodate worst-case variations in processing demands.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the software architecture, along with the software high level design, to verify that the task scheduling mechanism supports the task scheduling timing required by the VCF. All critical timing requirements are identified with margin to time deadlines captured for WCET for real-time throughput analysis.

(2) Ensure protections are in place to verify frame coherent data across all redundant processing elements.

Analysis.

(1) Analysis of the VCF design is performed to determine test cases for WCET assessment. Analysis includes assessment of all VCF functions including nominal VCF modes as well as failure scenarios. WCET is supported through measurement of all time critical margins during system development and software test.

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(2) Margin is established and justified based on a VCF worst-case analysis, which includes confidence bounds on worst-case timing. The minimum margin should be no less than the margin established in Section 15.

Test.

(1) Lab test: Real-time WCET measurements are taken during formal system and software test to demonstrate required margins are met.

6.2.18.4 Common mode failure accommodation.

Standard: The VCF software architecture and design provides mitigation for an in-flight, CMF resulting in an all-channel exception or reset. Mitigations may include an in-flight "fast start" mechanism or other design approach to handle processor exceptions. Vehicle control function transient and recovery analysis is provided to support the recovery time requirement. This requirement is flowed and captured in the VCF software requirements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the software architecture, along with the software high level design, to verify mitigations are in place for an in-flight, CMF resulting in an all-channel exception or reset.

Analysis.

(1) Vehicle control function transient and recovery analysis is performed to support the recovery time requirement. Ensure the analysis captures the state of the control system during the recovery (e.g., actuator control held at last position or free floating).

Test.

(1) Lab test: Failure modes and effects testing demonstrates VCF recovery response to a VCF software CMF.

(2) Lab test: Test cases are developed, and measurements taken, to verify that the software in all redundant channels can recover within the time constraints flowed from the VCF system design.

6.2.18.5 Security.

Standard: Implementation of security features does not degrade VCF performance or prohibit critical VCF functions. For UAS, this includes C2 communication security implementation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the security provisions requirements specified in the Program Protection Implementation Plan air vehicle/weapon system specification, and component hardware and software specifications.

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Analysis.

(1) Conduct analysis of security provision's potential to degrade or disrupt the VCF.

Test.

(1) Lab test: Conduct component and SIL tests to verify implementation of security requirements and provisions does not degrade VCF performance or prohibit critical VCF functions. Tests should include VCF performance during and after the activation of any security protection features.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct demonstration in flight test to verify VCF performance with implemented security measures.

6.2.19 Inertial and navigation sensors.

Inertial sensors including accelerometers and rate gyroscopes are frequently used in the VCF to provide a high-rate, high-quality feedback for the VCF. The inertial sensors can be packaged into a single unit with software to provide an inertial navigation unit. Further integration with other sensors such as GPS can improve the overall performance of the inertial navigation system by periodically correcting for drift in the inertial navigation solution. Integration of the sensors into the vehicle requires placement to minimize the effect of vehicle structural dynamics coupling on the sensors and degrading air vehicle ASE stability.

Inertial sensors are often augmented/aided with alternate sensors (e.g., GPS, EGI) to provide navigation grade performance in support of the VCF. These alternate sensors are often not designed or developed to flight-critical standards, which drives the VCF to provide high integrity mitigations to accommodate the reduced integrity of the alternate sensors.

It is desirable that sensor signals directly supporting air vehicle inner loop stability (typically 6-DOF angular rates and linear accelerations) are sourced from a dedicated, free inertial sensor source with no GPS aiding. The integration of flight-critical sensor data with non-flight critical sensor data, which may be required for outer loop control functions, can lead to situations where errors or defects in non-flight-critical systems result in errors in flight-critical systems. In these cases, monitors designed to detect errors in data inputs to a flight-critical system from a non-flight-critical system should be developed to flight-critical standards and hosted on flight-critical systems and there is residual risk associated with the use of a monitor.

6.2.19.1 Inertial sensors.

Standard: Free inertial sensor data cannot be corrupted by air vehicle structural dynamics or non-flight-critical sensor data. The VCF includes protections to preclude anomalous GPS data from corrupting VCF performance or causing loss of control when using GPS aided parameters in the control function. Free inertial sensor data, to include angular rates, linear accelerations, and attitudes (if required by the design), is provided to the VCF for stability and control.

Inertial sensors are located so that there is minimal impact from air vehicle structural modes. Corrections for the installation location do not interfere with SMI filters.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Inspection of VCF designs shows inertial sensor data is not mixed with non-flight-critical sensor data.

(2) Structural analysis identifies structural nodes and anti-nodes. The results of the analysis informs the sensor placement in the air vehicle and the corresponding structural filters to support ASE stability requirements.

(3) Conduct simulation analysis to verify anomalous GPS data cannot corrupt VCF performance or cause loss of control.

Test.

(1) Ground test: Ground vibration test verifies that sensor locations minimize the impact of structural vibrations.

(2) Ground test: SMI testing verifies that sensor integrations are immune to, or compensate for, structural modes.

6.2.19.2 Mixed criticality sensors.

Standard: Flight-critical sensors are connected directly to flight-critical computing. Inputs from sources not developed to flight-critical standards have dedicated monitors for health and validity which are developed to flight-critical standards and hosted on flight-critical hardware. Missing or erroneous sensor inputs from sources not developed to flight-critical standards do not result in exceeding air vehicle transient limits nor degrade the VCF below Operational State III.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspection of the VCF architecture shows flight-critical sensors connect directly to flight-critical computing without passing through non-flight-critical equipment.

(2) Inspection of the VCF architecture shows monitors of inputs from sources not developed to flight-critical standards are hosted on flight-critical hardware.

Analysis.

(1) Air vehicle transient analysis utilizing finalized monitor threshold and persistence values demonstrates transients are within air vehicle established limits.

Test.

(1) Lab test: Conduct FMET SIL testing to verify monitor performance and associated transients.

(2) Lab test: Pilot-in-the-loop simulations show that missing or erroneous data from non-flight-critical sources does not degrade the VCF below Operational State III.

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6.3 Internal aerodynamics.

Compliance with these standards ensures all propulsion integration effects are accounted for in the air vehicle thrust-drag accounting system, and that the integrated propulsion system (airframe, inlet, engine, nozzle) is free from instabilities caused by non-uniform inlet flow that would result in loss of air vehicle, loss of SCFs, or loss of flying qualities.

Examples of such instabilities include expulsion of shocks during inlet unstart, engine stall or flameout caused by inlet flow distortion, fan or compressor blade flutter caused by inlet flow distortion, or afterburner flameout caused by nozzle aerodynamics.

Inlet flows that may initiate propulsion system instabilities include subsonic planar wave distortion, supersonic inlet buzz, inlet total pressure and temperature distortion, inlet swirl distortion, and variable geometry resonant oscillation.

Total pressure or temperature distortion may be present in the inlet, but the distortion cannot cause propulsion instabilities. Inlet distortion is typically the largest destabilizing effect on the engine stability limit and will be included in the deterministic engine stability audit.

6.3.1 Airframe-inlet compatibility.

Standard: The integrated propulsion system is free of instabilities resulting from flow disturbances entering the inlet from the airframe and external sources.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Identify potential sources of non-uniform inlet flow caused by external sources such as forebody boundary layers, air data booms, pitot-static probes, weapons bay doors, external stores, pods, munitions, landing gear, catapult steam, wake from other air vehicles, exhaust from own or other air vehicles, ice accretion, hot de-icing air, armament gas, and debris. Consider all test and operational flight conditions including ground operations.

(2) Identify potential impacts to propulsion system stability caused by any changes to airframe and integrated propulsion system geometry between ground-tested, flight-tested, and production air vehicles.

Analysis.

(1) Estimate the magnitude and extent of potential non-uniform inlet flow using CFD. Use steady-state CFD to quantify inlet recovery, quantify distortion trends, and to identify regions where unsteady CFD is needed. Use unsteady CFD to estimate time-dependent effects of distortion. For processing and validation of CFD-based time-dependent distortion, refer to SAE AIR6345.

(2) Conduct analysis to identify airflow and air vehicle attitude limitations for high-speed inlet (supersonic or hypersonic) unstart.

(3) Conduct analysis using wind tunnel database and CFD to determine ground-to-flight geometry and Reynolds number scaling for measured inlet database. For methodology and instrumentation details refer to SAE AIR5687.

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(4) Conduct analysis to quantify thrust, power, or torque impacts caused by inlet performance degradation and integration effects.

Test.

(1) Lab test: Conduct sub-scale wind tunnel tests to measure inlet flow nonuniformities and to validate pre-test CFD. Test instrumentation includes distributed surface static pressures (or pressure sensitive paint), boundary layer rakes, and inlet rakes. Inlet rakes include steady state and dynamic instrumentation. For methodology and instrumentation details refer to SAE AIR1419, SAE ARP1420, and SAE ARP6420.

(2) Lab test: For high-speed inlets, conduct sub-scale wind tunnel tests to verify inlet unstart boundaries.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(3) Flight test: Conduct flight tests to verify inlet performance, to validate ground-to-flight database correlations, to validate propulsion system stability, and to assess effects of any changes between ground and flight test articles. Instrument flight test air vehicles with an AIP rake. For methodology and instrumentation details refer to SAE AIR1419, SAE AIR5867, SAE ARP1420, and SAE ARP6420.

6.3.2 Engine-inlet compatibility.

Standard: The integrated propulsion system is free of instabilities resulting from flow disturbances generated by the inlet, inlet duct, and secondary air systems located in the inlet.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review integrated propulsion system geometry to identify potential sources of total pressure distortion, total temperature distortion, swirl distortion, and planar waves.

Analysis.

(1) Verify that the inlet can provide the total mass flow demanded by the engine, bleeds, ECS, bay ventilation, and any other mass flow required by the air vehicle. The inlet should be able to provide mass flow to all secondary airflow systems that use airflow from the inlet, throughout the ground and flight envelope, without creating instability-inducing distortion or airflow deficit to the engine.

(2) Characterize dynamic total pressure inlet distortion, steady-state swirl distortion, temperature ramp and spatial distortion, and planar wave distortion and include in the engine stability audit with the corresponding engine sensitivities.

(3) For configurations with observed planar wave content (supersonic inlet buzz and subsonic planar wave distortion), quantify planar wave airflow limits from ground test data. For methodology and instrumentation details refer to SAE AIR5866 and SAE ARP6420.

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Test.

(1) Lab test: Conduct sub-scale wind tunnel tests to measure inlet dynamic total pressure, swirl, and planar wave distortion. For methodology and instrumentation details refer to SAE AIR1419, SAE ARP1420, and SAE ARP6420.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Lab test: For swirl distortion, compliance can be shown by examination if the engine is equipped with inlet guide vanes. For swirl methodology and instrumentation details, refer to SAE AIR5686 and SAE ARP6420.

(3) Flight test: Conduct flight tests to verify inlet operability and propulsion system stability. Instrument flight test air vehicle with an AIP rake. For methodology and instrumentation details, refer to SAE AIR1419, SAE AIR5867, SAE ARP1420, and SAE ARP6420.

6.3.3 Engine-inlet-nozzle compatibility.

Standard: The integrated propulsion system is free from instabilities caused by integration effects such as engine or nozzle control modulation caused by inlet distortion, inlet, engine, and nozzle airflow matching, and nozzle integration with the airframe.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify the inlet can operate through the range of engine and nozzle-induced backpressures. Inlets should maintain positive flow throughout the operating range including the effects of throttle transients, nozzle scheduling, fuel scheduling, and afterburner transients.

(2) For highly integrated or embedded nozzles, conduct analysis to verify that the nozzle and afterbody aerodynamic flow path integrity is not compromised by hot streaks and quantify any impacts to propulsion system stability. At a minimum, assess the effects of changes in nozzle aerodynamic throat, flow separation, nozzle thrust coefficient, and discharge coefficient on integrated propulsion system performance.

Test.

(1) Lab test: Conduct nozzle afterbody sub-scale test to evaluate throttle-dependent forces and moments produced by the integrated aft-body geometry. For highly integrated systems, the effects of inlet spillage on the throttle-dependent afterbody loads should be evaluated as part of the test.

(2) Lab test: Conduct ground test to evaluate the effects of active or passive thrust vectoring on inlet, engine, and nozzle operability.

Prior to entering operational test or operations, verify the standard by the following method(s):

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Test.

(3) Flight test: Conduct flight tests to verify integrated propulsion system thrust. For in-flight thrust determination methodology, refer to SAE AIR1703.

6.4 External aerodynamics and performance.

Aerodynamic and performance AW standards address the performance of the air vehicle over the full flight envelope covering all phases of flight and operating conditions and considers failure conditions impacting air vehicle performance. To conduct mission planning and flight activities safely, the flight crew needs performance information, which is clearly communicated, comprehensive, and accurate. The furnished performance information accounts for relevant attributes and limits of other air vehicle subsystems, degraded and emergency configurations, methods of employment, and minimum SOF performance margins where applicable. Takeoff and landing performance data accounts for runway condition (e.g., dry, wet, snow, ice) and surface type (e.g., paved, semi-prepared). The performance information includes impacts resulting from airframe and engine icing, limitations that would impact air vehicle performance, as well as failure conditions that would impact air vehicle performance (e.g., loss of engine, flaps, speed brake). Aerodynamic, installed propulsion, and mass properties databases form the basis of the performance information. Developmental allowances or uncertainty factors on end performance, aerodynamics, propulsion, and mass properties may vary with the maturity of performance information verification. Compliance with these standards ensures that the information furnished to the flight crew (in flight manuals, personal electronic devices, flight management systems) provides the performance information needed to conduct mission planning and flight activities.

6.4.1 Performance information.

Standard: The performance information provided to the flight crew is IAW MIL-DTL-7700, Section V – Operating Limitations, Section VI – Flight Characteristics, Section VII – Adverse Weather Operation, Performance Data Appendix/Manual or NAVAIR ACO NATOPS Style Guide, Chapter 4 – Operating Limitations, Part IV – Flight Characteristics and Control Procedures, Part VI – All-Weather Operation, Part XI – Performance Data allowing for mission planning and flight activities within the flight envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation of the most recent aerodynamic, installed propulsion, and mass properties databases used to develop air vehicle performance. Aerodynamic data covers the full flight envelope for all configurations and loadings. Wind tunnel aerodynamics are corrected for scale effects, support system interference, and geometric differences. Propulsion data is corrected for losses and efficiencies for the air vehicle installed configuration and represents a government approved engine status. Mass properties are representative of all air vehicle configurations and loadings. Databases cover the full up and away flight envelope and takeoff and landing configurations (in and out of ground effect).

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(2) Review performance information provided to the flight crew to ensure data is available for all phases of flight related to planning and operation (e.g., launch, takeoff, climb, cruise, endurance, maneuver, hover, in-flight refueling, descent, approach, landing, and recovery).

(3) Review performance information provided to the flight crew to ensure the data covers the full range of mass properties, operating conditions, and atmospheric conditions for all air vehicle variants, configurations, and loadings within the flight envelope. This includes performance impacts due to icing.

(4) Review documentation of the performance analysis, calculations, ground rules, assumptions, and plotting techniques that went into the development of the performance information provided to the flight crew to ensure it complies with MIL-STD-3013, Definitions and General Requirements sections for fixed-wing air vehicles and JSSG-2001, Appendix D, Level III data requirements for rotary-wing air vehicles.

(5) Review preliminary performance information provided to the flight crew in other electronic forms (e.g., personal electronic devices, flight management systems, ground control stations, mission planning calculators) to ensure it meets the intent of MIL-DTL-7700, Section V – Operating Limitations section, Section VI – Flight Characteristics section, Section VII – Adverse Weather Operation section, and Performance Data Appendix/Manual section or NAVAIR ACO NATOPS Style Guide, Chapter 4 – Operating Limitations, Part IV – Flight Characteristics and Control Procedures, Part VI – All-Weather Operation, Part XI – Performance Data and complies with MIL-STD-3013, Definitions and General Requirements for fixed-wing vehicles and JSSG-2001, Appendix D, Level III data requirements for rotary-wing air vehicles.

(6) Review preliminary performance information provided to other persons responsible for operation of the air vehicle (e.g., Landing Signal Officer, Catapult Officer) to ensure it complies with MIL-STD-3013, Definitions and General Requirements for fixed-wing air vehicles and JSSG-2001, Appendix D, Level III data requirements for rotary-wing air vehicles.

Analysis.

(1) Conduct empirical and CFD analysis to build the aerodynamic databases and aerodynamic increments required to determine air vehicle performance. Analysis includes alternate mission equipment, external stores, other configuration variations, impacts due to icing, and other degraded states as applicable.

(2) Conduct performance analysis to gather mission, maneuver, airfield takeoff and landing, or ship-based launch and recovery performance information required by the flight crew.

Test.

(1) Ground test: Conduct all wind tunnel tests necessary to build aerodynamic databases and aerodynamic increments required to determine air vehicle performance. Testing includes alternate mission equipment, external stores, other configuration variations, impacts due to icing, and other degraded states as applicable.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Examination.

(7) Review flight test reports to assess any updates to the aerodynamic and installed propulsion databases and quantify the resulting air vehicle performance characteristics throughout the full flight envelope for all tested configurations and loadings.

(8) Review flight manuals to ensure all performance information is available to the flight crew for all phases of flight related to planning and operation (e.g., launch, takeoff, climb, cruise, endurance, maneuver, hover, in-flight refueling, descent, approach, landing, and recovery).

(9) Review flight manuals provided to the flight crew to ensure the data covers the full range of mass properties, operating conditions and atmospheric conditions for all air vehicle variants, configurations, and loadings within the flight envelope. This includes performance impacts due to icing and other degraded states as applicable.

(10) Review the performance analysis, calculations, ground rules, assumptions, and plotting techniques that went into the development of the performance information provided in the flight manual to ensure it complies with MIL-STD-3013, Definitions and General Requirements for fixed-wing air vehicles and JSSG-2001, Appendix D, Level III data requirements for rotary-wing air vehicles.

(11) Review performance information provided to the flight crew in other electronic forms (e.g., personal electronic devices, flight management systems, ground control stations, mission planning calculators) to ensure it meets the intent of MIL-DTL-7700, Section V – Operating Limitations section, Section VI – Flight Characteristics section, Section VII – Adverse Weather Operation section, and Performance Data Appendix/Manual section or NAVAIR ACO NATOPS Style Guide, Chapter 4 – Operating Limitations, Part IV – Flight Characteristics and Control Procedures, Part VI – All-Weather Operation, Part XI – Performance Data and complies with MIL-STD-3013, Definitions and General Requirements.

(12) Review performance information provided to other persons responsible for operation of the air vehicle (e.g., Landing Signal Officer, Catapult Officer) to ensure it complies with MIL-STD-3013, Definitions and General Requirements for fixed-wing air vehicles and JSSG-2001, Appendix D, Level III data requirements for rotary-wing air vehicles.

Test.

(2) Flight test: Conduct flight testing to verify and update the aerodynamic and performance characteristics throughout the full flight envelope. Testing should include any alternate mission equipment, external stores, and other configuration variations. Compare flight testing data with modeling analysis and reconcile model with actual flight characteristics. Model reconciliation considers the flight test data quality and instrumentation measurement uncertainty.

6.4.2 Performance limits.

Standard: Flight performance limitations that affect operation of the air vehicle during normal, degraded, and emergency operating conditions are accounted for in the performance information, and identified and documented IAW MIL-DTL-7700 , Section V – Operating Limitations section, Section VI – Flight Characteristics section, Section VII – Adverse Weather Operation section, and Performance Data Appendix/Manual section or NAVAIR ACO NATOPS Style Guide, Chapter 4 – Operating Limitations, Part IV – Flight Characteristics and Control Procedures, Part VI – All-Weather Operation, Part XI – Performance Data. These include, but

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are not limited to, weight, CG, acceleration, speed, dynamic pressure, altitude, stall, buffet, engine operability, propulsion system limits, rate-of-climb, rate-of-descent, maneuverability, structural load limit, landing gear, brake energy, store carriage, temperature, wind, runway condition, and icing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review preliminary flight manuals and any additional operating limits to ensure flight performance limitations during normal, degraded, and emergency operating conditions are identified and documented for all required configurations and loadings.

(2) Review mass properties documentation to ensure weight and CG limitations are identified and documented for all required configurations and that all required loadings fall within these limits.

(3) Review documentation of other air vehicle attributes (e.g., structural and flutter limits, control surface scheduling, landing gear and brake energy capabilities, transmission limits, fuel system requirements) to ensure that these limits are properly considered in the performance information presented to the flight crew.

Analysis.

(1) Conduct empirical and CFD analysis to build the aerodynamic databases and aerodynamic increments required to determine air vehicle performance limits. Analysis is conducted up through the expected limits of the air vehicle (e.g., speed, Mach number, stall, buffet, AOA, AOS, braking energy) for all configurations and loadings.

(2) Conduct performance analysis to gather the performance information required to determine air vehicle limits.

Test.

(1) Ground test: Conduct wind tunnel tests to build the aerodynamic databases and aerodynamic increments required to determine air vehicle performance limits. Conduct testing up through the expected limits of the air vehicle (e.g., speed, Mach number, stall, buffet, AOA, AOS) for all configurations and loadings.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Review flight test reports to verify and update any flight performance operating limitations for all configurations and loadings.

(5) Review flight manuals to ensure flight performance limitations for normal, degraded, and emergency operating conditions are identified and documented for all configurations and loadings.

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Analysis.

(3) Conduct performance analysis, where flight testing is not possible, to verify and update the aerodynamic and performance limits throughout the full flight envelope for all configurations and loadings.

Test.

(2) Flight test: Conduct flight testing to verify and update the aerodynamic and performance limits throughout the full flight envelope for all configurations and loadings.

6.4.3 Failures affecting performance.

Standard: The performance information provided to the flight crew characterizes the impacts of propulsion system and other performance-relevant failures IAW MIL-DTL-7700, Section V – Operating Limitations section, Section VI – Flight Characteristics section, Section VII – Adverse Weather Operation section, and Performance Data Appendix/Manual section or NAVAIR ACO NATOPS Style Guide, Chapter 4 – Operating Limitations, Part IV – Flight Characteristics and Control Procedures, Part VI – All-Weather Operation, Part XI – Performance Data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review performance information provided to the flight crew to ensure the impact of propulsion system failure(s) (e.g., engine, transmission, driveshaft, propeller, thrust reversers, rotor) have been characterized and operational procedures and limitations have been documented and are properly modeled in the performance information provided to the flight crew.

(2) Review performance information provided to the flight crew to ensure the impact of performance-relevant failures (e.g., loss of flaps, spoilers, speed brakes, wheel braking) have been characterized and operational procedures and limitations have been documented and are properly modeled in the performance information provided to flight crew.

Analysis.

(1) Conduct empirical and CFD analysis to determine aerodynamic effects of failures that impact air vehicle performance.

(2) Conduct performance analysis to quantify the air vehicle performance impacts of propulsion system and other performance-relevant failures.

Test.

(1) Ground test: Conduct wind tunnel tests to determine aerodynamic effects for failures that impact air vehicle performance. Testing includes failure conditions that would impact the performance of the air vehicle (e.g., engine out, loss of flaps, loss of spoilers).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Examination.

(3) Review flight test reports to assess any updates to the aerodynamic and installed propulsion databases and other performance-relevant failures to quantify impacts to air vehicle performance for the tested failure states.

(4) Review flight manuals to ensure the impact of propulsion system failure(s) (e.g., engine, transmission, driveshaft, propeller, thrust reversers, rotor) have been characterized and operational procedures and limitations have been documented and are properly modeled in the performance information provided to the flight crew.

(5) Review flight manuals to ensure the impact of performance-relevant failures (e.g., loss of flaps, spoilers, speed brakes, wheel braking) have been characterized and operational procedures and limitations have been documented and are properly modeled in the performance information provided to the flight crew.

Analysis.

(3) Conduct performance analysis, where flight testing is not possible, to verify and characterize any propulsion system failure and performance-relevant failure to establish failure operational procedures and limitations.

Test.

(2) Flight test: Conduct flight testing to verify and update the aerodynamic and performance characteristics with propulsion system and other performance-relevant failures.

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7. PROPULSION

The following standards and MoC represent the minimum criteria necessary to establish, verify, and maintain an airworthy design for propulsion systems for manned and unmanned air systems and whether intended for use from the ground or aboard ships. These standards and MoC are divided into six sections: Propulsion System Safety and Data, Propulsion Performance and Operability, Propulsion Structural Integrity, Propulsion Components and Subsystems, Propeller and Propeller Drive Systems, and Other Propulsion Systems.

The standards and MoC for Propulsion System Safety and Data apply to all propulsion systems and subcomponents regardless of the propulsion system type (e.g., turbojet, turbofan, turboshaft, propeller driven systems, reciprocating engines, atypical propulsion systems).

The standards and MoC for Propulsion Performance and Operability, Propulsion Structural Integrity, and Propulsion Components and Subsystems apply to the physical and functional design of gas turbine engine systems and components of any gas turbine engine powered propulsion system (e.g., turbofans, turboprops, turboshafts).

The standards and MoC for Propeller and Propeller Drive Systems apply to the physical and functional design of propeller and propeller drive system components for propeller driven systems (e.g., turboprops).

The standards and MoC for Other Propulsion Systems apply to the physical and functional design of reciprocating engines and atypical propulsion systems (e.g., rotary, electric, pulse detonation engines).

Execution of the AW process requires artifacts be provided for substantiation of compliance. Examination of documents and reports are the basis of the AW assessment process. The analysis, demonstration, and test MoCs do not explicitly refer to the delivery of data and reports; however, to satisfy these MoCs, the accompanying data and report artifacts are to be provided to the assessors as source data for verification.

7.1 Propulsion system safety and data.

These criteria apply to all propulsions systems and subcomponents thereof. This section should be evaluated for any design or modification affecting the propulsion system, manufacturing or repair processes, material substitutions, propulsion system installation physical or functional characteristics (e.g., extractions, inlet or exhaust characteristics, mounts, loads or vibrations), air vehicle usage changes (e.g., maneuvers, icing operations, drag or gross weight), or other changes.

7.1.1 Extrinsic hazard controls.

Standard: Interface conditions, characteristics, and reliability requirements for propulsion system hazard controls (components, systems, and functions) that are not part of the propulsion system type design are defined and documented. Examples of such hazard controls may include bleed isolation valves, insulation, shielding for uncontained failures, ice detectors, fuel system fuses, leak detection systems, fire suppression systems, shaft shear sections, ventilation, mount fuses, instrumentation, and engine control system input or communication signals that are part of the propulsion system installation but not part of the propulsion system type design.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of ICD or other equivalent documentation verifies that interface conditions, characteristics, and reliability of required extrinsic hazard controls (not native to the propulsion system type design but required as part of the propulsion system installation) are defined.

7.1.2 Propulsion technical data.

Standard: Propulsion technical data are current, accurate, and consistent with verified propulsion design analysis and test results. The following items are documented in the identified technical data and technical manuals to enable continued AW:

Published Air Vehicle Flight Manual:

a. Propulsion system operating limit.
b. Normal operating procedures.
c. Emergency procedures.
d. Operating restrictions.
e. Installed performance at critical performance conditions of the air vehicle IAW 7.2.

Published Propulsion System Technical Orders and Technical Manuals:

a. Propulsion system maintenance requirements and procedures.
b. Propulsion system inspection requirements and procedures.
c. Propulsion system serviceability and repair limits.
d. Propulsion system life-limited components and associated replacement requirements.
e. Post-maintenance propulsion system test requirements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manual and propulsion system TOs and technical manuals to verify required content is present and is consistent with engineering source data.

7.1.3 Material and process characterization.

Standard: The performance, functional, and mechanical behavior of all propulsion system materials, including abradables, seals, coatings, and of manufacturing, joining, and repair processes used in the design are fully characterized. Fully characterized means all as-produced material behavior is understood, and properties required to support the full engine life cycle have been defined to a statistical basis and data pedigree, compatible with intended function and operating environment, and component classification or criticality assessment.

Allowable properties used in design reflect the effects of the design environment (e.g., load, temperature, time, chemicals) and the selected manufacturing, repair, quality assurance, and non-destructive evaluation processes including inherent process variability and allowable defects.

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All mechanical properties used in design except fracture toughness and crack growth (threshold and rate) are based on minimum material capabilities as defined by one of the following or equivalent:

a. Minus three sigma (-3σ) - 50% confidence level.
b. Minus two sigma (-2σ) - 95% confidence level.
c. B0.1 - 50% confidence level.
d. A-Basis as specified in MMPDS.

Note: Average (B50) properties are used for fracture toughness, crack growth threshold, and crack growth rate.

Physical properties, such as density, thermal expansion coefficient, elastic modulus, and Poisson's ratio, reflect mean values.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of documentation for materials and allowables historically characterized, including characterization via component testing, verifies full characterization, applicability to the system being certified, and compliance with statistical standards. Existing data obtained through historical tests can be used only when equivalence of the manufacturing processes are substantiated (e.g., same methods of producing billets, forgings, heat-treat processes, machining, surface treatment).

(2) Review of documentation verifies the extent of full characterization activities (e.g., rub rigs, weld trials, fluid exposure, defect studies).

Analysis.

(1) Conduct analysis to define allowable properties under damage states (e.g., fretting) using data collected through a combination of laboratory specimens, sub-element and component testing, and material damage models which are validated against databases and supplemented with historical data covering the range of potential damage states, or databases covering the properties under damage states.

Test.

(1) Lab test: Component material properties, established by test, are based on specimens fabricated from "as produced" parts, from parts produced by equivalent practices, or from parts similar in processing and size, since critical structural properties are dependent upon the manufacturing and post-production processes (e.g., heat treatment, stress relief, surface processes). Property variability in different regions of the same part will be characterized via test.

Damage states in parts, which may occur during field usage, are verified by test for their potential effect (e.g., HCF life).

The number of tests conducted for each curve or condition is sufficient to establish minimum material properties used in design or to establish the correlation between the data obtained from specimens cut from parts and the data base within the calibrated design methodology.

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7.2 Propulsion performance and operability.

These criteria apply to the functional performance, operability, and interface characteristics of gas turbine engines.

Critical performance conditions are phases of flight at which insufficient engine performance may result in hazardous air system effects (e.g., launch, takeoff, OEI, engine windmilling, climb, cruise, endurance, maneuver, hover, in-flight refueling, descent, approach, landing, recovery).

This section should be evaluated for any design or modification affecting engine internal environment conditions, engine control schedules, engine operating characteristics, engine hardware, propulsion system installation functional characteristics (e.g., extractions, inlet or exhaust characteristics), air vehicle usage changes (e.g., maneuvers, icing operations, drag or gross weight), or other changes.

7.2.1 Installed performance.

Standard: Minimum engine thrust or power, and fuel consumption are characterized with representative installation effects over the range of flight conditions and operational environments expected and are shown to meet the critical performance conditions of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., ICDs, environmental criteria documents) to verify that operational environments and installation effects have been defined and utilized in the characterization of performance.

Analysis.

(1) Analyses performed with a validated, configuration-controlled, installed-engine performance model calibrated to measured test data for characterization of performance.

Test.

(1) Lab test: Ground-based sea level and altitude cell tests to characterize and verify performance.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: In-flight tests to verify performance.

7.2.2 Degraded performance.

Standard: Degraded installed engine thrust or power is characterized and meets the critical performance conditions of the air vehicle. Degraded engine performance includes performance:

a. In backup control mode(s).
b. At field removal limits.

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c. After ingestion event (e.g., bird, ice, hail, water, sand, dust, ash).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., ICDs, environmental criteria documents) to verify that degraded performance environments have been defined and utilized in the characterization of degraded performance.

Analysis.

(1) Analyses performed with a model based on measured test data for characterization of degraded performance at conditions that have not been tested.

Test.

(1) Lab test: Ground-based sea level and altitude cell tests to characterize and verify degraded performance.

7.2.3 Stability margin.

Standard: Deterministic stability assessment, considering all destabilizing effects, shows positive engine stall margin at all flight conditions. Stability assessment follows guidelines outlined in, SAE AIR1419, SAE AIR5686, SAE AIR5866,SAE AIR5867, SAE ARP1420, and SAE ARP6420.

To have a closed stability margin assessment, stability margin is evaluated using the same control schedules as used in the evaluation of 7.2.4 and 7.4.3.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis is conducted via a validated stability assessment.

Test.

(1) Lab test: Compressor (e.g., fan, low pressure compressor, high pressure compressor) rig and ground-based sea level and altitude cell engine tests to measure stall lines.

(2) Lab test: Compressor (e.g., fan, low pressure compressor, high pressure compressor) rig and ground-based sea level and altitude cell engine tests to verify sensitivities to inlet flow distortion and other destabilizing influences.

(3) Lab test: Inlet model wind tunnel testing to verify the inlet performance, inlet flow distortion, and stability.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: In-flight testing to verify the inlet performance, distortion, and stability.

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7.2.4 Transient operation.

Standard: Thrust or power response times meet air system mission performance requirements during all maneuvers (e.g., land and ship approaches, AR, and quick stops; for rotary-wing air vehicles, unmask and remask, and nap of the earth ridgeline crossings) without surge or stall.

To have a closed transient operation solution, transient operation is evaluated using the same control schedules as used in the evaluation of 7.2.3 and 7.4.3.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyses performed with a model based on measured test data for characterization of transient operation at conditions that have not been tested.

Test.

(1) Lab test: Ground-based sea level and altitude cell tests to characterize and verify transient operation.

7.2.5 Air start.

Standard: Defined air start envelopes (e.g., spool-down, windmill, cross-bleed, starter assisted) meet air vehicle's intended operational needs. Air start capability is documented in the flight manual.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of documentation (e.g., ICD) verifies defined air start envelopes meet air vehicle's intended operational needs.

(2) Review of flight manual validates air start envelopes.

Test.

(1) Lab test: Ground-based altitude cell tests verify air start capability meets envelopes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: In-flight tests verify capability meets envelopes.

7.2.6 Stall recoverability.

Standard: The engine or air vehicle control system can detect and self-recover from an engine stall without commanded idle or shutdown and without exceeding thermal or structural limits.

Single engine applications possess an automatic relight system for recovery from combustor blowout unless it has been verified that automatic relight offers no improvement in engine recoverability.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Bench tests and ground-based sea level and altitude cell tests verify control system detection.

(2) Lab test: Ground-based sea level and altitude cell tests verify that the engine self-recovers from an instability induced by external influences after the external influence is removed.

7.2.7 Inlet compatibility.

Standard: Inlet functions at all expected induced and natural environmental conditions do not adversely affect the engine beyond what satisfies 7.2 standards, do not result in exceeding defined engine operating limits, and do not result in detrimental permanent deformation or damage when operated under steady state and transient operating conditions, throughout the flight and ground envelope per 7.3.1 for the stated life documented in engine technical data.

Inlet flow distortion characterization follows guidelines outlined in SAE AIR1419, SAE AIR5686, SAE AIR5687, SAE AIR5866, SAE ARP1420, and SAE ARP6420.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyses performed with a model calibrated to measured test data for characterization of inlet function at all environmental conditions not tested.

(2) For icing, analysis is performed with a model calibrated to measured test data at icing conditions not tested.

Test.

(1) Ground or Lab test: Installed engine testing to characterize and verify inlet function for environmental conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Installed engine testing to characterize and verify inlet function for environmental conditions.

(3) Flight, Ground, or Lab test: For icing environments, icing tunnel tests or installed ground icing tests or flight tests verify ice accretion and level of shedding is compatible with the tolerance of the engine(s).

7.2.8 Exhaust gas impingement.

Standard: Under all environmental, ground, and flight conditions (e.g., wind, power settings), exhaust plume temperatures and trajectories are characterized (e.g. temperature and velocity

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profiles), satisfy interaction limits, and do not result in loss or serious damage to the air vehicle, personal injury, or loss of life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of documentation (e.g. ICD, flight manual, and maintenance manual) shows exhaust plume interaction meets interface requirements and ground crew keep-out zones are documented.

Analysis.

(1) Analyses performed with a model calibrated to measured test data for characterization of exhaust plume interaction at all conditions not tested.

Test.

(1) Ground test: Ground testing to verify exhaust plume interaction with structure, fluid and vapor discharge, and personnel.

7.2.9 Thrust reverser or thrust vectoring.

Standard: Thrust reverser or thrust vectoring operation does not adversely affect engine beyond what satisfies 7.2 standards without exceeding defined engine operating limits and without detrimental permanent deformation or damage when operated under steady state and transient operating conditions, throughout the flight and ground envelope per 7.3.1 for the stated life documented in engine technical data.

Forces and moments and DR from the thrust vector are quantified, documented, and compatible with the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of documentation (e.g., ICD) shows operation of the reverser or vectoring is compatible with air vehicle flying qualities interface requirements.

Analysis.

(1) Analysis of reverser or vectoring flow field patterns verifies compatibility relative to impingement, inlet ingestion (e.g., propulsive, ECS, ventilation, auxiliary power system), FOD, and sand and dust generation.

Test.

(1) Ground or Lab test: Ground-based sea level cell tests or ground tests verify reverser or vectoring safety features and compatibility with engines and air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Demonstration.

(1) During flight testing, demonstrate reverser deployment.

Test.

(2) Flight test: Flight testing to characterize reverser or vectoring operation.

7.3 Propulsion structural integrity.

These criteria apply to gas turbine engines: the engine as a whole, and the structural components and assemblies which comprise it. This includes turbomachinery (e.g., ducts, cases, frames, blades, vanes, disks, rotors, spacers, seals, shrouds, housings, augmentors, and exhaust nozzles exclusive of air vehicle structure) and controls, accessories, and externals structural hardware (e.g., actuators, tubing, brackets, gears, shafts, housings, pumps, gearboxes, oil tanks). General standards for gas turbine engine controls, accessories, and externals are covered in 7.4. In addition to propulsion system and component structural integrity, 7.3 addresses structural interface characteristics between the engine and air vehicle, propellers, or drive systems.

This section should be evaluated for any design or modification affecting whole-engine physical characteristics, engine internal environment conditions, engine structural hardware, manufacturing or repair processes, material substitutions, propulsion system installation physical or functional characteristics (e.g., extractions, inlet or exhaust characteristics, mounts, loads or vibrations), air vehicle usage changes (e.g., maneuvers, icing operations, drag or gross weight), or other changes.

7.3.1 Design usage/duty cycle.

Standard: Design usage, duty cycles addressing all sources of repeated loads, and environmental exposure applicable to the type design being certified are defined and validated. All source material (e.g., military specifications, contract requirements, program documents) is identified, and any modification of source material for the type design being certified is substantiated.

Design usage definitions and environments include the following:

a. Mission definition and percent utilization.
b. Engine usage statistics (e.g., EFH, engine mission hours, engine operating time, throttle cycles, time at high power, dwell time, augmentor lights and time).
c. Time at flight envelope extremities (e.g., 10 hours at any point in envelope).
d. Thrust vectoring usage.
e. Engine operating envelope.
f. Engine attitude limits and exposure times.
g. Ambient temperature distributions and percent utilization.
h. Engine extractions and extraction duty cycle (e.g., bleed air, loaded accessory pads, PTO pads).
i. Engine deterioration characteristics and percent utilization (e.g., 1/3 new, 2/3 fully deteriorated).
j. Externally-applied forces (g's, rotational rates and accelerations).
k. Installed temperature, pressure, and vibration environment.
l. Acoustic noise environment (installed and uninstalled).
m. Icing environment.

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n. Sand and dust environment.
o. Corrosion environment.
p. Non-operating and transportation environment.
q. Windmilling envelope.
r. Cross-wind envelope.
s. Electromagnetic environment.

The damage content of simplified or composite duty cycles defined for analysis and test are shown to be equivalent to, or more severe than, that of the full mission usage set.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine design documentation or other source material to verify that all design usage, duty cycles, and environments necessary for the type design being certified are defined and valid. Sources are cited and modification of any source material is substantiated.

Analysis.

(1) Analysis defines usage and duty cycles for design, analysis, and test. Damage equivalency of resultant duty cycles is substantiated.

7.3.2 Loads.

Standard: Load factors are applied to loads that occur within the engine operating envelope IAW 7.3.1 to establish combined limit load conditions and combined ultimate load conditions.

Limit Load Factors:

a. 1.0 LF for externally-applied loads (including ground and flight loads, and external flow field loads).
b. 1.0 LF for thrust (including vectored thrust) and thermal loads.
c. 1.0 LF for internal pressure loads on engine cases and pressure vessels.
d. 1.5 LF for internal pressure loads on controls, accessories, and externals components.

Ultimate Load Factors:

a. 1.5 LF for externally-applied loads (including ground and flight loads, and external flow field loads).
b. 1.0 LF for thrust (including vectored thrust) and thermal loads.
c. 1.5 LF for internal pressure loads on engine cases, and pressure vessels.
d. 2.0 LF for internal pressure loads on welded regions of engine cases, and pressure vessels exposed to compressor discharge pressure. This pressure load is applied singly, not in combination with other loads.
e. 2.0 LF for internal pressure loads on controls, accessories, and externals components.

For all castings an additional 1.33 safety factor is applied to the resulting limit and ultimate load cases unless the cast material has been fully characterized IAW 7.1.3.

Design load conditions include abnormal operation such as surge, stall, afterburner blowout, and emergency or failure conditions such as rotor seizure, crash, and blade out:

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a. Rotor seizure loads are ultimate load conditions derived using seizure loads combined with limit externally-applied loads.
b. Crash loads are ultimate load conditions derived using the following minimum load components in any combination applied at the engine CG:
   1. 9g's forward to 1.5g's aft.
   2. 1.5g's left or right.
   3. 4.5g's down to 2g's up.
c. Blade loss loads are unique ultimate load conditions defined in 7.3.12. Additional loads to be combined for blade out load conditions include normal engine operating loads at takeoff power setting and symmetric takeoff loads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyses are accomplished to define design load conditions using a whole-engine structural model that includes air vehicle mounting provisions and stiffness per the ICD, the engine cycle model, and design criteria.

7.3.3 Non-operating loads.

Standard: The non-operating engine, including all engine mounted equipment and accessories, components, and operating fluids, tolerates a 4g axial, 2g lateral, and 3g vertical inertial limit load acting in any combination at the engine CG without deformation of ground handling mounts or damage to the engine; normal engine function is unaffected after the engine experiences this load case. Ultimate load case is 1.5X limit load case; ground handling mounts do not fail under this load case.

Non-operating engine, engine modules and LRUs/weapons replaceable assemblies, when packaged in their respective transportation support equipment or containers, withstand the externally applied forces and vibrational loads induced during transportation within, and external to, the DoW transportation system IAW 7.3.1 without impact to structural integrity or engine performance. In the absence of other direction MIL-STD-810, Method 514.6 Annex C and Method 516.6, may be used for defining appropriate transportation vibration and shock environments and transportation sequences.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of engine design documentation verifies that non-operating load cases have been defined.

Analysis.

(1) Analyses are accomplished to verify strength and deflection capability, and system-level impacts.

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Test.

(1) Lab test: Component static strength tests are conducted to verify limit load and ultimate load capability.

(2) Lab test: Engine test in a ground-based sea level test cell confirms no impact to structural integrity or engine function following exposure to engine transportation environment while using transportation support equipment defined in technical data.

7.3.4 Strength.

Standard: The engine withstands limit and ultimate load conditions as follows:

Limit Load Conditions:

Engine structure does not exhibit detrimental permanent set, damage, reduction of life limits, or temporarily deflect to the extent that operation or performance is impaired when operated to limit load within the engine operating envelope.

Ultimate Load Conditions:

Engine structure does not fail and enables controlled spool down of rotors under ultimate load conditions. The engine does not experience uncontained failure or uncontained fire under ultimate load conditions. For ultimate load cases involving limit engine-generated loads and ultimate externally applied loads (e.g., maneuvers) the engine also maintains performance enabling successful recovery of the air vehicle. Blade out and crash are unique ultimate load conditions. Standards for blade out are IAW 7.3.12. Standards for crash conditions are:

a. No failure of engine mounts resulting in liberation of the engine from the mount system.
b. No uncontained failures or uncontained fires.

For all strength assessments, positive margins of safety exist for the range of manufacturing tolerances and operational conditions.

For turboprop and turboshaft engines, the maximum allowable steady-state delivered shaft torque (mechanical) limit is at least 20% greater than the rating value reported in technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyses are accomplished using defined load conditions per 7.3.2 to verify strength and deflection capability, and system-level impacts.

(2) For turboprop and turboshaft engines, analysis verifies output shaft overtorque capability.

Test.

(1) Lab test: Component and engine static frame testing is accomplished to verify analytical results. Component tests are conducted at the maximum operating temperature or at a test pressure adjusted to account for the differences between operating and test temperatures. Strain gauges and other instrumentation are used during tests to validate analysis methods. Tests are conducted progressively to limit or ultimate load conditions. Engine static frame tests include means of imparting gyroscopic loads from the rotor system.

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(2) Lab test: Engine test in a ground-based sea level test cell is accomplished to verify analytical results (e.g., surge, stall, afterburner blowout).

(3) Lab test: For turboprop and turboshaft engines, conduct engine overtorque test in a ground-based sea level test cell accomplished per the conditions and success criteria as specified in 14 CFR 33.84.

7.3.5 Gyroscopic moments.

Standard: Exposure to rotational velocities and accelerations (externally-applied forces) for the stated life documented in engine tech data does not adversely affect the engine beyond the capability established in 7.2 standards, nor result in exceeding defined engine operating limits, detrimental permanent deformation, or damage when the engine is operated at maximum allowable steady-state engine speed throughout the engine operating envelope IAW 7.3.1.

Two additional gyroscopic conditions are assessed. Defined accelerations and rates for these conditions are applied at, or about, the air vehicle CG; compatible engine loads are derived based on installed engine position in air vehicle:

a. 1.4 rad/s about any axis in a plane perpendicular to the engine rotor axis combined with any LF conditions occurring within the flight envelope for 10^7 cycles. Engine structure does not exhibit detrimental permanent set, damage, or life reduction, or temporarily deflect to the extent that operation or performance is impaired. If this condition is tailored based on rational assessment of installed air vehicle characteristics and flight data then, at a minimum, rotor structure will still be evaluated for the 1.4 rad/s condition as stated to ensure HCF capability under maneuver conditions.
b. 3.5 rad/s about any axis in a plane perpendicular to the engine rotor axis combined with a ±1g vertical maneuver load for a period of 15 seconds. Engine structure does not fail, there is no uncontained failure or fire, and the engine maintains performance enabling successful recovery of air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyses are accomplished to verify strength, deflection, and fatigue capability, and system-level impacts.

7.3.6 Blade and disk deflection.

Standard: Blades and disks do not contact any static parts of the engine other than seals, shrouds, or any other intentional contact surfaces when operating at all points within the engine operating envelope, including surge and stall occurrences, and bowed rotor starts. Materials intended as rub couples are fully characterized across the range of operational conditions and incursion rates. Seals and clearances remain effective under all internal and external loads, manufacturing tolerances, cold and hot day operation including transient thermal conditions. Blade failure and blade or disk rubbing does not result in uncontained fire.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examination of material rub couples verifies capability compatible with engine operation and incursion rates.

Analysis.

(1) Analysis verifies that positive clearances, both axial and radial, and effective sealing exist under all operational conditions and maneuver load conditions.

(2) Rotor dynamics and thermal containment analyses verify capability under blade out and rub conditions.

Test.

(1) Lab test: Engine tests in a ground-based sea level and altitude test cell verify clearance predictions under steady state and transient operation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight tests verify maneuver effects. Instrumentation and dimensional inspections before and after testing are used to confirm analytical predictions.

7.3.7 Engine mounts.

Standard: The engine mounts retain the engine (including all engine mounted equipment and accessories, components, and operating fluids) in the air vehicle at all flight, takeoff, landing, and ground conditions.

The engine mounts withstand limit load conditions without detrimental permanent set, damage, reduction of life limits, or temporary deflection to the extent that operation or performance is impaired. The engine mounts withstand ultimate load conditions, including crash loads, rotor seizure loads in combination with limit maneuver loads, and blade out loads in combination with limit takeoff loads, without failure. Any air vehicle loads supported through the engine structure and the two additional gyroscopic conditions defined in 7.3.5 are included.

Physical (e.g., dimensions, growth, location, attachment method) and functional (e.g., stiffness, loads) details of the air vehicle-to-engine mount interface are documented in the ICD.

Any unique features required for mount fire-tolerance capability (e.g., shielding, blankets) are described in the ICD.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of the ICD substantiates that the mount interface is fully defined, including configuration details required for mount fire-tolerance capability.

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Analysis.

(1) Analyses are accomplished to define all relevant load cases and to verify strength and deflection capability.

Test.

(1) Lab test: Component or assembly testing of mount capability to limit and ultimate load conditions is accomplished to validate analytical models and predictions. Testing is taken to mount failure to verify margin.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Engine installation and flight testing demonstrates mount interface compatibility and functionality per design intent.

Test.

(2) Flight test: Instrumented flight test verifies installed mount load analysis.

7.3.8 Inlet overpressure.

Standard: Worst-case engine stall loads (e.g., hammer shock) are within air inlet system limits as documented in the engine-to-air-vehicle ICD.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of the ICD verifies that engine air inlet component structural overpressure limits or requirements are defined.

Analysis.

(1) Analysis is conducted to define maximum surge pressure. Loads are verified to be within ICD limits.

Test.

(1) Lab test: Engine test in a ground-based sea level test cell is conducted to calibrate surge overpressure predictions.

7.3.9 Overspeed and overtemperature.

Standard: The engine is capable of continued operation, without exceeding serviceable limits, during and after the following conditions:

a. Normal operation throughout the engine operating envelope including maximum allowable transients up to the overspeed protection limit provided by the engine control
b. Overspeed: Rotor speeds of 105% of maximum allowable steady-state speed for the high-pressure rotors, 107% of maximum allowable steady-state speed for the low-pressure

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rotors, and for turboprop and turboshaft engines, 115% of the output shaft rated speed reported in technical data, at maximum allowable turbine gas temperature for 5 minutes.

c. Overtemperature: Gas temperatures 75 °F above the maximum allowable gas temperature (which includes adders for maximum transient temperature and uncertainties in temperature due to sensors and analytical determinations), at maximum allowable steady-state rotor speed for 5 minutes. Alternate expressions of the overtemperature condition (e.g., 50 °F above maximum allowable high pressure turbine blade metal temperature) may be proposed but will be substantiated and approved.

The engine retains functional and structural capability to enable successful recovery of air vehicle after experiencing the maximum achievable instantaneous engine rotor speeds associated with any single control system failure. The engine or hardware does not need to maintain serviceable condition but enables successful recovery of the air vehicle as described above.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Engine overspeed limits, and control system thresholds and actions, are established and documented in technical data and design analysis documents.

Analysis.

(1) An analysis of the overspeed, and overtemperature capability of the engine is provided. Plastic growth, and its effects, are determined for these conditions.

(2) Control system failures are analyzed to identify the single failure resulting in highest overspeed or greatest impact of rotor growth. Engine performance and operability following most limiting control system failure are analyzed, and resultant air vehicle recovery capability is reported.

Test.

(1) Lab test: Engine overspeed test is conducted in a ground-based sea level test cell to substantiate or correlate analysis. All rotors are subjected to engine operation for a stabilized period of at least five minutes duration at the defined overspeed conditions. Following the test, parts and assemblies are within serviceable limits with no evidence of imminent failure. For engines with more than one rotor system, the test may be performed separately for each rotor system. Rig tests may be performed with justification. If a cold spin pit is used for hot flow components, speed is added to compensate for temperature effects.

(2) Lab test: Engine overtemperature test is conducted in a ground-based sea level test cell to substantiate or correlate analysis. Upon successful completion of the overspeed test, the same engine and hardware are operated at the defined overtemperature conditions for five minutes. Following the test, parts and assemblies are within serviceable limits and with no evidence of imminent failure.

7.3.10 Shaft and disk failure controls.

Standard: The engine incorporates design features or architecture to protect against catastrophic shaft and disk failures (e.g., uncontained failure, uncontained fire).

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a. An engine main shaft bearing or lubrication system failure does not cause parting or decoupling of shaft(s).
b. Hazard controls are incorporated to achieve a probability of disk failure caused by shaft decoupling (loss of load) not greater than 10^{-4} events over the DSL of the engine (e.g., $2.5x10^{-8}$ events/EFH for a 4000 EFH engine DSL). Hazard controls account for all failure causes and failure modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of product definition data and models, and engine configuration description verifies presence of design features or architecture supporting fail-safe design practices

Analysis.

(1) Conduct analysis, including assessment of test and service history, to determine disk failure rates through aggregation of hazard controls.

Demonstration.

(1) Demonstrate effectiveness of any active hazard controls included in design (e.g., overspeed trip or fuel shutoff systems and associated BIT functions, Fault Detection and Accommodation actions).

7.3.11 Disk burst speed.

Standard: The minimum loaded disk burst speed is at least 115% of the maximum steady-state speed (122% for output shaft on turboprop and turboshaft engines) or 5% above the maximum attainable transient overspeed associated with the worst single-point engine system failure whose probability of occurrence is greater than 10^{-8}, whichever condition results in the higher speed. Worst-case thermal conditions are applied. Burst speed is the minimum of the speed that results in actual disk rupture, or that which results in plastic growth and damage that could lead to a hazardous condition (e.g., uncontained fire, uncontained failure, engine mount failure, loss of shutdown control).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis is accomplished to determine the engine system failure case that results in the maximum attainable transient overspeed.

(2) Burst analysis is conducted for each disk at the most limiting condition.

Test.

(1) Lab test: Disk burst lab testing is conducted on the most limiting rotor (disk with the minimum burst capability) of each module. Strain data are collected to calibrate burst analysis. Maximum test speed is sufficient to verify that a minimum tensile strength component (-3σ) can meet the burst margin requirement based on the specific ultimate strength capability of the test

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component. These conditions are maintained for a minimum of 30 seconds. The test is considered successfully completed if there is no evidence of imminent failure.

7.3.12 Blade out.

Standard: Subsequent to blade failure at maximum allowable transient rotor speed, including allocations for transient overshoot, control tolerance, and measurement tolerance, the engine does not experience or cause:

a. Failure of:
   1. Primary engine structure (e.g., cases, frames, ducts, struts, links).
   2. Rotor structure or shafts.
   3. Main shaft bearings, bearing supports, or bearing housings.
   4. Engine mounts.
b. Engine rotor overspeed.
c. Loss of ability to shut down the engine.
d. External leakage of flammable fluid.
e. Uncontained fire.
f. For multi-engine air vehicles, loading exceeding limit conditions in the other engines.
g. Secondary failures that result in catastrophic air vehicle damage or loss of air vehicle control.

The combined load case for blade out includes dynamic imbalance loads from blade loss plus normal engine operating loads at takeoff power setting and symmetric takeoff maneuver loads.

Blade loss loads for conventional blades are based on the imbalance equivalent to fracture in two adjacent blade attachments at the minimum neck section above the outermost retention feature. For fan blades only this total imbalance may be reduced based on substantiated experience on a similar configuration; one full blade imbalance minimum.

Blade loss loads for integrally bladed rotors are based on the imbalance equivalent to liberation of two airfoils at the outer tangency point of the blade root fillet or, for welded or bonded airfoils, at the bond plane. For fan blades only this total imbalance may be reduced based on substantiated experience on a similar configuration; 0.5-blade imbalance minimum.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis is conducted using a whole-engine rotor dynamics model to determine blade out loads throughout the engine and at interfaces (e.g. engine mounts), and to determine the limiting blade out condition for use in an engine-level blade out test. In determining the limiting blade out condition, blade loss in the fan, compressor, and turbine sections of the engine is considered.

(2) Analysis is accomplished using the resultant blade out loads to verify blade out capability and system-level impacts.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(1) Lab test: Engine test in a ground-based sea level test cell is accomplished at the limiting blade out condition to verify engine blade out capability and calibrate analytical predictions.

7.3.13 Containment.

Standard: The engine completely contains a fan, compressor or turbine blade failure. No uncontained fires, or leakage of flammable fluid result and the engine contains all parts damaged and released by the failure of a single blade. Uncontained failures will not cause fire or catastrophic damage to engine external systems or air vehicle systems, or injury to personnel.

"Single blade" is defined consistent with 7.3.12 criteria:

a. For conventional blades, fracture in the blade attachments at the minimum neck section above the outermost retention feature.
b. For integrally bladed rotors, fracture at the outer tangency point of the blade root fillet or, for welded or bonded airfoils, at the bond plane.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine drawings or system configuration descriptions to verify that external engine and air vehicle systems (e.g., fuel, hydraulic lines) are routed to reduce exposure and limit secondary damage.

Analysis.

(1) Blade containment analysis which relates the released blade kinetic energy to the energy required for containment is accomplished for each rotor stage.

(2) For uncontained failures, trade studies and failure risk assessments (e.g., FMECA, SSHA) are accomplished jointly with the airframer to substantiate containment strategy decisions.

Test.

(1) Lab test: The analysis is substantiated or correlated with rig or full engine containment tests of the most limiting stage in a ground-based sea level test cell.

7.3.14 High cycle fatigue.

Standard: Engine components are free of damaging vibration, damaging resonance conditions, or other damaging DR, regardless of propulsion system excitation source, when operated, at all engine-rotational speeds and thrust or power settings under installed or uninstalled steady state and transient conditions, at maximum allowable field imbalance of the engine rotor systems, IAW 7.3.1 for the stated life documented in engine technical data.

a. Maximum vibratory stress is below minimum Goodman allowable limits for the following cyclic endurance capability:
    1. Titanium 10^9 cycles.
    2. Other Metallic Materials 10^7 cycles.
    3. Composite or Other Materials 3 X 10^7 cycles.

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Maximum vibratory stress for flow-path components and components subject to high variability or uncertainty in forcing, dynamic characteristics, coupling, or damping does not exceed 60% of the Goodman limit (40% margin). Such components comprise rotating components, rotating assemblies, rotating seals, shafts, airfoils (e.g., blades, vanes, stators, nozzles), struts, fuel nozzles, and control system probes that protrude into the turbomachinery flow path (e.g., T2 sensor). Reduced vibratory stress margin for other components will be justified.

b. Self-excited vibrations such as flutter, separated flow vibration, or other non-integral vibrations do not occur within the operating range of the engine. Any required operation or installation limits are defined.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Operating limits to prevent damaging response or component failure are defined in engine technical data.

(2) Installation requirements to prevent damaging response or component failure are defined in the ICD.

Analysis.

(1) Analysis is accomplished to determine vibratory characteristics of engine components and assemblies (e.g., frequencies, mode shapes, damping, coupling, mistuning, coincidence). Analysis is calibrated to component, rig, and engine test results. Models are used to evaluate sensitivity to influence parameters (boundary conditions, geometry, manufacturing, operating conditions). Campbell diagrams are prepared to verify margin to potential resonances (e.g. low-order crossings, upstream and downstream counts) particularly at sustained power operating ranges. Models are used to guide instrumentation layout for testing. Model quality is assessed using HCF Protocol described in JSSG-2007, Appendix A, Verification section, Component Vibrations paragraph.

(2) Goodman diagrams, with maximum and minimum stress truncated to yield strength, are prepared using calibrated models and test results. Models and test results are used to project maximum responses throughout the operating range and envelope.

(3) Analysis of rig and engine test results are conducted to determine operating limits for self-excited vibrations and to verify that responses do not occur in the engine operating range.

Test.

(1) Lab test: Bench and rig tests are accomplished to determine component and assembly vibratory characteristics (e.g. frequencies, mode shapes, damping, coupling, mistuning, coincidence), to determine peak vibratory response, and to inform instrumentation application for subsequent engine tests. Multiple components and assemblies are tested to determine variability across a range of influence parameters.

(2) Lab test: Instrumented core and engine tests are conducted in sea-level and altitude test cells using production configuration hardware and control schedules. Aeromechanical stress surveys are accomplished to verify vibratory stresses and stability margins, including flutter. Test

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conditions for aeromechanical tests include exposure to a range of influence parameters (e.g., rotor imbalance, off-nominal vane schedules, air vehicle inlet conditions, engine inlet total pressure and total temperature, expected inlet distortion and separation, accessory loads, stalls, bleed, multiple engines) per the HCF Protocol described in JSSG-2007, Appendix A, Verification section, Component Vibrations paragraph to verify robustness to expected variation, and also aerodynamic and aeromechanical factors that might induce or influence flutter.

(3) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell and includes HCF stairsteps in 50-60 RPM increments and dwell times IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph.

(4) Ground test: Ground test with an instrumented engine is conducted using production configuration hardware and control schedules, and production inlet configuration to verify installed vibratory and flutter response. Static crosswind conditions are included in the evaluation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Flight test with an instrumented engine is conducted using production configuration hardware and control schedules, and production inlet configuration to verify installed vibratory and flutter response.

7.3.15 Engine system vibration.

Standard: The engine is free of damaging vibration, damaging resonance conditions, or other damaging DR, regardless of propulsion system excitation source, when operated, at all engine-rotational speeds and thrust or power settings under installed or uninstalled steady state and transient conditions, at maximum allowable field imbalance of the engine rotor systems, IAW 7.3.1 for the stated life documented in engine technical data.

a. Maximum engine mechanical vibration limits, for both installed and uninstalled operation, are defined for each measurement location and direction as functions of frequency, and engine order.
b. Damaging critical speeds for engine rotor systems or rotating assemblies do not occur in the rotor speed range from 20% below idle speed to 20% above maximum operating speed. Damping and balancing allow any critical speed that exists from rotor rest to 20% above maximum operating speed to be traversed with smooth engine operation and without engine damage.
c. The natural frequencies of the mounting system with the engine installed are at least 20% below idle shaft rotational speed(s) in all damaging modes of vibration which can be excited by rotor imbalance.
d. Windmilling operation does not result in damaging vibration to the engine or air vehicle. Windmilling following a blade out event does not result in any condition not allowed under the 7.3.12 standard.
e. All rotor areas that could puddle oil incorporate drains to limit potential imbalance.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Rotating component and rotating assembly, balance limits and procedures are defined for production and sustainment in engine technical data.

(2) Engine vibration limits are defined in engine technical data.

(3) Review of product definition data and models verifies that rotor oil drain features are present.

Analysis.

(1) Analysis is accomplished to determine vibratory characteristics of the complete engine system installed in the air vehicle. A dynamic model of the total engine system including all relevant mass, stiffness, damping, and boundary conditions (e.g., rotor(s), bearings, supports, dampers, frames, cases, mounts, air vehicle interface) is developed to analyze critical speeds and associated DRs. Analysis and models are calibrated to engine test results. Calibrated models are used to evaluate sensitivity to influence parameters (e.g., boundary conditions, imbalance distribution and phase, thermal bow, misalignment, operating conditions).

(2) Analysis is accomplished to determine the range of windmilling speeds, considering installation effects, for each rotor throughout the engine operating envelope.

Test.

(1) Lab test: Resonance search testing (ping testing) is accomplished on fully assembled and instrumented ground test engine.

(2) Lab test: Instrumented engine testing is accomplished in sea level and altitude test cells to verify critical speed predictions, imbalance sensitivity, case deflections, and vibration amplitude. The full engine speed range up to 105% scheduled physical speed is tested. Start tests examine sub-idle operation and effects of bowed rotor. Effects of rotor imbalance up to the maximum allowable field limit, using worst-case phase and mass placement as determined by analysis of imbalance sensitivity, are verified by test.

(3) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown inspection of hardware is accomplished to document condition. Minimum endurance test duration is twice the planned flight test duration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown inspection of hardware is accomplished to document condition. Minimum endurance test duration is the scheduled depot inspection interval or half the cold part DSL.

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(5) Flight test: Vibration testing with an instrumented engine is conducted to verify installed vibratory environment and response.

7.3.16 Air vehicle induced vibration.

Standard: Air vehicle induced engine vibration (e.g., armament firing, inlet or exhaust vibration, AMAD or accessory excitation, APU operation, landing gear transmitted vibration, drive system excitation, propeller or rotary-wing operation/blade passage), including effects of engine mount damping and isolation systems if present, does not exceed engine vibration limits when the engine is operated, at any thrust or power setting (including engine off) under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Engine mounting provisions, engine-to-air-vehicle physical interfaces, and engine vibration limits are verified by examination of the ICD.

Analysis.

(1) Design analysis is accomplished to verify vibration transmissibility and engine vibratory response across the engine mount interfaces, and other interfaces capable of transmitting vibration, for any air vehicle-generated forced vibration and impulses.

Test.

(1) Ground test: Ground test confirms that engine vibration limits are not exceeded.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight test confirms that engine vibration limits are not exceeded.

7.3.17 Pressure balance.

Standard: At all allowable attitudes, and under steady state and transient engine operation throughout the engine operating envelope, the engine pressure balance system prevents:

a. Bearing damage or reduction in stated bearing life from null loading, bearing crossover, skid, or other operating characteristics.
b. Damaging vibration and DR for both installed and uninstalled engine operation.
c. Vibration or DR that inhibits the flight crew from effectively accomplishing required functions during installed operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Analysis results indicate that loads imposed on the engine bearings assure bearing operation without damage. Bearing loads are checked across the full range of engine operating modes, operating conditions, power settings, and flight conditions.

Test.

(1) Lab test: Bearing rig tests confirm individual bearing load capability and durability.

(2) Lab test: Full-up instrumented (bearing loads and engine vibration) engine testing ensures engine bearing radial and thrust loading is within design limitations and ensures operation of the bearing and rotor support system. Engine test is conducted in an altitude test cell to simulate altitude and ram conditions representative of operational use.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Vibration testing confirms that vibration characteristics are non-damaging and do not inhibit the flight crew from effectively accomplishing required functions, and that vibration magnitude does not exceed defined engine or air vehicle limits.

7.3.18 Low cycle fatigue.

Standard: Engine hardware has positive LCF margin to the stated life documented in engine tech data when operated, under steady state and transient conditions, IAW 7.3.1.

Part-specific critical material, processing, and manufacturing characteristics, and associated controls, are established for LLPs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of LLP product definition data (e.g., models, drawings) verifies that part-specific critical material, processing, and manufacturing characteristics, and associated controls, are defined.

Analysis.

(1) Analysis is conducted to determine minimum LCF life, and to define mature and repeatable material, processing, and manufacturing control parameters required to achieve that life for each LLP.

Test.

(1) Lab test: Component tests (e.g., spin pit, pressure cycling, cyclic fatigue) of LLPs are conducted. Test conditions include thermal stress, or a simulation thereof, boundary conditions, and interface constraints. Substantiation of the number of test articles and cycles per article required to validate design analysis or verify minimum life is provided.

(2) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph.

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Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown inspection of hardware is accomplished to document condition. Minimum endurance test duration is twice the planned flight test duration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown inspection of hardware is accomplished to document condition. Minimum endurance test duration is the scheduled depot inspection interval or half the cold part DSL.

7.3.19 Acoustic noise.

Standard: Exposure to the acoustic noise environment for the stated life documented in engine tech data does not adversely affect the engine beyond the capability established in 7.2 standards, nor result in exceeding defined engine operating limits, detrimental permanent deformation, or damage when the engine is operated, at all thrust or power settings under installed or uninstalled steady state and transient conditions, IAW 7.3.1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the definition of the installed and uninstalled acoustic noise environment used for final design analysis. The basis for the design curve is stated (e.g., test on actual system, correlation from another system).

Analysis.

(1) Structural analysis determines hardware tolerance to defined acoustic noise environment. Non-operating conditions are considered if applicable (e.g., supplementary propulsion systems, multi-mode propulsion systems).

Test.

(1) Lab test: Acoustic noise environment is characterized via measurement during engine test in a ground-based sea level test cell. Full power range is characterized (typical peak responses are at highest throttle setting). Instrumentation includes microphone arrays.

(2) Lab test: Component durability is verified by measured dynamic stress or acceleration and via post-test inspection of engine-level durability test (AMT) hardware.

(3) Ground test: Ground test is conducted to verify engine bay and external noise environment.

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7.3.20 Creep.

Standard: Creep capability of static and rotating parts is such that the engine capability established in 7.2 standards is not adversely affected, engine operating limits are not exceeded, and detrimental permanent deformation or damage does not occur when the engine is operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analytical prediction of creep and component growth and percent stress rupture life, as a function of design life, is accomplished on each creep-critical component. Flight conditions at extremities of engine operating envelope and deteriorated conditions are included in creep and stress rupture design analysis. Design operating stresses are established based on experience (contractor design system) that indicates a high probability that creep and stress rupture life can be achieved (e.g., 0.2% plastic creep life, specified diametral rim growth, 50% stress rupture life).

Test.

(1) Lab test: Component tests validate analytical predictions.

(2) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Testing validates analytical predictions. Inspection requirements for test engines include measurement of critical dimensions (e.g., snap and rim diameters, bolt circles, bores) prior to test and at each scheduled inspection interval. Verification of dimensional growth as a function of test time is conducted and included as a part of the interpretation and verification of test results. Minimum endurance test duration is twice the planned flight test duration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Testing validates analytical predictions. Inspection requirements for test engines include measurement of critical dimensions (e.g., snap and rim diameters, bolt circles, bores) prior to test and at each scheduled inspection interval. Verification of dimensional growth as a function of test time is conducted and included as a part of the interpretation and verification of test results. Minimum endurance test duration is the scheduled depot inspection interval or half the cold part DSL.

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7.3.21 Corrosion susceptibility.

Standard: Corrosion capability of static and rotating parts is such that the engine capability established in 7.2 standards is not adversely affected, engine operating limits are not exceeded, and detrimental permanent deformation or damage does not occur when the engine is operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data. Corrosion capability includes all forms of environmental attack (e.g., stress corrosion cracking, hot corrosion, sulfidation, intergranular corrosion).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine CPCP that describes activities throughout the life cycle to establish and maintain corrosion resistance of engine hardware to ensure content is consistent with MIL-STD-1568, General Requirements and CPCP paragraphs.

Analysis.

(1) Review analysis of the materials, coatings, and material interfaces (material pairs) used in the engine to ensure they are not corrosion prone or corrosion sensitive and that corrosion protective measures are effective.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: Substantiate corrosion resistance and effects through engine testing in a sea-level test cell with a corrosive environment as defined in JSSG-2007, Appendix A, Verification section, Corrosive Atmosphere paragraph. Turboshaft engine corrosion susceptibility test is preceded by two hours of sand ingestion if design usage includes significant time in sand and dust environments.

7.3.22 Damage tolerance for fracture critical components.

Standard: When operated, under steady state and transient conditions, IAW 7.3.1, fracture-critical engine parts resist failure caused by the presence of material, manufacturing, processing, maintenance, and service-induced defects, retaining residual strength to carry maximum limit stress, for twice the inspection interval defined in engine technical data without repair. Residual strength load conditions account for expected control system overspeed and engine deterioration. Fracture-critical parts are safety-critical or mission-critical components that achieve damage tolerance IAW MIL-STD-3024, Controlled Crack Growth (Fracture Analysis) paragraph.

Initial surface flaw sizes are those detectable at the minimum demonstrated reliability (POD/CL) of the NDI methods used during manufacture and in-service maintenance. Initial embedded defect sizes and occurrence rates are characterized by exceedance curves (e.g., intrinsic material defect distributions) and controlled by material and processing controls and complementary NDI utilized during material processing and forming. Minimum flaw size detection POD/CL for all NDI methods is:

a. Manual inspections: 90%/95%.

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b. Fully automated inspections 90%/50%.

Material and processing controls, inclusive of the following, are established:

a. Material and manufacturing operation specifications throughout the procurement and production chains.
b. Allowable limits and controls for material property variation.
c. Critical characteristics and defect sizes, locations, and orientations on product definition data.
d. In-process and finished-part (production and in-service) NDI requirements and acceptance standards.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) For each fracture-critical part, review design documentation, engine life-management documentation, sustainment technical data, and product definition data that verifies compliance with material and processing control elements of this standard.

Examples of such documentation include Durability and Damage Tolerance Control Plan, Engine Life Management Plan, PSIP Master Plan and Report, Technical Orders, NDI requirement definition, drawings and models, and material and processing specifications.

Analysis.

(1) Controlled crack growth (fracture mechanics) analysis is conducted for each component classified as fracture critical. Analyses account for repeated and sustained stresses, vibratory stress (including threshold value), and include the effects of load interactions. Stress intensity is based on the structural geometry and flaw geometry. The critical stress intensity is based on the required residual strength load level or vibratory stress threshold value, and temperature conditions that exist at the component location being analyzed. Analysis initial flaw sizes are defined by the demonstrated production and sustainment NDI capability, or by exceedance curves for embedded defects, established IAW this standard. Limit loads used for residual strength analysis are defined for each fracture-critical part. Embedded defect analysis can be based on probabilistic methods that account for the distribution of input variables.

Demonstration.

(1) Production NDI demonstrations are performed IAW MIL-HDBK-1823 to ensure that flaws larger than the analysis initial flaw size will not occur in finished components to the reliability level required by this standard.

Test.

(1) Lab test: Fracture-critical components are selected for verification by test (e.g., cyclic spin test, pressure cycling test). These components are either pre-flawed or contain natural flaws and are cycled to verify flaw growth characteristics.

(2) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests

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are conducted using engine control software or schedules of the type design. Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics and ability to inspect IAW technical data. Minimum endurance test duration is twice the planned flight test duration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) In-service NDI demonstrations are performed IAW MIL-HDBK-1823 to ensure that flaws larger than the corresponding analysis flaw size will not occur in released components to the reliability level required by this standard. In-service inspection flaw size detection capability is demonstrated on parts with prior engine operation time.

Test.

(3) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics and ability to inspect IAW technical data. Minimum endurance test duration is the scheduled depot inspection interval or half the cold part DSL.

7.3.23 Damage tolerance for non-fracture critical components.

Standard: When operated, under steady state and transient conditions, IAW 7.3.1, non-fracture-critical engine parts resist failure caused by the presence of material, manufacturing, processing, maintenance, and service-induced defects per a damage tolerance approach covered by the MIL-STD-3024, Fracture Screening through Additively Manufactured Components paragraphs. Non-fracture-critical parts are safety-critical or mission-critical components for which LCF-driven crack growth is not a primary failure mechanism, parts that are not highly loaded, or non-structural parts.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis is conducted per the damage tolerance approach covered by the MIL-STD-3024, Fracture Screening through Additively Manufactured Components paragraphs to verify damage tolerance capability.

Test.

(1) Lab test: Components are selected for verification by test (e.g., impact testing, damping testing, leak-before-burst testing, proof testing, defect characterization testing) per the damage tolerance approach covered by the MIL-STD-3024, Fracture Screening through Additively Manufactured Components paragraphs.

(2) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph unless exempted by the damage tolerance approach covered by the MIL-STD-3024, Fracture

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Screening through Additively Manufactured Components paragraphs. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics. Minimum endurance test duration is twice the planned flight test duration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph unless exempted by the damage tolerance approach covered by the MIL-STD-3024, Fracture Screening through Additively Manufactured Components paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics. Minimum endurance test duration is the scheduled depot inspection interval or half the cold part DSL.

7.3.24 Life management.

Standard: Life management actions required to maintain engine structural integrity and functionality (e.g., component inspection, repair, and replacement requirements) are defined and are based on 7.3.1.

Individual component maintenance intervals are based on tracked usage parameters that contribute to life degradation (e.g., TACs, hours, augmentor lights), or on validated, configuration-controlled usage-based-lifing algorithms (e.g., advanced lifing algorithms).

Components requiring usage tracking including safety-critical, fracture critical, CSI, and life limited components are identified. A usage tracking system, comprising the following functions, is implemented based on the unique tracking requirements of each tracked component:

a. Analysis procedures.
b. Serialization.
c. Usage or damage data collection.
d. Flight and ground software.

The tracking system provides visibility of component-specific usage data required to maintain tracked components IAW propulsion technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of the Engine Life Management Plan, service management plan, and applicable maintenance manuals (e.g., TOs, Naval Aviation Technical Information Products) verifies that tracked components, required life management actions, and that the component life tracking program description and details are defined and documented.

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Analysis.

(1) Analysis is accomplished to determine the appropriate life tracking parameter and to validate algorithms for usage-based-lifing-tracked components. Usage-based lifing algorithms remain valid when exposed to expected or typical in-service usage variation and reproduce damage consistent with design system predictions for defined usage inputs.

Test.

(1) Lab test: Component testing is accomplished to validate usage-based lifing algorithm predictions.

(2) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown and NDI of hardware verifies suitability of usage-based lifing algorithms. Minimum endurance test duration is twice the planned flight test duration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Full engine durability test or AMT/ASMET is conducted in a ground-based sea level test cell IAW JSSG-2007, Appendix A, Verification section, Endurance Tests paragraph. Tests include product-representative inlet conditions and extractions, and rotor imbalance. Tests are conducted using engine control software or schedules of the type design. Post-test teardown and NDI of hardware verifies suitability of usage-based lifing algorithms. Minimum endurance test duration is the scheduled depot inspection interval or half the cold part DSL.

7.3.25 Foreign object and domestic object damage.

Standard: The engine is capable of operating for one inspection interval without repair, after ingestion of foreign or domestic objects which produce damage equivalent to a Kt of three.

For fan and compressor airfoil leading and trailing edges, and other components that protrude into the flowpath (e.g., sensors, probes, rakes), stresses remain under the 100% Goodman allowable for a Kt=3 equivalent notch or a validated probabilistic analysis verifies a failure rate below 10^{-7}.

Unrepaired FOD/DOD in the root fillet area (e.g., airfoil/rotor interaction zone) of integrally bladed rotors results in airfoil fracture rather than rotor fracture.

The engine is capable of controlled shutdown after ingestion of FOD/DOD.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Serviceable and repairable limits for FOD/DOD and inspection intervals are defined in propulsion technical data (e.g., maintenance manuals, service and repair manuals, overhaul manuals).

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Analysis.

(1) High cycle fatigue analysis of airfoil aeromechanical test results, and flowpath component vibratory test results verifies FOD/DOD capability.

(2) Analysis verifies crack trajectory of FOD/DOD in the root fillet area (e.g., airfoil/rotor interaction zone) to show compliance.

(3) Analysis verifies that the engine can be shut down after ingestion of FOD/DOD.

Test.

(1) Lab test: Conduct component and rig test with damaged airfoils to determine fatigue sensitivity (e.g., Kf, q) for damage equivalent to Kt=3. Otherwise, notched fatigue properties from standard material specimens are used.

7.3.26 Bird ingestion.

Standard: Critical ingestion parameters affecting engine power loss and damage are defined for the independent Medium bird ingestion and Large bird ingestion conditions described below. Critical ingestion parameters include the following effects and are uniquely defined for static and rotating hardware capable of being struck by birds or bird debris:

a. Bird ingestion speed.
b. Critical target location(s), including entry into core.
c. Rotor speed or engine power setting.

Note: Critical bird ingestion speeds reflect the most critical conditions within the range of airspeeds used for flight operations, including low-level, high-speed operation and powered lift operation if applicable, up to 6000 ft AGL.

Medium bird ingestion

Following ingestion of one bird weighing 2.5 lbs at the determined critical ingestion parameters, the engine:

a. Does not experience non-recoverable flameout or stall (IAW 7.2.6).
b. Retains functional and structural capability to enable successful recovery of the air vehicle.

Large bird ingestion

Ingestion of one bird weighing 4.4 lbs at the determined critical ingestion parameters does not cause or result in:

a. Catastrophic failure of:
    1. Primary engine structure (e.g., cases, frames, ducts, struts, links).
    2. Rotor structure and shafts.
    3. Engine mount system leading to inadvertent engine separation.
b. Uncontained failure or fire.
c. External leakage of flammable fluid.
d. Engine rotor overspeed.
e. Loss of ability to shut down engine.
f. Catastrophic damage to the air vehicle or adjacent engines.

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For inlet throat areas greater than 2092 in^2 follow 14 CFR 33.76, Large Bird Weight Requirements table for large bird ingestion weight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis is conducted to determine the critical ingestion conditions affecting engine power loss and damage.

(2) Impact analysis is conducted for rotating and static hardware to assess strain, strain rate, deflection, deformation, and potential clashing or interaction with other components.

(3) Dynamic analysis is conducted to assess effect of impact and component damage on imbalance, vibration, torque, deflection, and system-level effects.

Test.

(1) Lab test: Full engine test in a ground-based sea level test cell is necessary to ensure all interactive and system-level effects of a bird strike are properly accounted for (e.g., imbalance, high vibration, rotor-to-stator contact, main shaft torque, progressive damage, performance or operability effects (see 7.2.2)).

7.3.27 Ice and hail ingestion.

Standard: The engine can independently ingest sheet ice and hailstones as described below without:

a. Non-recoverable flameout or stall (per section 7.2.6).
b. Violating engine operating limits.
c. Impairment of any functional capability.
d. Damage that requires corrective action prior to next maintenance opportunity.

Sheet ice:

a. 7 to 9 pieces of ice of varying size and shape. 5 lbs total weight of all ice pieces with no single piece weighing more than 0.75 lb. Ice size, shape, and thickness is informed by inlet ice accretion characteristics.
b. Ingestion velocity simulates ice from inlet lip being sucked into the engine or 50 ft/s, whichever is more damaging.
c. Ingestion occurs at the most damaging thrust or power setting at takeoff, cruise, and descent operating conditions.

Hailstones:

a. 30, 1-inch diameter hailstones, maximum of 10 at a time.
b. Ingestion velocity is 750 ft/s, maximum cruise true airspeed up to 15000 ft, or maximum true airspeed for any defined low-level, high-speed mission, whichever is most damaging.
c. Ingestion occurs at the most damaging power setting at takeoff, cruise, and descent operating condition, or at the defined power setting for low-level, high-speed mission.

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For inlet capture areas greater than 2100 in^2 follow 14 CFR 33.77, Minimum Ice Slab Dimensions table for quantity, size, and associated weight of sheet ice, and 14 CFR 33.78, All Engines section for number and size of hailstones.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis is conducted to determine the critical ingestion conditions affecting engine power loss, operability, and damage.

(2) Impact analysis is conducted for rotating and static hardware to assess strain, strain rate, deflection, deformation, and potential clashing or interaction with other components.

Test.

(1) Lab test: Full engine test in a ground-based sea level test cell is necessary. Test includes critical ingestion conditions determined by analysis (or component and rig test if relevant). For hail ingestion engine test, one-third to one-half of the hailstones should be aimed at the critical inlet face area with the remainder randomly distributed over the inlet face area. The hailstones should be ingested in a rapid sequence to simulate hailstorm encounter.

7.3.28 Sand and dust ingestion.

Standard: Exposure to the sand and dust environment for the stated life documented in engine tech data does not adversely affect the engine beyond the capability established in 7.2 standards, nor result in exceeding defined engine operating limits, impairing functional capability (e.g., cooling and thermal management), detrimental permanent deformation, or damage when the engine is operated, under steady state and transient conditions, IAW 7.3.1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine drawings or system configuration descriptions to verify incorporation of specific design features, components, or arrangements that enable engine sand and dust tolerance and capability (e.g., sump pressurization and ventilation inlets, buffer air particle separators, bleed paths).

Analysis.

(1) Conduct analysis to ensure proper sizing and shaping of cooling passages to prevent blockage or clogging from sand and dust.

(2) Conduct analysis to determine components susceptible to damage or deterioration from sand and dust and verify that material and coating selections are tolerant to the defined sand and dust environment.

(3) Conduct specific analyses defined in JSSG 2007, Appendix A, Verification section, Compressor Airfoil Erosion Coating paragraph if such coatings are part of the type design.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: Full engine test in a sea-level test cell is necessary to determine the effect of sand ingestion on the performance, operability, lube system contamination, bleed air quality, internal air-cooling system (e.g., secondary flow circuits), compressor erosion coatings, integrated heat exchanger effectiveness, and durability. Specific test conditions (e.g., sand types, sand concentration, exposure duration, engine power settings) are selected to simulate mission exposure conditions including powered lift, hover, brown out using JSSG 2007, Appendix A, Verification section, Sand and Dust Ingestion paragraph as guidance. Include inlet particle separator during sand ingestion test if part of type design.

(2) Lab test: Conduct specific rig and engine tests defined in JSSG 2007, Appendix A, Verification section, Compressor Airfoil Erosion Coating paragraph if such coatings are part of the type design.

7.4 Propulsion components and subsystems.

These criteria apply to engine controls, accessories, and externals. Control system includes electronic controls, hydromechanical controls, fuel system components, variable geometry, and sensors. Accessories include the gearbox, lubrication, electrical, ignition, anti-ice and de-ice, and engine monitoring systems. Externals include tubing, plumbing, cables, and brackets.

This section is evaluated for design or modifications impacting the engine controls, accessories and externals, operating characteristics, engine to air vehicle interfaces or functional characteristics (e.g., drive system extractions, fluid or electrical interfaces, clearances, loads, vibratory or acoustic excitations), and air vehicle usage changes (e.g., maneuvers, icing operations, drag, or gross weight).

This section does not evaluate the engine EMI, engine electronic hardware and software; these are evaluated in Section 13 and Section 15.

7.4.1 Engine control system.

Standard: Engine control system maintains engine operation IAW capability established in 7.2 standards without exceeding defined engine operating limits and without detrimental permanent deformation or damage when operated, under steady state and transient conditions. The architecture accommodates all control mode operations, including failure conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., engine control system design documentation, hardware drawings, system configuration descriptions, installed hardware, materials and manufacturing records, design data records, inspection reports, or other substantiating records and reports) to verify the architecture accommodates both steady state and transient engine operation, including failure conditions, across the entire flight and ground envelope.

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(2) Review FMECA of the control system to verify the impact of engine control, sensor and effector failures, or malfunctions on engine operation.

(3) Review System Safety Assessment of the control system to identify faults or failures, and their predicted failure rates, that result in a change in thrust or power, transmission of erroneous data, or an effect on engine operability producing a surge or stall.

Analysis.

(1) Analysis based on other military systems or previous civil AW certification support documentation are conducted to verify design of the engine control system.

Test.

(1) Lab test: Closed loop bench testing, using production qualified components, ensures that the system can properly interact with all other systems and components on the engine.

(2) Lab test: Closed loop fault injection bench testing ensures the control system can correctly identify and accommodate known critical failures.

(3) Lab test: Engine ground-based sea level and altitude cell testing verifies the control system's ability to maintain speed, temperature, pressure, and fuel flow throughout the engine operating envelope.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Flight testing provides required levels of engine performance and operability IAW 7.2.

7.4.2 Isolation of subsystems.

Standard: Engine controls and subsystems are physically and functionally isolated to avoid cascading failures due to any single or common cause.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, installed hardware, materials and manufacturing records, design data records, inspection reports, photographs, FMECA, or other substantiating records and reports) to verify both the physical and functional isolation.

7.4.3 Control system stability.

Standard: Control system maintains both stable engine operation and response during all steady state and transient operations. Engine control loops maintain a minimum of 6 dB gain margin and 45 degrees of phase margin. The engine provides stable thrust and power levels in response to flight crew commands using the entire range of power setting inputs and transients.

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For turboshaft applications, the engine control system stability has gain and phase margin to prevent dynamic coupling with the rotor system or propeller during steady state and transient operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., ICD, design data records, and other substantiating records and reports) and verify that the design meets phase and gain stability margins.

Analysis.

(1) Design analysis verifies phase and gain stability margins.

Test.

(1) Lab test: Closed loop bench tests verify phase and gain stability margins.

(2) Lab test: Ground-based sea level cell tests verify the engine phase and gain stability margins.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing verifies the engine's ability to respond to flight crew commands.

7.4.4 Fault accommodation.

Standard: The engine control system fault accommodation detects control system failure(s) that will result in fail-operational, fail-degraded, or fail-safe operation of the engine. After any single failure or combination of failures of the engine control system which have a probability of occurrence greater than 10^{-7} per flight hour, the engine maintains a level of performance not worse than 7.2.2. For multi-engine (three or more engines) an engine shutdown is permissible if the shutdown does not result in loss of air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., design data records, hardware drawings, system configuration descriptions, FMECA, or other substantiating records and reports) and verify any failure of the engine controls and supporting subsystems results in a fail-operational, fail-degraded, or fail-safe condition.

Test.

(1) Lab test: Closed loop and fault injection bench testing ensures the engine control system identifies and accommodates faults and failures, and the engine maintains a level of performance not worse than 7.2.2.

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(2) Lab test: Engine ground-based sea level and altitude cell tests ensure the control system identifies and accommodates faults and failures, and the engine maintains a level of performance not worse than 7.2.2.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing for degraded engine control modes (e.g., reversionary, backup, engine out) verifies the engine maintains a level of performance not worse than 7.2.2 and that engine control failures do not propagate.

7.4.5 Fuel system.

Standard: All engine fuel system components (e.g., pumps, regulators, carburetors, flow metering valves, check valves, nozzles, spray bars, tubing) are sized to simultaneously provide the necessary fuel flows, pressures, and temperatures to the main combustor, augmentor, heat exchangers, cooling systems, and all variable geometry fueldraulic subsystems. All engine fuel system components are IAW capability established in 7.2 standards without exceeding defined engine operating limits when operated, under steady state and transient conditions, throughout the engine operating envelope.

In-line filtration systems include cleaning, replacement, and a bypass indication (manual or electronic) provision(s).

The engine fuel system supplies the required amount of fuel at the required pressure and temperatures across the maximum and minimum values for high vapor to liquid ratios, temperature ranges, contamination, lubricity, and suction lift for primary, alternate, and emergency use fuels.

Engine fuel system pressure vessels and lines can withstand 1.5X (proof) maximum normal operating pressures, without performance degradation, permanent deformation or leakage, and 2X (burst) maximum normal operating pressures without fracture.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, or other substantiating records, and reports) to verify design of the fuel system and in-line filtration system.

Analysis.

(1) Analysis is accomplished to ensure the fuel system and fuel components are sized to provide the required fuel flows, pressures, and temperatures for both steady state and transient conditions throughout the engine operating envelope.

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Test.

(1) Lab test: Wet rig bench testing ensures that the fuel system produces required flows, pressures, and temperatures.

(2) Lab test: Component proof and burst tests ensure the fuel system pressure vessels and lines meet engine design limits.

(3) Lab test: Fuel system filter flow and contamination test ensure that the filter cleans debris from the fuel, maintains flow and pressure, and activates bypass when the fuel filter delta pressure limit is exceeded.

(4) Lab test: Fuel system performance tests (e.g., suction lift, cavitation, vapor to liquid ratio, lubricity) are conducted to verify design of the fuel system.

(5) Lab test: Engine ground-based sea level cell test verifies that the fuel system produces required flows, pressures, and temperatures.

7.4.6 Ignition system.

Standard: The engine ignition system ignites the main combustor and augmentor throughout the ground and air start envelopes and under carbon and water fouling conditions.

The engine control system detects a flameout and activates the ignition system (auto-relight) without flight crew involvement, or the flight crew can manually activate the main and augmentor ignition systems. Ignition system activation will not result in mechanical damage or inhibit a successful flight crew-initiated restart.

For single engine applications, the main combustor ignition system should be redundant and provide a minimum of two separate igniters and two independent ignition exciter output circuits.

External cabling is insulated to prevent inadvertent conduction and shock hazard.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, or other substantiating records and reports) to verify that each engine is equipped with a redundant main ignition system and external cabling is insulated.

Analysis.

(1) Analysis verifies that the ignition system ignites the main combustor and augmentor throughout the ground and air start envelopes.

Test.

(1) Lab test: Engine ground-based sea level cell test verifies the control system's ability to ignite the main combustor and augmentor throughout the ground and air start envelopes and correctly identify an engine flameout and automatically activate the ignition system without flight crew action.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Demonstrate the flight crew's ability to identify an engine flameout indication and activate the ignition system, throughout the ground and air start envelopes.

Test.

(2) Flight test: Flight testing verifies the control system's ability to ignite the main combustor and augmentor throughout the ground and air start envelopes and correctly identify an engine flameout and automatically activate the ignition system without flight crew action.

7.4.7 Anti-ice and de-ice systems.

Standard: Engine anti-ice and de-ice systems prevent damaging ice buildup or provide non-damaging ice removal at all engine speeds and power levels and will not result in heat-induced damage to the engine's structure.

Moisture cannot accumulate and freeze on critical control system components (e.g., sensors, lines) that result in control system malfunction. Anti-ice systems prevent ice from accumulating on the engine structure that could result in ingestion and subsequent mechanical damage to internal rotating components. De-ice systems remove existing ice accumulations before they can be ingested and cause mechanical damage to internal rotating components.

For automatic anti-ice and de-ice systems, the flight crew can override the engine control system and manually operate the anti-ice or de-ice systems. Anti-ice and de-ice system operational temperatures are monitored, and the systems are automatically turned off or adjusted to prevent engine damage. Notification is provided to the flight crew in the event of anti-ice or de-ice system malfunction.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, or other substantiating records and reports) to verify design of the anti-ice and de-ice systems.

Analysis.

(1) Analysis is accomplished to verify anti-ice and de-ice systems' ability to prevent or remove ice while operated at icing conditions as determined by the Critical Point Analysis.

Demonstration.

(1) Bench testing of the control system demonstrates ice detection and activation of the anti-ice and de-ice system.

Test.

(1) Lab test: Bench testing of the anti-ice and de-ice plumbing, valves, and sensors verify the system's ability to prevent or remove ice prior to it damaging the engine.

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(2) Lab test: All anti-ice and de-ice system controls are tested to ensure that the flight crew can override the control system and manually operate the anti-ice and de-ice system.

(3) Lab test: Fault injection testing of the anti-ice and de-ice systems verifies the ability to properly recognize temperature exceedances and initiate system function shutdown.

(4) Lab test: Climatic ground-based cell testing verifies the anti-ice and de-ice systems' ability to prevent or remove ice prior to damaging the engine.

7.4.8 Cooling and thermal management.

Standard: Cooling and thermal management systems are sized to remove excess heat from the engine and subsystems (e.g., electronic controls, sensors, lubrication system). Cooling and thermal management systems function without exceeding defined engine operating limits and without detrimental permanent deformation or damage when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Engine and air vehicle cooling and thermal management systems are integrated and function together to ensure thermal load dissipation for the entire air vehicle, to include post-shutdown conditions such as engine soak back.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, design data records, inspection reports, ICD, or other substantiating records and reports) to verify design of the engine cooling and thermal management systems.

Analysis.

(1) Analysis and modeling of engine components is accomplished to determine their thermal loading and heat rejection characteristics. Results are used to verify the engine components' ability to ensure thermal management systems are properly sized and integrated.

Test.

(1) Lab test: Engine ground-based sea level and altitude cell testing are used to verify thermal management system functions and capability.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing is used to verify thermal management system functions and capability.

7.4.9 Variable geometry systems.

Standard: Variable geometry system components (e.g., pumps, actuators, bleed valves, plumbing, and mechanical cables) operate with a full range of motion and force margins to control the engine variable geometry systems under both steady state and transient conditions throughout the engine operating envelope. Variable geometry system components maintain full

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functional capability when exposed to the maximum static and dynamic loads, temperatures, and flows. Variable guide vane control is maintained under fixed fuel flow failure conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, design data records, inspection reports, or other substantiating records and reports) and verify design of the engine variable geometry system.

Analysis.

(1) Analysis of components are conducted to verify design of the variable geometry system.

Test.

(1) Lab test: Bench tests of the variable geometry system components are conducted to verify design of the variable geometry system.

(2) Lab test: Engine ground-based sea level and altitude cell tests of the variable geometry systems are conducted to verify design of the variable geometry system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight tests of the variable geometry system are conducted to verify the variable geometry system design.

7.4.10 Lubrication system operation.

Standard: Engine lubrication system provides oil supply, scavenge, cooling, filtration, and de-aeration when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

The engine operates in a low or no lubrication condition for specified periods (e.g., operation attitudes or dynamic conditions of flight). Lubrication system temperature, pressure and quantity information is monitored (e.g., appropriate sensor, gage, or manual means) and has features for overfill protection. Lubrication system debris is monitored (e.g., electric or magnetic chip detectors, quantity debris monitors and Joint Oil Analysis Program). A system deaerator removes entrained air from the oil.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine drawings and lubrication system configuration description to verify design of the oil supply, scavenge, cooling, filtration, de-aeration, and monitoring systems.

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Analysis.

(1) Analysis of the lubrication supply and scavenge system are conducted to verify design of the oil supply, scavenge, cooling, filtration, and de-aeration system.

Test.

(1) Lab test: Lubrication system simulator test verifies operational attitudes and ensures all components maintain proper lubrication and scavenging.

(2) Lab test: Oil filter flow and contamination test ensure the filter's ability to clean debris, maintain oil flow and pressure, and activate bypass. Inspection of the oil filter determines its capabilities for required maintenance.

(3) Lab test: Oil deaeration test ensures the system deaerator removes entrained air from the oil.

(4) Lab test: Engine ground-based sea level and altitude cell testing ensures the lubrication system pressures, temperatures, and flows.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Flight testing verifies engine operation in low or no lubrication condition.

7.4.11 Lubrication system non-combustion.

Standard: Lubrication system and bearing compartments (e.g., tanks, lines, gearboxes, sumps) do not allow the collection or buildup of materials that initiates or supports combustion. Components that are exposed to both fuel and oil (e.g., heat exchangers, fuel lubricated oil pumps) do not allow engine fuel flow to enter the lubrication system, bearing compartments or gearboxes. Fuel and oil carrying components do not allow mixing of the two systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, materials and manufacturing records, design data records, inspection reports, photographs, or other substantiating records and reports) and verify components cannot mix and lubrication systems will not initiate and support combustion.

Analysis.

(1) Analysis verifies fuel and oil carrying components cannot mix and lubrication systems will not initiate and support combustion.

Test.

(1) Lab test: Bench testing verifies fuel and oil components cannot mix and lubrication systems will not initiate and support combustion.

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7.4.12 Propulsion monitoring system.

Standard: Propulsion monitoring system detects, isolates, and records engine faults that affect or could affect continued operation of the air vehicle. The flight crew is immediately notified of any safety faults that affect continued operation of the air vehicle. Propulsion monitoring system failure or malfunction does not cause a failure of any other safety-critical system. Faults are recorded and are maintained for post-flight maintenance download.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the ICD and flight manual to ensure flight crew are alerted of all engine safety-critical faults.

Analysis.

(1) Analysis is conducted to verify design of the propulsion monitoring system.

Test.

(1) Lab test: Fault injection bench testing verifies that the propulsion monitoring system detects, isolates, records, and sends faults for display to the flight crew.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight test ensures fault detection notifies the flight crew of faults.

(3) Flight test: Post-flight testing ensures the faults are recorded and maintained for download.

7.4.13 Engine bleed air system.

Standard: Engine bleed air system provides bleed air extraction at the required temperatures and airflows when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data. For high pressure bleed systems, failures resulting in impingement of high-pressure air do not cause damage or malfunction that may lead to loss of air vehicle. For air vehicles with more than one engine, bleed air requirements are specified and managed through the ICD IAW capability established in 7.2 standards.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine drawings and the ICD to verify design of the engine bleed air system.

Analysis.

(1) Analysis is conducted to verify design of the engine bleed air system.

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Test.

(1) Lab test: Engine ground-based sea level and altitude cell bleed air tests are conducted, in conjunction with engine performance tests, to verify the design of the engine bleed air system.

7.4.14 Controls and subsystems rotating components.

Standard: Engine controls and subsystem rotating components (e.g., oil pumps, fuel pumps, turbochargers, starters, generators, alternators) are designed to be damage tolerant. Components are designed with protections against damaging neighboring critical systems and components.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, or other substantiating records and reports) and verify the design of the controls and subsystems rotating components.

Analysis.

(1) Analysis of components verifies damage tolerant capability. Analysis verifies the components are designed with protections against damaging neighboring critical systems and components, and with protections from an uncontained failure of neighboring components.

Test.

(1) Lab test: Component testing (e.g., design verification testing, qualification and certification testing) verifies damage tolerant capability.

7.4.15 Tubing and plumbing routing.

Standard: External hardware (e.g., engine mounted tubing, manifolds, clamps) are mounted and routed on the engine such that there is no interference or contact with neighboring components or engine structure and that no wear or chafing conditions exist. The orientation and routing of tubes and lines carrying any combustible fluids provide separation from all potential sources of temperatures or ignition such that they may operate without any exceedance of the fluid combustion temperature. Positive clearances are documented in the ICD.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, engine ICD or other substantiating records and reports) to verify design of the engine external hardware mounting and routing.

Analysis.

(1) Analysis is conducted to verify design of the engine external hardware mounting and routing.

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7.4.16 Engine mounted components vibratory response.

Standard: Engine mounted components, tubing, manifolds, and clamps do not contain natural (resonant) frequencies within the engine or air vehicle (e.g., gearboxes, driveshaft(s), transmissions) operating range or have damping provisions to prevent resonances, damage, or failure. Damaging vibration and resonances do not occur in the operating speed range from 20% below idle speed to 20% above maximum operating speed. Engine mounted components, tubing, manifolds, and clamps withstand an engine full blade out vibration excitation without failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, materials and manufacturing records, design data records, inspection reports, or other substantiating records and reports) and verify design of the engine mounted components, tubing, manifolds, and clamps.

Analysis.

(1) Analysis verifies engine mounted components, tubing, manifolds, and clamps are free of critical resonant conditions at all speeds in the operating range.

(2) Analysis verifies engine mounted components, tubing, manifolds, and clamps withstand excitations resulting from a full blade out condition.

Test.

(1) Lab test: Vibration surveys (ping testing) and vibration (shaker table) testing verifies the natural frequencies of the engine mounted components, tubing, manifolds, and clamps are outside the engine and air vehicle operating range.

(2) Lab test: Engine ground-based sea level cell testing verifies engine mounted components, tubing, manifolds, and clamps withstand excitations resulting from a full blade out condition.

7.4.17 Externals strength.

Standard: Externals fluid carrying components (e.g., pressure vessels, tubes, fittings, and manifolds) have design margin for their maximum operating conditions and required duty cycles encountered during normal engine operation. Tubing and lines meet damage tolerance capability (e.g., leak before burst). Pressure vessels and lines withstand 1.5X normal operating pressures (proof test), without performance degradation or leakage and 2X maximum operating pressures (burst test), without permanent deformation or leakage.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, or other substantiating records and reports) and verify that externals fluid carrying components meet the engine design.

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Analysis.

(1) Analysis is accomplished to ensure damage tolerance capability (e.g., leak before burst).

Test.

(1) Lab test: Bench top proof and burst testing ensures the components meet damage tolerance capability.

(2) Lab test: Bench top pressure cycle testing ensures the components and lines do not leak or rupture.

7.4.18 Flammable fluid systems.

Standard: Fuel carrying systems (e.g., components, manifolds, lines, and fittings) are fire resistant for at least 5 minutes at a minimum flame temperature of 2000 °F.

Lubrication carrying systems and hydraulic fluid carrying systems (e.g., components, manifolds, lines, and fittings) are fireproof for at least 15 minutes at a minimum flame temperature of 2000 °F.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, videos, or other substantiating records and reports) and verify fire resistance or fireproof capability of the hardware.

Analysis.

(1) Analysis is conducted to verify the flammable fluid systems are fire resistant or fireproof.

Test.

(1) Lab test: Fire testing verifies that all fuel components, fuel lines, oil carrying components, and oil lines are fire resistant or fireproof, respectively.

7.4.19 Gearboxes.

Standard: Propulsion gearboxes support mounted components, provide mechanical speed, power, and torque to all mounted components, and transmit torque to the engine for starting. Internal gears, shafts, and rotating hardware are free from damaging resonances at all speeds up to the maximum overspeed condition. Gearboxes function without exceeding defined engine operating limits and without detrimental permanent deformation or damage when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review documentation (e.g., hardware drawings, materials and manufacturing records, design data records, inspection reports, ICD or other substantiating records and reports) and verify design of the gearbox.

Analysis.

(1) Analysis is accomplished to verify gearbox supports all mounted components.

(2) Analysis is accomplished to ensure that the gears, shafts, and rotating hardware are free of damaging resonance conditions at all speeds in the operating range to include at least 20% above the maximum steady-state operating speed and 20% below idle speed.

(3) Analysis is accomplished to ensure the gearbox is capable of operation when 1.25X the maximum permissible torque or power rating specified for each individual drive is applied simultaneously to all drives.

Test.

(1) Lab test: Bench testing verifies the gearbox ability to support all mounted components.

(2) Lab test: Vibration testing verifies internal gears, shafts, and rotating hardware are free of damaging resonance conditions at all speeds in the operating range to include at least 20% above the maximum steady-state operating speed and 20% below idle speed.

(3) Lab test: Endurance testing verifies the gearbox is capable of operation when 1.25X the maximum permissible torque or power rating specified for each individual drive is applied simultaneously to all drives.

(4) Lab test: Engine ground-based sea level and altitude cell testing verifies the gearbox ability to support all mounted components.

7.4.20 Gearbox mounted component failures.

Standard: The gearbox and mounted components (e.g., oil pumps, fuel pumps, starters, generators, PTO driveshaft) allow disengagement (e.g., shear sections) prior to causing damage to the gearbox or other components. Components, whose continued operation is required to maintain air vehicle operation, do not contain shear or failure sections.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, materials and manufacturing records, design data records, inspection reports, photographs, ICD or other substantiating records and reports) and verify design of the gearbox and the gearbox mounted components.

Analysis.

(1) Analysis of the gearbox and mounted components verifies disengagement provisions have been incorporated into the design.

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7.4.21 Electrical power.

Standard: The engine driven alternator and generator is sized to provide uninterruptible electrical power where the engine is the primary or alternate power source for the engine control system, engine ignition system, and any other engine electrical accessory. Transfer between primary and alternate power sources does not interrupt engine operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, design data records, inspection reports, or other substantiating records, reports, ICD) and verify design of the engine driven alternator and generator.

Analysis.

(1) Analysis of the engine's electrical consumption establishes the power generated by the primary or alternate power source.

Demonstration.

(1) Demonstrate during ground testing, the transfer between primary and alternate power sources does not interrupt engine operation.

Test.

(1) Lab test: Bench testing verifies the electrical power generation.

(2) Lab test: Engine ground-based sea level and altitude cell testing verifies the electrical power generation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Demonstrate during flight testing, the transfer between primary and alternate power sources does not interrupt engine operation.

Test.

(3) Flight test: Flight testing verifies the electrical power generation.

7.4.22 Electrical components.

Standard: Electrical components are affixed and routed on the engine such that positive clearance between adjacent components and air vehicle structure is present so that no wear or chafing conditions exist. The orientation and routing of electrical components provide separation from all potential sources of extreme temperatures or ignition (e.g., tubes and lines carrying combustible fluids, and hot air bleed lines). Electrical components are designed for electrical bonding, ground isolation, insulation resistance, damage pin, EMI, lightning, dielectric strength, and explosion proof capabilities.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, design data records, inspection reports, ICD or other substantiating records and reports) to verify design of the engine electrical components.

Analysis.

(1) Analysis is conducted to verify design of the engine electrical components.

Test.

(1) Lab test: Bench testing is conducted to verify design of the engine electrical components.

7.4.23 Electrical connectors and cables.

Standard: Electrical connectors contain redundant, visually verifiable, locking features and keyed interfaces to prevent misalignment and improper installation. Connectors and cables are protected from water intrusion. Cables are routed and affixed to prevent wear and chafing. At cold atmospheric temperatures, the electrical cables maintain flexibility without damage, and the connectors are connected and disconnected using normal maintenance procedures. Separation exists between cables, combustible fluids, and ignition sources to minimize the potential of fire and ignition.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, ICD, or other substantiating records and reports) and verify design of the engine electrical cables and connectors.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(1) Analysis is conducted to verify design of the engine electrical connectors and cables.

Demonstration.

(1) Demonstrate on the installed engine, the connector's redundant locking features and keyed interfaces to prevent misalignment and improper installation.

(2) Demonstrate, on an installed engine, at cold atmospheric temperatures, the electrical cables maintain flexibility without damage, and the connectors are connected and disconnected using normal maintenance procedures.

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7.4.24 Electrical cables vibratory response.

Standard: Electrical cables are designed to ensure their natural frequencies are outside the engine and air vehicle operating range. Cables are routed and affixed to provide damping, to prevent damaging resonances. Damaging vibration and resonances do not occur in the operating speed range from 20% below idle speed to 20% above maximum operating speed.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, ICD, or other substantiating records and reports) to verify the design of the electrical cables.

Analysis.

(1) Analysis verifies electrical cables natural frequencies are outside the engine and air vehicle operating range.

(2) Analysis verifies electrical cables withstand excitations resulting from a full blade out condition.

Test.

(1) Lab test: Vibration surveys (ping testing) and vibration (shaker table) testing verifies the natural frequencies of the engine cables are outside the engine and air vehicle operating range.

(2) Lab test: Engine ground-based sea level cell testing, with vibration measuring instrumentation verifies the natural frequencies of the engine cables are outside the engine and air vehicle operating range.

7.4.25 Physical interface loads.

Standard: Engine to air vehicle physical interfaces (e.g., mechanical, fluid, and electrical connections) remain connected when subjected to maximum loading conditions (e.g., bending, torsional loads, pressures, temperatures, vibratory, misalignment) when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the ICD to verify engine to air vehicle interface loads are fully defined.

Analysis.

(1) Analysis verifies engine to air vehicle interface loads.

Test.

(1) Ground test: Installed engine ground testing verifies engine to air vehicle interface loads.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing verifies engine to air vehicle interface loads.

7.4.26 Engine to air vehicle clearances.

Standard: Positive clearance between the engine to air vehicle (including associated components, plumbing, and harnesses) is maintained when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, ICD, or other substantiating records and reports) to verify the engine to air vehicle clearances.

Analysis.

(1) Analysis is conducted to verify the engine to air vehicle clearances.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Flight testing is conducted to verify the engine to air vehicle clearances.

7.4.27 Drains and ventilation systems.

Standard: Engine drains and ventilation systems are sized to have flow and volume capacities when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data. Expulsion of fluids and vapors do not create an AW hazard.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, ICD, or other substantiating records and reports) to verify design of the engine drains and ventilation systems.

Analysis.

(1) Analysis is conducted to verify design of the engine drains and ventilation systems.

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Test.

(1) Lab test: Engine ground-based sea level and altitude cell endurance tests are conducted to capture the fluid quantity and source; verifying design of the engine drains and ventilation systems.

7.4.28 Installed engine accessibility.

Standard: Propulsion systems are designed to allow servicing, inspections, and maintenance activities. Access accommodates the maintenance equipment, maintainer limitations, and environmental conditions (e.g., chemical protective gear, gloves).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, photographs, ICD, or other substantiating records and reports) to verify the installed engine accessibility.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(1) Virtual models or physical mock-ups verify the installed engine accessibility.

Demonstration.

(1) Demonstrate accessibility on the installed engine.

7.4.29 Functional compatibility.

Standard: Engine to air vehicle interfaces maintain functional compatibility when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data. The engine control system does not propagate any functional failures to the air vehicle. Failure(s) of the air vehicle and propulsion communication system results in a fail-safe capability that allows continued engine operation without exceeding engine operating limits. For multi-engine (three or more engines) air vehicles, an engine shutdown is permissible if the shutdown does not result in loss of air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., interface control and design documents, hardware drawings, design data records, inspection reports, photographs, or other substantiating records and reports) to verify the functional capability of the engine to air vehicle interfaces.

Analysis.

(1) Analysis is conducted to verify the functional capability of the engine to air vehicle interfaces.

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Test.

(1) Ground test: Installed engine ground testing is conducted to verify integrated system functional compatibility of the engine to air vehicle interfaces.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing is conducted to verify integrated system functional compatibility of the engine to air vehicle interfaces.

7.4.30 Extractions.

Standard: The engine supplies extractions (e.g., bleed air, horsepower, electrical power) when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the ICD to verify extraction interfaces are fully defined.

Analysis.

(1) Analysis verifies the engine extraction demands are supplied.

Test.

(1) Lab test: Bench and rig testing verifies the engine extraction demands are supplied.

(2) Lab test: Engine ground-based sea level and altitude cell testing verifies the engine extraction demands are supplied.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing verifies the engine supplied extractions.

7.4.31 Bleed air contamination.

Standard: Engine generated substances contained in the customer bleed air are below the TLVs set by the American Conference of Governmental Industrial Hygienists when the engine(s) are operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review materials, coatings, and material interfaces used in the engine to identify engine-generated bleed air particles and contaminates.

Analysis.

(1) Analysis verifies bleed air contamination levels are below TLV.

Test.

(1) Lab test: Engine ground-based sea level and altitude cell testing verifies bleed air contamination levels are below TLV.

(2) Ground test: Installed engine ground testing verifies bleed air contamination levels are below TLV.

7.4.32 Engine shutdown.

Standard: Air vehicle initiated fuel shutoff does not result in immediate or subsequent exceedance of any engine limits, cause engine damage that jeopardizes the air vehicle, or result in a post-shutdown fire.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies an air vehicle initiated fuel shutoff performs an engine shutdown without exceeding limits, damaging the air vehicle, or resulting in a fire.

Test.

(1) Lab test: Rig testing verifies an air vehicle initiated fuel shutoff performs an engine shutdown without exceeding limits, damaging the air vehicle, or resulting in a fire.

(2) Lab test: Engine ground-based sea level cell testing verifies a test facility-initiated fuel shutoff performs an engine shutdown without exceeding limits, damaging the air vehicle, or resulting in a fire.

(3) Ground test: Installed engine testing verifies an air vehicle initiated fuel shutoff performs an engine shutdown without exceeding limits, damaging the air vehicle, or resulting in a fire.

7.4.33 Engine bay and nacelle cooling and ventilation.

Standard: Engine bay or nacelle cooling, and ventilation maintain the temperatures of engine components, structures, engine fluids, and bay or nacelle equipment within the temperature limits established for these components and fluids when operated, under steady state and transient conditions, IAW 7.3.1 for the stated life documented in engine technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review design documentation to verify temperature limits are defined for engine components, structures, engine fluids, and bay or nacelle equipment.

Analysis.

(1) Thermal analysis and modeling of engine components, engine fluids, and bay or nacelle equipment verifies system thermal performance and that no temperature limits are exceeded.

Test.

(1) Ground test: Installed engine ground testing verifies engine bay or nacelle cooling, and ventilation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing verifies engine bay or nacelle cooling, and ventilation.

7.4.34 Crew station compatibility.

Standard: Crew station provides the propulsion controls and information to independently start and stop each engine, control and set thrust for each engine, and assess engine operating conditions. Faults are recorded and are provided to the flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation and analyses (e.g., FMEA, FMECA, sneak circuit, common cause, interface control design, software design, and hardware design documents) to verify that the crew station contains the engine controls. Faults are recorded and are provided to the flight crew.

Test.

(1) Ground test: Installed engine ground testing verifies the crew station capability. Faults are recorded and are provided to the flight crew.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing verifies the crew station capability. Faults are recorded and are provided to flight crew.

7.5 Propeller and propeller drive systems.

These criteria apply to the physical, functional, performance, and interface characteristics of propellers, propeller drive systems, and propeller controls regardless of engine type. The term “propeller system” is used throughout; the propeller system includes the propeller and all elements of the propeller drive system (e.g., propeller mount/hub, spinner, propeller drive shaft,

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fluid-pressurized components, pitch control components, reduction gearbox, torque measurement system, negative torque system, propeller brake, and mechanical over-speed governor).

Critical performance conditions are phases of flight at which insufficient propeller system performance may result in hazardous air system effects (e.g., launch, takeoff, OEI, climb, cruise, endurance, maneuver, hover, in-flight refueling, descent, approach, landing, recovery).

This section should be evaluated for any design or modification involving the propeller system, propeller control system and components, propeller operating characteristics, manufacturing or repair processes, material substitutions, propeller system installation physical or functional characteristics (e.g., drive system extractions, fluid or electrical interfaces, mounts, clearances, loads or vibrations), air vehicle usage changes (e.g., maneuvers, icing operations, drag or gross weight), or other changes.

7.5.1 Performance and operability.

Standard: Propeller steady-state performance, transient performance, and operability is characterized with representative installation effects over the range of flight conditions and operational environments expected and is shown to meet the critical performance conditions of the air system without exceeding defined engine operating limits and without detrimental permanent deformation or damage when operated, under steady state and transient conditions, IAW 7.5.2 for the stated life documented in technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., ICDs, environmental criteria documents) to verify that operational environments and installation effects have been defined and utilized in the characterization of performance and operability.

Analysis.

(1) Analyses performed with a validated, configuration-controlled, steady state and transient, propeller performance model calibrated to measured test data for characterization of performance and operability.

Test.

(1) Lab test: Whirl stand tests verify sea level static performance of the propeller.

(2) Lab test: Engine-propeller system combination ground-based cell tests characterize and verify steady state and transient performance and operability.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Engine-propeller system combination flight tests characterize and verify the installed steady state and transient performance and operability.

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7.5.2 Design usage/duty cycle.

Standard: Design usage, duty cycles addressing all sources of repeated loads, and environmental exposure applicable to the type design being certified are defined and validated. All source material (e.g., military specifications, contract requirements, program documents) is identified, and any modification of source material for the type design being certified is substantiated.

Design usage definitions and environments include the following:

a. Mission definition and percent utilization.
b. Usage statistics.
c. Time at flight envelope extremities (e.g., 10 hours at any point in envelope).
d. Reverse pitch and feathering usage.
e. Propulsion system operating envelope.
f. Attitude limits and exposure times.
g. Ambient temperature distributions and percent utilization.
h. Reduction gearbox extractions and extraction duty cycle (e.g., loaded accessory pads).
i. Externally-applied forces (g's, rotational rates and accelerations).
j. Installed temperature, pressure, and vibration environment.
k. Acoustic noise environment (installed and uninstalled).
l. Icing environment.
m. Sand and dust environment.
n. Corrosion environment.
o. Non-operating and transportation environment.
p. Windmilling envelope.
q. Cross-wind envelope.
r. Electromagnetic environment.

The damage content of simplified or composite duty cycles defined for analysis and test are shown to be equivalent to, or more severe than, that of the full mission usage set.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine design documentation or other source material to verify that all design usage, duty cycles, and environments necessary for the type design being certified are defined and valid. Sources are cited and modification of any source material is substantiated.

Analysis.

(1) Analysis defines usage and duty cycles for design, analysis, and test. Damage equivalency of resultant duty cycles is substantiated.

7.5.3 Loads and strength.

Standard: The propeller system withstands limit and ultimate load conditions as follows:

Load Combinations:

Load factors are applied to loads that occur within the entire flight and ground envelope IAW 7.5.2 to establish combined limit load conditions and combined ultimate load conditions. For all

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castings an additional 1.33 safety factor is applied to the resulting limit and ultimate load cases unless the cast material has been fully characterized IAW 7.1.3.

Limit Load Conditions:

Propeller system structure, when operated to limit loads within the entire flight and ground envelope, does not exhibit detrimental permanent set, damage, or reduction of life limits; and does not temporarily deflect to the extent that operation or performance is impaired, or clashing between propeller and engine, or propeller and air vehicle occurs. Limit LFs are as follows:

Limit Load Factors:

a. 1.0 LF for externally-applied loads (including ground and flight loads, and external flow field loads).
b. 1.0 LF for thrust and thermal loads.
c. 1.0 LF for internal pressure loads on cases and pressure vessels.
d. 1.5 LF for internal pressure loads on controls, accessories, and externals components.

Ultimate Load Conditions:

Propeller system structure does not experience failure or deform to the extent of creating adverse propeller forces or unbalance, cause an uncontained engine failure or uncontained fire, or cause an engine mount system failure under ultimate load conditions. For ultimate load cases involving limit operating loads and ultimate externally-applied loads (e.g., maneuvers) the propeller system also maintains performance enabling successful recovery of the air vehicle. Ultimate LFs are as follows:

Ultimate Load Factors:

a. 1.5 LF for externally-applied loads (including ground and flight loads, and external flow field loads).
b. 1.0 LF for thrust and thermal loads.
c. 1.5 LF for internal pressure loads on cases and pressure vessels.
d. 2.0 LF for centrifugal loads at maximum rated propeller rotational speed applied singly to blade hubs, blade retention systems, counterweights, and blade features associated with transition to retention features (e.g., composite blade bonded to metallic retention features).
e. 2.0 LF for internal pressure loads on controls, accessories, and externals components.

The non-operating propeller, including all mounted equipment and accessories, components, and operating fluids, tolerates a 4g axial, 2g lateral, and 3g vertical inertial limit load acting in any combination at the propeller CG without deformation of ground handling features or damage to the propeller; normal propeller function is unaffected after experiencing this load case. Ultimate load case is 1.5X limit load case; ground handling features do not fail under this load case.

Propeller system is capable of further operation without maintenance action after operation at the maximum overspeed and overtorque limits documented in technical data.

For all strength assessments, positive margins of safety exist for the range of manufacturing tolerances and operational conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Analyses are accomplished to define design load conditions.

(2) Analyses are accomplished using defined load conditions to verify strength and deflection capability, and system-level impacts.

(3) Analyses are accomplished to verify propeller system overspeed and overtorque capability.

Test.

(1) Lab test: Centrifugal load tests are accomplished per the conditions and success criteria as specified in 14 CFR 35.35 to verify analytical results. Tests are conducted at the maximum operating temperature or at test conditions adjusted to account for the differences between operating and test temperatures. Strain gauges and other instrumentation are used during tests to validate analysis methods. Tests are conducted progressively to limit or ultimate load conditions. Inspection of all test articles before, during, and after test verifies no unexpected interactions, failures or wear.

(2) Lab test: Hydraulic component proof and burst tests are accomplished per the conditions and success criteria as specified in 14 CFR 35.43 to verify analytical results. Tests are conducted at the maximum operating temperature or at test conditions adjusted to account for the differences between operating and test temperatures. Strain gauges and other instrumentation are used during tests to validate analysis methods. Tests are conducted progressively to limit or ultimate load conditions.

Inspection of all test articles before, during, and after test verifies no unexpected interactions, failures or wear.

(3) Lab test: Overspeed and overtorque tests are accomplished on a whirl stand IAW JSSG-2009, Propeller Subsystem, Whirl Stand Tests, Overspeed Run paragraph or per the conditions and success criteria as specified in 14 CFR 35.41 to verify analytical results. Inspection of all test articles before, during, and after test verifies no unexpected interactions, failures or wear.

7.5.4 Durability.

Standard: The durability of propeller system hardware is such that the propeller system capability established in 7.5.1 standards is not adversely affected, propeller system operating limits are not exceeded, and detrimental permanent deformation or damage does not occur when the propeller system is operated, under steady state and transient conditions, IAW 7.5.2 for the stated life documented in technical data. Durability includes resistance to fatigue, cracking, delamination, creep, permanent set, engine and propeller fluids, ultraviolet radiation, wear, and tolerance to environments defined in 7.5.2.

Part-specific critical material, processing, and manufacturing characteristics, and associated controls, are established for PCPs IAW 14 CFR 35.16.

Operating limitations and life management actions required to maintain propeller system structural integrity and functionality (e.g., component inspection, repair, and replacement requirements) are based on 7.5.2 and are recorded in the technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examination of PCP product definition data (e.g., models, drawings) verifies that part-specific critical material, processing, and manufacturing characteristics, and associated controls, are defined.

(2) Examination of the service management plan, and applicable maintenance manuals (e.g., TOs, Naval Aviation Technical Information Products) verifies that PCPs, and associated life management actions, are defined and documented.

Analysis.

(1) Analysis is conducted to determine minimum fatigue life and mandatory replacement times, and to define mature and repeatable material, processing, and manufacturing control parameters required to achieve that life for each PCP. Scatter factors used to establish life limits are substantiated.

(2) Analysis shows all critical vibratory frequencies, mode shapes, and stresses as a function of blade angle and rpm. Analysis is calibrated to test results. Campbell diagrams are prepared to verify margin to potential resonances, particularly at sustained operating ranges.

Test.

(1) Lab test: Propeller components including the blades and barrel, pitch changing mechanism, pitch lock, negative torque signal, control unit, and ice control system are durability tested to establish their capability to perform their function for the operational life defined in technical data. Testing includes fatigue evaluation under combined steady and vibratory loads as needed to determine failure locations, failure modes, vibratory capability, and life limits. A complete teardown inspection is conducted at the conclusion of the test. Records are made of failures, wear, distortion, and other unusual conditions.

(2) Lab test: Whirl stand testing, including vibratory stress surveys and endurance tests, of the propeller system are conducted to verify durability. A complete teardown inspection is conducted at the conclusion of the test. Records are made of failures, wear, distortion, and other unusual conditions.

(3) Lab test: The following propeller-and-engine combination tests are conducted on a sea-level test stand. An instrumented vibratory stress survey is conducted to determine stress characteristics of the propeller and engine combination. Endurance tests are conducted IAW JSSG-2009, Propeller Subsystem, Propeller and Engine Test Stand Tests, 50-Hour Test, Cyclic Tests paragraph or per the conditions and success criteria as specified in 14 CFR 35.39. Post-test teardown inspection of hardware is accomplished to document condition. Minimum endurance test duration is 50 hours.

(4) Ground test: Installed ground testing of the integrated propeller system/engine/air vehicle is conducted on the air vehicle being certified. Instrumented ground vibratory stress surveys are conducted on the propeller at each air vehicle nacelle. Static crosswind conditions are included in the evaluation. Documented propeller operating limitations are confirmed. Inspection of all test articles before, during, and after test verifies no unexpected interactions, failures, or wear.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(5) Lab test: Endurance testing of the propeller-and-engine combination is conducted IAW JSSG-2009, Propeller Subsystem, Propeller and Engine Test Stand Tests, 150-Hour Test paragraph or per the conditions and success criteria as specified in 14 CFR 35.39. Post-test teardown inspection of hardware is accomplished to document condition. Minimum endurance test duration is 150 hours.

(6) Flight test: Installed flight testing of the integrated propeller system/engine/air vehicle is conducted on the air vehicle being certified. Instrumented flight vibratory stress surveys are conducted on the propeller at each air vehicle nacelle. Crosswind and maneuver conditions are included in the evaluation. Documented propeller operating limitations are confirmed. Inspection of all test articles before, during, and after test verifies no unexpected interactions, failures or wear.

7.5.5 Damage tolerance.

Standard: When operated, under steady state and transient conditions, IAW 7.5.2, PCPs IAW 14 CFR 35.16, resist failure caused by the presence of material, manufacturing, processing, maintenance, and service-induced defects, retaining residual strength to carry maximum limit stress, for at least twice the inspection interval defined in technical data without repair. Residual strength load conditions account for changes in vibratory characteristics during usage.

Initial surface flaw sizes are those detectable at the minimum demonstrated reliability (POD/CL) of the NDI methods used during manufacture and in-service maintenance. Initial embedded defect sizes/rates are characterized and controlled by material and processing controls and complementary NDI utilized during material processing and forming. Minimum flaw size detection POD/CL for all NDI methods is 90%/90%.

Material and processing controls, inclusive of the following, and PCP usage tracking plans, are established:

a. Material and manufacturing operation specifications throughout the procurement and production chain.
b. Allowable limits and controls for material property variation.
c. Critical characteristics and defect size, locations, and orientations on product definition data.
d. In-process and finished-part (production and in-service) NDI requirements and acceptance standards.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) For each PCP, review design documentation, propeller certification documentation, sustainment technical data, and product definition data that verifies:

a. Compliance with material and processing control elements of this standard.
b. Tracking plans.

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Examples of such documentation include TOs, NDI requirement definition, drawings and models, material and processing specifications.

Analysis.

(1) Controlled crack growth (fracture mechanics) analysis is conducted for each PCP. Analyses account for repeated and sustained stresses, vibratory stress (including threshold value), and include the effects of load interactions. Stress intensity is based on the structural geometry and flaw geometry. The critical stress intensity is based on the required residual strength load level or vibratory stress threshold value, and temperature conditions that exist at the component location being analyzed. Analysis initial flaw sizes are defined by the demonstrated production and sustainment NDI capabilities, or by exceedance curves for embedded defects, established IAW this standard. Limit loads used for residual strength analysis are defined for each PCP.

Demonstration.

(1) Production NDI demonstrations are performed IAW MIL-HDBK-1823 to ensure that flaws larger than the analysis initial flaw size will not occur in finished components to the reliability level required by this standard.

Test.

(1) Lab test: Propeller critical parts are selected for verification by test. These components are either pre-flawed or contain natural flaws and are cycled to verify flaw growth characteristics, strength, failure location, and failure mode and mechanism.

(2) Lab test: Endurance testing of the propeller-and-engine combination is conducted on a sea level test stand IAW JSSG-2009, Propeller Subsystem, Propeller and Engine Test Stand Tests, 50-Hour Test, Cyclic Tests paragraph or per the conditions and success criteria as specified in 14 CFR 35.39. Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics and ability to inspect IAW technical data. Minimum endurance test duration is 50 hours.

(3) Ground test: Installed ground testing of the integrated propeller system/engine/air vehicle is conducted on the air vehicle being certified. Static crosswind are included in the evaluation. Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics and ability to inspect IAW technical data.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) In-service NDI demonstrations are performed IAW MIL-HDBK-1823 to ensure that flaws larger than the corresponding analysis flaw size will not occur in released components to the reliability level required by this standard. In-service inspection flaw size detection capability is demonstrated on parts with prior operation time.

Test.

(4) Lab test: Endurance testing of the propeller-and-engine combination is conducted on a sea level test stand IAW JSSG-2009, Propeller Subsystem, Propeller and Engine Test Stand Tests, 150-Hour Test paragraph or per the conditions and success criteria as specified in 14 CFR 35.39.

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Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics and ability to inspect IAW technical data. Minimum endurance test duration is 150 hours.

(5) Flight test: Flight testing of the integrated propeller system/engine/air vehicle is conducted on the air vehicle being certified. Crosswind and maneuver conditions are included in the evaluation. Post-test teardown and NDI of hardware verifies predicted damage tolerance characteristics and ability to inspect IAW technical data.

7.5.6 Propeller interfaces.

Standard: All physical and functional interfaces between the engine, air vehicle, and the propeller system are established and controlled to ensure compatibility.

The allowable range of characteristics of the propeller system at engine and air vehicle interfaces is specified and documented.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of documentation (e.g., schematics, drawings and models, ICDs, installation manuals) verifies that detailed descriptions of the propeller physical and functional interfaces with associated engine and airframe and all related modules and components are established and controlled (e.g., physical arrangements and position; functional relationships and characteristics; interface loads and moments; mass properties; direction of rotation; fluid, drain, electrical, and data interfaces; operating clearances).

Analysis.

(1) Analysis is conducted to define installation interface characteristics or to verify that the specific propeller system, engine, and air vehicle combination being certified complies with allowable range of characteristics already documented (e.g., in installation manuals or ICD).

Test.

(1) Ground test: The propeller system and engine installation, in the air vehicle being certified, is checked in the static condition to verify the functional characteristics of the control linkages, voltages, continuity of control circuitry and typical preflight checks. The relationship between all lever positions and signal levels and the propeller blade angle is confirmed for the complete range of lever and signal excursions for increasing and decreasing power conditions.

7.5.7 Vibration and balancing.

Standard: The propeller system is free of damaging vibration, damaging resonance conditions, or other damaging DR, regardless of excitation source, when operated, at all rotational speeds and thrust or power settings under installed or uninstalled steady state and transient conditions, IAW 7.5.2 for the stated life documented in technical data.

a. Maximum mechanical vibration limits, for both installed and uninstalled operation, are defined for each measurement location and direction.
b. Maintenance balance procedures are defined in technical data and eliminate vibration that could cause equipment to operate outside specified requirements, cause equipment

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damage, or that inhibits the flight crew from effectively accomplishing required functions during installed operation.

c. Damaging critical speeds for propeller system do not occur in the speed range from 20% below the minimum operating speed to 20% above maximum allowable transient shaft rotational speed in both forward and reverse thrust modes. Damping and balancing allow any critical speed that exists from rest to 20% above maximum operating speed to be traversed with smooth operation and without damage.
d. The propeller is free from flutter in both forward and reverse thrust modes under conditions up to 120% of maximum rated engine speed and at power settings up to the standard day maximum take-off power rating of the engine.
e. Windmilling operation does not result in damaging vibration to the propeller system, engine, or air vehicle.
f. Propeller system areas that could puddle oil or fluid have drains.
g. No damaging vibration is transmitted to or from the engine through the propeller system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Propeller system vibration limits are defined in technical data.

(2) Propeller system balance limits and procedures are defined for assembly and maintenance in technical data.

(3) Examination of flight manuals and maintenance manuals verifies that vibration-driven or flutter-driven operating limitations are defined.

(4) Review of product definition data and models verifies that propeller system oil and fluid drain features are present.

Analysis.

(1) Analysis is accomplished to determine vibratory characteristics of the complete propeller system installed in the air vehicle. A dynamic model of the system including all relevant mass, stiffness, damping, and boundary conditions is developed to analyze critical speeds and associated DRs. Analysis and models are calibrated to test results.

(2) Flutter modeling and analysis is performed to verify stable operation of the propeller system and components under all operating conditions within the flight operating envelope of the air vehicle on which the propeller system is to be used. Analysis of test results is conducted to determine operating limits for self-excited vibrations and to verify responses do not occur in the operating range.

(3) Analysis is accomplished to determine the range of windmilling speeds, considering installation effect throughout the entire flight and ground envelope.

Test.

(1) Lab test: Component bench tests are accomplished to validate blade natural frequencies and mode shapes.

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(2) Lab test: Instrumented vibration surveys are accomplished in a whirl stand to verify the stress characteristics of the hub and blade and the flutter characteristics of the blade and propeller system.

(3) Lab test: Instrumented vibration stress survey of the propeller installed on the engine in a sea level test stand covering all appropriate conditions of engine operation verifies the stress characteristics of the engine and propeller system and verifies measured stresses for any vibratory response within the operating range are within the allowable material limits. Propeller and engine vibration monitoring confirms no damaging vibration is transmitted to or from the engine.

(4) Ground test: Instrumented ground vibratory stress survey of the propeller system on all nacelles of the air vehicle being certified verifies the stress characteristics of the propeller when operated in the air vehicle environment, verifies measured stresses for any vibratory response within the operating range are within the allowable material limits, and verifies no flutter occurs within defined operating limits. Static crosswind conditions are included in the evaluation. Installed vibration testing in air vehicle confirms no vibration that could cause equipment to operate outside specified requirements, cause equipment damage, or that inhibits the flight crew from effectively accomplishing required functions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Instrumented flight vibratory stress survey of the propeller system on all nacelles of the air vehicle being certified verifies the stress characteristics of the propeller when operated in the air vehicle environment, verifies measured stresses for any vibratory response within the operating range are within the allowable material limits, and verifies no flutter occurs within defined operating limits. Crosswind and maneuver conditions are included in the evaluation. Installed vibration testing in air vehicle confirms no vibration that could cause equipment to operate outside specified requirements, cause equipment damage, or that inhibits the flight crew from effectively accomplishing required functions.

7.5.8 Bird strike.

Standard: The propeller blades and spinner can withstand the impact of a four-lb bird at the propeller and spinner critical location(s) (e.g., highest stress, most damaging vibratory locations) over the range of flight conditions expected of the air vehicle without causing a structural failure or inability to control the propeller throughout the flight and ground envelope per the 7.5.2 for the stated life documented in technical data.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., propeller control system design documentation, hardware drawings, system configuration descriptions, design data records, test reports, or other substantiating records and reports) to verify the design.

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Analysis.

(1) Analysis is conducted to determine the critical impact conditions affecting propeller structural integrity and control, including both impact location and operating conditions. Structural analysis verifies the structural integrity and controllability.

Test.

(1) Lab test: Component and rig tests, including test at the determined critical impact conditions, verify the structural integrity and controllability.

7.5.9 Reversing and pitch controls.

Standard: Overspeed during propeller reversal is compatible with engine overspeed limits. The primary features of propeller pitch control system function independently of the engine oil system or the air vehicle electrical system. The propeller control system includes a mechanical pitch lock that engages in the event of overspeeding or loss of hydraulic pressure or similar failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Propeller test, engine-propeller combination test, and air system flight tests demonstrate overspeed mechanical pitch lock and mechanical low pitch stop functionality.

Test.

(1) Lab test: Whirl stand test verifies compatibility with engine overspeed limits.

(2) Lab test: Propeller test and engine-propeller combination testing verifies propeller pitch control system functionality and full reversal.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Air vehicle flight test verifies propeller pitch control system functionality and full reversal.

7.5.10 Feathering system.

Standard: Feathering propellers, when commanded, will feather and unfeather from all flight conditions and rotational speeds, to include emergency operations (e.g., windmilling dives), under steady state and transient conditions, IAW 7.5.2 for the stated life documented in technical data. When commanded to feather, the propeller pitch traverses to the full feather position at a rate compatible with the asymmetric thrust on a multi-engine air vehicle and without loss of flight energy due to drag on a single engine air vehicle.

An emergency means of initiating feathering is incorporated which is independent of the normal means for such operations. Propeller pitch control systems that use engine oil to feather, incorporate a method to allow the propeller to feather if the engine oil system fails.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., propeller control system design documentation, hardware drawings, system configuration descriptions, design data records, ICD, test reports, or other substantiating records/reports) to verify the design.

Analysis.

(1) Steady state, transient, and emergency operating conditions are analyzed to verify the control of the propeller feathering system.

Test.

(1) Lab test: Propeller test stand testing provides the control response to steady state, transient, and emergency operating conditions to verify the control and feathering of the propeller.

(2) Lab test: Engine-propeller system combination ground-based cell testing provides the control response to steady state, transient, and emergency operating conditions to verify the control and feathering of the propeller.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight test of the engine-propeller combination provides the control response to steady state, transient, and emergency operating conditions to verify the control and feathering of the propeller.

7.5.11 Propeller control.

Standard: Propeller control system maintains propeller system operation without exceeding defined operating limits and without detrimental permanent deformation or damage when operated, under steady state and transient conditions, IAW 7.5.2 for the stated life documented in technical data. The propeller control system does not adversely affect the capability established in 7.5.1. The architecture accommodates all control mode operations (e.g., automatic, manual, or emergency) including all failure conditions (e.g., loss of oil pressure, control linkage failure, loss of electrical voltage or continuity, overspeed, overtorque) that result in loss of thrust to maintain level flight or non-recoverable in-flight shutdown.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., propeller control system design documentation, hardware drawings, system configuration descriptions, installed hardware, design data records, inspection reports, or other substantiating records/reports) to verify the architecture accommodates both steady state and transient engine operation, including failure conditions, across the entire flight and ground envelope.

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Test.

(1) Lab test: Engine-propeller system combination ground-based sea-level testing verifies stability of the control system and the control response to power and speed changes throughout the range of operation.

(2) Ground test: Installed engine-propeller system combination testing verifies stability of the control system and the control response to power and speed changes throughout the range of operation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Air vehicle flight testing verifies stability of the propeller control system and the propeller control response to power and speed changes throughout the range of operation.

7.5.12 Condition monitoring.

Standard: The propeller drive system condition monitoring detects, isolates, and records faults. Faults indicating a hazardous condition result in immediate notification to the flight crew. The drive system condition monitors and sensors (e.g., speeds, oil debris, lubrication system pressures, temperatures, and quantities, control operating modes, usage) are configured and integrated into a diagnostic system that provides accurate information to the flight crew and ground crew. Faults are recorded and are maintained for post-flight download.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., system design documentation, ICD, hardware drawings, system configuration descriptions, installed hardware, design data records, inspection reports, or other substantiating records and reports) to verify that the monitoring system detects, isolates, notifies, and records faults.

Test.

(1) Lab test: Closed loop, fault injection testing verifies the system identifies, isolates, records, and displays faults and failures.

(2) Ground test: Air vehicle ground testing verifies the system identifies, isolates, records, and displays faults and failures.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Fight test: Air vehicle flight testing verifies the drive system condition monitoring system notifies flight crew of faults.

(4) Flight test: After flight testing is completed the faults are recorded and maintained for flight crew and ground crew download.

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7.5.13 Ice control system.

Standard: Propeller system ice control system (e.g., anti-ice, de-ice) prevents damaging ice buildup or provides non-damaging ice removal, in the defined icing environment IAW 7.5.2, at all speeds and power levels. The ice control system operation and ice removal does not result in damage to the propeller system structure, engine, or other air system subsystems/components.

For an automatic ice control system, flight crew can override the control system and manually operate the ice controls system. Ice control system operational temperatures are monitored, and the systems are automatically turned off to prevent damage. Notification is provided to the flight crew in the event of ice control system malfunction.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, materials and manufacturing records, design data records, inspection reports, or other substantiating records and reports) to verify the ice control system prevents damaging ice buildup or provides non-damaging ice removal.

Analysis.

(1) Analysis of the operational and environmental conditions defines the propeller system's critical icing conditions and verifies that the ice control system prevents damaging ice buildup or provides non-damaging ice removal.

Demonstration.

(1) Bench testing of the control system demonstrates ice detection and activation of the ice control system.

Test.

(1) Lab test: Bench testing and propeller system ground testing of ice control system components verify the system's ability to prevent or remove ice prior to it damaging the propeller system.

(2) Lab test: All ice control system controls are tested to ensure that the flight crew can override the control system and manually operate the ice control system.

(3) Lab test: Fault injection testing of the ice control system verifies the ability to properly recognize exceedances and initiate system function shutdown.

7.6 Other propulsion systems.

These criteria apply to the physical and functional design of non-gas turbine engines.

This section should be evaluated for any design or modification involving the non-gas turbine engines, their control system and components, their operating characteristics, manufacturing or repair processes, material substitutions, physical or functional installation characteristics, air vehicle usage changes (e.g., maneuvers, icing operations, drag, or gross weight). Tailoring is required for all atypical propulsion systems.

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7.6.1 Reciprocating engines.

Standard: Reciprocating engine systems meet 14 CFR 33, Subpart C and D Airworthiness Standards. Validate the design margin of the propulsion systems and components, under all intended operational conditions, IAW the approved design, mission, and usage limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspection of the type certificate, if certification exists, against engine specification, mission requirements, and verification results. Original type certificate data is required to verify the operational and environmental conditions.

Analysis.

(1) When the type certification does not meet the military operational and environmental requirements, conduct additional analysis to validate that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

(2) When the type certification does not exist, conduct analysis to determine that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

Test.

(1) Lab test: When the type certification does not meet the military operational and environmental requirements, conduct additional ground-based test cell tests to validate that systems and components can perform the intended functions in all intended environmental and operating conditions.

(2) Lab test: When the type certification does not exist, conduct ground-based test cell tests to determine that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: When the type certification does not meet the military operational and environmental requirements, conduct additional flight tests to validate that systems and components can perform the intended functions in all intended environmental and operating conditions.

(4) Flight test: When the type certification does not exist, conduct flight tests to determine that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

7.6.2 Atypical propulsion systems.

Standard: Other propulsion system configurations (e.g., rotary, electric, pulse detonation engines) meet 14 CFR 33 or Section 7. Validate the design margin of the propulsion systems and

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components, under all intended operational conditions, IAW the approved design, mission, and usage limits. Tailoring is required for all atypical propulsion systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspection of the type certificate, if certification exists, against engine specification, mission requirements, and verification results. Original type certificate data is required to verify the operational and environmental conditions.

Analysis.

(1) When the type certification does not meet the military operational and environmental requirements, conduct additional analysis to validate that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

(2) When the type certification does not exist, conduct analysis to determine that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

Test.

(1) Lab test: When the type certification does not meet the military operational and environmental requirements, conduct additional ground-based test cell tests to validate that systems and components can perform the intended functions in all intended environmental and operating conditions.

(2) Lab test: When the type certification does not exist, conduct ground-based test cell tests to determine that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: When the type certification does not meet the military operational and environmental requirements, conduct additional flight tests to validate that systems and components can perform the intended functions in all intended environmental and operating conditions.

(4) Flight test: When the type certification does not exist, conduct flight tests to determine that the propulsion systems and components can perform the intended functions in all intended environmental and operating conditions.

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8. AIR VEHICLE SUBSYSTEMS

This section describes the primary subsystems commonly found on air vehicles. These subsystems include hydraulic and pneumatic, environment control, fuel, fire and hazard protection, landing gear and deceleration, auxiliary power, emergency power, mechanisms, and rotary-wing drive. Additional subsystems for rotary-wing air vehicles addressed in this section include external cargo hook, external rescue hoist, and FRIES.

8.1 Hydraulic and pneumatic systems.

8.1.1 Redundant hydraulic system operation.

Standard: The loss of any one hydraulic system will not result in flying qualities worse than Level 2 or loss of any SCF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review hydraulic system drawings and schematics for redundant sources of hydraulic power to ensure SCFs (e.g., flight control actuation, deceleration, and landing gear extension) are maintained in the event of the loss of one hydraulic system.

Analysis.

(1) Conduct FMECA and hazard analyses (SSHA and SHA) to identify failures and their consequences. Verify that a loss of a single hydraulic system does not result in loss of SCFs.

(2) Conduct analysis of the hydraulic system to verify that in the event of a single system failure, pressures and flows of the redundant (normal or backup) hydraulic system meet demands for continued operation of SCFs.

Test.

(1) Lab test: Conduct FMET using HITL (e.g., iron bird) which simulates the complete installation of critical hydraulic, flight control, landing gear and door components, and line lengths to introduce system faults and verify flying qualities and SCFs are maintained.

(2) Ground test: Conduct on air vehicle testing to validate analysis and laboratory test results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify that the normal, backup, and emergency hydraulic systems' performance throughout the operational envelope prevents the loss of a SCF (e.g., handling qualities, flight control actuation, and emergency landing gear extension). Hydraulic system performance is monitored throughout the flight test program to document system performance if unplanned failures occur. Flight test data is examined and compared against previous analysis, ground, and lab test data, and reconciled accordingly to show redundancy is provided to maintain SCFs.

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8.1.2 Single point failures.

Standard: Any hydraulic and pneumatic system(s) single point of failure will not result in catastrophic or critical consequences. Failures include those caused by external damage in a location where multiple hydraulic systems are in close proximity as specified by SAE AS5440B, System Separation paragraph, or feed the same component (e.g., a location where separate systems have components close together, or transfer valves, switching valves, and actuators are fed by more than one hydraulic system).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system drawings and schematics to identify if two or more systems are in close proximity (e.g., switching valves, actuators) and whether a single point failure will result in the loss of two or more hydraulic systems. Verify system separations prevent failures from physical damage (e.g., battle damage, bird strike, and air vehicle component failures) which would result in a loss of two hydraulic systems.

Analysis.

(1) Verify FMECA and hazard analyses (SSHA and SHA) identify the consequence of each single point failure.

(2) Conduct analysis, including modeling the hydraulic system, to assess the consequence and impacts to system performance (e.g., pressures and flows) of identified single point failures in the hydraulic systems (see Analysis MoC (1) results) to verify that the results are not catastrophic or critical.

8.1.3 Power levels and transition.

The hydraulic and pneumatic systems maintain power to meet air vehicle demands during normal, backup, and emergency operation for all conditions. Transition between normal and backup or emergency systems is predictable and manageable.

8.1.3.1 Power sources and sizing.

Standard: The hydraulic and pneumatic system power sources and lines are sized to supply hydraulic and pneumatic power (flow rates and pressures) during normal, backup, and emergency operating modes to meet the demands of the air vehicle across its design envelope (ground and flight environmental conditions). The total hydraulic fluid volume, including reserves, is sized to provide for system fluid exchanges, simultaneous hydraulic system actuation, compressibility, thermal effects, and leakage. Reservoirs are designed and sized IAW SAE AS5586A, Reservoir Capacity paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review system design documentation (e.g., system specifications, ICDs, and drawings) to verify hydraulic and pneumatic system design requirements (e.g., operating pressure, fluid type, system capacity, allowable leakage rates, and startup time) are defined.

(2) Inspect system description documentation to verify which hydraulically and pneumatically powered subsystems are powered during normal (e.g., flight controls, weapon delivery), backup (e.g., landing gear extension), and emergency (e.g., landing gear extension blow down and flight controls) operating modes.

Analysis.

(1) Conduct steady state and dynamic performance analysis to determine that system flow and pressure demands are met throughout all normal and backup modes of operation during critical phases of flight, under worst-case flow and pressure demands generated by all combinations of multiple interfacing systems (e.g., flight controls, landing gear and weapons systems) operating simultaneously including a cold soaked system.

(2) Conduct steady state and dynamic performance analysis to verify that emergency hydraulic and pneumatic power systems (e.g., pumps, blow down bottles, accumulators) meet system flow and pressure demands of flight-critical systems to maintain stable flight and landing capability for the time it takes to land the air vehicle under all environmental and worst-case operating conditions (e.g., carrier landing which is a high system demand) as defined by the air vehicle requirements.

(3) Conduct analysis to verify fluid reservoirs are sized to contain the fluid volume required to meet the hydraulic systems' demands, including leakage rates, during all operating modes, and under worst-case flow and pressure demands generated by all combinations of multiple interfacing systems (e.g., flight controls, landing gear, and weapons systems) operating simultaneously including a cold soaked system.

Test.

(1) Lab test: Conduct HITL (e.g., iron bird) testing to verify that power demands are met for air vehicle SCFs (e.g., flight controls movement, landing gear bay doors and extension) during steady state and dynamic (transient) operations, under worst-case flow and pressure demands generated by all combinations of multiple interfacing systems (e.g., flight controls, landing gear, and weapons systems) operating simultaneously. Simultaneous operation of multiple subsystems verifies there are no resulting failure modes, anomalies or issues that impact stability of flight controls or safety-critical utility systems.

(2) Lab test: Conduct HITL (e.g., iron bird) testing to verify that emergency hydraulic and pneumatic systems meet power demands for flight-critical systems to maintain stable flight and landing capability for the time it takes to land the air vehicle under worst-case operating conditions as defined by the air vehicle requirements. Exercising the system verifies there are no known failure modes, anomalies, or issues that would cause degradation or failure of SCFs of the air vehicle.

(3) Lab test: Conduct HITL (e.g., iron bird) testing to verify that fluid reservoirs are sized to contain the fluid volume required to meet the system(s) demands during all operating modes, under worst-case flow and pressure demands generated by all combinations of multiple

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interfacing systems (e.g., flight controls, landing gear, and weapons systems) operating simultaneously.

(4) Ground test: Conduct on air vehicle testing to verify that hydraulic and pneumatic normal and backup systems meet power demands of the air vehicle under worst-case flow and pressure demands generated by all combinations of multiple interfacing systems (e.g., flight controls, landing gear, and weapons systems) operating simultaneously including a cold soaked system.

(5) Ground test: Conduct on air vehicle testing to verify that emergency hydraulic and pneumatic systems meet power demands for flight-critical systems for the time it takes to land the air vehicle when operating under worst-case conditions as defined by the air vehicle requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(6) Flight test: Conduct flight testing to verify that normal and backup hydraulic and pneumatic systems meet the power demands of the air vehicle throughout the design envelope of the air vehicle under worst-case flow and pressure demands generated by all combinations of multiple interfacing systems (e.g., flight controls, landing gear, and weapons systems) operating simultaneously including a cold soaked system.

(7) Flight test: Conduct flight testing to verify that emergency hydraulic and pneumatic systems meet the power demands for flight-critical systems for the time it takes to land the air vehicle when operating under worst-case conditions as defined by the air vehicle requirements.

8.1.3.2 Transition of power to backup and emergency systems.

Standard: The air vehicle has the means to transition from the normal power system to the backup and emergency power system. Transition may include isolation of the system (automatically or manually) in the event of a failure in the normal hydraulic or pneumatic system to identify the location of a leak. Transition of power from the normal system to the backup and emergency system maintains stable flight throughout the transition (e.g., no detrimental pressure spikes or drop in power). This applies to all phases of operation throughout the design envelope of the backup or emergency system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspect system architecture documents (e.g., system description document, schematics, and models) to verify logic and methods of transition from normal to backup and emergency systems are defined and emergency procedures are provided for the flight crew.

Analysis.

(1) Conduct analysis to verify that transition from normal to backup and emergency systems occurs in time to meet system operational requirements and that no power drop or pressure spikes causes uncommanded or unresponsive flight control inputs or loss of SCFs.

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Test.

(1) Lab test: Conduct HITL (e.g., iron bird) testing to verify that transition from normal to backup and emergency systems occurs in time to meet system operational requirements and that no power drop or pressure spikes causes uncommanded or unresponsive flight control inputs or loss of SCFs.

(2) Ground test: Conduct on air vehicle test to verify that transition from normal to backup and emergency systems occurs in time to meet system operational requirements and that no power drop or pressure spikes causes uncommanded or unresponsive flight control inputs or loss of SCFs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to verify that transition from normal to backup occurs in time to meet system operational requirements and that no power drop or pressure spikes causes uncommanded or unresponsive flight control inputs or loss of SCFs.

(4) Flight test: Conduct flight test to verify that transition from normal to emergency systems occurs in time to meet system operational requirements and that no power drop or pressure spikes causes uncommanded or unresponsive flight control inputs or loss of SCFs.

8.1.3.3 Interfaces.

Standard: All interfaces (inputs and outputs) with the hydraulic and pneumatic system are fully defined. Hydraulic and pneumatic systems are powered by inputs from external power sources (e.g., gearboxes, air bottles, batteries, and ram air turbines). Power and drive system interfaces meet the demands of the hydraulic and pneumatic systems. Hydraulic and pneumatic systems provide output power to other interfacing systems (e.g., flight controls, landing gear, and weapon bay doors). The hydraulic and pneumatic systems provide the required power (pressure and flow) to meet the requirements of all interfacing components and systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review ICDs and system specification documentation to verify that interfaces with other systems are defined.

Analysis.

(1) Conduct analysis to verify that the static and dynamic performance of the air vehicle's hydraulic and pneumatic systems meet the demands of interfacing systems.

Test.

(1) Lab test: Conduct HITL testing to verify static and dynamic performance of the system meets the demands of interfacing systems.

(2) Ground test: Conduct on air vehicle testing to verify static and dynamic performance of the system meets the demands of interfacing systems.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify static and dynamic performance of the system meets the demands of interfacing systems.

8.1.4 Fluid operating temperatures.

Standard: The air vehicle maintains hydraulic fluid operational temperatures within the minimum and maximum limits IAW SAE AS5440B, Fluid Temperature Limitations paragraph throughout the design envelope. The hydraulic system is designed to operate within the specified fluid temperature limits and contains features (e.g., relief valves, shut-off valves, and fuses) to prevent system failure, degraded system performance, or leakage which could create a fire hazard condition. Thermal conditioning functions (e.g., heat exchangers, warming circuits, warming orifices, and temperature transducers) are incorporated to maintain specified fluid operating temperatures to prevent high fluid temperature effects to include degradation of seals, over pressurization, and leakage as well as low fluid temperature effects to include high fluid viscosities, restricted flow, leakage, and filter bypass.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documents (e.g., system and fluid specifications, drawings, and design review data) to verify system temperature limits are defined. Inspection of drawings verifies features are included for thermal conditioning.

Analysis.

(1) Conduct analysis of hydraulic systems to verify cold startup, steady state, and dynamic system operation, under worst-case flow and pressure demands generated by all combinations of multiple interfacing systems (e.g., flight controls, landing gear, and weapons systems) operating simultaneously and dormant systems activating during normal and worst-case operating environmental temperatures of the air vehicle. Hydraulic fluid temperatures are verified to meet interfacing system requirements and are maintained within fluid property limits and requirements. Cold soak and hot soak analysis verifies hydraulic system and components will function as intended during operational use. Thermal conditioning solutions are assessed for proper system sizing and ability to maintain fluid temperature limits.

Test.

(1) Lab test: Conduct HITL (e.g., iron bird) testing to verify analysis showing hydraulic fluid temperatures are maintained within limits during cold startup, steady state, and dynamic system operation under worst-case flow and pressure demands generated by all combinations of multiple interfacing systems (e.g., flight controls, landing gear, and weapons systems) operating simultaneously and dormant systems activating. Testing also verifies features (e.g., relief valves, shut-off valves, and fuses) and thermal conditioning solutions perform as required to meet the demands of the air vehicle.

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(2) Ground test: Conduct on air vehicle testing to verify analysis showing hydraulic fluid temperatures are maintained within limits during cold startup, steady state, and dynamic system operation, including multiple interfacing systems operating simultaneously and dormant systems activating during normal and worst-case operating environment temperatures of the air vehicle. Cold soak and hot soak testing verify hydraulic systems and components will function as required during operational use. Testing also verifies features (e.g., relief valves, shut-off valves, and fuses) and thermal conditioning solutions perform as required to meet the demands of the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to verify hydraulic fluid temperatures are maintained within limits during steady state and dynamic system operation throughout the design envelope of the air vehicle. Flight testing verifies in-flight dormant systems (e.g., weapons bay doors, refueling slipway, and cargo ramp) will function as intended during operational use. Testing also verifies thermal conditioning solutions perform as required to meet the demands of the air vehicle.

8.1.5 Flight crew displays.

Standard: The air system provides the means at crew station(s) for flight crew to indicate the hydraulic and pneumatic system parameters (e.g., fluid quantity, pressure, and temperature) and hydraulic system malfunctions and faults for each system IAW SAE AS5440B, System Pressure Indication paragraph, System Low-Pressure Warning Light paragraph, and Reservoir Fluid Level Indication paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings, flight crew interfaces, and flight manual(s) to verify hydraulic and pneumatic system information is indicated to the flight crew.

Analysis.

(1) Conduct analysis, such as a SSHA and FMECA, to verify that system failures and corrective actions are identified.

Test.

(1) Lab test: Conduct laboratory testing (e.g., HITL and FMET) or air vehicle simulations to verify that hydraulic and pneumatic system status, malfunctions, and faults are indicated to the flight crew.

(2) Ground test: Conduct on air vehicle testing under both normal system operation and with simulated failures to verify that hydraulic and pneumatic system status, malfunctions, and faults are indicated to the flight crew.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(3) Flight test: Verify that the hydraulic and pneumatic system status, malfunctions, and faults are indicated to the flight crew.

8.1.6 Technical manuals.

Standard: Flight manuals include hydraulic and pneumatic system normal, backup, and emergency procedures, system parameters, system status indications, WCAs, restrictions and air vehicle operating limitations. Maintenance manuals include hydraulic and pneumatic systems life limits, inspections, limitations, and servicing procedures. The manuals' information is substantiated and reflects the as designed or modified system. The technical manuals incorporate unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, qualification testing findings).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manuals to verify that the flight crew is provided with information addressing WCAs, system status indications, limits of operation, normal, backup, and emergency procedures.

(2) Review technical manuals to verify they incorporate unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, qualification testing findings).

Analysis.

(1) Conduct supporting analyses to establish the basis for the published operating procedures, flight manual limits, and the system operating limits.

(2) Conduct supporting analyses to establish the basis for the published maintenance manual life limits, inspections, limitations, and servicing procedures including unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, qualification testing findings).

Test.

(1) Ground test: Conduct on air vehicle testing to validate and verify flight manual normal and emergency procedures for hydraulic and pneumatic systems.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct on air vehicle testing to validate and verify flight manual normal and emergency procedures for hydraulic and pneumatic systems.

8.1.7 Hydraulic and pneumatic plumbing installations.

The installation of the system components, hoses, fittings (including repair fittings), tubing, and mounting hardware accounts for operation under combined conditions and environments. Examples of conditions include vibration, thermal expansion, platform bending, structural

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loading, acceleration, and shock. The criteria below address both hydraulic and pneumatic systems as both system types utilize similar standards. The DoW recognizes SAE AS5440B requirements referenced in this subsection as acceptable for meeting both hydraulic and pneumatic criteria.

8.1.7.1 Clearance and separation.

Standard: Multiple systems are physically separated IAW SAE AS5440B, System Separation paragraph, to prevent loss of SCFs due to damage. Hydraulic and pneumatic lines will have a minimum separation of 0.25-inch, IAW SAE AS5440B, Tubing Clearance paragraph, with adjacent structure, tubing, and other installations. Positive separation from wiring is IAW SAE AS50881H, Gas and Fluid Carrying Lines and Tubes paragraph. Clearances are maintained under all operating conditions, environments, tolerance stack-up, and installation preloads to avoid problems such as binding, chafing or jamming. Hydraulic lines (including hoses, drains, and vents) are separated and routed below all ignition sources (e.g., exhaust stacks, electrical wiring, avionics, and insulating materials) to prevent risk of fire from line leakage.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system-level drawings or CAD models to verify that specified tubing and hose clearances are designed into the system.

(2) Review results of a quality assessment or a government in-person sampling of physical inspection(s) (e.g., 30-60-90% of build inspections) of the air vehicle to verify minimum hydraulic and pneumatic line separation is maintained.

Demonstration.

(1) Conduct air vehicle functional checkout of tubing and flexible hose installations to verify that clearances are maintained throughout the whole range of motion and under worst-case environmental conditions (e.g., cold soak, hot soak, and air vehicle vibrations and loading). The effects of actuation, loading, and line movement will also be observed as system is exercised to verify line clearances are maintained to prevent chafing of lines and kinking of hoses.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) Conduct visual inspection of hoses, tubing, and any adjacent lines, components, or structure after flight test to verify there is no evidence of contact (e.g., chafing).

8.1.7.2 Clamping.

Standard: Clamping of hydraulic and pneumatic tubing and hoses is IAW SAE AS5440B, Tubing Support paragraph and Hose Support paragraph. This includes clamping types, location of clamps, spacing, and identifying natural frequencies of installed tubing. Tubing and its supporting hardware are not used to provide support of other air vehicle installations, such as wiring, other air vehicle tubing (e.g., fuel and air lines), or similar installations. Where hydraulic tubes of different diameters are connected, spacing of the smallest tube diameter will be used.

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Marriage or butterfly clamps attached between hydraulic or pneumatic tubing may be used to maintain clearance, but not for support of such tubes. Marriage or butterfly clamps attached between hydraulic or pneumatic tubing and other system lines, such as wiring, other air vehicle tubing, or similar installations are prohibited.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system-level drawings, processes (manufacturers process specifications), and CAD models to verify that the hydraulic and pneumatic system design meets specified clamp spacing, has provisions for tubing expansion and contraction, and lines are not attached via clamps to other tubing, wiring, or components.

(2) Review results of a quality assessment or a government in-person sampling of physical inspection(s) (e.g., 30-60-90% of build inspections) of the air vehicle to verify hydraulic and pneumatic lines are clamped IAW the standard.

Analysis.

(1) Review analysis to verify natural frequencies of tube installations are identified and that no resonant frequencies exist with surrounding air vehicle structure and systems.

Test.

(1) Ground test: Conduct RAP testing on installed tubing for critical systems to verify the analysis and that no resonant frequencies exist with surrounding air vehicle structure and systems.

8.1.7.3 Design for irreversibility.

Standard: Line fittings and component connections are designed such that they are unable to be installed in reverse or inadvertently cross connected with different systems. Hydraulic and pneumatic lines are permanently marked IAW SAE AS5440B, Tubing and Fitting Identification paragraph, to indicate direction of flow, the system designation, and component operated.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system-level drawings, processes (manufacturer's process specifications), and CAD models to verify that the hydraulic and pneumatic systems are designed to prevent reverse installation of lines or components, and that lines are properly marked.

(2) Review results of a quality assessment or a government in-person sampling of physical inspection(s) (e.g., 30-60-90% of build inspections) of the air vehicle to verify hydraulic and pneumatic lines and components are properly marked.

8.1.8 Hydraulic and pneumatic pressure.

Over pressurization can cause detrimental effects to tubing and components that could result in unexpected failures and potential loss of flight-critical and SCFs. Therefore, it is important that

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adverse pressure fluctuations are precluded from the system. Back pressure can cause inadvertent activation of system components (e.g., internal actuator locks and brakes). Insufficient pump head pressure can cause cavitation, which can cause rapid wear, premature component failure, and loss of hydraulic system function.

8.1.8.1 System relief valves.

Standard: System pressure relief valves and thermal expansion relief valves are provided to prevent sustained excessive pressures which may cause component structural failures IAW SAE AS5440B, Relief Valves, System, and Thermal Expansion paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of drawings verifies that pressure relief valve(s) and thermal expansion valve(s) are in place to prevent the hydraulic and pneumatic system from exceeding peak pressure limits.

(2) Review component specification documents to verify that the relief valve(s) cracking and reseating pressures are within hydraulic and pneumatic system peak and rated pressure requirements. Hydraulic system pressure requirements are IAW SAE AS5440B, Surge Pressure paragraph and SAE AS5440B, Pressure Limitations paragraph.

Analysis.

(1) Conduct analysis to verify that the relief valves have a capacity equal to or greater than the rated flow of the largest pump when two or more pumps have a common pressure line IAW SAE AS5440B, System Relief Valves paragraph.

Test.

(1) Lab test: Conduct HITL (e.g., iron bird) testing to verify system pressure is maintained within system limits.

8.1.8.2 Back pressure.

Standard: The hydraulic systems are designed such that proper functioning of components (e.g., internal actuator locks and brakes) and other subsystems are not affected by the dynamic back pressure in the return system. The hydraulic systems are designed such that failure of a component in the system does not prevent the proper functioning of other subsystems, emergency systems, or alternate systems due to back pressure in the return side of the system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct flow analysis of each system's return pressure to determine maximum back pressure levels to verify interfacing components are not inadvertently actuated (e.g., brakes).

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Demonstration.

(1) Conduct ground demonstration on air vehicle to verify each system's return pressure does not inadvertently actuate system components (e.g., brakes).

Test.

(1) Lab test: Conduct HITL (e.g., iron bird) testing to measure maximum return back pressure to verify flow analysis.

(2) Lab test: Conduct HITL (e.g., iron bird) testing to verify that back pressures do not adversely affect SCFs IAW SAE AS5440B, Back Pressure paragraph and SAE AS5440B, Brakes paragraph.

(3) Ground test: Conduct on air vehicle testing to measure maximum return back pressure to verify flow analysis.

8.1.8.3 Pressure spikes.

Standard: Hydraulic and pneumatic systems withstand pressure spikes across the design envelope of the air vehicle. System peak pressures do not exceed the limits for their respective pressure class IAW SAE AS5440B, Surge Pressure paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify peak pressures during system actuation of components do not exceed 135% of nominal system operating pressure. For 8000 psi systems, do not exceed 120% of nominal system operating pressure.

Test.

(1) Lab test: Conduct HITL (e.g. iron bird) testing to verify peak pressures during any phase of the system operation do not exceed 135% of the nominal system operating pressure, when systems and components are exercised. For 8000 psi systems, do not exceed 120% of nominal system operating pressure.

(2) Ground test: Conduct on air vehicle test(s) to verify peak pressures during any phase of system operation does not exceed 135% of the nominal system operating pressure, when various systems and components are exercised. For 8000 psi systems, do not exceed 120% of nominal system operating pressure.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Monitor peak pressures during flight testing to verify peak pressures during any phase of system operation does not exceed 135% of the nominal system operating pressure, when various systems and components are exercised. For 8000 psi systems, do not exceed 120% of nominal system operating pressure.

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8.1.8.4 Pump pulsation.

Standard: Hydraulic pump pulsation (ripple) generated by high-speed rotating fluid equipment (e.g., engine driven pumps, electrically driven pumps, and ground cart pumps) does not induce resonant vibrations causing failure in the air vehicle system's tubing, components, and system supports (e.g., clamps and brackets) across the complete speed range (including start and stop), pressures, and flows of the pump IAW SAE AS5440B, Pump Pulsation paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct modeling or analysis to determine pump ripple frequencies and to verify they do not cause resonant frequencies or fatigue, resulting in failure of adjacent tubing and components.

Test.

(1) Lab test: Conduct air vehicle representative HITL (e.g., iron bird) testing to verify pump ripple frequencies and to verify they do not cause resonant frequencies or fatigue, resulting in life reduction of adjacent tubing and components.

(2) Ground test: Conduct on air vehicle testing to verify each approved ground cart's pump ripple frequencies and to verify they do not cause resonant frequencies or fatigue, resulting in life reduction of adjacent tubing and components.

(3) Ground test: Conduct on air vehicle testing to verify each air vehicle pump's ripple frequency and to verify they do not cause resonant frequencies or fatigue, resulting in life reduction of adjacent tubing and components.

8.1.8.5 Pump head pressure.

Standard: Minimum head pressures are maintained for all hydraulic system pumps during normal, backup, and emergency operation to prevent pump cavitation across the design envelope of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of pump inlet pressure to verify that minimum head pressure is supplied to pump(s) across the design envelope of the air vehicle.

Test.

(1) Lab test: Conduct HITL (e.g., iron bird) testing of pump(s) inlet pressure to verify that minimum head pressure is supplied to pump(s) while operating multiple systems simultaneously.

(2) Ground test: Conduct on air vehicle testing of pump(s) inlet pressure to verify that minimum head pressure is supplied to pump(s) while operating multiple systems simultaneously.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(3) Flight test: Conduct flight testing of pump(s) inlet pressure to verify that minimum head pressure is supplied to pump(s) across the design envelope of the air vehicle.

8.1.9 Impurities.

Buildup of particulates can cause performance degradation of a system, clogged orifices, abrasion, and wear which can lead to premature failure of components. Contaminating fluids (e.g., water or non-hydraulic fluids) or air in the system can cause degraded system performance which could affect controllability and safety of the air vehicle. Sampling is used to determine contamination levels, both particulate and contaminating fluids, and bleed ports are used to remove air and contaminating fluids.

8.1.9.1 Particulate filtration and fluid cleanliness.

Standard: The hydraulic and pneumatic filtration systems will maintain a particulate contamination level that complies with the air vehicle specifications for system cleanliness during flight, ground, and filling operations. If no cleanliness level is specified, the hydraulic filtration system will maintain a particulate contamination level of Class 8 (or cleaner) IAW SAE AS4059G, Contamination Levels paragraph and Maximum Contamination Limits for Differential Particle Counts (particles/100 mL) table, or Class 5 (or cleaner) IAW NAVAIR 01-1A-17, Hydraulic Fluid Contamination chapter and Navy Standard for Particulate Contamination table. The level of water in each hydraulic system is less than 250 parts per million. The level of chlorine is less than 100 parts per million. The levels of other contaminants are defined and meet the requirements of the air vehicle specification. Filters are fully qualified or obtained from a QPL of prior approved filters against a military specification (e.g., MIL-DTL-8815) and meet requirements for the air vehicle system in terms of pressure drop and flow rate through the filter assembly.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review QPL to verify that the filter element has been fully qualified.

(2) Inspect documentation such as system design specifications and industry specifications to verify that contamination (e.g., particulate, chlorine, water, and barium) levels are defined and documented.

(3) Inspect hydraulic ground cart system specifications to verify that the cart's filtering capability meets the required air vehicle minimum cleanliness level.

Test.

(1) Lab test: For unqualified parts or parts not on the QPL, conduct lab testing to verify that the filter is qualified to QPL specifications.

(2) Lab test: Conduct HITL (e.g., iron bird) testing to verify that filtration system performance (e.g., pressure drop, flow rate, and cleanliness) meets air vehicle requirements during static and dynamic system operation.

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(3) Ground test: Conduct on air vehicle testing to verify that filtration system performance (e.g., pressure drop and flow rate) meets air vehicle requirements during static and dynamic system operation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct fluid sampling on each system prior to flight test to establish a baseline cleanliness level and sample each system after first 20 hours of flight test to verify the system(s) maintains the required cleanliness level.

8.1.9.2 Fluid contaminant sampling.

Standard: Provisions exist for taking representative fluid samples and removal of particulates and contaminating fluids (e.g., water or non-hydraulic fluids) via bleed ports from the hydraulic system. Sampling locations are kept free from external contaminants.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of documentation such as system design drawings and specifications to verify that sampling ports exist and are protected from external contaminants.

(2) Review of documentation such as system design drawings and specifications to verify the location of fluid bleed ports.

(3) Review of maintenance procedures to verify that methods for sampling and fluid contaminant removal are documented.

8.1.9.3 Entrapped air removal.

Standard: Provisions exist for bleeding free air from each hydraulic system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of documentation such as system design drawings and specifications to verify the location of air bleed ports.

(2) Review of maintenance procedures to verify that methods for air removal are documented.

Demonstration.

(1) Conduct on air vehicle ground demonstration to verify air removal procedures.

8.1.10 Qualification.

Qualification testing demonstrates that all hydraulic system components (including lines and fittings) function properly throughout their service life in their intended environment. Air vehicle flight testing can be started once a defined amount of qualification data is obtained from lab

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testing (e.g., component SOF testing) and has been determined to meet minimum SOF requirements to permit initial flight test with proper restrictions and life limits in place.

8.1.10.1 Component qualification.

Standard: All hydraulic and pneumatic components, either individually or as part of a subsystem, have passed all required qualification tests (e.g., proof, burst, impulse, endurance, EMI, containment, over-speed, acceleration, pressure cycling, temperature cycling, and fluid compatibility) IAW the applicable paragraphs of SAE AS8775. Components meet qualification requirements as identified in individual component specifications (e.g., pumps, valves, actuators, reservoirs) in SAE AS5440B, Applicable Documents paragraph. Proof and burst pressures are tested to the values IAW SAE AS5440B, Pressure Limitations paragraph. Life limits and restrictions are defined when components are not fully qualified.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review qualification test plans to verify that qualification test requirements are fully defined.

(2) As an alternative to Examination MoC (1), review SOF testing requirements to verify minimum levels of testing are defined in the event full qualification is not achieved before first flight.

(3) Review flight and maintenance manuals to verify life limited parts and restrictions are documented.

Analysis.

(1) Conduct analysis for components that are qualified by similarity to verify that they address differences in qualification test parameters, component design, materials, installation, integration, and intended usage and environment from the originally qualified component, to meet current qualification requirements.

Test.

(1) Lab test: Conduct full qualification testing to verify the ability of hydraulic subsystem components to perform their intended function and meet specified requirements within the installed operating environment.

(2) Lab test: As an alternative to Test MoC (1), conduct SOF testing to verify components meet minimum testing requirements in the event full qualification is not achieved before first flight.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Complete full qualification testing of any components not fully qualified prior to flight test in MoC (2) to verify the ability of the hydraulic system and components to perform their intended function and meet specified requirements within the installed operating environment.

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8.1.10.2 Lines and fitting qualification.

Standard: Hydraulic and pneumatic systems utilize either previously qualified lines (tubing and hoses) and fittings from a QPL (maintained by DoW or industry) or new materials are fully qualified. Tubing is designed and qualified IAW SAE AS5440B, Tubing Material paragraph. Fittings are designed and qualified IAW SAE AS5440B, Fittings and Boss Fittings paragraphs. Hoses are designed and qualified IAW SAE AS5440B, Hose Assemblies paragraph. Proof and burst pressure for lines and complete system(s) are tested to the values IAW SAE AS5440B, Pressure Limitations paragraph. Impulse testing is conducted IAW SAE AS603B. Flexure testing is conducted IAW SAE ARP1185C.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspect drawings and documentation to verify lines and fittings (including repair fittings) are from a QPL.

Test.

(1) Lab test: Conduct lab testing of any tubing, hoses, and fittings not on a QPL to verify they meet all required qualification tests.

(2) Ground test: Conduct on air vehicle proof pressure test on complete hydraulic system(s) to verify proper installation of tubing, hoses, and fittings and that leaks are within limits.

8.2 Environmental control systems.

An airworthy ECS starts with good systems engineering practices as identified in Section 4 to ensure the as-designed ECS incorporates all applicable air vehicle system safety requirements to include failure mode management and can be installed and operated as intended providing hazard free operation and system interactions. The design should encompass the intent of provisions from applicable military specifications prescribing methods of design and construction, selection of materials and components, and failure provisions.

8.2.1 Equipment and subsystem interfaces.

Standard: The ECS meets safety-critical interface requirements (e.g., temperature, pressure, airflow, humidity) for equipment and subsystems under normal and emergency operating conditions throughout the air vehicle flight envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review ICDs and subsystem specifications to establish safety-critical interface requirements for the ECS.

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Test.

(1) Lab test: Conduct system-level testing in an integration laboratory to verify that the ECS meets all safety-critical equipment and subsystem interface requirements (e.g., temperature, pressure, airflow, and humidity) in the installed environment.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that the ECS meets safety-critical equipment and subsystem interface requirements throughout the flight envelope.

8.2.2 Pressurization.

Standard: The ECS provides a pressure environment for occupants and critical equipment, as required by the air vehicle specification, under normal and emergency operating conditions throughout the design envelope. The pressurization subsystem is free of pressure instabilities under all flight conditions, and the ECS controls pressurization rates to prevent pressure surges in occupied compartments. Check valves or restrictions prevent sudden pressure loss from occupied compartments to the ambient environment or ECS ducting in the event of ECS shutdown or failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine schematics and installation drawings to verify that the location and orientation of components, plumbing, safety provisions, and ambient sense ports are compatible with the air vehicle flight profile and performance envelope.

(2) Review air vehicle specification to verify pressurization schedules are defined.

Analysis.

(1) Conduct analysis to verify that the pressurization system design and components are properly selected for the air vehicle size, performance, and ECS flow rates, and provide pressurization IAW with the defined pressure schedule over the air vehicle flight envelope.

Test.

(1) Lab test: Conduct subsystem-level lab testing to verify that the defined air vehicle pressure schedule and rates of change are maintained throughout the air vehicle flight envelope.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight testing to verify the analysis and that pressure schedules and rates are maintained as specified by air vehicle pressurization requirements across the air vehicle flight envelope.

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8.2.3 Degraded system operation.

Standard: The ECS maintains safety-critical functionality during degraded operation, including thermal conditioning of safety-critical equipment and maintaining occupied compartment ventilation long enough to enable successful recovery of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct FMECA and SHA to verify that loss of ECS function does not result in SOF hazards.

(2) Perform subsystem-level analyses (e.g., thermal analysis, pressurization) to verify that safety-critical functionality is maintained until termination of flight during degraded operation.

(3) Perform analysis to verify that emergency ventilation provisions meet ventilation requirements after loss or degradation of the ECS.

Test.

(1) Lab test: Perform component qualification testing to verify that safety-critical equipment function is maintained until termination of flight following loss or degradation of the ECS.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that safety-critical equipment does not exceed maximum operating temperatures during degraded operation.

8.2.4 Technical manuals.

Standard: Flight manuals, checklist, and other flight crew technical documents contain ECS normal and emergency procedures, operating modes, controls and displays, WCAs, restrictions and operating limitations, system descriptions, and other engineering data. Maintenance manuals contain ECS servicing, maintenance procedures, and component life limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manuals to verify accuracy and completeness of ECS normal and emergency procedures, limitations, system descriptions, operating modes, controls and displays, WCAs, and other relevant engineering data.

(2) Review maintenance manuals to verify that manual(s) include inspections and safety-critical component life limits to address system design performance issues and incomplete qualification testing status.

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Analysis.

(1) Conduct supporting analyses to establish the basis for published operating procedures, flight manual limits (e.g., emergency time to land in worst-case conditions), and system operating limits.

(2) Conduct supporting analyses to establish the basis for published maintenance manual procedures, inspections, and component life limits due to incomplete qualification status, field inspection results, or field service deficiency reports.

Test.

(1) Ground test: Conduct ground test to validate and verify flight manual normal and emergency procedures.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight test to validate and verify flight manual normal and emergency procedures.

8.2.5 Flight crew interfaces.

8.2.5.1 Environmental control system status and fault indications.

Standard: The air system provides the means at the flight crew station to indicate ECS parameters, system faults, degraded system performance, and flight-critical equipment temperature limit exceedances.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings and flight crew interfaces to verify that ECS information is provided to the flight crew to monitor ECS performance and to notify of ECS failures, degradation, or flight-critical equipment temperature limit exceedances.

Analysis.

(1) Conduct analysis, such as a SSHA and FMECA, to verify that ECS failures are identified.

Test.

(1) Lab test: Conduct lab testing (e.g., HITL, FMET) or air vehicle simulations to verify that ECS information and notifications of ECS failures, degradation, and flight-critical equipment temperature limit exceedances are provided to the flight crew.

8.2.5.2 Crew station environmental control system controls.

Standard: Crew station allows the flight crew the ability to configure the ECS as required to maintain SCFs (e.g., manual cabin pressure release, selection and shutoff of bleed sources, and cabin temperature) over the air vehicle flight envelope.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings, flight crew interfaces, and operating procedures to verify that crew station allows for the flight crew to configure the ECS as required to maintain SOF throughout the flight envelope.

Demonstration.

(1) Conduct on air vehicle demonstration to verify that the crew station allows the flight crew to configure the ECS as required to maintain SCFs.

Test.

(1) Flight test: Conduct flight testing to verify that crew station allows the flight crew to configure the ECS as required to maintain SCFs.

8.2.6 Personnel accommodation.

8.2.6.1 Occupied compartment environment.

Standard: The ECS provides thermal conditioning and humidity control to all occupied compartments of the air system throughout the design envelope. Temperature limits are IAW MIL-STD-1472, Heating, Ventilating, and Air Conditioning (HVAC) for Manned Spaces paragraph. Humidity control is IAW MIL-E-18927, Humidity Control paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct thermal analysis to verify that the ECS maintains thermal conditioning and humidity levels within established limits throughout the flight envelope.

Test.

(1) Ground test: Conduct ground testing to verify that the ECS maintains thermal conditioning and humidity levels within established limits throughout the ground envelope.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that the ECS maintains thermal conditioning and humidity levels within established limits throughout the flight envelope.

8.2.6.2 Protective flight garment air supply.

Standard: The ECS supplies air at the pressure, flow, temperature, and humidity compatible with protective flight garment supply systems (e.g. pressure suits, anti-g garments, or ventilation garments).

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify that the ECS meets protective flight garment interface requirements under all air vehicle operating conditions.

Test.

(1) Ground test: Conduct ground testing to verify that the ECS meets protective flight garment interface requirements under air vehicle ground operating conditions.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that the ECS meets protective flight garment interface requirements under all air vehicle flight operating conditions.

8.2.7 Personnel air quality.

The ECS protects the flight crew's breathing air from contamination in all forms, including oil leakage in the engine and other harmful contaminants ingested by the engine, by providing air ventilation and air exchange rates.

8.2.7.1 Ventilation.

Standard: Ventilation airflow of the occupied spaces and equipment compartments during normal system operating conditions is IAW MIL-E-18927, Airflow paragraph. Fresh air that has not been recirculated is introduced into occupied compartments by the ECS.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify that the ECS meets the minimum ventilation airflow rates under normal operating conditions throughout the flight envelope.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(1) Flight test: Conduct flight testing to verify the ECS meets the minimum ventilation airflow rates under normal operating conditions throughout the flight envelope.

8.2.7.2 Isolation of occupied compartments.

Standard: Crew station allows the flight crew to shutoff air flow to occupied compartments to prevent introduction of smoke, fumes, toxic gases, or other such contaminants into the occupied compartments breathing air (when the source of contamination is the ECS).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review schematics, drawings, and flight manuals to verify that crew station and procedures are in place to allow the flight crew to shutoff airflow to occupied compartments.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(1) Conduct air vehicle level demonstration to verify the ability that the crew station allows the flight crew to shutoff air flow to occupied compartments.

8.2.8 Environmental protection.

8.2.8.1 Environmental protection systems.

Standard: The ECS provides environmental protection as required for SOF throughout the air vehicle flight envelope. Subsystems provide for removal of rain, snow, ice, and fog from transparencies, and protection of critical flight surfaces and engine inlets from ice accretion. Ice shed from air vehicle surfaces and propulsion components (propellers and spinners) does not present a hazard from ingestion into engines or impact on the airframe. No single environmental protection subsystem failure including control station SCFs results in loss of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to define the environmental conditions that drive protection requirements across the air vehicle flight envelope.

(2) Conduct analysis to verify the capability of the environmental protection subsystems to maintain SOF under all environmental conditions throughout the air vehicle flight envelope.

(3) Conduct FMECA and SHA for the air system to verify that no single point failure of the environmental protection subsystems results in loss of the air vehicle.

Test.

(1) Lab test: Conduct lab testing of critical flight surfaces and engine inlets in an icing tunnel under simulated flight conditions to verify the results of analyses used to establish environmental protection requirements.

(2) Ground test: Conduct ground testing in a climatic chamber under simulated flight conditions to verify that environmental protection subsystems meet the required performance.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(3) Flight test: Conduct flight testing to verify that environmental protection subsystems function as intended and meet the required performance to maintain SOF under all environmental conditions throughout the air vehicle flight envelope.

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8.2.8.2 Icing environment.

Standard: A means exists of monitoring the flight environment for icing conditions (atmospheric conditions of visible moisture and temperature that are conducive to airframe and inlet icing) or incipient icing by the flight crew or an automated system throughout the air vehicle flight envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documents to verify that methods of monitoring the flight environment for potential icing environments are provided.

Test.

(1) Lab test: Conduct system-level testing in an environment laboratory to verify that flight environment monitoring provisions detect potential icing conditions prior to accumulation of ice on the air vehicle.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight testing with a chase air vehicle to verify that flight environment monitoring provisions detect potential icing conditions prior to accumulation of ice on the air vehicle.

8.2.8.3 Ice detection.

Standard: A means exists of monitoring external surfaces and detecting ice accumulation by the flight crew or an automated system throughout the air vehicle's flight envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documents, flight manuals, and technical data to verify that limits for ice accumulation or ice accretion rates are established to allow the flight crew or an automated system to actuate environmental protection systems as required for SOF.

Analysis.

(1) Conduct analysis to verify that ice detection systems or methods provide an indication of ice accretion before accumulation on critical surfaces and engine inlets results in loss of engine power or degradation of handling qualities beyond Level 2.

Test.

(1) Lab test: Conduct laboratory testing in an integration lab or icing tunnel to verify that ice detection indications (e.g., WCAs or an ice detection signal to automated ice protection systems) are initiated before critical accretion thresholds are exceeded.

Prior to entering operational test or operations, verify the standard by the following method(s):

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Test.

(2) Flight test: Conduct flight testing to verify that ice detector indications occur before accretion on critical surfaces affects SOF.

8.2.9 Bleed air.

8.2.9.1 Bleed air ducting and components.

Standard: Bleed air ducting and components operate within the design operating temperatures and pressures without material or functional degradation. Bleed air ducting and component surface temperatures do not present a SOF hazard for nearby structure, wiring, flammable fluids and materials, and equipment. Installation prevents impingement and chafing of ducting under all air vehicle flight conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspect drawings to verify that standoff distances and clamping prevent chafing and impingement of ducting and components.

Analysis.

(1) Conduct analysis to determine worst-case bleed air pressure and temperature combinations under normal operating and failure conditions.

(2) Conduct thermal analysis to verify that bleed air component surface temperatures do not result in any SOF hazard to nearby structure, wiring, flammable fluids, or equipment.

Test.

(1) Lab test: Conduct component-level testing to verify that bleed air ducting and components meet the structural integrity requirements IAW JSSG-2009, Appendix D, Proof Pressure paragraph.

(2) Lab test: Conduct component-level testing to verify that bleed air ducting and components meet the structural integrity requirements IAW JSSG-2009, Appendix D, Burst Pressure paragraph.

8.2.9.2 Leak monitoring and detection.

Standard: A leak monitoring and detection system is employed for bleed air or other compressed air sources as required for SOF. A WCA notifies the flight crew when a bleed air leak occurs. Leak detection system sensors maintain functionality following exposure to a leak, and functionality can be verified.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review drawings and design documents to verify that a leak detection system is provided as required and that leak detection sensors are located where leakage of high-temperature compressed air would create an ignition source or damage critical structure, wiring, or equipment.

Analysis.

(1) Conduct analysis to verify that bleed air leak alarm thresholds meet temperature requirements for critical structure, flammable fluid auto-ignition temperatures, and nearby flight-critical equipment temperature limits.

Demonstration.

(1) Conduct a ground demonstration to verify that leak detector functionality can be verified by the flight crew.

Test.

(1) Lab test: Conduct component-level qualification testing to verify that leak detection sensors are actuated at the established alarm threshold temperatures.

(2) Lab test: Conduct system-level laboratory testing to verify that a bleed air leak WCA is annunciated in the event of a bleed air leak detection.

(3) Lab test: Conduct system-level laboratory testing to verify that bleed air leak detection sensors recover functionality following a leak detection after the source of the leak has been secured.

8.2.9.3 Bleed air shutoff.

Standard: Provisions exist for bleed air shutoff in order to provide the air vehicle with secure means for preventing bleed air from jeopardizing SOF. No single point bleed air system failure causes an uncontrollable flow of high temperature bleed air into the interior of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review installation drawings and CAD models to verify that shutoff provisions prevent leakage of bleed air from creating a fire hazard or jeopardizing critical structure, wiring, or equipment.

Analysis.

(1) Conduct FMECA and SHA to verify that no single point bleed air system failure results in an uncontrolled flow of high temperature bleed air into the interior of the air vehicle or at the bleed source.

Test.

(1) Lab test: Conduct system-level testing to verify that bleed air shutoff and isolation provisions stop the flow of bleed air as intended, and that the timing and redundancy of bleed air shutoff functionality meets system requirements.

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8.2.10 Equipment thermal management.

Standard: The ECS meets all SOF thermal conditioning requirements of the equipment specifications throughout the flight envelope. Mass flow and delivery temperature of thermal management media (e.g. bleed air, ram air, vapor cycle refrigerant, polyalphaolefin) meet thermal requirements of flight-critical equipment during continuous operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct thermal analysis to verify that ECS thermal conditioning meets the thermal management requirements of the air vehicle and maintains flight-critical equipment temperatures within operating limits across the flight envelope.

Test.

(1) Lab test: Conduct system-level testing to verify that the ECS meets the thermal management requirements of flight-critical equipment in the installed environment.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Flight test: Conduct flight testing to validate the thermal analysis and to verify that the ECS meets the thermal management requirements of flight-critical equipment throughout the flight envelope.

8.2.11 Smoke and fume removal.

Standard: The ECS provides a means for removal of smoke and fumes from all occupied compartments. Smoke, fumes, and noxious gases are cleared with sufficient rapidity to prevent asphyxiation, incapacitation, and loss of cockpit visibility.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review ECS design documentation and installation drawings to verify the availability of smoke and fume evacuation provisions.

(2) Review flight manuals to verify the availability of smoke and fume evacuation procedures.

Analysis.

(1) Conduct analysis to verify that smoke and fumes are cleared via fresh air exchange or by compartment pressure release.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(1) Flight test: Conduct flight testing to verify the effectiveness of smoke and fume removal provisions and procedures.

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8.3 Fuel system.

The air vehicle fuel system ensures a reliable supply of fuel to the engine(s) under all operating conditions. It typically includes fuel tanks, fuel lines, vent lines, pumps, valves, fuel quantity measurement systems, and system monitoring equipment. Larger air vehicles may have multiple fuel tanks, typically in the wings and fuselage, with pumps for each engine and transfer systems for managing fuel distribution. Fuel systems also include components for refueling, defueling, and maintaining proper air vehicle CG.

For the purposes of these criteria, an approved fuel includes the base fuel itself, plus any additives that are required as part of the specification for that fuel. Changing the combination of base fuels and additives is considered a different fuel.

8.3.1 Integration.

Standard: The fuel system is functionally and physically compatible with other air vehicle systems (e.g., propulsion, hydraulic, electrical, pneumatic, inerting, display, AR and other unique interfaces) and other ground support equipment (e.g., fuel trucks, hydrant delivery systems, closed circuit refueling).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation (e.g., ICDs) to verify interface requirements of the fuel system are established between air vehicle systems.

Analysis.

(1) Conduct analysis of interface requirements between the fuel system and other systems to verify functional and physical compatibility.

Test.

(1) Lab test: Conduct component and fuel system integration testing to verify the functional and physical compatibility of the fuel system with other air vehicle systems.

(2) Ground test: Conduct fuel system function checkout tests to verify physical and functional compatibility between the fuel system and other systems.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to show compatibility between the fuel system and other systems while in flight.

8.3.2 Qualification tests.

Standard: All components, either individually or as part of the fuel system, have completed and passed qualification testing (e.g., durability). Life limits and restrictions are defined because of qualification testing.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of SOF testing documents or qualification test reports to verify ability of the fuel system and subsystem components to operate and perform their intend function, within the installed operating environment.

(2) Review flight and maintenance manuals to verify life limits and procedures are documented due to incomplete qualification test or failures which occurred during qualification testing.

Analysis.

(1) Conduct component SOF and qualification analysis to verify requirements, capabilities, and limitations.

Test.

(1) Lab test: Conduct qualification testing to verify the ability of the fuel system components to perform their intended function and meet specified requirements within the installed operating environment.

(2) Lab test: As an alternative to Test MoC (1), conduct SOF testing to verify components meet minimum testing requirements in the event full qualification is not achieved before first flight.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Complete full qualification testing of any components not fully qualified prior to flight test in Test MoC (2) to verify the ability of the fuel system and subsystem components to perform their intended function and meet specified requirements within the installed operating environment.

8.3.3 Failure modes and effects.

Standard: No single failure or likely combination of failures of the fuel system result in loss of air vehicle, cause additional damage or failure, or reduce fuel supplied to any system to below what is required for continued flight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct FMECA and SSHA to identify failures and their consequences.

(2) Conduct analysis of fuel system performance with single failure or likely combination of failures to show failure will not cause loss of air vehicle, reduce fuel flow to the engine below what is required for continued flight, or cause further system failures. Any associated limitations or restrictions needed for continued flight are identified.

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Test.

(1) Lab test: Conduct lab HITL (iron-bird) testing to introduce failure modes, evaluate system responses, and validate analysis results.

(2) Ground test: Conduct ground tests to validate analysis and lab test results, including failure modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to verify fuel system performance under normal and simulated failure conditions.

8.3.4 Operating pressures.

Standard: The system is designed such that fuel system operation, including single failures or likely combinations of failures, do not result in fuel pressures that exceed positive and negative pressure limits for any component, plumbing, or tank structure. Fuel system operations include valve closures, simultaneous valve closures, operation of fuel pumps, AR operations, fueling, defueling, fuel feed and transfer, and any other air vehicle operation. Steady-state pressures do not exceed normal operating pressure limits. Transient pressures do not exceed proof pressure limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct system analysis that steady-state pressure limits are not exceeded throughout steady state air vehicle operation.

(2) Conduct system analysis that verifies proof pressure limits are not exceeded throughout air vehicle operation.

Test.

(1) Ground or Lab test: Conduct simulator or ground surge testing that verifies proof pressure limits are not exceeded during normal and single failure conditions.

(2) Lab test: Conduct fuel system simulator tests that verify negative pressure limits are not exceeded.

(3) Ground test: Conduct fuel system ground tests that verify negative pressure limits are not exceeded.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing that verifies analysis and previous testing that shows steady-state pressures do not exceed normal operating pressure limits during normal steady-state system operations.

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(5) Flight test: Conduct flight testing that verifies analysis and previous testing that shows proof pressure limits are not exceeded during normal system operations, including transients.

8.3.5 Pressure capability.

Standard: Operating, proof (limit), and burst pressure requirements are defined for all plumbing and components of the fuel system (not including fuel tanks) as completely assembled and installed within the air vehicle:

a. Can withstand exposure to the specified proof pressure (2X operating) for five minutes without resulting in fuel leakage, system performance degradation, or decrease in life limit.
b. Can withstand exposure to the specified burst pressure (3X operating) without external fuel leakage.
c. Can withstand negative pressure of one atmosphere (14.7 psi) without collapse or damage of the fuel system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation (e.g., system specification and system design descriptions) to verify operating, proof, and burst pressures are defined for plumbing and components of the fuel system.

Analysis.

(1) Conduct static analysis to verify each component in the fuel system can withstand operating, proof, burst, and negative pressures.

Test.

(1) Lab test: Conduct laboratory testing of component and plumbing to verify the capability to withstand operating, proof, burst, and negative pressures.

(2) Ground test: Conduct air vehicle ground proof tests verify the capability of the system to withstand proof pressures.

8.3.6 Tank pressures and venting.

Standard: Tank pressure does not exceed tank structural or functional limits due to a single failure under normal operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine requirements documentation to verify fuel tank and vent system pressure requirements are defined.

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Analysis.

(1) Perform system analysis that shows relief valves and vent paths maintain tank pressures within limits during maximum climb and descent rates, AR, jettison, refueling, defueling, fuel transfer activities, and Onboard Inert Gas Generating System operation, including with single failures.

(2) Conduct analysis to show thermal expansion of fuel does not create a hazard such as excessive fuel venting or spills.

Test.

(1) Lab test: Conduct component tests that show pressure relief components meet performance requirements to control tank pressures.

(2) Ground test: Conduct ground tests that show the vent system maintains tank pressures within design limits for all air vehicle operations and conditions, including single failure modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight tests that show the vent system maintains tank pressures per requirements for all air vehicle operations and conditions and validates lab and ground test results.

8.3.7 Fuel flow to engine.

Standard: The fuel system supplies the engine(s) with the required amount of fuel IAW engine interface requirements (e.g., pressure, flow, and temperature) during all operations and phases of flight (e.g., afterburner use, inverted flight, fuel jettison, and AR).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to verify fuel delivery requirements to engine(s) are defined.

Analysis.

(1) Conduct a fuel feed system flow analysis and engineering modeling to verify the fuel system provides the engine with fuel IAW engine interface requirements under all operational conditions and maneuvers.

Test.

(1) Ground test: Conduct ground tests to verify the performance of the fuel feed system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight tests to verify the performance of the fuel feed system.

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8.3.8 Fuel transfer.

Standard: The fuel system transfers fuel to meet air vehicle functional requirements under all operational conditions (e.g., CG management, AR, thermal management, fuel jettison, and engine feed). Transfer rates into engine feed tanks will be at least as high as the maximum fuel consumption rate of the engine.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine documentation to show fuel transfer requirements are identified and documented.

Analysis.

(1) Perform analysis to show fuel transfer rates meet air vehicle requirements under all operational conditions.

(2) Perform analysis to show that the fuel transfer subsystem is not affected by operation of the fuel jettison system.

(3) Perform analysis to show fuel jettison functions do not jettison fuel below minimum fuel state requirements.

Test.

(1) Lab test: Perform tests to show the fuel transfer subsystem meets performance requirements under all operational conditions.

(2) Lab test: Perform tests to show that the fuel transfer subsystem is not affected by operation of the fuel jettison system.

(3) Ground test: Perform tests to show that the fuel transfer subsystem is not affected by operation of the fuel jettison system.

(4) Ground test: Perform tests to show the fuel transfer rates meet system requirements.

(5) Ground test: Perform tests to show fuel jettison functions do not jettison fuel below minimum fuel state requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(6) Flight test: Perform flight tests to validate analysis and ground test results that show fuel transfer rates meet performance requirements under all operational conditions.

(7) Flight test: Perform flight tests to verify that the fuel transfer subsystem is not affected by the operation of the fuel jettison system.

8.3.9 Refueling and defueling.

Standard: The air vehicle can be refueled and defueled between any two fuel states without creating an AW hazard (e.g., over and under pressurization, fuel spills, fuel venting, static

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discharges) for both normal and single failure conditions, for all required refueling and defueling methods and modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to show the fuel system can be refueled and defueled without creating a hazard using all available refueling methods and modes, in all internal and external tank configurations, at all expected flight and ground conditions.

Test.

(1) Ground test: Conduct ground test to verify the capability of the air vehicle to be refueled and defueled between any two fuel states without creating an AW hazard, including to final high level shut off for both internal and external fuel tank configurations.

(2) Ground test: Conduct ground test to verify the refueling sequencing of tanks will stay within CG and structural limitations.

(3) Ground test: Conduct ground test to verify all methods and modes of intended refueling such as pressure, gravity, hot pit, and AR (if applicable), do not create AW hazards.

(4) Ground test: Conduct ground test to verify refueling at ground attitude limits do not create an AW hazard such as spills or venting fuel.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight test to verify the results of analysis and ground test performance.

8.3.10 Fuel quantity.

Standard: The fuel system measures and calculates fuel states to the accuracy and precision required for all systems that use fuel data, for all approved fuel and additive combinations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documents to verify the fuel quantity accuracy and precision required for all systems that use fuel data are defined.

Analysis.

(1) Perform analysis that shows the fuel quantity measuring system is capable of meeting design requirements for all fuel and additive types, across all operational conditions.

Test.

(1) Lab test: Conduct component and system tests to show the fuel quantity system meets accuracy and precision requirements using simulated inputs.

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(2) Ground test: Conduct ground tests to verify fuel quantity measurements and calculations meet system requirements for accuracy and precision for all testable configurations and attitudes.

(3) Ground test: Conduct ground tests that verify empty, drained, usable, and trapped fuel configuration weights and CGs are within predicted tolerances.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight test to verify fuel quantity measurements and calculations validate analysis and ground test performance.

8.3.11 Secondary barriers.

Standard: Dual vapor and liquid-proof barriers are installed between fuel and potential ignition sources (e.g., avionics bays, sensor bays) and inhabited areas. The fuel system will detect fuel leakage across the primary barrier and drain this leakage to a location that will not create an additional hazard.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documents to show and verify the presence of two vapor and fluid-proof barriers between fuel tanks and plumbing, and other zones that contain potential ignition sources.

Analysis.

(1) Conduct analysis of design documents to show and verify there is a means to detect a leak in the primary barrier, and to remove leaked fluid and vapors to a location that will not create an additional hazard.

Test.

(1) Ground test: Conduct ground tests to show and verify the performance of the dual barriers, leak detection, and removal of leaked fluids and vapors.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Post-flight inspections determine if any leaks occurred, and if the leak detection system functioned properly to detect the leak.

8.3.12 Explosive atmosphere.

Standard: The fuel system is designed and arranged to prevent the ignition of fuel or fuel vapors within the systems. Fuel system components are explosive atmosphere qualified for the category of fire hazard zone they reside in, and operate without creating a fire or explosion hazard, including single failure conditions. Components inside fuel tanks are intrinsically safe (power levels too low to be ignition sources, low surface temps, and hermetically sealed) including

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failure conditions or will be qualified as explosion proof. Fuel pumps are not ignition sources in failure or run-dry conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation to verify all air vehicle compartments containing fuel components are properly zoned.

(2) Review of fuel system components installations verifies components are properly bonded and grounded.

Analysis.

(1) Conduct a thermal analysis to show all fuel system surfaces do not exceed auto-ignition temperatures.

(2) Conduct thermal, containment, and ignition energy analysis of fuel system components to show them to be intrinsically safe.

(3) Conduct failure analysis of components to show they will not become ignition sources during single failure conditions, including electrical faults.

Test.

(1) Lab test: Conduct explosive atmosphere testing IAW MIL-STD-810, Method 511 Explosive Atmosphere that shows fuel system components are explosive atmosphere qualified for the zone they are located in.

(2) Ground test: Conduct ground testing that shows fuel system bonding and grounding measurements meet design requirements.

8.3.13 Fuel system strength.

Standard: Fuel components, plumbing, tanks, and support structures withstand all expected loading conditions, including all operations for static, dynamic, durability related loads in combination with all internal pressure loads, and ground loads. This includes loads resulting from fuel slosh, vibration, relative movement from fuel lines and fittings, fuel system operation, expected pressure changes, and thermal stress.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct structural analysis that shows the fuel system and support structures will not fail under all operating conditions, including all static, dynamic, and durability related loads, in combination with all internal pressure loads and ground loads.

Test.

(1) Ground test: Conduct air vehicle system-level proof testing to verify installation integrity.

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(2) Ground test: Conduct ground tests that validate analysis that shows the fuel tanks and components can withstand all ground and flight conditions and environments.

(3) Ground test: Conduct ground tests to show that pressure and thermal relief features maintain pressures within design limits.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct post-flight inspections to verify required clearance, support, and installation integrity have been maintained.

(5) Flight test: Conduct flight test to show that the fuel tanks can withstand all flight conditions and environments.

(6) Flight test: Conduct flight tests to verify that pressure and thermal relief features maintain pressures within design limits.

8.3.14 Compatibility with approved fuels.

Standard: The fuel system will function as required under all operating conditions and environments when using approved primary fuels. Alternate, restricted, and emergency fuels, and their limitations or restrictions are identified and documented.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of system requirements verify that primary, alternate, restricted, and emergency fuels and additives are identified.

(2) Review of flight and maintenance manuals verify limitations for the use of alternate, restricted, and emergency fuels and additives are provided to the flight and ground crew.

Analysis.

(1) Conduct analysis that shows air vehicle material compatibility with all required fuels and additives.

(2) Conduct analysis of required fuel and additive types to determine what fuel and additive combinations need to be flight tested, and which combinations verify compatibility by similarity analysis to tested combinations.

Test.

(1) Lab test: Conduct lab tests to verify individual components and sub-assemblies of the fuel system meet performance requirements and are compatible with each fuel type and additive combination required for use in flight test.

(2) Ground test: Conduct ground tests to verify the air vehicle meets performance requirements with each fuel type and additive combination required for use in flight test.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to verify the air vehicle meets performance requirements with each fuel type and additive combination determined as required for flight test by Analysis MoC (2).

8.3.15 Fuel system displays and controls.

Standard: Controls and displays provide the information and control functions needed to operate the fuel system for all phases of flight and operational conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct examination of design document to show interface requirements for the fuel system are defined and the information and controls needed to monitor and operate the fuel system are provided.

Analysis.

(1) Verify FMECA and SHA have determined which failures require flight crew action, and the corresponding failure information is correctly presented to the flight crew so that the correct actions can be taken.

Test.

(1) Lab test: Conduct lab testing to verify that the information and controls provided to the flight crew allow for operation of the fuel system under all operational conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that the information and controls provided to the flight crew allow for operation of the fuel system under all operational conditions.

8.3.16 Technical manuals.

Standard: Flight manuals include fuel system normal, alternate, emergency procedures, system parameters, indications, WCAs, restrictions, and air vehicle operating limitations. Maintenance manuals include fuel system life limits, and servicing procedures. The manuals' information is substantiated and reflects the as-designed or modified system. The technical manuals incorporate unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, qualification testing findings).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review flight manuals to verify that the flight crew is provided with information addressing WCAs, indications, limits of operation, normal, emergency, and alternate procedures.

(2) Review maintenance manuals to verify that manual(s) include inspections and safety-critical component life limits to address system design issues and incomplete qualification testing status.

Analysis.

(1) Conduct supporting analyses to establish the basis for the published operating procedures, flight manual limits, and the system operating limits for the fuel system.

(2) Conduct supporting analyses to establish the basis for the published maintenance manual inspections and component life limits due to incomplete qualification status, field inspection results, and field service deficiency reports.

Test.

(1) Lab test: Conduct lab testing to validate and verify flight manual normal and emergency procedures for the fuel system.

(2) Ground test: Conduct on air vehicle testing to validate and verify flight manual normal and emergency procedures for the fuel system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify flight manual procedures for the fuel system.

8.3.17 Fuel system center of gravity.

Standard: Air vehicle CG is maintained within limits during any fuel system or air vehicle function, including release of stores, AR (if applicable), fuel transfer, fuel dumping operations, wing sweep operations, catapult launches, arrested landings, thermal management, and engine feed. Single crewmember operated air vehicles will maintain CG within limits without action by the crewmember.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that the flight manual defines operational CG limits and includes any fuel system procedures required to prevent any combination of functions and conditions from exceeding any CG limit.

Analysis.

(1) Perform analysis of fuel system procedures and air vehicle operations to show that the air vehicle CG is maintained within limits for all functions and operating conditions.

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Test.

(1) Lab test: Conduct lab tests of the fuel system to show that the fuel system performs as required to maintain air vehicle CG within limits for all functions and operating conditions.

(2) Ground test: Conduct air vehicle ground tests to verify performance of the fuel system for CG management functions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to verify fuel system functions for CG management matches analysis and ground test results.

8.3.18 Contamination.

Standard: The fuel system tolerates specified contamination without degradation, and includes design features (e.g., drains, screens, and filters) and procedures for detecting and removing any excessive contamination (including water) that would degrade fuel system performance or service life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review air vehicle requirements to verify the maximum contamination types and amounts that the fuel system can tolerate are defined.

(2) Review maintenance manuals to verify the required inspections and procedures required to detect and remove excessive contamination are included.

Analysis.

(1) Perform analysis that shows fuel system components are designed to operate with required levels of contamination without performance or service life degradation.

(2) Perform analysis that shows the fuel system is designed to control and purge contaminants to maintain them below required levels.

Test.

(1) Lab test: Conduct individual component qualification testing to verify components can operate with required levels of contamination without performance or service life degradation.

(2) Ground test: Processes and procedures for detecting and removing contaminants are verified.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Monitor flight test to verify that the fuel system controls and removes excessive contaminants.

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8.3.19 Diagnostics.

Standard: Built-in-test and fault isolation provisions identify fuel system faults and failures and accurately inform the flight crew so that appropriate actions may be taken to prevent further damage, correct faults, and continue flight and recovery of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Perform analysis of design to determine the faults and failures of the fuel system that require BIT monitoring and fault isolation to prevent hazardous conditions from developing from single failures or likely combinations of failures.

(2) Perform analysis to determine flight crew actions required for each fault condition are identified and documented.

Test.

(1) Lab test: Perform lab test to verify BIT, fault isolation, and flight crew notification performance for components and fuel subsystem.

(2) Ground test: Perform ground test to verify BIT, fault isolation, and flight crew notification performance for the air vehicle fuel system meets requirements and to confirm the analysis and lab results.

(3) Ground test: Conduct ground test to verify flight crew actions required to troubleshoot, correct faults, prevent further damage or system degradation, continue flight, and recover the air vehicle are correct.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight test to verify BIT, fault isolation, and flight crew notification performance for the fuel system and to confirm the analysis, lab, and ground test results.

(5) Flight test: Conduct flight test to verify flight crew actions required to troubleshoot, correct faults, prevent further damage or system degradation, continue flight, and recover the air vehicle are correct.

8.3.20 Electrical and electromagnetic effects.

Standard: The fuel system prevents fire or explosions due to static electricity discharges, lightning strikes, or the introduction of electrical power into fuel tanks for all flight and ground operations, AR, and fuel leak conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review design requirements to show fuel tank and component sealing design criteria are defined.

Analysis.

(1) Perform a FMECA to verify risks are mitigated to required levels.

(2) Perform analysis to show fuel tank and component sealing and sealant integrity will meet performance and life cycle requirements.

(3) Perform analysis that show E3 on the fuel system are IAW MIL-STD-464, Detailed Requirements section.

(4) Perform analysis that shows lightning protection features will prevent the creation of ignition sources during lightning strikes.

Test.

(1) Lab test: Conduct tests that show fuel tank and component sealing integrity meet performance requirements.

(2) Ground test: Conduct air vehicle inspection and measurements to verify compliance with the air vehicle bonding and lightning protection requirements.

(3) Ground test: Conduct lightning ground tests to verify adequacy of the protection designs.

(4) Ground test: Conduct leak checks to confirm integrity of final assembly.

8.4 Fire and hazard protection.

Fire and explosion protection in air vehicles centers around prevention, detection, suppression, and containment. Prevention includes separation of fuels from ignition sources and oxidizers, drainage, and ventilation. Detection systems are needed in zones with a higher risk of developing a fire, or areas where it is impractical or impossible to separate combustibles and ignition sources, such as engine bays. Suppression is used to extinguish fires that are detected, and typically uses an extinguishing agent discharged into the area of the fire to cool, smother, or inhibit the chemical process of combustion. Containment is used to limit the damage done by a fire and prevent it from spreading to other areas of the air vehicle. Determining the correct approach for these concepts is supported by zoning each area of the air vehicle based on the fire hazards and failure modes present in each area. However, accomplishing this all starts with ensuring as part of the systems engineering process that air vehicle requirements include and integrate FPSs and features into the air vehicle design based on the type of air vehicle, intended use, the potential for fire hazards, and design features present. The fire protection requirements and design are derived from system-level operational needs, interfaces with other air vehicle systems and usage cases, including detection, crewmember action, and air vehicle recovery timelines.

8.4.1 Fire hazard analysis.

Standard: Each compartment of the air vehicle and area surrounding the air vehicle is zoned as a fire zone, flammable zone, flammable leakage zone, ignition zone, or support equipment zone,

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according to the fire and explosion hazards present during normal and single failure conditions. Fire protection is provided for each zone according to the hazards present.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine design documentation to verify that a hazard zone analysis is complete, and each compartment is assigned a fire protection hazard zone type.

Analysis.

(1) Conduct analysis that shows flammable fluids, vapors, or other materials in each compartment are identified.

(2) Conduct analysis that shows potential ignition sources caused by normal or failure modes in each compartment are identified.

(3) Conduct analysis that shows single point failures or likely combinations of failures in each compartment involving flammable materials or ignition sources are identified.

(4) Conduct analysis that shows each compartment is zoned as a flammable zone, flammable leakage zone, ignition zone, support equipment zone, or fire zone.

(5) Conduct analysis that shows the air vehicle includes fire protection in each zone to mitigate fire and explosion hazards identified for each zone.

8.4.2 Fire hardening.

In areas of the air vehicle where it has been determined that fire, explosion, or overheat hazards can occur, fire hardening (fireproof or fire resistant) provisions are used to protect flight-critical and safety-critical components or systems which if damaged by fire, explosion, or overheat conditions, will result in increased or additional hazard or propagation of the hazard to other air vehicle compartments. This includes but is not limited to flight and engine controls, critical structure, flammable fluid and oxidizer tanks or lines, and the FPS itself.

8.4.2.1 Fireproof.

Standard: Flammable fluid and oxidizer lines and tanks, structure, flight and engine controls, fire walls and fire barriers, and any other flight-critical components are fireproof and able to handle a worst-case fire created by the flammable materials present for a minimum of 15 minutes (e.g., JP-8 at 2000 °F and 10 BTU/s/ft^2).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Flight-critical, safety-critical, and fire zone items are identified, and hazard analysis determines the level of fire proofness needed for each item.

(2) Analysis shows that flight-critical, safety-critical, and fire zone items as well as the FPS itself can survive worst-case fire and heat conditions that may be encountered while continuing to

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perform intended function for the time needed (minimum of 15 minutes) to be fireproof as determined by the hazard analysis.

Test.

(1) Lab test: Flight-critical, safety-critical, and fire zone items required to be fireproof are shown to survive worst-case fire and heat conditions that may be encountered while continuing to perform intended function for the time needed (minimum of 15 minutes) to be fireproof as determined by the hazard analysis.

8.4.2.2 Fire resistant.

Standard: Material or components that function to isolate a fire and configure the air vehicle to fight a fire, such as valves, actuators, airflow controls, and powerplant (e.g., engine, APU) installation components such as fluid-carrying line, flammable fluid system components, drainage and ventilation collection systems, wiring, air ducts, fittings, and powerplant controls are fire resistant and able to function under worst-case conditions for a minimum of five minutes (e.g., JP-8 at 2000 °F and 10 BTU/s/ft^2).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Safety-critical or fire zone items are identified, and hazard analysis determines the level of fire resistance needed for each item.

(2) Analysis shows safety-critical and fire zone items can survive worst-case fire conditions that may be encountered while continuing to perform intended function for the time needed (minimum of five minutes) to be fire resistant as determined by the hazard analysis.

Test.

(1) Lab test: Safety-critical or fire zone items are shown to survive worst-case fire conditions that may be encountered while continuing to perform intended function for the time needed (minimum of five minutes) to be fire resistant as determined by the hazard analysis.

8.4.3 Drainage.

Standard: Drains are arranged to remove all flammable or reactive fluid leakage to a location outside the air vehicle. All compartments containing flammable fluid components, and areas surrounding fuel tanks, are drained. Drains are sized to accommodate all leakage, including failure cases. Drained fluids do not re-enter the air vehicle or create any additional hazard. Drain lines are not interconnected where the return of any fluid or vapor may create a fire hazard or damage any interconnected components. Interconnected component drains (e.g., engine combustor drains, pressurizing and dump valve drains, and afterburner or tailpipe drains) are not connected with each other or any other drain and flammable fluid drains are not manifolded with the fluid drains from ignition zones. If drain lines for different fluids are interconnected, fluids do not cause a hazard through contact or mixing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Inspect design documentation to verify drainage features are provided in locations where flammable fluids are present during normal or failure conditions.

(2) Conduct inspection of design documentation that shows flammable fluid drains are not connected to any zones or components that may be ignition sources.

Analysis.

(1) Conduct analysis that show drain locations and drain paths do not allow flammable fluids to accumulate inside the air vehicle or drain lines for all steady-state flight and allowable ground attitudes.

(2) Conduct analysis that shows drains are sized to accommodate the flow rate of possible leakage, including failure cases, of the area or compartment it is expected to drain.

(3) Conduct analysis that shows drained fluids do not re-enter the air vehicle or create additional hazards after exiting the air vehicle.

(4) Conduct analysis that shows pressure differentials due to airflow do not inhibit the performance of drains or limit the ability of fluids to exit the air vehicle.

(5) Conduct analysis of design documentation that shows flammable fluid drains are not interconnected in a way that will allow fluids to create a fire hazard or to damage interconnected components.

(6) Perform analysis that shows fluids in any interconnected drains are compatible and will not cause a hazard when mixed.

Test.

(1) Ground test: Conduct ground test that shows flammable fluids do not accumulate inside the air vehicle at allowed ground attitudes and drain system performance matches analysis.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test that shows flammable fluids are drained from the air vehicle under all steady-state flight conditions, do not re-enter the air vehicle or create an additional hazard, and drain system performance matches analysis.

8.4.4 Ventilation.

Standard: Ventilation systems prevent flammable vapors from exceeding lower explosive limit concentrations and remove all hazardous quantities of flammable or reactive vapors from the air vehicle to a location outside the air vehicle. Vapors do not re-enter the air vehicle or create additional hazards. Airflow reversals or no-flow conditions are not present under any ground or flight conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Inspect design documentation to verify ventilation features are provided in locations where flammable fluids or vapors are present during normal or failure conditions.

Analysis.

(1) Conduct analysis that determines the ventilation and airflow required to prevent flammable vapor concentrations from exceeding the lower explosive limit for any possible flammable vapor present, considering possible leak rates and amounts, under all flight and ground conditions.

(2) Conduct analysis that shows ventilation provisions are able to maintain the airflow required to prevent flammable vapor concentrations from exceeding the lower explosive limit for any possible flammable vapor present, considering possible leak rates and amounts, under all flight and ground conditions.

(3) Conduct analysis that shows ventilation provisions remove all flammable and reactive vapors to a location outside the air vehicle, and those vapors do not re-enter the air vehicle or create additional hazards.

(4) For ground conditions that may generate flammable vapors prior to air vehicle power on or engine start, conduct analysis that shows alternate ventilation methods (e.g., opening panels, using forced air from ground carts) lower flammable vapor concentrations to below lower explosive limits prior to power-on or engine start.

Test.

(1) Ground test: Conduct ground test that shows the ventilation system is capable of providing the airflow required by Analysis MoC (1), Analysis MoC (2), and Analysis MoC (3) for ground conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test that shows the ventilation system provides the airflow required by Analysis MoC (1) and Analysis MoC (2) for flight conditions.

8.4.5 Ignition sources.

Standard: Potential ignition sources are separated from flammable fluids and vapors by two fluid and vapor proof barriers. Equipment in flammable leakage zones and fire zones are shown to not be ignition sources under normal operating conditions. Surface temperatures in flammable and flammable leakage zones are below auto-ignition temperatures, and surface temperatures in fire zones are below hot surface ignition temperatures. Equipment in flammable zones are shown to be explosion proof (containment) under failure conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of design documents to verify that two vapor tight barriers exist between ignition sources and flammable fluids or vapors.

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Analysis.

(1) Conduct analysis that shows a component is intrinsically safe if intrinsic safety (e.g., possible internal ignition sources are hermetically sealed by the component) is being used instead of explosive atmosphere testing.

Test.

(1) Lab test: Conduct lab tests that show that all equipment in flammable leakage zones and fire zones are explosive atmosphere qualified to MIL-STD-810, Method 511, Procedure I, or are intrinsically safe.

(2) Lab test: Conduct lab tests that show that all equipment in flammable zones are both explosive atmosphere qualified to MIL-STD-810, Method 511, Procedure I, and explosion containment qualified to MIL-STD-810, Method 511, Procedure II, or are intrinsically safe.

(3) Lab test: Conduct lab tests that show that all equipment and surfaces in flammable or flammable leakage zones do not have a worst-case surface temperature above the auto-ignition temperature of any flammable fluid or vapor in that zone.

(4) Lab test: Conduct lab tests that show that all equipment and surface temperatures in fire zones do not have a worst-case surface temperature above the hot surface ignition temperature of any flammable fluid or vapor in that zone.

(5) Ground test: Conduct ground tests that show that all equipment and surfaces in flammable or flammable leakage zones do not have a worst-case surface temperature above the auto-ignition temperature of any flammable fluid or vapor in that zone for all ground conditions.

(6) Ground test: Conduct ground tests that show that all equipment and surface temperatures in fire zones do not have a worst-case surface temperature above the hot surface ignition temperature of any flammable fluid or vapor in that zone for all ground conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(7) Flight test: Conduct flight tests that validate results of Test MoC (3), to show all equipment and surfaces in flammable or flammable leakage zones do not have a worst-case surface temperature above the auto-ignition temperature of any flammable fluid or vapor in that zone for all flight conditions.

(8) Flight test: Conduct flight tests that validate the results of Test MoC (4), to show that all equipment and surface temperatures in fire zones do not have a worst-case surface temperature above the hot surface ignition temperature of any flammable fluid or vapor in that zone for all flight conditions.

8.4.6 Single failures.

Standard: Single failures of the FPS that impact the ability of the system to properly function, meet fire protection requirements, or resulting hazards are identified and eliminated by FPS design change or documented procedures and inspections.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of FPS FMECA to show failure modes and criticalities from single point failures are identified and eliminated.

(2) Conduct inspection of flight and maintenance manuals to show inspections and procedures required to eliminate FPS FMECA failure mode hazards are identified and implemented.

Analysis.

(1) Conduct a FPS FMECA that show single failure hazards behave as predicted, and hazards from single point failures are eliminated as predicted.

Test.

(1) Lab test: Conduct lab tests that show single failure hazards behave as predicted, and hazards from failures are eliminated as predicted.

(2) Ground test: Conduct ground test that validates Analysis MoC (1) and Test MoC (1) by showing single failure hazards behave as predicted, and hazards from failures are eliminated as predicted.

8.4.7 Detection and control of overheating.

Standard: The air vehicle detects, and controls overheat conditions from normal or single failure conditions to prevent potential fire and explosion hazards during all flight and ground operations. Compartment temperatures are maintained below the auto-ignition temperature for the possible flammable fluids or vapors present, and surface temperatures are maintained below the hot surface ignition temperature for the possible flammable fluids or vapors present.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of design documents that show performance requirements of the overheat protection systems (e.g., set temperature, activation and reset time, clearance signal time, repeatability, false alarm rate) are defined and based on operational use and overall air vehicle design approach for preventing fires and explosions.

Analysis.

(1) Conduct thermal analysis that validates performance requirements and operational environment of the overheat protection systems.

Test.

(1) Lab test: Conduct lab tests that verify the overheat protection systems meet its intended function and maintains temperatures to prevent potential fire and explosions.

(2) Ground test: Conduct ground tests that show the overheat protection system meets its intended function and maintains temperatures to prevent potential fire and explosions.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to validate Analysis MoC (1), Test MoC (1), Test MoC (2) results for system performance in the installed environment and flight envelope.

8.4.8 Equipment separation.

Standard: Flammable fluids and oxidizers are separated from electrical wiring by at least 0.5-inch. Electrical wiring is routed above flammable fluid lines so that any leakage does not impinge on the wiring.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspections of design drawings to verify that appropriate minimum clearances of at least 0.5-inch are provided between electrical wiring and flammable fluid carrying lines under all operational conditions, and electrical lines are routed above flammable fluid lines.

(2) Conduct inspections of air vehicle as built to verify conformity to design drawings for clearances between the electrical wiring and flammable fluid carrying lines.

(3) Conduct inspections of design drawings to verify oxygen equipment is not installed in a fire zone and flammable fluid lines and oxygen lines are not routed together or in proximity to each other without proper isolation design.

Analysis.

(1) Conduct analysis to show minimum clearances between electrical wiring and flammable fluid carrying lines are maintained under all operational conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Conduct post-flight inspections to verify minimum clearances between electrical wiring and flammable fluid carrying lines are maintained after exposure to operational loading conditions.

8.4.9 Fire zone provisions.

Standard: Sufficient cooling air mass flow through fire zones is provided to ensure that the temperature limits for all items within, including engines or APUs, accessory equipment, the supporting structure, and the compartment itself within the zone, are not exceeded under any operating condition. Surface temperatures are maintained below the hot surface ignition temperature of any flammable fluids present, and compartment air temperature is maintained below the auto-ignition temperature for any flammable vapors present.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct design analysis and thermal model analysis to show the heat rejection and cooling needed to prevent hot surface and auto-ignition sources are provided under all flight and ground operating conditions.

Test.

(1) Lab test: Conduct lab tests to verify thermal analysis of predicted surface temperatures of components.

(2) Ground test: Conduct ground tests to validate analysis that shows the heat rejection and cooling needed to prevent hot surface and auto-ignition sources is provided under all flight and ground operating conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight tests to validate analysis that shows the heat rejection and cooling needed to prevent hot surface and auto-ignition sources is provided under all flight and ground operating conditions.

8.4.10 Fire protection system displays and controls.

Standard: Flight crew station information, including any WCAs and controls, are available to notify the flight crew of the system operating conditions, and allow operation of the FPS. Flight crew are able to check the functioning of each fire detector circuit during all flight and ground conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of design documents to show interface requirements for the FPS displays, controls, and information (including any WCAs) needed to monitor and operate the system to prevent, suppress, extinguish, or mitigate the effects of a fire are defined.

(2) Conduct inspection of design documents to show the controls, displays, and information (including any WCAs) provided to the flight crew allows for the operation of the FPS to prevent, suppress, extinguish, or mitigate the effects of a fire.

(3) Conduct inspection of design documents to show there are means to allow the flight crew to check the functioning of each fire detector circuit during all flight and ground conditions.

Analysis.

(1) Conduct analysis to show the controls, displays, and information (including any WCAs) provided to the flight crew allows for the operation of the FPS under all operational conditions to prevent, suppress, extinguish, or mitigate the effects of a fire.

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Test.

(1) Ground test: Conduct ground test to show the controls, displays, and information (including any WCAs) provided to the flight crew allows for the operation of the FPS to prevent, suppress, extinguish, or mitigate the effects of a fire.

(2) Ground test: Conduct ground test to show there are means to allow the flight crew to check the functioning of each fire detector circuit during all flight and ground conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to show the controls, displays, and information (including any WCAs) provided to the flight crew allows for the operation of the FPS to prevent, suppress, extinguish, or mitigate the effects of a fire.

8.4.11 Protection of potential fire zones.

Standard: Fire extinguishing systems are provided in APU compartments, engine compartments, and other identified fire zones. Detection systems provide timely and accurate indications so the flight crew may perform necessary actions to extinguish or suppress fires. Fire extinguishing and automated fire suppression systems extinguish or contain fires before they spread to another area of the air vehicle or create additional hazards (e.g., thermal failure of components outside the fire zone).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of requirement documents to verify fire detection, extinguishing, and automated fire suppression requirements are defined for each fire zone, and under what conditions the extinguishing and automated fire suppression systems are expected to operate.

(2) Conduct inspection of design documents to verify fire detection, extinguishing, and automated suppression systems are provided for each fire zone.

(3) Conduct inspection of design documents to show that fireproof protective devices are provided to isolate a fire within a defined fire zone from any portion of the air vehicle where a fire can create a hazard.

Analysis.

(1) Conduct analysis for each potential fire location that shows that the fire detection system provides timely and accurate indications of all potential fire types so the flight crew may perform necessary actions to extinguish a fire without it spreading to another area of the air vehicle or creating another hazard.

(2) Conduct analysis that determines the extinguishing agent concentrations and duration levels required to extinguish any potential fire under all flight and ground conditions the fire extinguishing system is expected to operate.

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(3) Conduct analysis that shows that the automated fire suppression system provides the required agent concentrations and duration levels required to extinguish any potential fire under all flight and ground conditions the automated fire suppression system is expected to operate.

Test.

(1) Lab test: Conduct laboratory testing that verifies Analysis MoC (1) and shows performance of the fire detection system to provide timely and accurate indications of potential fires without false alarms.

(2) Lab test: Conduct laboratory testing that validates Analysis MoC (2) and Analysis MoC (3) and shows extinguishing agent concentrations and duration levels are provided by the fire extinguishing system or automated fire suppression system under all conditions the fire extinguishing system or automated fire suppression system is expected to operate.

(3) Ground test: Conduct ground test to show the fire detection system meets fire detection activation parameter requirements and provides timely and accurate indications to the flight crew for all potential fires during ground conditions.

(4) Ground test: Conduct ground test that validates Analysis MoC (2), Analysis MoC (3), and Test MoC (2), and shows extinguishing agent concentrations and duration levels are provided by the fire extinguishing system or automated fire suppression system under all conditions the fire extinguishing system or automated fire suppression system is expected to operate.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight test that validates analysis, lab, and ground testing, and shows extinguishing agent concentrations and duration levels are provided by the fire extinguishing system or automated fire suppression system under all conditions the extinguishing system or automated fire suppression system is expected to operate.

8.4.12 Protection of unoccupied compartments.

Standard: Compartments not continuously occupied by crewmembers (e.g., cargo compartments, lavatories, galleys, partitioned-off crewmember rest areas) include fire detection and extinguishing and automated suppression system features required to mitigate the risk of fire presented by the design and approved cargo types in each compartment. Fires are detected and flight crew are notified within 60 seconds of the fire starting.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspect design requirements to verify approved cargo types and crewmember requirements to determine occupiable compartments that do not have a crewmember present to monitor for smoke/fire.

(2) Inspect design documentation to verify fire detection and fire extinguishing or automated fire suppression requirements have been determined for each occupiable compartment that does not have a crewmember present to monitor for smoke/fire.

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(3) Inspect design and operational documents to show that firefighting procedures have been determined for each occupiable compartment that does not have a crewmember present to monitor for smoke/fire.

(4) Inspect design documents to verify fire detection and fire extinguishing or automated fire suppression systems are installed in each occupiable compartment that does not have a crewmember present to monitor for smoke/fire.

Analysis.

(1) Perform analysis that shows the fire detection features of the system are able to detect fires in each occupiable compartment that does not have a crewmember present to monitor for smoke/fire within 60 seconds so that flight crew actions may be performed to mitigate the fire before it causes damage to flight-critical systems or structure, or spreads to additional areas or cargo.

(2) Perform analysis that shows fire extinguishing or automated fire suppression systems provide required agent concentration in all compartment locations for the required time to extinguish fire for typical and worst-case cargo loading configurations and conditions.

Test.

(1) Lab test: Perform lab testing that shows the fire detection features of the system are able to detect fires within 60 seconds so that flight crew actions may be performed to mitigate the fire before it causes damage to flight-critical systems or structure, or spreads to additional areas or cargo.

(2) Lab test: Perform lab testing that shows fire extinguisher or automated fire suppression systems provide required agent concentration in all not continuously occupied compartment locations for the required time to suppress fire for typical and worst-case cargo loading configurations and conditions.

(3) Ground test: Perform ground test to show each fire detection circuit can be checked for function on the ground.

(4) Ground test: Perform ground testing that shows the fire detection features of the system are able to detect fires within 60 seconds so that flight crew actions may be performed to mitigate the fire before it causes damage to flight-critical systems or structure, or spreads to additional areas or cargo.

(5) Ground test: Perform ground testing that shows fire extinguishing and automated fire suppression systems provide required agent concentration in all not continuously occupied compartment locations for the required time to suppress fire for typical and worst-case cargo loading configurations and conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(6) Flight test: Perform flight test to show each fire detection circuit can be checked for function in flight.

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(7) Flight test: Perform flight testing that shows the fire detection features of the system are able to detect fires within 60 seconds so that flight crew actions may be performed to mitigate the fire before it causes damage to flight-critical systems or structure, or spreads to additional areas or cargo.

(8) Flight test: Perform flight testing that shows fire extinguishing or automated fire suppression systems provide required agent concentration in all compartment locations for the required time to suppress fire for typical and worst-case cargo loading configurations and conditions.

8.4.13 Protection of inhabited areas.

Standard: Provisions exist to prevent a SOF hazard or personnel injury hazard from smoke, vapors, extinguishing agents, flames, or fumes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of design documentation to show provisions exist to prevent smoke, vapors, extinguishing agents, flames, or fumes from potential fire locations from entering inhabited areas or creating additional hazards.

Analysis.

(1) Conduct airflow analysis to show smoke, vapors, extinguishing agents, flames, or fumes from potential fire locations are prohibited from entering inhabited areas or creating additional hazards.

(2) Conduct analysis that shows extinguishing agent discharge in areas that may contain crewmembers or passengers will not create a personnel injury hazard.

Test.

(1) Ground test: Conduct ground test to show smoke, vapors, extinguishing agents, flames, or fumes from potential fire locations are prohibited from entering inhabited areas or creating additional hazards.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to show smoke, vapors, extinguishing agents, flames, or fumes from potential fire locations are prohibited from entering inhabited areas or creating additional hazards.

8.4.14 Fire zone isolation.

Standard: Each fire zone includes features to shut-off all flammable fluids and de-energize all electrical ignition sources entering the fire zone in the event of a fire during all ground and flight operations.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspect design documents to show shut-off valves are present that will prevent flammable fluids or vapors from entering the fire zone during a fire.

(2) Inspect design documents to show all electrical ignition sources in the fire zone are able to be de-energized during a fire.

Analysis.

(1) Conduct analysis to show the activation of fluid shut-off or electrical isolation for one fire zone does not adversely affect any remaining propulsions systems, if present.

Test.

(1) Ground test: Conduct ground tests to show all flammable fluid or vapor lines to a fire zone are able to be closed without causing any additional hazard or adversely impacting any additional propulsion systems, if present.

(2) Ground test: Conduct ground tests to show all electrical equipment in the fire zone are able to be de-energized in the event of a fire without causing any additional hazard.

8.4.15 Electrically powered fire protection.

Standard: Electrical power is provided to all electrically powered fire protection equipment (e.g., fire detection, isolation, suppression) during all phases of operation, including EPS failure conditions, engine start, and battery operations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of electrical design documentation that shows electrical power is provided to all electrically powered fire protection equipment during all phases of operation, including power failure conditions, engine start, and battery operations.

Analysis.

(1) Conduct analysis that shows required electrical power is provided to all electrically powered fire protection equipment during all phases of operation, including power failure conditions, engine start, and battery operations.

Test.

(1) Lab test: Conduct lab test that shows required electrical power is provided to all electrically powered fire protection equipment during all phases of operation, including power failure conditions, engine start, and battery operations.

(2) Ground test: Conduct ground test that shows that required electrical power is provided to all electrically powered fire protection equipment during all ground conditions and operations, including power failure conditions, engine start, and battery operations.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test that shows required electrical power is provided to all electrically powered fire protection equipment during all flight conditions and operations, including power failure conditions, engine start, and battery operations.

8.4.16 Fire zone containment.

Standard: Containment is provided for all fire zones to prevent the fire from spreading to other compartments. Barriers are provided to withstand worst-case temperatures and heat flux for the expected flammable fluids and operating conditions (e.g., for JP-8, 2000 °F fire with a heat flux of 10 BTU/s/ft²) for the time required to take required emergency actions (historically, 15 minutes).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of design documents to verify fire barriers are provided for each fire zone to prevent any fire from spreading to any other compartments.

Analysis.

(1) Conduct analysis that shows fire barriers will prevent the spread of fire outside the fire zone for the time required to perform emergency actions.

Test.

(1) Lab test: Conduct lab testing that shows fire barriers will withstand worst-case temperature and heat flux for expected flammable fluids and prevent the fire from spreading to any other compartment for the time required to perform emergency actions.

8.4.17 Interior finishes and materials.

Standard: Interior finishes and materials are not combustible, and do not release any toxic by-products when exposed to fire or high temperatures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of design documents to show non-combustible materials are used in the interior of the air vehicle.

Analysis.

(1) Conduct analysis that shows any toxic by-products that are created during a fire or exposure to high temperatures.

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Test.

(1) Lab test: Conduct lab testing that shows new materials are non-combustible and do not produce toxic by-products when exposed to fire or high temperatures.

8.4.18 Hand fire extinguishers.

Standard: For air vehicles where crewmembers are expected to manually fight fires, the following minimum number of handheld fire extinguishers are provided:

a. At least one hand fire extinguisher is conveniently located in each occupied compartment.
b. At least one readily accessible hand fire extinguisher is available for use in each cargo compartment that is accessible to crewmembers in flight.
c. At least one hand fire extinguisher is located in, or readily accessible for use in each galley.
d. For air vehicles with passenger compartment capacities greater than 30 passengers, the number of hand fire extinguishers are IAW the table in 14 CFR 25.851.
e. Each hand fire extinguisher is approved for the types of fires and environment expected.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of design drawings to show hand fire extinguishers are provided as IAW the standard to mitigate fire hazards in each location.

Analysis.

(1) Conduct analysis that shows the number and type of hand fire extinguishers in each compartment are capable of mitigating the predicted fire hazards in each area.

Test.

(1) Ground test: Conduct ground test that shows fire extinguishers are accessible and easily accessed by crewmembers and passengers in each area.

8.5 Landing gear and deceleration systems.

The landing gear and deceleration systems provide the air vehicle with the capabilities of holding position, towing, taxi, takeoff, landing touchdown, balanced field, critical field length abort, directional control, and arresting gear systems compatibility. The landing gear and deceleration systems include shock absorbers, landing gear and door actuation mechanisms, nose gear steering system, wheels, brakes, tires, drag chutes, arresting hook systems, anti-skid control, brake control, brake actuation mechanisms, and mooring rings. In addition, specialized systems are included such as kneeling, crosswind positioning, skis, and skids.

The structural aspects of the landing gear and arresting hook systems such as load cases, static strength, and fatigue are covered under Section 5 standards and MoCs. A major part of landing gear and deceleration systems AW is for the components to successfully pass full qualification testing or a minimum of SOF testing before first flight. To achieve full compliance multiple standards will have to be complied with. Those standards and MoCs are contained in Section 4

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for the environmental testing (i.e., MIL-STD-810), Section 5 for structural verification, Section 8.4 for explosive atmosphere, Section 8.5 unique subsection testing, and Section 13 for EMI.

The technical manual standard 8.5.15.5 for landing gear and arresting hook systems include MoCs that are in support of assessments conducted in Section 4 for flight manuals and Section 16 for maintenance manuals.

8.5.1 Landing gear general arrangement.

8.5.1.1 Ground flotation.

Standard: The wheeled landing gear or skid landing system will prevent the air vehicle from exceeding the bearing strength of all airfield and landing surfaces (rigid, flexible or bare soil) on which the air vehicle is to operate.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis of the wheeled landing gear configuration of tire loads, tire spacing, tire pressures, and worst-case design limits of air vehicle weights and CGs, verifies the air vehicle does not exceed the bearing strength of required airfield and landing surfaces. The Aircraft Classification Number/Pavement Classification Number or Aircraft Classification Rating/Pavement Classification Rating analyses are used for rigid and flexible surfaces. Bare soil analysis is conducted using approved methods (e.g., ASD-TR-68-34, Pavement-Transportation Computer Assisted Structural Engineering).

(2) Analysis of the skid landing system configuration of loads, spacing, pressures, and worst-case design limits of air vehicle weights, and CGs, verifies the air vehicle does not exceed the bearing strength of required airfield and landing surfaces.

Test.

(1) Lab test: Conduct tire footprint area measurements IAW MIL-DTL-5041, Footprints paragraph at rated tire pressure and tire pressure for bare soil operations, varying tire pressures and air vehicle weights to support flotation analysis.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) For air vehicles which operate on low California Bearing Ratios surfaces (i.e., semi-prepared and unprepared), conduct an air vehicle ground demonstration to verify the number of passes the air vehicle can make on the specified surfaces.

8.5.1.2 Ground clearances.

Standard: For all operational and servicing conditions (e.g., multivariate landing, taxi, takeoff, ski jumps, all stores loading configurations, engine runs, flat gear with flat tire, and deflated strut situations), the landing gear system maintains ground clearance between any part of the air vehicle and air vehicle components and the ground (e.g., service equipment, air vehicle

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arrestment systems, runway lights, catapult shuttle, and deck obstructions). Clearance will also be maintained between landing gear components, stores, and any part of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Clearance analysis verifies the landing gear system maintains ground clearance for all operational conditions and clearance to stores and parts of the air vehicle as specified in MIL-STD-1289, Store to Aircraft Clearance, Minimum Ground Clearance, and Landing Gear Clearance paragraphs.

Demonstration.

(1) Conduct taxi and turning demonstrations to verify analysis.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Conduct flight demonstration to verify landing gear system maintains ground clearance during normal takeoff and landing operations.

8.5.1.3 Wheel well clearances.

Standard: For retractable gear, a 0.5-inch minimum clearance is maintained within the wheel well between all landing gear components, air vehicle structure, and air vehicle systems including linkages, wiring, hydraulics, fuels, weapons systems, and ECSs during all flight conditions, and during both normal and emergency operation. A minimum clearance of 1 inch is provided between all stores and any portion of the landing gear. Clearance will be maintained over the landing gear system's life including adverse wear effects.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of drawings or model verifies clearances are present to allow the landing gear to move through the various stages of gear retraction and extension for all design, assembly, and manufacturing tolerances.

Analysis.

(1) Kinematic analysis verifies the landing gear system and mechanism clearances for all combinations of free play, installation tolerances, and rigging are maintained through full range of landing gear and door motion and under all design loading conditions (e.g., air loads and temperature ranges) to allow the gear to reach the commanded position.

(2) Clearance analysis verifies the landing gear system maintains a 1-inch clearance with stores throughout the landing gear retraction and extension sequence.

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Demonstration.

(1) Conduct landing gear extend and retract system checkout on the air vehicle during production to verify clearances and no unintended interference conditions exist within wheel well.

Test.

(1) Lab test: Conduct dynamometer testing to verify the thrown and grown tire envelopes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Conduct flight demonstrations to verify clearances (e.g., no chafing or rubbing) are maintained during takeoff, flight, and landing operations.

8.5.1.4 Protection of equipment in wheel wells.

Standard: The effects of a bursting tire, flailing tire tread, and wheel brake temperatures do not degrade or cause loss of SCFs provided by components and structure located on the landing gear and in the wheel wells.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to verify safety-critical components, structure, or shielding can withstand force of tire failures to prevent loss of a SCF.

(2) Conduct thermal analysis to verify safety-critical components and structure can withstand brake temperatures with gear retracted from undetected break failures (e.g., dragging brake) or protected with heat shields.

8.5.1.5 Air vehicle stability and control on the ground.

Standard: The air vehicle will maintain dynamic stability and control, and have no adverse dynamics (e.g., shimmy, gear walk, porpoising, and yaw skids), during taxi, takeoff roll, and landing roll. The landing gear system design, for new and worn conditions, will suppress all divergent loads and forces at all operational ground speeds. Divergent loads and forces caused by runway roughness, tire imbalance or design, brake vibrations, and landing gear natural response will be controlled by either active or passive means to prevent oscillations from becoming detrimental.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Stability and dynamic analysis verify the stability and control of the air vehicle IAW the standard.

(2) Parametric shimmy analysis of both nose and main landing gears verifies natural or augmented dampening is present so that the amplitude of any landing gear oscillations after three

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cycles is reduced to 1/3 or less of the original disturbance. Parametric analysis includes effects of free play, wear, friction variations, allowable mis-service strut and tire conditions, brake and antiskid control, and tire imbalances.

Test.

(1) Lab test: Conduct component bench testing to verify required damping characteristics of any active damper installed using validated input rates and loads.

(2) Ground test: Perform on air vehicle taxi and turning tests at various weights and speeds to verify stability and control of the air vehicle, no adverse dynamics, and no shimmy. If analysis and testing show tendency to shimmy or insufficient damping, specific induced excitation testing (e.g., runover bumps) is required to show no adverse effects to air vehicle (e.g., excessive load cycles on nose landing gear, flight crew physical operations, and controllability).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify stability and control of the air vehicle and no adverse dynamics during takeoff and landing operations.

8.5.1.6 Air vehicle turnover.

Standard: For all possible landing gear configurations, the turnover loads do not exceed a 0.5g load at the critical CG location. Taxi and towing operations include all expected turn speeds and turn radii, all possible strut and tire conditions at all air vehicle weights, and all ground configurations including crowned and sloped surfaces.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct turnover analysis to verify lateral position of the main landing gear keeps the turnover angle from exceeding 54 degrees for ship-based air vehicles and 63 degrees for land-based air vehicles.

(2) Conduct turning analysis to verify landing gear layout allows the air vehicle to conduct turns up to 0.5g.

(3) Conduct turning radius analysis to verify the air vehicle has ability to maneuver from taxiways to runway and to depart runway to taxiway after flight.

Test.

(1) Ground test: Conduct ground test to verify air vehicle's stability in turns.

8.5.1.7 Air vehicle tip back.

Standard: Tip back will not occur for the aft-most CG, including all stores loading, under both static and dynamic conditions such as when maximum braking (air vehicle or tow vehicle) is applied. Placement of the landing gear produces a tip back angle of not less than 15 degrees. Tip

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back will not occur while air vehicle is traveling in the aft direction at 5 miles per hour, on a 3-degree slope or a designated speed and slope based on mission requirements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies tip back does not occur during reverse braking and towing operations.

(2) Conduct analysis to verify placement of the landing gear produces a tip back angle of not less than 15 degrees.

Demonstration.

(1) Conduct air vehicle demonstration to verify tip back does not occur during reverse braking and towing operations.

8.5.1.8 Kneeling.

Standard: For air vehicles equipped with landing gear kneeling capability, the control systems will enable lowering and raising of the air vehicle in a controlled and predictable manner that allows the landing gear to be returned to a configuration required for landing and flight, without sudden or adverse movements that may cause damage or result in an AW hazard. The kneeling system controls will be clearly identifiable, indicate gear position or status, and remain accessible to both ground and flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of drawings and inspection of air vehicle verifies controls are accessible.

(2) Review of technical publications verifies proper procedures and sequencing for kneeling operations.

Analysis.

(1) Analysis verifies the kneeling system is sized to raise and lower air vehicle, maintain air vehicle in the final positions to allow loading and unloading, and be configured for flight. All kneeling movement is controllable.

(2) Static strength analysis verifies air vehicle and landing gear structural components have positive strength margins for DLLs and ultimate loads (1.5X DLLs). Yielding does not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate loads. If support equipment is used, positive strength margins of 3X DLL are maintained for support equipment.

Demonstration.

(1) Conduct demonstration on the air vehicle to verify kneeling capability during required operational and environmental conditions, and the kneeling control system indicates the landing gear is in the commanded position.

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Test.

(1) Lab test: Conduct testing of hardware for kneeling to verify the strength analysis of air vehicle and landing gear structural components strength margins.

(2) Ground test: Conduct ground testing of structural components to verify load and stresses are within design margins.

8.5.1.9 Landing gear hiking and dehiking system.

Standard: Landing gear hiking system positions and supports the air vehicle at the AOA needed for the air vehicle to takeoff within the specified conditions and runway length. The landing gear hikes or dehikes automatically when the air vehicle is in a weight off wheels status or a gear up command is issued to prevent the gear from being damaged or the landing gear from jamming in the wheel well. The position of a hiked landing gear on a manned or UAV is indicated to the flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of drawings and inspection of air vehicle verifies flight crew controls are accessible.

(2) Review of flight manual verifies proper procedures, sequencing for hiking and dehiking operations, and indication to the flight crew.

Analysis.

(1) Conduct sizing analysis to verify the hiking actuation system can raise the air vehicle to the specified AOA.

(2) Conduct static strength analysis to verify the hiking system structural components have positive strength margins for DLLs and ultimate loads. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate loads. Ultimate load is 1.5X DLL.

Test.

(1) Lab test: Conduct testing of hardware to verify hiking system structural components meet limit and ultimate load strength margins.

(2) Ground test: Conduct ground testing of hiking actuation system and structural components to verify hiking system's ability to maintain air vehicle's AOA during takeoff roll, to hike or dehike, and to provide indication to the flight crew of gear position.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing on air vehicle to verify the hiking actuation system and structural component's ability to maintain air vehicle's AOA up to takeoff, to hike or dehike, and to provide indication to the flight crew of gear position.

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8.5.2 Landing gear structure.

8.5.2.1 Prevention of penetration of occupied areas and fuel tanks.

Standard: Structural failure of any landing gear system component(s) will not result in any penetration of occupied areas, fuel tanks, and fuel components. Any penetration damage to hydraulics, electrical systems, and ECS does not cause a catastrophic failure. The failed landing gear does not prevent crewmember and passenger egress.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of landing gear system drawings verifies that expected structural failures of the landing gear system does not result in penetration of occupied areas, fuel tanks, and components.

Analysis.

(1) Verify SHA is completed to show expected structural failures of the landing gear system does not result in penetration of occupied areas, flammable fluid carrying lines, and fuel tanks. Any penetration does not cause a catastrophic failure.

(2) Conduct modeling to verify that for expected structural failures of the landing gear system and resulting collapse of gear does not result in penetration of occupied areas and fuel tanks.

8.5.2.2 Landing gear energy absorption.

Standard: Landing gear energy absorption capability supports the air vehicle for landings at all operational weights and environments. The landing gear can absorb two successive landings within the time to meet air vehicle requirements with no degradation in capability. Landing gear static and dynamic loads generated during landing are considered.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review structural criteria document to verify landing load cases for nose and main landing gear.

Analysis.

(1) Conduct multivariate dynamic analyses (e.g., 6-DOF landing simulation models) to verify approach speeds and landing sink rate conditions to establish the development landing load cases.

(2) Conduct analyses using methods IAW JSSG-2006, Land-based and Ship-based Aircraft Ground Loading Parameters paragraph, MIL-A-8863, or MIL-STD-3063 to define landing ground and deck load cases (e.g., level, tail down, and multivariate) including reserve energy for all USAF fixed-wing air vehicles and USAF and USN rotary-wing air vehicles.

(3) Conduct dynamic analysis of the shock absorber to verify damping characteristics and establish landing limit load cases (e.g., level, tail down, and multivariate) under normal serviced

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shock struts. The analysis includes a reserve energy drop to define landing ultimate load case for USAF air vehicles. Analysis identifies worst-case loads.

Test.

(1) Lab test: Conduct full scale drop testing of nose (or tailwheel) and main landing gear assemblies with wheels spun up in a reverse direction to reflect landing speed at worst-case load conditions to verify metering and damping effectiveness, loads (e.g., spin up and spring back), clearances, strut frictions, stroking, and bottoming characteristics of the landing shock absorbers. The landing gear assembly includes shock absorber, wheels, tires, brakes, actuators, locks, and other structural members as used on the air vehicle. Conduct testing as follows:

a. For USAF air vehicles, testing is conducted IAW SAE AS6053A, Design Landing Tests, Miscellaneous Landing Tests, Equipment Servicing, and Reserve Energy paragraphs.
b. For USN air vehicles, drop testing cases can be defined based on nominal and worst-case landing conditions developed from Analysis MoCs (1) and (2). Resulting loads correlate with analytical predictions.
c. For rotary-wing air vehicles, drop testing is based on a limit sink speed of 12 ft/s in combination with 1g rotor lift and other landing cases including all combinations of the following: all horizontal speeds from 30 kts rearward to the best forward speed for minimum rate of descent in autorotation, air vehicle pitch attitudes (ranging from 15-degrees nose down to 15-degrees tail down), combinations of sinking speed and roll angle lying on and within the envelopes, and brakes locked and unlocked operations for VLs. A reserve energy drop of 14.7 ft/s is performed.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify loads and accelerations from operations are within design envelopes.

8.5.2.3 Shock strut mis-servicing allowables.

Standard: No damage or functional impairment (e.g., dynamic energy absorption) occurs to the landing gear system or air vehicle structure during taxi, takeoff, and landing events with the mis-servicing levels listed in TABLE V.

TABLE V. *Shock strut mis-servicing allowances.*

Air Vehicle	Gas Pressure	Oil Volume	Tire Pressure
Fixed-wing	±10%	100% fluid level minus 0.5-inch of oleo stroke	
Fixed-wing Aircraft Carrier	±15%	±15%	±20%
Rotary-wing	±20%	±10%	±20%

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct dynamic analysis of the shock absorber to verify damping characteristics and establish landing limit load cases (e.g., level, tail down, and multivariate) under mis-serviced conditions. Analysis identifies worst-case loads.

Test.

(1) Lab test: Conduct full scale drop testing of nose and main landing gear assemblies with wheels spun up in a reverse direction to reflect landing speed at worst-case load conditions to verify metering and damping effectiveness, loads (e.g., spin up and spring back), clearances, strut frictions, stroking, and bottoming characteristics of the landing shock absorbers at mis-serviced conditions. The landing gear assembly includes shock absorbers, wheels, tires, brakes, actuators, locks and other structural members as used on the air vehicle. Conduct testing as follows:

a. For USAF air vehicles, drop testing of mis-servicing cases are conducted IAW SAE AS6053A, Miscellaneous Landing Tests paragraph.
b. For USN air vehicles, drop testing cases can be defined based on the nominal and worst-case landing conditions developed from Analysis MoC (1). Resulting loads correlate with analytical predictions.
c. For rotary-wing landing gear is at a limit sink speed of 12 ft/s in combination with 1g rotor lift and address landing cases including all combinations of the following: all horizontal speeds from 30 kts rearward to the best forward speed for minimum rate of descent in autorotation, air vehicle pitch attitudes (ranging from 15-degrees nose down to 15-degrees tail down), combinations of sinking speed and roll angle lying on and within the envelopes, and brakes locked and unlocked operations for VLs.

8.5.2.4 Rebound.

Standard: For nose, main, and tail landing gear, the rebound and dynamic characteristics for takeoff and landing conditions (e.g., design sink rates and landing weights) limit bouncing, rebounding, and damage to the landing gear which causes the air vehicle to become uncontrollable and results in a mishap.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a static strength analysis to verify that each landing gear can sustain rebound loading conditions without gross section yielding or detrimental deformations at or below 115% DLL. The DLL is based on a LF of 20 acting on the unsprung mass of the landing gear when the gear is fully extended and not in contact with the ground after landing impact. No rupture allowing release of piston occur at or below ultimate loads. Ultimate load is 1.5X DLL.

(2) Conduct dynamic analysis to verify nose and main landing gear rebound characteristics on catapult launch.

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Test.

(1) Lab test: Conduct sudden extension tests at various gear stroke levels to verify rebound characteristics. Torque arms retain the piston and struts control gear extension to prevent overloading the gear components air vehicle remains controllable after initial landing impact.

(2) Lab test: Conduct drop testing to verify rebound characteristics.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Landing tests at various sink rates and air vehicle weights verify landing gear's ability to control rebounding to keep air vehicle controllable.

(4) Flight test: Conduct catapult or ski jump testing to verify nose gear rebound characteristics for shipboard usage.

8.5.2.5 Strut stiction.

Standard: The shock strut design allows for controlled vertical movement during loading by minimizing strut friction to prevent binding of struts, damage to air vehicle or prevent weight on wheels interlocks activation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of load versus stroke and strut friction to verify the landing gear struts do not stick, preventing shock absorber from controlling vertical movement of air vehicle.

Demonstration.

(1) Conduct ground demonstrations on air vehicle to verify struts move without binding.

Test.

(1) Lab test: Lab test verifies the load versus stroke properties for the landing gear struts.

8.5.3 Tire load and speed ratings.

Standard: The tire design will support all air vehicle performance requirements (e.g., taxi, turning, braking, takeoff and landing operations, wet and dry runway conditions, and arresting gear cable rollover and strikes) as specified in MIL-DTL-5041, Dynamic Durability paragraph. Tire design accounts for all critical weights up to maximum gross weight, critical and worst-case CG conditions, braking, flotation requirements, and air vehicle velocities so that the tires fully support the air vehicle's ground load envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct performance analysis, to include 6-DOF modeling, to define the maximum expected tire load and speed profiles for all operational requirements. For air vehicles which land on arresting cables, conduct analysis to determine loading on tires.

Test.

(1) Lab test: Conduct tire testing on dynamometer as specified in MIL-DTL-5041, Dynamic Durability Test paragraph, at maximum expected tire loads, distances, and speeds, to verify tire load and speed rating, and pressure limits.

(2) Lab test: Conduct lab testing to verify the tire can withstand a minimum burst pressure of 3X the rated pressure for rotorcraft, 3.5X for land-based air vehicle, and 4X rated pressure for ship-based air vehicle.

(3) Lab test: Conduct lab testing to verify the amount tire deflects under maximum side load and maximum yaw angle as defined from the design multivariate load (e.g., drift, yaw, pitch, and roll sink rate) and pressure conditions including load in which tire bottoms.

(4) Lab test: Conduct tire tests IAW MIL-DTL-5041, Cable Bruise Test paragraph, to verify the tires can withstand the minimum cable bruise load. Testing is done for air vehicles which land directly on cables (e.g., carrier-based air vehicles).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight testing to verify tire design integrity (e.g., tread adhesion, chunking, air retention, load support, cable bruising, and deceleration) and verify the design loads and speeds used in lab qualification testing of the tire are not exceeded during all air vehicle operational requirements.

8.5.4 Wheel loads.

Standard: The wheel design will support normal and emergency ground operations at all air vehicle gross weights and flight configurations. The wheel has the capability to roll on rim without the wheel fragmenting for flat or blown tires.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct load analysis as specified in SAE ARP1493C, Wheel Capacity paragraph, to establish each wheel's load capability.

(2) Conduct a static analysis to verify the wheel has positive strength margins to meet worst-case wheel loads.

(3) Conduct fatigue analysis to verify wheels meet minimum roll life distance of 3,000 miles.

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Test.

(1) Lab test: Conduct static testing IAW SAE ARP1493C, Radial Load Test and Combined Load Test paragraphs and subparagraphs, of the wheel and tire assembly to verify strength capability of the wheel. Design landing radial load test will be conducted with arresting cable for carrier-based air vehicle or other air vehicle which land directly on arresting cable.

(2) Lab test: Conduct lab testing, as specified in SAE ARP1493C, Burst Test paragraph, to verify wheel can withstand the minimum burst pressure.

(3) Lab test: Conduct dynamometer testing of the wheel and tire assembly, as specified in SAE ARP1493C, Roll Test paragraph, to verify the wheel can roll for a minimum of 3,000 miles.

(4) Lab test: Conduct lab testing to verify the wheel can roll on rim at a speed of no less than 10 mph under a load equal to the wheel rated static load for 15,000 ft. During test, no fragmentation of the wheel is permitted. For shipboard catapulted fixed-wing air vehicles, the nose wheel with a tire installed and inflated to carrier pressure should be taxied on a dynamometer flywheel rotating at launch speeds until tire blowout occurs. When tire blowout occurs, the wheel may be unlanded. The wheels with demountable flanges should be angled with respect to the flywheel surface so that the outboard flange is closest to the flywheel. The applied load should be the design limit catapult load perpendicular to the flywheel surface. The tire may be locally abraded to weaken the carcass so that blowout is accelerated.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight testing to verify wheel design integrity and verify that the simulated loads used for laboratory qualification of the wheel are not exceeded during all air vehicle operational requirements.

8.5.5 Wheel and tire burst protection.

Standard: The braked wheels will incorporate thermal sensitive pressure release devices (e.g., fuse plugs) so that tire pressure will be released prior to tire or wheel failure resulting from temperature induced structural degradation. All wheels will have overinflation protection devices designed to release tire pressure before material of wheel or tire is degraded or tire pressure reaches explosive levels.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of drawings verifies overheat and over-pressurization devices are incorporated in wheel and tire system design.

Analysis.

(1) Thermal analysis verifies proper temperature is established to release pressure before wheel or tire material degrades or tire pressure reaches explosive levels.

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Test.

(1) Lab test: Conduct dynamometer testing to verify performance of thermal relief devices during the wheel, brake, and tire assembly testing for the rejected takeoff brake energy tests.

(2) Lab test: Conduct inflation testing of the wheel and tire assembly to verify overinflation protection device releases pressure and at a rate faster than the maximum allowed inflation rate.

8.5.6 Wheel brakes.

8.5.6.1 Brake energy capability.

Standard: The braking system has the energy, torque, and distance performance capability to stop the air vehicle during all operational requirements (e.g., landings and RTOs) and on all specified runway surfaces and conditions. Achieving maximum brake energy will not result in catastrophic failure modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a braking performance analysis to define brake energy capacities to meet operational requirements for normal landings, maximum weight landings, maximum weight RTO, and integrated combat turns. The analysis includes factors such as ambient temperatures, runway altitudes, flight crew reaction times, engine thrust, braking distribution between wheels (e.g., 50/50 split), wing lift, coefficients of friction, and deceleration devices. The use of reverse thrust in RTO analysis is not allowed.

(2) Conduct a thermal analysis of tire, wheel, and brake assembly to verify stopping performance is maintained for all flight operations (e.g., RTOs). Brake temperatures do not cause fires in the assembly, premature release of tire pressure, and loss of torque.

Test.

(1) Lab test: Conduct laboratory dynamometer testing IAW SAE ARP1493C, Wheel Brake Capacity Requirements table, to verify the brakes meet the energy capacities as established by Analysis MoC (1) and stopping performance, at critical weights and speeds, throughout the defined wear range of the brake. Brakes for USN air vehicles are salt water tested as part of this testing. Acceptable performance includes:

a. Minimum average deceleration rates of 10 ft/sec^2 (6 ft/sec^2 for rotary-wing air vehicles) are maintained.
b. No damage to axle or brake attachment points.
c. No damage to brake occurs performing the land plane and maximum landing gross weight stops. Fuse plugs will not release.
d. Deceleration rates are not impacted with a wet brake (i.e., salt water for USN air vehicles).
e. Rejected takeoff testing:
    1. Does not result in a catastrophic over pressurization of tire and wheel assembly.
    2. Tire pressure release occurs at a minimum of one minute after parking brake application.

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3. No condition occurs (or has occurred during the stop), including fire associated with the tire, and wheel and brake assembly, that could damage the air vehicle and prevent evacuation of the air vehicle until five minutes after the conclusion of the stop. The parking brake is applied for a minimum of 3 minutes.

(2) Ground test: Conduct ground testing, using a buildup approach to air vehicle weights and speeds, to verify braking system energy levels, brake temperatures, and deceleration rates compared to laboratory test results. Air vehicle speed is limited to prevent ability of the air vehicle to attain flight.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify braking system energy levels, brake temperatures, and deceleration rates at critical weights and speeds under air vehicle operations (rejected takeoffs and landings) will not result in catastrophic failure modes (e.g., wheel structural failure, failure to release tire pressure, loss of torque, premature tire pressure release, and sustained fires within five minutes after the stop). A maximum gross weight rejected takeoff with fully worn brakes, at specification conditions verifies ability of brakes to stop the air vehicle under worst-case operational conditions.

8.5.6.2 Brake redundancies.

Standard: Single point failures (e.g., electrical, pneumatic, hydraulic or mechanical connecting or transmitting element fails, or if any single source of brake operating energy supply is lost) in the brake system do not adversely affect the brake's capability to produce torque. An alternate means of powering the brakes is available to stop within the operational runway lengths and conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system schematics and drawings to verify alternative power sources and redundancies are provided for braking.

Analysis.

(1) Conduct FMECA and SSHA to identify all single point failures in the braking system that could result in failure to stop the air vehicle, and mitigations are in place to prevent loss of brakes.

(2) Conduct static sizing analysis to verify the brake maintains torque to stop the air vehicle under single point failures.

Demonstration.

(1) Conduct on air vehicle demonstration to verify alternative power source automatically becomes active under single point failures to the primary power source.

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Test.

(1) Lab test: Conduct lab testing (e.g., HITL or iron bird) to verify that brake system redundancies are capable of stopping the air vehicle within the operational runway lengths and conditions.

8.5.6.3 Brake torque.

Standard: Braking system is sized to provide holding torque sufficient to prevent wheel rotation with engines at run-up thrust and for ship-based air vehicles to hold air vehicle on sloped decks IAW MIL-B-8584, Design paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Static analysis verifies piston size, disk area, and applied brake pressure produces sufficient torque to prevent wheel rotation with engines at run-up thrust or slope.

Demonstration.

(1) Conduct on air vehicle demonstration to verify brake system torque sizing prevents wheel rotation

Test.

(1) Lab test: Conduct lab testing to verify brake system holding torque capability.

8.5.6.4 Brake adjusters.

Standard: The brakes maintain running clearances throughout the wear life of the brake heat stack including thermal effects to allow the air vehicle to meet stopping distances, avoid collisions, and provide the flight crew with consistent brake application characteristics to maintain air vehicle control.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Conduct lab testing to verify the brakes maintain running clearances when brake command is removed and as the brake stack wears.

8.5.7 Brake control and antiskid.

8.5.7.1 Brake application characteristics.

Standard: Right and left brakes are separately actuated and the flight crew and autonomous system brake input commands are proportional to brake control torque output allowing directional control of air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review of brake control schematic verifies independent control of left and right brakes are provided.

(2) Review of the brake pedal input (deflection and force) versus brake pressure or torque curve shows flight crew brake inputs are proportional to brake pressure or torque output.

Test.

(1) Lab test: Conduct lab testing to verify that input commands and output braking pressures or torques are proportional as designed.

(2) Ground test: Conduct ground testing to verify the air vehicle system operation and integration is within acceptable limits of lab testing results.

8.5.7.2 Braking control (manned air vehicle).

Standard: Brake pedal characteristics (e.g., force, deflection, brake pedal shape, brake pedal travel, and foot position) allow the flight crew to apply varying input commands from brakes off to initial brake rotors contact pressure to maximum brake pressure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Conduct lab testing to verify system force, travel, and response assessments for the specified size range of flight crew for manned air vehicles.

(2) Ground test: Perform on air vehicle ground testing to verify controllability of the braking system during taxiing, differential braking and turning operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify controllability of the braking system during all ground operations including the transition to and from flight.

8.5.7.3 Autonomous braking.

Standard: Autonomous and non-cockpit braking system can control and stop the air vehicle under all taxi, runway lengths, and conditions (e.g., wet, dry, and roughness).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review control logic command modes are present and allow different deceleration rates to stop air vehicle under varying surface conditions.

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Test.

(1) Lab test: Conduct HITL and simulator testing to verify control modes stop the air vehicle to meet required conditions (e.g., runway length, directional control, different runway coefficient of friction).

(2) Ground test: A buildup ground test verifies the software control algorithm can maintain required air vehicle speed and brake pressure and torque rates to allow taxiing and stopping of air vehicle between waypoints.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: A buildup flight test verifies the software control algorithm can maintain required air vehicle deceleration and brake pressure and torque application rates to perform rejected takeoffs and landings.

8.5.7.4 Release of brakes.

Standard: The braking system prevents a dragging brake condition. When input braking command is removed, the brake actuation system returns to the brakes off position. The brakes release time prevents heat buildup of brake stack or skidding of the tire.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Conduct lab testing to verify that brake pressure or clamping force is removed and running clearances are maintained when input commands are removed. Brake system response prevents dragging brakes.

(2) Ground test: Conduct ground test to verify that brake pressure or clamping force is removed when brake command is removed.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to verify that brake pressure or clamping force is removed when brake command is removed.

8.5.7.5 Brake control system redundancies.

Standard: An alternate and independent brake control system provides a means of stopping and controlling the air vehicle when the normal means are unavailable. The level of control and stopping performance are equal to that provided by the normal system. On air vehicles with antiskid control, an equivalent level of antiskid protection is provided.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct a FMECA to verify brake control system failures (including wheel speed sensors) have been mitigated such that level of control and stopping performance is maintained.

(2) Performance analysis verifies the availability of a redundant and alternate means to provide equivalent control and stopping performance.

Test.

(1) Lab test: Brake system simulator testing confirms that there is a separate and independent method of stopping the air vehicle when the primary means is not available. The control system notifies the flight crew when system failures occur and allows for either manual or automatic switching to the alternate system with no loss of braking.

8.5.7.6 Automatic braking selected by flight crew.

Standard: A flight crew initiated automatic braking system will have a means to arm and disarm the system. The system allows the flight crew to override the system at any time with manual braking.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review control panel to verify ability to select braking modes, deceleration rates, power system on and off.

Test.

(1) Lab test: Conduct simulator testing to verify control panel operation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing verifies control system maintains selected braking modes, deceleration rates, and ability of flight crew to arm and disarm the system.

8.5.7.7 Parking brake torque.

Standard: Parking brake static torque levels prevent the air vehicle from rolling at specified operational conditions and ship-based air vehicle the torque levels are IAW SAE AS8584B, Parking Brake System paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Static analysis shows brake produces sufficient torque and addresses timing values if needed to hold air vehicle at required specified operational conditions.

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Demonstration.

(1) Conduct ground demonstration to verify the parking brake ability to hold air vehicle under required engine thrust levels or the ship-based torque levels with no tire slippage on the wheel.

(2) For rotary-wing and VTOL, a flight demonstration is conducted to verify performance on maximum slope landings.

Test.

(1) Lab test: Static torque lab test verifies the brake produces the required torque levels to hold air vehicle under required engine thrust levels or the ship-based torque levels with no tire slippage on the wheel.

8.5.7.8 Park brake monitoring.

Standard: The parking brake system prevents inadvertent activation by the flight crew. The on-off status of the parking brake is indicated to the flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review cockpit controls and placement to verify inadvertent activation is prevented.

Demonstration.

(1) Conduct ground demonstration to verify indication system informs the flight crew if system is activated. Parking brake engages and disengages when flight crew input command is given or automatically based on control system inputs (e.g., advancing throttles above set point).

Test.

(1) Lab test: Simulator testing verifies system design logic, indication, and function of parking brake.

8.5.7.9 Antiskid system redundancies.

Standard: The braking and decelerations system responds to any single antiskid malfunction or power interruptions without compromising the ability to maintain control of the air vehicle. The antiskid self-correcting features switch to alternate braking modes in a predictable and controllable manner. The controlling function provides for safe recovery of the air vehicle from any single antiskid failure state to bring the air vehicle to a stop.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review of the antiskid system's FMECA verifies antiskid system failure modes have been identified and mitigated to maintain braking and directional control.

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Demonstration.

(1) Conduct ground demonstration to verify proper functioning of the antiskid system as integrated within the brake control system.

Test.

(1) Lab test: Conduct simulator testing to verify the modes of operation as identified in the antiskid system's FMECA under single point failures.

(2) Lab test: Conduct HITL lab testing, utilizing all brake and antiskid system hardware (e.g., control box) including representative brake line lengths and pressure drops, antiskid software, air vehicle model, landing gear damping model, and wheel speed sensor simulator model, to verify control law state changes which resulted from failure conditions (e.g., loss of wheel speed sensor), ability to adapt to full variety of runway conditions, stopping efficiency, and stopping distance performance under specified environmental design conditions (i.e., coefficient of friction).

8.5.7.10 Locked wheel prevention.

Standard: The antiskid system controls each braked wheel's rotation to decelerate the air vehicle while preventing flat spotting of tires and does not permit completely locked wheels within the antiskid control speed range for all normal operating speeds (e.g., takeoff and landing) and on touchdown.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Simulation of brake control system and algorithms verifies the ability to detect tire skidding and ability to modulate or remove brake pressure or clamping force to maximize braking force. System prevents deep skids which impact tire wear and locked wheel condition to prevent blowing of tires.

Test.

(1) Lab test: Lab testing of wheel speed sensor verifies accurate sensing of wheel speed in the installed environment and demonstrates the cutout speed for the wheel speed sensor within the antiskid control speed range.

(2) Lab test: Conduct HITL lab testing utilizing brake and antiskid system hardware (e.g., control box), representative brake lines (lengths and pressure drops), antiskid software, air vehicle model, landing gear damping model, and wheel speed sensor model. Testing verifies skid levels and locked wheel protection under specified runway coefficients of friction. The antiskid system prevents deep tire skids or locking of the wheels while meeting stopping distance performance requirements under specified environmental design conditions (e.g., coefficients of friction).

(3) Ground test: Conduct ground testing to verify proper measurement of wheel speed in installed environment and ability to modulate or remove brake pressure or clamping force to maximize braking force (minimize stopping distance), limiting deep skidding, and locked wheels.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing to verify proper measurement of wheel speed in installed environment and ability to modulate or remove brake pressure or clamping force to maximize braking force (minimize stopping distance), limiting deep skidding, and locked wheels.

8.5.7.11 Brake control modes.

Standard: The brake control system prevents the loss of braking, locked wheels, and deep skidding of tires during brake control mode transitions and changes (e.g., hydroplaning, despin, paired wheel and antiskid to pressure control) to stop the air vehicle within the required stopping distance.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Conduct HITL lab testing, utilizing all brake and antiskid system hardware (e.g., control box) including representative brake line lengths and pressure drops, antiskid software, and air vehicle model, landing gear damping model, and wheel speed sensor simulator model, to verify transitions between control modes do not cause locked wheel(s) and air vehicle maintains directional control under these modes.

(2) Ground test: Conduct ground testing to verify transitions between control modes do not cause locked wheel(s), loss of braking, and air vehicle maintains directional control under these modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify transitions between control modes do not cause locked wheel(s), loss of braking, and air vehicle maintains directional control under these modes.

8.5.7.12 Antiskid control system in environmental conditions.

Standard: The integrated brake control, antiskid and brake systems function in the installed environment and heat buildup does not cause locked wheels, loss of brake function during braking events.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Conduct ground testing on air vehicle to verify that the brake control, antiskid and brake systems are compatible and can function in the installed environment without impacting the stopping performance (e.g., locked wheels and loss of braking) of the air vehicle.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that the brake control, antiskid and brake systems are compatible and can function in the installed environment without impacting the stopping performance (e.g., locked wheels and loss of braking) of the air vehicle.

8.5.7.13 Antiskid coupling.

Standard: The brake system in conjunction with the antiskid system does not induce any adverse loads to landing gear and airframe structure (e.g., landing gear walking) and does not induce adverse air vehicle oscillations (e.g., directional control, pitching, and wing rocking).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Dynamic analysis verifies no adverse gear loadings and undesirable dynamics due to antiskid control cycles during all phases of brake operations.

Test.

(1) Lab test: Conduct HITL testing to verify landing gear fore and aft stability during antiskid operations.

(2) Ground test: Conduct ground testing to verify no adverse loading due to antiskid operations during all phases of brake operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify no adverse dynamic loading and adverse oscillation due to antiskid operations during all phases of brake operations including the transition to and from flight.

8.5.7.14 Stopping performance.

Standard: The brake system, in conjunction with the antiskid system, will stop the air vehicle for the required runway length, width and surface conditions (e.g., dry, wet, snow, ice, semi-prepared, unprepared, and mats) from maximum braking speed to the lowest ground speed of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Performance analysis verifies stopping performance capability for all specified operations, speeds, conditions, and locations.

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Test.

(1) Lab test: Conduct dynamometer testing of the brake to determine brake torque characteristics (e.g., effect of temperature and velocity on the values of developed torque, brake torque versus pressure or clamping force, and brake torque hysteresis) for antiskid simulation and HITL testing.

(2) Lab test: Conduct laboratory testing on the individual components of the brake control system to develop the characteristics to support the overall simulation of the brake control system.

(3) Lab test: As an alternative to Test MoC (4), conduct HITL lab testing, utilizing all brake and antiskid system hardware (e.g., control box) including representative brake line lengths and pressure drops, antiskid software, air vehicle model, landing gear damping model, and wheel speed sensor simulator model, to verify control law state changes which resulted from failure conditions (e.g., loss of wheel speed sensor), ability to adapt to full variety of runway conditions, stopping efficiency, and stopping distance performance under specified environmental design conditions (i.e., coefficient of friction).

(4) Lab test: As an alternative to Test MoC (3), conduct simulator or dynamometer testing of the brake, brake control system, and algorithms to verify stopping performance, control law state changes, skid detection, locked wheel protection, and component failure accommodations. Testing covers all aspects in Test MoC (3).

(5) Lab test: Conduct SIL test or air vehicle simulation testing to verify the total system integration with flight crew control or autonomous system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(6) Ground test: Perform on air vehicle ground braking tests to verify the system's stopping distance for different runway conditions, control of air vehicle (cornering, deceleration and directional control), and operations on the specified runway conditions. If air vehicle is to operate on mats, semi-prepared or unprepared surfaces, testing against each required soil type is conducted to verify drag, stopping performance, and runway condition reading values.

(7) Flight test: Conduct flight testing on both dry and wet runways to verify the system's stopping distance for different runway conditions, control of air vehicle (cornering, deceleration and directional control), and operations on the specified runway conditions. If air vehicle is to operate on mats, semi-prepared or unprepared surfaces, testing against each required soil type is conducted to verify drag, stopping performance, and runway condition reading values.

8.5.8 Ground directional control.

8.5.8.1 Steering capability.

Standard: The steering system allows the flight crew to perform all required ground and shipboard maneuvering, operations during parking, taxiing, takeoff, and landing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Static torque analysis is conducted to verify actuators or motors are sized to turn the nose tire(s) without air vehicle movement on a surface with a 0.8 coefficient of friction.

(2) Analysis verifies steering system meets minimum turning radius needed for air vehicle ground operations.

Demonstration.

(1) Conduct ground demonstrations to verify flight crew's ability to capture the centerline using steering only, have direction control and perform other ground movements (e.g., minimum turn radius, 90-degree turns, enter and transit taxiways and runways, and align air vehicle to catapult equipment).

Test.

(1) Lab test: Conduct laboratory testing of steering components to verify torque capability, rates (degrees/sec) and damping characteristics.

8.5.8.2 Steering control system.

Standard: The steering control system protects against uncommanded steering events, including hard-over commands, to prevent loss of control of the air vehicle. The steering system will disengage or correct the command to the extent that the air vehicle remains in its original commanded direction or switches to a controllable configuration (e.g., dynamically stable, free caster, self-centering). The control system accommodates high and low speed operations by integrating steering system responses with the FCS by limiting turn angle to prevent loss of control.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review of the steering system's FMECA and SSHA to verify that all steering and directional control failures, including hard-overs and uncommanded events, are identified and risks mitigated.

Test.

(1) Lab test: Conduct component failure lab testing to verify performance of all components and switching logic within the steering system.

(2) Lab test: Conduct simulator testing to verify flight crew's ability to laterally control the air vehicle with the blending of maximum turn angle versus air vehicle speed through takeoff and landing rollouts.

(3) Lab test: Conduct FMET to verify control system can maintain directional control or have nose wheel revert to free caster or safe state after a single point failure without loss of control.

(4) Ground test: Conduct ground testing to verify flight crew can maintain lateral control of the air vehicle for the design conditions (cross winds, air vehicle speeds, and runway surface coefficient of frictions).

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight testing to verify flight crew can maintain lateral control of the air vehicle for the design conditions (cross winds, air vehicle speeds, and runway surface coefficient of frictions).

8.5.8.3 Steering system monitoring.

Standard: The on-off, failures, and mode status of the steering system are indicated to the flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review displays to verify system status is provided.

Demonstration.

(1) Conduct ground demonstration to verify indication system informs the flight crew when system is engaged, disengaged, and mode status. Steering system engages and disengages when flight crew input command is given or automatically based on control system inputs.

Test.

(1) Lab test: Simulator testing verifies system design logic and indication of the steering system status.

8.5.8.4 Steering engagement.

Standard: Control of the air vehicle will be maintained during engagement or disengagement of the steering control function for all operational speed ranges and conditions, whether the event occurs from flight crew commanded or system commanded actions. The air vehicle maintains its previously commanded directional heading or defaults to a controllable configuration (e.g., dynamically stable, free caster, self-centering).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review of the steering system's FMECA and SSHA to verify control of the air vehicle is maintained with the steering system.

(2) Review design and failure analysis to verify performance of the steering system engagement and disengagement for all loading conditions.

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Demonstration.

(1) Conduct ground demonstration to verify control of the air vehicle is maintained with transitions from steering-on to steering-off and changes from low to high gain (maximum steering angle).

Test.

(1) Lab test: Conduct lab testing to verify control for the expected modes of operation and engagements for all loading conditions.

8.5.8.5 Backup for directional control.

Standard: Ground directional control of the air vehicle is maintained at all times, including under single point failure conditions. The primary and alternate (secondary or emergency) systems will provide continued directional control of the air vehicle during ground operations. Systems that impact directional control of the air vehicle include steering, brakes, flight control surfaces, and propulsion.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review of the steering and brake systems' FMECA verifies that single point failure conditions and degraded modes of operation are identified, and mitigations are in place to allow continued directional control.

Test.

(1) Lab test: Conduct simulator testing to verify that the directional control system functions in normal and degraded modes.

(2) Ground test: Conduct ground demonstration to verify proper functioning of the primary and alternate (secondary or emergency) directional control systems.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify proper functioning of the primary and alternate (secondary or emergency) directional control systems.

8.5.9 Landing gear retraction and extension.

8.5.9.1 Landing gear retraction and extension operation.

Standard: Retraction and extension dynamics and mechanical interface of the landing gear and door, including locking system, are established for all modes of operation. Landing gear door sequencing during normal and emergency operations ensures that the landing gear can be extended during all designed for and approved operational conditions.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to determine hinge moments, inertial loads, and actuator loads required to open and close the doors and retract and extend the landing gear against air loads.

(2) Power (hydraulic, pneumatic or electrical) requirement analyses are conducted to establish system interface requirements for actuation power and rates. The analysis addresses the ground and flight temperature environments.

Demonstration.

(1) Conduct ground demonstration (e.g., full scale mockup or air vehicle level) to verify proper functioning of the landing gear throughout the gears' full range of motion and operational modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Conduct flight test to verify retract and extend operations under air and inertial loads and ensure no unintended contact and functionality of actuation with air vehicle structure or other components within the wheel well.

8.5.9.2 Prevention of landing gear extension.

Standard: Landing gear extension capability is not prevented by a loss of a door(s) or single point failure in the actuation system. The emergency extension system is independent of the landing gear primary power source(s).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify the landing gear actuation and door system's FMECA and SSHA is complete and identifies all failures and operational events that prevent landing gear extension.

(2) Analysis of the landing gear, hydraulic system, control logic, electrical system, and door system verifies sufficient capability to fully deploy the landing gear, either by normal or emergency means.

Demonstration.

(1) Conduct ground demonstration to verify operation of the emergency extension system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Conduct flight testing to verify operation of the emergency extension system under air and inertial loads.

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8.5.9.3 Reversal of landing gear command.

Standard: Changes in position command while the gear is in transit does not cause the gear system to jam, stall, or prevents the gear from achieving the last commanded position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) For normal system actuation, an analysis of the landing gear, hydraulic system, control logic, electrical system, and door system verifies capability of the landing gear to achieve the last commanded position.

Demonstration.

(1) Conduct ground demonstration with air vehicle on jacks to verify landing gear will go to last commanded position even if commanded position is reversed in midcycle for the normal gear retract and extension system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Conduct flight demonstration to verify landing gear will go to last commanded position even if commanded position is reversed in midcycle for the normal gear retract and extension system.

8.5.9.4 Gear status indication.

Standard: Flight crew has indication of each landing gear's position and any disagreement between commanded and actual gear position (i.e., up and locked and down and locked status).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies that all modes of landing gear operation and position are indicated to the flight crew during all normal and emergency conditions and addresses all failure events as defined by the landing gear system's FMECA.

Demonstration.

(1) Conduct ground demonstration to verify that proper gear status indications are given to the flight crew.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Conduct flight demonstration to verify that proper gear status indications are given to the flight crew.

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8.5.9.5 Position warning system.

Standard: Visual and aural landing gear position warnings are provided to the flight crew indicating when the air vehicle is close to the ground and close to landing speeds without the landing gear extended. Warnings are provided to the flight crew in time to extend and confirm the gear is in the down and locked position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies that warnings are provided for all normal and emergency conditions and addresses all failure events as defined by the FMECA.

Test.

(1) Lab test: Conduct testing in the air vehicle simulator to verify flight crew receives a warning if the landing gear is not down and locked when air vehicle is close to the ground and close to landing speed.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to verify correct functioning of the warning and indication system to prevent landing without the landing gear down.

8.5.9.6 Gear down limit speed.

Standard: For takeoff and go-around conditions, the landing gear is retracted and doors closed before its gear down design limit speeds are reached under maximum takeoff acceleration and that prior to landing, the landing gear can extend and lock at maximum gear down speed within specified time limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies the landing gear retract and extend times are compatible with air vehicle performance and operating conditions (e.g., temperature environment and air loads).

Test.

(1) Ground test: Conduct ground test to verify the retract and extend times.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to verify the retract and extend times under air loads for all operational performance speeds and accelerations within specified time limits.

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8.5.9.7 Emergency extension times.

Standard: Emergency landing gear extension achieves its final down and locked position at gear down limit speeds within the specified time limits before power is lost or interrupted and prior to touchdown. The emergency extension can be completed under all flight and ground thermal environmental envelopes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies the emergency extension times are compatible with the air vehicle landing requirements and thermal environments.

Test.

(1) Ground test: Conduct ground tests to verify the emergency extension performance.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight tests to verify the extend times and extension performance under maximum design limit air and inertial loads.

8.5.9.8 Gear position restraint.

Standard: Means are provided to positively lock the gear in the final commanded position (gear up and gear down), including with and without the primary power source, for all ground, landing, and flight conditions. Passive means prevent the gear from unlocking during power failure or no power applied, or hydraulic leakage or deflection under allowed maximum free play worn conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies gear locking mechanism maintains gear position under all expected mission loading including vibration and maneuvering loads without primary power source. Analysis verifies locking mechanism does not unlock from landing dynamic load conditions (vertical, drag, spin up and spring back).

Demonstration.

(1) Conduct ground demonstration to verify locking mechanism maintains gear position under all expected ground and towing mission loading and maneuvering without primary power source.

Test.

(1) Lab test: Conduct lab testing of locking system on full scale nose and main landing gear assemblies or on HITL (e.g., iron bird) to verify dynamic latching loads into uplocks for gear retraction and down locks for gear extension.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Conduct flight test to verify locking mechanism maintains gear and door positions under all expected mission loadings, store configurations, maneuvering, and landing loads (within the flight envelope) without primary power source.

8.5.9.9 Release of uplocks provisions.

Standard: Landing gear position uplocks have redundant provisions (e.g., redundant power sources) or shear capability to prevent jamming or failure modes which would result in loss of gear extension capability.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct the landing gear actuation system's FMECA to verify the performance of the gear position restraints with and without failures.

(2) Verify shear capability by conducting strength analysis to verify actuation system's ability to overcome uplocking mechanism to allow the landing gear extension.

Demonstration.

(1) Conduct ground demonstration to verify the gear position restraints functionality with alternate or emergency power.

Test.

(1) Lab test: Conduct function testing of the locks in a HITL (i.e., iron bird) or full-scale mockup to ensure no single point failure mode prevents the landing gear from extending.

(2) Lab test: Conduct lab testing of uplocks to verify the latches shear capability at the design actuation load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Conduct flight demonstration to verify the gear position restraints functionality with alternate or emergency power.

8.5.9.10 Gear position restraint locks for ground operations.

Standard: Locking device(s) and interlocks are provided to prevent inadvertent gear and door actuation while the air vehicle is on the ground (including towing and takeoff roll). The locking device(s) and interlocks prevent the gear and doors from moving for all expected ground configurations and ground operations even if landing gear is inadvertently retracted. A visual warning indicator is provided when a locking device is installed.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis verifies the effectiveness of the ground locking device installation to prevent inadvertent gear and door retraction while the air vehicle is on the ground.

(2) Static analysis verifies locking device strength is designed to handle the full retraction power and load without permanent deformation or damage to air vehicle interfaces and landing gear structure.

Demonstration.

(1) Conduct ground demonstration with air vehicle jacked to verify position restraint and load carrying capability of the ground locking device and verifies the visual indication when a locking mechanism is installed.

Test.

(1) Lab test: Conduct lab testing in a SIL to verify functionality of interlocks.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Conduct ground demonstration of the interlocks to verify the flight crew cannot prematurely retract the landing gear while air vehicle has weight on wheels.

8.5.10 Arresting hook systems.

This section addresses both emergency and carrier-based (i.e., operational) arresting hook systems as defined by MIL-STD-18717. The term “operational” describes an arresting hook system that is routinely used, versus emergency use only. Arresting hook system refers to equipment installed on the air vehicle. Arresting gear system refers to the ground based or carrier-based equipment (e.g., E-28, Mark 7, AAG, and BAK-12). Emergency arresting systems are used infrequently to stop the air vehicle which experienced a failure (primary braking system is unavailable, fire, or control issue). An emergency arresting hook system is also where the arresting gear system is not immediately engaged on landing (e.g., F-16 and F-15). The air vehicle is in a three-point attitude before the cable is engaged, and no brakes are applied. A carrier-based arresting hook system allows immediate engagement and operational usage (i.e., every landing).

8.5.10.1 Air vehicle arrestment performance (loads).

Standard: The arresting hook system (which engages a cable) is capable of stopping the air vehicle at all design conditions (e.g., RTOs, in flight engagements, emergency landings, and brake overruns) without damage to the air vehicle or the arresting hook system. Arresting gear system engagements are defined by energy level, hook loads, and air vehicle speeds and weights. Engagement limits, for both on-center and off-center conditions (distance from runway/deck centerline) and multivariate (yaw, pitch and roll) engagements, are established for the air vehicle and type of arresting gear system.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analyses to define engagement parameters such as speeds, air vehicle weights, and arresting energy levels for the operational environmental conditions (e.g., ambient temperatures and runway altitudes) to establish the arresting system loads. For emergency arresting hook systems as used on USAF air vehicles, the arresting gear system is not engaged on landing impact. The air vehicle has all wheels on the runway before engagement and no braking allowed for approach end engagements. Braking on departure end can be used before air vehicle traverses the cable.

(2) Conduct analysis to determine compressive loads from the initial arresting hook impact (e.g., pogo) on touch down or reverse loading conditions.

(3) Conduct analysis based on Analysis MoC (1) engagement conditions and Analysis MoC (2) to defined arresting hook system design loads for the arresting gear systems being engaged addressing both on-center and off-center engagements.

(4) Conduct static strength stress analysis to verify arresting hook system and air vehicle structure can support design limit and ultimate loads defined in Analysis MoC (2). Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate loads. Ultimate load is 1.5X DLL.

Test.

(1) Lab test: Conduct lab component level static strength testing to verify arresting hook system meets limit and ultimate loads.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Review of flight manuals verifies the air vehicle engagement limits (on-center, off-center, and in-flight engagements) for each type of arresting gear system (e.g., BAK-12, Mark 7, AAG, and E-28) specified.

Test.

(2) Ground test: Conduct air vehicle engagements with the arresting gear system to verify arresting hook system successfully engages cable, hook loads do not exceed design loads, and cable dynamics does not cause air vehicle damage.

(3) Flight test: Conduct runway operations including approach end engagements after landing to verify arresting hook system successfully engages cable, hook loads do not exceed design loads, and cable dynamics does not cause air vehicle damage.

(4) Flight test: For carrier-based arresting hook systems, conduct shipboard operations to verify arresting hook successfully engages cable, hook loads do not exceed design loads, and cable dynamics does not cause air vehicle damage.

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8.5.10.2 Arresting hook system's ability to engage the cable.

Standard: The arrestment hook system engages the arresting cable and prevents damage to the air vehicle or stores. The arresting hook system design includes provisions which:

a. Minimizes the arresting hook from contacting the runway or deck on landing in a direction that causes damaging compression loads to be developed in the hook shank and in the air vehicle structure.
b. Minimizes the arresting hook point bouncing on landing impact, as the hook point drags along the landing surface, or hitting a runway/deck obstruction by incorporating damping to prevent a missed engagement.
c. Takes into account cable dynamics due to tire rollover effects.
d. Allows the hook to engage the cable with an air vehicle that has a flat nose tire and strut.
e. Allows the hook to engage the cable under air vehicle dynamic bounce conditions. Hook point bounce that occurs during landing roll out, not caused by hook point contact with the runway/deck, but by the air vehicle's compressed main landing gears rebounding as the nose gear compresses.
f. Prevents collateral damage to the air vehicle due to a missed engagement, hook and cable from damaging the air vehicle after cable engagement due to cable dynamics, and hook slap due to a missed engagement.
g. Allows lateral movement of arresting hook to conduct off-center (distance from runway/deck centerline) and multivariate (yaw, pitch and roll) engagements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review installation drawings to verify arresting hook assembly can engage the cable/deck pendant at all air vehicle attitudes expected during carrier and field operations and aborted takeoffs.

Analysis.

(1) Conduct a dimensional and geometric analysis to verify the arresting hook system ability to engage the cable.

(2) Conduct dynamic analysis or simulation of air vehicle and arresting gear system to ensure the arresting hook system can engage the cable.

Test.

(1) Lab test: Conduct lab testing on damper to verify damping level minimizes hook bounce to prevent missed engagements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Ground test: Conduct ground testing to verify the installation, functionality, and performance capability of the air vehicle arresting hook system to engage arresting gear systems (e.g., BAK-12, Mark 7, AAG, and E-28).

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(3) Flight test: Conduct flight testing to verify the arresting hook system's ability to engage arresting gear system without damaging the air vehicle and arresting hook system.

8.5.10.3 Hook actuation.

Standard: The arrestment hook system can be extended and retracted (if required) by the flight crew or autonomous system in time to meet all normal and emergency conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system schematics to verify redundant power sources are present for operational arresting hook systems.

Analysis.

(1) Review of the arresting gear's FMECA and SSHA analysis identifies all failures that prevent extension and retraction (if required) of the hook.

(2) Conduct actuation load analysis to verify the arresting hook can be extended and retracted in time to meet all normal and emergency conditions (e.g., air loads and inertia).

Demonstration.

(1) Conduct ground demonstration to verify the installation and functionality of the arresting hook actuation system for normal and emergency conditions.

Test.

(1) Lab test: Conduct lab testing of actuation system to verify the arresting hook system's actuation components (e.g., actuators or motors) ability to meet limit and ultimate load conditions.

(2) Lab test: Conduct lab endurance testing to four lifetimes for emergency arresting hook systems and two lifetimes for carrier-based arresting hook systems on the actuation system (e.g., actuators and motors) to verify it meets the DSL.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing up to gear down limit speeds to verify the performance capability of the hook actuation system to extend and position the arresting hook system for engagement.

8.5.10.4 Arresting hook position status.

Standard: Arresting hook system deployed position status is indicated to the flight crew. Status is provided for all positions (e.g., stowed, locked up, and locked down).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review of flight manual and crew station design documentation verifies the indication system is provided to the flight crew.

Demonstration.

(1) Conduct arresting hook system actuations to verify indication system matches arresting hook position.

8.5.10.5 Arresting gear system and tire failure considerations.

Standard: For air vehicles equipped with arresting gear, no part of the landing gear, air vehicle, or stores will snag the arresting cable for all air vehicle configurations when air vehicle is rolling on wheels with a flat tire. Minimum landing gear wheel diameter size is greater than six inches.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify cable rollover capability during tire failure and roll on wheel condition for all operational requirements and cable dynamics.

8.5.10.6 Air vehicles not equipped with arresting hook.

Standard: For air vehicle which do not have arresting hook installed, the air vehicle can traverse arresting gear cables, including support structures, at takeoff and landing speeds without causing loads exceedance on landing gear structure, damage to tire and wheels, and no part of the air vehicle including landing gear, doors, antennas, brakes, hydraulic lines, and stores will contact the arresting cable for all air vehicle configurations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a static loads analysis to verify the air vehicle's capability to rollover the arresting gear cable and impact of cable supports do not exceed the landing gear structural strength.

(2) Conduct a dynamics analysis to verify the air vehicle does not cause the arresting gear cable to snag on any part of the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Ground test: Conduct ground test to verify air vehicle can rollover the arresting gear cable without overloading landing gear structure (e.g., tire contact with cable supports), cause the cable to contact any part of the air vehicle, and not damage the tires.

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8.5.11 Parachute deceleration systems.

8.5.11.1 Drag and spin chutes.

Standard: The drag or spin chute design, attachments, and deployments meet all specified performance requirements without causing adverse loading or damage to the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of flight manual verifies air vehicle speeds for deployment and drag performance of the chute are available to the flight crew and information aligns with qualified design.

Analysis.

(1) Conduct sizing analysis to verify the drag chute aids in decelerating the air vehicle to meet stopping distance requirements and required runway conditions (e.g., dry, wet and icy surfaces).

(2) Conduct sizing analysis to verify the spin chute stabilizes the air vehicle to recover from a spin in flight.

(3) Conduct loads analysis to determine shock and sustained loads from chute deployment.

(4) Conduct static strength stress analysis of chute mechanisms, landing gear and air vehicle structure to verify various components are sized for the chute loads. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No chute panel blow out, rupture of structure and risers, premature release of chute occurs at or below ultimate loads. Ultimate load is 1.5X DLL.

(5) Conduct trajectory analysis to show that chute canopy and lines deployment minimizes contact with the air vehicle to allow full deployment of chute.

Test.

(1) Ground test: Conduct testing on air vehicle to verify the installation, functionality, and performance capability of the drag or spin chute system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct testing on air vehicle to verify the installation, functionality, and performance capability of the drag chute system during flight, approaches, landings, and RTOs.

8.5.11.2 Drag or spin chute release.

Standard: The drag or spin chute system has provisions to manually release the chute when deployed to prevent loss of air vehicle control. A frangible device prevents overload conditions in the case of inadvertent deployment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review of flight manual verifies flight and ground envelopes where the flight crew can deploy the drag or spin chute.

Analysis.

(1) Conduct stress analysis to verify frangible device (e.g., D-ring) and chute releases at specified load condition.

Demonstration.

(1) Conduct ground demonstration to verify flight crew's ability to manually release the chute.

Test.

(1) Lab test: Conduct laboratory load testing of mechanisms (e.g., mechanical fuse) to verify that the frangible device fails, and chute is released at specified load condition.

8.5.11.3 Whole air vehicle parachute operation.

Standard: The parachute system reliably lowers an entire air vehicle to the ground including in the event of loss of control, failure of the air vehicle structure, or other in flight emergencies. The parachute landing system deployment method prevents the parachute from getting entangled with the air vehicle (e.g., flight control surfaces, propulsion system) or causing damage to the parachute preventing a successful landing. For the operational envelope, the parachute landing system controls the rate of descent, glide rate, and oscillations to allow recovery of the crewmembers, air vehicle, and onboard equipment with no damage beyond acceptable operating limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct sizing analysis to verify the parachute can descend the air vehicle to touchdown without causing harm to crewmembers or damaging the air vehicle.

(2) Conduct trajectory analysis to show that parachute canopy and lines deployment minimizes contact with the air vehicle to allow full deployment of parachute within deployment envelope.

Demonstration.

(1) Conduct ground demonstration on air vehicle to verify the installation, functionality, and performance capability (e.g., deployment time and parachute extraction characteristics) of the parachute system. The parachute landing system sacrificial elements (e.g., installation covers, doors) and extractive devices (e.g., mechanical, pyrotechnics) do not impact or cause damage to the air vehicle or parachute upon deployment that would prevent a successful landing.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(1) Flight test: Conduct flight testing to verify parachute deployment does not result in entanglement with the air vehicle (e.g., flight control surfaces, propulsion system) and decent characteristics result in a controlled landing of the air vehicle.

(2) Flight test: Conduct parachute system testing to verify a demonstrated reliability of not less than 90% at a 90% confidence level at the system level.

8.5.11.4 Parachute load factors.

Standard: The air vehicle and parachute landing system can sustain limit and ultimate loads due to the parachute generated forces and accelerations. Parachute components are designed with a 1.5 factor of safety for ultimate load.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct initial loads analysis of parachute deployment loads based on the initial parachute sizing determined in standard 8.5.11.3.

Test.

(1) Lab test: Conduct static load testing to verify that the parachute system and components (i.e., bridles, lanyards, harnesses, activation cables, and etc.) can sustain design loads and maintain functionality for all expected conditions without deformation or damage beyond operational limits.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(2) Verify ultimate load by conducting parachute strength Test MoC (2) or Test MoC (3).

(3) Conduct static strength analysis of chute, chute attachment mechanisms and air vehicle attachment structure using the ultimate load factor of safety of 1.5 to verify components are designed for chute deployment loads.

Test.

(2) Flight test: As an alternative to Test MoC (3), conduct a minimum of three successful drop tests on the air vehicle of the parachute assembly under ultimate load conditions to demonstrate the parachute's strength. The maximum parachute opening force measured in the three tests will be the ultimate parachute opening load. The test parameters are:

a. Minimum test weight is 1.25X air vehicle maximum gross takeoff weight.
b. Minimum test speed is 1.1X air vehicle's maximum intended parachute deployment speed.

(3) Flight test: As an alternative to Test MoC (2), conduct a minimum of three successful parachute strength tests with a dead weight. The maximum parachute opening force measured in the three tests will be the ultimate parachute opening load. The test parameters are:

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a. Minimum test weight is air vehicle maximum gross takeoff weight.
b. Minimum test speed is air vehicle's maximum intended parachute deployment speed.

8.5.11.5 Landing impact attenuators.

Standard: Landing impact attenuation systems (e.g., shock absorbers, landing air bags, and intentional crushable structure) in conjunction with the parachute landing system minimizes damage to air vehicle, allowing reuse of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct loads analysis based on the deceleration rate determined IAW 8.5.11.3, to verify sizing and performance parameters.

(2) Conduct dynamic simulations to verifying attenuation system can absorb landing impact loads.

Test.

(1) Flight test: Conduct flight testing of a representative installation to verify the attenuation systems produces the landing loads developed in Analysis MoC (1) and Analysis MoC (2).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify the attenuation systems prevents loss of the air vehicle.

8.5.11.6 Aborted landing.

Standard: When a parachute landing system is the primary means to recover the air vehicle after every flight, the air vehicle flight performance and handling qualities allow the transition from the landing approach phase back to normal flight in the event of aborted parachute deployment. The speeds, altitudes, and environmental conditions needed for the landing approach phase, abort, and continued flight are defined.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manual to verify that approach and abort procedures are defined.

Analysis.

(1) Conduct parametric analysis to define speeds, altitudes, and environmental conditions needed for the landing approach phase, abort, and continued flight.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(1) Flight test: Conduct flight testing to verify that landing can be aborted just prior to deployment of the parachute landing system and that the air vehicle can maintain controlled flight subsequent to the abort.

8.5.11.7 Minimization of dragging.

Standard: Parachute landing system provides a means to release the parachute(s) after touchdown to minimize dragging of the air vehicle across the landing surface (e.g., ground and water) during all operating conditions (e.g., high winds and currents).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system design documents to verify parachute release system is incorporated.

Demonstration.

(1) Conduct ground demonstration to verify that the air vehicle is not dragged by the parachute landing system across the recovery surface after landing.

8.5.11.8 Environmental exposure.

Standard: Parachute landing system storage compartment and component design should protect packed parachute system from adverse internal and external environmental conditions (e.g., flight loads, EMI, rain, humidity, and snow). For parachute landing system components located near the propulsion system, cooling, shielding, and material selection prevents degradation of the parachute landing system that could prevent successful landing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review material data sheets to verify material selection can function in the design environmental conditions.

Analysis.

(1) Conduct thermal analysis to verify parachute landing system can function and is not damaged in its installed environment.

Test.

(1) Lab test: Conduct lab testing to verify the absence of adverse effects (e.g., failure to deploy and material degradation) due to the operating environment.

(2) Ground test: Conduct ground test to verify the absence of adverse effects due to the operating environment.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(3) Flight test: Conduct flight test to verify the absence of adverse effects due to the operating environment.

8.5.12 Auxiliary deceleration systems.

Standard: Auxiliary deceleration systems such as thrust reversers and speed brakes (alone or in combination with other deceleration systems as the main wheel braking) meet the stopping performance requirements and do not cause adverse loading, dynamics, or structural damage to the air vehicle. Directional control is maintained when auxiliary deceleration devices are used for all operations, conditions, and environments.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a performance analysis to verify the auxiliary deceleration system is sized to provide the drag or thrust needed to decelerate the air vehicle to meet stopping distance requirements.

(2) Conduct analysis to verify the auxiliary deceleration system does not cause adverse loading, dynamics, or structural damage to the air vehicle.

Test.

(1) Ground test: Conduct ground test to verify that the auxiliary deceleration system meets the predicted performance (drag or thrust) to stop the air vehicle. The air vehicle maintains directional control and is not subjected to adverse loading, dynamics, or structural damage.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that the auxiliary deceleration system meets the predicted performance (drag or thrust) to stop the air vehicle. The air vehicle maintains directional control and is not subjected to adverse loading, dynamics, or structural damage.

8.5.13 Specialized systems.

Standard: Specialized systems (e.g., skis, skids, crosswind positioning, tire inflation and deflation, rock deflectors, net capture systems and floats) perform their intended functions and do not interfere with ground operation or control of the air vehicle (both in flight and on the ground), throughout the design envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to verify specialized system performance and operations, for all expected missions including all specified environments and surfaces, does not have any adverse impact to ground operation or control of the air vehicle.

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Test.

(1) Ground test: Conduct ground test to verify the specialized system performance and operations, for all expected missions including all specified environments and surfaces, does not have any adverse impact to ground operation or control of the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to verify the specialized system performance and operations, for all expected missions including all specified environments and surfaces, does not have any adverse impact to ground operation or control of the air vehicle.

8.5.14 Ground handling.

8.5.14.1 Towing to taxi air vehicle.

Standard: The landing gear system allows towing of the air vehicle at all mission weights, in all directions, and on all operational surfaces. The airframe and landing gear system accommodates connection to tow vehicles or tow bars. Towing procedures (including failures of towbar/fuse pins or towing device) and landing gear design features (e.g., torque arm disconnects) prevent damage to air vehicle and controls air vehicle movement.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review landing gear drawings to verify that any towing limitations are identified on landing gear (e.g., maximum tow angle).

(2) Review braking schematics to verify air vehicle brakes can be applied in the event of towbar failure.

(3) Review maintenance procedures to verify towing procedures (e.g., speeds, turning angles, and braking techniques) and limitations are documented to prevent possible landing gear overloading or loss of control of air vehicle.

Analysis.

(1) Conduct load analysis to determine maximum tow loads due to tow bar interface tolerances, braking inputs, and tow speeds.

(2) Conduct sizing analysis to verify there is sufficient internal power to apply the air vehicle brakes to stop the air vehicle for towing emergencies.

Demonstration.

(1) Conduct air vehicle demonstration to verify emergency tow brake function operation.

(2) Conduct ground testing to verify towing compatibility with towing equipment and loads do not exceed landing gear design limits.

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8.5.14.2 Ship-based air vehicle mooring.

Standard: Ship-based air vehicle mooring provisions demonstrate damage free operation of the landing gear, air vehicle, and surrounding structure for all operational conditions such as heavy weather, deck pitch, roll, and design wind speed. The placement of mooring provisions ensures full capability to extend, retract, and store the landing gear.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review mooring locations and patterns to verify compatibility with deck fitting positions and positioning of restraints does not damage or interfere with the landing gear operation.

Analysis.

(1) Conduct a loads analysis to define limit and ultimate loads envelope needed to secure the air vehicle.

(2) Conduct static strength analysis to verify mooring provisions and supporting landing gear structure can sustain limit and ultimate loads. Permanent deformations do not occur at DLL. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct a ground demonstration to verify the location of mooring provisions allow for tie down to ship deck fittings and no damage to air vehicle.

8.5.14.3 Ground foreign object damage.

Standard: The landing gear design limits engine inlet exposure by establishing acceptable spray patterns and thrown object trajectories. The location of the main and nose gear tires minimizes the throwing of debris, dirt, water, snow, and slush to prevent damage to the air vehicle and engines (e.g., loss of thrust, engine FOD ingestion, fuel tank rupture, and structural damage to flight-critical components).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a trajectory analysis to verify that the landing gear configuration does not direct runway contaminants (e.g., water and slush) or debris into the engines or damage the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Demonstration.

(1) Conduct ground demonstration to verify that the landing gear configuration does not direct runway contaminants (e.g., water and slush) or debris into the engines or damage the air vehicle.

8.5.15 Landing gear and deceleration systems overarching standards.

8.5.15.1 Damage tolerance and durability approaches.

Standard: Landing gear and deceleration system components (e.g., secondary load path items, wheels, linkages, hydraulics, and electrical) which perform SCFs are capable of performing such functions in the presence of undetected flaws, defects, and maintenance or service induced damage so that the SCF is not lost. Damage tolerance and durability approaches can be achieved by the application of fracture mechanics, fail-safe, safe life, leak-before-burst principles, redundancy, or in the case of composite parts, by showing tolerance to impact damage.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review list of SCFs and safety-critical components to verify proper classification of components to ensure components are designed to be damage tolerant.

(2) Review engineering schematics of landing gear and deceleration actuation systems for redundancy level.

Analysis.

(1) Conduct a fracture mechanics analysis on pressure vessels to verify that a through crack in a fluid container leaks fluid before burst by demonstrating tolerance of a through crack 2X and 10X the wall thickness.

(2) As an alternative to Analysis MoC (3), conduct damage tolerance analysis to verify the ability of components to perform in the presence of material, manufacturing, processing, or handling or operationally-induced damage for the minimum required maintenance-free period of unrepaired usage.

(3) As an alternative to Analysis MoC (2), for materials which are not damage tolerant, conduct a durability analysis using the safe life approach with material scatter factors to verify required operational service life.

Test.

(1) Lab test: As an alternative to Test MoC (2), conduct damage tolerance tests to verify the damage tolerance characteristics of safety-critical components to meet DSL.

(2) Lab test: As an alternative to Test MoC (1), for components whose materials are not damage tolerant, conduct a durability test using the safe life approach with material scatter factors to verify required operational service life.

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8.5.15.2 Single point failures.

Standard: No single component or function failure of the landing gear system results in the loss of any one of the following functions: ability to extend landing gear, ability to extend an operational arresting gear, decelerating air vehicle, air vehicle ground loading support, and ground directional control.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a landing gear system FMECA and SSHA to identify all single point failures which result in a loss of a landing gear SCF and any mitigations to those failures.

Test.

(1) Lab test: Conduct FMET to verify SCFs can be maintained under single point failure conditions.

8.5.15.3 Landing gear interfaces.

Standard: All interfaces (e.g., structural mounting, backup structure, electrical, and hydraulic) between the landing gear and air vehicle are defined and controlled to ensure that the landing gear and deceleration systems can perform their SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review ICDs and system specification documentation to verify that interfaces with other systems are defined.

(2) Conduct review of engineering drawings and specifications to verify interface compatibility with wheels, brakes, and tire data.

(3) Conduct review of engineering drawings and specifications to verify interface compatibility with landing gear attachment hardware (e.g., trunnion pins) to backup structure.

(4) Conduct review of engineering drawings and specifications to verify interface compatibility with hydraulic tubing connections, electrical connectors, and mounting provisions and clearance.

Analysis.

(1) Conduct modeling of landing extension and retraction systems to verify power interface inputs and outputs.

(2) Conduct modeling of the braking and brake control systems to verify power interface inputs and outputs.

Test.

(1) Ground test: Conduct a ground test to verify interfacing systems provide power (e.g., hydraulic, pneumatic, and electric) to complete landing gear and deceleration system SCFs.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct a flight test to verify that interfacing systems provide power (e.g., hydraulic, pneumatic, and electric) to complete landing gear and deceleration system SCFs.

8.5.15.4 Flight crew interface.

Standard: Warnings, cautions, and advisories are provided to the flight crew as a means to assess the landing gear and deceleration systems operational condition and the state of functionality.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system design documents to verify provisions for the necessary monitoring of the landing gear and deceleration systems operation and health are provided.

(2) Conduct review of flight manuals to verify WCAs provide system status information, health, and actions required to flight crew.

Test.

(1) Lab test: Conduct lab integration testing, including FMET, to verify compatibility of landing and deceleration systems with the controls and health monitoring systems.

(2) Ground test: Conduct ground tests of the installed systems to verify flight crew receives WCAs based of fault tolerances.

8.5.15.5 Technical manuals.

Standard: Flight manuals will include landing gear and deceleration systems normal, backup, and emergency procedures, system parameters, indications, WCAs, restrictions, and air vehicle operating limitations. Maintenance manuals will include landing gear and deceleration systems life limits, limitations, and servicing procedures. The manuals' information is substantiated and reflects the as designed or modified system. The technical manuals incorporate unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, qualification testing findings).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manuals to verify that the flight crew is provided with information addressing WCAs, indications, limits of operation, normal, emergency, and backup procedures.

(2) Review technical manuals to verify they incorporate unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, qualification testing findings).

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Analysis.

(1) Conduct supporting analyses to establish the basis for the published operating procedures, flight manual limits, and the system operating limits.

(2) Conduct supporting analyses to establish the basis for the published maintenance manual life limits, inspections, limitations, and servicing procedures including unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, qualification testing findings).

Test.

(1) Ground test: Conduct on air vehicle testing to validate and verify flight manual normal, emergency, and backup procedures for landing gear and deceleration systems.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to validate and verify flight manual normal, emergency, and backup procedures for landing gear and deceleration systems.

8.5.15.6 Qualification testing.

Standard: All components, either individually or as part of the landing gear and deceleration subsystem, have completed and passed qualification testing (e.g., environmental, proof, burst, dynamometer, static, fatigue and durability). Life limits and operating restrictions are defined for components which did not fully pass qualification testing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of SOF testing documents or qualification test reports to verify ability of the landing gear and deceleration systems and subsystem components to operate, perform their intend function, and meet specified requirements within the installed operating environment.

(2) Review flight and maintenance manuals to verify life limits, operating restrictions, and procedures are documented due to incomplete qualification test or failures which occurred during qualification testing.

Analysis.

(1) Conduct component SOF and qualification analysis to verify requirements, capabilities, and limitations.

Test.

(1) Lab test: As an alternative to Test MoC (2), conduct qualification testing to verify the ability of the landing gear and deceleration subsystem components to perform their intended function and meet specified requirements within the installed operating environment.

(2) Lab test: As an alternative to Test MoC (1), conduct SOF testing to verify components meet minimum testing requirements in the event full qualification is not achieved before first flight.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Complete full qualification testing of SOF components to verify the ability of the landing gear and deceleration subsystem components to perform their intended function and meet specified requirements within the installed operating environment.

8.6 Auxiliary power systems and emergency power systems.

This section covers APUs (both ground and in flight use applications), airframe accessory gearboxes, engine starting system components, PTO shafts, emergency power systems, and ram air turbines. Examples of emergency power systems may include, but are not limited to, APU, jet fuel starters, and ram air turbines.

A major part of APS and emergency power systems AW is for the components to successfully pass full qualification testing or a minimum of SOF testing before first flight. To achieve full compliance multiple standards will have to be complied with. Those standards and MoCs are Section 4 for environmental qualification (i.e., MIL-STD-810), 8.4 for explosive atmosphere, 8.6 for unique subsystem testing, and Section 13 for EMI.

8.6.1 Qualification of components.

Standard: All components, either individually or as part of the APS and emergency power system, have completed and passed qualification testing (e.g., environmental, durability, acceleration, gyroscopic moments, proof, and burst). Life limits and restrictions are defined for components which did not achieve full qualification.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of SOF testing documents or qualification test reports verifies the ability of the APS, emergency power system, and subsystem components to operate, perform their intended function, and meet specified requirements within the installed operating environment.

(2) Review flight and maintenance manuals to verify life limits and procedures are documented due to incomplete qualification test or failures which occurred during qualification testing.

Analysis.

(1) Conduct component SOF and qualification analysis to verify requirements, capabilities, and limitations.

Test.

(1) Lab test: As an alternative to Test MoC (2), conduct qualification testing to verify the ability of the APS, emergency power system, and subsystem components to perform their intended function and meet specified requirements within the installed operating environment.

(2) Lab test: As an alternative to Test MoC (1), conduct SOF testing to verify components meet minimum testing requirements in the event full qualification is not achieved before first flight.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: Complete full qualification testing of SOF components to verify the ability of APS, emergency power system, and subsystem components to perform their intended function and meet specified requirements within the installed operating environment.

8.6.2 System operations.

8.6.2.1 System operational power.

Standard: The power, torque, speed, bleed pressure, and temperature provided by the APS and emergency power system are within specified design limits of the control system. No single point failure or combination of failures will result in loss of APS and emergency power system as designed to be controlled by APU control system and no failure will have a probability of occurrence greater than 10^{-7} per flight hour that can propagate to impact other emergency power requirements (e.g., loss of a hydraulic pump does not also cause loss of electrical generator).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review control system, APS, and emergency power system design specification documents to verify that performance parameters and critical system limits such as power, torque, speed, bleed pressure, and temperature are established.

Analysis.

(1) Conduct analysis to verify required power levels are provided and maintained for continued operation of flight-critical equipment throughout the design operating envelope of the APS and emergency power system (e.g., maximum flight ceiling to ground, negative g, attitude, stall, surge, and flameout).

(2) Conduct FMECA analysis to verify continued system operation or termination following any combination of failures that have a probability of occurrence greater than 10^{-7} per flight hour, or a single APU control system failure.

Test.

(1) Lab test: Conduct a 150-hour endurance test IAW TSO-C77b, Endurance Test paragraph followed by an acceptance test procedure to verify that the APU system meets the required power demands and endurance capability.

(2) Lab test: Conduct integration testing of the APS and emergency power system controller software with hardware (e.g., APU and APU controller) to verify that the system operates as intended for startup, runtime, and system shutdown throughout the APS and emergency power system design operating envelope.

(3) Ground test: Conduct air vehicle ground testing to verify that the APS and emergency power system meets air vehicle design requirements for start time, from deployment to full loading of the system.

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(4) Ground test: Conduct air vehicle ground testing on the APS and emergency power system to verify the system operates as intended and meets power demands of its interfacing systems (e.g., hydraulic, electrical, and pneumatic).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct air vehicle flight testing on the APS and emergency power system to verify the system operates as intended and meets power demands of its interfacing systems (e.g., hydraulic, electrical, and pneumatic) throughout the APS and emergency power system design operating envelope (e.g., maximum flight ceiling to ground, negative g, and attitude).

8.6.2.2 Safety features shutdown.

Standard: The APU has shutdown features to protect the equipment against hazardous malfunctions and conditions such as overspeed, overtemperature, and inadvertent activation. Shutdown features can be overridden to prevent loss of SCFs during emergency scenarios.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of system design document identifies and describes system hardware and function of protective shutdown.

Analysis.

(1) Conduct system safety analyses (e.g., SSHA) to verify continued system operation or termination following any single failure or combination of failures (e.g., overspeed and overtemperature) that would lead to critical or catastrophic consequences. The analysis also identifies that loss of a shutdown feature (e.g. exhaust gas temperature thermocouple failure, overtemperature at run, overtemperature at start) does not inhibit SCFs during in flight emergency scenarios.

Test.

(1) Lab test: Conduct software V&V testing (e.g., FMET) to verify that the system enables the shutdown features to prevent damage to the APS and emergency power system.

(2) Lab test: Conduct system HITL to verify all protective shutdown features are verified to function as intended via simulated inputs.

(3) Ground test: Conduct air vehicle integration testing to verify intended operation of all APS and emergency power system protective shutdown features.

8.6.3 Functional and physical compatibility.

Standard: The APS and emergency power system do not induce hazardous conditions to the interfacing subsystems (e.g., hydraulic and fuel pumps, generators, starting mechanisms, and ECS ducting). Fault tolerance or fail-safe measures are implemented in safety-critical interfaces.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify that during normal operating conditions and worst-case single failure operating and loading conditions, interfacing subsystems can maintain operation within specified limitations of the system (e.g., bending, torsional and gyroscopic loads, pressures, temperatures, and vibration).

(2) Conduct FMECA to verify that fault tolerance or fail-safe measures implemented in safety-critical interfaces, prevent additional system damage.

Test.

(1) Lab test: Conduct lab testing (e.g., FMET) to verify fault tolerance and fail-safe measures for the operating system function as intended.

(2) Lab test: Conduct integrated system lab testing to verify integrated system is compatible and functions as intended.

(3) Ground test: Conduct ground testing (e.g., stress, thermal, pressure, or vibration surveys) to verify critical analysis assumptions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing (e.g., stress, thermal, pressure, or vibration surveys) to verify critical analysis assumptions.

8.6.4 Containment of failed high speed rotating parts.

8.6.4.1 Damage tolerance and containment.

Standard: High speed rotating components are damage tolerant for minimum of 2X their inspection interval, even in the presence of material, manufacturing, processing, and handling defects for the DSL and design usage specified in the model specification. If design is not damage tolerant, rotating components have provisions in place to contain failed parts, preventing any potential damage to surrounding SCIs, SOF components, and injury to flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of material characterization data validates material properties used in failure mechanics analysis.

Analysis.

(1) Conduct thermal analysis to verify the impact of worst-case operating temperatures on the material properties on the rotating component.

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(2) Conduct stress analysis to verify the stress on the critical location on the rotating component.

(3) Conduct design analysis of complete operational environment of rotating components to verify burst strength using minimum material properties to set RPM limits.

(4) Conduct design analysis of complete operational environment of rotating components to assess minimal LCF life margin using minimum material properties.

(5) Conduct a damage tolerance analysis to determine residual FCG life, safe crack growth periods based on inspection intervals to support two inspections during FCG life.

(6) Conduct trajectory and size analysis in an installed configuration and environment (e.g., air vehicle vibrations) to verify that probability of loss of safety-critical systems and components is 10^{-8} or less in the event of an uncontained failure.

(7) As an alternative to Analysis MoCs (1-5), conduct an analysis of maximum energy burst (three pieces, 120-degree segments) to verify containment of fragments and includes assessment of the various failure modes that may result in axial movement of the rotating group.

Test.

(1) Lab test: Conduct component-level testing to verify thermal analysis assumptions.

(2) Lab test: Conduct component-level testing to verify stress analysis assumptions.

(3) Lab test: Conduct a spin pit test at the component level to verify the component burst strength.

(4) Lab test: Conduct an AMT at the APS and emergency power system level to verify that the life of the rotating components is 2X its inspection interval and validates Analysis MoC (4) and Analysis MoC (5).

(5) Lab test: As an alternative to Test MoCs (1-4), conduct a maximum energy hub burst test at the worst associated operating temperatures and stresses and at a minimum speed of 105% of the normal overspeed shutdown value to verify analysis and verify containment of hazardous fragments. All fragments stay within rotating component housing or within the secondary containment structure.

(6) Lab test: As an alternative to Test MoCs (1-4), conduct a containment test of blade separation at a minimum speed of 105% of the normal overspeed shutdown with the rotor and static structure at the worst associated operating temperatures and stresses to verify analysis and verify containment of hazardous fragments.

8.6.4.2 Power take-off anti-flail.

Standard: Containment or other provisions will prevent a failed PTO coupling assembly or driveshaft from causing secondary damage (due to flailing or whipping) to safety-critical components and systems, including fuel, hydraulic, and ECS lines.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct a FMECA analysis to verify a failed coupling does not cause damage to surrounding safety-critical systems or structure.

(2) In the case a PTO design does not have anti-flail provisions, conduct a zonal analysis to verify there are no safety-critical systems that could be damaged in the surrounding area by a whipping or flailing PTO shaft. The zonal analysis may be part of the FMECA assessment.

(3) Conduct static and dynamic analyses to verify anti-flail provisions have positive strength margins to handle containment in the event of a PTO shaft failure.

Test.

(1) Lab test: Conduct lab testing on PTO system to verify anti-flail capability of the PTO assembly. The anti-flail device will be expected to provide containment for a duration to be specified during air vehicle system development. For a non-containment solution, safety analysis and supporting tests validate analysis that personnel and safety-critical systems will not be damaged or injured in the event of a flailing or whipping shaft.

8.6.5 Vibration.

This section addresses vibrations that are caused by the APS and emergency power system on itself and the surrounding environment.

8.6.5.1 System induced vibrations.

Standard: The installed APS and emergency power system's natural (resonant) frequencies do not cause damage to the APS and emergency power system and surrounding installed environment (e.g., APU bay) at all operating conditions or has damping provisions to prevent resonance damage or failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct simulation or analysis to verify that the system (compressor, turbine, shafts, gear trains, and other highly stressed parts) is free from vibration stresses that damage the system or other air vehicle systems throughout the operating envelope.

Test.

(1) Ground test: Conduct installed ground testing to show that vibration levels of APS and emergency power system do not exceed qualified vibration levels of APS and emergency power system equipment and other surrounding equipment. Inspection of equipment after test shows the system is free of any damage that could compromise structural integrity and function of equipment.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Flight test: Conduct instrumented flight testing to show that vibration levels of APS and emergency power system do not exceed qualified vibration levels of APS and emergency power system equipment and other surrounding equipment.

8.6.5.2 Power take-off shaft alignment.

Standard: Operating the PTO system (e.g., shaft, clutch, flex coupling, interfacing AMAD, and EMAD stub shafts) with the maximum shaft imbalance and installation misalignment does not induce excess vibration or accelerated wear of system components during worst-case operating conditions. Installation procedures identify the maximum misalignment allowed and ensure installation does not allow alignment greater than the defined maximum misalignment condition.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspection of design criteria and maintenance publications verify that requirements for balance and alignment of the PTO shaft tolerances are established.

Demonstration.

(1) Demonstrate the installation of the PTO shaft to verify that installation procedures maintain alignment of PTO shaft within tolerances specified in engineering drawings when installed on the air vehicle.

Test.

(1) Lab test: Conduct endurance testing in a government approved test fixture to verify sufficient shaft power is maintained for operation of the AMAD at the maximum allowable imbalance and misalignment without failure and detrimental vibrations. The test conditions will be based on worst-case operating conditions (e.g., worst-case speed, torque, misalignment) and misalignment of the PTO shaft after testing is maintained within the specified misalignment tolerances.

8.6.6 Response to failures.

Standard: The APS and emergency power system can respond to external failures while providing levels of bleed air, shaft, and electrical and hydraulic power in sufficient time to maintain stable flight and landing of the air vehicle during an emergency.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct performance-based modeling and analysis to verify that the APS and emergency power system has the power capacity to maintain sufficient pneumatic, shaft, electrical and hydraulic power for the duration of time needed to perform established emergency procedures and land while maintaining flight stability.

(2) Conduct analysis to verify sufficient power levels are provided to deploy or start-up flight-critical equipment in the event of loss of main power systems (e.g., APU start-up in flight, main

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engine restart capability by the APS and emergency power system, deployment of ram air turbine, and emergency hydraulic pumps).

Test.

(1) Lab test: Conduct altitude chamber testing to verify that the APS and emergency power system can start in time to provide full loading of systems.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Lab test: Conduct wind tunnel tests to verify deployment and performance of the emergency system (ram air turbine and APU) inlet and exhaust features within its operating environment.

(3) Flight test: Conduct flight testing to verify that the APS and emergency power system meets air vehicle design requirements for start time (e.g., from system startup to main engine start or full loading) and is capable of providing full loading of the APS and emergency power system throughout its design operating envelope and various flight conditions (e.g., maneuvers, attitudes, negative g, and shock), from ground to maximum service ceiling of system (e.g., maximum operating celling of 22,000 ft of electrical power, 16,000 ft for maximum pneumatic power).

8.6.7 Safety considerations.

Safety considerations unique to an APS and emergency power systems include structural mounting and proper cooling and ventilation of a system. As the mounting of an APS and emergency power system is considered a secondary structure, it should be evaluated to determine the mounts that can endure the air vehicle and APS and emergency power system loads experienced throughout the flight envelope. Mounts are imperative for ensuring a system (e.g., APU) remains stationary in flight, not impinging or straining system hardware (e.g., hydraulic tubes and bleed air ducting) connected to it. Several APS and emergency power systems (e.g., hydraulic motors, APU, and gearboxes) may be utilized during ground operations, backup and emergency operations in flight. These systems will produce heat as a biproduct during operation. Thus, it is important to ensure proper cooling and ventilation is provided to these systems to prevent overtemperature conditions that would degrade performance and system operation over time.

8.6.7.1 Structural mounting.

Standard: Auxiliary power system and emergency power system mounts withstand maximum combination of static and dynamic loading throughout the design envelope of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to define static and dynamic load cases for the mounts.

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(2) Conduct structural FEA or modeling to verify the ability to withstand specified limit loads without permanent deformation and ultimate loads without failure throughout the design envelope of the air vehicle.

(3) Conduct analysis to verify that maximum static and dynamic mount loads are not exceeded during imbalance under a failed blade condition, and the critical vibration amplitudes and frequencies transmitted by the APU and EPU from the mounting points to the airframe through the normal operating range of the APU and EPU.

Test.

(1) Ground test: Conduct ground test surveys (e.g. stress, thermal, pressure, and vibration) to verify critical analysis assumptions made.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Review of design criteria to verify maximum static and dynamic mount loads including flight loads and loads that result from APS and emergency power system seizure, imbalance under a failed blade condition, and the critical vibration amplitudes and frequencies transmitted by the APS and emergency power system mounting points to the airframe.

8.6.7.2 Cooling and ventilation of system and components.

Standard: APS and emergency power system compartment cooling and ventilation provisions maintain the temperatures of system components, fluids, and structure within the temperature limits established for these components and fluids, across the APS and emergency power system's operating conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of drawings and system specification documents verify that cooling provisions are in place to ensure continuous cooling of the APS and emergency power system compartment and system across its operating conditions.

Analysis.

(1) Conduct an analysis to show that cooling provisions for the APS and emergency power system maintain system components, fluid, and compartment operating temperatures are below established limits when operated across the design envelope of the system.

Test.

(1) Lab test: Conduct hot and cold environment lab testing (e.g., McKinley Climatic Lab) in mockup compartment or tail cone to verify that APS and emergency power systems have thermal conditioning across its design envelope, in simulated ground and flight operations.

(2) Ground test: Conduct ground testing in temperature extremes to verify the APS and emergency power system components, fluid and compartment temperature limits maintains operating temperatures below established limits.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify that APS and emergency power system components, fluid, and compartment temperature limits are not exceeded across the design envelope of the system.

8.6.8 Flammable fluid ingestion and exhaust gas impingement.

Standard: APS and emergency power system inlet and exhaust characteristics (e.g., velocities, temperatures, acoustics, and exhaust by-products) do not cause damage to air vehicle subsystems and air vehicle structure. Exhaust plumes (gases) are transported off the air vehicle and do not impinge on air vehicle structure and equipment, such that maximum operating and structural temperatures are exceeded. Exhaust plumes do not impinge on flammable fluids or vapor discharge to the extent that auto ignition temperature limits of fluids or gases are exceeded. APS and emergency power system are not susceptible to leakage from flammable fluid lines, fitting, or components entering the inlet air stream.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct inspection of drawings or CAD models to verify that leakage and drainage of flammable fluid lines, fittings, or components cannot enter the intake air stream or exhaust plume of the APS and emergency power system.

Analysis.

(1) Conduct CFD modeling or analysis of various air vehicles orientations and wind conditions to verify that exhaust gases do not impinge on air vehicle structure or equipment, exceeding structural integrity or maximum operating temperatures.

(2) Conduct design analysis or CFD modeling to verify there is no plume attachment to the air vehicle during in flight operation.

Test.

(1) Ground test: Conduct ground testing to verify exhaust plume does not impinge on air vehicle, causing damage to structure or equipment when operated in various air vehicle orientations and wind directions.

(2) Ground test: Conduct ground testing to verify that the exhaust system prevents leakage of exhaust gas into the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight test to verify air vehicle skin temperature limits near exhaust plume are not exceeded and there is no noticeable equipment or structural damage or scarring of air vehicle skin due to plume attachment.

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8.6.9 Accessory drive system compatibility.

Standard: The accessory drive system's compatibility with the air vehicle accessories and engine drive system is designed to withstand torsional vibrations and system, static and dynamic loads, as well as possible misalignment of accessories (e.g., electrical drive generator, hydraulic pump, and fuel pump).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of installation procedures identify the maximum misalignment allowed for accessories and define the maximum allowable misalignment condition.

Analysis.

(1) Conduct design analysis to verify strength margins throughout the drive train and that the drive train is capable of withstanding torsional loading from AMAD or EMAD interfaces.

(2) Conduct design analysis to verify that rotating interfaces (e.g., hydraulic pump, electrical drive generators, and air turbine starters) are capable of withstanding accelerations from the vibrating modes produced by AMAD or EMAD.

(3) Conduct flight load analysis to determine the maximum allowable misalignment of accessories interfacing with the AMAD or EMAD.

Test.

(1) Lab test: Conduct lab testing of the accessory drive system with simulated or physical PTO shaft and accessory loading (e.g., electrical drive generator, hydraulics pump, and clutch) to verify installed performance of the system and torsional loads and vibrations do not exceed qualified limits of the accessory gearbox and its accessories.

(2) Ground test: Conduct installed ground testing to verify torsional vibrations do not exceed qualified limits of the accessory gearbox and its accessories and there are no resonant frequencies observed during system operation.

8.6.10 Flight crew interfaces.

8.6.10.1 Flight crew displays.

Standard: The air system provides a means for the flight crew to monitor APS and emergency power system performance and status. APS and emergency power system faults are recorded and indicated to the flight crew for system failures, degraded performance, and flight-critical equipment temperature limit exceedances.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings, flight crew interfaces and flight manual(s) to verify APS and emergency power system information is provided to the flight crew to monitor APS and emergency power

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system performance and to notify of failures, degradation, or flight-critical equipment temperature limit exceedances.

Test.

(1) Lab test: Conduct lab testing (e.g., HITL, FMET) or air vehicle simulations to verify that APS or emergency power system information and notifications of failures, degradation, flight-critical equipment temperature limit exceedances are provided to the flight crew.

(2) Ground test: Conduct testing on air vehicle of APS and emergency power system under both normal system operation and with simulated failures to verify that APS and emergency power system status information and faults are recorded and provided to the flight crew.

8.6.10.2 Flight crew manual controls.

Standard: Crew station controls (e.g., manual on-off-start and emergency shutdown, and ram air turbine deployment) are provided to allow the flight crew to configure the APS and emergency power system to maintain SCFs across the design envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings, flight crew interfaces, and flight manual(s) to verify APS and emergency power system crew station controls have been provided.

Demonstration.

(1) Conduct on air vehicle demonstration of APS and emergency power system controls under both normal system operation and with simulated failures to verify that APS and emergency power system controls function as designed.

Test.

(1) Flight test: Conduct flight testing or air vehicle simulations to verify that APS and emergency power system controls function as designed.

8.6.11 Technical manuals.

Standard: Flight manuals will include AMAD and APS and emergency power system normal, backup, and emergency procedures, system parameters, indications, WCAs, restrictions, and air vehicle operating limitations. Maintenance manuals will include AMAD and APS and emergency power system life limits, inspections, limitations, and servicing procedures. The manuals' information is substantiated and reflects the as designed or modified system. The technical manuals incorporate unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, and qualification testing findings).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review flight manuals to verify that the flight crew is provided information addressing WCAs, indications, limits of operation, normal, emergency and backup procedures.

(2) Review technical manuals to verify they incorporate unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, and qualification testing findings).

Analysis.

(1) Conduct supporting analyses to establish the basis for the published operating procedures, flight manual limits (e.g., emergency time to land in worst-case conditions), and the system operating limits.

(2) Conduct supporting analyses to establish the basis for the published maintenance manual life limits, inspections, limitations, and servicing procedures including unique actions required to substantiate AW compliance (e.g., mitigation of system design limitations, and qualification testing findings).

Test.

(1) Ground test: Conduct on air vehicle testing to validate and verify flight manual normal and emergency procedures for AMAD and APS and emergency power system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight test to validate and verify flight manual normal and emergency procedures for AMAD and APS and emergency power system.

8.6.12 Failure of lubrication to the airframe mounted accessory drive.

Standard: Airframe mounted accessory drive gearbox can function at the maximum power level for a minimum of 30 minutes with complete loss of lubricant from the lubrication system and is in a condition that the AMAD gearbox is still capable of transmitting the required power to maintain SOF. The operational conditions are such that the loss of lubricant occurs at the most severe power condition and that the air vehicle can transition to cruise and land at the end of the 30-minute period. If an emergency lubrication system is used, any resulting attitude limitations during loss of lubricant operation are defined.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Conduct two separate 30-minute lab tests on two AMAD gearboxes with the lubrication system starved at the system's supply side (downstream from the pump), continuing to scavenge, and at the most severe power condition to verify that the AMADs are still able to function and transfer power to meet interfacing system demands for maintaining SOF. The AMADs are not required to be in a serviceable condition after each test.

Creditable run time starts at the point at which the cockpit low oil pressure warning would be displayed. For non-pressurized AMAD gearboxes, creditable run time starts when the oil being

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drained from the gearboxes ceases to flow in a steady stream. The AMAD gearbox is configured in an air vehicle attitude simulating the cruise power condition. For a VTOL air vehicle, the test spectrum and attitudes are commensurate with expected field use. For AMAD's that contain a clutch, a 30-minute loss-of-lubrication overrunning test consistent with the loss-of-lubricant test spectrum above demonstrates the ability of continued operation.

8.6.13 Failure of airframe mounted accessory drive mounted components.

Standard: Failure or seizure of any individual accessory mounted to the AMAD gearbox does not prevent the AMAD gearbox from driving remaining flight-critical equipment. Examples of flight-critical equipment powered by the AMAD include electrical drive generators, hydraulic pumps, and fuel pump.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct an inspection of product definition data, such as drawings, models, and system descriptions, to verify a method is in place to isolate the failed individual accessory mounted equipment from the AMAD. A typical approach to meet this requirement incorporates a shear section into the shaft of accessories driven by the AMAD gearbox.

Analysis.

(1) Conduct a SCF analysis to verify flight-critical equipment, that operates by power transfer through the AMAD, is identified.

(2) When using spline shear sections, conduct analysis of the shear section feature to verify that the spline shears as intended under the load of the AMAD when a failed component seizes up (inhibits rotation).

(3) Conduct analysis on the design features which isolate a seized or failed accessory from the AMAD to ensure the AMAD is still capable of transmitting power to other accessories mounted to AMAD.

Test.

(1) Lab test: Conduct integrated level testing to verify that failure of flight-critical equipment does not seize the AMAD, and the gearbox continues to provide power to the other flight-critical equipment attached to the AMAD.

8.7 Mechanisms.

Mechanisms include equipment involved in the securing, fastening, and mechanizing of air vehicle doors, hatches, canopies, ramps, and weapon launchers. This equipment includes items such as locks, latches, bearings, hinges, linkages, indicators, and actuators. Mechanical actuation subsystems that provide motion and position locking functions for stowable and deployable surfaces such as folding wing panels, folding rotor blade systems, folding tail rotors, air scoops, air vents, and weapons bay doors are also included. Equipment that is mechanical in form, fit, and function, but not covered by any other system-level requirements should be included herein.

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8.7.1 Functionality.

Standard: All SOF critical mechanisms perform their allocated air vehicle functions under their specified operating environments and load conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The component sizing and operating characteristics are determined analytically based on specified missions.

Demonstration.

(1) Checkouts on the air vehicle verify installation, rigging, normal and emergency functionality, and performance.

Test.

(1) Lab test: Test selection, test methods, and test conditions reflect the expected operating environment(s) for individual components and subsystems. Mechanical systems are designed for safe and durable operation IAW the damage tolerance and durability requirements of the cited air vehicle structures specification.

(2) Lab test: Full scale functional test mock-up duplicating the air vehicle installation and interfaces, and system control verify system performance, rigging, installation, and indication requirements.

(3) Lab test: System components are laboratory tested to life and performance requirements.

8.7.2 Effects of jam loads.

Standard: Latches, locks, linkages, and the rigging have the strength and rigidity to withstand all actuator stall loads resulting from a jam load condition, without adverse deflection or deformation. The system is designed to prevent jamming or blocking by cargo, baggage, or foreign objects in the open or closed position. The design prevents system damage that causes subsequent improper operation. When the system function is linked to the lock system, any damage to the system actuation linkage or support structure does not cause the system and locks to lose their related synchronization.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The sizing and operating characteristics are determined analytically. Analysis is used to determine conditions where the mechanical linkage exerts the greatest force due to the available mechanical advantage on the critical parts of the system.

Test.

(1) Lab test: Integration of the mechanical systems tested in a mock-up simulating the expected loads and structural attachments. Testing verifies that the system will withstand jam loads at any point in the system without detrimental deformation. All jam load tests are conducted on the full

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scale article in conjunction with the latch and lock jam load tests. Jam load tests are accomplished on every critical part of the system.

8.7.3 Effects of mechanism failure.

Standard: The failure of any mechanism does not cause the loss of control of the air vehicle or prevent landing. Failure of mechanism (e.g., loss of a door, failure of mechanical system) does not rupture hydraulic lines, tear electrical cables, or disable any additional systems which results in loss of the air vehicle or prevents landing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the air vehicle drawings and analysis of the failed mechanism verify that damage will not occur to the flight-critical components such as control surfaces and engines.

Analysis.

(1) Conduct a FMECA to verify the consequence of hydraulic, electrical, or mechanical failures or loss upon door separation or mechanical failure.

8.7.4 Independence from flight controls.

Standard: The inadvertent loosening or opening of air vehicle doors, door latches, locks, or fasteners does not restrict the operation of any FCS. In flight operation of any mechanical system and doors does not interfere with or affect the operation of any flight control surface, nor compromise the ability to maintain flight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of the air vehicle drawings to verify the independence of flight controls from the mechanical systems.

Analysis.

(1) Conduct a FMECA to verify the consequences of inadvertent loosening or opening of the mechanical components.

8.7.5 Fail-safe latching.

Standard: No single failure allows any door latch to open inadvertently. The latching system holds the doors closed in the event of a single failure within the system, to include hinges. The latching system is fail-safe. Maximum possible relative deflection between the air vehicle structure, doors, and the latching system do not cause unlatching under any ultimate design loading condition. Locking and latching systems are separated so that a single malfunction cannot cause the hazardous actuation of both the locking and latching actuators.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Failure modes of the latching system are determined by analysis. Critical loading conditions are identified by structural analysis.

Demonstration.

(1) Demonstrations are conducted to ascertain that no door opening will occur subsequently to a single latch system or hinge failure.

Test.

(1) Lab test: Structural tests are conducted to ascertain that no door opening will occur subsequent to a single latch system or hinge failure.

8.7.6 Interrelation of latching and locking systems.

Standard: Any locking system is incapable of locking or indicating it is locked unless all the latches are properly latched in the fully secured position. The latching system is independent of the locking system so that any inadvertent unlatching attempt does not cause unlocking. There is no possibility of an out of sequence door locking action producing a false indication of a locked door. There is no indication of a locked condition other than when the locking action is fully completed.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify by analysis that the latching system is independent of the locking system so that any inadvertent unlatching attempt will not cause unlocking. Verify by analysis that the locking system is independent of the latching system and that inadvertent unlocking does not cause unlatching. Verify by analysis that inadvertent latching system activation does not cause unlocking or unlatching with the locking system engaged.

Demonstration.

(1) Conduct a demonstration to verify that the latching system is independent of the locking system so that any inadvertent unlatching attempt will not cause unlocking. Conduct a demonstration to verify that the locking system is incapable of locking or indicating it is locked unless the latches are in a fully secured position and the locking action is completed. Conduct a demonstration to verify that the locking system is independent of the latching system and that inadvertent unlocking does not cause unlatching. Conduct a demonstration to verify that inadvertent latching system activation does not cause unlocking or unlatching with the locking system engaged.

8.7.7 Door pressurization interlock.

Standard: All air vehicle doors, whose inadvertent opening would present a probable hazard to continued flight and landing, have provisions to prevent depressurization (or inadvertent

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pressurization) of the air vehicle to an unsafe level if the doors are not fully closed, latched, and locked. The door latches cannot be unlocked unless the air vehicle has been depressurized. The latching system incorporates an interlock that senses fuselage air pressure differential to prevent unlocking. The system also guards against pressurization before doors are latched and completely locked.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to verify that the locking system is incapable of unlocking at unsafe pressurization levels.

Demonstration.

(1) Conduct a demonstration to verify that the locking system is incapable of unlocking at unsafe pressurization levels. Demonstration verifies that the system prevents depressurization when all doors are secured and compartments are pressurized. Demonstration verifies that the system prevents pressurization before doors are secured.

Test.

(1) Lab test: Conduct testing on a full scale test model to verify that a completely assembled lock and latching system will not unlock under unsafe pressurization levels, even when the locking system receives a command to unlock.

(2) Ground test: All probable failure modes are tested on a full scale test article. Testing ensures that the pressurization prevention system will hold the locks in the closed position when the fuselage is pressurized, and the locks are energized to open.

8.7.8 Door status monitoring and indication.

Standard: The lock or unlocked status of the doors is continuously presented to the flight crew and ground crew. The system also assists in trouble shooting in the event of a malfunction. Indicator sensors directly sense the position of doors, latches, and locks without the use of sensor targets. Mechanical indicators use positive mechanical linkage for extension and retraction. The indication system is designed such that the deflection of the air vehicle structure under all ground and flight load conditions does not cause false indications.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of air vehicle drawings to verify that the sensors will only respond to actual door system components position.

Analysis.

(1) Conduct an analysis to verify that each indication system will continuously monitor and provide an unsafe indication when either the door, latching, or locking system is unsecured and will provide a safe indication when the systems are secured. All malfunctions that could give a

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false indication are included. Particular emphasis is placed on those malfunctions that could give a safe indication for an unsafe condition.

Demonstration.

(1) Conduct a demonstration to verify that each indication system will continuously monitor and provide an unsafe indication when either the door, latching, or locking system is unsecured and will provide a safe indication when the systems are secured. All malfunctions that could give a false indication are included. Particular emphasis is placed on those malfunctions that could give a safe indication for an unsafe condition.

Test.

(1) Ground test: Conduct ground tests to verify that the indication system will not report false indications due to the deflection of the air vehicle structure under all ground load conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight tests to verify that the indication system will not report false indications due to the deflection of the air vehicle structure under all flight load conditions.

8.7.9 Emergency door operation.

Standard: Door control systems are designed for emergency operation by means of manual actuation of the door and drive sequence. The door system can be manually operated in the event of a control or primary power source failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to verify that the system can be manually sequenced or opened in an emergency.

Demonstration.

(1) Conduct a demonstration on the full scale test article or production air vehicle to verify that the system can be manually sequenced or opened in an emergency. Demonstrations include the event where normal system operation opens the doors partially and manual operation is required to complete the operating cycle.

8.7.10 Door seals.

Standard: All door seals prevent rain or water leakage into the air vehicle during all flight and ground operations and while the air vehicle is parked and depressurized under storm conditions. All seals are airtight when pressurized and protect against water entry when the air vehicle is unpressurized. The seals exclude water at all times with the doors or canopy closed.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct an analysis to verify that the seals will have satisfactory performance when pressurized and unpressurized.

Demonstration.

(1) Conduct a demonstration on the air vehicle to verify under rainstorm conditions that there are no leaks in the air vehicle when pressurized and unpressurized.

8.7.11 Environmental conditions.

Standard: All actuation subsystems are able to be locked and unlocked, provide for folding, unfolding, and deploying, and can be folded, unfolded, and deployed within a wind environment that encompasses atmospheric and weather-induced conditions, wind-over-deck from carrier vessel movement, and downwash and jetwash conditions caused by other air vehicles expected in the operational ground and deck environment. The mechanical subsystem is able to operate and complete all its functional actions (to include locking) under all expected mission and environmental conditions. Worst-case conditions include operations in the air and on the ground.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to verify locks can withstand the environmental conditions without opening.

Test.

(1) Lab test: Conduct component and system load testing to verify operation at worst-case conditions.

(2) Ground test: Conduct testing on air vehicle to verify operation of locks at worst-case conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct testing on air vehicle to verify operation of locks at worst-case conditions.

8.7.12 Locking of structural load path mechanisms.

Standard: Mechanisms which provide a SOF structural load path incorporate redundant means of locking the mechanism in position. Mechanical systems are designed IAW with the damage tolerance and durability requirements of the cited air vehicle structures specification.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct static analysis to verify the locking function has sufficient strength margins for limit and ultimate load conditions.

Test.

(1) Lab test: Conduct testing to verify the locking function ability to support limit and ultimate load conditions.

8.7.13 Locked or unlocked condition.

Standard: The locked or unlocked condition of mechanisms with sensors is displayed in the cockpit, on air vehicle, and in all remote flight crew control locations during ground operations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a FMECA to verify probable failures and system indications associated with the failures.

Test.

(1) Lab test: Conduct testing on the full scale article and control station, as applicable, to verify that the conditions of mechanisms with position sensors are properly displayed. Testing includes all the malfunctions that could give a false indication as determined by the analyses. Particular emphasis is placed on those malfunctions that could give a safe indication for an unsafe condition.

8.7.14 Actuation power and control.

Standard: A means is provided for controlling utility actuation. The actuation power and control for moving the mechanism to its commanded position is independent from the power and control used to hold and lock the mechanism in its initial and final commanded state.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The design and operating characteristics are verified by analysis.

Demonstration.

(1) Demonstrations on the air vehicle validate mechanical system installation, rigging, functionality, and performance. Normal and emergency functions are demonstrated and verified.

Test.

(1) Lab test: System components are laboratory tested to the performance requirements.

(2) Lab test: Full scale functional tests duplicating the system power and system control verify system performance, rigging, installation, and indication requirements.

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8.7.15 Actuation time.

Standard: All actuation subsystems are able to perform their specified function within the specified time and cycle. Also specify allowable intervals between actuation cycles as well as total cycles expected during the application lifetime. The time between initiation of the command to the completion of the action is within the design allowable. The time between successive operations of the same cycle is not degraded and can be consistently repeated, for the DSL.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to verify the actuation rates, cycle times, and interval times to be completed as required.

Test.

(1) Lab test: Conduct component and system tests to verify actuation rates, cycle times, and interval times to be completed as required.

(2) Lab test: Conduct full scale checkouts on a simulator to verify system installation, rigging, control, and functional performance.

(3) Ground test: As an alternative to Test MoC (2), conduct on the air vehicle testing to verify system installation, rigging, control, and functional performance.

(4) Ground test: Conduct ground tests to demonstrate compatibility with air vehicle performance requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct air vehicle flight tests to demonstrate compatibility with air vehicle performance requirements.

8.7.16 Actuation without damage.

Standard: Utility actuation subsystems incorporate some means to prevent damage to adjacent movable surfaces (e.g., flaps) during folding and unfolding operations. The actuation system control has interlocks or logic that will prevent actuation power or movement when other mechanical surfaces or flight control surfaces are in a position to be damaged or compromised.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of components and systems to verify actuation logic and control to the extent that no actuation power is applied, or movement occurs when other surfaces are in the way.

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Demonstration.

(1) Conduct checkouts on the air vehicle verify system interlock and control logic meet design requirements.

8.7.17 Actuation subsystem attachment location.

Standard: The actuation subsystem attachment is not an integral part of the air vehicle structure, such as a wing rib, but is a replaceable attachment designed so that, in case of an overload or fatigue failure, the attachment fails in lieu of a structural component failure on the primary air vehicle. No structural failure of the mechanical system actuation system and its attachment fitting causes damage to air vehicle structure. Replacement of mechanical system attachment does not require the replacement of major structure (e.g., bulkhead, and keel beams).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings to verify that mounting and attachment fittings are designed to be replaceable without major structural effects.

Analysis.

(1) Conduct an analysis to verify that mounting and attachment fittings are designed to be replaceable without major structural effects.

(2) Conduct static analysis to verify positive strength margins for design limit and ultimate loads.

Test.

(1) Lab test: Conduct laboratory testing to verify that system components meet DLLs and ultimate loads.

8.7.18 Mechanism clearances.

Standard: Clearance is provided in the deployed or stowed position and during the deployment operation to prevent damage to the surface, attached equipment, and to other areas of the air vehicle. When stationary or in motion, clearance is maintained between the mechanism and all other air vehicle structure for all expected operational conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of components and systems, including worn states, to verify loading and natural and induced environmental conditions of various components.

Test.

(1) Lab test: Conduct component and system tests to verify clearances between air vehicle structure and systems.

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(2) Lab test: Conduct full scale checkouts on a simulator to verify system installation, rigging, and clearances.

(3) Ground test: Conduct checkouts of the air vehicle to verify system installation, rigging, and clearances.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct air vehicle flight tests to demonstrate compatibility with air vehicle clearance requirements.

8.7.19 Aborted and resumed operation of controls.

Standard: The controls automatically stop the sequence in the event a malfunction or failure occurs part way through a cycle, and in the event of a power interruption, doors do not change position, and the door controls go to the stop position and remain in the stop position upon resumption of power. In the event of a power interruption, the doors go to the stop position and remain in the stop position upon resumption of power. The door controls do not require reprogramming upon resumption of power.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review air vehicle documents to verify that the door controls are provided such that the flight crew can operate the controls quickly and safely without danger of damaging the air vehicle.

Analysis.

(1) Conduct a FMECA to verify the consequence of hydraulic, electrical, pneumatic or mechanical failures.

Demonstration.

(1) Conduct a ground demonstration to verify that the door controls are provided such that the flight crew can operate the controls quickly and safely without danger of damaging the air vehicle. Conduct a demonstration of the door control system capability to react to the failure situations identified in Analysis MoC (1).

8.7.20 Crewmember sequencing of controls.

Standard: The controls are capable of stopping or reversing door movement at any time in the cycle at the option of the crewmember by selecting the appropriate control option. The door will move to the last commanded position at any time in the control sequence. Commanded changes in door position during the cycling sequence do not cause jamming nor prevent the door from obtaining the last commanded position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct a FMECA to verify operation for all commanded door positions.

(2) Conduct an analysis to verify that jamming will not occur.

Demonstration.

(1) Conduct ground demonstration on the air vehicle to verify that door movement matches commanded position.

8.7.21 In-flight actuation prevention for ground only systems.

Standard: Locking mechanisms incorporate a means of operational command interrupt to prevent in-flight actuation of ground-only operating systems, that all mechanical and powered locks and actuators are designed to prevent undesired surface positioning in flight, and that the "fold" sequence requires two separate, deliberate actions in the case of flight-critical surfaces. The locking features are positive, do not depend on any power source to remain engaged, and are not accessible during flight conditions. A means of operational command interrupt is incorporated to prevent actuation of ground-only systems while in flight. The "fold" sequence requires two separate deliberate crewmember actions in the case of flight-critical surfaces such as wings and rotor blades.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review air vehicle documents to verify that the ground-only systems cannot be operated in-flight.

Analysis.

(1) Conduct an analysis to verify that the ground-only systems cannot be operated in-flight.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight demonstrations to verify that the ground-only systems cannot be operated in-flight.

8.7.22 Prevention of inadvertent actuation.

Standard: Positive locking provisions are incorporated to prevent inadvertent actuation of a mechanism after a safety device such as a fuse or thermal switch is activated, and which do not depend on any power source to remain engaged. Positive locking provisions, once engaged, remain engaged even after a safety device is activated or when a power source is removed.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review drawings to verify positive locking provisions.

Analysis.

(1) Conduct analysis to verify strength of the positive locking provisions.

Demonstration.

(1) Conduct ground demonstration of the positive locking provisions.

8.7.23 Strength of removable surface securing devices.

Standard: Removable surface securing devices, if specifically permitted by the government, have strength equal to or exceeding that of the air vehicle. Removable surface securing devices are only used in lieu of integral locks when specifically authorized by the government. These devices are capable of withstanding rough handling without damage and can secure the movable surface in winds up to 60 kts from any direction.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings to verify that mounting and attachment fittings are designed to be replaceable without major structural effects.

Analysis.

(1) Conduct an analysis to verify that mounting and attachment fittings are designed to be replaceable without major structural effects.

(2) Conduct an analysis to verify the system components design loads and ultimate loads.

Demonstration.

(1) Conduct on air vehicle demonstrations to verify installation, functionality, and performance.

Test.

(1) Lab test: Conduct laboratory testing to verify that system components have positive strength margins for DLLs and ultimate loads.

8.7.24 Performance of bearings.

Standard: Airframe bearings are capable of joining mechanical elements; transmitting design loads through the full range of the system operating parameters; permitting rotation, misalignment, or both while maintaining a specified dimensional relationship between the joined elements; and reducing friction and wear. Airframe bearings' strength meets the static loading conditions without deformation and fracture. The airframe bearings are designed to operate under the dynamic loading conditions over the intended range of motions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Conduct an analysis to verify bearing's ability to operate under the static and dynamic loading conditions over the intended range of motions.

Test.

(1) Lab test: Conduct testing on bearings to verify their ability to operate under the static and dynamic loading conditions over the intended range of motions.

8.7.25 Life limit of bearings.

Standard: Each flight safety-critical bearing has defined life limits established to operate under load in worst-case environmental conditions. A life limit ensuring a probability of failure less than 10^{-7} per flight hour is established.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to verify the worst-case environmental conditions.

(2) Conduct and analysis to establish a baseline life under a representative load and motion spectrum.

Test.

(1) Lab test: Conduct component testing to verify the integrity and durability of the bearings. One uncontaminated bearing is tested to establish a baseline life under a representative load and motion spectrum. A sample of bearings is tested with that same spectrum and worst-case environmental contamination conditions for that type of bearing (e.g., plain lined, rolling element, or elastomeric). Results of testing are analyzed to determine bearing safe life.

8.7.26 Endurance of mechanisms.

Standard: All SOF mechanisms have positive durability margins to maintain residual strength and endurance throughout the specified service life under all expected mission, environmental extremes, and contaminated conditions simultaneously (e.g., cycles, vibrations, loads, temperature, contamination).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to show positive margin to meet for 4X the endurance life limit.

Test.

(1) Lab test: Conduct testing on mechanism to verifies successful completion of 2X the endurance life limit without degradation of performance or wear exceeding wear tolerances.

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8.7.27 No binding or jamming of flight-critical mechanisms.

Standard: All SOF critical mechanisms do not cause binding or jamming with surrounding structure or any portion for the system under any combination of temperature effects, air loads, landing loads, structural deflections, tire condition, landing gear condition, buildup of manufacturing tolerances or wear. Sufficient clearance is maintained to prevent jamming of mechanisms with surrounding structure and system. The following minimum clearances are used after all combined adverse effects are accounted for:

a. 0.25-inch between elements which move in relation to each other, and which are connected to or are guided by different structural or equipment elements.
b. 0.50-inch between elements and air vehicle structure and equipment to which the elements are not attached.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis to show that the minimum clearance in worst-case conditions is not exceeded.

Test.

(1) Lab test: Conduct ground testing to verify critical clearances are actively monitored during endurance testing and maintained.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify critical clearances are actively monitored and maintained.

8.7.28 Canopy deployment power.

Standard: Deployment power is available under normal and emergency conditions and there is no interference with manual actuation of the canopy when air vehicle or external power is not available. An alternate or secondary power source is provided that will operate the canopy. A manual system is provided for ingress and egress with all air vehicle power off and for canopy operation when primary actuation system is not available. An external means is provided to enable a ground rescue crew to manually open the canopy.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Conduct demonstration on the air vehicle to verify deployment power availability and manual capabilities.

Test.

(1) Lab test: Conduct lab testing to verify deployment power availability of the actuation system.

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8.8 Rotary-wing drive system.

The rotary-wing drive section includes the primary mechanisms which provide power to the rotors. The main components are shafts, coupling, and gearboxes.

8.8.1 Design usage and duty cycle.

Standard: The rotary-wing drive system usage accommodates all intended operating environments. Design usage also includes air vehicle missions and mission mix parameters to determine rotor drive systems loading and usage, flight envelopes, externally applied forces, engine attitude limits, ambient temperature distribution, icing environmental conditions, corrosive atmosphere conditions, noise environment, and engine performance retention characteristics. Based on these conditions a duty cycle for the rotor and drive systems in the air vehicle is determined from the loads and environmental spectra representative of the required service life to meet life requirements in the air vehicle specification.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., system configuration descriptions, system layout, system ICD, and rotor drive system loads derivations) to verify environmental and mission load spectra and duty cycle are defined.

Analysis.

(1) Conduct analysis of drive system materials, design, and operating conditions to verify the system architecture supports all intended operating conditions.

(2) Conduct analysis of air vehicle flight envelope, torque requirements, maneuver loads, and other air vehicle induced loading and determine the duty cycle requirements of rotor and transmission drive systems for the air vehicle to meet service life requirements.

Test.

(1) Ground test: Conduct ground testing to verify air vehicle loads, excursions, and usage are compliant with the environmental and load spectra and duty cycle defined to meet the air vehicle specification requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify rotary-wing drive system loads and usage estimates developed from Analysis MoC (1) and Analysis MoC (2) are valid under the thermal, icing, and noise environments as specified in the air vehicle specification requirements.

8.8.2 Drive system design.

Standard: The rotary-wing drive system (housings, gears, bearings, shafts, couplings, and clutches) supports the power, torque, and strength for operation throughout the designed flight envelope. The strength margins of the drive system components include application of

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appropriate limit and ultimate LFs as specified in the air vehicle specification. The rotary-wing drive subsystem is of a robust design capable of operating beyond its maximum rated condition for those instances where excursions may occur such as autorotation, other emergency conditions, and defined transients. Excursion capabilities are defined as:

a. An applied torque of at least 20% greater than the system input maximum continuous rating without degradation of the component's performance or life.
b. An output shaft speed of at least 20% greater than the maximum operating speed of the power absorber without degradation of transmission system, component performance, or life.

Transient capability of the power drive system is defined IAW 8.8.1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, design data records, inspection reports, ICD, test reports, or other substantiating records and reports) to verify that maximum torque and speed limitations are defined IAW 8.8.1.

Analysis.

(1) Conduct loads analysis to verify the expected levels of power produced by the rotary-wing air vehicle and its associated components and systems.

(2) Conduct stress analysis based on the loads developed in Analysis MoC (1) to verify the rotary-wing drive system has positive strength margins of safety to meet the maximum rated torque, speed, and excursion capabilities.

Test.

(1) Lab test: Conduct component-level testing on shafts to verify the structural integrity, performance, and capability to run at speeds and torques which are at least 20% greater than the system input maximum continuous rating.

(2) Lab test: Conduct bench testing to verify the structural integrity, performance, and capability of the drive system to run at speeds and torques which are at least 20% greater than the system input maximum continuous rating.

(3) Ground test: Conduct ground test to verify the structural integrity, performance, and capability of the drive system to run at speeds and torques which are at least 20% greater than the system input maximum continuous rating.

8.8.3 Structural design.

Standard: The rotary-wing drive systems components are structurally capable of supporting operations of the air vehicle throughout its flight envelope. All drive system components are able to operate without any degradation to durability throughout the air vehicle and engine envelopes. Durability margins, as developed through ASIP process or system specification requirements, of the drive system components include design allowables utilizing appropriate material knockdown factors for fatigue life evaluations. The drive system is capable of operating beyond its maximum rated torque conditions without degradation to the systems durability for those

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instances where excursions may occur such as autorotation, other emergency conditions, and defined transient conditions. Excursion capabilities are defined as:

a. An applied torque of at least 20% greater than the subsystem (e.g., gearbox, shafting) input maximum continuous rating.
b. An output shaft speed of at least 20% greater than the maximum operating speed of the power absorber. System load limits are established.

Each gearbox of the drive system and associated components are rated at the most severe input power condition (torque and speed) for all allowed operating modes exclusive of transient conditions. Transient capability of the drive system is defined by the contractor relative to the specific air vehicle application. The rating is based on the durability, DR, and structural integrity requirements specified in the air vehicle specification.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., design reports, hardware drawings, material and manufacturing records, design data records, inspection reports, or other substantiating records and reports) to verify derivation of design allowables, maximum rated torque, speed, and transient conditions the drive system will be capable of meeting.

Analysis.

(1) Conduct a fatigue analysis to verify durability margins for all components in the drive system meets the expected life of the air vehicle IAW 8.8.1.

(2) Conduct gear tooth bending fatigue analysis to verify DSL (e.g., unlimited).

Test.

(1) Lab test: Conduct gear tooth bending fatigue bench testing of a production representative gearbox design at elevated loads (e.g., 125%, 140% of maximum continuous power) for unlimited life in bending fatigue.

(2) Lab test: Conduct Integrity and Overstress bench test of gearboxes to verify the need for any design configuration changes. Testing should consist of 200 hours with at least 100 hours at 120% rated gearbox power and 60 hours at 100% rated gearbox power. Testing completed at 120% rated gearbox power include periods of operation at overspeed conditions.

(3) Lab test: Conduct Production Configuration Verification bench testing to verify the final production configuration and the changes made as a result of the Test MoC (2). Testing should consist of 200 hours at the representative load spectrum for the air vehicle and incorporate all production and design changes made to drive system components occurring after the integrity bench test. An intermediate inspection of hardware should be completed at 60 hours for assessment of gearbox condition and resolution of discrepancies prior to continuation of testing.

(4) Lab test: Conduct system-level pre-flight acceptance test to verify the need for required inspections, limitations, and restrictions for flight release. A 50-hour system-level test should be completed without component replacement before first flight using a missionized test cycle that

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simulates the projected use of the air vehicle. The configuration of the drive system components should be equivalent to the flight test air vehicle configuration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Lab test: Conduct 200-hour Military Qualification Test of each gearbox of the drive system to verify that the gearbox is operating as designed and tear-down inspections do not reveal failure, impending failure, or undue wear which would compromise SOF or reliability. A 200-hour test should be run without component replacement using the pre-flight acceptance test cycle and test hours should accrue at a rate to lead the flight hours of the lead flight test vehicle by at least a 2 to 1 ratio until that high time air vehicle attains 100 flight hours.

8.8.4 Dynamic coupling.

Standard: The rotary-wing drive system is free of damaging dynamic coupling modes when the engine, engine accessories, rotor system, and all dynamic transmission components are operated as a system. The engine control system ensures positive gain and phase margins are present to avoid torsional instabilities. Sub-critical shafting connecting components within the drive systems are 20% below the first critical speeds. Super-critical shafting, when used in the drive system, has positive strength margins, damping, and resonance separations 20% above and below any critical speeds to ensure ability to operate.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., system layout, hardware drawings, system configuration descriptions, and ICD) to verify defined transmission system and engine interfaces. Vibration characteristics, limits, and torsional stability requirements are defined within the air vehicle specification.

Analysis.

(1) Conduct torsional stability analysis demonstrating no dynamic instabilities are present. Torsional stability analysis considers the engine control system interfaces with the drive subsystem throughout the flight envelope. The analysis defines all drive system spring constants, inertia, and damping coefficients utilized in the assessment with substantiation data to demonstrate by analysis the absence of dynamic coupling modes that would impact the air vehicle in all ground and flight conditions.

(2) Critical speed analysis is provided for all drive system shafting to identify the critical speed margins in relation to expected operating speeds. All sub-critical shafting is 20% below the first critical speed and super-critical shafting analysis shows damping provisions to prevent stress and deflection amplitudes from exceeding material design allowables.

Test.

(1) Lab test: Conduct bench test of all shafting to verify shaft critical speeds are within predicted values.

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(2) Ground test: Conduct bench test to verify stresses are within material design limits for all shafting operating at any critical speeds.

(3) Lab test: Conduct RAP testing on all shafting to verify modal analysis predictions (resonant frequencies).

(4) Ground test: Conduct restrained ground testing of air vehicle to verify no damaging dynamic coupling modes exist between the engine and drive system once installed in the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight testing of air vehicle to verify no damaging dynamic coupling modes exist between the engine and drive system once installed in the air vehicle.

8.8.5 Rotor blade passage resonant frequencies.

Standard: The rotary-wing drive system, including housings, gearing, shafting, couplings, and air vehicle mounting locations are designed to be free from damaging resonant frequencies at all operating speeds and powers throughout the flight envelope of the air vehicle including any effects of high-energy, low frequency vibrations generated by the main and tail rotor systems. Components should be free of damaging resonant frequencies at designed operating speeds and a minimum of ±20% of those operating speeds.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documents (e.g. system configuration descriptions, design data records, hardware drawings, inspection reports, and ICD requirements) to verify vibration characteristics requirements are defined in the drive system design specifications.

Analysis.

(1) Conduct resonant frequency and modal analysis of all transmission system components (e.g., housings, gearing, shafting, couplings, and air vehicle mounting locations) throughout the envelope of rated torque and speeds for the air vehicle to substantiate no damaging resonance modes exist.

(2) Conduct analysis of air vehicle rotor system vibrations for impacts to transmission system components and substantiate no damaging vibration modes exist.

(3) Conduct a stress response analysis based on the frequency responses of Analysis MoC (1) and Analysis MoC (2) to determine if the drive system has positive stress margins to address vibrational impacts.

Test.

(1) Lab test: Conduct component-level RAP testing of components to confirm modal predictions in analytical design.

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(2) Lab test: Conduct a gear resonance frequency test to verify vibratory modes. The dynamic stress levels in each gear are measured in locations sensitive to all significant vibratory modes. A speed scan from zero to the speed of maximum overspeed is performed with:

a. Minimum load.
b. Approximately 50% load.
c. Maximum load.

(3) Ground test: Conduct ground air vehicle testing to verify absence of resonant frequencies in the drive system.

8.8.6 Misalignment.

Standard: The drive shaft couplings can accommodate the maximum permissible misalignment of the drive shafts for all possible combinations of torque and speed, including airframe deflections.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., component specification, hardware drawings, system configuration descriptions, ICD, rotor drive system test reports, or other substantiating reports) to verify the couplings' maximum misalignment allowance is defined.

Analysis.

(1) Conduct analysis of the airframe deflections, coupling interfaces, and manufacturing tolerances to define amount of allowable misalignment.

(2) Conduct vibration analysis to verify natural frequencies and acceleration loads at maximum misalignment identified in Analysis MoC (1).

(3) Conduct a stress analysis to verify the drive shafts and coupling mechanism components are capable of operating at 120% maximum torque at maximum permissible misalignment.

(4) Conduct fatigue analysis to verify design life. Analysis is based on minus 3-sigma, 50% confidence, material curves. The analysis includes prediction of the range of values for steady, cyclic, and vibratory stresses, and the life predictions relative to the analysis. Analysis ensures the loading reflects the influence of the environmental and operational factors on the life calculation methods of the drive coupling fatigue sensitive components.

(5) Conduct damage tolerance analysis based on worst case combinations of misalignment, torque and speed throughout the flight and ground envelope IAW 8.8.1 to verify crack growth rates within at least 3X the normal inspection interval.

Test.

(1) Lab test: Conduct component-level testing at required torque and speed with shafts at maximum misalignment to verify drive system has positive strength margins of safety.

(2) Lab test: Conduct system-level testing to verify modal response.

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(3) Lab test: Conduct endurance fatigue testing at maximum permissible misalignment on two production quality couplings to verify the coupling fatigue life.

(4) Lab test: Conduct damage tolerance testing on a bench setup to verify ability to transmit torque. If a flexible coupling is used, testing is conducted with a flexure failure on one side of the coupling. Testing is completed on one of the two endurance test couplings. Prior to the test, the coupling is intentionally damaged by scratching or punching the flexure surface at the most limiting and critical location. Testing validates continued operation from initial failure indication to complete loss of function per the standard, and visual and audible inspections of the specimens, performed during the testing, validate any inspection intervals documented in technical data. The vibration characteristics of the coupling are monitored and recorded during testing to validate vibration analysis. Fault tolerance testing ensures continued operation from initial failure indication to complete loss of function for a duration of at least 3X the normal inspection interval.

8.8.7 Drive shaft containment.

Standard: The coupling does not allow flailing of the drive shaft in the event of a component failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of drawings verify anti-flail protection is installed.

Analysis.

(1) Conduct anti-flail analysis to verify drive shaft containment at maximum torque and speed conditions.

Test.

(1) Lab test: Conduct bench test with a failed coupling to verify continued operation for 30 minutes.

8.8.8 Rotor securing.

Standard: Rotors can be held from rotating in winds up to 45 kts, from any direction, during engine power up and at ground idle to prevent inadvertent liftoff under heavy winds. The system is capable of being operated from the cockpit by the flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions) to verify rotor brake(s) can be operated by the flight crew from the cockpit.

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Analysis.

(1) Conduct a torque sizing analysis to verify ability of brake(s) to secure rotors up to winds of 45 kts.

(2) Conduct static strength analysis to verify braking components meet limit and ultimate torque conditions. Ultimate load is defined as 2X limit torque.

Demonstration.

(1) Conduct ground demonstration to verify the capability to keep the rotors from rotating under 45 kts winds during engine power up and at ground idle.

Test.

(1) Lab test: Conduct lab test to verify brake(s) can produce and hold required torque.

8.8.9 Rotor brake shutdown.

Standard: The rotor braking system(s), consisting of aerodynamic rotor drag and subsequent mechanical braking, are capable of stopping the rotor, from 100% speed, after engine shutdown. The rotor braking system, at a minimum:

a. Can accomplish two emergency stops from 100% rotor speed within the shortest duration possible (the brake is permitted to be nonrepairable after the second emergency stop). The minimum stopping time is based on a structural analysis to protect the drive subsystem gears and components from overloads due to sudden stops.
b. Prevents rotation of rotor, under various ground and shipboard conditions, when the rotor brake is applied.
c. Has no critical vibratory modes or torsional dynamic instabilities from 0 to 120% of maximum operating speed.
d. Can stop the rotor system at a specified engagement speed for a minimum of 400 cycles.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., system layout, hardware drawings, system configuration descriptions, and ICD) to verify power systems and interfaces are defined.

Analysis.

(1) Conduct thermal analysis to verify rotor has mass to absorb heat and not lose torque stopping capability. Temperatures generated do not result in brake fires or loss of brake component structural integrity.

(2) Conduct a torque sizing analysis to verify ability of brake(s) to stop rotors for emergency stops.

(3) Conduct wear analysis to verify ability of brake(s) to stop the rotors for 400 engagements.

(4) Conduct static strength analysis to verify braking components meet limit and ultimate torque conditions. Ultimate load is defined as 2X limit torque.

(5) Conduct fatigue analysis to verify rotor braking system meets DSL.

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Test.

(1) Lab test: Component bench testing verifies the capability to keep the rotors from rotating, and the ability of the brake to stop the rotors for both normal and emergency stops in the required time as defined in the air vehicle specification.

(2) Lab test: Conduct wear tests to verify brake is capable of stopping the rotor 400X, without replacement of any part. Test cycles based on a mix of the normal operational and emergency rotor speeds as defined in the air vehicle specification.

(3) Lab test: Conduct static torque lab testing to verify brake(s) can produce and hold required torque.

(4) Lab test: Conduct lab testing to verify brake system ability to secure the rotors during repeated single engine startup cycles, at the specified power, without failure and verify the brake system cannot be engaged during flight.

(5) Ground test: Conduct ground testing on the air vehicle to verify the ability to stop rotors from the maximum rotor speed as defined in the air vehicle specification.

8.8.10 Control interlocks.

Standard: Control interlocks (e.g., engine speed and WOW) prevents inadvertent actuation of the rotor braking system to prevent overloading of brakes and loss of rotor speed if air vehicle is airborne.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of control interlocks logic to verify in ability of the brakes to applied as required.

Analysis.

(1) Conduct a FMECA to verify single point failures do not prevent interlocks from allowing the brakes to be engaged with engine speed above ground idle and with air vehicle is off ground.

Demonstration.

(1) Conduct an air vehicle ground demonstration to verify the brakes to applied as required.

Test.

(1) Lab test: Conduct SIL test of control interlocks to verify system does not cause application of braking under required conditions.

8.8.11 Free wheel unit (engagement/disengagement of load absorbers).

Standard: The clutch assembly for rotary-wing vehicles permits automatic engagement and disengagement of the power drive system from the engines under the following conditions: autorotation, startup and shutdown conditions, and OEI. The number of engagements without losing the ability to transmit the required power (torque and speed) is IAW 8.8.1.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documents (e.g., hardware drawings, system configuration descriptions, design data records, inspection reports, and ICD) to verify conformance to design requirements and dimensional controls.

Analysis.

(1) Conduct a loads analysis based on the rated torque and speed from engine interface requirements during an engagement and determine worst-case pressure velocity operating conditions during overrun.

(2) Conduct a static strength analysis to verify that the clutch can drive the load absorbers under normal operations.

(3) Conduct a fatigue analysis of clutch assembly to verify design has positive life margins.

Test.

(1) Lab test: Conduct static torque test to verify the torsional spring rate (angular deflection of the outer race relative to the inner race) of the clutch and ability to meet rated torque. After the rated torque test, static torque is gradually increased to a minimum of 200% maximum rated torque (no slippage allowed).

(2) Lab test: Conduct fatigue (stroking) tests in the lab to verify the clutches' ability to meet usage and load spectrum.

(3) Lab test: Conduct overrunning test in the bench test on an integrated clutch assembly (e.g., bearings, seals, and oil jets) to verify capability of extended operation without damage to operational surfaces. The overrunning clutch test is conducted in two parts, for two hours each. The first is a differential overrunning test at 100% differential speed (the clutch driving member stationary and the driven member at 100% speed). The second overrun test is to the worst-case engagement element pressure velocity.

(4) Lab test: Conduct cold temperature engagement tests per performance specification requirements on the clutch assembly to verify clutch operates properly.

(5) Lab test: Conduct clutch durability test on two of each clutch configurations of the power drive subsystem for a minimum of 2,400 clutch engagements. In each engagement, the clutch is loaded to rated torque and speed after engagement. The clutch engagements include a minimum number of dynamic engagements. A dynamic engagement is defined as a condition where the clutch engages a rotating shaft in a manner that simulates how it will be used in service. After completion of the specified number of engagements, the static torque test is conducted to verify component condition.

8.8.12 Rotor meshing.

Standard: Operation of externally phased intermeshing-rotor systems cannot occur if the rotors become dephased. Positive mechanical and electrical interlocks are provided to prevent the operation of rotors unless they are locked in phase. Means are included for flight crew indication that the rotors are locked in phase.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, system configuration descriptions, design data records, inspection reports, ICD, test reports, or other substantiating reports) and confirm that interlock prevents the engine from being started if rotors are not in phase.

Demonstration.

(1) Conduct demonstration on the air vehicle of the positive mechanical and electrical interlocks to verify that engine cannot be started if rotors are not in phase and indications are provided to the flight crew when rotors are in phase and locked.

8.8.13 Wear and chafing.

Standard: A minimum of 0.25-inch clearance is provided between rotating and static hardware to prevent wear and chafing while the drive system is operating.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review hardware drawings or digital models to verify clearance is provided.

(2) Conduct inspection of production air vehicles during the build process to verify clearance is provided.

Analysis.

(1) Conduct dimensional analysis (e.g., CAD simulations) to verify clearance is provided.

Test.

(1) Ground test: Conduct ground testing on the air vehicle to verify that clearances are maintained and no wear or chafing on the drive system.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify that drive system maintains clearance to prevent wear and chafing.

8.8.14 Protection from external cleaning hazards.

Standard: The rotary-wing drive components are designed such that approved external cleaning procedures and compounds will not breach the sealing of those components and do not result in the introduction of contaminants into these components.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review documentation (e.g., hardware drawings and system configuration descriptions) to verify the drive system is protected from cleaning procedures and compounds.

Analysis.

(1) Conduct dimensional and layout modeling of drive system design and configuration to verify that the protective provisions have been incorporated to prevent cleaning fluids, compounds, and other contaminants from penetrating the transmission and drive system components.

Demonstration.

(1) Conduct air vehicle demonstration (e.g., water washing) to verify that protective provisions have been incorporated to prevent contaminants from penetrating the rotary-wing drive system components during the cleaning procedures.

8.8.15 Contamination.

Standard: Components are designed to prevent contaminants and fluids from puddling or becoming trapped internally on rotating components, on external gearbox surfaces, or around seals without the ability of the fluids to run off or be removed through drainage features to minimize corrosion.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings, materials and manufacturing records, design data records, inspection reports, photographs, borescope video, or other substantiating records and reports) and inspect part characteristics and surface conditions to determine if any areas could trap contaminants and fluids.

Analysis.

(1) Conduct an analyze of rotary-wing drive system design, configuration, and models to determine fluid sources, flows, leakage paths, and areas that could trap contaminants and fluids. The analysis of rotary-wing drive system design confirms drainage features are sized to remove or prevent puddling of contaminants and fluids.

Demonstration.

(1) Conduct a demonstration on the air vehicle to verify that the drainage features are sized to remove or prevent puddling of contaminants and fluids.

8.8.16 Accessibility.

Standard: The rotary-wing drive systems are designed to allow servicing, inspections, and maintenance activities. Access accommodates the maintenance equipment, maintainer's anthropometric dimensions, strength limitations, and environmental conditions (e.g., chemical protective gear, gloves).

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings and system configuration descriptions) to verify the installed drive system accessibility.

Analysis.

(1) Conduct 3-D modeling (e.g., CAD simulation) and physical mock-ups to verify the installed drive system is accessibility.

Demonstration.

(1) Conduct ground demonstration to verify accessibility of the installed drive systems.

8.8.17 Controllability.

Standard: The rotary-wing drive system provides the response to maintain control of the air vehicle without exceeding operating limits and without detrimental permanent deformation or damage when operated under steady state and transient operating conditions, throughout the flight and ground envelope, including loss of lubricant, OEI, and autorotations, IAW 8.8.1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify the response levels to maintain air vehicle operation at all air vehicle design conditions.

(2) Conduct a torsional stability analysis to verify design response behavior between propulsion drive system and rotary-wing drive system.

Test.

(1) Lab test: Conduct rig testing to verify that the rotary-wing drive system provides the expected response.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct air vehicle testing to verify that the rotary-wing drive system properly responds to maintain control of the air vehicle and torsional stability margins between propulsion drive system and rotary-wing drive system.

8.8.18 Gearbox lubrication systems.

Standard: The gearbox lubrication system operates and provides oil pressure and flow rates throughout the flight and ground envelopes, uses breathers to maintain pressure balance, removes heat generated by gear meshes and bearings, and provides an oil film to reduce wear between sliding elements (e.g., gears, bearings, and splines). A means for setting gearbox internal

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pressure and flow rates is provided. For pressurized lubrication systems, the design provides the ability to set pressures and flow rates during steady-state ground operations. Lubrication pump pulsation (e.g., ripple) is accommodated for and is not detrimental to the life of lubrication system components. Breathers are equipped with filtration devices which remove airborne particles of 10 microns or larger. Breathers are arranged to prevent loss of oil from the gearbox under all operating conditions and gearbox attitudes. Gearboxes operate at oil temperatures below the maximum specified temperature allowed under all combinations of gearbox power level, ambient conditions, gearbox attitudes, and within the operational lubricant levels.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review functional description documentation and schematics to verify the limits of the lubrication system with respect to high and low temperature environments, air vehicle flight envelope limits (attitude and altitude), and cooling system provisions for gearbox and transmission system components.

(2) Review schematics to verify minimum lubrication flow rates to each oil jet to gearbox components.

Analysis.

(1) Conduct a heat balance analysis to verify sizing the lubrication cooling system to accommodate the highest ambient air condition, the minimum gearbox oil flow, the maximum allowable oil temperatures, and minimum cooling airflow.

(2) Conduct analysis to verify the gearbox lubrication system meets design requirements (e.g., oil jetting for mesh cooling, lubrication of bearings and seals).

(3) Conduct sizing analysis of the breathers to verify gearbox pressures remain within design requirements.

(4) Conduct modeling or analysis to determine pump ripple frequencies and to verify they do not cause resonant frequencies or fatigue, resulting in failure of adjacent tubing and components.

Test.

(1) Lab test: Conduct lab test on each gearboxes' pressurized lubrication system to verify the ability to set pressures and flow rates during steady-state ground operations, maintain pressurization control to prevent loss of lubrication, and ability to cool oil to maintain oil properties.

(2) Lab test: Conduct sampling of lubrication oil to verify breather filtration system prevents contamination from entering the gearbox.

(3) Lab test: Conduct lab test of the oil jetting system to verify targeting of intended gearbox components.

(4) Lab test: Conduct air vehicle representative component level testing on each gearboxes' pressurized lubrication system to verify pump ripple frequencies and to verify they do not cause resonant frequencies or fatigue, resulting in life reduction of adjacent tubing and components.

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8.8.19 Loss of lubrication.

Standard: Gearboxes function for at least 30 minutes after complete loss of the lubricant from the primary lubrication system and are in a condition that the gearbox is still capable of transmitting the required power.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., hardware drawings and system configuration descriptions) for emergency or auxiliary lubrication provisions to verify attitude and power restrictions are defined for loss of lubricant operation.

Test.

(1) Lab test: Conduct loss of lubrication test on the two gearbox system units with final gearbox configuration. Transmission and gearbox lubrication systems are starved at the system's supply side (downstream from the pump) and continue to scavenge oil. The 30-minute operating period is as follows:

a. 2 minutes at rated power to simulate hover.
b. 26 minutes at a power condition to simulate cruise.
c. 2 minutes at a power condition simulating VL.

Creditable run time starts at the point at which the cockpit low oil pressure warning is displayed. For non-pressurized gearboxes, creditable run time starts when the oil being drained from the gearboxes ceases to flow in a steady stream. Gearboxes are able to transmit the required power at the conclusion of testing but are not required to be in a serviceable condition.

8.8.20 Oil filtration system.

Standard: An oil filtration system is provided to remove fine particles whose size (e.g., 3 microns for first stage and 40 microns for the second stage) can cause physical damage or failure in the contacting and sliding elements of the rotary-wing drive system. The filtration system includes a bypass function if the filter becomes clogged to ensure gearbox continues to function.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design drawings to verify the installation of filters.

Test.

(1) Lab test: Conduct a lab test of the filtration system to verify contamination removal and the ability to bypass the filter if clogged. For full qualification, the lab test includes:

a. Filter element multi-pass filtration ratio.
b. Maximum particle passed.
c. Filter bypass performance.
d. Impending bypass indicator performance.

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e. Cold start and temperature lockout.
f. Filter assembly clean pressure drop.
g. Filter assembly proof pressure.
h. Filter housing burst test.
i. Filter element bubble point.
j. Filter element collapse pressure.
k. Pressure build-up (dirt capacity).

8.8.21 Control system stability.

Standard: Rotary-wing engine system control system maintains both stable operation and response during all steady state and transient operations of the flight envelope. Rotary-wing system control loops maintain a minimum of 6 dB gain margin and between 30 and 60 degrees of phase margin. Drive train resonant peaks are attenuated by a minimum of 6 dB below unity gain. Control systems maintain required margins for both single and multi-engine air vehicle operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review documentation (e.g., design data records, test reports, and other substantiating records and reports) and verify that the design meets phase and gain stability margins in the standard.

Analysis.

(1) Torsional stability analysis verifies phase and gain stability margins meet the standard. Analysis includes linearized models of the engine control loops and rotor system for predicting performance throughout the flight envelope of the air vehicle.

Test.

(1) Lab test: Conduct open and closed loop bench testing to verify control loops maintain a 6 dB gain margin and between 30 and 60 degrees of phase margin stable operation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify stable control response to flight crew commands.

8.8.22 Health monitoring system.

Standard: The rotary-wing drive and lubrication monitoring system senses (e.g., speeds, oil debris, control operating modes and usage), detects, isolates, and records rotary-wing drive system faults that affect or could affect continued operation of the air vehicle. Operating oil temperature, oil pressure, and operating limits are continuously monitored. The bypass status of the oil filtration system is indicated to the flight crew and ground crew. Warning, caution, and advisories are provided to alert the flight crew of system failures which affect continued operation of the air vehicle and degraded system performance.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review flight manuals and drawings to verify health, vibration, usage, and rotary-wing drive and lubrication condition monitoring systems are incorporated into the air vehicle design and requirements for communicating the status of systems is incorporated in the air vehicle specification.

Test.

(1) Lab test: Conduct laboratory testing (e.g., FMET) to verify that the rotary-wing drive and lubrication condition monitoring system identifies, isolates, and displays faults, failures, and system status.

(2) Flight test: Conduct laboratory testing to verify debris monitoring system ability to detect debris of the size, shape, and material specified, and the characteristic of debris considered abnormal and its insensitivity to normal wear.

(3) Lab test: Conduct laboratory testing for health monitoring systems to determine diagnostic indicators capable of isolating mechanical component faults and thresholds.

(4) Ground test: Conduct ground testing on air vehicle to verify that the rotary-wing drive, filtration and lubrication condition monitoring system notifies flight crew and ground crew of WCAs and system status.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight testing on the air vehicle to verify that the rotary-wing drive system and lubrication condition monitoring system notifies flight crew of WCAs and system status.

(6) Flight test: Conduct flight testing on the air vehicle to verify in-flight monitoring of the drive system operational usage (power and time).

8.8.23 Accessory drives.

Standard: Accessories (e.g., lubrication oil pumps, oil cooler fans, and generators) driven by a gearbox are driven anytime the rotor system is rotating, including autorotation. Failure or seizure of the accessories does not cause damage to the mating gearbox(s) and damage to the rotary drive subsystem components. Accessory drive splines are protected with spline inserts. Accessory drive train systems do not have any resonances which affect the strength of the components when operated under steady state and transient operating conditions, throughout the flight and ground envelope IAW 8.8.1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review documentation (e.g., hardware drawings, materials and manufacturing records, design data records, inspection reports, or other substantiating records and reports) to verify that accessory drives meet the standard.

Analysis.

(1) Conduct a stress analysis to verify the accessory drive shaft will fail in shear before damage occurs to the gearbox.

(2) Conduct vibrational analysis on the accessory drive gears to verify that there are no damaging vibrational modes.

(3) Conduct a stress response analysis based on the modal and frequency responses of Analysis MoC (2) to verify that the accessory drive gears have positive strength margins of safety to accommodate any vibrational impacts.

Test.

(1) Lab test: Conduct component-level testing to verify shear strength capability of the accessory drive components.

(2) Lab test: Conduct a gearbox system-level bench testing to verify no damaging resonance vibrational modes.

(3) Ground test: Conduct a gearbox system-level testing on the air vehicle to verify no damaging resonance vibrational modes.

8.9 External cargo hook systems.

8.9.1 Cargo hook system strength.

Standard: The cargo hook system has a limit LF of 2.5X the rated or working load capacity unless otherwise stated in the air vehicle specification. The ultimate load is 1.5X the limit load. The cargo hook system is capable of withstanding the rated load capacity to 15 degrees with the vertical forward and lateral and 30 degrees with the vertical to the rear.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a loads analysis to define the rated or working load capacity.

(2) Conduct a static strength stress analysis to verify the cargo hook system has positive strength margins to support limit load and ultimate load cases. Permanent deformations do not occur at DLL. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate load.

Test.

(1) Lab test: Conduct lab test on cargo hook system to verify ability to support limit and ultimate load cases. Permanent deformations do not occur at DLL. Gross section yielding or detrimental

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deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify loads and stresses based on Analysis MoC (1) and Analysis MoC (2).

8.9.2 Cargo hook system operation.

Standard: The operation of the external cargo hook system, in normal, automatic, and emergency modes, allows external loads to be transportable throughout the operating envelope without creating an AW hazard to the air vehicle, crewmembers, or ground crew during normal flight conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of flight manuals and cargo loading manuals verifies that the maximum and minimum loads for external cargo movement and flight envelope operating limits (e.g., maneuvering, CG, weight, altitude, bank limits, and airspeed) are defined.

Analysis.

(1) Conduct dynamic and stability analyses to verify the external loads do not create an AW hazard for flight operations. External cargo reactions (e.g., bounce, swing, hook off-loading and release, and vibrations), do not affect the air vehicle, crewmembers, or ground crew during hover with hookup and release, and transition into flight, hover, and flight.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Conduct flight testing with the external cargo load to verify that all transitional operations (air-to-ground and ground-to-air) and in flight operations do not have adverse air vehicle, external cargo, or coupled vibration effects causing damage, loss of control, or external cargo instability effects due to external cargo reactions including bounce, swing, and hook off-loading.

8.9.3 Securing of cargo.

Standard: The external cargo hook allows the load to be secured to the cargo hook and prevents inadvertent release of the cargo.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review of external cargo hook system documentation (e.g., drawings and system descriptions) to verify design features are in place to secure and retain the external load.

Analysis.

(1) Conduct kinematic analysis to verify retention mechanism ability to secure and retain the external load.

Test.

(1) Lab test: Conduct laboratory testing to verify the endurance of the external cargo hook ability to secure and retain the external load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing of the cargo hook system to verify ability to secure and retain the external load.

8.9.4 Release of external cargo.

Standard: The external cargo hook system has controls which allows for the flight crew to release cargo (not at the pendant) as commanded for automatic and emergency operations for the loading envelope. The release system consists of a primary and a backup release (e.g., manual release) subsystems that are isolated from one another. The status (safe, unsafe, system malfunction) of the cargo system is continuously displayed to the flight crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of external cargo hook system documentation (e.g., drawings, system descriptions, system design analyses, and flight manual) to verify design features are in place to control release of external cargo and indicate status of the system.

Analysis.

(1) Conduct FMECA and SSHA to identify all single point failures which results in a loss of jettison capability and any mitigations to those failures.

(2) Conduct kinematic analysis to verify release mechanism ability to jettison the external load.

Demonstration.

(1) Conduct on air vehicle demonstration to verify the release subsystems operate the external cargo hook as intended and air vehicle provides status indications.

Test.

(1) Lab test: Conduct laboratory endurance testing on the release mechanism for the external cargo hook system to verify ability to release loads throughout the service life.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify the release subsystems operate the external cargo hook as intended (e.g., at maximum bank angle and speed) and air vehicle provides status indications.

8.9.5 External cargo hook service life.

Standard: The external cargo hook system (e.g., attaching hardware, release mechanism, carrying device structural systems and their attachments) has positive endurance margins to maintain residual strength throughout the specified service life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a loads analysis to define the usage spectrum used to determine the external cargo hook endurance life.

(2) Conduct fatigue analysis on the external cargo hook system using the usage spectrum developed in Analysis MoC (1) to verify endurance capability to support loading throughout the specified service life.

Test.

(1) Lab test: Conduct laboratory endurance testing on external cargo hook system components to verify ability to support usage spectrum loads and release capability throughout the service life.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify loads used in the usage spectrums established in Analysis MoC (1).

8.10 External rescue hoist.

8.10.1 Rescue hoist system operation.

Standard: The operation of the rescue hoist system allows personnel to be raised and lowered throughout the operating envelope without creating an AW hazard to the air vehicle and without causing harm to crewmembers, passengers, or ground crew.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of flight manuals verifies that the maximum and minimum loads and flight envelope operating limits (e.g., maneuvering, CG, weight, altitude, bank limits and airspeed) are defined.

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Analysis.

(1) Conduct dynamic and stability analyses to verify the operation of the rescue hoist system (personnel loadings and cable deployment) does not create an AW hazard impacting flight operations. Rescue hoist system reactions including spinning, oscillating, bouncing, swinging, trolling, deploying, and retracting do not affect the air vehicle during hover and transition into flight or hover.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: Conduct lab testing of rescue hoist and attachment hardware to verify rescue hoist system installation, rigging, clearances, control, functions, and performance to raise and lower the load.

(2) Flight test: Conduct flight testing of the rescue hoist system to verify that all transitional operations (air-to-ground and ground-to-air) and in flight operations do not create air vehicle, personnel, or coupled vibration effects causing damage to air vehicle systems, loss of control of the air vehicle, harm to crewmembers, passengers, or ground crew, or instability effects due to spinning, oscillating, bouncing, swinging, trolling, deploying, and retracting of the external load.

8.10.2 Rescue hoist strength.

Standard: The rescue hoist has a limit LF of 3.5X the rated or working load capacity unless otherwise stated in the air vehicle specification. The rescue hoist has an ultimate LF of 1.5X the limit load acting within a cone of action of 30 degrees to the vertical.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a loads analysis to define the rated or working load capacity.

(2) Conduct a static strength stress analysis to verify the rescue hoist system has positive strength margins to support limit load and ultimate load cases. Permanent deformations do not occur at DLL. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failure occur at or below ultimate load.

Test.

(1) Lab test: Conduct lab testing on rescue hoist system to verify ability to support limit and ultimate load cases. Permanent deformations do not occur at DLL. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify loads and stresses based on Analysis MoC (1) and Analysis MoC (2).

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8.10.3 Rescue hoist system service life.

Standard: The rescue hoist system (e.g., attaching hardware, cable release, carrying device structural systems and their attachments) has positive durability margins to maintain residual strength and endurance throughout the specified service life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a loads analysis to define the usage spectrum used to determine the rescue hoist fatigue and endurance life.

(2) Conduct fatigue analysis on the rescue hoist system using the usage spectrum developed in Analysis MoC (1) to verify durability capability to support loading throughout the specified service life.

(3) Conduct analysis to verify the rescue hoist can operate for four lifetimes with the usage spectrum developed in Analysis MoC (1).

Test.

(1) Lab test: Conduct laboratory fatigue testing on rescue hoist system, which is for a fixed hoist (not reeling), to verify service life and component life limits.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Lab test: Conduct endurance testing to verify the rescue hoist can operate during the reel in and out cycles for two lifetimes without degradation of performance or wear exceeding wear tolerances.

(3) Flight test: Conduct flight testing to verify the loads used in the usage spectrum.

8.10.4 Jettison capability for cable.

Standard: The rescue hoist system has a means to jettison the cable (e.g., cut the cable), prevent inadvertent jettison of the cable, and secure the remaining cable to prevent whipping around the cabin and door as the load is released throughout the operating envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Conduct review of rescue hoist system documentation (e.g., drawings, system descriptions, system design analyses, and flight manual) to verify design features are in place to jettison cable and prevent inadvertent jettison of the cable (e.g., guarded switch) throughout the operating envelope.

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Analysis.

(1) Conduct kinematic analysis to verify the release mechanism ability to jettison the cable with an external load.

(2) Conduct an analysis to verify mechanism ability to prevent accidental jettison.

(3) Conduct an analysis to verify mechanism ability to prevent whipping of cable no matter when cable is jettisoned throughout the operating envelope.

Test.

(1) Lab test: Conduct laboratory testing to verify functioning of cable jettison and ability to prevent whipping of cable.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct a demonstration on the air vehicle to verify ability to prevent accidental cable jettison, functioning of cable jettison, and ability to prevent whipping of cable throughout the operating envelope.

8.11 Fast rope insertion and extraction systems.

8.11.1 Fast rope insertion and extraction system operation.

Standard: The FRIES allows for insertion and extraction of personnel throughout the operating envelope without creating an AW hazard to the air vehicle and without causing harm to crewmembers, passengers, or ground crew throughout the operating envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review of flight manuals verifies that the maximum and minimum loads and flight envelope operating limits (e.g., maneuvering, CG, weight, altitude, bank limits and airspeeds) are defined.

Analysis.

(1) Conduct dynamic and stability analyses to verify the operation of the FRIES (personnel insertion and extraction and rope deployment) does not create an AW hazard impacting flight operations. Fast rope insertion and extraction systems load reactions from insertion, extraction, swinging, and trolling, do not affect the air vehicle during hover, and transition into flight or hover.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Conduct flight testing of FRIES to verify that all operations do not create effects causing damage to air vehicle systems, loss of control of the air vehicle, harm to crewmembers, passengers, or ground crew, or instability effects due to insertion, extraction, swinging, and

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trolling. The FRIES operating envelope and limits (e.g., maneuvering, CG, weight, altitude, bank limits and airspeeds) are verified.

8.11.2 Fast rope insertion and extraction systems strength.

Standard: The FRIES has a limit LF of 3.5X the rated or working load capacity unless otherwise stated in the air vehicle specification. The FRIES has an ultimate LF of 1.5X the limit load acting within a cone of action of 30 degrees to the vertical.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a loads analysis to define the rated or working load capacity.

(2) Conduct a static strength stress analysis to verify the FRIES has positive strength margins to support limit load and ultimate load cases. Permanent deformations do not occur at DLL. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failure occur at or below ultimate load.

Test.

(1) Lab test: Conduct lab test on FRIES to verify ability to support limit and ultimate load cases. Permanent deformations do not occur at DLL. Gross section yielding or detrimental deformations do not occur at or below 115% DLL. No rupture, buckling, or collapsing failures occur at or below ultimate load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct flight testing to verify loads and stresses based on Analysis MoC (1) and Analysis MoC (2).

8.11.3 Fast rope insertion and extraction systems service life.

Standard: The FRIES (e.g., attaching hardware, quick release, and structural systems and their attachment to air vehicle) has positive durability margins to maintain residual strength throughout the specified service life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a loads analysis to define the usage spectrum used to determine the FRIES fatigue life.

(2) Conduct fatigue analysis on the FRIES using the usage spectrum developed in Analysis MoC (1) to verify durability capability to support loading throughout the specified service life.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(1) Lab test: Conduct laboratory testing on the FRIES components to verify service life or component life limits.

8.11.4 Quick release.

Standard: The FRIES has a means to jettison the rope and prevent accidental release by incorporating redundant retention features throughout the operating envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct kinematic analysis to verify jettison mechanism ability to prevent accidental release of the rope with and without an external load.

(2) Conduct kinematic analysis to verify jettison mechanism ability to release the rope with and without an external load.

Test.

(1) Lab test: Conduct laboratory testing to verify jettison mechanism can prevent accidental release of the rope with and without an external load.

(2) Lab test: Conduct laboratory testing to verify jettison mechanism ability to release the rope with and without an external load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct a flight test to verify jettison mechanism can prevent accidental jettison with and without an external load.

(4) Flight test: Conduct a flight test to verify jettison mechanism ability to release the rope with and without an external load.

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9. CREW SYSTEMS

Section 9 defines the AW criteria in the crew systems areas of life support systems, crew stations, control stations, air vehicle interiors, air system lighting, human performance, ejection systems, emergency egress and bailout, crash survivability, and transparency integration.

Airworthiness in each of these areas primarily relates to the protection of humans from injury because of adverse conditions (e.g., altitude and g protection) or emergency conditions (e.g., crash protection). Airworthiness within crew systems also relates to well-designed human interfaces that allow for the performance of crewmember tasks to control the air vehicle throughout the flight envelope (e.g., the task workload remains at acceptable levels throughout the flight envelope).

Throughout this section, different categories of air vehicles are addressed. The categorization is as follows:

a. Cargo air vehicles are defined as an air vehicle (e.g., C-17, C-130, KC-135) that is designed or converted for the carriage of cargo (e.g., goods, freight).
b. Multi-place air vehicles have multiple crewmembers (e.g., USAF E-3 Sentry, AC-130, B-52).
c. Tactical air vehicles have a fighter or close air support capability (e.g., F-15, F-16, F-18, A-10).
d. Transport air vehicles are configured to carry passengers (e.g., C-21, C-40, VC-25, C-17 with passengers, C-130 with passengers).
e. Overwater transport air vehicles (air vehicles primarily designed or developed for the overwater missions) have missions over water for extended periods of time or land on aircraft carriers (e.g., P-3, P-8, C-2).

9.1 Life support systems.

The life support systems are designed to sustain crewmembers and passengers (collectively referred to as “occupants”) throughout the flight envelope and provide a means of dealing with emergencies in flight. Life support systems also support escape, descent, provisions for survival, and recovery in emergency situations.

Life support systems often include supporting equipment, such as threat and survival equipment (e.g., LEP, anti-exposure, cold water immersion, CBRN). This equipment should be evaluated with respect to the integration impact to the air vehicle.

Key sections related to life support system interfaces are 4.3, 7.2, 8.2, 8.4, 8.5, 9.4, 12.2, 15.3, and 16.1.

9.1.1 Altitude protection.

The primary high altitude hazard is reduced oxygen partial pressure due to ambient pressure changes which leads to insufficient oxygen for the occupants and can result in hypoxia. Occupant altitude protection is multifaceted and depends on pressurization of the air vehicle cabin as the primary altitude protection mechanism along with providing proper oxygen concentrations to the occupants (see 8.2.1). Depending on the type of air vehicle and mission, additional protection may be provided by supplemental oxygen, pressure breathing for altitude, and emergency oxygen.

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9.1.1.1 Crewmember breathing systems.

Standard: The complete primary and backup breathing systems will be in compliance with MIL-STD-3050, Detailed Requirements paragraphs, throughout all phases of flight (e.g., taxi, takeoff, cruise, flight maneuvers, approach, landing), air vehicle altitudes and g loads, as well as during all applicable operational environments (e.g., CBRN).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the breathing system specifications to ensure all phases of flight are accounted for in the oxygen system hardware.

(2) Review FMECA to verify that all breathing system components are covered.

(3) Examine component hardware configuration documents to confirm compliance with hardware drawings.

Analysis.

(1) Verify that oxygen system safety analysis is included in the SSA and shows that failures across the system are detected, annunciated, and have automatic responses.

Test.

(1) Lab test: Conduct system-level testing of all breathing system components (e.g., AFE, regulator, and air-plumbing) together to determine their functional performance within the breathing system, to include failure mechanisms, to ensure all failure modes are identified and reliability is understood.

(2) Lab test: Conduct system-level testing to characterize breathing system performance within the intended air vehicle envelope, to include failure modes and mechanisms as well as proper detection and annunciation of alerts. The entire breathing system will be tested as indicated in MIL-STD-3050, Contractor Verification and Validation of ACBS and Government Verification and Validation of ACBS sections.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(3) Flight test: Flight test: Flight test data confirms conformance with MIL-STD-3050, Detailed Requirements paragraphs.

9.1.1.2 Supplemental oxygen.

Standard: Air vehicles flying with cabin pressure altitudes above 10,000 feet MSL will provide supplemental oxygen to all occupants or restrict flight time above 10,000 feet IAW TABLE VI. Flight Durations Where Supplemental Oxygen is not Required. The duration of the supply is sufficient for all occupants for the entire duration of flight at cabin pressure altitudes above 10,000 ft MSL. Air vehicles operating above 14,000 feet MSL always require supplemental oxygen.

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TABLE VI. *Flight durations where supplemental oxygen is not required.*

Not to exceed 1 hour between 10,000 feet MSL and 12,500 feet MSL
Not to exceed 30 minutes between 12,500 feet MSL and 14,000 ft MSL.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review FMECA to verify that all supplemental oxygen systems are covered.

(2) Inspect component hardware configuration documents to confirm compliance with hardware drawings.

(3) Inspect air vehicle drawings to verify the air vehicle supplemental oxygen system and individual dispensing units are available and accessible to every occupant.

Analysis.

(1) Verify analysis of the supplemental oxygen system concentration, flow, and duration shows adequate supply to crewmembers and passengers.

Test.

(1) Lab test: Conduct testing of supplemental oxygen system under MIL-STD-3050, Test Condition 4 at altitudes up to the maximum rated altitude of the air vehicle to verify a mean alveolar oxygen partial pressure of no less than 60.8 mmHg is maintained.

9.1.1.3 Emergency oxygen - crewmember.

Standard: Emergency oxygen is provisioned and accessible to crewmembers for protection against loss of pressurization, smoke and fumes, and ejection (if applicable). The emergency oxygen systems need to be accessible by users from their seated locations. The emergency oxygen supplies a minimum 94% oxygen concentration at a volume and pressure sufficient to ensure crewmember performance to descend from the air vehicle maximum ceiling to an altitude below 10,000 feet MSL.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review FMECA to verify that all emergency oxygen systems are covered.

(2) Inspect component hardware configuration documents to confirm compliance with hardware drawings and location to crew stations.

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Analysis.

(1) Verify Emergency Oxygen System Safety Analysis shows that failures across the system are annunciated and have automatic responses.

Demonstration.

(1) Demonstrate that crewmembers are able to access the emergency oxygen masks and don them within five seconds at each crew station.

Test.

(1) Lab test: Conduct testing with appropriate facial and mask hardware to ensure the functional performance of the emergency oxygen system allows crewmembers to perform all flight-critical tasks.

(2) Lab test: Conduct testing of automated breathing system switching to verify uninterrupted transition to emergency system.

(3) Lab test: Conduct testing to verify emergency oxygen subsystem performance is sufficient for descent from air vehicle ceiling to 10,000 feet MSL.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Conduct review of materials, connectors, wiring, and subsystem assemblies to ensure compatibility with the air vehicle.

9.1.1.4 Emergency oxygen - passenger.

Standard: Air vehicles flying above 10,000 feet MSL are capable of providing emergency oxygen from the air vehicle, or from a stand-alone system, to each passenger until the air vehicle is able to descend below 10,000 feet. The passenger emergency oxygen system, such as automatic dispensing units and portable oxygen generation packs, will be in compliance with 14 CFR 25.1441, 25.1443, 25.1445, 25.1447, and 25.1449. The supply of oxygen is sufficient for the duration of time that passengers may be exposed to cabin altitudes up to 40,000 feet while breathing up to 30 liters per minute, Body Temperature Pressure Saturated, and with a tidal volume of 1.1 liters.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings or digital mockups to verify emergency oxygen provisions.

a. Ensure that the total number of portable emergency oxygen generators or dispensing units in passenger sections exceeds the number of seats by at least 10%.
b. Ensure that at least two oxygen dispensing units are connected to oxygen terminals in each lavatory.

(2) Inspect air vehicle and oxygen system specifications to ensure all operational phases are accounted for in the altitude protection system hardware.

(3) Review FMECA to verify that all emergency oxygen systems are covered.

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Analysis.

(1) Verify Oxygen System Safety Analysis shows that failures across the system are annunciated and have automatic responses.

(2) Conduct analyses to determine the life limits for components within the emergency oxygen system based upon reliability and durability data.

(3) Conduct review of materials, connectors, wiring, and subsystem assemblies to ensure compatibility with the air vehicle.

Demonstration.

(1) Using a mockup or production air vehicle, demonstrate that passengers are able to access the emergency oxygen system and don them within five seconds at each location.

(2) On air vehicle with automatic dispensers, demonstrate that the units are deployed before the cabin pressure altitude exceeds 15,000 feet.

9.1.1.5 Emergency oxygen - ejection.

Standard: A sufficient emergency oxygen quantity to each crewmember is available for use during escape altitudes above 14,500 feet. Emergency oxygen flow is automatically initiated and supplies a minimum of 94% oxygen to crewmembers within 0.5 seconds of exposure to atmosphere. The duration is sufficient to allow the crewmember to descend from the maximum air vehicle altitude. The system also provides an indication to the crewmembers that there is no longer enough oxygen for escape from maximum ejection altitude.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of the oxygen system concentration, flow, and quantity to ensure adequate supply to crewmembers.

(2) Conduct analysis that predicts structural and functional integrity of the system in the escape system induced environment.

Test.

(1) Lab test: Conduct lab tests in an altitude chamber of system to verify pressure and flow rates with simulated human breathing through altitude schedules expected during escape and descent to 14,500 feet.

(2) Lab test: Conduct human testing in mockups including altitude chamber to verify that flow rates and pressure support crewmember consciousness.

(3) Lab test: Conduct windblast testing to verify that the AFE required (e.g., oxygen mask, hoses, and connectors) continues to deliver emergency oxygen during the ejection, retains structural integrity, and is functional post-test.

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9.1.1.6 Pressure suit systems.

Standard: The full or partial pressure suit and associated plumbing, regulator, and tubing used for altitude protection is in compliance with AIR-STD-1052 Ed 2 v1, Physiological Criteria paragraph for the normal and emergency envelopes of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Inspect air vehicle and oxygen system specifications to ensure all operational phases are accounted for in the altitude protection system hardware.

(2) Examine component hardware configuration documents are inspected to confirm compliance with hardware drawings.

Test.

(1) Lab test: Test the man-mounted altitude protection components to ensure their functional performance in conjunction with the breathing system, including failure mechanisms, to ensure all failure modes are identified and to establish reliability.

(2) Lab test: Test the altitude protection of the pressure suit throughout the air vehicle envelope in an altitude chamber to verify the approved pressure schedule and include failure modes and mechanisms to verify alert annunciation.

(3) Lab test: Accomplish testing in an altitude chamber with breathing machine and head forms and human subjects, using the air vehicle's breathing system.

(4) Lab test: Test pressure suits IAW the approved pressure schedules and verify that the lower portion of the altitude garment always inflates in advance or concurrently with the upper portion of the altitude garment.

(5) Lab test: Verify that the pressure suits provide adequate ventilation and thermal management.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(6) Flight test: Conduct flight tests to verify compliance with the approved pressure schedule.

9.1.2 G-induced loss of consciousness protection.

Standard: Crewmembers in air vehicles that can exceed 4.5g require physiological protection during high-g maneuvers. G protection may include but is not limited to the following: lower body coverage (anti-g suit or trousers), g sensing valve (to inflate the suit), PBG breathing regulator, optional upper body coverage, and occipital bladder on the helmet. G protection provisions are compatible with AFE, and pressures are IAW FIGURE 1 and FIGURE 2. If upper counterpressure garment is used, it is to inflate prior to the lower pressure garment. Inflations will be properly coupled with the ACBS.

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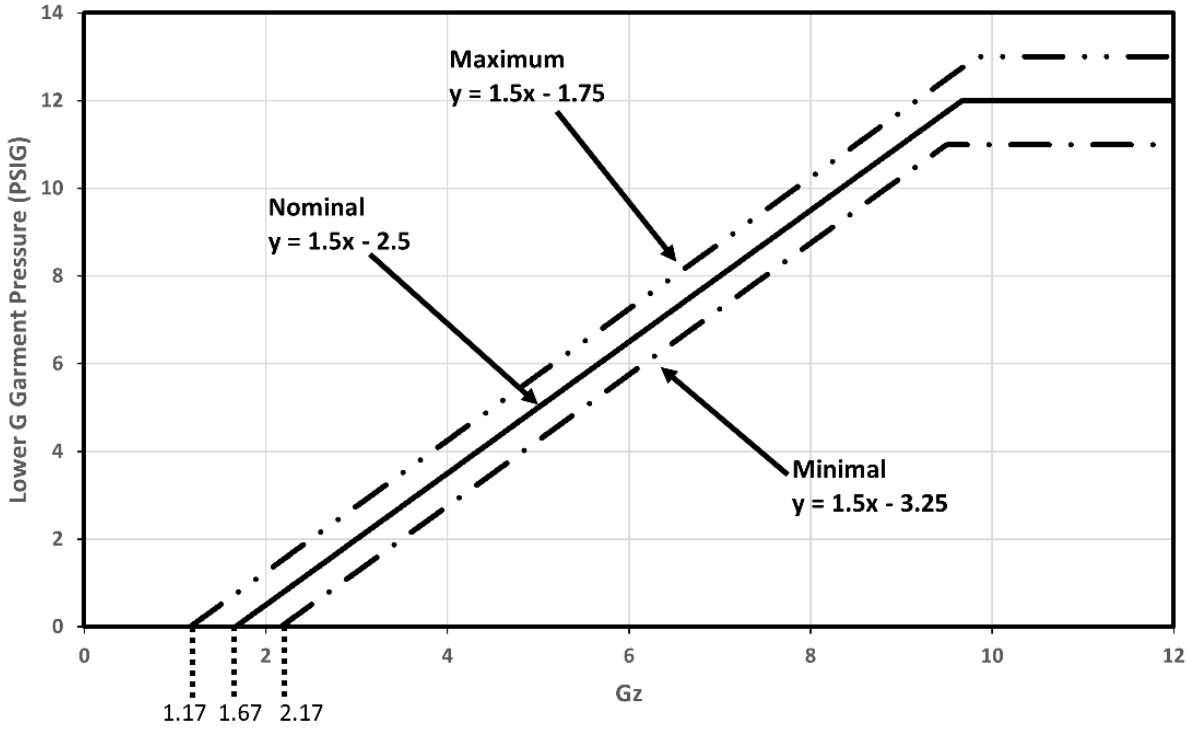


FIGURE 1. G-suit pressure schedule.

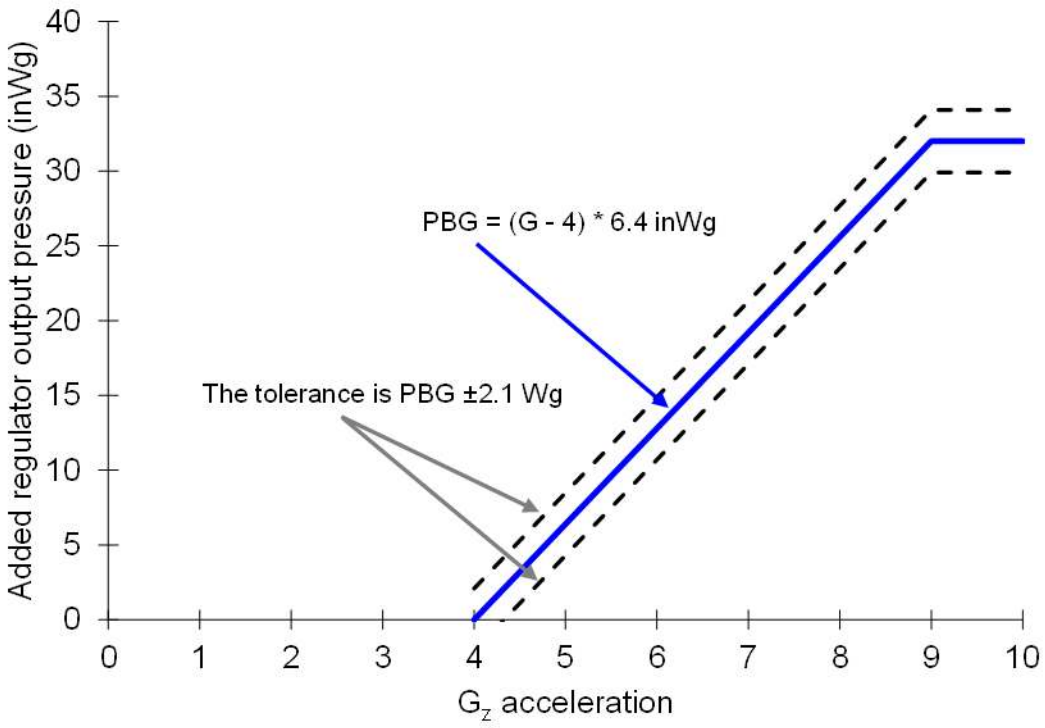


FIGURE 2. PBG pressure schedule.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: G system pressure regulation - Test the g protection system in an unmanned state to verify pressure regulation capability IAW FIGURE 1 and FIGURE 2. This testing will verify that over pressurization of the system cannot occur.

(2) Lab test: Conduct qualification testing in a centrifuge environment. Test the system first unmanned to verify system performance, followed by manned testing to assess functional operation and crewmember tolerance and endurance, and that the system complies with FIGURE 1 and FIGURE 2.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(3) Flight test: Conduct a flight test program to assess relevant environmental and operational conditions. Results of the flight test data will be evaluated to confirm conformance to the g pressure schedule and PBG schedule as applicable.

9.1.3 Emergency and survival equipment.

All air vehicles are equipped with emergency and survival equipment. Emergency equipment are devices intended to aid crewmembers and passengers during inflight and ground emergencies. Survival equipment are devices intended to ensure survival of crewmembers and passengers after an emergency event occurs such as a crash or ditching. The types and quantity of emergency and survival equipment can vary based on air vehicle type and intended operating environment. This equipment may include fire extinguishers, smoke and fumes protection, megaphones, medical kits and supplies, automatic external defibrillators, means for emergency evacuation, specialized tools or fracturing equipment, life rafts, life preserver, survival kits, and anti-exposure equipment and portable lights. Type certificates for the air system do not cover ensuring the proper functionality of the emergency and survival equipment. The Type Certificate ensures that the air system has the provisions needed for this equipment during flight operations.

9.1.3.1 Emergency and survival equipment - general.

Standard: Provisions for emergency and survival equipment will be equipped as required by flight operations to deal with inflight and ground emergencies, ditching, and ejection, and this equipment should not interfere with operation of the air system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings or digital mockups to verify emergency and survival equipment provisions.

9.1.3.2 Crewmember smoke and fume protection.

Standard: All crewmembers will have access to smoke and fume protection devices.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings or digital mockups to verify provisions for smoke and fume protection devices and that they are readily accessible.

(2) Review drawings or digital mockups to verify there is a one for one correlation of smoke and fume protection devices to crewmembers.

Demonstration.

(1) Using a mockup or production air vehicle, demonstrate for each crew station that the intended anthropometric range for the platform population can access the devices and don them within five seconds without any interference from other already donned AFE.

9.1.3.3 Flotation devices - general.

Standard: For air vehicles with overwater missions, one life preserver per occupant is installed on all air vehicles without escape and ejection systems. This includes one life preserver for each occupant.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Demonstrate on air vehicle or in an air vehicle mockup that the greatest capacity of fully encumbered occupants can access the flotation devices.

9.1.3.4 Flotation devices - rotary-wing.

Standard: For rotary-wing and tiltrotor air vehicles, the flotation device will remain functional through the air vehicle's water entry. The device needs to be manually inflatable using only one hand, operable from both the left and right. Additionally, the device has a dual chamber with an oral inflation capability.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(1) Lab test: The flotation device should be included during crash and fall tests up to the air vehicle's maximum crash energy.

(2) Lab test: Verify that the flotation device functions in a pool using test subjects of various sizes of occupants for the intended air vehicle user population. Demonstration includes egress from submerged air vehicles, inflation, measurement of freeboard, and boarding of life raft.

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9.1.3.5 Flotation devices - air vehicles with escape and ejection systems.

Standard: For air vehicles with ejection seats, the flotation device (if used) will remain functional and automatically inflate when the ejected crewmember enters seawater. The device will be dual chambered with an oral inflation capability.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: The flotation device will be included during windblast tests of the ejection system and AFE up to the maximum rated velocity of the ejection system. Verify that the system remains intact during the windblast test and properly inflates when exposed to seawater.

(2) Lab test: The floatation device will be included during sled testing of the ejection system and AFE up to the maximum rated velocity of the ejection system. Verify that the system remains intact during the ejection sled testing and properly inflates when exposed to seawater.

9.2 Crew stations, control stations, and air vehicle interiors.

The crew stations, control stations, and air vehicle interiors section encompasses crew or control station layout, controls functionality, emergency markings, stowage for emergency equipment, interior finishes, surface touch temperatures, communications systems, crewmember alerting systems, and mass notification systems. The intended anthropometric range is defined by each air vehicle platform. Crewmembers and passengers are collectively referred to as “occupants” in this section.

9.2.1 Crew station and control station layout.

The crew station and control station layout accommodates the intended anthropometric range, crewmember equipment, and crewmember motions.

All combinations of required crewmember equipment and anticipated crewmember activities should be assessed to ensure adequate integration with the crew station or control station layout. Crewmember equipment can vary for each air vehicle platform. Crewmember equipment includes but is not limited to helmet mounted devices, life support equipment, survival equipment, and combinations of this equipment.

9.2.1.1 Controls and display readability.

Standard: Displays and other crew station or control station components (e.g., indicators, panel labeling) are located so that they are readable from designated crewmember positions at design eye point, under all operating conditions of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Conduct a demonstration to verify control and display readability from the design eye point in daytime operating conditions up to full sun (10,000 fc). The demonstration should include all displays and illuminated indicators and be done in a production representative mock-up or on the

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air vehicle. Participants will wear equipment that is representative of equipment worn during daytime operations (e.g., helmet and tinted visor).

(2) Conduct a demonstration to verify control and display readability from the design eye point in nighttime operating conditions in an environment as dark as possible (e.g., a hangar with the doors shut, lights out, at nighttime, an engine hush house) IAW MIL-STD-3009, System Level Verification paragraph. The demonstration should include all displays and crewmember station components (e.g., illuminated indicators, panel labeling) and be done in a production representative mock-up or on the air vehicle. Participants will wear equipment that is representative of equipment worn during nighttime operations (e.g., helmet and clear visor).

(3) Conduct a demonstration to verify control station control and display readability from the design eye point in minimum and maximum lighting conditions including ambient lighting and external artificial lighting sources. The demonstration should include all displays and illuminated indicators and be done in a production representative mock-up or in the control station. Participants will wear required crewmember equipment.

9.2.1.2 Transparency interference.

Standard: The transparency system shape does not interfere with crewmember motions while wearing required crewmember equipment during normal and emergency conditions.

Crewmember motions consider any movement in which crewmembers might come into contact with the transparency (e.g., normal ingress and egress, emergency egress, check-six, landing, use of specialized cockpit equipment, transferring equipment from one crew member to another, inertial reactions to accelerations, AR) as required by the air vehicle's operations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to identify any potential interference with crewmember positions and motions while wearing required crewmember equipment during normal and emergency conditions.

Demonstration.

(1) Demonstrate in a representative mockup or on the air vehicle that participants representative of the intended anthropometric range can accomplish normal and emergency procedures while wearing required crewmember equipment.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Demonstrate during flight test that transparency system shape does not interfere with crewmember positions and motions while wearing required crewmember equipment during representative operating conditions.

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9.2.1.3 Interior field of view.

Standard: The interior FOV is unobstructed for all safety-critical controls and displays for the intended anthropometric population with crew mounted equipment (e.g., helmet, oxygen mask, survival vest).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis from the design eye point of each crewmember position to identify any controls or displays that are obscured, the crewmember population affected, the impact to SOF, and the degree of the obstruction.

Demonstration.

(1) Demonstrate unobstructed visual access of controls and displays in a representative mockup or the air vehicle with participants wearing the required crewmember equipment and the intended anthropometric population.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Flight test assessments verify internal visibility needed to accomplish all flight tasks.

9.2.2 Controls functionality.

Controls are components that allow the crewmember to provide input to an air vehicle. This includes, but is not limited to, controls for avionic systems, dedicated controls, and multifunction controls to include touchscreens.

9.2.2.1 Accommodation for controls.

Standard: Controls can be fully actuated without range of travel restrictions by the intended anthropometric population under all operating conditions while wearing required crewmember equipment. Controls should be located IAW MIL-STD-203, Location of Controls paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground or Lab test: Conduct human-in-the-loop evaluation with a group of participants, representative of the range of anthropometric dimensions of the intended user population, to verify ability to operate controls through the full range of travel. This evaluation should be in a production representative mock-up or on the air vehicle with participants wearing required crewmember equipment. Measure clearances and reaches while participants are appropriately restrained based on the reach zone requirements of the controls at the extreme seat positions.

Prior to entering operational test or operations, verify the standard by the following method(s):

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Test.

(2) Ground test: Verify the crew station or control station accommodation envelope by measuring the clearances and reaches on a group of participants, representative of the range of anthropometric dimensions of the intended user population, in a production representative air vehicle. Participants will wear required crewmember equipment. Measure clearances and reaches while participants are appropriately restrained based on the reach zone requirements of the controls at the extreme seat positions. Use measurements and participant body dimensions to determine the accommodation envelope of the air vehicle, the minimum and maximum anthropometric dimensions required to fly the air vehicle.

9.2.2.2 Emergency control accidental activation.

Standard: Provisions are in place to guard against accidental activation of emergency controls, as specified in MIL-STD-1472, Emergency Shutoff Controls and MIL-STD-1472, Prevention of Accidental Actuation paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review cockpit or crew station or control station layout drawings to verify that emergency controls are designed with provisions to prevent accidental activation as specified in MIL-STD-1472, Emergency Shutoff Controls and MIL-STD-1472, Prevention of Accidental Actuation paragraphs.

Demonstration.

(1) Demonstrate in a representative mockup or on the air vehicle that emergency controls have appropriate provisions to prevent accidental activation to include ingress and egress.

9.2.2.3 Escape system initiation controls.

Standard: For an air vehicle with an escape system, escape system initiation controls (automatic, manual, or both) can be activated with either hand by the defined anthropometric population of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review cockpit or crew station layout drawings to verify that escape system initiation controls are located at each crew station in a position that it is possible to be operated with either hand.

Test.

(1) Ground or Lab test: Test in a representative mock-up or in the crew station to ensure that escape system initiation controls can be activated with either hand by the full expected range of user anthropometric reach and strength values in MIL-STD-1472, Anthropometric

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Accommodation paragraph and MIL-STD-1472, Static Muscle Strength table in Push and Pull Forces paragraphs.

9.2.3 Emergency markings.

Emergency controls serve an emergency function. Emergency controls require rapid identification and activation in emergency situations. Emergency markings only designate emergency controls. Emergency markings should be consistent throughout the air vehicle and control stations. Controls designated as emergency controls can vary by platform. Examples of typical emergency controls may include, but are not limited to, in flight escape system controls, emergency brake controls, and engine fire extinguisher controls.

9.2.3.1 Emergency control operation markings.

Standard: Controls involving emergency or critical operations, such as an escape hatch, are clearly marked as to the method of operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings to verify that emergency controls are marked with the method of operation.

(2) Examine the air vehicle and verify that emergency controls are marked with the method of operation.

9.2.3.2 Emergency control marking of functional groups.

Standard: Panels with functional groups involving emergency or critical controls should be outlined with a 5mm (0.1875in) red border, IAW paint color 21136 of AMS-STD-595. Alternatively, functional groups of emergency controls may be designated by contrasting color pads or patches for critical and non-critical functional areas.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings to verify that emergency controls are properly marked.

(2) Examine the air vehicle and verify that emergency controls are properly marked.

9.2.3.3 Emergency control immediate action marking.

Standard: Immediate action striping on NVIS compatible panels should be IAW MIL-DTL-7788, Immediate Action Striping-NVIS Compatible paragraph. Immediate action striping on non-NVIS compatible panels should be IAW MIL-DTL-7788, Immediate Action Striping-Non-NVIS Compatible paragraph. Emergency controls that are not on a panel are coded with orange-yellow and black immediate action striping IAW with MIL-STD-1472, Immediate Action Controls paragraph.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings to verify that emergency controls are properly marked.

(2) Examine the air vehicle and verify that emergency controls are properly marked.

9.2.3.4 Emergency control marking for red compartment lighting.

Standard: When red compartment lighting is used, an orange-yellow, IAW paint color 23538 of AMS-STD-595, and black-striped, IAW paint color 27038 of AMS-STD-595, border should be used to outline functional groups involving emergency controls.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review drawings to verify that emergency controls are properly marked.

(2) Examine the air vehicle and verify that emergency controls are properly marked.

9.2.3.5 Emergency and survival control marking - passenger air vehicle.

Standard: Controls for emergency and survival equipment are marked for identification and method of operation. Illustrations and pictorial representations should be used to convey operation of emergency controls where passenger language abilities vary or are unknown.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(1) Review drawings to verify that approved emergency and survival equipment controls are marked for identification and method of operation.

(2) Examine the air vehicle and verify that the approved emergency and survival equipment controls are marked for identification and method of operation.

(3) Verify illustrations and pictorial representations are used to convey operation of critical safety controls where passenger language abilities vary or are unknown.

9.2.3.6 Equipment storage marking - passenger air vehicle.

Standard: Each stowage provision, such as a locker or compartment, for required emergency and survival equipment are marked to identify the contents and facilitate the easy removal of the equipment.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

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Examination.

(1) Review drawings to verify that each stowage provision for emergency and survival equipment is marked for identification.

(2) Examine the air vehicle and verify that approved emergency and survival equipment is marked for identification.

Analysis.

(1) A human factors analysis verifies the ability of different sized passengers to easily remove emergency equipment from their storage locations.

9.2.4 Stowage for emergency equipment.

Standard: Stowage provisions for required emergency equipment are furnished and arranged so that the equipment is accessible without using tools and stowage location is visible without having to move other items.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review cockpit or crew station layout drawings or digital mockups to verify stowage provisions for emergency equipment.

Demonstration.

(1) Demonstrate in a mockup or on the air vehicle that stowage provisions are visible and accessible without having to move other items.

9.2.5 Interior finishes, components, and equipment.

Standard: The material of all interior finishes, components, and equipment will be flame resistant and have low smoke generation. This includes, but is not limited to lavatories, galleys, and areas that are not continuously occupied.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Conduct a lab test for flammability of interior materials to verify that the material of all interior finishes, components and equipment will be flame resistant IAW 14 CFR Appendix F to Part 25 Parts I and II or other approved test requirements.

(2) Lab test: Conduct a lab test for low smoke generation testing of interior materials to verify that the material of all interior finishes, components and equipment will be IAW 14 CFR Appendix F to Part 25 Part V or other approved test requirements.

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9.2.6 Surface touch temperatures.

Standard: Touch temperatures of surfaces in the crew station or control station that impact the crewmember's ability to attain, sustain, and complete flight safely do not exceed the temperature limits in the Temperature Exposure Limits table in MIL-STD-1472.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Measure the surface temperatures once the temperatures have reached a steady state using thermal imaging in a heat-soaked air vehicle, cooled using air vehicle power without an ECS cart, to verify that they comply with MIL-STD-1472, Thermal Contact Hazards paragraphs.

9.2.7 Communication systems.

Communications systems permit flight crew to alert additional crewmembers; support, launch and recovery personnel; and military or civilian airspace controllers. Additionally, they allow flight crew to give feedback of events, conditions, and situations which could present a hazard to the safety of the occupants, endanger human life, or cause substantial damage to the air vehicle. One-way systems include public or passenger address systems. Two-way systems include radios and intercom systems.

An aided communication path is an electrically or electronically enhanced, wired or wireless, real-time, analog or digital voice communications pathway. The communication paths can include: within an air vehicle, air vehicle to another air vehicle, and air vehicle to ground.

9.2.7.1 Speech intelligibility - two-way systems.

Standard: Each critical aided voice communication path provides a means for intelligible two-way communication, defined as 80% or better mean score on the MRT or 0.41 STIPA. The 0.41 STIPA only applies if the system does not have vocoders or center clipping.

Critical aided voice communication paths may include:

a. Pilot communications with ATC.
b. Among flight crew.
c. Crewmember communications with ground personnel or other crewmembers.
d. Flight crew with another external air vehicle flight crew.
e. Flight crew to launch, recovery, and deck personnel.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) For systems that do not have vocoders or center clipping, compute the STIPA IAW IEC 60268-16 Measurement Procedures, Post-processing of Data and Applications section from measurements of modulation reductions of the standard STIPA signal through critical aided audio communication paths from talker mouth to listener ear locations, in conjunction with acoustic noise at the talker mouth and listener ear, using modulation reductions in those octave

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bands which are within the bandwidth of the audio system. When the bandwidth of the transmission path is less than the audio range (88-11,300 Hz) of the STIPA signal, correct the STIPA score by omitting octave bands which cannot contain speech from the computation (combined effective bandwidth of the system). Only use raw STIPA scores (e.g., directly from a meter) when bandwidth of the transmission path encompasses the entire range of human speech (e.g., unaided communication between adjacent crew members). A STIPA score of at least 0.41 is considered sufficient to meet the standard. Analysis can also verify speech intelligibility by similarity when changes are made to subsystems or components which transmit speech, produce, or attenuate noise.

(2) Identify combinations of crew equipment configurations, flight conditions, and listening locations where speech intelligibility scores are below 80% MRT or 0.41 STIPA, as applicable.

Test.

(1) Ground or Lab test: Measure the bandwidth of the communications system from each type of talker microphone with the signal source (not the STIPA signal) at mission representative distance, to each type of listener headset.

(2) Ground test: Measure on air vehicle the STIPA IAW IEC 60268-16 Measurement Procedures, Post-processing of Data and Applications section, along the transmission paths from each talker location to at least 50% of listener locations, during quiet conditions (with only equipment necessary to perform the test powered). The STIPA signal should be annunciated at a sound pressure level 10 dB louder than the speech interference level for the worst-case noise at the speaking location but no louder than a “loud” voice as defined by ANSI S3.5, at an operational representative distance from the microphone sound port. Verify the input signal level by measuring the voltage across the microphone input. The loudspeaker used to provide input to the microphone should have directional characteristics of the human mouth, and a flat frequency response across the audio range of the system. Measurement locations for both noise and STIPA signals should be IAW MIL-STD-1474, Appendix D Measurement Locations paragraphs.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(3) Flight or Ground test: Conduct an MRT, IAW ANSI/ASA S3.2, to test speech intelligibility for configurations, flight conditions, and listening locations in which STIPA could not be obtained or did not meet the 0.41 value. Modified rhyme test scores cannot be combined for different personnel positions in the communication paths. If the MRT score range between participants is greater than 10 percentage points, then the scores may be more indicative of the test condition, variations and test conditions or noise at listeners ears, than the system under test itself. In these cases, repeat the MRT with additional participants to increase the statistical value of the data. Hearing protection utilized will be the same for each test participant and at a minimum the least effective hearing protection, as determined by analysis, permitted for operational use.

9.2.7.2 Speech intelligibility - one-way systems.

Standard: For passenger carrying air vehicles, a STIPA of at least 0.41 can be achieved at the ear of all seated and standing personnel in habitable locations at which the occupant is not expected to be connected to the intercom system and is wearing hearing protection permitted for

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operational use (if any) during any flight segment, air vehicle configuration, and equipment configuration in which critical communications are expected and the passengers are not restrained in crashworthy seating. For one-way systems, critical communications are communications are from the pilot or flight crew to passengers in an emergency condition.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(1) Compute the STIPA IAW IEC 60268-16 Measurement Procedures, Post-processing of Data and Applications section, for every combination of talker location, passenger location, air vehicle configuration, and flight condition, using modulation reductions in those octave bands which are within the bandwidth of the audio system, in conjunction with noise at passenger locations and noise entering via the talker microphone.

Test.

(1) Ground or Lab test: Measure the bandwidth of the audio system from each type of talker microphone with the signal source (not the STIPA signal) at mission representative distance, to each type of loudspeaker at 1 meter in an anechoic chamber.

(2) Ground test: Measure the STIPA IAW IEC 60268-16 Measurement Procedures, Post-processing of Data and Applications section, along the transmission paths from each talker location to at least 50% of listener locations, during quiet conditions (typically with only equipment necessary to perform the test powered) in a production representative air vehicle. The STIPA signal should be annunciated at a sound pressure level 10 dB louder than the Speech Interference Level for the worst-case noise at the speaking location but no louder than a “loud” voice as defined by ANSI S3.5, at a mission representative distance from the microphone sound port. The input signal level should be verified by measuring the voltage across the microphone input. The loudspeaker used to provide input to the microphone should have directional characteristics of the human mouth, and a flat frequency response across the audio range of the PA system. Measurement locations should be IAW MIL-STD-1474 Appendix D, Measurement Locations paragraphs.

9.2.7.3 Intercom accessibility.

Standard: An intercom system is accessible for immediate use at any crew station and provides two-way communication between crewmembers in all compartments.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review engineering drawings or digital models to verify that controls to activate the intercom system are located at each crew station in a position so that the intended anthropometric range of crewmembers can activate the system.

(2) Examine crew stations to verify that controls to activate the intercom system are located at each crew station in a position so that the intended anthropometric range of crewmembers can activate the system.

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Demonstration.

(1) Demonstrate in a production representative mock-up or on the air vehicle that the intercom system provides two-way communication between crewmembers in all compartments.

9.2.8 Crewmember alerting systems.

Crewmember alerting systems provide feedback of events, conditions, and situations which could result in an AW hazard to the occupants or cause damage to the air vehicle. These systems can include integrated visual and aural WCA systems. Crewmember alerting systems have alerts categorized into different alert levels depending on their urgency of response. Alert signals, advisory signals, caution signals, and warning signals are described further in MIL-STD-1472 and MIL-STD-411.

The following criteria focus on how the alerting system information is conveyed to the crewmembers. The avionics architecture aspects of the alerting system are in 11.3.1.

9.2.8.1 Crewmember alerting system functionality.

Standard: The crewmember alerting system provides the crewmembers with information of all events, conditions, and situations which could present a hazard to the occupants, cause damage to the air vehicle, or allow for the identification of non-normal operation or system conditions and the determination of the appropriate actions to take. Crewmember alerting system prioritization is IAW with MIL-STD-411, Message Presentation paragraphs and MIL-STD-411, Message Format paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Provide a WCA Matrix or alerting philosophy that outlines alert definitions, impacts to operations, cascading and inhibited alerts, autonomous air vehicle actions, priority rationale, and expected crewmember responses.

(2) For UAS, describe any external vehicle status lighting signals tied to WCAs.

(3) Conduct a FMECA to provide data on subsystems failures and failure reporting processes to the crewmembers.

Demonstration.

(1) For UAS, conduct a ground demonstration to verify that any external vehicle lighting signals tied to WCAs perform as designed.

Test.

(1) Lab test: Conduct iterative build-up of hardware and software testing (e.g., FMET, software verification testing, and FQT) from unit up to system level to verify that crewmember alerting system reports subsystem failures or states and the correct visual and aural annunciations occur. This testing should verify cascading alerts.

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(2) Lab test: Conduct a lab test to verify the crewmember alerting system prioritization is IAW MIL-STD-411, Message Presentation paragraphs and MIL-STD-411, Message Format paragraphs.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(3) Ground test: Review ground test results to verify the functionality of the crewmember alerting system.

(4) Flight test: Review flight test results to verify the functionality of the crewmember alerting system.

9.2.8.2 Verbal warning signals.

Standard: All verbal warning signals are IAW MIL-STD-1472, Verbal Warning Signals paragraphs, and the annunciation levels are 6 dB above the level of communications audio at all volume settings, but not to exceed 110 dB.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify that the verbal warning signals are IAW MIL-STD-1472, Verbal Warning Signals paragraphs.

(2) Verify the audio signal characteristics (e.g., frequency, duration, intensity) of the verbal warning signals are IAW MIL-STD-1472, Verbal Warning Signals paragraphs.

(3) Verify that the verbal warning signals are 6 dB above the level of communications audio at all volume settings.

Demonstration.

(1) Conduct a demonstration to verify the effectiveness of the verbal alerts to include the perception and recognition by the crewmembers of the alerting condition during possible other concurrent audio events and correct discrimination by the crewmembers of alerting signals in representative noise conditions. The demonstration will be conducted in an environment that incorporates all normal operational stressors to which crewmembers will be exposed in the context of a representative flight profile. The demonstration should include cascading failure scenarios. This evaluation should be in a production representative mock-up.

Test.

(1) Ground or Lab test: Record audio of verbal warning signals and voice communications at all combinations of master and channel volume settings.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Ground test: Review ground test results to validate the results from analysis, lab testing, and demonstration.

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(3) Flight test: Review flight test results to validate the results from analysis, lab testing, and demonstration.

9.2.8.3 Non-verbal aural alerts.

Standard: All non-verbal aural alerts are IAW MIL-STD-1472, Characteristics of Audio Warning Signals paragraphs and MIL-STD-1472, Signal Characteristics in Relation to Operational Conditions and Objectives paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify that non-verbal aural alerts are IAW MIL-STD-1472, Characteristics of Audio Warning Signals paragraphs and MIL-STD-1472, Signal Characteristics in Relation to Operational Conditions and Objectives paragraphs.

Demonstration.

(1) Conduct a demonstration to verify the effectiveness of the non-verbal aural alerts to include the perception and recognition by the crewmembers of the alerting condition during possible other concurrent audio events and correct discrimination by the crewmembers of alerting signals in representative noise conditions. The demonstration will be conducted in an environment that incorporates all normal operational stressors to which crewmembers will be exposed in the context of a representative flight profile. The demonstration should include cascading failure scenarios. This evaluation should be in a production representative mock-up.

Test.

(1) Ground or Lab test: Record the characteristics of the signals at listener ear positions in quiet conditions if possible.

(2) Lab test: Verify the audio signal characteristics (e.g., frequency, duration, intensity) of the aural alerts are IAW MIL-STD-1472, Characteristics of Audio Warning Signals paragraphs, MIL-STD-1472, Signal Characteristics in Relation to Operational Conditions and Objectives paragraphs.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(3) Ground test: Review ground test results to validate the results from analysis, lab testing, and demonstration.

(4) Flight test: Review flight test results to validate the results from analysis, lab testing, and demonstration.

9.2.8.4 Visual alerts - display characteristics.

Standard: Warning, caution, and advisory signals are visually detectable, distinguishable, and legible under all ambient lighting conditions.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Conduct a human-in-the-loop demonstration to verify the effectiveness of the visual alerts, to include the accurate perception and recognition by the crewmembers of the alerting condition(s) and corrective action(s) to take, in all ambient lighting conditions while wearing all possible combinations of gear included in the optical train. Conduct the demonstration in an environment that incorporates normal operational stressors to which crewmembers will be exposed in the context of a representative flight profile. The demonstration should include cascading failure scenarios. This evaluation should be in a production representative mock-up or in the crew station.

Test.

(1) Lab test: Verify WCA signals are IAW the luminance requirements in MIL-STD-411, Luminance paragraphs under the Warning Signals, Caution Signals, Advisory Signals, and the Mechanical Visual Signals sections.

(2) Lab test: For NVIS compatible cockpits or crew stations, verify radiance for WCA signals are IAW MIL-STD-3009, NVIS Radiance Requirements table.

(3) Lab test: For NVIS compatible cockpits or crew stations, verify chromaticity for warning, cautions, and advisory signals are IAW MIL-STD-3009, Chromaticity Requirements table.

(4) Lab test: Verify contrast of WCA signals is IAW MIL-L-85762, Illuminated Visual Signals paragraphs in the Daylight Legibility and Readability section.

(5) Lab test: Verify WCA signals are color coded IAW MIL-STD-411, Color Coding paragraph.

(6) Lab test: Verify that visual signals that flash, are IAW MIL-STD-411, Visual Signal Flash Rate paragraph.

(7) Ground test: Verify WCA signals are IAW the luminance requirements in MIL-STD-411, Luminance paragraphs under the Warning Signals, Caution Signals, Advisory Signals, and the Mechanical Visual Signals sections.

(8) Ground test: For NVIS compatible cockpits or crew stations, verify radiance for warnings, cautions and advisories are IAW MIL-STD-3009, NVIS Radiance Requirements table.

(9) Ground test: For NVIS compatible cockpits or crew stations, verify chromaticity for warnings, cautions and advisories are IAW MIL-STD-3009, Chromaticity Requirements table.

(10) Ground test: Verify WCA signals contrast is IAW MIL-L-85762, Illuminated Visual Signals paragraphs in the Daylight Legibility and Readability section.

(11) Ground test: Verify WCA signals are color coded IAW MIL-STD-411, Color Coding paragraph.

(12) Ground test: Verify that visual signals that flash, are IAW MIL-STD-411, Visual Signal Flash Rate paragraph.

Prior to entering operational test or operations, verify the standard by the following method(s):

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Test.

(13) Ground test: Review ground test results to verify the display characteristics of the crewmember alerting system.

(14) Flight test: Review flight test results to verify the display characteristics of the crewmember alerting system.

9.2.8.5 Master warning and master caution location.

Standard: Master cautions and master warnings are located within a 30-degree cone forward-facing centered on the design eyepoint.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review cockpit or crew station drawings or digital model to verify placement of the master caution and warning with respect to the design eye point.

Test.

(1) Ground or Lab test: Measure the angle from the design eye point to the master caution and master warning indicators to verify the location is within the 30-degree cone.

9.2.9 Mass notification systems.

Mass Notification Systems provide a means to communicate urgent and emergency information to all personnel on board in all crewmember seats, passenger seats, lavatories, workstations and other accessible compartments. These systems can be visual, aural, or both. Public address systems provide one-way communication to passengers and crewmembers not wearing communications headsets. Intercom systems normally used to provide aided, two-way communication among crewmembers may also be a means of mass notification.

Pre-crash warnings are the first element in the survival chain of events. They alert crewmembers and passengers to take actions to maximize their chances of survival and may be provided by aural, visual, or tactile means. If they are nonverbal, the visual or tactile indications are explained prior to takeoff to be effective in an emergency. Pre-crash warnings are less essential when personnel are secured in crashworthy seats at all times.

Intelligibility of mass notification systems will be assessed under 9.2.7.

9.2.9.1 Mass notification systems performance.

Standard: Mass notifications will be recognized and understood at all crewmember seats, passenger seats, lavatories, workstations and other accessible compartments under all anticipated ambient noise conditions.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

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Demonstration.

(1) Conduct a demonstration in a production representative air vehicle or control station to verify that the mass notifications are recognized and understood on first presentation from all crewmember seats, passenger seats, lavatories, workstations and other accessible compartments under all anticipated ambient noise and lighting conditions and with crewmember equipment. The demonstration will be in flight for manned air vehicles.

9.2.9.2 Pre-crash warning system operation.

Standard: For passenger carrying air vehicles, a pre-crash warning system provides a warning method or system that enables pilots and crewmembers, in the event of a potential mishap, to quickly and clearly convey a message to all air vehicle occupants. The pre-crash warning annunciations will be distinct from other alarms and alerting signals. Pre-crash warning systems can have either visual or auditory annunciations or include both.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(1) Conduct a demonstration to verify that the pre-crash warning system conveys a warning indication at all passenger seats, lavatories, and crewmember seats, and workstations under all operating and environmental conditions of the air vehicle. The demonstration will also verify that the pre-crash warning annunciations are distinct from other aural and visual annunciations. The demonstration should be done in a production representative mock-up or on the air vehicle.

9.2.9.3 Pre-crash warning system accessibility.

Standard: Means of activating or utilizing the pre-crash warning system are accessible for immediate use from each crew station in the cockpit without crewmembers leaving their seated position.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Examination.

(1) Review engineering drawings or digital models to verify that controls to activate the pre-crash warning system are located at each crew station in the cockpit in a position so that the intended anthropometric range of crew members can activate the system from a seated position.

(2) Examine the air vehicle and verify that controls to activate the pre-crash warning system are located at each crew station in the cockpit in a position so that the intended anthropometric range of crew members can activate the system from a seated position.

9.2.9.4 Public address system operation.

Standard: The PA system is accessible for immediate use from each of two stations in the cockpit IAW 14 CFR 25.1423f. The system is capable of operation within three seconds from the time a microphone is removed from its stowage IAW 14 CFR 25.1423b. The system is designed so that no unused, unstowed microphone will render the system inoperative IAW 14 CFR

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25.1423d. The system is capable of functioning independently of any required crewmember interphone or intercom system and is readily accessible, IAW AC-25-17A, to the crewmember designated to make announcements IAW 14 CFR 25.1423e and 14 CFR 25.1423g. Failure or inoperability of any PA loudspeaker does not degrade operation of other loudspeakers.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(1) Demonstrate on production representative mockup or air vehicle that the PA system is accessible for immediate use from each of two stations in the cockpit.

(2) Demonstrate on production representative mockup or air vehicle that an unstowed microphone will not render the PA system inoperative.

(3) Demonstrate on production representative mockup or air vehicle that the PA system is capable of functioning independently from the integrated communication system.

(4) Demonstrate on production representative mockup or air vehicle that the PA system is readily accessible from all required crewmember stations.

(5) Demonstrate on production representative mockup or air vehicle that the PA system is not degraded if any of the PA loudspeakers is inoperable.

Test.

(1) Ground test: Conduct a test of the PA system to verify that it is capable of operation within 3 seconds from when a microphone is removed from its stowage.

9.2.9.5 Public address system power.

Standard: A PA system is operable when the air vehicle is in flight or stopped on the ground, after the shutdown or failure of all engines and auxiliary power units, or the disconnection or failure of all power sources dependent on their continued operation, for at least 10 minutes IAW 14 CFR 25.1423a.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(1) Conduct an analysis of the system to verify that the PA system is operable for at least 10 minutes IAW 14 CFR 25.1423a.

Test.

(1) Ground test: Conduct a test of the PA system to verify that it is operable for at least 10 minutes IAW 14 CFR 25.1423a.

9.2.9.6 Public address system feedback.

Standard: Operation of the PA system from any speaking location and orientation does not induce audible feedback, ringing or echo at any listening location, including within any intercom network capable of monitoring the PA channel.

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Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(1) Verify on the air vehicle by activating the talker microphone at all operationally relevant locations throughout the air vehicle, and at relative orientations (e.g., forwards, sideways) of the talker to the microphone and the nearest loudspeaker. A swept tone (160-5000 Hz, repeating every 1.5 seconds) acoustically annunciated from a small portable speaker near the input microphone may be used as a stimulus to reduce the onset time of feedback.

9.3 Air system lighting.

This section encompasses interior lighting, emergency lighting systems, and visible mode and covert mode external lighting to include AR probe and drogue lights.

For external lights, colors have the International Commission on Illumination chromaticity coordinates for Aviation Red, Aviation Green, and Aviation White as follows:

a. Aviation Red:
   1. y is not greater than 0.335.
   2. z is not greater than 0.002.
b. Aviation Green:
   1. x is not greater than 0.440−0.320y.
   2. x is not greater than y−0.170.
   3. y is not less than 0.390−0.170x.
c. Aviation White:
   1. x is not less than 0.300 and not greater than 0.540.
   2. y is not less than x−0.040” or yc−0.010, whichever is the smaller.
   3. y is not greater than x + 0.020 nor 0.636−0.400x.

Note: Where yc is the y coordinate of the Planckian radiator for the value of x considered.

9.3.1 Interior lighting.

Interior lighting systems include all components that emit or transmit light within the air system. This includes, but is not limited to, lights for illuminating consoles, displays, transilluminated control panels, bezels and push buttons, inspection lights, utility lights, overhead lighting, and floor lighting.

Lighting criteria for crewmember alerting system indicators are covered in 9.2.8.4.

9.3.1.1 General lighting.

Standard: The interior lighting system provides lighting for crewmembers, passengers, and ground crew in occupied areas (e.g., crew stations, control stations, cargo compartment, loading and ramp areas, passageways, passenger seating area(s), and avionics bays). Lighting will be IAW MIL-STD-3009, General Lighting for Crew Stations and Compartments table.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Test.

(1) Ground test: Test lighting to ensure that illumination meets MIL-STD-3009, General Lighting for Crew Stations and Compartments table.

9.3.1.2 Reflections, glare, and halation.

Standard: Interior lighting does not cause glare, reflection, or halation across the range of lighting settings. If reflections, glare, and halation exist on the canopy, windshields, and windows they will be minimized. Reflections, glare, and halation that affect the outside vision of the crewmembers will not impact the ability to fly the air vehicle. Reflections, glare, and halation resulting from lighting sources do not obscure the displayed flight-critical information when viewed from the crewmembers' design eye point.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Show that interior lighting does not cause glare, reflections, or halation which could interfere with their interior or exterior vision and impact the ability to fly the air vehicle. Demonstration will take place under no moon or equivalent environment.

9.3.1.3 Lighting for flight-critical and safety-critical tasks.

Standard: The crew station lighting enables crewmembers to perform all flight-critical and safety-critical tasks under all anticipated lighting conditions. Luminance uniformity is maintained throughout the entire range of luminance control. Panel and associated controls (e.g., bezels, knobs, dials) luminance criteria are specified in MIL-DTL-7788, Luminance of Markings section. Within a lighting circuit, provide luminance such that the average luminance ratio between lighted components is not greater than two to one. All displays that present flight-critical and safety-critical information will meet the following:

a. For day operation, high ambient contrast for graphic symbols and alphanumerics will be IAW MIL-L-85762, High Ambient Daylight Contrast Requirements table.
b. For night operations, the minimum luminance adjustment will be less than or equal to 0.03 fL and the maximum luminance adjustment will be 4.4-8.8 fL.
c. Provide luminance, within each display, such that the luminance variation between the brightest and dimmest areas is not greater than 30%.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Measure luminance of installed components to ensure compliance with standard.

(2) Ground test: Test luminance uniformity across the crew station throughout the entire range of luminance control. For displays, at any given luminance level, the average luminance variation will not exceed 30%.

(3) Ground test: Test luminance uniformity across the crew station throughout the entire range of luminance control. For panels and associated controls (e.g., bezels, knobs, dials), at any given

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luminance setting, the average luminance ratio between other lighted components within the crew station is not greater than two to one.

(4) Ground test: Test high ambient contrast of displays IAW MIL-L-85762, Monochrome Electronic and Electro-optical Display Inspection paragraph, MIL-L-85762, Minimum Contrast Requirements paragraph, MIL-L-85762, Reflected Luminance Measurements paragraph, MIL-L-85762, Diffuse Reflected Luminance Measurements paragraph, and MIL-L-85762, Specular Reflected Luminance Measurements paragraph.

9.3.1.4 Night vision imaging system compatible lighting.

Standard: Interior lighting does not interfere with NVDs and does not impact the ability to maintain flight. Interior lighting does not cause glare, reflection, or halation that interferes with NVDs. Night vision imaging system lighting is compliant with MIL-STD-3009, Chromaticity Requirements table and NVIS Radiance Requirements Using English Units table.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Night vision imaging system spectral radiance is tested IAW MIL-STD-3009, Spectral Radiance Limits section.

(2) Ground test: Night vision imaging system chromaticity is tested IAW MIL-STD-3009, Chromaticity Measurements section.

(3) Ground test: Night vision imaging system compatibility is tested IAW MIL-STD-3009, System Level Verification section.

9.3.2 Emergency lighting.

The emergency lighting system provides illumination and guidance for passenger and crewmember emergency evacuation. This includes illumination of each exit to include interior doors and associated signage, and the air vehicle's exterior surrounding.

For CDA, the emergency lighting system conforms to 14 CFR 25.812 Emergency Lighting for transport category air vehicles, or 14 CFR 29.812 Emergency lighting for rotary-wing air vehicles.

9.3.2.1 Emergency lighting systems.

Standard: The emergency lighting system will provide illumination and guidance for passenger and crewmember emergency evacuation IAW MIL-L-85762, General Lighting for Crew Stations and Compartments table, MIL-PRF-85676, Fixed and Rotary Wing Emergency Egress Lighting System Description paragraph, and MIL-PRF-85676, General Emergency Illumination paragraph. This includes illumination of each exit to include interior doors and associated signage, and the air vehicle's exterior surroundings. The luminance of each lighted component is IAW MIL-PRF-85676, Rotary Wing and Tilt-rotor Aircraft paragraph. Exit signs are IAW MIL-PRF-85676, Exit Signs paragraphs and their subparagraphs. The battery supply to emergency lighting allows for continuous operation for at least 10 minutes after an emergency landing.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: For all military air vehicles, emergency lighting system illumination conforms with MIL-L-85762, General Lighting for Crew Stations and Compartments table, MIL-PRF-85676, Fixed and Rotary Wing Emergency Egress Lighting System Description paragraph, and MIL-PRF-85676, General Emergency Illumination paragraph.

(2) Ground test: For military rotary-wing air vehicles, emergency lighting system lighted component luminance conforms with MIL-PRF-85676, Rotary Wing and Tilt-rotor Aircraft paragraph.

(3) Ground test: For military fixed-wing air vehicles, exit signs conform with MIL-PRF-85676, Exit Signs paragraphs and their subparagraphs.

(4) Ground test: For all military air vehicles, verify that the battery supply to emergency lighting allows for continuous operation for at least 10 minutes after an emergency landing.

9.3.2.2 Night vision imaging system compatible emergency lighting.

Standard: Night vision imaging system compatible emergency lighting conforms to MIL-STD 3009, Chromaticity Requirements table and NVIS Radiance Requirements Using English Units table.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Night vision imaging system spectral radiance is tested IAW MIL-STD-3009, Warning, Master Caution, Caution Signal, and Emergency Exit Lighting Radiance paragraph.

(2) Ground test: Night vision imaging system chromaticity is tested IAW MIL-STD-3009, Chromaticity Requirements table for NVIS green A/B or NVIS white.

9.3.3 External lights (visible mode) - anticollision lights.

Air systems need to conform to the external lighting criteria for the air vehicle to operate in the National Airspace System without restriction. The external lighting criteria are directly derived from the FAA requirements.

9.3.3.1 Anticollision light - general.

Standard: The anticollision light system consists of one or more approved anticollision lights located so that their light will not impair the flight crew's vision or detract from the conspicuity of the position lights. The system will consist of enough lights to illuminate the vital areas around the air vehicle. For fixed-wing air vehicles, the field of coverage extends in each direction within at least 75 degrees above and 75 degrees below the horizontal plane of the air vehicle, except that there may be solid angles of obstructed visibility totaling not more than 0.5 steradians. For rotary-wing air vehicles, the field of coverage extends in each direction within at

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least 30 degrees above and 30 degrees below the horizontal plane of the air vehicle, except that there may be solid angles of obstructed visibility totaling not more than 0.5 steradians.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Anticollision light locations are verified by examining drawings, digital mockups, or the physical air vehicle.

Analysis.

(1) Light beam spread and overlap is verified by analysis.

9.3.3.2 Anticollision light - flash rate.

Standard: The arrangement of the system, that is, the number of light sources, beam width, flash rate or speed of rotation, and other characteristics, gives an effective flash frequency of not less than 40, nor more than 100, cycles per minute. The effective flash frequency is the frequency at which the air vehicle's complete anticollision light system is observed from a distance and applies to each sector of light including any overlaps that exist when the system consists of more than one light source. In overlaps, flash frequencies may exceed 100, but not 180, cycles per minute.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Flash characteristics are tested during ground test.

9.3.3.3 Anticollision light - chromaticity.

Standard: Each anticollision light is Aviation Red or Aviation White for fixed-wing air vehicles or Aviation Red for rotary-wing air vehicles.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Chromaticity of the anticollision lights is verified as installed on the air vehicle.

9.3.3.4 Anticollision light - intensity.

Standard: The minimum light intensities in all vertical planes, measured with the red filter (if used) is expressed in terms of effective intensities. The following relation is assumed:

$$I_e = \frac{\int_{t_1}^{t_2} I(t)dt}{0.2 + (t_2 - t_1)}$$

Ie = effective intensity(cd)

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I(t) = instantaneous intensity as a function of time

t2 – t1 = flash time interval

Each anticollision light effective intensity is equal or exceeds the applicable values in TABLE VII.

TABLE VII. *Minimum effective intensity.*

Angle Above or Below the Horizontal Plane	Effective Intensity (cd) Fixed-Wing	Effective Intensity (cd) Rotary-Wing
0 to 5 degrees	400	150
5 to 10 degrees	240	90
10 to 20 degrees	80	30
20 to 30 degrees	40	15
30 to 75 degrees	20	N/A

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Effective intensity is measured in ground testing.

9.3.4 External lights (visible mode) - position.

Compliance with the external lighting criteria allows the air vehicle to operate in commercial airways without restriction.

Intensities are determined with the light source operating at a steady value equal to the average luminous output of the source at the normal operating voltage of the air vehicle.

Dihedral angles Left (L), Right (R), and Aft (A) for the horizontal plane are shown on FIGURE 3 and the intensity distribution in the vertical plane is shown on FIGURE 4.

The dihedral angles are defined as follows:

a. Dihedral angle L (left) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the air vehicle, and the other at 110 degrees to the left of the first, as viewed when looking forward along the longitudinal axis.
b. Dihedral angle R (right) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the air vehicle, and the other at 110 degrees to the right of the first, as viewed when looking forward along the longitudinal axis.
c. Dihedral angle A (aft) is formed by two intersecting vertical planes making angles of 70 degrees to the right and to the left, respectively, to a vertical plane passing through the longitudinal axis, as viewed when looking aft along the longitudinal axis.

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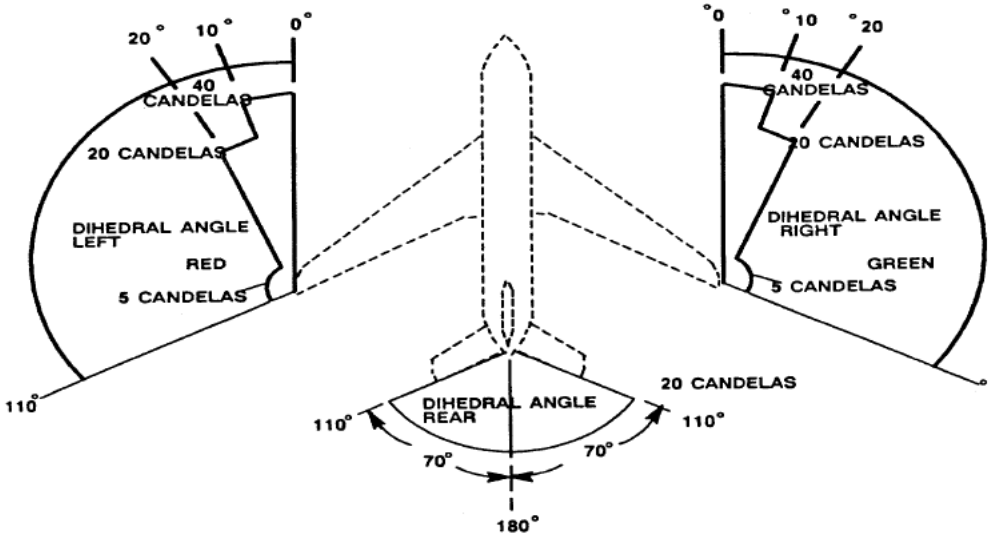


FIGURE 3. Position light dihedral angles and intensity distribution in the horizontal plane.

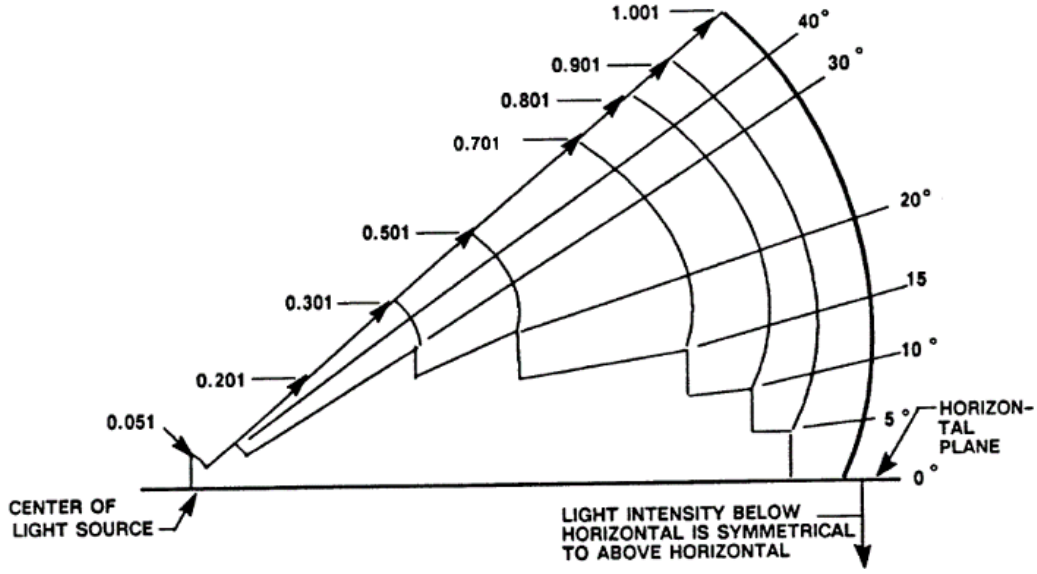


FIGURE 4. Position light intensity distribution in the vertical plane.

9.3.4.1 Position lights - general.

Standard: Position lights meet the following:

a. Forward position lights: Forward position lights consist of an Aviation Red and an Aviation Green light spaced laterally as far apart as practicable and installed forward on the air vehicle so that, with the air vehicle in the normal flying position, the red light is on the left side and the green light is on the right side. Each forward position light shows unbroken light within the dihedral angles shown on FIGURE 3.

b. Rear position light: For rotary-wing air vehicles, the rear position light consists of an Aviation White light mounted as far aft as practicable. For fixed-wing air vehicle, the rear

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position light consists of an Aviation White light mounted as far aft as practicable on the air vehicle tail or on each wing tip. Each rear position light shows unbroken light within the dihedral angles shown on FIGURE 3. If the rear position light, when mounted as far aft as practicable, cannot show unbroken light within dihedral angle A, a solid angle or angles of obstructed visibility totaling not more than 0.04 steradians is allowable within that dihedral angle, if such solid angle is within a cone whose apex is at the rear position light and whose elements make an angle of 30 degrees with a vertical line passing through the rear position light.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Forward and rear position light installation locations are verified by examination of the light installation on the air vehicle.

Test.

(1) Ground test: Chromaticity of the position lights is verified as installed on air vehicle.

9.3.4.2 Position lights - intensities.

Standard: The light intensities of forward and rear position lights, with covers and color filters installed (if equipped), is expressed in terms of minimum intensities in the horizontal plane and minimum intensities in any vertical plane within dihedral angles L, R, and A, and meets the following requirements:

a. Intensities in the horizontal plane: Each intensity in the horizontal plane (the plane containing the longitudinal axis of the air vehicle and perpendicular to the plane of symmetry of the air vehicle) is equal to or exceed the values in TABLE VIII.

TABLE VIII. *Horizontal plane intensities.*

Dihedral angle (light included)	Angle from right or left of longitudinal axis, measured from dead ahead	Intensity (candles)
L and R (forward red and green)	0 to 10 degrees	40
	10 to 20 degrees	30
	20 to 110 degrees	5
A (rear white)	110 to 180 degrees	20

b. Intensities in any vertical plane: Each intensity in any vertical plane (the plane perpendicular to the horizontal plane) is equal to or exceeds the appropriate value in TABLE IX, where I is the minimum intensity prescribed for the corresponding angles in the horizontal plane.

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TABLE IX. *Vertical plane intensities.*

Angle above or below the horizontal plane	Intensity, I
0 degrees	1.00
0 to 5 degrees	0.90
5 to 10 degrees	0.80
10 to 15 degrees	0.70
15 to 20 degrees	0.50
20 to 30 degrees	0.30
30 to 40 degrees	0.10
40 to 90 degrees	0.05

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Intensities will be measured at various angles IAW TABLE VIII and TABLE IX.

9.3.4.3 Position lights - maximum intensity overlap.

Standard: Intensities in the overlap regions do not exceed the values in TABLE X except for when the main beam intensities are substantially greater than the minima specified in TABLE VIII, and TABLE IX, and if the overlap intensities do not adversely affect the signal clarity of the main beam. When the peak intensity of the forward position lights is greater than 100 cd, the maximum overlap intensities between them may exceed the values in TABLE X provided that the overlap intensity in Area A is not more than 10% of peak position light intensity and the overlap intensity in Area B is not more than 2.5% of peak position light intensity.

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TABLE X. *Maximum intensity overlap.*

Overlap	Maximum Intensity	
	Area A (cd)	Area B (cd)
Green in dihedral angle L	10	1
Red in dihedral angle R	10	1
Green in dihedral angle A	5	1
Red in dihedral angle A	5	1
Rear white in dihedral angle L	5	1
Rear white in dihedral angle R	5	1

Area A includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 10 degrees but less than 20 degrees.

Area B includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 20 degrees.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Maximum intensities in overlapping beams of forward and rear of position lights are verified by measurements of lights installed on the air vehicle.

9.3.5 External lights (visible mode).

Compliance with the external lighting criteria allows for operation in commercial airways without restriction. In Section 10, criteria 10.1.5.2, 10.1.5.3, and 10.1.5.4 evaluate the effectiveness of external AR lighting in assisting the functions of AR via qualitative crewmember assessments during flight test that verify all AR tasks can be accomplished in the full range of environmental and operational conditions.

9.3.5.1 Landing lights.

Standard: Landing lights have a minimum intensity value of 300,000 cd with a minimum beam spread of 8 degrees in the vertical plane and 14 degrees total in the horizontal plane measured from the centerline of the air vehicle. The beam spread is defined as the beam angle where the intensity drops to 10% of the maximum intensity. Landing lights are free of dangerous glare or halation to the pilots. The chromaticity for landing lights will be Aviation White.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Landing light(s) coverage, aimability, and intensity are verified by geometric and photometric analysis.

Demonstration.

(1) Conduct a demonstration on the air vehicle to verify the landing light(s) aimability and that the design of the landing light(s) minimizes glare to the pilots.

Test.

(1) Lab test: Conduct component-level testing of the landing light test article to measure chromaticity, beam spread, and intensity.

(2) Ground test: Perform testing on the air vehicle to measure the light intensity at required beam spread angles as measured from the air vehicle centerline.

9.3.5.2 Taxi lights.

Standard: Taxi lights have a minimum intensity value of 50,000 cd with a minimum beam spread of 10 degrees in the vertical plane and 40 degrees total in the horizontal plane as measured from the centerline of the air vehicle. The beam spread is defined as the beam angle where the intensity drops to 10% of the maximum intensity. Taxi lights are free of dangerous glare or halation to the pilots. The chromaticity for taxi lights is Aviation White.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Taxi light(s) coverage, aimability, and intensity are verified by geometric and photometric analysis.

Demonstration.

(1) Conduct a demonstration on the air vehicle to verify the taxi light(s) aimability and that the design of the landing light(s) minimizes glare to the pilots.

Test.

(1) Lab test: Conduct component-level testing of the taxi light test article to measure chromaticity, beam spread, and intensity.

(2) Ground test: Perform testing on the air vehicle to measure the light intensity at required beam spread angles as measured from the air vehicle centerline.

9.3.5.3 Dual purpose landing and taxi lights.

Standard: Dual purpose landing and taxi lights have a minimum intensity value of 300,000 cd with a minimum beam spread of 8 degrees in the vertical plane and 14 degrees total in the horizontal plane measured from the centerline of the air vehicle. The beam spread is defined as the beam angle where the intensity drops to 10% of the maximum intensity. Dual purpose landing and taxi lights are free of dangerous glare or halation to the pilots. The chromaticity for landing lights is Aviation White.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Taxi light(s) coverage, aimability, and intensity are verified by geometric and photometric analysis.

Demonstration.

(1) Conduct a demonstration on the air vehicle to verify the dual purpose landing and taxi light(s) aimability and that the design of the light(s) minimizes glare to the pilots.

Test.

(1) Lab test: Conduct component-level testing of the dual purpose landing and taxi light test article to measure chromaticity, beam spread, and intensity.

(2) Ground test: Perform testing on the air vehicle to measure the light intensity at required beam spread angles as measured from the air vehicle centerline.

9.3.5.4 Riding light.

Standard: Riding (or anchor) light(s) required for seaplanes or amphibious air vehicles is installed so that it will show a white light for at least two miles under clear atmospheric conditions and show maximum unbroken light when moored or drifting.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Riding light performance is verified by geometric analysis.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct a flight demonstration to verify the riding light performance at a minimum range of two miles when moored or drifting.

9.3.5.5 AR probe and drogue lights.

Standard: AR probe and drogue lights conform to SAE ARP5825, Requirements for Probe/Drogue Style Refueling Lights paragraph, MIL-A-19736, Light paragraph, JSSG-2010-5, Aerial Refueling Lights section, ATP-3.3.4.2, Probe and Drogue section, and JSSG-2009, Probe Illumination section.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(1) Ground test: Test installed lighting on the air vehicle and verify conformance to SAE ARP5825, Requirements for Probe/Drogue Style Refueling Lights paragraph, MIL-A- 19736, Light paragraph, JSSG-2010-5, Aerial Refueling Lights section, ATP-3.3.4.2, Probe and Drogue section, and JSSG-2009, Probe Illumination section.

9.3.6 Covert external lights.

This section covers lighting requirements for covert operations. This type of lighting is not visible to the naked eye. In Section 10, criteria 10.1.5.2, 10.1.5.3, and 10.1.5.4 evaluate the effectiveness of external AR lighting to assist the functions of AR in covert mode via qualitative crewmember assessments during flight test that verify all covert AR tasks can be accomplished in the full range of environmental and operational conditions.

9.3.6.1 Covert anti-collision lights.

Standard: Covert anti-collision lighting conforms to SAE ARP5825, NRI Limits for the Anti-Collision Covert Light paragraph.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Ground test: Test installed covert anti-collision lights for conformance to SAE ARP5825, NRI Limits for the Anti-Collision Covert Light paragraph.

9.3.6.2 Covert landing lights.

Standard: Covert landing lights conform to SAE ARP4392, Covertness paragraph under the Landing Lights section, SAE ARP5825, Covert Radiance Limits for Landing/Taxi Lights paragraph, and SAE ARP5825, NRI Limits for the Landing/Taxi Covert Lights paragraph.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Ground test: Test installed covert landing lights for conformance to SAE ARP4392, Covertness paragraph under the Landing Lights section, SAE ARP5825, Covert Radiance Limits for Landing/Taxi Lights paragraph, and SAE ARP5825, NRI Limits for the Landing/Taxi Covert Lights paragraph.

9.3.6.3 Covert taxi lights.

Standard: Covert taxi lights conform to SAE ARP4392, Covertness paragraph under the Landing Lights section, SAE ARP5825, Covert Radiance Limits for Landing/Taxi Lights paragraph, and SAE ARP5825, NRI Limits for the Landing/Taxi Covert Lights paragraph.

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Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Ground test: Test installed covert taxi lights for conformance to SAE ARP4392, Covertness paragraph under the Landing Lights section, SAE ARP5825, Covert Radiance Limits for Landing/Taxi Lights paragraph, and SAE ARP5825, NRI Limits for the Landing/Taxi Covert Lights paragraph.

9.3.6.4 Covert position lights.

Standard: Covert position lights conform to SAE ARP5825, Covert Radiance Limits for Position Lights paragraph, SAE ARP5825, NRI Values for Covert Position Lights paragraph, SAE ARP5825, NRI Limits for Right Wingtip Covert Lights that are Supplemental to Position Lights that are not Installed on the Wingtip paragraph, and SAE ARP5825, NRI Limits for Left Wingtip Covert Lights that are Supplemental to Position lights that are not Installed on the Wingtip paragraph.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Ground test: Test installed covert position lights for conformance to SAE ARP5825, Covert Radiance Limits for Position Lights paragraph, SAE ARP5825, NRI Values for Covert Position Lights paragraph, SAE ARP5825, NRI Limits for Right Wingtip Covert Lights that are Supplemental to Position Lights that are not Installed on the Wingtip paragraph, and SAE ARP5825, NRI Limits for Left Wingtip Covert Lights that are Supplemental to Position lights that are not Installed on the Wingtip paragraph.

9.3.6.5 Covert AR probe and drogue lights.

Standard: Covert AR probe and drogue lights conform to SAE ARP4392, Covertness paragraph under the Refueling Probe Lights section. Installed covert AR lighting NRI matches the NVIS friendly NRI requirements in SAE ARP5825, NRI Limits for Refueling Lights paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Verify that the installed covert AR lighting conforms to SAE ARP4392, Covertness paragraph under the Refueling Probe Lights section.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(2) Ground test: Verify that installed covert AR lighting NRI matches the NVIS friendly NRI requirements in SAE ARP5825 NRI, Limits for Refueling Lights paragraph.

9.4 Human performance.

This section encompasses the means for crewmembers to monitor and control the air system's flight path management, navigation, alerting systems, communications, identification, propulsion, and mission systems. The presentation of normal and emergency checklists and procedures are addressed, as well as the location and arrangement of the PFD suite, crewmember workload, and situational awareness.

The HEDAD-O is an artifact that typically provides information that satisfies some of the MoC in this section. A recommended format for the HEDAD-O can be found in data item description DI-HFAC-80746.

Criteria related to flight symbology are specific to AW only, and do not fulfill any service specific requirements to obtain a PFR endorsement (e.g., USAF PFR endorsement IAW AFMAN 11-202V3, sections related to PFR endorsement).

9.4.1 Crewmember functional operations and workload.

Standard: All normal and emergency procedures can be accomplished within the limits defined by a government accepted workload assessment method (e.g., Bedford Workload Rating Scale). Workload assessments include evaluating crewmember situational awareness. Collect objective measures of performance (e.g., speed, accuracy, and heading) to corroborate the subjective scores. Crewmember tasks are documented and can be accomplished within the capabilities of the personnel.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) All crewmember tasks are defined and documented by functional and task analysis.

Test.

(1) Lab test: Workload and situational awareness assessments, including objective and subjective measures, are accomplished with human-in-the-loop simulation using trained and qualified crewmembers. This assessment includes all normal procedures and a subset of worst-case emergency procedures. The emergency procedures assessed are associated with WCAs. Assessments are conducted in a simulator, using a high-fidelity simulation representative of the air system's operating environment, using government accepted methods.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Workload and situational awareness test results for normal procedures are verified during flight test, using qualified crewmembers. Crewmembers are familiarized with the system under test.

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9.4.2 Primary flight displays.

Standard: Primary flight displays, including WCAs, provide the necessary information to the pilots to enable performance of all flight maneuvers, in both normal and emergency conditions without endangering the air vehicle. All crew stations used by pilots to control an air system have access to at least one complete PFD. The PFD provides continuous presentation of primary flight information IAW with MIL-STD-1787, Aircraft Display Symbology all sections (except Missile Launch Zone and Attack Steering Depictions) and MIL-STD-411, Integrated Alert Displays section, or civilian best practices (e.g., FAA AC 25-11).

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: New PFD designs (or significant deviations from baseline designs) and a selection of supplemental PFDs are tested in simulation or flight test. Tests represent the full set of flight maneuvers (e.g., unusual attitude recoveries, basic flight maneuvers, and mission demonstrations).

9.4.3 Flight manuals.

Standard: The flight manuals have documented emergency procedures that are clear, with corrective actions that can be accomplished without death, injury, loss of situational awareness, or loss of the air vehicle. Workload is IAW 9.4.1.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Examine the flight manuals and verify that the information and relevant procedures are documented, complete, and accurate.

(2) Examine the flight manuals and verify that the flight manuals are certified and complete IAW service-applicable procedures (e.g., AFI 11-215, CNAF M-3710.7).

9.4.4 External visibility and transmitted visual indications.

These criteria ensure that the air vehicle has external visibility and transmitted visual indications that allow pilots to maintain flight, conduct all necessary flight tasks, avoid ground or flight obstacles, and operate within military airspace and the National Airspace System.

The FOV criteria include references to establishing the clear vision areas. The clear vision area is the area of transparent material through which vision is unobstructed by structure, edge bonding material, or any material which prohibits clear vision. These areas are kept clear of obstructions to vision (e.g., misting, rain, icing, and insects).

The exterior FOV from the design eye box allows pilots to perform all primary operations in all normal and emergency operating conditions. The exterior FOV is free of any obstructions that could impair the pilot's view (e.g., structural components, equipment). The standards for

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minimum angles of unimpaired vision that are available from the design eye point are described in 9.4.4.2-9.4.4.6.

9.4.4.1 External visibility.

Standard: The transparency is free of distortion, optical defects, reflection, refraction, and transmissivity that result in visual aberrations. The optical properties of the entire optical path (e.g., transparency, HUDs, helmet mounted displays, and visors) from the outside world to the eye enable the pilot to perform any maneuvers within the approved operating limitations of the air vehicle, including but not limited to, taxiing, takeoff, approach, and landing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the primary pilot operations as they relate to the optical path and establish the regions of the transparency where primary operations occur.

Demonstration.

(1) Perform a demonstration on the physical air vehicle and show that the optical characteristics allow pilots to maintain visual capability. The demonstration is accomplished under day, night, and with NVDs under NVIS lighting conditions (if applicable).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Verify in flight that the optical characteristics provide the pilots with a clear, and undistorted view, and enable them to perform all maneuvers within the approved operating limitations of the air vehicle.

9.4.4.2 Exterior field of view - Air vehicles with single or tandem pilots.

Standard: No windscreen structure forward of the canopy bow exceeds 2.0 inches in width when projected onto a plane perpendicular to a line between the structure and the design eye position.

The blind spot caused by the structure between the top of the windscreen and the canopy does not exceed 35 degrees horizontally nor 7 degrees vertically.

The following minimum angles of unimpaired vision are available to the pilot in the forward position, from the design eye positions shown in TABLE XI.

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TABLE XI. *Exterior FOV - Air vehicles with single or tandem pilots - forward position.*

At 0° azimuth, 11° down and 10° up.
At 20° azimuth, left and right, 20° down.
At 30° azimuth, left and right, 25° down.
At 90° azimuth, left and right, 40° down.
At 135° azimuth, left and right, 20° down.

For the aft position, at 20 degrees azimuth, a minimum of 5 degrees vision down is provided, measured from the aft design eye box with the forward seat in the neutral seat position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze vision plots and verify that the total envelope (plus and minus 180 degrees in azimuth and plus and minus 90 degrees in elevation) from each pilot's design eye box provides the required clear vision areas required by the standard. Vision plots are completed for each pilot's position for all their primary operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Demonstrate in flight that the external FOV allows the pilots to perform all their primary operations.

9.4.4.3 Exterior field of view - Air vehicles with side-by-side pilots.

Standard: The following are the minimum angles of unimpaired vision available to the pilot from the design eye positions in TABLE XII.

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TABLE XII. *Exterior FOV - Air vehicles with side-by-side pilots.*

At 0° azimuth, at least 13° down and 12° up.
0° through 70° azimuth – at least 40° down and 40° up.
Between 70° and 110° azimuth, provide 40° down and 90° up.
135° azimuth, at least 20° down.
In the cockpit area inboard of each pilot's centerlines, maximum vision with a minimum of obstruction should be provided.
90° left azimuth, and right azimuth provide at least 70° downward with the pilot leaning outboard from their respective design eye positions.

Canopy bow vision obstruction within the area shown in TABLE XII does not exceed 2.0 inches in width when projected onto a plane perpendicular to a line between the structure and the design eye position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze vision plots and verify that the total envelope (plus and minus 180 degrees in azimuth and plus and minus 90 degrees in elevation) from each pilot's design eye box provides the required clear vision areas as described herein. Vision plots are completed for each pilot's position for all their primary operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Demonstrate in flight that the external FOV allows the pilots to perform all their primary operations.

9.4.4.4 Exterior field of view - Multi-place air vehicles with side-by-side pilots.

Standard: This standard applies to multi-place air vehicles with side-by-side pilots and anti-submarine and extended overwater patrol missions.

The following is the minimum angles of unimpaired vision available to the pilot from the design eye position in TABLE XIII.

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TABLE XIII. *Exterior FOV - Multi-place air vehicles with side-by-side pilots.*

At 0° through 30° azimuth, at least 17° down and 20° up.
Between 30° and 70° left azimuth, external vision should increase from at least 17° down and 20° up to at least 35° down and 40° up.
Between 70° and 110° azimuth, provide at least 35° down and 40° up.
To the right of the pilot centerline the angular vision downward and upward may decrease from 17° and 20° specified due to the increased distance from the design eye position to the windshield.
The transparent area should extend to 135° azimuth at eye level with vertical angles less than at 110°.
At 90° azimuth, provide at least 70° downward with the pilot leaning outboard from their respective design eye positions.

The entire transparent area between 20 degrees right and 30 degrees left is free of windscreen structure or posts.

No windscreen post or structure exceeds 2.5 inches in width when projected onto a plane perpendicular to a line between the post or structure and the pilot design eye position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze vision plots and verify that the total envelope (plus and minus 180 degrees in azimuth and plus and minus 90 degrees in elevation) from each crewmember's design eye box provides the required clear vision areas as described in the standard. Vision plots are completed for each pilot's position for all their primary operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Demonstrate in flight that the external FOV allows the pilots to perform all their primary operations.

9.4.4.5 Exterior field of view - Rotary-wing air vehicles with side-by-side pilots.

Standard: The following are the minimum angles of unimpaired vision available to the pilots from the design eye position in TABLE XIV.

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TABLE XIV. *Exterior FOV - Rotary-wing with side-by-side pilots.*

At 0° azimuth, 25° down and 20° up.
From 10° left azimuth to 10° right azimuth, downward vision should increase from 20° to 30°.
Between 10° right azimuth and 135° right azimuth, 50° of downward vision should be available.
From 0° to 80° right azimuth, upward vision should increase from 20° to 40°.
From 80° right azimuth to 100° right azimuth, upward vision should be 40°.
From 100° right azimuth to 135° right azimuth, upward vision may decrease gradually from 40° to 20°.
From 10° left azimuth to 100° left azimuth, 20° up and 20° down should be available.

No vertical obstruction between 20 degrees right and 20 degrees left of the longitudinal axis relative to the design eye position.

No horizontal obstructions in the area extending 15 degrees above the horizon from 135 degrees right to 40 degrees left and decreasing to a point 10 degrees above the horizon at 100 degrees left.

Mounting structures for the transparency sections do not exceed 2.0 inches in width when projected onto a plane perpendicular to a line between the structure and the design eye position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze vision plots and verify that the total envelope (plus and minus 180 degrees in azimuth and plus and minus 90 degrees in elevation) from each crewmember's design eye box provides the required clear vision areas as described in the standard. Vision plots are completed for each pilot position for all their primary operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Demonstrate in flight that the external FOV allows the pilots to perform all their primary operations.

9.4.4.6 Exterior field of view - Rotary-wing air vehicles with single or tandem pilots.

Standard: The following are the minimum angles of unimpaired vision available to the pilots from the design eye position in TABLE XV.

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TABLE XV. *Exterior FOV - Rotary-wing air vehicles with single or tandem pilots.*

0° azimuth, at least 25° down and 70° up.
20° azimuth left and right, 25° down and 70° up.
30° azimuth left and right, 30° down and 70° up.
90° azimuth left and right, 50° down and 70° up.
135° azimuth left and right, 34° down and 70° up.

Mounting structures for the transparency sections do not exceed 2.0 inches in width when projected onto a plane perpendicular to a line between the structure and the design eye position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze vision plots and verify that the total envelope (plus and minus 180 degrees in azimuth and plus and minus 90 degrees in elevation) from each crewmember's design eye box provides the required clear vision areas as described herein. Vision plots are completed for each flight crew position for all their primary operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Demonstrate in flight that the external FOV allows the pilots to perform all their primary operations.

9.4.4.7 Enhanced vision.

Standard: Enhanced vision system and EVFS imagery are free of interference, distortion, and glare that would adversely affect pilot's performance and workload. For flight-critical functions, the imagery does not provide the pilot with misleading information regarding detection, accurate identification and avoidance of terrain, obstacles, and other flight hazards. Imagery displayed on the HUD or equivalent display conforms to the real-world and is appropriate for the system's intended function, accounting for possible air vehicle attitudes, turbulence, and wind effects. If PFD information is displayed, it is accurate, and the horizon is conformal and aligned with the imagery. For other EVS characteristics refer to FAA AC 20-167A Airworthiness Approval of Enhanced Vision System, Synthetic Vision System, Combined Vision System, and Enhanced Flight Vision System Equipment, sections - EVS or SVS (or CVS when EVS and SVS combined) – General and EVS Specific Requirements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Demonstration.

(1) Demonstrate that the EVS image, as installed in the air vehicle, is free of interference, distortion, and glare that would adversely affect pilot performance and workload (e.g., reduced ability to accomplish tasks or degraded workload scores) for all intended flight conditions.

(2) Demonstrate that the imagery allows the pilot to detect, identify, and avoid terrain, obstacles, and other flight hazards for all intended flight conditions.

(3) Demonstrate that the imagery is conformal if the imagery is shown on a HUD or equivalent display for all intended flight conditions.

(4) Demonstrate that the PFD information agrees with other flight displays and that the horizon is aligned with the imagery and conformal for all intended flight conditions.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(5) Demonstrate in flight that the EVS image is free of interference, distortion, and glare that would adversely affect pilot performance and workload.

(6) Demonstrate in flight that the imagery accurately detects, identifies and allows for the avoidance of terrain, obstacles, and other flight hazards.

(7) Demonstrate in flight that the imagery is conformal if the imagery is shown on a HUD or equivalent display.

(8) Demonstrate in flight that the PFD information is accurate, and that the horizon is conformal and aligned with the imagery.

9.4.5 User interface error handling.

Standard: All user interfaces are designed IAW MIL-STD-1472, Human Performance and associated subparagraphs. Error handling is IAW MIL-STD-1472, Error Management and all associated subparagraphs.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(1) Review and analyze the HEDAD-O or flight manuals to verify that all user interface error handling processes conform to the standards.

9.4.6 Automation and autonomy user interfaces.

This element involves defining the levels of human interaction with autonomous functions and defining the interface characteristics for each level. The level of user interaction may change during mission execution depending on the varying demands of the user, changes in mission goals or expectations with respect to user insight, oversight, and override capabilities. Interface characteristics described below align with several sources to include DoDD 3000.09 Autonomy in Weapon Systems. Fundamental characteristics of the interface are defined in TABLE XVI.

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TABLE XVI. *Automation and autonomy user interfaces.*

Characteristic	**Definition**	**Corresponding MIL-STD-1472 References**	**Insight**	**Oversight**	**Override**
Function Allocation	Clear distinction between the human and system responsibilities regardless of dynamic or static allocation in autonomous or automatic functions.	Changing roles and responsibilities - 5.12.3.3.4 (Allocation) Analysis 5.12.3.3.5 (Function) Allocation	X	X	X
Predictability	Understanding current and future actions or states.	Current situation - Off-nominal situations. Limitation. Predictability -	X	X	X
Observability	Provides information for the user to support shared understanding of the state, task progression, and progress toward goals. Provides the needed level information about sources and calculations that contribute to the recommendations and predictions provided to the user.	Situational awareness Current situation Audio displays as part of the user interface Interface	X	X	X

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TABLE XVI. *Automation and autonomy user interfaces - Continued*

Characteristic	Definition	Corresponding MIL-STD-1472 References	Insight	Oversight	Override
Directability	Ability to easily direct and redirect resources, activities, and priorities. To the extent that the user is accountable, the user should be able to stop processes, change course, toggle between levels of autonomy, or override and assume manual control when necessary.	Off-nominal situations. Recovery. Override capability. Unmanned vehicles		X	X
Error Handling	Method by which the system identifies errors, informs the user, and provides a way for the user to correct.	Error management Human-supervised autonomous weapon systems Unmanned, remotely operated, automated, and autonomous weapon systems	X	X	X

The user insight, oversight, and override capabilities are defined below.

Insight:

User Insight (e.g., operator on-the-loop) is typically characterized by minimal interaction between the user and the air system and is generally observational only. User permission is not required for the system to conduct operations within the scope of mission intent. User does not have control capabilities at this level.

Oversight:

User Oversight (e.g., operator in-the-loop) is typically characterized by increased levels of interaction between the user and the air system, to include user permissions for actions. User permission may be required for system to conduct operations within the scope of mission intent. Users typically have control capabilities at this level.

Override:

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User Override is typically characterized by the operator having direct control of the air system.

9.4.6.1 User insight.

Standard: The interface to the autonomous systems provides users with awareness of the system or subsystem states, modes, and functions. The interface provides the users with data to understand the current system state, the surrounding environment, and predicted future states.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that the characteristics of predictability, observability, and error handling have been correctly provisioned in the air system design.

Analysis.

(1) Analyze the design and verify that the functions necessary to maintain flight and accomplish flight critical tasks are documented and clearly differentiated between the automated or autonomous system and the user.

(2) Analyze the user interfaces and show that the required characteristics of predictability, observability, and error handling are present in the user interface.

Test.

(1) Lab test: Test the implementation in a high-fidelity simulation or prototype and verify that the characteristics of predictability, observability, and error handling are correctly implemented in the user interface.

9.4.6.2 User oversight.

Standard: The characteristics for user oversight include the user interfaces required to interact with the system as necessary to accomplish flight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the air system design documentation (e.g., software documentation, digital mockups) and verify that the required characteristics of predictability, observability, directability, and error handling have been accurately provisioned in the air system design.

Test.

(1) Lab test: Test the implementation in a high-fidelity simulation or prototype and verify that users can oversee the air system operation and accomplish oversight.

9.4.6.3 User override.

Standard: Ensure that the air system includes the capability for the user to take manual control of the system or subsystem states, modes, and functions, to include flying operations if required.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the air system design documentation (e.g., software interface descriptions, drawings, digital mockups) and verify that the required characteristics of functional allocation, predictability, observability, directability, and error handling have been accurately allocated in the air system design.

Analysis.

(1) Analyze the air system design documentation (e.g., drawings, digital mockups) and verify that the required characteristics functional allocation, predictability, observability, directability and error handling are present in the design.

Test.

(1) Lab test: Test the implementation in a high-fidelity simulation or prototype and verify that users can override the air system operation when required.

9.5 Escape systems.

This section includes criteria for escape systems installed in air vehicles. Escape systems encompass the AW starting at the initiation of the escape system to touchdown of occupant on the ground. For test MoCs that include a manikin, the manikins used represent the intended anthropometric range and are equipped with the representative crewmember clothing and equipment.

Crash loads for escape systems can be found in 9.7.1. The coordinate system used for the standards throughout this section is shown on FIGURE 5.

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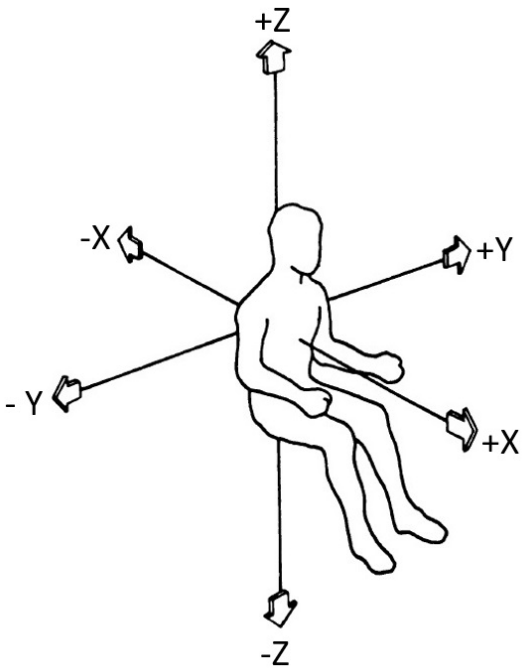


FIGURE 5. Human coordinate system.

Throughout 9.5, for standards requiring a test, analysis can be used instead of lab test when it can be shown that an item is equivalent or identical in design to another item which has been verified to the same or more stringent criteria.

9.5.1 Initiation to touchdown.

The criteria in this subsection apply from initiation to touchdown. Initiation is defined as when a signal to the escape system is initiated. Touchdown is defined as when the occupant physically touches the ground post-escape event.

9.5.1.1 Thermal energy exposure.

Standard: Thermal energy exposure, (e.g., thermal energy from operation of the escape system energy transmission system, propellant- or cartridge-actuated devices, the escape path clearance subsystem, and any rocket blast or rocket plumes associated with the escape process) is less than levels causing second degree burn injuries. Burn depth is less than 100 µm into the skin. Burns to the emergency and survival equipment do not prevent the equipment from performing the necessary functions (e.g., life preserver unit inflation or immersion suit protection).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the thermal data using BURNSIM, or other DoW approved model, and verify the corresponding burn depth as it relates to the heat flux.

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Test.

(1) Lab test: System-level tests are accomplished to verify the thermal energy exposure. Manikins are equipped with thermal gauge heat-flux transducers to determine the heat flux. Do not use temperature probes or thermocouples to determine heat flux due to excessive measurement error.

(2) Lab test: If damage to emergency or survival equipment is found, verify that the emergency and flight equipment still functions as intended.

9.5.1.2 Impulse noise - chest.

Standard: Impulse noise levels measured at the chest at each crew station are less than 195 dBP IAW MIL-STD-1474, Impulsive Noise Non-auditory Effects Criteria paragraph, unless specific analysis is performed to assess the risk of blast lung (and other internal organ) injury (e.g., analysis IAW 9.5.1.4).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Sound pressure levels are measured during system-level tests IAW MIL-STD-1474, Impulsive Noise Non-auditory Effects Criteria paragraph.

9.5.1.3 Impulse noise - head.

Standard: Impulse noise levels resulting from the escape system, measured at the ear canal (inside the helmet) at each crew station, are less than 140 dBP. For noise exposures where the peak pressure level is 140 dBP or greater, as measured at the ear canal (inside the helmet) or external to the helmet using the helmet impulsive peak insertion loss at each crewmember position, the $L_{IAeq8hr}$ does not exceed steady-state levels of 85 dBA or 140 dBP, as calculated IAW MIL-STD-1474, Compute the L_{IAeq} for a Single Impulse paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The $L_{IAeq8hr}$ does not exceed steady-state levels 85 dBA or 140 dBP, as calculated IAW MIL-STD-1474, Compute the L_{IAeq} for a Single Impulse paragraph.

Test.

(1) Lab test: Testing verifies the sound pressure level measurements at each crewmember position IAW the standard.

9.5.1.4 Blast pressure.

Standard: Blast pressure from the escape system, including canopy, hatch jettison, fracturing, and cutting systems is verified. Blast pressure does not exceed the injury threshold calculated using the injury threshold limit for blast pressure equation per (Bass, et al. 604-615).

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$$P = P^{*}(1 + a\,T^{b})$$

Where P = pressure (psi), P* = 6.0308, a = 6.68, T = duration, and b = 0.8333

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: Blast pressures are recorded during system-level tests at the manikin torso and chest under ambient temperature conditions. Blast pressures from canopy clearance systems are measured during system-level dynamic sled tests.

9.5.1.5 Probability of concussion.

Standard: The escape system prevents the probability of concussion (P concussion) from exceeding 5%. The P concussion is defined in the probability of concussion equation (Rowson and Duma).

$$P_{concussion} = \frac{1}{1 + e^{-(-10.2+.0433a+0.000873\alpha-0.00000092a\alpha)}}$$

Where 'a' is the observed peak linear acceleration and 'α' is the observed peak rotational acceleration as a function of time.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Acceleration and rate data are used to calculate P concussion. Use the maximum angular resultant acceleration within a ±30ms time window centered around each resultant linear acceleration at each recorded time step.

Test.

(1) Lab test: Full system-level dynamic ejection sled testing verifies that the system meets the standard for concussion. Linear acceleration is measured using linear accelerometers. Angular acceleration is measured using angular rate sensors (preferred) or can be measured using angular accelerometers with sufficient frequency response. Sensors are mounted within the CG of the manikin head during each test.

9.5.1.6 Individual axis chest accelerations.

Standard: The maximum accelerations experienced by an occupant during escape do not exceed the maximum chest accelerations in TABLE XVII.

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TABLE XVII. *Maximum chest accelerations.*

Direction	Maximum Allowable Acceleration
+ gx	35 / 15 g*
- gx	35 g
± gy	15 g
+ gz	25 g
- gz	20 g
*Note: +35gx chest prior to seat/occupant separation and +15 gx after.	

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Chest accelerations are verified through full system-level dynamic ejection sled testing.

9.5.1.7 Resultant chest accelerations.

Standard: The maximum resultant acceleration measured at the chest do not exceed 25g for ejection speeds up to 450 KEAS and linearly scaled up to 35g at 600 KEAS. For speeds below 600 KEAS, exceedances of the resultant acceleration limit may be permitted if they are less than 35g and are predominantly +gx acceleration. This is defined as the ±gy and ±gz components of the resultant being less than half their respective limits in 9.5.1.6.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: Resultant chest accelerations are verified through full system-level dynamic ejection sled testing.

9.5.1.8 Terrain clearance.

Standard: Escape systems recover crewmembers from zero altitude AGL (in level flight) up to the absolute ceiling of the air vehicle and from zero airspeed up to the maximum velocity of the air vehicle or 600 KEAS whichever is less. Escape systems meet user escape system terrain clearance requirements. Capability above 600 KEAS may be required for escape systems that enclose the crewmembers (e.g., an escape capsule).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Analysis of the ejection sled test data is used to validate a DoW approved trajectory model.

(2) The validated dynamic trajectory model shows that the escape system recovers crewmembers across the escape envelope.

Test.

(1) Lab test: System-level dynamic ejection sled testing demonstrates trajectories in all three dimensions, across the speed range and occupant weights for all air vehicle seat positions. This data is used to validate dynamic trajectory models used to predict terrain clearance.

9.5.1.9 Neck injury.

Standard: Neck injury criteria apply to all phases of the ejection with and without helmet mounted devices (e.g., night vision goggles or helmet mounted cueing systems). Deployable head and neck protection systems show compliance to the injury standard whether or not the head is in contact with the deployable systems.

For crewmembers with a nude weight of 245 lbs:

a. The upper neck injury risk is limited to 5% at speeds up to and including 600 KEAS.
b. The lower neck injury risk is limited to 5% at speeds up to and including 600 KEAS.

For crewmembers with a nude weight of 103 lbs:

a. Upper neck injury risk is limited to 5% up to and including 450 KEAS.
b. Lower neck injury risk is limited to 5% up to and including 450 KEAS.
c. For speeds greater than 450 KEAS, the upper and lower neck injury risk limits increase linearly as a function of speed to a maximum of 50% at 600 KEAS.

The upper and lower neck injury limits for crewmember nude weights between 103 and 245 lbs are linearly interpolated.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Data is collected from full system-level dynamic ejection sled testing using instrumented manikins. The neck injury standard applies separately to upper and lower neck injury.

9.5.2 Initiation to air vehicle separation.

The following criteria are applicable from escape system initiation to escape system-air vehicle separation.

Initiation is defined as when a signal is initiated manually by a crewmember or automatically by the air vehicle.

Escape system-air vehicle separation is defined as when escape system is physically separated from the air vehicle and the air vehicle no longer constrains the system to pitch, yaw, or roll relative to the air vehicle.

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9.5.2.1 Accelerations and angular velocities.

Standard: Seat accelerations are IAW INFO PUB ASM 4048 Ed 2v2, Multiaxial Dynamic Response Criteria (MDRC) Limits paragraph using the moderate limits in the INFO PUB ASM 4048 Ed 2v2, DR Limit Values paragraph.

The seat angular velocities are IAW the low limits in INFO PUB ASM 4048 Ed 2v2, Ejection Acceleration Limits, Short Duration Angular Velocity Limits.

The acceleration time history generated during subsystem catapult testing, with representative ejection occupant and system weights, at a pre-ignition temperature of 165 ºF, result in DRIz values less than 22.0.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Subsystem catapult testing is used to verify that DRIz values are less than 22.0 at 165 °F pre-ignition temperatures.

(2) Data from system-level dynamic ejection sled testing is used to verify the MDRC, DR, and Angular Velocity limits at ambient temperatures.

9.5.2.2 Torso restraint.

Standard: The occupant is kept in position with respect to support of the seating system under both inertial and applied loads (e.g., aerodynamic pressure). The restraint system prevents "submarining" or translation of the pelvis under the lap belt. The shoulders are restrained to align the thoracic and lumbar spine with the seat back. The restraint system ensures that ejection posture is maintained up to seat occupant separation. Torso displacement, as measured from the base of the neck (cervical vertebrae C7/T1) to the back tangent line, is less than 3.0 inches.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Subsystem dynamic load tests verify restraint mechanical integrity, seat back alignment, and torso displacement.

(2) Lab test: System-level dynamic ejection sled tests verify full escape system restraint capability.

9.5.3 Escape system-air vehicle separation to ejection seat-occupant separation.

The following criteria are applicable from escape system-air vehicle separation to ejection seat-occupant separation.

Escape system-air vehicle separation is defined as the escape system is physically separated from the air vehicle and the air vehicle no longer constrains the system to pitch, yaw, or roll relative to the air vehicle.

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Seat-occupant separation is defined as decoupling between the ejection seat and occupant. This can be defined using event channels or when the restraint system has positive distance from the seat.

The individual and resultant chest accelerations apply to the whole ejection sequence, including the escape system-air vehicle separation to seat-occupant separation phase, and can be found in 9.5.1.6 and 9.5.1.7.

9.5.3.1 Accelerations and angular velocities.

Standard: Seat accelerations are IAW the moderate limits within INFO PUB ASM 4048 Ed 2v2, Ejection Acceleration Limits for MDRC up to 450 KEAS. The MDRC limit may increase linearly from 1.0 at 450 KEAS to 1.7 at 600 KEAS.

The seat angular velocities are IAW the low limits in INFO PUB ASM 4048 Ed 2v2, Ejection Acceleration Limits, Short Duration Angular Velocity Limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: System-level dynamic ejection sled testing verifies the MDRC and angular velocity at ambient temperatures.

9.5.3.2 Limb flail.

Standard: Air vehicle escape systems ensure limbs do not flail. Legs do not move laterally beyond the sides of the seat. Rearward motion of the arms is limited to prevent injury.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the limb flail for all other anthropometry cases that will not be verified by lab testing.

Test.

(1) Lab test: System-level dynamic ejection sled testing verifies that the escape system prevents limb flail.

9.5.4 Recovery system deployment to touchdown.

The criteria in this section are applicable between recovery system deployment to just prior to occupant touchdown.

Recovery system deployment is defined as when extraction of the primary parachute from the parachute container begins.

Touchdown is defined as when the occupant physically touches the ground post-escape event.

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Check the source to verify that this is the current version before use.

9.5.4.1 Resultant acceleration.

Standard: The resultant chest acceleration does not exceed 20g for ejections above 8000 ft MSL. The resultant chest acceleration does not exceed 25g for ejections below 8000 ft MSL. The individual axis acceleration limits found in 9.5.1.6 also apply to this phase of ejection.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Validated simulations show chest accelerations do not exceed 20g for at altitudes greater than 8000 ft MSL.

Test.

(1) Lab test: Resultant chest accelerations are verified using full system-level dynamic ejection sled test with full-body manikins representing the intended anthropometric crew range across the escape envelope.

9.5.4.2 Parachute descent rate and steering.

Standard: Recovery parachute average descent rates do not exceed a vertical velocity of 24 ft/s at sea level on a standard day for the maximum applicable crewmember suspended weight. The recovery parachute provides a selectable gliding and steering capability that is optionally initiated by the ejected crewmember only. The recovery parachute limits horizontal velocity to less than 10 ft/s unless gliding has been selected by the crewmember. The parachute is automatically released upon crewmember entry into seawater.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) The automatic release capability is demonstrated in seawater conditions.

Test.

(1) Lab test: Steady-state descent velocity and steering is verified through full parachute system-level tests.

9.5.5 Escape path clearance.

Escape path clearance relates to the ability of the escape system, in conjunction with other air vehicle systems, to initiate and provide a clear path for the ejected crewmember. Proper sequencing is important to the escape path clearance because critical steps like jettisoning canopies or hatches, or fracturing canopies need to occur in a specific order. Adequate physical clearance between the ejection seat and the air vehicle is necessary so that the escape sequence is not disrupted.

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9.5.5.1 Escape path clearance system modes.

Standard: The escape path subsystem consists of a primary automatic mode, a backup automatic mode, and a manual mode. The failure of any of the three modes does not affect the performance of the remaining modes.

Systems incorporating a canopy or hatch jettison system, in primary automatic mode, incorporate a mechanism that prevents the escape system from colliding with the canopy or hatch before jettison.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) The escape system modes are verified by examination of engineering drawings or digital models and confirming:

a. The primary mode is initiated automatically through dual redundant paths when an ejection is initiated.
b. The system has a means to automatically clear the escape path when the primary system fails.
c. There is a means to manually initiate the escape path clearance system without initiating an ejection.
d. A mechanism is present in the design in canopy or hatch jettison systems to prevent collisions in primary automatic mode.

Analysis.

(1) Analysis of the trajectory of the canopy or hatch show that the delay timing provides clearance for the all occupant and seat configurations.

Test.

(1) Lab test: Breadboard testing of the mode selection and sequencing system shows that the system meets the timing and energy requirements for the operating environment.

9.5.5.2 Escape path clearance envelope.

Standard: The air vehicle escape path clearance subsystem ensures a clear escape path from the air vehicle for the crewmembers across the defined escape system envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the ejection seat escape path clearance envelope to ensure that there is no interference between the air vehicle, other occupants (if the air vehicle is multi-place), and debris that would prevent operation of the ejection system.

(2) Validated trajectory simulations show clearance of seat, occupant with the cockpit and air vehicle in the intended operating environments (e.g., temperature, altitude).

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Test.

(1) Lab test: Full system-level ejection sled tests are performed to verify escape path clearance.

9.5.5.3 Multi-crewmember escape path clearance.

Standard: For multi-crewmember air vehicle escape systems, escape path clearance, clearance between individual escape systems and occupants, and debris (e.g., hatches and canopies) is maintained after air vehicle separation throughout the escape system envelope for worst case crewmember size combinations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine escape system configuration and verify that the multi-crewmember escape systems incorporates a trajectory divergence function to prevent collisions between crewmembers.

(2) Examine the escape system configuration and verify that the sequencing logic and timing is maintained when automatic mode is selected.

Analysis.

(1) Validated dynamic trajectory simulations show that air vehicle escape path clearance is maintained for multi-crewmember escape systems throughout the escape system envelope.

Test.

(1) Lab test: Full multi-crewmember ejection sled testing shows that escape path clearance is maintained throughout for the escape system speed envelope for straight and level flight with worst case combinations of occupant size and weight.

9.5.5.4 Canopy and hatch jettison.

Standard: Jettisonable components (e.g., canopies and hatches) and other debris do not collide with the air vehicle, crewmembers, or the escape system within the escape system performance envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the system using a validated model and verify the trajectory of jettisonable components. Analysis shows a clear escape path for all roll, pitch, and yaw angles possible within the escape system performance envelope.

(2) Analysis shows that jettisonable components do not collide with an ejectee, escape system, or air vehicle during an escape event.

Test.

(1) Lab test: Test the escape system during full system-level ejection sled testing and verify that jettisonable components do not collide with the air vehicle, crewmembers, or escape system.

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9.5.5.5 Ejection seat air vehicle physical clearance.

Standard: There is no encroachment of a spherical volume with a 4.0 ft radius, centered about the ejected mass CG position, following this minimum clearance being attained for the first time, post air vehicle separation.

Ejection seat components (e.g., deployed stabilization system) maintain a minimum clearance of 12 inches from any air vehicle or canopy structure following this minimum clearance being attained for the first time.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Ejection testing will be performed to verify the trajectory of the ejection seat and ejection seat components.

9.5.5.6 Multi-place and inter-seat sequencing systems.

Standard: For systems that allow for one or a portion of the crewmembers to use the escape system independently, the ability for remaining crewmembers to subsequently use the escape system is not precluded. Inter-seat systems operate IAW the selected modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the air vehicle configuration and verify that the systems allow remaining crewmembers to use the escape system to leave the air vehicle.

Test.

(1) Lab test: Inter-seat sequencing system breadboard testing shows that system meets the defined modes, logic timing, and signal energy requirements.

(2) Lab test: For functions controlled by software, software testing shows that the system meets its defined logic and timing.

9.5.6 Fragment injury.

This section encompasses protection from penetrating or blunt force injury to the crewmembers during an ejection.

9.5.6.1 Penetrating injury.

Standard: Any fragments, particles, or debris (e.g., transparency or hatch fragments, blast debris from fracturing system) do not cause injury to crewmembers during an ejection from an air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) COMPUTERMAN, a DoW accredited fragment penetration injury model, is used to calculate the survivability based on the fragmentation testing. The overall survivability of the crewmembers calculated by COMPUTERMAN due to penetrating fragments is not less than 98%.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: System-level testing is accomplished to verify that crewmembers are not injured by fragments during an ejection from an air vehicle. The test is accomplished using a representative mockup of the air vehicle with manikins equipped with ballistic gel or foam. The ballistic gel or foam is calibrated so that the energy of any shrapnel that penetrates the manikins can be calculated.

9.5.6.2 Blunt force injury.

Standard: Any material that is loosed in the airstream during operation of the escape system (e.g., transparencies, hatches, fragments, blast debris) do not cause blunt force injury to the crewmember.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Full system-level dynamic ejection sled testing is accomplished to verify that debris from the ejection does not cause blunt force injury to the crewmember.

9.5.7 Escape system reliability.

Escape systems reliability relates to the ability of the escape system to reliably perform the appropriate functions as designed. The demonstrated reliability of escape systems is accomplished before first flight. Escape system reliability is a separate characteristic from the system performance attributes (e.g., how well the escape system protects the crewmembers over the escape system envelope).

9.5.7.1 Escape system design reliability.

Standard: The minimum design reliability of the air vehicle escape system is at least 98%. The design reliability of the escape system is calculated from the reliability data of the elements and components in the operating environment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Reliability analysis shows that the escape system meets the standard using design analysis and data for all escape system functional subsystems and components in the intended operating environment.

9.5.7.2 CAD/PAD and propulsion subsystem reliability.

Standard: Cartridge activated devices/Pyrotechnic actuated devices and propulsion component reliability are not less than 99.9% at the 90% confidence level in the intended operating environment.

Cartridge activated devices are IAW MIL-C-83124, Requirements section and Quality Assurance Provisions section.

Pyrotechnic actuated devices are IAW MIL-C-83125, Requirements section and Quality Assurance Provisions section.

Rocket catapults, propellent actuated catapults, and rocket motors are IAW MIL-P-83126, Requirements section and Quality Assurance Provisions section.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Reliability for all CAD/PAD, and propulsion components are verified IAW the standard.

9.5.7.3 Escape system demonstrated reliability.

Standard: Ejection seats (including encapsulated seats) have a minimum demonstrated reliability of at least 90% with a 90% confidence interval at the system level. Air vehicle integration shows a minimum demonstrated reliability of at least 75% with 90% confidence interval.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: For new systems, full system-level ejection sled testing with a minimum of 22 consecutive system-level ejection seat tests with zero failures of critical functions applicable to the final ejection seat configuration.

For air vehicle integration of qualified systems, a minimum of eight consecutive system-level tests completed with all components and subsystems (e.g., ejection seat, escape path clearance system, helmet, life support system, and AFE) present and representative of the final configuration without any failures of critical functions.

9.5.8 Backup through-canopy ejection.

Backup through-canopy injury criteria apply during initiation to seat-air-vehicle separation at all speeds from 0-600 KEAS where a backup ejection mode is required (e.g., a through-canopy ejection using seat canopy breakers). The criteria apply with and without helmet mounted

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devices (e.g., night vision goggles and helmet mounted cueing systems). The criteria apply to systems with and without deployable head and neck protection devices and whether the devices are or are not in contact with the pilots' head or helmet.

9.5.8.1 DRIz - backup through-canopy ejection.

Standard: The seat pan acceleration time history generated in system-level ejection sled or in-flight tests at ambient conditions, do not yield a DRIz value in excess of 22.0 IAW the limits within INFO PUB ASM 4048 Ed 2v2, Ejection Acceleration Limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Data from full system-level dynamic ejection sled testing is analyzed to show compliance with the standard.

Test.

(1) Lab test: Full system-level dynamic ejection sled testing is used to show compliance with the standard.

9.5.8.2 Neck injury - backup through-canopy ejection.

Standard: The maximum neck injury risk during a through-canopy backup mode ejection is 15%.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Data is collected from full system-level dynamic ejection sled testing using instrumented manikins.

9.5.8.3 Probability of concussion - backup through-canopy ejection.

Standard: Standard: The maximum P concussion during a through-canopy backup mode ejection is 15% when calculated using the probability of concussion equation (Rowson and Duma).

$$P_{concussion} = \frac{1}{1 + e^{-(-10.2+.0433a+0.000873\alpha-0.00000092a\alpha)}}$$

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Acceleration and rate data is used to calculate P concussion. Calculations will use the maximum angular resultant acceleration within a ±30ms time window centered around each resultant linear acceleration at each recorded time step.

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Test.

(1) Lab test: Data is collected from full system-level dynamic ejection sled testing using instrumented manikins.

9.6 Emergency egress and bailout.

This section includes criteria related to the means for occupants to leave the air vehicle during in-flight, water, and ground emergencies. This section includes exits, explosive components, emergency ground ditching and water egress, and occupant entrapment. Provisions for bailout using a parachute are also addressed.

All night lighting conditions for egress demonstrations and tests are light levels exterior to the air vehicle no greater than 0.3 fc IAW 14 CFR 25, Appendix J subparagraphs (a) and (d).

9.6.1 Air vehicle exits.

This section relates to provisions for all openings, exits, doors, and hatches that allow occupants to enter and leave an air vehicle.

9.6.1.1 Emergency ground egress provisions.

Standard: Emergency exits are assessed IAW the following:

a. Each occupied compartment has at least one emergency egress exit, and areas with seating restraints have emergency egress exits to the exterior of the air vehicle.
b. Exits are operable from inside and outside the compartments for interior doors, and from inside and outside the air vehicle for exterior exits.
c. Emergency egress exits are located away from hazardous areas of the air vehicle (e.g., propellers, or jet engine inlets and outlets) while still meeting egress time IAW the timed egress criteria.
d. Emergency exits are not located in areas that will be blocked after an emergency gear up landing (e.g., doors can be opened and closed with landing gear up and locked).
e. Emergency exits are large enough to allow the full range of anthropometric sizes and worst-case mission equipment (if applicable) to egress within the time allowed IAW 9.6.3.5, 9.6.3.6, 9.6.3.7, or 9.6.3.8.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of engineering drawings, digital models, or high-fidelity mockups verifies that each occupied compartment has an exit.

(2) Examination of engineering drawings, digital models, or high-fidelity mockups verifies that doors are operable IAW the standard.

(3) Examination of engineering drawings, digital models, or high-fidelity mockups verifies that exit locations are outside of documented hazardous areas (e.g., keep-out zones around engines or propellers) of the air vehicle.

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(4) Examination of engineering drawings, digital models, or high-fidelity mockups show that emergency exits can be operated with the landing gear up.

Demonstration.

(1) Demonstrate egress times IAW 9.6.3.5, 9.6.3.6, 9.6.3.7, or 9.6.3.8.

9.6.1.2 Bailout hatches and doors.

Standard: Each air vehicle with bailout capability has at least one exit that can be operated by an occupant from inside the air vehicle under the anticipated operational environments. Egress exits and provisions allow for clearance away from hazardous areas of the air vehicle (e.g., near propellers, or jet engine inlets or outlets, or air vehicle structure) during bailout.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of engineering drawings or digital models verify that an air vehicle with a bailout capability has an external exit with a door that can be opened internally.

(2) Examination of the air vehicle configuration with a bailout capability has an external exit with a door that can be opened internally.

(3) Examination of engineering drawings or digital models verifies that exit locations and bailout provisions are away from hazardous areas around the air vehicle.

Demonstration.

(1) Verify the ability to operate doors internally on a representative mockup or air vehicle under anticipated operational environments by human subjects representing the intended anthropometric population.

9.6.1.3 Exit weight and actuation marking.

Standard: The means of opening and weight of each removable emergency hatch or door is clearly marked on the hatch or door.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of engineering drawings or models verifies that each removable emergency hatch or door is clearly marked with its means of opening and weight.

9.6.1.4 Exit weight and actuation.

Standard: The means of opening and weight of each removable emergency hatch or door can be accomplished by the entire intended anthropometric population.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Test.

(1) Ground or Lab test: Testing in a representative mock-up or on the air vehicle shows that emergency egress doors are operable from inside and outside the air vehicle by the full expected range of user anthropometric reach and strength values in MIL-STD-1472, Anthropometric Accommodation paragraph and MIL-STD-1472, Static Muscle Strength table in Push and Pull Forces paragraphs.

9.6.1.5 Exit locking mechanisms.

Standard: All exits are lockable by a crewmember trained to do so. All exits are uncomplicated to open so that no training is required for operation. All exits stay locked and closed in compartments when the air vehicle is inflight unless mission needs allow the opening and use of exits in flight. The design of all exits provide clear indication that the doors are fully locked in flight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of engineering drawings or digital models verifies that there is clear indication of a locked or unlocked condition.

(2) Examination of the physical air vehicle configuration verifies that there is clear indication of a locked or unlocked condition.

Demonstration.

(1) Trained crewmembers demonstrate that exits are lockable, simple to open, and stay locked in flight when not opened for mission needs.

9.6.1.6 Parachute bailout air vehicle clearance.

Standard: Air vehicles with bailout capability have a clear bailout path. Jettisonable components and other debris do not impact the air vehicle or create a collision potential with other personnel bailing out of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of engineering drawings, digital models, or the physical air vehicle configuration verifies bailout exit clearance dimensions.

Demonstration.

(1) Verify by ground demonstration that the intended anthropometric population of bailout parachute users can effectively clear the bailout exit.

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9.6.1.7 Bailout parachute resultant thoracic accelerations.

Standard: The maximum resultant accelerations measured at the chest during bailout parachute opening do not exceed 15g (resultant vector sum).

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Chest accelerations are verified through full system-level dynamic test with full-body manikins representing the intended anthropometric population across the bailout envelope.

(2) Flight test: Chest accelerations are verified through full system-level dynamic test with human subjects representing the intended anthropometric population across the bailout envelope.

9.6.1.8 Bailout parachute descent velocity.

Standard: Bailout parachute average descent rates do not exceed a vertical velocity of 24 ft/s at sea level on a standard day for the maximum applicable suspended weight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Flight test: Steady-state descent velocity is verified through full parachute system-level tests.

9.6.1.9 Bailout parachute demonstrated reliability.

Standard: Bailout parachutes have a minimum demonstrated reliability of 98% at the 90% confidence interval.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Demonstrated reliability is verified by full system-level live jumps.

9.6.2 Explosive components.

Standard: Explosive components for the purpose of removing doors, hatches, and canopies to allow for egress do not cause blast or penetrating injury to any personnel both internal and external to the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis will be used to verify that blast injuries are not incurred by personnel near egress exits upon initiation.

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Test.

(1) Lab test: Test will be used to verify that blast injuries are not incurred by personnel near egress exits upon initiation.

(2) Lab test: Test will be used to verify that penetrating injuries are not incurred by personnel near egress exits upon initiation.

9.6.3 Emergency ground, ditching, and water egress.

Ground emergency egress includes egress provisions for air vehicles with and without escape systems (e.g., ejection seats). Types of applicable air vehicles include multi-place, tactical, transport, overwater transport, and rotary-wing.

9.6.3.1 Emergency marking and instructions.

Standard: Emergency exits are indicated with markings and instructions IAW 14 CFR 25.811 Emergency Exit Marking.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine drawings or digital models and verify that emergency exit marking and instructions are IAW the standard.

(2) Examine the physical air vehicle and verify that the emergency exit instructions and markings are IAW the standard.

9.6.3.2 Emergency ground evacuation provisions.

Standard: Emergency ground evacuation provisions are provided as follows:

a. Provide a means to descend to the ground without injury for all occupants.
b. A means to descend is provided for each exit that is not over the wing and more than 6 ft above the ground when the air vehicle is on level ground with landing gear down.
c. For exits opening to wing areas, provisions are incorporated to assist occupants from the wing surface to ground level.

It is acceptable to require training for crewmembers. For passengers, no training is necessary to descend without injury.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of drawings or digital models verifies that all exits have a means for occupant descent.

(2) Examination of drawings or digital models verifies that exits more than 6 ft above the ground and non-overwing exits provide a means for occupant descent.

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(3) Examination of procedures verifies that descent devices can be used without passenger training but with the assistance of crewmembers.

Demonstration.

(1) For crewmembers, demonstrate the ability to exit and descend to the ground during an emergency egress test. Demonstrations are accomplished using personnel representative of the intended anthropometric population in representative equipment configurations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) For passengers, demonstrate the ability to exit and descend to the ground during an emergency egress test. Demonstrations are accomplished using untrained personnel representative of the intended anthropometric population.

9.6.3.3 Ground, ditching, or emergency egress assist devices.

Standard: Ground, ditching, or emergency egress assist devices (e.g., doors, slides, descent reels, life rafts, and handles) are usable by the intended anthropometric range. The egress assist devices meet the ground egress time requirements in relevant operating environments. The devices are protected against inadvertent actuation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of drawings or digital models verifies that the egress assist devices are usable by the number and intended anthropometric population of occupants.

Demonstration.

(1) Verify that all ground, ditching, or emergency egress devices can be used by participants representative of the intended anthropometric population.

(2) Verify that deployment handles and actuators are designed to prevent inadvertent actuation during normal operations and incidental contact by participants representative of the intended anthropometric population.

(3) Verify the ground egress time is met when using the ground emergency descent devices by participants representative of the intended anthropometric range IAW 9.6.3.5, 9.6.3.6, 9.6.3.7, or 9.6.3.8.

(4) Emergency ground egress demonstrations show that handles and actuators are accessible and operable by participants representative intended anthropometric range. Operation does not cause injury that would prevent egress.

9.6.3.4 Onboard emergency egress equipment.

Standard: Onboard emergency egress equipment (e.g., crash axe, canopy penetrator) are capable of creating necessary exits in transparencies and designated crewmember compartment surfaces. Onboard egress equipment should be in compartments occupied during takeoff and landing.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of drawings or digital mockups should verify onboard exit creating device provisions.

(2) Examination of the physical air vehicle should show that provisions exist for onboard exit creating devices.

Demonstration.

(1) Demonstrate emergency egress and rescue equipment to show that exits can be created in transparencies and designated crewmember compartments with onboard emergency egress equipment.

9.6.3.5 Escape exits and routes – tactical air vehicles.

Standard: Crewmembers can egress from tactical air vehicles within 30 seconds under day and night lighting conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground or Lab test: Show that the egress time meets requirements under both day and night lighting conditions. Testing is accomplished using participants representative of the intended anthropometric population wearing required worst-case crewmember equipment. Tests are accomplished using an air vehicle or high-fidelity mockup.

9.6.3.6 Escape exits and routes – multi-place and cargo air vehicles.

Standard: Multi-place and cargo air vehicle exits and sizes have twice the capacity for the maximum number of crewmembers to egress in the specified time. The occupied areas allow for complete abandonment within the required time during ground emergency egress or ditching of the air vehicle with the landing gear extended as well as retracted.

Egress times are as follows:

a. For overwater transport air vehicles, all crewmembers can egress within 30 seconds using only half the exits under day and night lighting conditions.
b. For all other multi-place and cargo air vehicle configurations, except for air vehicles with overwater missions, all crewmembers can egress within 60 seconds using only half the exits under day and night lighting conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the air vehicle design and show that crewmembers can egress the air vehicle from the combination of a selection of half the exits which results in the most restrictive air vehicle

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configuration. The analysis includes crewmembers and passengers from across the intended anthropometric range.

(2) Analysis shows that egress is not impeded with landing gear retracted.

Test.

(1) Ground or Lab test: Tests from the air vehicle or high-fidelity mockup show that the equipped crewmembers and passengers can egress the air vehicle from that combination of a selection of half the exits which results in the most restrictive configuration. Tests include participants representative of the intended anthropometric range. The test includes both day and night conditions. The air vehicle or high-fidelity mockup is configured to include representative cargo and unstowed equipment.

9.6.3.7 Escape exits and routes – passenger transport air vehicles.

Standard: All crewmembers and passengers can egress from the air vehicle within the specified time using only half the exits under day and night lighting conditions. Transport air vehicle exits and sizes have twice the capacity for the maximum number of crewmembers and passengers to egress in the specified time. The crewmember and passenger areas of the air vehicle allow for complete abandonment in the specified time during ground egress or ditching of the air vehicle, with the landing gear extended as well as retracted, considering the possibility of the air vehicle being on fire, and at maximum seating capacity.

For transport air vehicles the following egress times apply:

a. Overwater transport air vehicles: all crewmembers are able to egress within 30 seconds.
b. Air vehicles carrying 25 or fewer crewmembers and passengers: all crewmembers and passengers can egress the air vehicle within 60 seconds using only half the exits.
c. Air vehicles carrying more than 25 crewmembers and passengers: all crewmembers and passengers can egress the air vehicle within 90 seconds using only half the exits.

Tests are performed under day and night conditions.

The timed evacuation criteria do not apply to patients strapped to a litter during aeromedical evacuation missions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the air vehicle design and show that the escape exits and routes can be egressed by crewmembers and passengers with half the exits blocked. Crewmembers and passengers are outfitted with the most restrictive equipment configurations. Analysis includes crewmembers and passengers across the intended anthropometric range.

(2) Analysis shows that egress is not impeded when the landing gear are retracted.

Test.

(1) Ground or Lab test: Test the air vehicle or a high-fidelity mockup and verify that the equipped crewmembers and passengers can egress the air vehicle from that combination of a random selection of half the exits which results in the most restrictive configuration. Tests

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include participants across the intended user anthropometric range. Test both day and night conditions. The air vehicle or high-fidelity mockup is configured to include representative cargo and un-stowed equipment. Participants do not have prior practice or rehearsal for the test.

9.6.3.8 Escape exits and routes – rotary-wing.

Standard: All crewmembers and passengers can egress from the air vehicle within the specified time using only half the exits under day and night lighting conditions.

Rotary-wing exits and sizes have twice the capacity for the maximum number of occupants to egress in the specified time. The crewmember and passenger areas allow for complete abandonment in the specified time during ground egress or ditching of the air vehicle, with the landing gear extended as well as retracted, considering the possibility of the air vehicle being on fire, and at maximum seating capacity.

For air vehicles with non-crashworthy fuel cells: All crewmembers can egress in 20 seconds using only one-half the exits.

For air vehicles with crashworthy fuel cells: All crewmembers can egress in 30 seconds using only one-half the exits.

For air vehicles that carry passengers: All crewmembers and passengers can egress in 60 seconds using only one-half the exits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review analysis that shows that that the equipped crewmember and passengers can egress the air vehicle from that combination of a selection of half the exits which results in the most restrictive configuration. Analysis includes crewmembers and passengers across the intended anthropometric range.

Test.

(1) Ground or Lab test: Testing verifies that the appropriately equipped (mission representative) crewmembers and passengers can egress the air vehicle from that combination of a random selection of half the exits which results in the most restrictive configuration. Tests include participants across the intended user anthropometric range. An emergency egress test from the air vehicle or a high-fidelity mockup is used to verify egress capability and time required under both day and night lighting conditions.

9.6.3.9 Emergency egress exits and routes - documentation.

Standard: Technical manuals incorporate crewmember emergency egress procedures. Crewmembers are trained in emergency egress procedures. Passenger briefings are provided to crewmembers, and passenger emergency egress instructions are readily available to passengers.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examine relevant documentation (e.g., technical manuals, training manuals, and training syllabus) and verify egress and rescue procedures for crewmembers, passengers, and ground rescue personnel.

9.6.4 Occupant entrapment.

Standard: Internal air vehicle doors are designed to prevent any person from being trapped inside any internal occupiable compartments (e.g., lavatories, gallies, or crew rest compartments). Doors can be actuated by the intended anthropometric range. Locking and latching mechanisms can be unlocked and actuated from both the inside and outside of the compartment without the use of special tools.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine engineering drawings or digital models of occupiable compartments with latches and locking mechanisms and verify that the latching and locking mechanisms can be actuated from inside and outside the space without special tools.

Demonstration.

(1) Demonstrate on a representative mockup or on an air vehicle that doors can be opened and unlocked from inside and outside the space without special tools.

9.7 Crash survivability.

This section includes seating system load capabilities, restraint systems, occupant strike envelope, post-crash operational exits, items of mass, stowage compartment structure, occupiable volume reduction in rotary-wing air vehicles, and mobile crewmember restraint.

9.7.1 Seating system crash load capabilities.

Commercial fixed-wing and rotary-wing seats are seats that have been adapted for military use and are installed in fixed-wing and rotary-wing air vehicles. Choose the closest type in SAE 8049D, Seat Type Categories table (e.g., Transport, Rotorcraft, or Normal, Utility, and Commuter) for applicable criteria limits.

Military rotary-wing seats are seats developed primarily for military use (e.g., V-22, HH-60, H-53).

Palletized seating is seating attached to a pallet that can be installed or removed from an air vehicle (e.g., C-17). Palletized seats meet the standards for seats that would be installed in the associated air vehicle. Choose the closest type in SAE 8049D, Seat Type Categories table (Transport or Rotorcraft) for applicable criteria limits.

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9.7.1.1 Fixed-wing and rotary-wing seats.

Standard: The seating, restraint system, and attachment to the air vehicle restrain occupants and maintain structural integrity during static and dynamic loading. The seating and restraint system are designed IAW SAE AS8049D, Strength section.

The restraint system prevents “submarining” or translation of the pelvis and lower part of the body under the lap belt.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Test the ultimate static loads of the seat, restraint system, and attachment to the associated air vehicle structure considering a nude occupant weight not less than 250 lbs. Testing and verification is accomplished IAW SAE AS8049D, Static Qualification Tests section and Static Test – Pass/Fail Criteria section.

(2) Lab test: Test the seat and restraint system protection capability under dynamic conditions. Test the seat and restraint with associated air vehicle structure and a seated occupant with an ATD selected IAW SAE 8049D, Occupant Simulation section. Dynamic tests and verification are accomplished IAW SAE AS8049D, Dynamic Qualification Testing section and Dynamic Impact Test Pass/Fail Criteria section.

9.7.1.2 Energy attenuating seats.

Standard: The seating and restraint system restrains occupants during static and dynamic loading. The seating and restraint system, including structural attachment to the air vehicle, withstands static and dynamic loads defined in MIL-STD-1290, Pilot and Copilot paragraph and Cabin paragraph, with a maximum weight nude occupant of the representative anthropometric range not less than 250 lbs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: Static loads: Test that the seat and restraint system with associated air vehicle structure restrains seated occupants during static loading. Verify the static load capabilities of the seat and restraint system with associated air vehicle structure and a seated occupant per MIL-STD-1290, Pilot and Copilot paragraph and Cabin paragraph.

(2) Lab test: Dynamic loads: Test that the seat and restraint system with associated air vehicle structure restrains seated occupants during dynamic loading. Verify the dynamic load capabilities of the seat and restraint system with associated air vehicle structure and a seated occupant per MIL-STD-1290, Pilot and Copilot paragraph and Cabin paragraph.

9.7.1.3 Energy attenuating seats – stroke clearance envelope.

Standard: Ensure that the seat installation in the air vehicle provides the stroke volume necessary to allow energy attenuating functionality under air vehicle crash loads.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the dynamic seat test data to verify that the air vehicle design does not obstruct the seat stroke envelope.

Test.

(1) Lab test: Test the seat stroke volume via dynamic seat testing IAW 9.7.1.1 or 9.7.1.2, depending on air vehicle type.

9.7.1.4 Ejection seat crash loads.

Standard: The seat, associated air vehicle structure, and crewmember restraint system withstand air vehicle crash loads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Lab test: The seat and restraint system with associated air vehicle structure meet the standard with a seated occupant. Manikins representative of the expected crewmember anthropometric population with clothing and AFE are restrained in the seat during the test.

9.7.2 Restraint system release point – non-escape system.

Standard: All restraint systems have a single point release for the restraint system of each occupant. The release can be disengaged by the seat occupant with a single gloved hand in normal and inverted orientations. The release point is designed to prevent inadvertent actuation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of engineering drawings verifies that each seat and restraint system has a single point release system.

Demonstration.

(1) Demonstrate that occupants representing the intended anthropometric range can use the single point release while wearing all AFE.

(2) Demonstrate that the release does not inadvertently actuate under the anticipated environmental and operational conditions.

Test.

(1) Lab test: Testing shows that the single point release can be activated and meet MIL-STD-1472, Arm, Hand, and Thumb-finger Controls paragraph post-dynamic test for the intended user population.

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(2) Lab test: Verify that the single point release is operable in an inverted position, with the weight of the occupant supported by the restraint system.

(3) Lab test: Verify that the single point release is functional under all anticipated environmental and operational conditions.

9.7.3 Occupant strike envelope.

Standard: Objects in the occupant strike envelope do not cause injuries that would incapacitate or prevent egress from the air vehicle for restrained and seated occupants during design crash loads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) As an alternative to Test MoC (1), compliance can be shown by an accredited government model to evaluate the occupant strike envelope.

Test.

(1) Lab test: As an alternative to Analysis MoC (1), conduct test to show that objects in the occupant strike envelope do not cause injury that would incapacitate or prevent egress for the intended anthropometric population from all restrained seated positions during design crash loads.

9.7.4 Post-crash operational exits.

Standard: Air vehicle exits used for ground egress can be opened by occupants after exposure to the design crash loads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the air vehicle structure using computational analysis (e.g., FEA) and show that exits can be opened by occupants after experiencing design crash loads.

9.7.5 Loose object restraint in occupied compartments.

Standard: Restraint or stowage is provided for all loose objects (e.g., tablets, other portable electronic devices, beverage cart in a galley) in all occupied compartments. Restraint and stowage withstand air vehicle flight and crash loads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) As an alternative to Test MoC (1), analysis verifies that object restraint or stowage withstands flight and crash loads.

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Test.

(1) Lab test: As an alternative to Analysis MoC (1), testing verifies that object restraint or stowage withstands flight and crash loads.

9.7.6 Stowage compartment structure.

Standard: Stowage compartments contain the maximum weight of the contents under the air vehicle flight and crash loads. The contents do not shift and become a hazard to the air vehicle, become a projectile under flight and crash loads, or prevent egress during an emergency.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) As an alternative to Test MoC (1), structural analysis verifies that the stowage compartment retains the maximum mass under the specified flight and crash loads.

Test.

(1) Lab test: As an alternative to Analysis MoC (1), testing verifies that the stowage compartment retains the maximum mass under the specified flight and crash loads.

9.7.7 Occupiable volume reduction – rotary-wing.

Standard: The rotary-wing structure does not impinge on occupants when subjected to the design crash loads. The structural volume reduction is 15% or less and does not displace or strike occupants. High mass objects (e.g., engines, transmissions, fuel cells, rotor masts, rotor hubs) are designed to prevent their displacement into the occupied spaces.

High mass objects do not displace into the occupied volume during the following impact conditions:

a. Rollover in the air vehicle pitch or roll axes.
b. Main rotor obstacle strike within the outer 10% of the blade span assuming the obstacle is an 8-inch cylinder.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) As an alternative to Test MoC (1), structural analysis (e.g., FEA) verifies that the design meets occupant volume reduction criteria.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Lab test: As an alternative to Analysis MoC (1), crash load testing verifies that the design meets the occupant volume reduction criteria.

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9.7.8 Mobile crewmember restraint system.

Standard: For crewmembers that move throughout the air vehicle during flight with open doors or ramps, a mobile restraint system provides crash retention and fall protection. The mobile restraint system includes a torso harness and adjustable lanyard designed to distribute loads on the user during dynamic events. The mobile restraint system withstands a 5,000-lb static load. The mobile restraint system includes an emergency single-handed release that permits rapid release.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the results of the drop testing and verify that no portion of the harness or restraint contacts the neck of the ATD or body block.

(2) Examine the results of the drop testing and verify that the system retains the ATD or body block. Verify that adjusters do not slip more than 1 inch, and the emergency release functions after exposure to loading.

Test.

(1) Lab test: Verification of mobile crewmember restraint performance includes static testing, drop testing, and dynamic testing utilizing body blocks or ATDs in representative initial conditions.

(2) Lab test: Drop tests evaluate falls to full-lanyard length and dynamic crash tests verify the capability of the restraint to retain an occupant during velocity changes of at least 33 ft/s.

9.8 Transparency integration.

This section includes the design requirements of the transparency system related to bird strike, structural, and thermal capabilities. Transparency systems may consist of a flat transparency window, windscreen, or a canopy system.

The AW standard for transparency bird strike is determined by threat exposure analysis. The test airspeed is calculated from the air vehicle design airspeed and historical bird strike data with a 4-lb bird weight. If other methods are used to determine the test airspeed, they should have established precedence (e.g., USAF EN-SB-12-002 Methodology to Establish Bird Strike Design Criteria).

9.8.1 Transparency system bird strike - manned.

Standard: Air vehicle transparencies and all supporting structure withstands the impact of a 4-lb bird at an impact speed, in knots, without injurious deformation, penetration, or air vehicle loss. The transparency AW bird impact speed is determined for the air vehicle by a bird strike threat exposure. Impact speed calculations assume a 4-lb bird, 7000 ft AGL limit, and a probability of loss less than $1x10^{-7}$. The impact does not result in loss of the air vehicle due to mechanical failure, penetration and spalling into the cockpit space, secondary projectiles (e.g., HUDs, rear-view mirrors) and deflection or object contact with the crewmember.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Determine the transparency AW bird impact speed by accomplishing a transparency bird strike threat exposure analysis. Assume a 4-lb bird, 7000 ft AGL limit, and a probability of loss less than $1X10^{-7}$.

Test.

(1) Lab test: Conduct full-scale bird strike tests, IAW ASTM F330-21, at the transparency AW bird strike impact speed. Verify that the transparency and all supporting frame structure withstands impact without penetration, spalling, secondary projectiles and deformation impact to the crewmember, or loss of the air vehicle.

9.8.2 Transparency system structural and thermal capabilities.

Standard: Transparency systems do not fail when exposed to maximum thermal and structural load stresses experienced in operational conditions. The maximum cabin pressure differential loads, combined with critical aerodynamic pressure and temperature effects, are considered and are consistent with the structural design requirements of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct a structural analysis to verify that the maximum thermal and structural load stresses are within material allowables.

Test.

(1) Lab test: Coupon or full-scale transparency tests verify that material properties are consistent with the maximum thermal and structural load capabilities.

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10. AERIAL REFUELING

Aerial refueling is the process of passing fuel from one air vehicle to another in flight or the unique pairing of two air vehicles for the purpose of passing fuel in flight. Equipment involved in AR includes boom, receptacle, drogue, and probe AR systems, and the interfaces with other air vehicle systems. Air vehicle systems and components, along with the AR system(s) involved in conducting AR, are also included in AR assessment.

10.1 Aerial refueling compatibility.

To facilitate AR, air vehicles are physically, aerodynamically, mechanically, electrically, and operationally compatible with targeted air vehicles.

10.1.1 Physical compatibility.

The AR subsystem interface is designed to permit successful engagement and disengagement procedures with the interface of the targeted tanker/receiver AR subsystem without impacting surrounding structure, FOD generation, interference or binding, or damage that would prevent successful AR operations.

10.1.1.1 Interface dimensions.

Standard: The AR system interface meets dimensional requirements to enable engagement and disengagement procedures with the interface of the corresponding tanker/receiver AR subsystem without interference, binding, or damage that would prevent AR operations.

a. Boom Tanker. Boom nozzle dimensions conform to ATP-3.3.4.5, Nozzle Dimensions paragraph, and allow connection with targeted receiver receptacle interface.
b. Drogue Tanker. Reception coupling dimensions meet ATP-3.3.4.6, Mating Dimensions paragraph, and allow connection with targeted receiver probe interface.
c. Receptacle Receiver. Receptacle dimensions meet ATP-3.3.4.5, Receptacle Dimensions paragraph, and allow connection with targeted tanker boom interface.
d. Probe Receiver. Probe nozzle dimensions conform to ATP-3.3.4.6, Mating Dimensions paragraph, and allow connection with targeted tanker reception coupling interface.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of drawings, specifications, and installed subsystem verifies boom nozzle conforms to ATP-3.3.4.5, Nozzle Dimensions paragraph.
b. Drogue Tanker. Examination of drawings, specifications, and installed subsystem verifies reception coupling conforms to ATP-3.3.4.6, Mating Dimensions paragraph.
c. Receptacle Receiver. Examination of drawings, specifications, and installed subsystem verifies receptacle conforms to ATP-3.3.4.5, Receptacle Dimensions paragraph.
d. Probe Receiver. Examination of drawings, specifications, and installed subsystem verifies probe nozzle conforms to ATP-3.3.4.6, Mating Dimensions paragraph.

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Demonstration.

(1) Verify the standard by the following:

a. Boom Tanker.
   1. Boom nozzle engagement and disengagement with targeted receiver receptacle interface demonstrated in laboratory environment.
   2. Endurance demonstration in a laboratory environment verifies boom nozzle is capable of repeated contacts and disconnects with targeted receiver receptacle without degradation for a statistically relevant sample size defined by the MTBF of the boom system or generation of FOD.
   3. Ground demonstration verifies boom nozzle engagement and disengagement with multiple receptacle-equipped receiver air vehicles without interference, binding, or damage.

b. Drogue Tanker.
   1. Reception coupling engagement and disengagement with targeted receiver probe nozzle without interference is demonstrated in laboratory environment.
   2. Laboratory endurance demonstration of tanker reception coupling with targeted receiver probe nozzle verifies reception coupling can withstand repeated contacts and disconnects without degradation for statistically relevant sample sizes defined by the MTBFs of the probe and drogue systems or generation of FOD and without creating damage that could cause the drogue to become unstable in flight.
   3. Ground demonstration verifies reception coupling engagement and disengagement with multiple probe-equipped receiver air vehicles without interference, damage, or hang-ups.

c. Receptacle Receiver.
   1. Receptacle system permits boom nozzle engagement and disengagement without interference or damage in laboratory environment.
   2. Endurance testing in a laboratory environment of receptacle system demonstrates repeated targeted tanker boom nozzle contacts and disconnects without degradation or generation of FOD.
   3. Air vehicle ground demonstration verifies engagement and disengagement of targeted tanker boom nozzle without interference, binding, or damage.

d. Probe Receiver.
   1. Probe nozzle engagement and disengagement with targeted tanker reception coupling without interference, damage, or FOD generation is demonstrated in laboratory environment.
   2. Air vehicle ground demonstration verifies probe nozzle engagement and disengagement with targeted tanker reception coupling without interference, damage, or hang-ups.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Verify the standard by the following:

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a. Boom Tanker. Flight testing with targeted receptacle-equipped receivers verifies successful engagement and disengagement without interference, nozzle binding, or unpredictable boom motion upon disconnect. Required test points include worst-case combinations of tanker and receiver gross weights and CGs, at all extents of the boom disconnect envelope, wet and dry contacts, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions, straight and level flight, in turns, and while tobogganing, and tension disconnects.
b. Drogue Tanker. Flight testing with targeted probe-equipped receivers verifies successful engagement and disengagement without interference or damage. Required test points include straight and level flight, in turns, wet and dry contacts, receiver closure rates of up to 10 ft/s, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions.
c. Receptacle Receiver. Flight testing with targeted boom-equipped tankers verifies successful engagement and disengagement without interference, nozzle binding or unpredictable boom motion upon disconnect. Required test points include worst-case combinations of tanker and receiver gross weights and CGs, at all extents of the boom disconnect envelope, wet and dry contacts, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions, straight and level flight, in turns, and while tobogganing and tension disconnects.
d. Probe Receiver. Flight testing with targeted drogue-equipped tankers verifies successful engagement and disengagement without interference or damage. Required test points include wet and dry contacts, receiver closure rates of up to 10 ft/s, minimum Mach, maximum Mach, minimum dynamic pressure, and maximum dynamic pressure conditions.

10.1.1.2 Physical articulation.

Standard: The AR system has features to accomplish and maintain connection when mating systems are not aligned.

a. Boom Tanker. The boom nozzle deflects angularly up to a 60-degree cone angle about the longitudinal axis of the boom and returns to within a 1-degree cone angle after deflection when the nozzle is not in contact with a receiver. The boom nozzle can rotate axially clockwise and counterclockwise at least 20 degrees and return to within 3 degrees of the 0-degree position after rotation when the nozzle is not in contact with a receiver. The nozzle has the capability to deflect and rotate and then return to neutral throughout the boom deployed envelope (airspeed, Mach, altitude, pitch, azimuth or roll, and boom wet (defined as 10 psig fuel pressure, which represents the worst-case fuel head pressure at maximum boom extension and lower elevation limit without fuel pumping pressure) or dry).
b. Drogue Tanker. Reception coupling ball joint deflects within a 45-degree or greater cone angle and permits 360 degrees of axial rotation about the axial centerline of the coupling body.
c. Receptacle Receiver. Receptacle bore axis installation angle enables alignment at contact with boom nozzle, in the center of the contact envelope, for all targeted tanker air vehicles. The receiver receptacle bore axis installation angle with respect to the receiver air vehicle waterline is calculated by using the following equation:

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$$X = 30 - \frac{A + B}{2} + \frac{C + D}{2}$$

A = maximum receiver air vehicle pitch angle
B = minimum receiver air vehicle pitch angle
C = maximum tanker air vehicle pitch angle
D = minimum tanker air vehicle pitch angle

d. Probe Receiver. Probe nozzle tip deflects up to 22.5 degrees off-center without damage and can deflect until the tip contacts the sliding sleeve lip, without breaking.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of drawings verifies boom nozzle and ball joint design allows for 60 degrees of angular deflection and 20 degrees of rotation axially with appropriate components to facilitate a return to neutral position.
b. Drogue Tanker. Examination of drawings verifies reception coupling ball joint design allows for required deflection and axial rotation.
c. Receptacle Receiver. Examination of drawings and air vehicle calculated AOAs verifies receptacle installation angle adheres to required formula.
d. Probe Receiver. Examination of drawings verifies probe nozzle design is capable of 22.5 degrees of deflection.

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker. Laboratory testing verifies the boom nozzle can articulate in a 60-degree cone about the boom longitudinal axis.
b. Drogue Tanker. Laboratory testing verifies the reception coupling deflects angularly to 45 degrees or greater and permits 360 degrees of axial rotation.
c. Receptacle Receiver. Laboratory testing with targeted tanker boom nozzle and structure simulating the center of the contact envelope, verifies receptacle installation angle alignment.
d. Probe Receiver. Laboratory testing verifies probe nozzle tip deflects up to 22.5 degrees without damage and deflects until the tip contacts the sliding sleeve lip, without breaking.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the standard by the following:

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a. Boom Tanker. Flight testing with targeted receptacle-equipped receivers results in no damage, breakage, or inadvertent disconnects due to inability of the boom nozzle to articulate as needed.
b. Drogue Tanker. Flight testing with targeted probe-equipped receivers results in no damage, breakage, or inadvertent disconnects due to inability of drogue coupling to articulate as needed.
c. Receptacle Receiver. Flight testing with targeted boom-equipped tanker results in no damage, breakage, or inadvertent disconnects due to misalignment of the receptacle bore and boom centerline.
d. Probe Receiver. Flight testing with targeted drogue-equipped tanker results in no damage, breakage, or inadvertent disconnects due to the inability of the probe nozzle tip to articulate as needed.

10.1.1.3 Physical clearance.

Standard: Clearance is provided around the AR subsystem's interface to permit engagement and disengagement procedures with the interface of the targeted tanker/receiver AR subsystem without impacting surrounding structure, FOD generation, or damage to the tanker or receiver.

a. Boom Tanker. Boom physical dimensions do not exceed the allowable boom envelope of ATP-3.3.4.5, Maximum Nozzle Envelope figure and Maximum Boom Envelope figure.
b. Drogue Tanker. Reception coupling and drogue components do not exceed the allowable envelope defined in ATP-3.3.4.6, Clearance Envelope paragraph, subpart 2, and interior surfaces are free of obstruction.
c. Receptacle Receiver. No equipment or structure impinges on area around the receptacle as defined in ATP-3.3.4.5, Clearance Requirements section. All fastener heads around the receptacle are flush with the surrounding structural surface.
d. Probe Receiver. Probe installation geometry provides required clearance as defined in ATP-3.3.4.6, Clearance Envelope paragraph, subpart 1.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of drawings, specifications, and installed subsystem verify boom design does not exceed allowable envelope in ATP-3.3.4.5, Maximum Nozzle Envelope figure and Maximum Boom Envelope figure.
b. Drogue Tanker. Examination of drawings, specifications, and installed subsystem verify reception coupling and drogue design meet clearances of ATP-3.3.4.6, Clearance Envelope paragraph, subpart 2, and interior surfaces are free of obstruction.
c. Receptacle Receiver. Examination of drawings, specifications, and installed subsystem show receptacle design and surrounding structure meet clearances of ATP-3.3.4.5, Clearance Requirements section and structural fasteners are installed flush with the surrounding contact area.

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d. Probe Receiver. Examination of drawings, specifications, and installed subsystem show AR probe installation design meets clearances of ATP-3.3.4.6, Clearance Envelope paragraph, subpart 1.

Demonstration.

(1) Verify the standard by the following:

a. Boom Tanker.
   1. Boom nozzle engagement and disengagement with targeted receiver receptacle without impacting surrounding structure is demonstrated in laboratory environment.
   2. Ground demonstration verifies boom nozzle engagement and disengagement with targeted receptacle-equipped receiver air vehicles without impacting surrounding structure or equipment.

b. Drogue Tanker.
   1. Probe engagement and disengagement with targeted tanker reception coupling and drogue demonstrates clearances specified in ATP-3.3.4.6, Clearance Envelope paragraph, subpart 2, in a laboratory environment.
   2. Ground demonstration with targeted probe-equipped receiver air vehicles show drogue and reception coupling provide clearance to receiver canopy, structure, and equipment.

c. Receptacle Receiver.
   1. Receptacle system permits targeted tanker boom nozzle engagement and disengagement without impacting or damaging surrounding structure in laboratory environment.
   2. Demonstration in a laboratory environment verifies targeted tanker boom nozzle can slide along receptacle slipway without damage or hinderance.
   3. Air vehicle ground demonstration verifies engagement and disengagement with targeted tanker boom nozzle without impacting surrounding structure or equipment.

d. Probe Receiver.
   1. Air vehicle ground demonstration of engagement and disengagement with targeted tanker drogue and reception coupling without interference between probe mast, support structure, probe door, or other air vehicle structure and drogue.
   2. Air vehicle ground demonstration shows refueling probe is deployed and stowed (if retractable) without interference and provides clearance to air vehicle canopy and air vehicle structure.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight testing with receptacle-equipped receivers demonstrates successful engagement and disengagement without impacting surrounding structure or components. Required test points include worst-case combinations of tanker and receiver gross weights and CGs, at all extents of the boom disconnect envelope, wet and dry contacts, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions, straight and level flight, in turns, and while tobogganing.

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b. Drogue Tanker. Flight testing verifies hose and drogue can be deployed and stowed without impacting surrounding structure or components. Flight testing with targeted probe-equipped receivers shows engagement and disengagement without impacting surrounding structure or components. Test points include wet and dry contacts, receiver closure rates of up to 10 ft/s, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions.
c. Receptacle Receiver. Flight testing with boom-equipped tanker demonstrates successful engagement and disengagement without impact or damage to surrounding structure or components. Test points include worst-case combinations of tanker and receiver gross weights and CGs, at all extents of the boom disconnect envelope, wet and dry contacts, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions, straight and level flight, in turns, and while tobogganing.
d. Probe Receiver. Flight testing verifies probe can be deployed and stowed without impacting surrounding structure or components. Flight testing with targeted drogue-equipped tanker demonstrates successful engagement and disengagement without damage to surrounding structure or components. Test points include wet and dry contacts, receiver closure rates of up to 10 ft/s, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions and failure conditions, such as failure to fully extend or deploy.

10.1.1.4 Physical separation.

Standard: The AR system design and installation allow spatial separation between vehicles to avoid collision and negative impacts to engine performance, flight control and stability, or structure or components of either air vehicle.

a. Boom/Receptacle air vehicle. Boom/Receptacle system design and location allow receptacle receiver to maintain attitude, velocity, and position throughout the extents of the boom envelope without degradation to air vehicle or engine performance or collision with tanker or other receiver air vehicle. For simultaneous receivers, the separation distance between receivers is at least 0.25 wingspan of the largest winged receiver that will be simultaneously refueled.
b. Probe/Drogue air vehicle. Probe/drogue system installation allows probe receiver to make contact and refuel throughout the fuel transfer zone without degradation to air vehicle performance or collision with the tanker or other receiver air vehicle. The nose of the receiver is aft of the tail of the tanker when engaged in refueling position at least 15 ft forward of full trail. For simultaneous receivers, the separation distance between receivers is at least 0.25 wingspan of the largest winged receiver that will be simultaneously refueled.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker/Receptacle Receiver. Examination of air vehicle drawings and models verifies separation exists between tanker and receiver at receiver positions: precontact,

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contact, and the corners of the boom disconnect envelope. Required separation exists between receivers when simultaneously refueling multiple air vehicles (if applicable).

b. Drogue Tanker/Probe Receiver. Examination of air vehicle drawings and models verifies separation exists between tanker and receiver at receiver positions: precontact, contact, and inner and outer hose refueling limits. Separation exists between receivers when simultaneously refueling multiple air vehicles (if applicable).

Analysis.

(1) Verify the standard by the following:

a. Boom Tanker/Receptacle Receiver. Aerodynamic modeling verifies tanker engine exhaust does not impinge on receiver structure and tanker wake does not negatively impact receiver engine performance, flight controls, stability or structure at receiver positions: precontact, contact, and the corners of the boom disconnect envelope.

b. Drogue Tanker/Probe Receiver. Aerodynamic modeling verifies tanker engine exhaust does not impinge on receiver structure and tanker wake and downwash effects do not negatively impact receiver engine performance, flight controls, stability or structure at receiver positions: precontact, contact, and within the hose refueling range.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Verify the standard by the following:

a. Boom Tanker/Receptacle Receiver. Evaluations of air vehicle performance during AR flight test show no negative impacts to air vehicle response, stability and control, engine performance, ram air cooling, and air data system performance.

b. Drogue Tanker/Probe Receiver. Evaluations of air vehicle performance during AR flight test show no negative impacts to air vehicle response, stability and control, engine performance, ram air cooling, and air data system performance.

10.1.1.5 Mating equipment.

Standard: The AR system includes necessary mating equipment to facilitate engagement and disengagement procedures with targeted tanker/receiver AR subsystem without interference, binding or damage that would prevent AR operations.

a. Boom Tanker. Tanker boom nozzle design includes an induction coil that meets dimensions and positioning of ATP-3.3.4.5, Annex B, Boom Nozzle Induction Coil Position figure, and Boom Nozzle Induction Coil Centering and Gap figure.

b. Drogue tanker. The reception coupling has three equally spaced locking roller mechanisms.

c. Receptacle receiver. Receptacle includes an induction coil that meets the dimension and positioning of ATP-3.3.4.5, Annex B, Receptacle Induction Coil Requirements figure and Receptacle Induction Coil Dimensions table. Receptacle is equipped with latches that conform to the dimensions and positioning of ATP-3.3.4.5, Latch Dimensions paragraph. Receptacle incorporates rollers that conform to the dimensional and positioning requirements of ATP-3.3.4.5, Receptacle Rollers figure and Receptacle Rollers table.

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Receptacle includes a contact sensor, designed to sense a contact when the receptacle sliding valve travels between 0.5 in and 0.75 in, that meets the dimensions and positioning of ATP-3.3.4.5, Receptacle Contact Sensor paragraph.

d. Probe receiver. Probe nozzle incorporates two equally spaced sleeve latches.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of drawings, specifications, and installed subsystem verifies boom nozzle mating equipment conforms to the dimension and positioning of the induction coil.
b. Drogue Tanker. Examination of drawings, specifications, and installed subsystem verifies the reception coupling has three equally spaced locking roller mechanisms.
c. Receptacle Receiver. Examination of drawings, specifications, and installed subsystem verifies receptacle mating equipment conforms to the dimension and positioning of the induction coil, latches, rollers, and contact sensor.
d. Probe Receiver. Examination of drawings, specifications, and installed subsystem verifies probe nozzle has two equally spaced latches.

Demonstration.

(1) Verify the standard by the following:

a. Boom Tanker.
    1. Demonstration in a laboratory environment verifies tanker boom nozzle induction coil alignment with targeted receiver(s) receptacle induction coil.
    2. Ground demonstration verifies boom nozzle induction coil alignment with targeted receiver air vehicle(s) receptacle interface.
b. Drogue Tanker.
    1. Demonstration in a laboratory environment verifies probe engagement and disengagement with targeted tanker reception coupling without interference.
    2. Ground demonstration verifies reception coupling engagement and disengagement with targeted receiver air vehicle(s) probe nozzle without interference.
c. Receptacle Receiver.
    1. Demonstration, with targeted tanker boom nozzle, in a laboratory environment, verifies signal coil alignment and operation, engagement and disengagement without interference, and contact sensor operation.
    2. Air vehicle ground demonstration with targeted tanker boom nozzle verifies signal coil alignment and operation, engagement and disengagement without interference, and contact sensor operation.
d. Probe Receiver.
    1. Demonstration in a laboratory environment verifies probe engagement and disengagement with targeted tanker reception coupling without interference.
    2. Air vehicle ground demonstration verifies probe nozzle engagement and disengagement with targeted tanker reception coupling without interference.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight testing with receptacle-equipped receivers verifies signal coil alignment and operation. Test points include worst-case combinations of tanker and receiver gross weights and CGs, at all extents of the boom disconnect envelope, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions, straight and level flight, in turns, and while tobogganing.
b. Drogue Tanker. Flight testing verifies probe nozzle engagement and disengagement with tanker reception coupling without interference. Test points include wet and dry contacts, receiver closure rates of up to 10 ft/s, minimum Mach, maximum Mach, and minimum and maximum dynamic pressure conditions.
c. Receptacle Receiver. Flight testing with boom-equipped tankers verifies signal coil alignment and operation, boom nozzle engagement and disengagement without interference, and contact sensor operation. Test points include worst-case combinations of tanker and receiver gross weights and CGs, at all extents of the boom disconnect envelope, wet and dry contacts, minimum Mach, maximum Mach, minimum dynamic pressure and maximum dynamic pressure conditions, straight and level flight, in turns, and while tobogganing.
d. Probe Receiver. Flight testing with drogue-equipped tankers verifies probe nozzle engagement and disengagement without interference. Test points include wet and dry contacts, receiver closure rates of up to 10 ft/s, minimum Mach, maximum Mach, minimum dynamic pressure, and maximum dynamic pressure conditions.

10.1.1.6 Boom envelope protection.

Standard: For boom tanker systems only, boom system automatically commands a disconnect such that a disconnect occurs prior to exceeding boom disconnect limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of specifications and system description documents verifies design incorporates boom system auto disconnect function that disconnects prior to exceeding the boom disconnect limits.

Analysis.

(1) Analysis shows boom system design will automatically command and achieve a disconnect prior to exceeding the boom disconnect limits.

Test.

(1) Lab test: Hardware-in-the-loop laboratory testing verifies the boom system automatically commands and achieves a disconnect prior to exceeding the boom disconnect limits.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: AR flight test verifies boom system automatically commands and achieves a disconnect before receiver-specific boom disconnect limits are reached.

10.1.2 Aerodynamic compatibility.

Tanker and receiver air vehicles are safely controllable and maneuverable during AR operations. Tanker and receiver handling qualities and workload are assessed using MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system) for AR tasks. While air vehicle flying qualities are primarily assessed in 6.1, this section specifically defines AR flight performance which includes the precision of air vehicle control attained by the flight crew, and the associated workload (amount of effort and attention), both physical and mental, that the flight crew will provide to attain a given level of performance. Precision of control cannot be defined independently of the amount of effort and attention required of the flight crew.

Qualitative assessments, that could include but are not limited to flight crew comments, HQRs, PIO ratings, and Bedford workload ratings, are used to gage level of effort for refueling tasks, including air vehicle flight controls, available thrust, and ability to stabilize in and around the refueling positions. These assessments cover more than just stability and control characteristics, also considered are crew station interface (e.g., displays, controls), the air vehicle environment (e.g., weather conditions, visibility, turbulence), and stress (e.g., mental and physical). In addition to HQR and PIO ratings, quantitative performance levels are defined in terms of control authority and maneuvering precision.

Flight performance during AR is qualitatively assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system).

Testing for tanker systems evaluates receiver bow wave impacts to boom handling qualities and tanker handling qualities and auto-pilot response. Receiver wake survey testing at specific locations around the tanker evaluates airflow effects between the two air vehicles, impacts to handling qualities, and visual cues for maintaining position. Closure rate testing evaluations are critical for both receptacle-equipped and probe-equipped receivers.

10.1.2.1 Aerial refueling envelope.

Standard: Tanker and receiver AR envelopes, altitude and airspeed are defined. Intersection of tanker and receiver envelope is defined and is operationally employable.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of flight manuals and system specifications for tanker and receiver air vehicles verify AR envelopes are documented.

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Analysis.

(1) Analysis of published AR envelope ensures both air vehicles have an operationally employable altitude and airspeed envelope.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Flight testing verifies AR operations can be executed within the limits of the AR envelope of the pairing.

10.1.2.2 Air vehicle flight performance.

Standard: The air vehicle has acceptable flight performance and satisfactory workload during AR operations throughout the AR envelope.

a. Boom Tanker. Tanker flight performance during refueling operations is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent rating system) and receiver motion does not adversely impact tanker ability to maintain commanded attitude, altitude, or airspeed. AR maneuvers include straight and level flight, turns, toboggans, climbs, and breakaway maneuvers throughout the AR envelope.
b. Drogue Tanker. Tanker flight performance during refueling operations is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent rating system) and receiver motion does not adversely impact ability to maintain commanded attitude, altitude, or airspeed. AR maneuvers include straight and level flight, turns, toboggans, climbs, and breakaway maneuvers throughout the AR envelope.
c. Receptacle Receiver. Receiver flight performance during refueling operations is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent rating system), including closing and stabilizing in contact position, station keeping throughout the boom envelope, straight and level flight, turns, toboggans, climbs, and breakaway maneuvers throughout the AR envelope.
d. Probe Receiver. Receiver flight performance during refueling operations is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent rating system), including contacts at closure rates up to 10 ft/s, stabilizing post contact, maneuvering in and out of refuel zone, station keeping, straight and level flight, turns, climbs, and breakaway maneuvers throughout the AR envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis of aerodynamic environment and air vehicle performance shows that the air vehicle has power and control authority to execute AR maneuvers with targeted air vehicles.

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Test.

(1) Lab test: Ground based piloted simulation verifies the air vehicle has power and control authority to execute AR maneuvers with targeted air vehicles.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight evaluations of air vehicle performance, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and HQRs verify tanker flight performance is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent system) during AR operations. Test points include station keeping with targeted receiver air vehicles in contact and pre-contact positions in straight and level flight, turning, and toboggans, throughout the AR envelope, at worst-case combinations of tanker and receiver gross weights and CGs.
b. Drogue Tanker. Flight evaluations of air vehicle performance, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and HQRs verify tanker flight performance is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent system) during AR operations. Test points include straight and level flight, turning, and toboggans with targeted receiver air vehicles in contact and refueling positions, throughout the AR envelope, at worst-case combinations of tanker and receiver gross weights and CGs.
c. Receptacle Receiver. Flight evaluations of air vehicle performance, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and HQRs verify receiver flight performance is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent system) during AR operations. Test points include station keeping, with targeted tanker air vehicles, pre-contact and in contact positions throughout the boom envelope, in straight and level flight, turning, and toboggans, at worst-case combinations of tanker and receiver gross weights and CGs, throughout the AR envelope.
d. Probe Receiver. Flight evaluations of air vehicle performance, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and HQRs verify receiver flight performance is assessed as no worse than a rating of 6 from MIL-STD-1797, Cooper-Harper rating scale figure (or equivalent system) during AR operations. Test points include wake surveys at pre-contact, contact and at the inner refueling limit, contacts at closure rates up to 10 ft/s, stabilizing post contact, maneuvering in and out of refuel zone, station keeping in straight and level flight and turns, and breakaway throughout the AR envelope, at worst-case combinations of tanker and receiver gross weights and CGs.

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10.1.2.3 Flight performance impacts on targeted air vehicles.

Standard: Aerodynamic effects between tanker and receiver do not result in a flight performance rating worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system) or instabilities for targeted air vehicles performing AR operations nor induce an environment that destabilizes targeted refueling equipment (e.g., induced boom oscillations, negative contacts, or soft drogue contacts).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis of aerodynamic environment and air vehicle characteristics show near field flow environment around air vehicle does not cause instabilities or adversely affect targeted air vehicles.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Flight evaluations of air vehicle performance, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and HQRs verify flight performance is no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system) for targeted air vehicle in refueling positions. Flight testing verifies stability of targeted AR equipment during refueling operations.

10.1.2.4 Autopilot performance.

Standard: Autopilot and augmentation systems are assessed no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system) and perform in a consistent and adequate manner during AR operations. Autopilot and augmentation systems include autothrottle, stability augmentation system, CAS, stall inhibitor system, and stability enhancement function. System(s) response to targeted air vehicle motion and position does not impact ability to maintain commanded attitude, altitude, and airspeed.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis of autopilot and augmentation systems design and requirements verify systems account for aerodynamic effects of refueling operations with targeted air vehicle(s).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Verify the standard by the following:

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a. Boom Tanker. Flight evaluations of air vehicle performance during AR, with autopilot and augmentation systems on, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and handling qualities verify tanker flight performance is assessed as no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system). Test points include station keeping in contact and pre-contact positions in straight and level flight, turning, and toboggans with targeted receiver air vehicles, throughout the AR envelope.
b. Drogue Tanker. Flight evaluations of air vehicle performance during AR, with autopilot and augmentation systems on, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and handling qualities verify tanker flight performance is assessed as no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system). Test points include straight and level flight, turning, and toboggans with targeted receiver air vehicles, throughout the AR envelope.
c. Receptacle Receiver. Flight evaluations of air vehicle performance during AR, with autopilot and augmentation systems on, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and handling qualities verify receiver flight performance is assessed as no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system). Test points include station keeping in contact and pre-contact positions in straight and level flight, turning, and toboggans with targeted tanker air vehicle, throughout the AR envelope.
d. Probe Receiver. Flight evaluations of air vehicle performance during AR, with autopilot and augmentation systems on, including predictability of response, stability and control, control sensitivity, time delay, oscillations, PIO tendencies, undesirable motions, workload and handling qualities verify receiver flight performance is assessed as no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system). Test points include contacts at closure rates up to 10 ft/s, stabilizing post contact, maneuvering in and out of refuel zone, station keeping in straight and level flight and turns, and breakaway maneuvers throughout the AR envelope.

10.1.2.5 Aerial refueling system flight performance.

Standard: Flight performance of refueling equipment is assessed as no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system) and has satisfactory workload (if applicable) to allow AR operations with targeted receiver air vehicles.

a. Boom Tanker. Boom system flight performance is assessed as no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system) and has satisfactory workload when operated in proximity to receiver air vehicles. Boom flying qualities, response, stability and control, control sensitivity, time delay, oscillations, undesirable motions, workload and handling qualities are no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system) for refueling tasks.
b. Drogue Tanker. Hose and drogue are stable during extension, retraction, and at full trail throughout the AR envelope. Hose reel system provides catenary curve and drogue presentation angle which permit probe engagement throughout refueling envelope. In free

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air, disturbances of the refueling hose damp-out to one-third amplitude in three cycles and disturbances of the drogue damp-out to one-half drogue diameter.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify the standard by the following:

a. Boom Tanker. Computational modeling and analysis of boom flight control laws predict boom will meet required levels of controllability in proximity to receiver air vehicles. Modeling shows excitation disturbances of the boom damp-out.
b. Drogue Tanker. Analysis shows hose and drogue are stable in trail throughout refueling envelope, catenary curve and drogue presentation angle permit probe engagement throughout refueling envelope. Modeling of hose and drogue show disturbances and excitations damp-out.

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker.
    1. Wind tunnel testing verifies aerodynamic performance matches analytical values and shows boom is stable in the expected airflow environment and disturbances damp-out as required.
    2. Ground based simulation and SIL testing with operator in the loop shows the boom is stable and controllable throughout the expected airspeed and altitude envelope to the extents of the boom control envelope.
b. Drogue Tanker. Wind tunnel testing of trailing components verifies disturbances of the hose and drogue damp-out as required and the catenary curve and drogue presentation angle permits probe engagement.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight evaluations of boom performance during free air flight and AR operation, including predictability of response, stability and control, control sensitivity, tactile feedback, time delay, oscillations, PIO tendencies, undesirable motions, workload and handling qualities verify boom flight performance is assessed as no worse than 6 from MIL-STD-1797, Cooper-Harper pilot rating scale figure (or equivalent rating system). Test points include, free air maneuvering, receptacle tracking, contacts and disconnects, in straight and level flight, turns, and toboggans, with targeted receiver air vehicles, throughout the AR envelope at the extremes of the boom disconnect envelope, at worst-case combinations of tanker and receiver gross weights and CGs.
b. Drogue Tanker.

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1. Flight testing verifies hose and drogue are stable and can be trailed, rewound and stowed throughout the refueling envelope.
2. Flight testing verifies hose reel system provides catenary curve and drogue presentation angle which permits probe engagement throughout refueling envelope.
3. Flight testing verifies, in free air, disturbances of the hose damp-out to one-third amplitude in three cycles and disturbances of the drogue damp-out to one-half diameter.

10.1.3 Mechanical compatibility.

Structural loads generated during AR operations are accounted for in air vehicle structural and AR system design to prevent premature failure, preserve AR capability, and ensure continued safety. Some designs include frangible or weak links to control the point of failure if operating loads exceed design strength. In probe system design, the forward end of the probe mast is intended to be the weak link. Therefore, ultimate strength for the probe mast is normally set to a value between the probe DLL and surrounding structure ultimate strength (e.g., 133% of the probe DLLs). The back-up or surrounding structure ultimate strength is set to a higher value (e.g., 150% of probe DLLs), to ensure it does not fail prior to the probe.

10.1.3.1 Structural design limit loads.

Standard: The AR system interface, its attachment to airframe structure, and the structure surrounding the interface withstands the loads experienced during the AR process.

a. Boom Tanker. The loads used in the design of the boom refueling system include the maximum, minimum, and most critical combination of loads that can result from contact, disconnect, boom strike, tension disconnect, in-contact axial and radial loads, with and without boom operator input, throughout disconnect envelope, fuel transfer and aerodynamic loads during stow, deploy, and maneuvering throughout the airspeed and altitude refueling envelope, including a tension load of 9,411 lbs applied to the boom-nozzle ball joint when fully extended.
b. Drogue Tanker. The loads used in the design of the drogue refueling system include the maximum and most critical combination of loads that can result from hose extension, contact, engagement, hose response, disengagement, off-center engagement, off-center disengagement, fuel transfer, and aerodynamic loads when stowed, trailed, or in transition. The refueling hose design limit tension load is defined by the maximum aerodynamic drag of the hose, fuel reception coupling and drogue of the system at the maximum design operating dynamic pressure for the system or 1,500 lbs for up to a 2-inch inner diameter hose and 2,000 lbs for an over 2-inch diameter hose, whichever load is higher. The coupling DLLs are as follows:
    1. 1,500 lbs tension load in combination with a 1,500 lbs radial load.
    2. 2,500 lbs compression load with a 1,500 lbs radial load.
    3. 1,500 lbs tension load.
    4. 2,500 lbs compression load.
    5. 1,500 lbs radial load.
c. Receptacle Receiver. The loads used in the design of the receptacle refueling system include the maximum, minimum, and most critical combination of loads that can result from engagement, disengagement, tension disconnect, boom strike, fuel transfer, and

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aerodynamic loads when receptacle is open, closed, and in transition. Design receptacle loads are as follows and are applied within the refueling envelope and disconnect envelope for targeted tanker boom subsystems:

1. Tension: a load of 9,411 lbs applied to the boom-nozzle ball joint.
2. Compression: a load of 9,411 lbs applied to the boom-nozzle ball joint.
3. Air vehicle skin area 12 inches out from receptacle perimeter:1,800 lbs vertical combined with 750 lbs lateral.
4. Aerodynamic loads: Slipway and its doors in both the open and closed positions.

d. Probe Receiver. The loads used in the design of the probe refueling system include the maximum, minimum, and most critical combination of loads that can result from engagement, disengagement, off-center disconnect, hose tension, fuel transfer and aerodynamic loads when AR equipment is stowed, deployed, and in transition. Design limit probe loads are as follows:
   1. 1,500 lbs tension.
   2. 2,500 lbs compression.
   3. 1,500 lbs radial.
   4. 2,500 lbs compression with 1,500 lbs radial.
   5. 1,500 lbs tension with 1,500 lbs radial.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify the standard by the following:

a. Boom Tanker. Stress and CFD analyses define limit loads due to AR load cases and aerodynamic environment. Structural analysis verifies boom structure and its attachment to the air vehicle can withstand defined limit loads.
b. Drogue Tanker. Stress and CFD analyses define limit loads due to AR load cases and aerodynamic environment. Structural analysis verifies AR system, components, and attachment structure can withstand defined limit loads.
c. Receptacle Receiver. Stress and CFD analyses define limit loads due to AR load cases and aerodynamic environment. Structural analysis verifies AR receptacle and surrounding structure can withstand defined limit loads.
d. Probe Receiver. Stress and CFD analyses define limit loads due to AR load cases and aerodynamic environment. Structural analysis verifies AR probe and surrounding structure can withstand defined limit loads.

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker. Component qualification laboratory testing verifies boom components and structure can withstand application of DLLs without detrimental deformation. Laboratory testing of integrated boom subsystem demonstrates loads during telescope extension and retraction and contact and disconnect with representative receptacle interface do not exceed DLLs.

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b. Drogue Tanker. Laboratory testing demonstrates hose extension, hose retraction and hose response loads with representative receiver interface do not exceed limit loads. Component qualification laboratory testing demonstrates the refueling coupling and drogue can withstand DLLs.
c. Receptacle Receiver. Component qualification laboratory testing verifies receptacle system and representative support structure can withstand DLLs.
d. Probe Receiver. Component qualification laboratory testing verifies probe system and representative support structure can withstand DLLs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight test validates load cases match analytically developed DLLs.

10.1.3.2 Structural design ultimate loads.

Standard: The AR system interface, its attachment to airframe structure, and the structure surrounding the interface do not fail (experience permanent deformation) prior to the ultimate loads that can be generated during the AR process.

a. Boom Tanker. Boom ultimate design loads:
    1. 16,166 lbs tension, boom fully extended.
    2. 7,500 lbs compression, boom not fully retracted.
    3. 20,000 lbs compression, boom fully retracted.
b. Drogue Tanker. Refueling hose ultimate design load is 2.5X the DLL.
c. Receptacle Receiver. Receptacle ultimate design loads:
    1. Tension: a load of 14,000/(Cos A) lbs applied to the boom-nozzle ball joint, where the angle A may vary from 0 degrees to 30 degrees and is relative to the receptacle axis.
    2. Compression: a load of 20,000 lbs applied at the boom-nozzle ball joint anywhere within a 34-degree cone.
    3. Slipway and its doors in both the open and closed positions, 2,000 lbs laterally and 5,000 lbs vertically.
d. Probe Receiver. Probe attachment and surrounding structure withstand at least 1.5X probe DLLs. Probe tip and mast ultimate strength is greater than probe DLLs and less than support structure ultimate strength to ensure probe fails prior to attachment and support structure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify the standard by the following:

a. Boom Tanker. Structural analysis verifies boom structure and its attachment to the air vehicle do not fail prior to the required ultimate loads.

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b. Drogue Tanker. Structural loads analysis verifies refueling hose will not fail prior to the required ultimate loads.
c. Receptacle Receiver. Structural loads analysis verifies that the receptacle, installation, and surrounding structure do not fail prior to the required ultimate loads.
d. Probe Receiver. Structural loads analysis verifies that the probe tip and mast, attachment and surrounding structure do not fail prior to the required ultimate loads and the refueling probe fails prior to the attachment structure.

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker. Static load laboratory testing of the boom system verifies boom components and structure do not fail prior to ultimate loads.
b. Drogue Tanker. Static load laboratory testing verifies refueling hose does not fail prior to ultimate load.
c. Receptacle Receiver. Static load laboratory testing on representative receptacle and surrounding structure verifies receptacle, attachment, and supporting structure does not fail prior to ultimate load.
d. Probe Receiver. Static load laboratory testing on representative probe and attachment structure verifies attachment and surrounding structure does not fail prior to ultimate load and refueling probe fails prior to attachment structure.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight testing, including contacts, disconnects, pressure refueling, tension disconnects, and slipway assisted contacts with and without crewmember input, show boom system does not fail or deform during AR operations.
b. Drogue Tanker. Flight testing including hose extension and retraction, receiver contacts at closure rates up to 10 ft/s and disconnects throughout the airspeed and altitude refueling envelope verify hose reel system and refueling assembly (fuel reception coupling and drogue) do not fail or deform during refueling operations.
c. Receptacle Receiver. Flight testing including contacts, disconnects, pressure refueling, tension disconnects, and slipway assisted contacts with and without crewmember input show that the receptacle does not fail or deform.
d. Probe Receiver. Flight testing including contacts at closure rates up to 10 ft/s verify probe system and support structure do not fail or deform.

10.1.3.3 Durability.

Standard: The AR system and surrounding structure withstands the repeated impacts, and engagement and disengagement cycles experienced during the AR process over the life of the air vehicle.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of system specifications and design criteria verifies repeated impacts, and engagement and disengagement cycle usage are established.

Analysis.

(1) Analysis verifies AR system and surrounding structure design withstands all repeated impacts, and engagement and disengagement cycles over the life of the air vehicle.

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker. Laboratory testing of boom refueling system with representative AR receptacle verifies boom components and structure withstand repeated impacts and usage. Laboratory testing of boom refueling system verifies boom system withstands simulated in-contact refueling cycles.
b. Drogue tanker. Laboratory testing of fuel reception coupling at angular positions of up to 15 degrees off-center with a simulated probe refueling system, closure rates up to 10 ft/s, and simulated drogue drag load at maximum design dynamic pressure verifies coupling withstands repeated impacts and usage without permanent deformation or creation of FOD. Laboratory testing of refueling drogue with a simulated probe refueling system, closure rates up to 10 ft/s, and simulated drogue drag load at maximum design dynamic pressure verifies drogue withstands repeated impacts and usage without permanent deformation or creation of FOD. Laboratory testing of hose reel system with simulated drogue drag loads verifies hose reel system withstands hose trail, hose response, disconnect, and hose rewind cycles.
c. Receptacle Receiver. Laboratory testing with representative boom system verifies receptacle system withstands repeated impacts and usage, and engagement and disengagement cycles.
d. Probe Receiver. Laboratory testing with representative reception coupling verifies probe system withstands repeated impacts and usage, and engagement and disengagement cycles.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing verifies the AR system and surrounding structure withstands repeated impacts and usage and validates analysis.

10.1.3.4 Interface loads.

Standard: The AR system manages loads transmitted to targeted air vehicle.

a. Boom Tanker. While coupled to a receiver, the boom system limits axial loads according to ATP-3.3.4.5, Coupled Push Pull Loads paragraph and limits radial loads to less than 750 lbs.

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b. Drogue Tanker. Hose reel system maintains refueling hose tension such that no excessive slack develops and inadvertent disconnects are avoided. Reception coupling latch forces required to engage the coupling with an MA-2 receiver AR probe nozzle having a 50-lb sleeve force are not greater than 200 lbs with 10 psig static fuel pressure in the coupling. Reception coupling latch forces required to disengage the probe nozzle from the coupling are adjustable between 300 and 550 lbs with 0 psig fuel pressure in the coupling. The force required to separate the nozzle from the coupling, with 50 psig of fuel pressure, with a latch-force setting of 520 ±20 lbs, is not greater than 830 lbs when the disconnect angle of the swivel-ball joint is at the 22.5-degree position.
c. Receptacle Receiver. The receptacle system limits the force to disengage the targeted tanker boom nozzle from the receptacle when the receptacle latches fail to actively release from a minimum of 4,800 lbs to a maximum of 9,411 lbs.
d. Probe Receiver. The probe system limits the force to depress the nozzle sleeve to the fully open position to less than 45 ±5 lbs and requires 15 lbs or greater force to move the nozzle sleeve to the fully closed position.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of specifications shows boom actuation system is required to manage interface loads.
b. Drogue Tanker. Examination of specifications show hose reel and reception coupling are required to manage interface loads.
c. Receptacle Receiver. Examination of specifications and receptacle latch design shows expected brute force disconnect loads to be within the required range.
d. Probe Receiver. Examination of specifications and nozzle latch design shows latch sleeve operation meets standard.

Analysis.

(1) Verify the standard by the following:

a. Boom Tanker. Analysis and modeling show boom system maintains the required interface loads at all possible boom fuel pressures and extremes of boom envelope.
b. Drogue Tanker. Analysis and modeling show hose reel and reception coupling design prevent excess hose slack, minimize negative contacts, and manage interface loads to within the allowable levels.
c. Receptacle Receiver. Analysis and modeling show receptacle latch design brute force disconnect loads to be within the required range.
d. Probe Receiver. Analysis and modeling show nozzle latch design operates according to the standard.

Test.

(1) Lab test: Verify the standard by the following:

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a. Boom Tanker. Laboratory testing of boom system verifies interface loads are within specified values.
b. Drogue Tanker. Laboratory testing of hose reel system verifies interface loads are within specified values. Laboratory testing of coupling engagement and disengagement forces verify latch forces are within specified values.
c. Receptacle Receiver. Laboratory testing verifies force to disengage with receptacle latches failed closed is within the required range.
d. Probe Receiver. Laboratory testing verifies nozzle sleeve operates according to the standard.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight testing with targeted receptacle equipped receivers verifies the boom system allows telescoping in both extension and retraction at rates up to 6.5 ft/s and axial and radial loads are within allowable levels throughout the altitude and air speed AR envelope, boom envelope, and operating fuel pressures. Additional test points include pressure refueling and slipway assisted contacts.
b. Drogue Tanker. Flight testing with targeted probe equipped receiver(s) verifies hose tension is maintained, excessive hose slack does not develop, negative contacts are minimized, and inadvertent disconnects due to high hose tension do not occur throughout the altitude and airspeed AR envelope and operating fuel pressures.
c. Receptacle Receiver. Flight testing with targeted boom equipped tanker verifies controlled tension disconnects with receptacle latches closed result in a tension load between 4,800 lbs and 9,411 lbs. A minimum of 3 tension disconnects should be performed and result in consistent tension load values. If loads are not consistent, additional test points should be accomplished.
d. Probe Receiver. Flight testing with targeted drogue equipped tanker verifies engagement, disengagement, and fuel transfer without fuel leakage.

10.1.3.5 Prevention of overload.

Standard: The AR system has design features to prevent overloading (reaching ultimate loads) of components or structure due to binding, single point failures, or unexpected dynamic events.

a. Boom Tanker. The boom tanker system incorporates an IDS that facilitates an immediate disconnect from the receptacle system latches. The system is available at any time (unlimited uses).
b. Drogue Tanker. N/A
c. Receptacle Receiver. The receptacle latches fail under 11,500 ±1,000 lbs tension and no loose debris is created following the failure.
d. Probe Receiver. The probe system is designed to fail prior to failure of probe attachment and surrounding air vehicle structure (e.g., weak, fuse, or frangible component).

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of system and design specifications verifies the boom tanker system incorporates an IDS that is available at all times for an unlimited amount of uses.
b. Drogue Tanker. N/A
c. Receptacle Receiver. Examination of system and design specifications verifies receptacle latches fail under 11,500 ± 1,000 lbs tension and no FOD is generated during failure.
d. Probe Receiver. Examination of system and design specifications verify probe system incorporates a weak link or structure that fails at loads less than the ultimate load.

Analysis.

(1) Verify the standard by the following:

a. Boom Tanker. Reliability analysis verifies the IDS will be available throughout the life of the boom system.
b. Drogue Tanker. N/A
c. Receptacle Receiver. Analysis of design verifies latches will fail under the required load.
d. Probe Receiver. Analysis shows probe system incorporates a weak link or structure that fails at loads less than the ultimate load.

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker. Laboratory testing verifies IDS actuates when commanded, with tension load, and clears receptacle toggles.
b. Drogue Tanker. N/A
c. Receptacle Receiver. Laboratory testing verifies latches fail under static load of 11,500 ±1,000 lbs.
d. Probe Receiver. Laboratory testing verifies weak link or probe structure fails prior to ultimate load.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight test verifies IDS actuates when commanded, under load, and clears receptacle toggles allowing boom retraction. Test points include corners of the boom envelope and normal and override signal amplifier modes.
b. Drogue Tanker. N/A
c. Receptacle Receiver. Flight test results show no incidents of latch complete failure at loads less than the specified value.

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d. Probe Receiver. Flight test results show no incidents of probe supporting structure damage.

10.1.3.6 Service life.

Standard: The air vehicle structure and components can withstand the loads spectrum and AR aerodynamic environment (e.g., temperature, vibration, duration) generated between itself and the targeted tanker/receiver air vehicle(s) throughout the AR envelope over the DSL of the air vehicle. Life-limited parts are identified and replacement intervals (e.g., number of contacts, fuel pressurization cycles, flight hours in AR environment) are specified and documented in flight and maintenance manuals.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of requirement specifications and design documentation identify service life requirements, life-limited part definitions and plan for tracking life-limited components.

(2) Examination of draft maintenance manuals and maintenance plans verifies LLP replacement instructions are documented.

Analysis.

(1) Fatigue analysis verifies that the AR system meets the required service life and identifies life-limited AR components.

Test.

(1) Lab test: System endurance testing verifies AR system meets service life requirements. Qualification testing in a laboratory environment verifies component life and durability performance in expected environment.

(2) Ground test: Ground testing verifies load spectrum and usage are aligned with the analytically developed fatigue load spectrum.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) Examination of maintenance and flight manuals verify life-limited AR components are tracked and replacement intervals identified.

Test.

(3) Flight test: Flight testing verifies load spectrum and usage are aligned with the analytically developed fatigue load spectrum.

10.1.4 Electrical compatibility.

Electrical compatibility addresses AR specific AW attributes of electrical interface for signals through air vehicle booms and receptacles, electromagnetics between AR tanker and target air vehicle, electromagnetic effects on communication and static discharge on coupled pairs.

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Evaluations of criteria 10.1.4.2, 10.1.4.3, and 10.1.4.4 are in conjunction with Section 13 evaluations.

10.1.4.1 Signal coil electrical interface.

Standard:

a. Boom Tanker. The tanker boom system sends disconnect commands and voice signal communications and receives voice signal and receptacle latch position information when coupled with a receptacle-equipped receiver. The tanker AR system detects a pulse coming from the receptacle of a connected receiver when the receptacle coil is excited between 18 and 30 V during a minimum of 3 ms. The boom nozzle coil induces in the receptacle coil electrical signal pulses IAW ATP-3.3.4.5, Signal Advance Characteristics figure and Signal Advance table. The signal amplifier provides electrical signal pulse and voice communication equivalent to a MIL-DTL-38449 Class 4 miniaturized electrical receptacle with intercommunication push to talk, PN M38449-4.
b. Receptacle Receiver. The receiver receptacle system receives disconnect and voice signal communications and sends voice communications, receptacle latch position and disconnect signals when coupled with a boom-equipped tanker. The receptacle system detects a pulse coming from the nozzle when the nozzle coil is excited between 18 and 30 V during a minimum of 3 ms. The receptacle's coil induces in the nozzle coil electrical signal pulses IAW ATP-3.3.4.5, Signal Advance Characteristics figure and Signal Advance table.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of specifications and system design documentation show boom system is required to send and receive electrical signals pulses IAW the standard.
b. Receptacle Receiver. Examination of specifications and system design documentation show receptacle system is required to send and receive electrical signals pulses IAW the standard.

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker. Testing in a laboratory environment verifies the boom system sends and detects electrical signal pulses IAW the standard.
b. Receptacle Receiver. Testing in a laboratory environment verifies the receptacle system sends and detects electrical signal pulses IAW the standard.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the standard by the following:

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a. Boom Tanker. Flight testing verifies disconnect commands and voice signal communications are sent by the boom tanker and are received by the targeted receiver air vehicles and receptacle latch position information and voice signals are received from targeted receiver air vehicles in each signal amplifier mode, using each disconnect method (e.g., boom operator disconnect switch, receiver pilot disconnect switch, co-pilot disconnect switch).
b. Receptacle Receiver. Flight testing verifies receptacle system receives disconnect commands and voice signal communications sent by the boom tanker and sends voice communications, receptacle latch positions to targeted tanker air vehicles in each signal amplifier mode, using each disconnect method (e.g., boom operator disconnect switch, receiver pilot disconnect switch, co-pilot disconnect switch).

10.1.4.2 Electromagnetic interference source.

Standard: The air vehicle emitting equipment does not interfere with the performance or operation of targeted air vehicle systems or other systems installed on the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Electromagnetic environmental effects analysis of emitting equipment verifies emitting equipment, that is powered on during AR operations, does not interfere with targeted air vehicle or other systems installed on the air vehicle.

Test.

(1) Lab test: Laboratory testing of air vehicle emitting equipment verifies spectral characteristics which support E3 analyses.

10.1.4.3 Communication system electromagnetic interference.

Standard: Data and voice communication systems on the air vehicle required for AR are compatible with other systems, including the FCSs, and flight crew on both the air vehicle and the targeted air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of specifications verifies that data and voice communication systems are compatible with flight critical system on either air vehicle and do not present a hazard to flight crew.

Analysis.

(1) Electromagnetic environmental effects analysis verifies data and voice communication equipment required for AR does not interfere with other systems, including the FCSs, nor present a hazard to flight crew.

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Test.

(1) Lab test: Laboratory testing of data and voice equipment verifies spectral characteristics which support E3 analyses.

10.1.4.4 Controls for potentially hazardous equipment.

Standard: Any systems on the air vehicle (including but not limited to emitting equipment, defensive systems, and weapons), which if active during the AR process, could create a hazard to the targeted air vehicle or to the crewmembers of the targeted air vehicle, can be individually inhibited or disabled prior to entering the AR process.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of specifications and system design shows systems that could create a hazard to targeted air vehicles can be individually inhibited during AR.

Test.

(1) Lab test: Laboratory test verifies systems can be individually inhibited.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing verifies systems can be individually inhibited prior to AR operations.

10.1.4.5 Aerial refueling systems electromagnetic protection.

Standard: The AR system is protected from EMI from any other installed system or from the targeted air vehicle emitting systems.

a. Boom Tanker. The boom tanker AR system, including the boom nozzle and related components, boom control system, boom signaling system, and thru-the boom voice communication system, is protected from EMI from any of the air vehicle's own systems and from targeted air vehicle emitting systems.
b. Drogue Tanker. The drogue tanker AR system, including the control, status, and emergency function components, is protected from EMI from any of the air vehicle's own systems and from targeted air vehicle emitting systems.
c. Receptacle Receiver. The receiver AR system, including the receptacle and associated components, receptacle signaling system, and through-the boom voice communication system, is protected from EMI from any of the air vehicle's own systems and from targeted air vehicle emitting systems.
d. Probe Receiver. The receiver AR system is protected from EMI from any of the air vehicle's own systems and from targeted air vehicle emitting systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examination of specifications ensures AR systems components are protected from EMI.

Analysis.

(1) Electromagnetic environmental effects analysis of installed AR system verifies the amount of hardening or shielding for electrical equipment to protect from expected interference sources.

Test.

(1) Lab test: Aerial refueling system electrical component laboratory testing verifies equipment is qualified to perform in simulated electromagnetic environment.

(2) Ground test: Air vehicle ground testing verifies AR equipment operation in system-level electromagnetic environmental testing.

10.1.4.6 Static discharge.

Standard: The AR system can withstand the static discharge typically encountered during the engagement of tanker and receiver interfaces. AR system meets operational performance requirements when subjected to a 300-kilovolt discharge at the AR system interface.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis of system design verifies AR system components have static discharge protection provisions.

Test.

(1) Lab test: AR system electrical component laboratory testing verifies equipment is qualified to withstand required static discharge levels.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight test verifies no incidents of static discharge result in negative impacts to tanker or receiver, such as electronic equipment failure or damage to components.

10.1.5 Aerial refueling sensing, data, visual cues, and lighting systems compatibility.

This section addresses the signals, data, and visual cues that are necessary for AR operations, including air vehicle and AR system position, air vehicle contours, and AR system status and commands. Signals and data are provided to and received by the targeted air vehicle. The signals, data, and visual cues will vary based on type of refueling (boom/receptacle or probe/drogue) and the specific pairing of manned air vehicles and UAVs.

10.1.5.1 Sensing system.

Standard: Sensing system(s) (e.g., window, sensor, camera and display, sensor and detection algorithm) used during the AR process provides data, to flight crew or automated system, needed

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to conduct AR operations under mission-defined environmental conditions. Data needed to conduct AR includes but is not limited to targeted air vehicle relative position and rates of motion, refueling system location, and refueling system status. Mission-defined environmental conditions include day, night, dusk, twilight, with and without NVIS, ambient moonlight levels, diverse backgrounds (e.g., snow, ocean, desert, populated city lights, foliage dominant), above clouds, below clouds, and in clouds. Parameters that impact the performance of the sensing system include but are not limited to FOV, resolution, latency, integrity, availability, thermal signatures of air vehicles involved, bloom and glare effects, solar effects, operation with single failures, and overlaid information.

a. Boom Tanker. Tanker air vehicle sensing system(s) provides situational awareness data required to accomplish AR with receptacle-equipped receivers.
    1. Human Viewed System. Sensing system enables viewing of visual cues including boom nozzle, boom flight controls, boom structure, boom stability, boom movement (throughout boom envelope and during deployment and stowage), receiver receptacle, receiver stability, receiver movement, contours on top of receiver, receiver canopy, area around the receptacle, edges of slipway, edges of the receptacle, boom position relative to receiver, closure rate of receiver to contact position, antennas, wing leading edges, engine nacelles, horizontal/vertical stabilizers, receptacle bore, engagement of the boom in the slipway and receptacle, and receptacle doors. If sensing system is not direct view through a window, data acquisition system does not introduce misleading cues through depth distortions, insufficient contrast or extraneous visual artifacts.
    2. Unmanned System. Sensing system provides needed data including air vehicle relative and absolute locations and sensor generated pose estimation of receiver receptacle. Data is received and interpreted within required parameters (e.g., integrity, availability, accuracy) to accomplish AR.
b. Drogue Tanker. Tanker air vehicle sensing system(s) provides situational awareness data and visual cues needed to accomplish AR with probe-equipped receivers.
    1. Human Viewed System. Sensing system enables viewing of visual cues including hose exit tunnel, drogue (from deployment, through extension to full trail), hose stability, receiver probe, receiver stability, receiver engagement profile, receiver movement in contact, and hose response in contact.
    2. Unmanned System. Sensing system provides needed data including air vehicle relative and absolute locations. Data is received and interpreted within required parameters (e.g., integrity, availability, accuracy) to accomplish AR.
c. Receptacle Receiver. Air vehicle sensing system(s) provides situational awareness data and visual cues needed to accomplish AR with boom-equipped tankers.
    1. Human Viewed System. Sensing system enables viewing of visual cues including PDLs, refueling boom, formation lights, and formation markings.
    2. Unmanned System. Sensing system provides needed data including air vehicle relative and absolute locations and sensor generated tanker relative position data. Data is received and interpreted within required parameters (e.g., integrity, availability, accuracy) to accomplish AR.
d. Probe Receiver. Air vehicle sensing system provides situational awareness data and visual cues needed to accomplish AR with drogue-equipped tankers.

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1. Human Viewed System. Sensing system enables viewing of visual cues including signal lights, hose markings, drogue, coupling, exit tunnel, air vehicle reference features, tanker underbody markings, and formation lights.
2. Unmanned System. Sensing system provides needed data including air vehicle relative and absolute locations and sensor generated pose estimation of tanker drogue. Data is received and interpreted within needed parameters (e.g., integrity, availability, accuracy) to accomplish AR.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of system specifications and system design verifies sensing system provides data to enable AR operations in all air vehicle environmental and mission conditions.

Analysis.

(1) Task analysis determines visual cues for human operated systems needed to accomplish AR with targeted air vehicles. Analysis of sensing system design and installation verifies data is provided in all air vehicle environmental and mission conditions.

(2) Computer modeling of refueling tasks determines data and performance parameters needed to accomplish AR with targeted air vehicles. Analysis of sensing system design and installation verifies data is provided in all air vehicle environmental and mission conditions.

Demonstration.

(1) Air vehicle ground demonstration verifies sensing system provides and, if human viewed, accurately displays data and visual cues needed to conduct AR in simulated environmental and mission environments.

Test.

(1) Lab test: Hardware-in-the-loop laboratory testing verifies sensing system provides, and if human viewed, accurately displays, data and visual cues required to conduct AR in simulated environmental and mission environments.

(2) Ground test: Air vehicle ground testing verifies sensing system provides and, if human viewed, accurately displays data and visual cues needed to conduct AR in simulated environmental and mission environments.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Verify the standard by the following:

a. Boom Tanker.
    1. Human Operator. Flight testing with targeted air vehicles verifies situation awareness data and visual cues needed to accomplish AR within the system's mission defined environmental conditions are provided and, if not directly viewed, accurately displayed. Test points include various ambient lighting conditions (e.g., day, night,

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dusk), receiver join up, astern, contact position, receiver positions throughout the boom envelope, multiple receivers and if utilizing visual sensors (remote vision systems), high and low sun angles, various direction headings, and background conditions.

2. Unmanned System. Flight testing with targeted air vehicles verifies sensing system receives data within parameters needed to accomplish AR within the system's mission defined environmental conditions. Test points include receiver join up, astern, contact position, contact, and disconnect. Stressing test conditions depend on the method of data acquisition (e.g., differential GPS, lidar, visible cameras).

b. Drogue Tanker.
   1. Human Operator. Flight testing with targeted air vehicles verifies situation awareness data/visual cues needed to accomplish AR within the system's mission defined environmental conditions are provided and, if not directly viewed, accurately displayed. Test points include various ambient lighting conditions (e.g., day, night, dusk), receiver join up, pre-contact, contact, in-contact maneuvering, different receivers, and if utilizing visual sensors (remote vision systems), high and low sun angles, various direction headings, and background conditions.
   2. Unmanned System. Flight testing with targeted air vehicles verifies sensing system receives data within parameters needed to accomplish AR within the system's mission defined environmental conditions. Test points include receiver join-up, pre-contact, contact, and in-contact maneuvering. Stressing test conditions depend on the method of data acquisition (e.g., differential GPS, lidar, visible cameras).
c. Receptacle Receiver.
   1. Human Operator. Flight testing with targeted air vehicles verifies situation awareness data and visual cues required to accomplish AR within the system's mission defined environmental conditions are provided and, if not directly viewed, accurately displayed. Test points include various ambient lighting conditions (e.g., day, night, dusk), with and without NVIS, join up, astern, contact position, throughout the boom envelope, and if utilizing visual sensors (remote vision systems), high and low sun angles, various direction headings, and background conditions.
   2. Unmanned System. Flight testing with targeted air vehicles verifies sensing system receives data within parameters needed to accomplish AR within the system's mission defined environmental conditions. Test points include join up, astern, contact position, and throughout the boom envelope. Stressing conditions depend on the method of data acquisition (e.g., differential GPS, lidar, visible cameras).
d. Probe Receiver.
   1. Human Operator. Flight testing with targeted air vehicles verifies situation awareness data and visual cues needed to accomplish AR within the system's mission defined environmental conditions are provided and, if not directly viewed, accurately displayed. Test points include various ambient lighting conditions (e.g. day, night, dusk), with and without NVIS, join up, pre-contact, contact, in contact maneuvering, and if utilizing visual sensors (remote vision systems), high and low sun angles, various direction headings and background conditions.
   2. Unmanned System. Flight testing with targeted air vehicles verifies sensing system receives data within parameters needed to accomplish AR within the system's mission defined environmental conditions are provided and, if not directly viewed,

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accurately displayed. Test points include join up, pre-contact, contact, and in-contact maneuvering. Stressing conditions depend on the method of data acquisition (e.g., differential GPS, lidar, visible cameras).

10.1.5.2 Cues and data for targeted sensing system.

Standard: Cues and data (visual or equivalent) are provided on the air vehicle and can be received and interpreted by the targeted air vehicle sensing system and do not degrade targeted sensing system performance, under mission-defined environmental conditions to facilitate AR. Mission-defined environmental conditions include day, night, dusk, twilight, high and low sun angles, sun in and out of FOV, with and without NVIS, ambient moonlight (waxing and waning) in and out of FOV, diverse backgrounds (e.g., snow, ocean, desert, populated city lights, foliage dominant), above clouds, below clouds, and in clouds.

a. Boom Tanker. Cues (visual or equivalent) for cleared to contact, position keeping, and breakaway are provided by the boom tanker can be received by the targeted receptacle equipped-receiver air vehicle. For current AR operations, these include receiver positioning markings, AR boom markings showing inner and outer receiver contact limit, size and movement indicators and associated lighting, exterior lighting, including flood lights, wing illumination, underbody illumination, boom nozzle illumination, PDLs, boom marker lights, position lights, rendezvous lights, formation lights, and any other exterior lighting necessary to accomplish AR activities.
b. Drogue Tanker. Cues (visual or equivalent) for ready for contact, fuel system status, position keeping, and breakaway are provided by the drogue tanker can be received by the targeted probe-equipped receiver air vehicle. For current AR operations, these include receiver positioning markings, AR hose markings (including length intervals, inner and outer fuel transfer range and full trail exceedance), exterior lighting including flood lights, wing illumination, underbody illumination, wing pod illumination, engine nacelle illumination, drogue illumination, hose exit tunnel illumination, hose illumination, drogue system status lights, position lights, rendezvous lights, formation lights, and any other exterior lighting necessary to accomplish AR activities.
c. Receptacle Receiver. Cues (visual or equivalent) needed to enable AR including receptacle system status and location are provided by the receiver and can be received by the targeted boom tanker air vehicle. For current AR operations these include boom lead-in markings in front of the receptacle, receptacle illumination, slipway illumination, position lights, rendezvous lights, formation lights, and any other exterior lighting necessary to accomplish AR activities.
d. Probe Receiver. Cues (visual or equivalent) needed to enable AR including probe system status and location are provided by the receiver and can be received by the targeted drogue tanker air vehicle. For current AR operations these include probe illumination, position lights, rendezvous lights, formation lights, and any other exterior lighting necessary to accomplish AR activities.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examination of system specifications verifies necessary cues and data are provided for AR operations within the system's mission defined environmental conditions. Examination of sensing system design verifies necessary cues and data to conduct AR operations are provided.

Analysis.

(1) Task analysis determines visual cues needed by targeted human viewed systems to accomplish AR. Analysis of air vehicle design verifies required data and visual cues are provided within the system's mission defined environmental conditions.

(2) For unmanned systems, computer modeling of refueling tasks determines data and associated performance parameters required for targeted unmanned systems to accomplish AR. Analysis of air vehicle design verifies data is provided.

Demonstration.

(1) Air vehicle ground demonstration verifies data and cues needed to conduct AR operations with targeted air vehicles are provided in simulated mission defined environmental conditions.

Test.

(1) Lab test: Hardware-in-the-loop laboratory testing verifies cues and data (visual and equivalent) needed to conduct AR is provided in simulated mission defined environmental conditions is provided.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing with targeted air vehicles verifies data and visual cues needed to accomplish AR within the system's mission defined environmental conditions are provided and interpretable. Test points include various ambient lighting conditions (e.g., day, night, dusk), receiver join up, astern/pre-contact, contact position, multiple receivers, in contact maneuvering. A qualitative crewmember assessment will be used to verify data and visual cues for human operated systems.

10.1.5.3 Intensity and variability of exterior lighting.

Standard: Exterior lights utilized during AR are independently controllable such that the intensity of each exterior light, or light group, can be varied to accommodate targeted tanker/receivers in all environmental lighting conditions as well as tanker/receiver orientations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of system specifications verifies lighting utilized during AR operations are independently controllable.

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Demonstration.

(1) Air vehicle ground demonstration verifies exterior lights utilized during AR are independently controllable.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Flight testing verifies each exterior light, or light group, can be varied to accommodate target air vehicle in all environmental lighting conditions. For human operated systems, qualitative crewmember assessments verify exterior lights can be independently controlled to accomplish all AR tasks within the full range of mission defined environmental conditions. Test points include day, twilight, night, with and without NVIS, throughout all phases of the AR process.

10.1.5.4 Night vision imaging system compatibility of exterior lighting.

Standard: Exterior lights utilized during AR are NVIS compatible for both visual (naked eye visible light spectrum) and covert (not naked eye visible) modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of system specifications verifies exterior lighting utilized during AR operations is NVIS compatible.

Demonstration.

(1) Air vehicle ground demonstration verifies exterior lights utilized during AR are NVIS compatible.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Flight testing verifies exterior lighting is NVIS compatible. For human operated systems, qualitative crewmember assessments verify NVIS compatibility with visible and covert exterior lights to accomplish all AR tasks within the full range of the system's mission defined environmental conditions. Test points include receiver join up and echelon, astern, and contact positions.

10.1.5.5 Lighting and marking durability.

Standard: All exterior lights and markings used during the AR process withstand exposure to expected environmental conditions and fluid exposures (e.g., fuel, hydraulic fluid, air vehicle cleaning solvents) without degradation in performance or appearance that creates a hazard to air vehicle or personnel.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of system specification verifies AR lighting and markings withstand exposure to expected environmental conditions and fluid exposures.

Analysis.

(1) Analysis of materials ensures lighting and markings are compatible with aviation environments and fluids, including but not limited to rain, salt fog, sand and dust, fungus, jet fuel, hydraulic fluid, isopropyl alcohol, and de-icing fluid.

Test.

(1) Lab test: Component qualification testing verifies lighting and markings utilized for AR can withstand expected environmental conditions and fluid exposure.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight test results show no incidents of degradation of lighting or marking visibility or interpretability due to environment or fluid exposure.

10.1.5.6 Aerial refueling status and commands.

Standard: Required data is exchanged between the air vehicle and the targeted air vehicle(s) during the AR process. System used to exchange data is compliant with appropriate classified information transfer requirements. Unmanned systems send and receive data within allowable design parameters.

a. Boom Tanker. Tanker has a method to provide AR status, commands, and acknowledgements to receptacle-equipped receiver air vehicle. AR status and commands include but are not limited to disconnect, breakaway, receiver position and rate corrections (e.g., verbal corrections, PDLs or data package), clear to contact, lighting level changes, heading changes, altitude changes, airspeed changes, and EMCON signals.
b. Drogue Tanker. Tanker has a method to provide AR status, commands, and acknowledgements to probe-equipped receiver air vehicle. AR status and commands include but are not limited to disconnect, breakaway, clear to contact, lighting level changes, heading changes, and EMCON signals.
c. Receptacle Receiver. Receiver has a method to provide AR status, commands, and acknowledgements to boom-equipped tanker air vehicle. AR status and commands include but are not limited to contact, disconnect, breakaway, lighting level changes, heading changes, altitude changes, airspeed changes, and EMCON signals.
d. Probe Receiver. Receiver has a method to provide AR status, commands, and acknowledgements to drogue-equipped tanker air vehicle. AR status and commands include but are not limited to disconnect, breakaway, lighting level changes, heading changes, altitude changes, airspeed changes, and EMCON signals.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of system specifications verifies that necessary data can be sent to and received from targeted air vehicles during the AR process.

Analysis.

(1) Task and failure analyses identify communications, time constraints, and reliability requirements.

Test.

(1) Lab test: Hardware-in-the-loop laboratory testing with simulated targeted system verifies necessary data is sent and received within allowable design parameters.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing with targeted air vehicles in operational conditions including EMCON scenarios verify necessary data is exchanged between air vehicles to enable AR operations.

10.1.5.7 Aerial refueling system data.

Standard: Information is provided to the flight crew or automated system to facilitate AR operations.

a. Boom Tanker. Information necessary for boom refueling includes but is not limited to boom position, fuel transfer rates, fuel pressures, fuel temperature, load alleviation status (if applicable), hydraulic system pressure, boom FCS mode (if applicable), boom nozzle load (if applicable), functional mode (e.g., NORMAL or OVERRIDE), system status (e.g., READY, CONTACT, DISCONNECT), and abnormal states.
b. Drogue Tanker. Information necessary for drogue refueling includes but is not limited to: AR system operational status (e.g., ON/OFF, EXTEND, TRAIL, REWIND, STOWED), fuel transfer rates, fuel pressures, fuel temperature, ram air turbine speed (if applicable), and hydraulic system pressure (if applicable).
c. Receptacle Receiver. Information necessary for boom refueling includes but is not limited to indicators of AR system status (e.g., READY, CONTACT, and DISCONNECT, receptacle position (e.g., OPEN/CLOSED), and functional mode (e.g., NORMAL/OVERRIDE)).
d. Probe Receiver. Probe position and status (if applicable) is provided.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examination of design documentation verifies that the information necessary for AR operations is provided to the flight crew or automated system.

Test.

(1) Lab test: Hardware-in-the-loop laboratory testing verifies information to facilitate AR operations is provided to the flight crew or automated system in a usable format.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing with targeted air vehicles verifies the information to facilitate AR operations is provided to the flight crew or automated system.

10.1.6 Fuel system compatibility.

Fuel system compatibility considers unique AR AW attributes to include fuel system types, AR fuel system pressure limits, AR fuel delivery pressures, and provision for AR fuel surge pressure between tanker and target air vehicles

10.1.6.1 Identification of fuels.

Standard: Fuel types permitted to be transferred and received during AR are identified and documented and there is at least one fuel type that is compatible for both systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of air vehicle flight manuals or test procedures verifies approved fuels are documented for each air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Examination of AR compatibility assessment and air vehicle flight manuals verify approved fuels are documented for each AR pairing.

10.1.6.2 Design limit pressures.

Standard: The air vehicle refueling systems and components meet operational, proof, and ultimate pressure requirements.

a. Boom Tanker. At least 120 (-5) psi operational, 240 (-10) psi proof, 360 psi ultimate (burst).
b. Drogue Tanker. At least 120 (-5) psi operational, 240 (-10) psi proof, 360 psi ultimate (burst).

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c. Receptacle Receiver. At least 90 (-3) psi operational, 180 (-6) psi proof, 270 psi ultimate (burst).
d. Probe Receiver. At least 60 (-3) psi operational, 120 (-6) psi proof, 180 psi ultimate (burst).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of component and system specifications shows AR subsystem is designed to meet operational, proof, and ultimate pressures.

Analysis.

(1) Analysis of AR system design and component qualification reports verifies system is qualified to the proof and ultimate pressures.

Test.

(1) Lab test: Lab testing of components and fuel subsystem/test rig verifies AR components and subsystems can withstand pressure limits. The components, individually, and the system are capped and pressurized up to the positive proof pressure limit for five minutes and then depressurized. The system is inspected for any signs of structural deformation and leakage. The system is also proof pressure tested to its negative limit in the same fashion. If the system cannot be pressure tested as a whole, then it is acceptable to test the system in multiple parts ensuring the breaks in the system are not areas of concern.

(2) Ground test: Air vehicle ground testing of AR subsystems verifies AR components and subsystems can withstand pressure limits. The system is capped and pressurized up to the positive proof pressure limit for five minutes and then depressurized. The system is inspected for any signs of structural deformation and leakage. The system is performance tested. The system is also proof pressure tested to its negative limit in the same fashion.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing with targeted air vehicle verifies maximum pressures does not exceed fuel system proof pressures and shows no unexpected fuel leakage, no permanent deformation or failures of fuel systems or components of either air vehicle involved in the refueling process.

10.1.6.3 Delivery pressure.

Standard: Tanker fuel delivery pressure does not exceed 55 psi at any flowrate, from maximum to zero flow, to ensure pressures remain within tanker and receiver design limits for normal and single failure conditions during AR operations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examination of system specification verifies fuel delivery system is required to deliver fuel to receiver air vehicle at a maximum of 55 psi at all flowrates.

Analysis.

(1) Analysis of fuel system design and fuel flow simulation verifies fuel delivery pressure will not exceed 55 psi for any flowrate condition.

Test.

(1) Lab test: Lab test of the AR delivery system to a simulated receiver air vehicle, at all possible delivery flowrates, with and without single failures, verifies fuel delivery pressure does not exceed 55 psi at the receiver interface.

(2) Ground test: Ground test of the AR delivery system, including offloading from a tanker air vehicle to a receiver air vehicle or simulated receiver air vehicle, at worst-case delivery flowrates (including single failure conditions) verifies fuel delivery pressure does not exceed 55 psi at the receiver interface.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing of worst-case tanker fuel pump configuration (including single failure condition) with receiver air vehicle verifies delivery pressure does not exceed 55 psi.

10.1.6.4 Surge pressures.

Standard: The air vehicle design prevents surge pressures, generated during the AR process, from exceeding proof pressure limits of the AR and fuel systems of the air vehicles involved under all possible flowrates and delivery pressures, including single failure conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of specifications verifies design is required to limit surge pressures to below proof pressure. Examination of drawings show fuel system design incorporates features (e.g., pressure regulators, surge suppressors, return valves, auto disconnect functions) to prevent fuel surges during AR, resulting from pump start up and valve closures, with and without single failures of fuel system components, from exceeding allowable proof pressures.

Analysis.

(1) Fuel flow modeling and simulation of AR operations verifies pressure surges resulting from pump start up and valve closures during onload and offload, with and without single failures of fuel system components, will not exceed specified air vehicle proof pressures.

Test.

(1) Lab test: Lab testing of fuel subsystem test rig with simulated receiver or tanker verifies surge pressures do not exceed proof pressures, with and without single fuel system component

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failures. Instrumentation should be placed within 3 ft of the AR interface and at the worst-case pressure locations based on analysis (e.g., immediately upstream of refuel shutoff valves, dead ends). Measurements are made with a minimum 1,000 Hz sample rate, or as required to accurately capture peak surge pressures. Test conditions include pump start up, fuel flow shutoffs, including all possible shutoff combinations, individual tank valve closures, fill to full, flowing disconnects, and single failure of surge protection components (e.g., regulator, surge boot, surge suppressor).

(2) Ground test: Air vehicle ground testing with targeted air vehicle or representative substitute system verifies surge pressures do not exceed proof pressures, with and without single fuel system component failures. Instrumentation should be placed within 3 ft of the AR interface and at the worst-case pressure locations based on analysis (e.g., immediately upstream of refuel shutoff valves, dead ends). Measurements are made with a minimum 1,000 Hz sample rate, or as needed to accurately capture peak surge pressures. Test conditions include, pump start up, fuel flow shutoffs, including all possible shutoff combinations, individual tank valve closures, fill to full, flowing disconnects, and single failure of surge protection components (e.g., regulator, surge boot, surge suppressor).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing with targeted air vehicle verifies surge pressures do not exceed specified proof pressures of either system involved in the refueling process. Instrumentation should be placed within 3 ft of the AR interface and as necessary to ensure maximum fuel system pressures are recorded during flight tests based on ground test results. Measurements are made with a minimum 1,000 Hz sample rate, or as needed to accurately capture peak surge pressures. Test conditions include, pump start up, fuel flow shutoffs, flowing disconnects, fill to full, and fuel tank shutoff conditions that are possible during AR.

10.2 Necessary functions.

Standard: AR systems have functions to ensure AR operations occur without creating a hazardous condition.

a. Boom Tanker. Tanker has the ability to reduce commanded delivery pressure, to advance boom state to contact, to send a disconnect signal to the receiver when commanded by the operator or automated system, and to transition the boom into and out of the stowed position.
b. Drogue Tanker. Tanker has the ability to reduce commanded delivery pressure, to extend and retract the refueling hose and drogue, and to jettison the refueling hose and trailing components.
c. Receptacle Receiver. Receiver has the ability to disconnect from the boom and to send a disconnect signal to the tanker when commanded by the crewmember or automated system, to reset the receptacle system state from disconnect to ready, to bypass the receptacle signal amplifier (if applicable), and to transition the receiver into and out of refueling configuration (e.g., open and close refueling door(s), transition flight control law logic, fuel transfer logic).

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d. Probe Receiver. Receiver has the ability to transition into and out of refueling configuration (e.g., probe extension and retraction, transition flight control law logic, fuel transfer logic).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the standard by the following:

a. Boom Tanker. Examination of specifications and design documentation verifies tanker has functions to reduce commanded delivery pressure, to advance boom state to contact, to send a disconnect signal to the receiver when commanded by the crewmember or automated system, and to transition the boom into and out of the stowed position.
b. Drogue Tanker. Examination of specifications and design documentation verifies tanker has functions to reduce commanded delivery pressure, to extend and retract the refueling hose and drogue, and to jettison the refueling hose and trailing components.
c. Receptacle Receiver. Examination of specifications and design documentation verifies receiver has functions to disconnect from the boom and to send a disconnect signal to the tanker when commanded by the crewmember or automated system, to reset the receptacle system state from disconnect to ready, to bypass the receptacle signal amplifier (if applicable), to transition the receiver into and out of refueling configuration (e.g., open and close refueling door(s), transition flight control law logic, fuel transfer logic).
d. Probe Receiver. Examination of specifications and design documentation verifies receiver has functions to transition into and out of refueling configuration (e.g., probe extension and retraction, transition flight control law logic, fuel transfer logic).

Analysis.

(1) Verify the standard by the following:

a. Boom Tanker. Analysis of tanker design, AR software logic and associated models verify tanker has the capability to reduce commanded delivery pressure, to advance boom state to contact, to send a disconnect signal to the receiver when commanded by the crewmember or automated system, and to transition the boom into and out of the stowed position.
b. Drogue Tanker. Analysis of tanker design, AR software logic and associated models verify tanker has the capability to reduce commanded delivery pressure, to extend and retract the refueling hose and drogue, and to jettison the refueling hose and trailing components.
c. Receptacle Receiver. Analysis of receiver design, AR software logic and associated models verify receiver has the capability to disconnect from the boom and to send a disconnect signal to the tanker when commanded by the crewmember or automated system, to reset the receptacle system state from disconnect to ready, to bypass the receptacle signal amplifier (if applicable), and to transition the receiver into and out of refueling configuration (e.g., open and close refueling door(s), transition flight control law logic, fuel transfer logic).

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d. Probe Receiver. Analysis of receiver design, AR software logic and associated models verify receiver has the capability to transition into and out of refueling configuration (e.g., probe extension and retraction, transition flight control law logic, fuel transfer logic).

Test.

(1) Lab test: Verify the standard by the following:

a. Boom Tanker. Testing in a laboratory environment verifies fuel system delivers fuel at a reduced delivery pressure when commanded, boom state advances to contact when commanded and disconnect signal is sent via the signal coil when commanded by the crewmember or automated system.
b. Drogue Tanker. Testing in a laboratory environment verifies fuel system delivers fuel at a reduced delivery pressure when commanded.
c. Receptacle Receiver. Testing in a laboratory environment verifies receptacle latches release the boom nozzle and a disconnect signal is sent via the signal coil when commanded by the crewmember or automated system, the receptacle system state resets from disconnect to ready when commanded, receptacle signal amplifier can be bypassed (if applicable), and receiver transitions into and out of refueling configuration (e.g., open and close refueling door(s), transition flight control law logic, fuel transfer logic) when commanded.
d. Probe Receiver. Testing in a laboratory environment verifies receiver transitions into and out of refueling configuration (e.g., probe extension and retraction, transition flight control law logic, fuel transfer logic) when commanded.

(2) Ground test: Verify the standard by the following:

a. Boom Tanker. Air vehicle ground testing verifies fuel system delivers fuel at a reduced delivery pressure when commanded.
b. Drogue Tanker. On air vehicle ground testing verifies fuel system delivers fuel at a reduced delivery pressure when commanded.
c. Receptacle Receiver. Air vehicle ground testing verifies receiver transitions into and out of refueling configuration (e.g., open and close refueling door(s), transition flight control law logic, fuel transfer logic) when commanded.
d. Probe Receiver. Air vehicle ground testing verifies receiver transitions into and out of refueling configuration (e.g., probe extension and retraction, transition flight control law logic, fuel transfer logic) when commanded.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Verify the standard by the following:

a. Boom Tanker. Flight test verifies tanker fuel system delivers fuel at a reduced delivery pressure when commanded, boom state advances to contact when commanded, disconnect signal is sent to the receiver when commanded by the operator or automated system, and the boom transitions into and out of the stowed position when commanded.

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b. Drogue Tanker. Flight test verifies tanker fuel system delivers fuel at a reduced delivery pressure when commanded, the refueling hose and drogue are extended and retracted when commanded, and the refueling hose and trailing components are jettisoned when commanded.
c. Receptacle Receiver. Flight test verifies receptacle latches release the boom nozzle and a disconnect signal is sent to the tanker when commanded by the operator or automated system, the receptacle system state resets from disconnect to ready when commanded, receptacle signal amplifier can be bypassed (if applicable), and receiver transitions into and out of refueling configuration (e.g., open and close refueling door(s), transition flight control law logic, fuel transfer logic) when commanded.
d. Probe Receiver. Flight test verifies receiver transitions into and out of refueling configuration (e.g., probe extension and retraction, transition flight control law logic, fuel transfer logic) when commanded.

10.3 Failure modes.

Standard: Failure modes that impact AR operations are identified and characterized, including failures of the AR systems and other system failures that impact AR operations (e.g., engine failure, landing gear retraction failure, hydraulic system failure, electrical system failure). Instructions, WCAs, notes, and restrictions for refueling operations with failures are documented in flight manuals.

a. Boom Tanker. Failure modes, including but not limited to, signal amplifier and coil, automatic load alleviation system, AR lighting, sensing systems, boom hoist, and fuel delivery system are characterized and requirements for continued operations are documented.
b. Drogue Tanker. Failure modes, including but not limited to, hose response system, refueling hose, refueling coupling, drogue, guillotine (if applicable), jettison system, AR lighting, sensing systems, and fuel delivery system are characterized and requirements for continued operations are documented.
c. Receptacle Receiver. Failure modes, including but not limited to, receptacle door(s), receptacle latches, signal amplifier and coil, fuel system, and AR lighting are characterized and requirements for continued operations are documented.
d. Probe Receiver. Failure modes, including but not limited to, probe extension/deployment system, fuel system, and AR lighting are characterized and requirements for continued operations are documented.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of system specifications and design documentation identify failure modes and continued operation requirements.

(2) Examination of flight manuals verifies instructions for operations with failures is documented.

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Analysis.

(1) Analysis of system design, FMEA, and FMECA characterize continued refueling operation with failure cases.

Test.

(1) Lab test: Hardware-in-the-loop lab testing of failure modes validates predicted performance.

(2) Ground test: Air vehicle ground testing of failure modes validates predicted performance.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing verifies performance of continued refueling operations with failure conditions.

10.4 Built-In-Test.

Standard: Integrated functional testing, including start-up, continuous, and initiated, identify AR and related system faults, and inform crewmember or automated system. A test or check-out function (preflight or pre-use) is incorporated for systems required for AR operations, including, but not limited to, receptacle doors, probe extension, refueling system control surfaces, refuel valves, and AR sensors.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of specifications, schematics, and design documentation show system monitors and reports AR system and support systems faults that could cause hazardous conditions.

Analysis.

(1) Analysis of design, such as a FMEA and FMECA, determines critical faults and failures that need BIT monitoring and fault isolation.

Test.

(1) Lab test: Hardware-in-the-loop laboratory testing verifies that monitoring, diagnostic, and BIT systems identify and report critical faults of AR and related systems.

(2) Ground test: Air vehicle ground testing of simulated fault conditions verifies functional tests identify and report failures.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing of simulated fault conditions verifies integrated test functions identify and report failures. Actual faults that occur throughout flight test are accurately identified and reported.

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10.5 Technical manuals.

Standard: Flight and maintenance manuals document normal and emergency procedures for the AR systems including applicable limitations, restrictions, warnings, cautions, notes, and allowable environmental conditions. Air vehicles approved by Aerial Refueling Certification Agency to AR with the platform including associated clearance levels and defined AR envelopes are identified.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of draft flight and maintenance manuals and flight test procedures verify AR procedures, allowable conditions, and refueling envelopes are documented.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Examination of flight and maintenance manuals verify AR procedures, allowable conditions and refueling envelopes are documented.

Test.

(1) Flight test: Flight test verifies procedures are accurate and usable.

10.6 Component wear and damage levels.

Standard: Inspection criteria for wear and damage are defined and documented for AR system components. Common wear items include boom nozzles, refueling hoses, refueling couplings and para-drogues, probe nozzles, and receptacles.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of design documentation shows allowable physical dimensions and performance parameters for AR system and components.

Test.

(1) Lab test: Qualification testing in a laboratory environment verifies AR component performance and allowable wear levels.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Examination of maintenance plans and manuals verifies allowable wear and damage levels are identified and replacement instructions are provided.

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10.7 Aerial refueling electronic hardware, software, and firmware.

Standard: Electronic hardware, software, and firmware used to perform AR operations which support an SCF are identified and documented as safety-critical. Control actions and movements of an AR system, and monitoring of loads, pressures, states, and positions of AR components are examples of safety-critical functionality.

Note: All air system SCFs are first identified by Section 4 criteria. This standard provides the means to identify the SCF subset specific to AR operations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examination of development documentation verifies that electronic hardware, software, and firmware used to perform AR operations which support an SCF are identified as safety-critical.

Analysis.

(1) SCFTA identifies all electronic hardware, software, and firmware items that support AR related SCFs.

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11. AVIONICS

11.1 contains the standards and MoC for communication, navigation and identification subsystems. 11.2 contains the standards and MoC for avionics architecture. 11.3 contains the standards and MoC for the safety-critical displays and controls. 11.4 contains the standards and MoC for diagnostic systems. 11.5 contains the standards and MoC for the engineering data expected in the technical manuals. The criteria use a buildup approach based on design, analysis and test data to demonstrate compliance with the applicable subsystems and systems.

11.1 Communication, navigation, identification subsystems.

Communication, navigation, identification encompasses the systems and equipment used to effectively communicate between the air system and ATC, enable accurate navigation for flight paths, and reduces collision risks by enhancing situational awareness. Communication, navigation, identification criteria uniquely combine military and civil functionalities, with CNS/ATM (the civil functionalities) being statutory requirements. Together these functionalities contribute to AW by ensuring that the air system can reliably attain, sustain, and complete its mission within approved usage limits in controlled airspace. Air vehicles not meeting civil CNS/ATM standards may be subjected to ground delays waiting for clearance, directed to operate on less optimum routes, or denied civil airspace access.

Department of War AW authorities self-certify compliance with CNS/ATM requirements to enable operations in civil airspace. To ensure CNS/ATM compliance, each DoW service has specific processes to tailor CNS/ATM standards based on user-requested capabilities and the airspace they intend to access. Programs should tailor these standards and reference the tailored standards in the AW certification basis and verify them in the compliance report.

11.1.1 Communication.

This section covers off-board communication subsystems that transmit and receive safety-critical voice and data information. Internal air vehicle communication subsystems, such as intercom and PA systems, are covered in Section 9. Although voice intelligibility using MRT is required for 11.1.1.1, the MoC in 9.2.7.1 will be utilized to determine compliance.

11.1.1.1 Voice communications.

Standard: The availability, continuity, integrity, and intelligibility of voice communication systems (e.g., Civil and Military ATC) support safety-critical communications. Communication performance complies with national and worldwide civil airspace requirements in the intended area of operations for CNS/ATM capabilities such as VHF Voice, SATVOICE, and HF Voice.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the communication subsystem architecture and identify the safety-critical hardware and software item list.

(2) Determine the CNS/ATM impacts, tailor the requirements from airspace authorities such as the FAA and ICAO, and document these requirements in the AW certification basis by reference.

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Analysis.

(1) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(2) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10-7 per flight hour.

Test.

(1) Lab test: Conduct HITL lab testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

(2) Lab test: Conduct HITL lab testing to introduce system faults identified in analysis and evaluate the consequence of those failures.

(3) Ground test: Conduct air vehicle ground testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) Show compliance with tailored CNS/ATM performance requirements.

Test.

(4) Flight or Ground test: Conduct ground or flight MRT to demonstrate voice intelligibility.

(5) Flight test: Conduct flight testing to verify 5x5 voice intelligibility at maximum range required for safety-critical communications.

(6) Flight test: Conduct flight testing to verify performance requirements and tailored CNS/ATM requirements.

11.1.1.2 Data communications.

Standard: The availability, continuity, and integrity of data communication systems (e.g., Civil ATC datalinks, safety-critical telemetry datalinks) support safety-critical communications. Communication performance complies with national and worldwide civil airspace requirements in the intended area of operations for CNS/ATM capabilities such as VDL Mode 2, HFDL, SATCOM, AFN, ADS-C, and CPDLC.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the communication subsystem architecture and identify the safety-critical hardware and software item list.

(2) Determine the CNS/ATM impacts, tailor the requirements from airspace authorities such as the FAA and ICAO, and document these requirements in the AW certification basis by reference.

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Analysis.

(1) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(2) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10-7 per flight hour.

Test.

(1) Lab test: Conduct HITL lab testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

(2) Lab test: Conduct HITL lab testing to introduce system faults identified in analysis and evaluate the consequence of those failures.

(3) Ground test: Conduct air vehicle ground testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) Show compliance with tailored CNS/ATM performance requirements.

Test.

(4) Flight test: Conduct flight testing to verify performance requirements and tailored CNS/ATM requirements.

11.1.1.3 Datalink command and control.

Standard: The availability, latency, continuity, and integrity of off-board datalink C2 (e.g., automatic and semi-automatic man-in-the-loop guidance) support safety-critical communications.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the communication subsystem architecture and identify the safety-critical hardware and software item list.

(2) Document the minimum requirements for availability, latency, continuity, and integrity.

(3) Verify redundancy is incorporated into the system and document lost link contingency procedures.

Analysis.

(1) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(2) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10^{-7} per flight hour.

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(3) Accomplish a link margin analysis to ensure the reliability of safety-critical communications.

(4) Demonstrate that antenna patterns provide the necessary coverage to preclude signal loss and that the impact of antenna nulls is mitigated.

Test.

(1) Lab test: Conduct HITL lab testing to verify performance requirements of the representative hardware and software configuration.

(2) Lab test: Conduct HITL lab testing to introduce system faults identified in analysis and evaluate the consequence of those failures.

(3) Ground test: Conduct air vehicle ground testing to verify performance requirements of the representative hardware and software configuration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing to verify installed performance requirements, datalink switchover (if applicable), and lost link contingency procedures.

11.1.1.4 Communications antenna performance.

Standard: Antenna coverage, gain, and link margin for voice and data communications support safety-critical communications, and antenna co-site performance is compatible with other air vehicle RF systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the communication subsystem architecture and identify the safety-critical hardware and software item list.

(2) Document the minimum requirements for range, link margin, antenna coverage and antenna gain.

(3) Verify that antennas are compatible with paired LRUs and meet performance standards.

Analysis.

(1) Accomplish a link margin analysis to establish the reliability of safety-critical communications.

(2) Demonstrate that antenna patterns provide the necessary coverage to preclude signal loss and that the impact of antenna nulls is mitigated.

(3) Perform a co-site analysis to ensure that co-site interference is mitigated.

(4) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

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(5) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10^{-7} per flight hour.

Test.

(1) Ground test: Conduct ground testing to verify antenna performance requirements and validate the co-site analysis results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Document a frequency management plan based on co-site analysis and testing.

Test.

(2) Flight test: Conduct flight testing to verify antenna performance requirements and validate co-site analysis and testing.

11.1.2 Navigation.

11.1.2.1 Navigation performance.

Standard: The accuracy, continuity, and integrity of navigation systems support safety-critical navigation. Navigation performance complies with national and worldwide civil airspace requirements in the intended area of operations for CNS/ATM navigation capabilities such as RVSM, VNAV, RNP, RNAV, LPV/LP, Navigational Database Processing, and TAWS.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the navigation subsystem architecture and identify the safety-critical hardware and software item list.

(2) Document the minimum requirements for accuracy, continuity, and integrity of the navigation system.

(3) Determine the CNS/ATM impacts, tailor the requirements from airspace authorities such as the FAA and ICAO, and document these requirements in the AW certification basis by reference.

Analysis.

(1) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(2) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10^{-7} per flight hour.

Test.

(1) Lab test: Conduct HITL lab testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

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(2) Lab test: Conduct HITL lab testing to introduce system faults identified in analysis and evaluate the consequence of those failures.

(3) Ground test: Conduct air vehicle ground testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Show compliance with tailored CNS/ATM performance requirements.

Test.

(4) Flight test: Conduct flight testing to verify performance requirements and tailored CNS/ATM requirements.

11.1.2.2 Navigation antenna performance.

Standard: Antenna coverage and gain for the navigation subsystem support safety-critical navigation, and antenna co-site performance is compatible with other air vehicle RF systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the navigation subsystem architecture and identify the safety-critical hardware and software item list.

(2) Verify that antennas are compatible with paired LRUs and meet performance standards.

Analysis.

(1) Demonstrate that antenna patterns provide the necessary coverage to preclude signal loss and that the impact of antenna nulls is mitigated.

(2) Perform a co-site analysis to ensure that co-site interference is mitigated.

(3) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(4) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10^{-7} per flight hour.

Test.

(1) Ground test: Conduct ground testing to verify antenna performance requirements and validate the co-site analysis results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) Document a frequency management plan based on co-site analysis and testing.

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Test.

(2) Flight test: Conduct flight testing to verify antenna performance requirements and validate co-site analysis and testing.

11.1.3 Identification.

11.1.3.1 Air traffic control surveillance.

Standard: The availability, continuity, and integrity of ATC surveillance systems support safety-critical surveillance. Surveillance performance complies with national and worldwide civil airspace requirements in the intended area of operations for CNS/ATM capabilities such as Mode S, and ADS-B Out.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the surveillance subsystem architecture and identify the safety-critical hardware and software item list.

(2) Determine the CNS/ATM impacts, tailor the requirements from airspace authorities such as the FAA and ICAO and document these requirements in the AW certification basis by reference.

(3) Obtain AIMS box level certification.

Analysis.

(1) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(2) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10-7 per flight hour.

Test.

(1) Lab test: Conduct HITL lab testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

(2) Lab test: Conduct HITL lab testing to introduce system faults identified in analysis and evaluate the consequence of those failures.

(3) Ground test: Conduct AIMS ground testing with no noted safety related deficiencies.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Obtain AIMS platform level certification.

(5) Show compliance with tailored CNS/ATM performance requirements.

Test.

(4) Flight test: Conduct flight testing to verify performance requirements and tailored CNS/ATM requirements.

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(5) Flight test: Conduct AIMS flight testing with no noted safety related deficiencies.

11.1.3.2 Airborne collision avoidance systems.

Standard: The availability, continuity, and integrity of airborne collision avoidance systems support safety-critical surveillance. Surveillance performance complies with national and worldwide civil airspace requirements in the intended area of operations for CNS/ATM capabilities such as TCAS, MILACAS-FR, MILACAS-XR, ACAS X, E-TCAS, and DAA systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the surveillance subsystem architecture and identify the safety-critical hardware and software item list.

(2) Determine the CNS/ATM impacts, tailor the requirements from airspace authorities such as the FAA and ICAO, and document these requirements in the AW certification basis by reference.

Analysis.

(1) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(2) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10^{-7} per flight hour.

Test.

(1) Lab test: Conduct HITL lab testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

(2) Lab test: Conduct HITL lab testing to introduce system faults identified in analysis and evaluate the consequence of those failures.

(3) Ground test: Conduct air vehicle ground testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) Show compliance with tailored CNS/ATM performance requirements.

Test.

(4) Flight test: Conduct flight testing to verify performance requirements and tailored CNS/ATM requirements.

11.1.3.3 Passive surveillance systems.

Standard: The availability, continuity, and integrity of passive surveillance systems (e.g., ADS-B In, Mode 5 Level 2 In, Mode 5 Level 2B In) support safety-critical surveillance. Surveillance

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performance complies with national and worldwide civil airspace requirements in the intended area of operations for CNS/ATM capabilities such as ADS-B In applications.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Document the surveillance subsystem architecture and identify the safety-critical hardware and software item list.

(2) Determine the CNS/ATM impacts, tailor the requirements from airspace authorities such as the FAA and ICAO, and document these requirements in the AW certification basis by reference.

(3) Obtain AIMS box level certification.

Analysis.

(1) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(2) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10^{-7} per flight hour.

Test.

(1) Lab test: Conduct HITL lab testing to verify performance requirements and tailored CNS/ATM requirements of the representative hardware and software configuration.

(2) Lab test: Conduct HITL lab testing to introduce system faults identified in analysis and evaluate the consequence of those failures.

(3) Ground test: Conduct AIMS ground testing with no noted safety related deficiencies.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Obtain AIMS platform level certification.

(5) Show compliance with tailored CNS/ATM performance requirements.

Test.

(4) Flight test: Conduct flight testing to verify performance requirements and tailored CNS/ATM requirements.

(5) Flight test: Conduct AIMS flight testing with no noted safety related deficiencies.

11.1.3.4 Identification antenna performance.

Standard: Antenna coverage and gain for surveillance systems support safety-critical surveillance, and antenna co-site performance is compatible with other air vehicle RF systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Document the surveillance subsystem architecture and identify the safety-critical hardware and software item list is identified.

(2) Verify that antennas are compatible with paired LRUs and meet performance standards.

Analysis.

(1) Demonstrate that antenna patterns provide the necessary coverage to preclude signal loss and that the impact of antenna nulls is mitigated.

(2) Perform a co-site analysis to ensure that co-site interference is mitigated.

(3) Verify that the SSHA identifies hazards, their severity, along with any design and procedural mitigations.

(4) Verify failures that can result in critical or catastrophic severity hazards occur at a rate less than or equal to 10^{-7} per flight hour.

Test.

(1) Ground test: Conduct ground testing to verify antenna performance requirements and validate the co-site analysis results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(3) Document a frequency management plan based on co-site analysis and testing.

Test.

(2) Flight test: Conduct flight testing to verify antenna performance requirements and validate co-site analysis and testing.

11.2 Avionics architecture.

The avionics architecture section covers several key areas, including normal modes of operation, off-nominal conditions, installation and integration, and data buses. The standards for normal modes of operation focus on the avionics architecture in a fault-free state of operation, with a particular emphasis on deterministic operation, timing, and latency. Specific timing and latency for avionics control modes are addressed in 11.3.3.2. Timing and latency for the alerting system are addressed in 11.3.1.2. For timing and latency, end-to-end refers to the total time between the signal source (where the data is generated) and the consumption of data (where the data is used, processed, or displayed). Timing anomalies are deviations from the expected timing behavior of a system or component. These deviations can manifest as tasks taking longer to execute than their deadlines, data arriving later than expected, or events occurring out of sequence due to unexpected delays. Latency anomalies are unexpected variations or spikes in the time it takes for data or signals to travel through a system, manifesting as unpredictable delays that can disrupt real-time operations and degrade performance. Off-nominal conditions refer to situations in which the system experiences failures and responds to those failures. These failures can induce redundant operations, timing and latency failures, interface failures, and failure propagation. For avionics standards, interfaces include physical, functional, and logical interfaces. For 11.2.3.1,

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continuity is the ability to maintain operation without interruption or loss of data in the presence of failures or other adverse conditions. Data buses are the backbone of the avionics architecture, and its design needs to ensure reliability and integrity of data buses transporting safety-critical information.

11.2.1 Modes of operation.

11.2.1.1 Deterministic operation.

Standard: The avionics system operates in a deterministic manner to support all SCFs. Verified inputs into a system and subsystem produce expected outputs. Avionics real-time operation is defined and assessed to verify the performance of SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine system specification documentation to verify real-time operation and system determinism support SCFs.

Analysis.

(1) Analyze FMECA, FMEA, SHA, and other timing and design information to ensure the system or subsystem operates deterministically.

Test.

(1) Lab test: SIL testing verifies expected outputs throughout the integrated system and subsystems.

(2) Ground test: Review functional ground test results to verify performance of SCFs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Review functional flight test results to verify performance of SCFs.

11.2.1.2 Timing and latency (normal modes).

Standard: End-to-end timing and latency are defined and shown to support the integrity of all SCFs. For PFR information, 100 milliseconds is the standard, but a different system specified value can be used if analysis and testing show there is no adverse impact to the flight crew or flight control response times.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify design specifications, system requirements, and FMEA identify timing and latency for normal modes for operation of SCFs.

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(2) Analyze timing and latency IAW the specified values to ensure there are no impacts to flight crew or system response times affecting the integrity of the SCFs.

Test.

(1) Lab test: Conduct laboratory testing at worst-case operating conditions to verify the timing for normal modes are IAW specified values and there are no impacts to the flight crew or system response times affecting the integrity of the SCFs.

(2) Ground test: Air vehicle ground tests validate analysis and laboratory test results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to validate analysis, laboratory, and ground test results. Safety-critical function timing and latency performance is verified by flight test.

11.2.2 Off nominal conditions.

11.2.2.1 Redundancy.

Standard: Redundancy mechanisms and switching functions operate to ensure the avionics subsystem provides SCF integrity. The probability of loss or degradation of SCFs capable of resulting in a catastrophic or critical hazard should be 10^{-7} or less per flight hour.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Failure mode effects analysis, SCFTA, SHA, and other design information is analyzed to identify potential failure modes and ensure redundancy is incorporated with separation to support the integrity and reliability requirements. Analysis shows single point failures meet low probability, less than 10^{-7}, of HMI and loss or degradation of SCFs. Analysis shows failures affecting SCFs support the SCF integrity in the standard.

Test.

(1) Lab test: SIL testing and FMET verifies redundant operations for all associated failures. Testing verifies when operating in any redundant degraded mode that any degradation status is clear to the crewmember.

(2) Ground test: Air vehicle ground tests validate redundancy and lab test results through select test cases. Any other ground test failures impacting redundancy show the redundancy mechanization does not result in catastrophic or critical consequences.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(3) Flight test: Record any critical failures encountered related to redundancy of the avionics system. Noted failures demonstrate that any single failure does not result in the loss or degradation of SCFs.

11.2.2.2 Timing and latency anomalies.

Standard: Timing and latency anomalies are assessed and documented. The avionics system implements mitigation measures to ensure anomalies do not result in the loss or degradation of SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review FMEA documentation to identify latency and timing anomalies.

(2) Review test plans and procedures to identify the worst-case operating conditions.

Test.

(1) Lab test: Conduct laboratory testing under the worst-case operating conditions to verify the timing performance of SCFs. Timing and latency tested in the SIL match the predicted values in the system specifications.

(2) Ground test: Air vehicle ground tests will be used to validate laboratory test results. Any ground test issues impacting timing and latency do not impact performance of SCFs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Record any timing and latency anomalies encountered that impact the performance of SCFs. Noted anomalies do not result in the loss or degradation of SCFs.

11.2.2.3 Timing and latency (degraded modes of operation).

Standard: End-to-end timing and latency for backup and emergency modes are defined and shown to support the integrity of all SCFs. For PFR information in these modes, 100 milliseconds is the standard, but a different system specified value can be used if analysis and testing show there is no adverse impact to the flight crew or flight control response times.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review of FMEA, SCFTA, SHA, and other design information identifies all backup and emergency modes of operation for SCFs. Analyze timing and latency IAW the specified values to ensure there are no impacts to flight crew or system response times affecting the integrity of the SCFs.

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Test.

(1) Lab test: Conduct laboratory testing at worst-case operating conditions to verify the timing for backup and emergency modes are IAW specified values and there are no impacts to the flight crew or system response times affecting the integrity of the SCFs.

(2) Ground test: Air vehicle functional ground tests verify timing and latency performance of SCFs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight Test: Record any timing and latency issues encountered that are associated with backup or emergency modes and impact the performance of SCFs. Noted issues do not result in the loss or degradation of SCFs.

11.2.2.4 Interfaces and failure propagation.

Standard: Physical and functional interface failures do not result in HMI or the loss of SCFs. All failures for backup and emergency modes, including transition between modes, are defined, documented, and tested. Failures do not propagate through SCF threads.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify FMEA and SHA identify failures and their effects for interfaces and mode transitions to preclude loss or degradation of SCFs.

(2) Analyze SCFTA to identify possible points of failure propagation for interface failures along the safety-critical avionics threads.

Test.

(1) Lab test: Conduct subsystem- and system-level testing to verify there are no interface failures that could degrade SOF items or equipment. Conduct system-level FMET and monitor for propagation of failures along safety-critical threads.

11.2.3 Installation and integration.

11.2.3.1 Accuracy of safety-critical data.

Standard: The accuracy, continuity, and integrity of all avionics safety-critical parameters support SCF operation for the air vehicle over the entire flight envelope, profiles, and specified geographic locations. Safety-critical data is transmitted between systems and subsystems without the loss of accuracy and integrity. Avionics system hardware and primary flight data parameters listed in MIL-STD-1787, Primary Flight Data paragraph meet air vehicle accuracy, continuity, and integrity specifications.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the detailed design information specifies the required accuracy, continuity, and integrity of safety-critical parameters, primary flight data parameters, and avionics system hardware.

Analysis.

(1) System hazard analysis, FMEA, and FMECA verifies that critical information related failures do not result in loss of information or introduction of HMI.

(2) Conduct FTA of the avionics system's SCFs to ensure continuity and integrity of safety-critical parameters in the presence of failures.

Test.

(1) Lab test: Conduct laboratory testing to verify accuracy meets the specifications in the program detailed design information and does not result in loss of information or introduction of HMI for the air vehicle.

(2) Lab test: Conduct FMET to verify the continuity and integrity of the system.

(3) Ground test: Air vehicle ground tests will be used to validate analysis and laboratory test results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing to validate analysis, laboratory, and ground test results in normal modes.

11.2.3.2 Non-critical failures impacting safety-critical functions.

Standard: Non-safety-critical modifications, failures of non-safety-critical equipment, and test instrumentation do not result in the loss or degradation of SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review design artifacts, SCFTA, and FMECA to identify failures with non-safety-critical equipment that may impact SCFs.

(2) Review test instrumentation documentation, system design, and interface specifications to ensure test instrumentation does not adversely affect the air vehicle.

Test.

(1) Lab test: Conduct FMET and regression tests to verify failures produce expected behaviors and non-safety-critical failures do not adversely impact SCFs.

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(2) Ground test: Perform air vehicle flight regression testing of the modification to verify the performance of SCFs. Flight test results demonstrate non-SCFs and test instrumentation do not impact performance of SCFs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Perform air vehicle flight regression testing to verify the performance of SCFs. Flight test results demonstrate non-SCFs, and any test instrumentation do not impact performance of SCFs.

11.2.3.3 Electrical power.

Standard: Avionics subsystem equipment utilizes electric power without adverse impacts to SCFs, IAW MIL-STD-704, Detailed Requirements paragraph. The avionics subsystem equipment specification specifies the type of electric power to be utilized and the detailed performance required during normal, abnormal, emergency, starting, and transfer operation of the air vehicle electric system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system specifications to identify power performance requirements for avionics equipment during different air vehicle modes.

Analysis.

(1) Analyze system specifications to ensure they meet the requirements for the intended power environment IAW MIL-STD-704, General and Detailed Requirements paragraphs.

(2) Analysis for components that are qualified by similarity address differences in installation, integration, and intended usage from the originally qualified component to ensure an equivalent level of safety.

Test.

(1) Lab test: Verify proper operation of the avionics subsystem by test to demonstrate that the equipment provides required performance within the envelope of possible conditions present within the EPS. Tests of production or production representative equipment verify power quality, IAW MIL-STD-704, General and Detailed Requirements paragraphs. Tests should be performed at the equipment and integration levels for all modes of operation.

(2) Ground test: Ground tests validate avionics system power requirements under all operating conditions and load combinations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(3) Flight test: Flight test results validate avionics system power requirements and ensure no safety-critical deficiencies exist with respect to the compatibility of avionics and the air vehicle power environment.

11.2.4 Data bus.

11.2.4.1 Data bus reliability.

Standard: Data buses, through which safety-critical data transits, meet system specified physical failure rates and data error rates in support of avionics hardware reliability and data integrity. In the absence of system specified physical failure rates, the data bus design will use a threshold MTBF of 10^{-7} failures per flight hour. In the absence of system specified data error rates, the data bus design will use a threshold word error rate of 10^{-7} per word and an undetected BER of 10^{-12} per bit. Maximum allowable word error rates and BERs are normally defined in the standard cited for a data bus implementation. Data buses will meet specific failure probability threshold, such as 10^{-9} per flight hours for catastrophic failures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system specification and design documents that define data bus performance requirements to verify the physical redundancy and switchover mechanism are implemented in the event of a bus failure.

(2) Review system specification documents that define requirements for physical failure rates and data error rates in order to verify the error rates and error correction features.

(3) Review any military or commercial data bus standards and specifications being implemented to ensure that data bus performance adheres to the standards.

(4) For GOTS and COTS assemblies, review contractor technical data to verify data bus specifications and requirements.

Analysis.

(1) Verify hardware failure analysis documents such as FMECA and FMEA determine the physical failure rates are IAW the standard.

(2) Review data bus performance analysis documents for both normal and worst-case operating conditions to identify any issue in error rates and validate the effectiveness of error correction mechanisms.

(3) Review data bus performance simulation results to analyze the failure rates and error rates for data transmission.

Test.

(1) Lab test: Conduct laboratory testing to measure word error rate.

(2) Lab test: Conduct laboratory testing to measure BER.

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(3) Ground test: Conduct air vehicle ground testing to verify data bus failures are detected.

(4) Ground test: Conduct air vehicle ground testing to verify nominal data bus operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Conduct flight testing to verify the data bus operations.

11.2.4.2 Data bus timing.

Standard: Data buses, through which safety-critical data transits, meet the specified data transmission rates in support of system timing requirements. The timing requirements are provided in each data bus specification document. For MIL-STD-1553, the data transfer rates are 1 Mbit/s and for ARINC 429, the data transfer rates are 12.5 kbit/s (low speed) or 100 kbit/s (high speed). For other data bus standards, review the timing requirements IAW the specified commercial or military data bus specification.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system specification documents to define data bus performance requirements.

(2) Review any military or commercial data bus standards and specifications being implemented to verify the timing requirements.

Analysis.

(1) Review data bus performance analysis documents for both normal and worst-case operating conditions to identify any deviations or issues, and that the specified system can handle stress without significant performance degradation.

(2) Review data bus performance simulation results to ensure that the data transmission meets the standards and specific program requirements.

Test.

(1) Lab test: Conduct laboratory testing to ensure data bus transmissions rates are within specifications.

(2) Lab test: Conduct laboratory testing to measure normal and worst-case data bus timing to ensure data is transmitted within the maximum time allowed by the system specification.

(3) Ground test: Conduct air vehicle ground testing to verify data bus operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing to verify the performance of the data bus system operation.

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11.3 Safety-critical displays and controls.

The safety-critical displays and controls section includes the display of WCAs, other safety-critical parameters, and the reliability and timeliness of safety-critical controls. The following criteria focus on the avionics architecture responsible for conveying system information to the flight crew. The human factors aspects of these systems and information are in 9.2.8 and 9.4.2.

11.3.1 Warnings, cautions, and advisories.

11.3.1.1 Avionics architecture of alerting systems.

Standard: The avionics architecture responsible for supporting WCAs correctly senses the triggering event and outputs the expected WCA. The avionics architecture possesses the failure detection, throughput, determinism, and logic necessary for correct display of WCAs under worst-case operating conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Review FMEA, FMECA, and WCA design information to determine avionics failures, throughput constraints, and non-deterministic aspects that impact the avionics architecture's ability to provide correct WCA operation in the presence of worst-case operating conditions.

Test.

(1) Lab test: Lab testing (e.g., FMET, SIL) validates analysis and verifies correct performance of avionics architecture supporting WCAs in the presence of worst-case operating conditions.

(2) Ground test: Ground testing validates analyses and lab test results and verifies performance of SCFs related to the avionics architecture supporting WCAs.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight Test: Record any critical failures related to the ability of the avionics architecture to support the performance of the WCAs. Noted failures do not result in the loss or degradation of correct WCA annunciation.

11.3.1.2 Timeliness of alerting systems.

Standard: Timing of the alerting system is assessed, documented, and shown to support SCF operation. All safety-critical warnings should be reported within 100 milliseconds following a safety-critical failure, and if over 100 milliseconds, it should be determined via test that there is no delay long enough to impact the flight crew or system response time. All warning and caution messages should be displayed for a minimum of 3 seconds IAW MIL-STD-411, Minimum Message Duration paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review program requirements to verify response time of WCA system is documented.

Analysis.

(1) Verify all warning and caution messages have a minimum display time of 3 seconds IAW MIL-STD-411, Minimum Message Duration paragraph.

Test.

(1) Ground or Lab test: Conduct laboratory or ground testing to verify timing for WCAs meet program requirements or in the absence of system specified timing and latency requirements the end-to-end timing and latency should be no more than 100 milliseconds, and if over 100 milliseconds, it should be determined via test that there is no delay long enough to impact the flight crew or system response time.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing validates analyses and test results and verifies performance of SCFs related to the timeliness of WCA messaging.

11.3.2 Display of other safety-critical parameters.

Standard: All safety-critical parameters, other than PFR and WCAs, are displayed as required to ensure flight crew situational awareness of all safety-critical systems.

a. The system continuously and accurately displays any critical parameters as required. Critical parameters may include landing gear status, cabin pressure, hydraulic system pressure, oxygen status, propulsion system information, fuel information, as well as items specific to air vehicle type (e.g., swing wing position, tilt rotor position).
b. Display indications notify the flight crew of any impending abnormal or dangerous situations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to identify the display of safety-critical parameters and their associated attributes.

Analysis.

(1) Analyze FMECA and FMEA to ensure displayed critical parameters are available after any single point failure.

Test.

(1) Lab test: Conduct laboratory testing to verify the display of safety-critical parameters according to requirements specifications and the display of any abnormal situations.

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(2) Lab test: Conduct FMET to verify safety-critical parameters are available after single point failures.

(3) Ground test: Conduct air vehicle ground testing to validate laboratory test results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing to validate laboratory and ground test results.

11.3.3 Safety-critical controls.

11.3.3.1 Reliability.

Standard: Avionics controls reliably maintain control of all SCFs. Safety-critical avionics control modes will use a probability of failure of 10^{-7} unless a different value is specified and justified based upon the controlled function.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis to verify control functionality in design documentation and FMEA.

Test.

(1) Lab test: Conduct laboratory testing to verify reliability achieves a probability of failure of 10^{-7} or a different value if it is specified and justified based upon the function.

(2) Ground test: Ground testing validates analyses and lab test results and verifies reliability of safety-critical avionics controls.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Flight testing validates analyses and lab test results and verifies reliability of safety-critical avionics controls.

11.3.3.2 Timeliness.

Standard: Timing of controls and control feedback allows control of all SCFs to be maintained. Avionic subsystem controls for controlling avionic modes and system function have timeliness and latency requirements defined and documented or in the absence of system specified timing and latency requirements the end-to-end timing and latency should be no more than 100 milliseconds, unless determined via test there is no delay long enough to impact the flight crew or system response time.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examine program requirements for response time of controls.

Analysis.

(1) Verify timing of system controls is defined and considered in design documentation and FMEA.

Test.

(1) Lab test: Conduct laboratory testing to verify timing of controls meets program requirements or in the absence of system specified timing and latency requirements, the end-to-end timing and latency should be no more than 100 milliseconds, unless determined via test there is no delay long enough to impact the flight crew or system response time.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Flight testing validates analyses and lab test results and verifies timing of safety-critical avionics controls.

11.4 Diagnostics.

The following standards are needed to show compliance for the air system's diagnostics system and capabilities. Each subsequent standard requires completion of the previous standard in a buildup approach to compliance. All air system SCFs are first identified by Section 4 criteria. 11.4.1.1 provides the means to identify the SCF subset specific to avionics and diagnostics equipment identification necessary to conduct critical failure analysis and test.

11.4.1 Air system diagnostics ability to detect and report critical failures.

11.4.1.1 Identification of safety-critical functions and equipment.

Standard: All SCFs supported by avionics and diagnostic equipment, and all safety-critical avionics and diagnostic equipment, are identified.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) An analysis (e.g., SCFTA) is performed to physically and functionally decompose each SCF to identify all SCFs which are supported by avionics and diagnostic equipment. All safety-critical avionics and diagnostic equipment are identified and documented.

11.4.1.2 Identification of critical failure modes.

Standard: All failures which can impact SCF operation are identified and classified as critical failures.

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Method of Compliance:
Prior to entering flight test, verify the standard by the following method(s):
Analysis.

(1) Failure mode analysis (e.g., FMEA, FMECA) analyzes all failure modes of the air system and identifies the subset of failures which can impact SCF operation and classifies those failures as critical failures.

11.4.1.3 Critical failure detection methods.

Standard: All critical failures are detected to enable actions (by air system or flight crew) that prevent loss of the air vehicle or personnel injury.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation to identify critical failure detection methods.

Analysis.

(1) Analysis of the air system design and architecture verifies that all critical failures can be detected to enable actions (by air system or flight crew) that prevent loss of the air vehicle or personnel injury. For each critical failure, the detection method(s) (e.g., BIT, pre-flight checks, redundancy management) are identified.

Test.

(1) Lab test: Laboratory testing verifies detection of all critical failures to enable actions (by air system or flight crew) that prevent loss of the air vehicle or personnel injury.

(2) Ground test: Ground testing verifies detection of critical failures to enable actions (by air system or flight crew) that prevent loss of the air vehicle or personnel injury.

11.4.1.4 Critical failure (degradation) reporting.

Standard: The avionics and diagnostics systems provide notification of all critical failures or degradations to the flight crew or responding system within 100 milliseconds. A different system specified value can be used if analysis and testing show there is no impact to the flight crew or system response time.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review system and subsystem specifications to identify specified latency values for critical failures and degradation reporting.

(2) Review design documentation to identify critical failure and degradation reporting methods.

(3) Review FMEA, FMECA, SHA and SCFTA to identify critical failure and degradation reporting methods.

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Analysis.

(1) Conduct a timing analysis to verify critical failure reporting latency, to verify worst-case elapsed time from moment of failure to flight crew notification, is no greater than 100 milliseconds. A different specified value could be used if analysis and testing show there is no impact to the flight crew or system response time.

Test.

(1) Lab test: Conduct laboratory testing to validate the analysis and demonstrate critical failures and degradations are reported and meet the timing requirements.

(2) Ground test: Conduct air vehicle ground testing to verify critical failures are reported.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: During flight testing, record any critical failures encountered. The avionics and diagnostics system demonstrate accurate and timely reporting of those failures.

11.4.2 Diagnostics use of SOF parameters.

Safety-of-flight parameters are defined as those variables and metrics that directly impact the stability, control, and performance of an air system. The SOF parameter tolerance defines the specified limit or range IAW the standards, operating conditions, and design safety for the air system. There are certain systems that calibrate SOF parameters (e.g., Pitot Static System, Air Data Computer, and Inertial Navigation System).

11.4.2.1 Measuring and reporting of SOF parameters.

Standard: The diagnostic system measures and reports the safety-critical parameters within the specific tolerances in support of critical failure detection.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation and specifications to identify the SOF parameters, the SOF equipment in the air vehicle, and the ground diagnostic systems to ensure complete identification and reporting coverage of all SOF parameters.

(2) Review design documentation and specifications for SOF parameters measured by the diagnostics systems to ensure all tolerances are identified.

Analysis.

(1) Verify that the FMEA and FMECA identify all critical failure modes and their criticality for SOF parameters.

(2) Verify FMEA and FMECA include critical diagnostics system failures associated with an inability to measure and report safety-critical parameters within the specific tolerances.

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(3) Review test reports to verify program defined tolerances and tolerance ranges are established for each SOF parameter.

Test.

(1) Lab test: Conduct laboratory testing to verify the diagnostics system accurately measures SOF parameters within the required tolerances and provides accurate reporting to the crewmembers or system.

(2) Lab test: Conduct FMET on the diagnostics system to validate diagnostic system failure analysis.

(3) Ground test: Conduct air vehicle ground testing to verify the diagnostics operations are consistent with the lab operations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight test: Conduct flight testing of the air vehicle to verify operation of the diagnostics system and accurate critical failure reporting occurs.

11.4.2.2 Systems requiring calibration.

Standard: The SOF parameters requiring calibration are identified. The operation and calibration procedures of safety-critical items are defined and shown to provide the needed calibration for the SOF parameters being measured.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review the avionics equipment, TMDE, calibration equipment, and standard parameters to identify those systems requiring measurement or calibration support.

(2) Review design documentation, technical manual, and source material to identify the tolerances, accuracy, and test accuracy ratio of the SOF parameters.

(3) Review calibration procedures and operational procedures to verify that the operations of the systems requiring calibration are correct when utilizing the TMDE.

Analysis.

(1) Verify that the avionics system, subsystems, equipment calibration, and measurement requirements are identified and traceable to the approved National Institute of Standards and Tests standards or other DoW approved measurements requirements.

(2) Analysis of the Calibration Measurement Requirements Summary shows TMDE supporting the calibration of SOF parameters is defined.

(3) Analysis of Calibration Measurement Requirements Summary shows that common and peculiar calibration equipment, approved National Institute of Standards and Tests standards or other DoW approved measurements requirements, and TMDE are used for calibration and measurement of SOF parameters.

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11.5 Technical manuals.

The technical manuals provide comprehensive guidance on all required flight and non-flight operations pertaining to the diagnostics systems.

11.5.1 Standard for flight manuals.

Standard: The flight manual contains the engineering data to ensure diagnostic systems address SOF parameters. Flight manuals contain all required avionics limits to ensure flight operations, including all limitations established as a result of all other avionics systems AW criteria assessments. All avionics related emergency operations are documented.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review specifications for SCFs and SCIs. For COTS and GOTS systems, review the original equipment design and specifications.

(2) Review design documentation, operational manual, and supplemental data documentation to identify SCFs and SCIs.

Analysis.

(1) Review FMECA, SHA, FMEA, and FMET to analyze the critical functional failure modes and to demonstrate linkage to WCAs.

(2) Analysis of the flight manual provides the accurate and complete output of the diagnostics systems to the flight crew, shows failures reported are linked to WCAs, and details the corrective actions the flight crew have to take and any options they may have.

(3) Analysis of the flight manual verifies that all required SOF avionics limits are included.

Test.

(1) Ground or Lab test: When conducting the operations of SCFs and items, verify and demonstrate the accuracy and completeness of the flight manual. Conduct the operation of the normal and emergency procedures per specified program requirements.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: When conducting the operations of SCFs and items, verify and demonstrate the accuracy and completeness of the flight manual. Conduct the operation of the normal and emergency procedures per specified program requirements.

(3) Flight test: Analyze all incidents, hazards, or mishaps related to flight operations that result from flight test to ensure AW avionics hazards requiring flight crew awareness or action to mitigate have procedures, and WCAs included in the flight manual.

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11.5.2 Standard for mission crew manuals.

Standard: Mission crew manuals contain the engineering data to ensure that mission equipment address SOF and its operation. Mission crew manuals contain all the procedures for detection and flight crew or crewmember response to failures that could adversely impact SOF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review any mission equipment design requirements that may adversely impact SOF when integrated onto the air vehicle.

(2) Review design documentation, mission crew, and supplemental data documentation to identify mission equipment and its interoperability with SCFs and SCIs.

(3) Verify FMECA, FMEA, FMET, and SHA to analyze mission equipment failure modes.

(4) Ensure that the mission crew manuals contain the correct procedures and WCAs necessary to instruct flight crew or crewmember action in the event of a mission equipment failure with the potential to adversely impact SOF.

Analysis.

(1) Analysis of testing shows any problem reports are accounted for in the mission crew manual.

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12. ELECTRICAL SYSTEM

This section defines the standards and MoCs for all air systems. It represents the minimum criteria necessary to establish, verify, and maintain an airworthy design by substantiating the electrical power generation and electrical wiring systems within the air system and its intended operational environment. Criteria in this section address development or modification to electrical and electronic equipment (including form, fit, function updates; or diminishing manufacturing sources modifications), the electrical power generation system, or the electrical wiring system, including existing wiring installations. Stores, expendable and non-expendable items, internally or externally carried, will be treated as an extension of the air vehicle and assessed IAW any applicable standards of the air vehicle. Test systems, or equipment that contain electrical power or wiring components, and impact safety-critical systems, are also included.

12.1 Electrical power generation system.

For airborne, shipborne, ground applications, UAS fixed location or portable control stations, or for integrated stores the electric power generating system is defined as including electrical power sources, main power buses, associated control, and regulation. This section lays the foundation of the electrical power generating system architecture, ensuring operation during normal and emergency modes, and transition between modes, is consistent with the design for providing the required power, both quantity and quality, to safety-critical systems, thereby ensuring a robust power backbone for all other air systems. Limitations and operating procedures for the electrical power generating system are also defined, verified, and documented.

12.1.1 Power quantity.

Standard: For the air system, sufficient power capacity is available to meet the power requirements during all modes of operation, MPs, failure conditions, and malfunction recovery procedures. Electrical load demand for each mission requirement is defined both without and with critical failures. Power supply capacity exceeds load demand for all operating conditions, including transient and failure conditions to include multiple power source failures and other emergency conditions. Battery power is available for all safety-critical systems for the period of time it takes to land the air vehicle under worst case operating conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Failure conditions of electrical power sources are identified and analyzed in the FMECA.

(2) The ELA properly documents the power requirements and conditions anticipated on the air vehicle IAW MIL-E-7016, Requirements paragraphs, as tailored to each specific air vehicle.

(3) The battery analysis verifies battery power is available for all safety-critical systems for the period of time it takes to land the air vehicle under worst case operating conditions.

Test.

(1) Lab test: An electrical system test, using production or production representative power system components, verifies the ability of the electrical system to operate under the power

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requirements during all modes of operation, MPs, failure conditions, and malfunction recovery procedures.

(2) Ground test: Air vehicle ground testing substantiates that sufficient power capacity is available to meet air vehicle power requirements.

(3) Ground test: Power source capacity test verifies the system provides rated performance when loaded to 100% capacity.

(4) Ground test: Air vehicle level load measurements verify the loads are consistent with the ELA.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(5) Flight test: Air vehicle flight testing substantiates that sufficient power capacity is available to meet air vehicle power requirements.

(6) Flight test: Air vehicle level load measurements verify the loads are consistent with the ELA.

12.1.2 Operation of power sources.

Standard: Each electrical power generation system component (e.g., generators, converters, inverters, transformer rectifier units, batteries) meets performance requirements as identified in individual component specifications. Operation of the electric power sources and their component parts present no hazards to external systems and crewmembers. Provisions are included to allow a manual and automatic disconnect of these electrical power sources from the system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Design documentation (e.g., schematics, drawings, and models) verifies contactors are included in the design to disconnect electrical power sources from the electrical system.

Analysis.

(1) Failure conditions of electrical power sources are identified and analyzed in the FMECA to verify no hazards to external systems and crewmembers are present.

Test.

(1) Lab test: Electrical power source qualification testing ensures components meet all required air vehicle performance requirements.

(2) Lab test: Electrical system tests, using production or production representative power system components, verifies the manual and automatic disconnect of electrical power sources from the rest of the power system.

(3) Ground test: Electrical system air vehicle ground tests, using production or production representative power system components, verifies the manual and automatic disconnect of electrical power sources from the rest of the power system.

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12.1.3 Safe operation of integrated electrical power system.

Standard: For the air system, operation of the integrated EPS for normal and emergency modes operates IAW its designed states and modes. This includes use of actual or simulated drives and loads, all flight and control configurations, transition between modes, bus switching, load shedding, and fault condition operation (e.g., detection, clearing, and reconfiguration). No malfunction or failure of any electric power source, component, or bus impairs the ability of any remaining source or bus to supply power. The integrated EPS isolates fault conditions, localizing them to the lowest level protective device.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Failure conditions of electrical power sources are identified and analyzed in the FMECA to understand the impacts to the total EPS.

(2) Electrical power system analysis verifies the components of the EPS are appropriately specified for their intended operation, during all modes of operation, once integrated into the EPS.

Test.

(1) Lab test: Electrical system tests, using production or production representative power system components, verifies the operation of the integrated EPS during normal, transfer, abnormal, fault clearing, and emergency modes of operation.

(2) Ground test: Air vehicle ground testing substantiates operation of the integrated EPS during normal, transfer, abnormal, and emergency modes of operation.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Air vehicle flight testing substantiates operation of the integrated EPS during normal, transfer, and emergency modes of operation.

12.1.4 Power quality.

Standard: The EPS, for the air system, provides electric power quality to each safety-critical load circuit and load combination, defined in the ELA, under all operating conditions IAW MIL-STD-704, Detailed Requirements paragraphs. During the operation of other air vehicle systems, power quality is maintained IAW MIL-STD-704, Detailed Requirements paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) A power quality analysis verifies electrical power quality, IAW the interface requirements of MIL-STD-704, Detailed Requirements paragraphs, is provided at the input terminals of the safety-critical power utilization equipment under all operating conditions and load combinations.

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Test.

(1) Lab test: Electrical system evaluation lab tests, using production or production representative power system components, verifies power quality, IAW the interface requirements of MIL-STD-704, Detailed Requirements paragraphs, is maintained at the input terminals of the safety-critical power utilization equipment under all operating conditions and load combinations.

(2) Ground test: Air vehicle level ground test verifies power quality, IAW the interface requirements of MIL-STD-704, Detailed Requirements paragraphs, is maintained at the point of regulation under all operating conditions and load combinations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Air vehicle level flight test verifies power quality, IAW the interface requirements of MIL-STD-704, Detailed Requirements paragraphs, is maintained at the point of regulation under all operating conditions and load combinations.

12.1.5 Uninterruptible power.

Standard: For the air system, the EPS, in conjunction with power hold-up capacity and other characteristics of utilization equipment, provides continued electrical power to support the integrity of SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) An analysis shows that the uninterruptable power source, in conjunction with power hold-up capacity and other characteristics of utilization equipment, provides continued electrical power to support the integrity of SCFs during power interruptions.

Test.

(1) Lab test: Electrical system tests, using production or production representative power system components, verifies that, during transfer, abnormal, fault clearing and emergency modes of operation, uninterruptable power is provided for safety-critical loads.

(2) Ground test: Air vehicle level ground testing verifies that, during transfer and emergency modes of operation, uninterruptable power is provided for safety-critical loads.

12.1.6 Battery operation and installation.

This section ensures batteries that provide power to safety-critical devices, or a battery failure that has the potential to be serious or catastrophic, are reviewed from the perspectives of design, integration, and installation onto, the air system. This section applies to all active batteries, both primary (non-rechargeable) and secondary (rechargeable), as well as thermal and reserve batteries. Operation of a battery cell, while installed in its operating environment, should be covered under cell level testing, equipment qualification testing, or end item safety testing.

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Specific to lithium batteries, they have a high energy density and have the potential for thermal instability. Consequently, lithium batteries can present a serious or catastrophic safety hazard if improperly designed, tested, used, and integrated into air systems. Lithium battery failures can result in a thermal runaway event (self-sustaining uncontrolled increase in pressure and temperature leading to the possible rupture of the battery shell) and the possible venting of hazardous gases from the battery. The high temperatures caused by a thermal runaway may ignite the flammable gases vented from the battery and result in an air system fire. Because of their hazardous nature, and potential for a catastrophic or critical hazard severity, lithium batteries should always be reviewed for AW impact. Additionally, all equipment powered by lithium electrochemical power sources that utilize lithium metal, alloys, or compounds, including lithium-ion batteries, should be compliant to the below AW certification criteria.

There are some criteria within 12.1.6 that should be tailored based on the battery chemistry, size, installation location, similarity to an already certified battery, or whether it is a primary or secondary battery. Specifically, tailoring is required for 12.1.6.1. The MoC used to verify compliance will depend on what lithium battery is being used. The correct MoC to show compliance should be selected, while the remaining MoCs should be tailored out.

12.1.6.1 Lithium battery safety.

Standard: Lithium batteries meet safety requirements IAW NAVSEA TM-S9310-AQ-SAF-010, Lithium Battery System Design paragraphs, and Safety Assessment Testing paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) For batteries where a previous certification letter has been issued IAW NAVSEA S9310-AQ-SAF-010, a similarity analysis verifies the battery is compliant if it is used in the same device and environment.

(2) For batteries that meet the conditions of NAVSEA TM-S9310-AQ-SAF-010, Exceptions to Testing, Review, and Certification Requirements paragraphs, a battery safety analysis verifies that the conditions are adhered to.

Test.

(1) Lab test: Test data results, provided from an end item safety test IAW NAVSEA TM-S9310-AQ-SAF-010, Safety Assessment Testing, are assessed against the criteria within NAVSEA TM-S9310-AQ-SAF-010, Passing Criteria Consideration to determine battery safety compliance.

12.1.6.2 Battery installation.

Standard: Each battery installation has provisions to prevent any hazardous effect on structure, essential systems, or personnel during all conditions. Batteries, with or without embedded electronics, are capable of operating under all air vehicle power quality conditions and load combinations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Installation drawings verify corrosive fluids or gases that escape the battery or cell will not damage surrounding structures or adjacent essential equipment, to include wiring, fuel lines, and hydraulic lines.

(2) For primary batteries, design documentation (e.g., device schematics and device component description) verifies that there is a means to prevent charging of the battery, or any cell, from any air vehicle power (e.g., blocking diode or galvanically isolated).

Analysis.

(1) A toxicity analysis shows there is no accumulation of explosive or toxic gases, expelled from a battery or cell, in hazardous quantities within the air vehicle. The analysis considers the hazardous concentration thresholds of these byproduct gases from the National Technical Information Service Publication No. PB94-195047, and the potential impact from air vehicle integration (e.g., location and ECS).

(2) An analysis, using testing results, verifies each battery installation has provisions to prevent any hazardous effect on structure or essential systems caused by the maximum amount of heat the battery can generate during a short circuit of the battery or of its individual cells.

Test.

(1) Lab test: For secondary batteries, testing show that the battery complies with the power quality requirements IAW MIL-STD-704, DC Power Characteristics paragraphs, utilizing the test procedures in MIL-HDBK-704-1 General Information paragraphs.

12.1.6.3 Battery charging.

Standard: For secondary batteries that provide backup power to safety-critical systems, charging systems are in place to automatically charge the battery IAW the battery requirements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) The battery and changing system design documentation shows the charging system is compatible with the battery device.

Test.

(1) Lab test: Electrical system tests, using production or production representative power system components, verifies the charging system meets the installed battery requirements.

(2) Ground test: Air vehicle ground testing verifies the charging system meets the installed battery requirements.

12.1.7 Electrical system limitation.

Standard: Electrical system limitations that affect air vehicle normal or emergency operation are defined.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Technical manuals (e.g., Technical Orders, Flight Manuals, Operator's Manuals, NATOPS) review verifies electrical system limitations are described.

Analysis.

(1) Failure conditions, and limitations, of electrical power sources are identified and analyzed in the FMECA.

(2) The ELA properly documents power source capacity for each mode of operation and identifies load shedding scenarios.

12.1.8 Electrical system procedures.

Standard: For the air system, published normal and emergency operating procedures are defined in crewmember and maintenance manuals. Crewmember handbooks or manuals identify all applicable restrictions, warnings, cautions, advisories, and notes. These items are identified in such a manner as to attract attention and set them apart from normal text. When an unsafe condition is detected and annunciated, the appropriate manuals have clear and precise corrective procedures for handling the failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Technical manuals (e.g., Technical Orders, Flight Manuals, Operator's Manuals, NATOPS) review verifies they include correct operating criteria (e.g., altitude limitation, electrical loading, air vehicle battery operation and functionality) and describe normal and emergency operating procedures.

Analysis.

(1) The FMECA defines abnormal modes and that appropriate procedures are included in the flight manuals.

(2) The ELA properly documents power source capacity for each mode of operation and identifies load shedding scenarios. Scenarios which impact normal and emergency operating procedures are included within the proper manuals.

12.1.9 Power system fault detection and notification.

Standard: The electrical system, of the air system, will provide appropriate power-on, initiated, and continuous BIT, fault detection, indication, and fault alarm rates. Status and failure indication for each applicable power source is provided to crewmembers.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Design documentation identifies BIT, fault detection, indication, and fault alarm rates.

(2) The FMECA verifies that fault detection and isolation is included within system design, and indication of the system operation is provided to the crewmembers.

Test.

(1) Lab test: Electrical system tests, using production or production representative power system components, verifies appropriate power-on, initiated, and continuous BIT, electrical power generation system faults are detected, and system status is indicated to the crewmembers.

(2) Ground test: Air vehicle ground testing substantiates the operation of the EPS BIT.

12.1.10 Notification of battery discharge.

Standard: The electrical system, of the air system, is designed to provide a signal to allow crewmember notification for any system malfunction that results in an unplanned discharge or degraded state of health of any battery used in an emergency battery system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Design documentation describes the battery threshold for go or no-go flight to ensure that battery capability exists to power those loads which are essential for continued controlled flight and landing.

(2) Electrical schematics, ICDs, and flight manual show that indication exists to provide crewmember notification of unplanned discharge or degraded state of health of the battery.

Test.

(1) Ground test: Air vehicle ground testing substantiates that a signal is provided to allow for crewmember notification when induced with an unplanned discharge or degraded state of health of any battery used in an emergency battery system.

12.2 Electrical system wiring.

This section is comprised of criteria for EWIS design and architecture, installation, and bonding and grounding of electrical circuits, equipment, and stores integrated with the air vehicle. EWIS is defined as any wire, fiber optic link, wiring or fiber device, or a combination of these items (including terminations) installed in any area of the air vehicle for the purpose of transmitting electrical energy, signals, or data between two or more electrical end points. As a baseline of an AW assessment, the components of the EWIS need to be understood as well as the environments they are installed in and subject to. Reliance on the environmental control document will be required when selecting and assessing EWIS components.

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12.2.1 Selection of components.

Standard: Electrical wiring, electrical system components, and support devices are selected IAW SAE AS50881, Component Selection and Installation paragraphs and Selection of Parts and Materials paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) An installed equipment list(s) review (e.g., Bill of Materials or Parts Breakdown List) verifies standard parts and materials are selected from the DoD's QPL IAW SAE AS50881, Selection of Parts and Materials paragraphs. Usage of nonstandard parts is highlighted and justification (e.g., analysis and test data) is provided as rationale.

12.2.2 Ampacity wiring selection.

Standard: For current handling capability and voltage drop, air vehicle wire sizes, including new and impacted existing wiring, are IAW SAE AS50881, Current Carrying Capacity paragraphs and Voltage Drop paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings, and models) review verifies wiring sizes are identified.

Analysis.

(1) A wiring analysis verifies sizing of electrical conductors and selection of insulation types are IAW SAE AS50881, Current Carrying Capacity paragraphs and Voltage Drop paragraph.

12.2.3 Circuit protection.

Standard: Circuit protection is provided for wiring associated with power distribution throughout its entire run, including circuits contained in or exiting from any electronic enclosures performing intermediate power switching or distribution functions IAW SAE AS50881, Current Carrying Capacity paragraphs and MIL-STD-7080, Circuit Protective Devices paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings and models) review verifies circuit protection devices are identified and are IAW MIL-STD-7080, Circuit Protective Devices paragraphs, for the application.

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Analysis.

(1) Utilizing the ELA and wiring analysis, a circuit protective device analysis verifies each circuit protective device is IAW SAE AS50881, Current Carrying Capacity paragraphs including continuous operating and transient conditions.

(2) A circuit protection coordination analysis verifies a fault is isolated at the lowest level of the distribution system. Include coordination with the power system fault isolation and recovery system process.

Test.

(1) Lab test: Circuit protection coordination testing, using production or production representative power system and wiring system components, verifies a fault is isolated at the lowest level of the distribution system without propagation to other circuits or sources.

(2) Ground test: Air vehicle ground testing verifies the operation of the circuit protective devices.

12.2.4 Circuit isolation.

Standard: Redundant circuits and components are both physically and electrically isolated IAW SAE AS50881, Redundance Circuits paragraph and MIL-STD-7080, Vulnerability paragraph. This includes components related to EWIS and associated wire termination devices (e.g., connector contacts, relays, circuit breakers, and switches).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings, and models) review verifies redundant circuits are physically and electrically isolated, and redundant systems do not share the same EWIS components or grounding points.

12.2.5 Avoidance of single point failures.

Standard: The electrical system is designed to preclude single point failures which impact SCFs when redundant functions are integrated within a new or existing electronics enclosure (e.g., circuit breaker panel, relay box, junction box). This does not include EWIS inside electronic equipment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings, and models) review verifies single point failures related to EWIS do not exist with respect to redundant functions.

Analysis.

(1) Single point failures, which impact SCFs, are precluded in the FMECA.

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Test.

(1) Lab test: Testing of failure conditions within electronic enclosures, when related to redundant functions, verifies single point failures are precluded.

12.2.6 Sufficiency of electrical wiring interconnect system installation.

Showing compliance to the criteria in 12.2.6 starts with the understanding of the fundamental requirements, and guidance of basic practice for EWIS installation, outlined in SAE AS50881. The requirements for electrical bonding and grounding are contained within MIL-STD-464. Since the AW standards do not dictate the training requirements for those installing the EWIS, to ensure the installation meets these requirements, it is crucial that engineering drawings, or models, document them. A physical installation examination is conducted to verify that the requirements are being met after the installation due to differences between the installation drawings and the actual installation.

12.2.6.1 Prevention of ignition.

Standard: Wiring installed in flammable fluid leakage zones is protected IAW SAE AS50881, Wiring Inside Fuel Tanks paragraph. This protection is to prevent potential ignition sources which could result from installation, operational environment, and aging of the new or impacted existing wiring.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings, and models) review verifies wiring installed in flammable fluid leakage zones is protected.

(2) A physical installation examination through the means of a first article inspection, verification, or quality assurance review verifies wiring installed in flammable fluid leakage zones is protected IAW SAE AS50881, Wiring Inside Fuel Tanks paragraph.

12.2.6.2 Faults in safety-critical wiring.

Standard: Any failure (e.g., open circuit fault or short circuit fault) within a new or impacted existing wiring harness does not cause the loss of any SCF(s).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Any failure within a wiring harness is identified and analyzed in the FMECA to ensure there is no loss of any SCF(s).

Test.

(1) Lab test: Electrical system tests, using production or production representative power system components, verifies a failure within a wiring harness does not cause loss of any SCF(s).

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12.2.6.3 Wiring separation.

Standard: The wiring design and installation procedures maintain positive separation of new or impacted existing wiring from fluid or gas carrying lines, flight controls and other mechanical systems, high temperature equipment, and equipment and structural hardpoints IAW SAE AS50881, Wiring Installation paragraphs and Protection and Support paragraphs. Positive separation considers movement caused by dynamic G loading, thermal effect, and vibration. Electrical power wiring is routed separately from sensitive circuits, unprotected and multiple source wiring, and essential equipment wiring IAW SAE AS50881, Wiring Installation paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings, and models) review verifies wiring maintains positive separation from fluid or gas carrying lines throughout the installation.

(2) A wiring design documentation (e.g., schematics, drawings, and models) review verifies wiring maintains positive separation from flight controls and other mechanical systems throughout the installation.

(3) A wiring design documentation (e.g., schematics, drawings, and models) review verifies wiring maintains positive separation from high temperature equipment throughout the installation.

(4) A wiring design documentation (e.g., schematics, drawings, and models) review verifies wiring maintains positive separation from equipment and structural hardpoints throughout the installation.

(5) A wiring design documentation (e.g., schematics, drawings, and models) review verifies electrical power wiring is routed separately from sensitive circuits throughout the installation.

(6) A wiring design documentation (e.g., schematics, drawings, and models) review verifies electrical power wiring is routed separately from unprotected and multiple source wiring throughout the installation.

(7) A wiring design documentation (e.g., schematics, drawings, and models) review verifies electrical power wiring is routed separately from essential equipment wiring throughout the installation.

(8) A physical installation examination through the means of a first article inspection, verification, or quality assurance review verifies wiring installation IAW SAE AS50881, Wiring Installation paragraphs and Protection and Support paragraphs.

12.2.6.4 Wiring supports.

Standard: The wiring design provides primary and secondary support for the new or impacted existing wiring throughout the installation IAW SAE AS50881, Primary Support paragraphs and Secondary Support paragraphs.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings, and models) review verifies primary and secondary supports for the wiring are provided throughout the installation.

(2) A physical installation examination through the means of a first article inspection, verification, or quality assurance review verifies primary and secondary supports for the wiring throughout the installation IAW SAE AS50881, Primary Support paragraphs and Secondary Support paragraphs.

12.2.6.5 Avoidance of damage.

Standard: The wiring design provides for routing and installation to avoid damage to wiring IAW SAE AS50881, Routing paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) A wiring design documentation (e.g., schematics, drawings, and models) review verifies wires are routed and installed to avoid damage to wiring IAW SAE AS50881, Routing paragraph.

(2) A physical installation examination through the means of a first article inspection, verification, or quality assurance review verifies wiring routing and installation is IAW SAE AS50881, Routing paragraph.

12.2.6.6 Bonding and grounding.

Standard: The air vehicle EWIS is installed IAW the bonding and grounding requirements of SAE AS50881, Ground Return paragraphs. All equipment and equipment racks are designed for proper electrical bonding and grounding IAW MIL-STD-464, Electrical Bonding paragraphs and External Grounds paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) An equipment installation drawings and models review verifies all equipment and equipment racks are properly electrically bonded and grounded.

(2) A wiring design documentation (e.g. schematics, drawings, and models) review verifies all EWIS is properly electrically bonded and grounded.

Test.

(1) Lab test: Bonding and grounding testing, using production or production representative components, verifies electrical bonding and grounding is IAW MIL-STD-464, Electrical Bonding paragraphs and External Grounds paragraphs.

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(2) Ground test: Air vehicle bonding and grounding ground testing verifies electrical bonding and grounding is IAW MIL-STD-464, Electrical Bonding paragraphs and External Grounds paragraphs.

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13. ELECTROMAGNETIC ENVIRONMENTAL EFFECTS

This section defines the standards and MoC for all air systems and represent the minimum criteria necessary to establish, verify, and maintain an airworthy design by substantiating EMC within the air system and its intended operational EME. This section is applicable to new air systems, modifications to previously certified air systems that include changes to electrical/electronic equipment (including form, fit, function updates; or diminishing manufacturing sources modifications), and modifications to the OML that may impact lightning or p-static compliance. Electromagnetic environmental effects have the potential to cause functional anomalies or damage to electrical/electronic equipment and the air system.

The E3 requirements in MIL-STD-464 and verification approaches may be tailored based on engineering justification derived from the system's operational requirements and engineering analysis. Procuring activities should tailor early in the development of an air system, and the technical rationale substantiating the tailoring should be documented in a program deliverable, such as the Electromagnetic Environmental Effects Integration and Analysis Report. Procuring activities are advised that qualification failures do not constitute a valid technical justification for tailoring any of the requirements.

13.1 Component and subsystem electromagnetic environmental effects qualification.

Component and subsystem E3 qualification is a necessary step for compliance with system requirements. During EMI qualification, the equipment is tested with higher fidelity than during system E3 testing, and results can be used to tailor system test E3 plans to establish worst-case operational scenarios. Although MIL-STD-461 is the standard for DoW EMI qualification testing, other standards such as RTCA DO-160 may be utilized but have to be approved by the procuring activity and the AW authority, and the details tailored in the AW certification basis describing their equivalency to MIL-STD-461 limits and test levels. When comparing other standards to MIL-STD-461, the primary consideration should be potential impacts to 13.2 criteria. For example, an RF radiated susceptibility evaluation should consider onboard RF transmitters and the tailored external electromagnetic RF environment for the system. The EMI qualification test setup should be aligned with the system installation, taking into account critical aspects, such as grounding, harness shielding, electrical interfaces, and operating mode(s), to ensure accurate and representative results.

For non-safety-critical equipment, unique avionics architectures or modifications may allow for alternate means of compliance, such as analysis, or a reduced set of testing to be sufficient in showing that anomalous behavior or equipment damage does not culminate in a credible AW hazard of harm to the host air vehicle. Tailoring of the standard and MoC is encouraged in those instances. Typical factors that need analyzed to determine the appropriateness of this approach for non-safety-critical equipment include:

a. Equipment type, pedigree, and use.
b. Air vehicle installation: the extent the equipment is physically installed in a zone or bay shown not to create a physical or functional hazard.
c. Air vehicle interfaces: the extent the equipment is almost exclusively, or exclusively physically and functionally federated from connections to the host air vehicle.
d. Failure modes: the extent failure modes associated with "do no harm" to the host air vehicles are well defined, with validated mitigating provisions. For example, improper

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operation can discretely be shown through analysis not to cause adverse impacts to other equipment or the air vehicle; smoke and fire risk can be shown to pose a negligible level of risk due to mitigating provisions.

To support tailoring, the rationale is expected to provide technical justification and describe how the alternate method(s) are expected to show sufficiency for compliance. For an AW certification basis developed early in the lifecycle of the program, tailoring aspects will be addressed in an AW certification basis update when this information is known. This includes the following:

a. Design information showing equipment type and installation location.
b. Design information describing and showing connections to the host air vehicle.
c. Description of associated failure analysis and mitigating factors to show how hazards are precluded if there is an E3 induced event.

There may be other considerations in establishing suitable tailoring depending upon program particulars. It is recommended proposed tailoring be coordinated with E3 SMEs to ensure that appropriate tailoring of the MoCs for 13.1.2 are substantiated with a technical rationale.

13.1.1 Flight-critical and safety-critical equipment.

Standard: All flight-critical and safety-critical equipment comply with the EMI control requirements for emissions and susceptibility (considering air vehicle type, mission, equipment functionality, and installation) using the safety-critical limits as specified in MIL-STD-461, Detailed Requirements paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify that the equipment meets the applicable MIL-STD-461 test requirements and limits considering air vehicle type, mission, equipment functionality, and installation by reviewing and analyzing the EMI test procedures.

Test.

(1) Lab test: Electromagnetic interference tests verify requirements were met using proper test setup for the system application, appropriate modes of operation are executed during testing, and compliance to the applicable limits.

13.1.2 Non-flight and non-safety-critical equipment.

Standard: All non-flight-critical and non-safety-critical equipment comply with the EMI control requirements for emissions and susceptibility (considering air vehicle type, mission, equipment functionality, and installation) using the applicable limits as specified in MIL-STD-461, Detailed Requirements paragraphs, to ensure that the equipment does not adversely affect the safety of the air system or its crewmembers and passengers and does not inhibit SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Verify that the equipment meets the applicable MIL-STD-461 test requirements and limits considering air vehicle type, mission, equipment functionality, and installation by reviewing and analyzing the EMI test plan or procedures.

Test.

(1) Lab test: Electromagnetic interference tests verify emissions requirements were met using proper test setup for the system application, appropriate modes of operation are executed during testing, and compliance to the applicable limits.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(2) Lab test: Electromagnetic interference tests verify susceptibility requirements were met using proper test setup for the system application, appropriate modes of operation are executed during testing, and compliance to the applicable limits.

13.2 System-level electromagnetic environmental effects qualification.

System-level E3 qualification demonstrates that the air system is compatible within itself and compatible with the electromagnetic environment in which it will operate, to ensure flight safety. System-level qualification incudes Intra-system EMC, Inter-system compatibility, lightning, electrostatic environments, HERP, HERF and HERO. Box level EMI test results should be considered as part of the overall E3 compliance to effectively evaluate system-level EMC. Perform full system-level E3 qualification to verify compatibility with all electromagnetic environments that the air system will be required to operate in.

It is noteworthy that for the EMC and external EME criteria, the air system includes the control station for UAVs. The UAVs control stations can be mobile, ground or fixed facilities that comply with its tailored E3 standards, MoCs and system-level (EMC and external EME) testing will include the control station as part of the test configuration.

Equally important are the steps needed in the design and validation for meeting lightning requirements in MIL-STD-464. One of the early steps is to follow SAE ARP5414 for defining the air system lightning zones, which defines the levels in MIL-STD-464 for coupon testing of OML samples. Direct effects coupon test results should have a lightning hazard assessment for determining potential AW impacts following SAE ARP5577, Perform Lightning Hazard Assessment section. In addition, guidance for fuel system lightning protection is provided in the FAA AC 20-53.

13.2.1 Air system electromagnetic compatibility.

Standard: Intra-system EMC is required at the air system level to demonstrate that equipment and subsystems provide SOF in conjunction with other equipment and subsystems which are required to operate concurrently. This includes verification that flight and safety-critical RF receivers are not degraded, maintain safety of the air system or its crewmembers and passengers, and do not inhibit SCFs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Test.

(1) Ground test: System-level EMC test report that includes detailed description of the tested air system, pass-fail criteria, source-victim methodology with detailed description of the modes of operation used and any observations during testing with accompanying root cause analysis and corrective action (e.g., see MIL-STD-464, Intra-system Electromagnetic Compatibility paragraph).

(2) Ground test: Spectrum analyzer test of flight-critical and safety-critical receivers or antenna coupling analysis of MIL-STD-461 RE102 and CE106 test results for all electrical or electronic equipment.

13.2.2 Compatibility of air system to the external electromagnetic environment.

Standard: Verify that the air system equipment can safely operate in the appropriate external RF electromagnetic environment defined in MIL-STD-464, External RF EME paragraph.

Method of Compliance:

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(1) Ground test: Conduct testing to verify the standard by one of the following method(s):

a. Option 1: Air system high level RF testing and analysis of the test results that includes detailed description of the tested air system, pass-fail criteria, source-victim methodology (with detailed descriptions of the modes of operation used during testing), and any observations during testing. Provide root cause analysis and corrective action taken.
b. Option 2: Air system shielding effectiveness testing and analysis of the test results demonstrating that the EME is compatible with the equipment susceptibility qualification levels (CS114 and RS103) in the installed configuration, including test or analysis of RF equipment performance after EME exposure, and analysis of the overall test results.

13.2.3 Lightning.

13.2.3.1 Direct effects.

Standard: The air system withstands the direct effects of lightning when exposed to the external lightning environment defined in MIL-STD-464, Lightning paragraph or SAE ARP5412, Idealized Standard External Lightning Environment section, following RTCA DO-160, Lightning Direct Effects section by protecting against physical damage to air vehicle structures, systems, and components that does not result in handling qualities worse than Level 2 or adversely affecting SCFs. In addition, the fuel system design prevents the ignitions of fuel vapor from lightning direct effects by eliminating possible ignition sources or reducing flammability via active inerting of the fuel system.

Physical damage from Lightning Direct Effects can include:

a. Damage to the OML:
    1. Skin puncture to fuel boundaries or pressurized bulkheads.
    2. Foreign object debris to critical systems such as engine.
    3. Structural damage to flight control surfaces.

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b. Fuel system sparks that ignite fuel ullage:
   1. Sparking from skin fasteners and structural joints within fuel areas.
   2. Excessive electrical current through fluid transport tubing flexible couplers.
   3. Arcing from dielectric voltage breakdown between conductors.
c. Damage to electronics:
   1. Electrical transients on electrical wiring connected to externally exposed antennas, equipment, and harnesses subject to direct lightning attachment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Perform analysis to verify that laboratory testing does not result in damage that causes handling qualities to be worse than Level 2.

(2) Perform analysis to verify that the laboratory test results do not adversely affect SCF culminating in a catastrophic or critical event.

Test.

(1) Lab test: Full scale testing of radomes, canopies, externally mounted sensors, rotor blades and antennas.

(2) Lab test: Coupon testing to demonstrate no puncture of air vehicle skin/structure using representative samples that include features such as treatments, coatings, or fasteners.

(3) Lab test: For fuel systems design without active inerting, verify that ignition sources have been eliminated via high current testing of fuel system representative components following SAE ARP5416 Ignitable Mixture (Flammable Gas) Test Method or equivalent test methods such as RTCA DO-160 Ignitable Mixture (Flammable Gas) Test Method.

(4) Lab test: For fuel system design with active inerting, verify by measuring the oxygen content of the tanks that inerting is maintained throughout the flight envelope and ground operations.

13.2.3.2 Indirect effects.

Standard: The air vehicle withstands the indirect effects of lightning. The indirect effect coupling levels are determined by an air vehicle level analysis when the air vehicle is exposed to the external lightning environment defined in MIL-STD-464, Lightning paragraph or in SAE ARP5412, Idealized Standard External Lightning Environment section.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) As an alternative to Test MoC (1), determine air vehicle system-level lightning induced transient levels through air system level computational model and analysis utilizing validated models and methods with supporting documentation following guidance from SAE ARP5415, Verification by Analysis section.

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(2) Margin analysis to verify that the air vehicle components meet the established system-level lightning induced transient levels via equipment qualification testing MIL-STD-461 CS117 or RTCA DO-160, Lightning Induced Transient Susceptibility section.

Prior to entering operational test or operations, verify the standard by the following method(s):

Test.

(1) Ground test: As an alternative to Analysis MoC (1), determine system-level lightning induced transient levels through air vehicle tests using high level pulse or low-level CW techniques.

13.2.4 Electrostatic charge control.

Standard: The air vehicle is able to control and dissipate the build-up of electrostatic charges caused by particle impingement, fluid flow, air flow, and other triboelectric charge-generating mechanisms to avoid ordnance hazards, personnel shock hazards and to control p-static interference or damage to electronics. Structural surfaces have sufficient conductivity to dissipate the electrostatic charge in a timely manner that precludes arcing and sparking, that all components are electrically bonded, and that an electric path to earth ground is provided. The air vehicle electrostatic environments are defined in MIL-STD-464, Electrostatic Environments paragraphs or SAE ARP5672, Establish the Aircraft P-Static Charging Environment section.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Verification that the air vehicle dissipates the electrostatic charges in such a time to eliminate the potential for arcing, sparking and causing interference to flight and safety-critical sensitive receivers is accomplished by air vehicle level test and analysis of the test result.

(2) Lab test: P-static testing of OML samples to validate electrostatic charges are dissipated in a timely manner that prevents elevated static voltages that can culminate in arcing and sparking.

(3) Ground test: Bonding measurements or validated bonding assembly process specifications backed by empirical/test data.

13.2.5 Hazards of electromagnetic radiation.

Standard: Radio frequency transmitters pose no HERP, HERF, and HERO, and that the appropriate manuals include recommended safe separation distances from onboard and external transmitters to personnel and fuel sources.

a. Hazards of electromagnetic radiation to personnel: The criteria to protect personnel from the electromagnetic radiation from air system emitters is defined in IEEE STD C95.1™-2019, Exposure Limits section.
b. Hazards of electromagnetic radiation to fuel: Fuel cannot be inadvertently ignited by radiated electromagnetic fields from air system emitters or by the external RF electromagnetic environment defined in MIL-STD-464, External RF EME paragraph or an equivalent RF EME.
c. Hazards of electromagnetic radiation to ordinance: Electrically initiated devices used in ordnance and other parts and equipment of the air vehicle cannot be inadvertently actuated during or experience performance degradation after exposure to the radiated

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electromagnetic fields from air system emitters and by the external RF electromagnetic environment defined in MIL-STD-464, Hazard of Electromagnetic Radiation to Ordnance (HERO) paragraph or an equivalent RF EME.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Hazards of electromagnetic radiation to personnel: Verification is accomplished by measurements of the RF generated by the onboard emitters or safe separation distance analysis based on the limits in IEEE STD C95.1-2345™-2014, Exposure Limits section. MIL-HDBK-235-1, Calculations section and NAVSEA OP 3565 Volume 1, Calculation of Electromagnetic Fields appendix, provides the methodology for assessing RF hazards calculations.

(2) Hazards of electromagnetic radiation to fuel: Verification is accomplished via analysis based on the methodology of MIL-HDBK-235-1, Calculations section, and NAVSEA OP 3565 Volume 1, Calculation of Electromagnetic Fields appendix, for calculating hazard distance from RF emitters against areas where fuel vapors are expected (e.g., vents, receptacles). The peak power density limit is 5 W/cm^2 to calculate the safe separation distances from onboard RF transmitters.

(3) Hazards of electromagnetic radiation to ordinance: Verification is accomplished via analysis based on the methodology of MIL-HDBK-235-1, Calculations section, and NAVSEA OP 3565 Volume 1, Calculation of Electromagnetic Fields appendix, for calculating hazard distances from onboard RF emitters to HERO SAFE, HERO SUSCEPTIBLE and HERO UNSAFE EIDs. In addition, use HERO test results following Test MoC (1) to substantiate HERO certification for EIDs validating safe distances are appropriate once the items are integrated into the air system.

Test.

(1) Ground test: Hazards of electromagnetic radiation to ordinance: Verification is accomplished by testing of the EIDs and associated circuitry to the margins defined in MIL-STD-464, Margins paragraph; to the external RF electromagnetic environment defined in MIL-STD-464, Hazard of Electromagnetic Radiation to Ordnance (HERO) paragraph.

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15. COMPUTER SYSTEMS AND SOFTWARE

This section establishes the AW verification criteria (sometimes referred to as ‘criteria’) that are to be applied and assessed against the CS&S used within air systems for the purpose of certifying the AW of an air system. The set of standards and MoCs contained in this section establish the minimum verification needed to show compliance and achieve a baseline level of safety upon which further risk acceptance is not required.

This narrative is organized by the following topics: Scope of Applicability, Multicore Processing Environments, Definitions and Terminology Considerations, Modification Considerations, Section Assumptions, CS&S AW Methodology Compared to Other Methodologies Found in this Handbook, Numbering Scheme Used Within Standards and MoCs, and AI.

Scope of Applicability: The criteria of Section 15 are to be applied to all CS&S that support the execution and integrity of operations associated with SCFs. Additionally, some criteria are applicable to CS&S that do not support SCFs, however they interface or integrate with CS&S that do support SCFs and therefore can result in an AW hazard. The scope of Section 15 is intended to assess the design and integration of the CS&S of the air system. This entails assessing the design of the SPA and how it is integrated together, the analog and digital electronic hardware components, and the software including associated data files that are executing on the air system. Firmware and programmable logic devices are also included within the scope but are not explicitly mentioned in this section. Firmware is defined as the code and data files that are used to configure a programmable logic device to govern its functionality and operation. For the purposes of this section, firmware will be considered software. Programmable logic devices will be considered hardware.

Multicore Processing Environments: Section 15 has incorporated AW verification criteria to address the use of MCP environments that support the operation of SCFs. The MCP environment verification criteria incorporated into this section are derived from what was first published in the USAF AA-22-01.

Definitions and Terminology Considerations: Definitions have been developed for many of the terms used within Section 15, but they have not been incorporated into Section 3 of this handbook; instead, they have been documented in a unique definitions list. Additionally, some of the terms found in Section 3 of this handbook have been modified for Section 15 use and are included in the unique definitions list. Examples include safety-critical and flight-critical as well as other terms. It is highly recommended that applicants contact the respective Airworthiness Office of the military AW authority performing the assessment review, or the military service’s technical lead for Section 15 to obtain the list of definitions developed for this section.

a. One change that has been introduced in Section 15 for this update compared to the last version of the handbook is replacing the terminology of Safety Supporting Element (SSE) with Element Supporting Safety-Critical Function(s) (ESS). The new term means essentially the same as the previous term, where the identified element is supporting the operation of an SCF (or multiple SCFs) under expected nominal or off-nominal conditions over the defined envelope of operation. The change to this new terminology was made to convey a more intuitive meaning of what is intended for this section and eliminate the need for a specialized definition for SSE. The change in terminology also translates to the hardware -only and software-only subcategory of elements, so Safety Supporting Hardware Element (SSHE) is now expressed as Hardware Element

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Supporting Safety-Critical Function(s) (HESS) and Safety Supporting Software Element (SSSE) is now Software Element Supporting Safety-Critical Function(s) (SESS).

b. An additional terminology change in this version involves introducing the phrase "the integrity of SCF operation" which is used throughout Section 15 in criteria to indicate that the contributing aspects of the design that support SCF operation possess the design attributes to work as intended (i.e., integrity) to achieve a baseline level of safety upon which further risk acceptance is not required. When a verification criterion is evaluating a design attribute to ensure that the necessary qualities are present in the design to work as intended, the phrasing "ensure the integrity of SCF operation" or some variation thereof is used. When a verification criterion is evaluating a design attribute to ensure that the necessary qualities are present in the design to work as intended when coupled to items presenting a potential degradation to SCF operation, the phrasing "does not adversely impact the integrity of SCF operation" or some variation thereof is used.

Modification Considerations: When applying Section 15 criteria to an air system modification, expectations will be scoped to the AW impacts associated with the modification. Criterion 15.1.1 is an example. For a modification to an air system, 15.1.1 would not expect that a full list of SCFs supported by CS&S equipment be identified. Instead, the expectation would be that only those SCFs (supported by CS&S equipment) that are new or directly/indirectly impacted by the modification are identified (Note: this assumes that a full list of SCFs supported by CS&S equipment was provided and independently reviewed as part of a previous AW assessment of the air system; if not, then the full list of SCFs may be expected). Criterion 15.1.1 has been modified as part of this handbook update to clarify its expectations for modification efforts. The additional language in 15.1.1 does not signify that it is the only criterion within Section 15 that is applicable for a modification; all criteria in Section 15 will be assessed for applicability and tailoring to address air system modification scope. It is highly recommended that all applicants contact the respective Airworthiness Office of the military AW authority performing the assessment review, or the military service's technical lead for this section, for any questions regarding applicability of criteria or change in scope for modifications, particularly if no previous, independent assessment of the air system has been performed.

Section Assumptions: With regards to applying the criteria in Section 15 there are some assumptions that are inherent on how the language will be applied to the items in an air system.

a. When the SPA is identified within criteria, the definition of a SPA includes all related CS&S items in the air system. When considering a SPA in the context of a modification, the scope of application of a criterion related to the SPA should be limited to the portion of the SPA items that are directly or indirectly impacted by the modification, where directly impacted means the item is being modified, and indirectly impacted means items are not being modified but are physically or functionally coupled to the items that are being modified.
b. The use of the word "system" has a particular assumption associated with it when it is used to indicate the architectural level or levels within the air system architecture. The word "system" is often used to indicate one or more of the upper levels of the air system architecture. System can refer to the highest level of the air system (the air system itself), or a lower architectural level like the air vehicle level, a subsystem level within the air vehicle, or an element within a subsystem. The criteria containing this language will be evaluated to the proper scope of application and may include multiple levels of testing in

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order to fully accomplish “system” level verification. In principle, the lowest level of testing that can fully verify a criterion is what is generally expected to show compliance, but many developers choose to perform relevant verifications at other than the lowest level for cost or schedule efficiency reasons, so this is not mandated, but is an assumption that is associated with the use of the word “system”.

c. Another assumption behind the language of this section is that SESSs will be designated as CSCIs and that CSCIs are not contained within another CSCI. Thus, an item designated as a CSCI is at the lowest level of decomposition for the air system hierarchy of CIs.

d. Reference Architecture: Section 15 assumes that an air system will be architected in a hierarchical manner. When any portion of this section is applicable to an AW project, the applicant should consider the use of the reference architecture identified herein to communicate to the military AW authority at what level verifications are being performed within the system architecture. It is recommended that the applicant also clarify how their implemented architectural hierarchy differs from the reference architecture identified herein. Each level of the hierarchy below the top-most level is fully contained within the level above it (meaning a given level of the architecture does not exclude any portion of the level below it). Requirements, architectural, and design information is expected to be defined for each level of the reference architecture down to the Hardware Element and Software Element level, and as needed at lower levels. The top-most level of the hierarchy is the Air System level and consists of all aspects of the system relevant to AW. The next level down is the Segment level. Below that is the Subsystem level followed by the Element level. Elements are then decomposed to the next level which are the Hardware Element level, Firmware Element level, and Software Element level. Hardware Elements are then decomposed to the Electronic Hardware Component level, Firmware Elements are decomposed to the Firmware Component level, and Software Elements are decomposed to the Software Component level. The final level of the architecture includes Firmware Components decomposed to the Firmware Unit level, and Software Components decomposed to the Software Unit level. The language found in the criteria of this section do not always use the explicit terms used in this reference architecture (e.g., “unit level” is used in some cases instead of “software unit”). Pictorially the reference architecture is shown on FIGURE 6.

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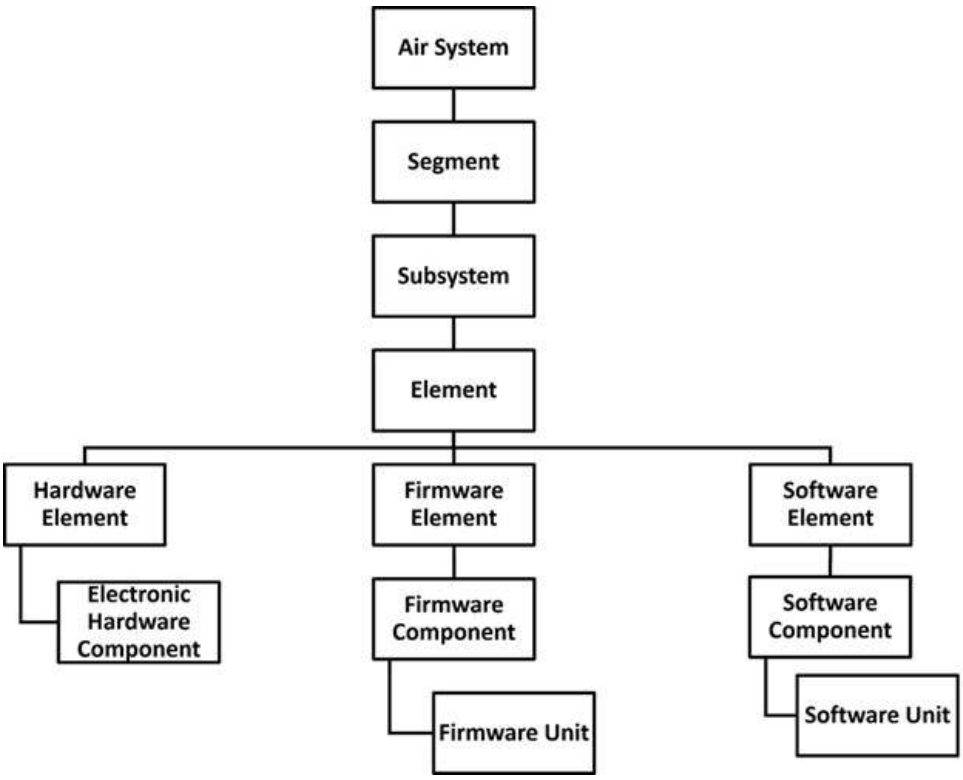


FIGURE 6. CS&S reference architecture.

e. All artifact evidence provided to show compliance to the criteria of this section are expected to be relevant to the configuration being assessed.
f. When analysis, test, or demonstration is identified as an MoC in this section, the expectation is that there will be an artifact provided (e.g., a report) containing the results of the verification activity or access provided to a database that contains the results.

Computer systems and software AW Methodology Compared to Other Methodologies Found in this Handbook: The AW criteria established in Section 15 utilize an underlying methodology that in some cases may differ from other methodologies. For example, the methodology used by this section to define the criticality of an item supporting one or more SCFs is based on the greatest functional severity of the SCF(s) supported. This methodology may differ from the methodologies employed to identify SCIs or CSIs. Thus, it should not be assumed that methodologies found outside of this section align with the expectations in this section.

Numbering Scheme Used Within Standards and MoCs: Within Section15, each standard and MoC has been separated into numbered line items “1” to “N”, where “N” corresponds to the total number of line items. For a given criterion, the number next to an MoC line item corresponds to the numbered line item in the standard being verified. For example, MoC line item “1” is verifying standard line item “1”, MoC line item “2” is verifying standard line item “2”, and so on. This numbering scheme is intended to provide traceability between an AW attribute within a standard and the corresponding method(s) for verifying it. All standards and MoCs have at least one numbered line item. When a standard or MoC contains more than one numbered line item, the additional line items “2” to “N” represent additional AW attributes and verification methods (not already addressed by line item “1”) that apply when a design utilizes MCP environments that could impact the integrity of SCF operations. These additional line items are clearly

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indicated by the phrase “In addition, for designs utilizing MCP environments:” which precedes line item “2” in the relevant standard or MoC. It is important to note that while line items “2” to “N” apply only when utilizing MCP environments, line item “1” is not limited to single core processing environments. When utilizing MCP environments that could impact the integrity of SCF operations, all line items in a standard and MoC apply.

Artificial Intelligence: Artificial intelligence is a machine-based technology that models a set of human-defined objectives, and utilizes data inputs (provided by machines, sensors, or humans) it receives to infer relationships which allow it to generate estimations, predictions, recommendations, decisions, or content that influence real or virtual environments. When utilizing AI in a way that could impact the integrity of SCF operations, the criteria in Section 15 apply. Some aspects of AI can be fully assessed utilizing the current Section 15 standards while other aspects of AI will need supplemental standards developed to establish the design pedigree for the AI technology and its integration and use within the air system to ensure the integrity of SCF operation. When utilizing AI in a way that could impact the integrity of SCF operations, applicants should coordinate its use with AW authority SMEs early in the air system’s development lifecycle. In addition, applicants should consult with their AW authority for any published guidance on this topic. Early coordination will assist applicants in addressing the necessary architectural and design attributes needed to support AW expectations, as well as criteria tailoring considerations.

The nature of Section 15 involves multiple disciplines (e.g., systems integration, avionics hardware, software) to fully address the verification of criteria associated with the CS&S of an air system. Additionally, in some cases, architectures and designs associated with CS&S can be very complex, and thus an applicant may find it challenging to know how to apply the criteria to an air system design. It is recommended that the criteria of this section be assessed for applicability, and the associated plans for execution be made as early in the project as possible. Planning should also include coordination with the technical staff of the AW authority for proper interpretation and application as needed.

15.1 System processing architecture.

15.1.1 Safety-critical functions.

Standard:

1. The SCFs (Section 4 verifies the identification of SCFs) which are supported by CS&S equipment to perform their intended function are identified. The subset of SCFs supported by CS&S equipment which are flight-critical are identified. For all SCFs which are supported by CS&S equipment, identify the SCFs which are either not implemented, inhibited, or partially implemented. In the case of an air system modification, new SCFs that are incorporated by the modification, which are supported by CS&S equipment, are identified and assessed to determine if they are flight-critical. Identify all SCFs supported by CS&S equipment that are directly or indirectly impacted by the modification. For all SCFs which are supported by CS&S equipment, identify the SCFs which are either not implemented, inhibited, or partially implemented.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes analysis. An analysis (e.g., SCFTA) is performed to functionally decompose each SCF and assess its allocation to equipment in the architecture to identify all SCFs supported by CS&S. Analysis identifies the subset of SCFs supported by CS&S equipment which are flight-critical. Analysis also identifies the SCFs which are either not implemented, inhibited, or partially implemented. For air system modifications, analysis identifies new SCFs incorporated by the modification that are supported by CS&S equipment and determines the SCFs which are flight-critical. A change impact analysis is performed, utilizing the SCFTA, to identify all SCF threads supported by CS&S equipment that are directly or indirectly impacted by the modification. Analysis identifies the SCFs which are either not implemented, inhibited, or partially implemented.

15.1.2 System processing architecture requirements.

Standard:

1. Verify that the SPA safety requirements are fully defined and documented. Safety and performance requirements are allocated to the architecture. The SPA is defined. An analysis of the SPA is performed to address attributes such as functional requirements, processing demands, timing criticalities, data flow, interfacing elements, and fault tolerance. Federated and integrated elements of the SPA are identified.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and examination. Verify the SPA is defined by examining design documentation. Verify that the analysis of the SPA is complete and is documented. Analysis determines if the SPA supports program safety and performance requirements. Ensure all technical and safety SPA risks are documented and appropriately corrected or mitigated.

15.1.3 System processing architecture redundancy.

Standard:

1. Verify that the SPA has employed redundancy and fault tolerance to preclude the loss of SCFs due to the occurrence of single point failures or data channel loss. Fault tolerant requirements for the SPA have been identified and documented. The SPA does not contain any items (e.g., digital hardware, interfaces, software components/modules, and operating system) whose failures defeat the redundancy mechanization.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

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1. Verification methods include analysis and testing of the SPA. Verify through analysis and testing, including FMET, that a single point failure does not result in the loss of SCF. Verify techniques such as the following are employed to preclude the existence of single point failures: voting schemes, cross channel data link mechanization, data bus management, and I/O signal management approach. Verify that the SPA fault tolerance and redundancy management mechanization is capable of handling all single failures.

15.1.4 Safety-critical function threads.

Standard:

1. Verify that all SPA supported SCF threads have been identified, documented, and completely traced. All SCF threads have been traced from the SPA to the component level. All elements supporting an SCF have been identified as part of the SCF thread and categorized as ESSs. Safety-critical function thread analysis also identifies any ENSSs whose integration with an ESS needs to be evaluated for potential adverse impact to SCF operation. Each SCF thread has been verified and validated at all levels of the design to ensure end-to-end coverage of the SCF in nominal and off-nominal conditions to ensure the integrity of SCF operation.

In addition, for designs utilizing MCP environments:

2. For each SCF identify all MCPs utilized, processing cores utilized, and chip related resources (e.g., processor interconnects, I/O management, controllers, memory, buses) supporting the operation and performance of the SCF.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes analysis. Verification includes functional thread analysis for each SCF from the SPA to the ESSs and components. Ensure ESSs and components are appropriately identified, labeled, and traced to each SCF they support. Ensure ENSSs whose integration with an ESS needs to be evaluated for potential adverse impact to SCF operation are identified and documented. Verify the sufficiency of coverage and repeatability of the functional thread analysis conducted on the SCFs. Safety-critical function V&V coverage analysis verifies that end-to-end coverage of each SCF is accomplished for nominal and off-nominal conditions to ensure the integrity of SCF operation.

In addition, for designs utilizing MCP environments:

2. The SCFTA identifies all MCPs, processing cores, and chip related resources supporting the operation and performance of each SCF.

15.1.5 Probability loss of control and hazard mitigations.

Standard:

1. Verify that the SPA is designed to meet PLOC, PLOA, SCF processing, hazard mitigations, and reliability requirements. Air system PLOC and PLOA numbers are specified in system documents and allocated to the SPA. The SPA requirements and allocation in support of PLOC are aligned with the PLOC criteria in Section 6 of this handbook (PLOC does not exceed 10^{-7} per

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flight hour). Proper levels of redundancy, sufficient cross checks, and hazard mitigations are incorporated in the SPA to ensure safety-critical components accommodate failures while achieving safety, PLOC, PLOA, fail-operational, and fail-safe requirements. SPA electronic hardware components are designed to meet specified reliability requirements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the Standard that is being verified:

1. Verification method includes examination, analysis, and test. Examination of requirements verifies PLOC and PLOA numbers are specified. Analyses cover the design of all hardware and software elements of the SPA per the standard, and all associated assumptions of the analyses have been validated through analysis and test.

15.1.6 System processing architecture interfaces.

Standard:

1. Verify that all ESSs of the SPA that interface (physically or functionally) with other elements (ESSs or ENSSs) continue safe operation in the event there is a data channel failure or data corruption with the interfacing elements. The SPA physically and functionally separates ESSs from ENSSs to maintain safe system operation. Failure of ENSSs or the loss or corruption of data channels does not affect ESSs. All interfaces which handle data associated with SCFs have been identified (baseline established and documented) as safety-critical. Element Supporting Safety-Critical Function(s) interfaces are designed to ensure data, calculation, and system timing dependencies do not impede system performance in any operational mode or degrade architectural safety coverage.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, interface testing, and examination of documentation. The documentation is examined for completeness in identifying all ESS and ENSS interfaces. Verify through analysis (e.g., common mode) that the SPA requirements and design physically and functionally separate ESSs from ENSSs. Verify through testing that ENSSs do not adversely impact the integrity of SCF operation. Testing of these interfaces addresses interface performance (e.g., interface loading; handshaking; protocols; timing, data, calculation, and system dependencies) for all operational modes and failure modes to ensure safe system operation.

15.1.7 Computer system integrity levels.

Standard:

1. Verify that all SCFs are fully allocated to elements within the SPA and that each element is assigned a CSIL based on the functional criticality of the SCF supported. The CSILs are defined. Each SPA element has an appropriately assigned CSIL and the corresponding process is applied

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to all components of the element. All SCFs have been allocated to ESSs. Computer system integrity levels are not lowered through alternative design techniques (e.g., the use of redundancy, monitoring, or other techniques is not an acceptable justification for lowering a CSIL).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, and examination of documentation. Examination of process documentation verifies that CSILs are defined. Examine all SCFs and SCFTA documentation. Verify that the analysis has identified all elements of the SPA and that the associated CSIL has been determined. Ensure all ESSs are assigned a CSIL based upon the functional criticality of the SCF supported.

15.1.8 Computer system integrity level processes.

Standard:

1. Every CSIL has a corresponding development process defined and applied and each process is adequate to support the safety requirements of the classification. Computer system integrity levels are identified and documented. For each CSIL, ensure the program process and planning documentation or applicable integrity program plans clearly define and document appropriate processes, and that those processes are applied. The defined development and test processes support safety requirements and CSILs. All ESSs, SESSs, and HESSs are designed, developed, integrated, and tested to appropriate CSILs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and examination of process documentation. Ensure CSILs are identified and documented. Verify through analysis the adequacy of the CSIL processes. Verify that appropriate CSIL processes are documented and followed.

15.1.9 Data flow and control flow.

Standard:

1. Verify that interfaces (control and data flow) supporting SPA ESSs are clearly defined and documented. All ESS interfaces which handle control and data associated with SCFs have been identified as safety-critical and are documented. Interface requirements for safety are defined and accounted for throughout the development process (e.g., drive specific test processes). Safety-critical interfaces are designed to ensure data, calculation, and system timing dependencies do not impede system performance in any operational mode.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and examination of documentation. The documentation is examined for completeness in identifying and defining all ESS and ENSS interfaces. Examine test process documentation to ensure safety-critical interfaces are fully verified. Analysis verifies interface dependencies do not impede system performance and safety.

15.1.10 Physical and functional separation.

Standard:

1. Verify that physical and functional separation between ESSs and ENSSs are accounted for in the SPA. Partitioning schemes and OSs utilized to separate ESSs from ENSSs are developed and tested at the highest CSIL supported. Partitioning schemes ensure partitioning breaches are prevented and that derived requirements are defined to ensure partitioning integrity is maintained.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and test. Analysis ensures the effectiveness of partitioning schemes. Analysis (e.g., FMEA, partitioning analysis) and testing (e.g., FMET) verify that techniques employed for separation are effective and safe. System integration tests are conducted on the SPA at the highest CSIL of the ESSs residing within. These tests verify that the ENSSs do not adversely impact the integrity of SCF operation. Ensure the testing of all elements, the SPA, and computer system interfaces is commensurate with their CSILs.

15.1.11 Notification of loss of critical processing.

Standard:

1. Verify that the flight crew are notified upon the loss of flight-critical processing capability or redundancy in flight-critical processing. Flight-critical processing systems have been identified. Fault detection and redundancy management design detects loss of redundant processing capabilities for flight-critical systems. Flight crew are alerted to the loss of flight-critical processing or the loss of redundancy in a flight-critical processing system through the notification of either the functional loss or degradation that results, or the loss of redundant processing capacity. The annunciation of loss or degraded processing is properly prioritized. The loss or degradation of flight-critical processing is detected and annunciated with functional response characteristics and latency that ensures the integrity of SCF operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

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1. Verification methods include analysis, test, demonstration, and examination. Examination of safety documentation identifies flight-critical processing systems. Analysis identifies the functional response characteristics and maximum acceptable latency necessary to ensure the integrity of SCF operation. Examination of requirements documentation verifies the maximum acceptable latency is established as a requirement for the design. Verify that proper annunciations have been designed into the system by examining requirements, design documentation, and flight manuals. Test and demonstration verify the annunciations and proper prioritization to the flight crew when loss or degradation of flight-critical processing or loss of redundancy occurs. Verify using FMET and system performance testing that the system fault detection and redundancy management design detects and annunciates loss or degradation of flight-critical processing with functional response characteristics and latency that ensures the integrity of SCF operation.

15.1.12 System processing architecture – electrical power integration.

Standard:

1. Verify that uninterruptable power capability is designed into SPA ESSs. Verify that the electrical power interface and internal power design of SPA ESSs are able to interface with the quality and quantity provided by the EPS to provide and maintain continuous SCF operation under nominal and off-nominal scenarios for all expected dynamic conditions. The SPA is designed such that the nominal and off-nominal (e.g., shorting conditions, overdraw) operation of elements cannot degrade the EPS (e.g., degradation of power quality) in a way that could adversely impact the integrity of SCF operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, testing, and examination.

   a. Verify through examination of design documentation the compatibility of the interface design between SPA elements and the EPS and that SPA ESSs have incorporated design features to support uninterruptable power. Verify through examination of design documentation that SPA elements incorporate design features to prevent degradation to the EPS that could adversely impact the integrity of SCF operation.
   b. Verify through analysis the SPA ESSs can receive and deliver uninterruptable power to enable continuous SCF operation under nominal and off-nominal scenarios for all expected dynamic conditions. Analysis should include identifying SPA load-shedding design, failure mode effects on electrical power design within each ESS, identification of all dynamic conditions each ESS electrical power design is expected to experience, and the ability of ESSs to maintain continuous SCF operation under nominal and off-nominal scenarios under all dynamic conditions. Verify through analysis that SPA element design features prevent degradation to the EPS that could adversely impact the integrity of SCF operation.
   c. Verify through test the SPA ESSs can receive and deliver uninterruptable power to enable continuous SCF operation under nominal and off-nominal scenarios for all expected dynamic conditions. Verify through test that SPA element design features

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prevent degradation to the EPS that could adversely impact the integrity of SCF operation.

15.2 Design and functional integration of system processing architecture elements.

15.2.1 Functional coupling.

Standard:

1. Verify that ESSs avoid unnecessary coupling. The SPA design mechanization accounts for the dependencies of data and parameters utilized by ESSs within the SCF threads. The ESS interdependencies (e.g., data, control, functional, and hardware) of the SCF thread satisfy the safety requirements (e.g., latency, processing, and data). Data coupled to these threads are defined in terms of criticality and requirements are in place to avoid unnecessary coupling. Only ESS data and control required to satisfy the SCF processing requirements are coupled between the ESSs. Unnecessary coupling is avoided by eliminating non-essential elements and data in an SCF thread.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, demonstration, and examination. Examination of SCF threads is performed to ensure that unnecessary coupling does not exist. Process documentation is reviewed to verify that processes are in place to ensure unnecessary coupling is eliminated. Examination verifies that all data requirements for ESSs are defined. Essential coupling, processing rates, associated allowable latencies, and transient limitations required for safety-critical data are analyzed, tested, and demonstrated.

15.2.2 Functional autonomy and critical data sources.

Standard:

1. Verify that safe functional autonomy is achieved by ensuring that ESSs do not rely on single source safety-critical data or non-safety-critical data to support SCFs. All data sources for ESSs are identified. Safety-critical data are identified. The ESSs are designed to preclude reliance on single source safety-critical data or from non-safety-critical sources for safety-critical processes and processing tasks.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and test. Failure mode effects analysis and FMET verify that loss of non-safety-critical data or a safety-critical source will not cause loss of SCF.

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15.2.3 Integration methodology.

Standard:

1. Verify that the integration methodology used for the SPA ESSs is defined, documented, and provides complete verification coverage of SCFs at all levels, for each flight configuration release. The entire SPA is developed, integrated, and verified using a defined, documented, and proven process which includes complete test coverage (requirements, functions, failure conditions, mission validation) at all levels. Verification methodology includes end-to-end testing of SCF threads. ESSs, along with ENSSs, are developed and tested individually; then integrated to form a SPA. Each build released for flight operations has all the functionality necessary to ensure integrity of SCF operation. The integration methodology identifies conditions (adequacy of partitioning, decision points, areas of testing) for which a subset of testing is permissible in lieu of complete testing. Regression process addresses adequacy of partitioning, the areas of change, other areas affected by the change(s), and core regression testing required independent of the change(s). The integration methodology adequately accounts for unique aspects of the technologies utilized (e.g., MCPs).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes examination of integration planning and test documentation. Examination of integration plans and process documentation verifies that the integration methodology has been implemented and provides complete verification coverage. Verify that the ESSs of the SPA are clearly identified such that any dependencies that safety-critical systems (e.g., vehicle management system) have on other systems are addressed. Ensure integration plans show a reasonable build up approach. Verify that the test documents reflect all levels of testing throughout all levels of the air system architecture. Verify the plans and results of end-to-end SCF testing. Verify that the integration methodology adequately accounts for unique aspects of the technologies utilized (e.g., MCPs).

15.2.4 Critical discrepancies.

Standard:

1. Verify that discrepancies impacting safety-critical hardware and software are identified and corrected or mitigated. The system development process identifies and applies a discrepancy reporting process for software, hardware, system, laboratory, air vehicle, control station, and support system problems, which documents and appropriately corrects or mitigates all safety-critical faults at all testing levels. Typical areas to include: software anomalies, hardware anomalies, test case inconsistencies, simulation anomalies, and lab and tool problems. A discrepancy review board is established which evaluates problem reports for safety effect, prioritization and urgency of correction required, and documents approved discrepancy mitigations and resolutions. If used for problem mitigation, operational safety restrictions (e.g., flight envelope, operating modes, fleet grounding) are documented in the flight clearance, flight manual, or equivalent documentation.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes examination of discrepancy reporting processes, discrepancy documentation, and technical manuals (e.g., TOs, flight manuals, operator's manuals, NATOPS). Ensure that discrepancies impacting safety-critical hardware or software are documented, corrected or mitigated, and appropriate restrictions are imposed when necessary. Details of corrective or mitigation actions are reviewed.

15.2.5 Simulations, models, and tools.

Standard:

1. Verify that simulators, models, and tools used in the development, integration, and testing of software and hardware supporting SCFs have been appropriately qualified and validated. All simulators, models, and tools used for design, development, integration, and test have been identified and a qualification and validation approach has been defined. Each simulator, model, and tool is identified as off-the-shelf (commercial or government), modified, or developed for the application, has been adequately qualified and validated for its intended use to the appropriate qualification level to include showing equivalent performance to the actual air system configuration as needed for the V&V being performed. Verification tools do not mask safety issues within products they are used to verify.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination. Verify that the qualification and validation approach is defined and documented. Examine tool documentation to ensure all simulators, models, and tools have been identified, highlighting how they are used and if they are off-the-shelf, modified, or newly developed. Analyses and tests verify the equivalent performance (accuracy and fidelity) of all simulators, models, and tools to the actual air system configuration, as needed for the V&V being performed. Analyses and tests ensure verification tools do not mask product problems.

15.2.6 Safety interlocks.

Standard:

1. Verify that safety interlocks provide safe engagement or disengagement of systems, subsystems, modes, or functions for flight and ground operations. All SPA safety interlocks have been identified and defined. The specifics of how each safety interlock is used by each ESS is documented. Safety interlocks safely control mode engagement based on ground or flight parameters and prevent engagement of incompatible modes.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, demonstration, and examination. Examination of design documentation ensures that safety interlocks have been identified and defined. Verify that safety interlocks prevent the engagement or disengagement of systems, subsystems, modes, or functions that are incompatible or could create a hazard or undesirable situation. Verification includes end-to-end testing of the interlock design through laboratory testing and demonstration, ground testing, and flight testing.

15.2.7 Single event effect susceptibility.

Standard:

1. Verify that the SPA incorporates mechanisms to mitigate SEEs to prevent a catastrophic or critical severity mishap. Single event effect susceptibility of ESSs and other equipment that could contribute to a catastrophic or critical mishap is identified based on planned usage and operating environment. Detection, correction, and prevention techniques are employed to ensure the integrity of SCF operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and test. Analysis determines SEE susceptibility of ESSs and other equipment that could contribute to a catastrophic or critical mishap and ensures an acceptable level of protection across the design. Testing ensures fault detection mechanisms can detect the effects of SEEs and the system can employ mitigating corrections. Testing verifies the effectiveness of SEE shielding techniques.

15.2.8 Security techniques.

Standard:

1. Verify that implemented security techniques do not adversely impact the integrity of SCF operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and test. Verify that security techniques have been analyzed and tested to ensure that they do not result in adverse impacts to the integrity of SCF operation.

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15.3 Hardware elements.

15.3.1 Hardware element supporting safety-critical function(s) performance.

Standard:

1. Verify that all HESSs are safe. System processing architecture HESSs are identified. HESS performance meets processing requirements. Hardware elements supporting SCF(s) are of mature and proven technology with safety-critical application heritage. Hardware elements supporting SCF(s) are designed to detect failures. These HESSs may include but are not limited to the following: circuit cards, internal and external buses, signals, discretes, interfacing items, feedback loops, I/O, CCDLs, processors, field programmable gate arrays, application specific integrated circuits, backplanes, data storage devices, connectors, and power supplies. Hardware elements supporting SCF(s) are designed with reliable components and meet system reliability requirements.

In addition, for designs utilizing MCP environments:

2. The processor family, model, and architectural and performance details (e.g., clock rate, total number of processing cores) of each MCP have been documented including known processor errata and deficiencies. Any known uses of the MCP in safety-critical applications have been documented to include known issues with its use in the given applications. Documented MCP errata, deficiencies, and issues have no adverse impact to the integrity of SCF operation or have been mitigated and shown to be safe.

3. The processing cores and other resources that are hardware enabled, and hardware disabled in the MCP are identified. Methods used for hardware disabling are identified. Hardware enabled processing cores and other resources that are functionally deactivated by MCP settings maintain the configured state throughout system operation. Mechanisms are in place to detect and accommodate inadvertent state changes to maintain the integrity of SCF operations.

4. For processing cores supporting SCFs, the lowest level cache is dedicated in its physical entirety to its respective processing core. Additionally, higher levels of cache are dedicated to processing cores either in their physical entirety or through use of a partitioning approach that allocates portions of cache to individual processing cores. Cache partitioning approaches do not unacceptably degrade SCF performance.

5. Data coherency is implemented, maintains the integrity (value and timing) of data associated with safety-critical processing across the MCP environment, and ensures the integrity of SCF operation.

6. Unused processing cores within an MCP are hardware disabled or functionally deactivated. Processing core activation and deactivation mechanisms ensure functionally deactivated processing cores remain functionally deactivated and functionally activated processing cores remain functionally activated. The MCP environment performs periodic monitoring during operation to ensure processing core activation and deactivation scheme does not change. Fault handling techniques maintain the integrity of SCF operation in the presence of changes in activation state (e.g., functionally deactivated processing core becomes functionally activated).

7. The failure modes of all processing cores, chip resources, and components of an MCP, whether hardware enabled, hardware disabled, functionally activated, or functionally

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deactivated, have been identified, are understood, and mitigations are in place to prevent adverse impacts to SCF operation in the event of their occurrence.

8. The MCP hardware contribution to the processing of SCFs is shown to be safe, deterministic, and in compliance with latency and performance requirements.

9. Interface management approaches, between processing cores and external to the MCP, are identified for each processing core to include rates, prioritization, routing, error management, interference resolution, and methods for ensuring consistency across shared memory.

10. Interference channels and areas of possible resource contention have been identified with their associated mitigations necessary to prevent adverse impacts to the integrity of SCF operation. Processing conditions that can increase the likelihood of interference or resource contention are identified.

11. Failure modes of the MCP have been identified along with their detectability. Critical failures have been identified and are detectable. Detectable failure modes are shown to be detected during PICFO and safely managed by the system. The system provides a safe response to MCP failure modes (single or combination) IAW fault tolerance requirements.

12. A safe approach is used for managing the features and settings of each processing core and resource in an MCP. All configurable features and settings of the MCP are identified and the flight configuration of the MCP settings are documented. The deactivation of processing cores and resources within an MCP through the use of features and settings does not adversely impact the integrity of SCF operation. Mechanization of MCP settings ensures settings remain persistent during operation. Fault conditions that can result in changes to the MCP settings are detectable and safely accommodated. All dynamically modifiable performance features of the MCP (e.g., dynamic modification of clock rates), to include all features that can impact deterministic processing, are identified and how each will be configured to eliminate or minimize processing changes in order to support SCF processing. Features of the MCP that have the potential to adversely impact the integrity of SCF operation (e.g., result in non-deterministic operation, create interference channels or resource contention that cannot be characterized) are disabled or safely mitigated to ensure no adverse impact to the integrity of SCF operation.

13. The MCP is designed to fully support an AMP or BMP scheme for processing cores processing a SESS. The MCP does not permit (by design or settings) the dynamic allocation of a process or VM to another processing core processing a SESS. Processing schemes used on processing cores not processing a SESS do not have any nominal or failure mode impacts to SESS processing.

14. Microcode update mechanisms are identified and a control plan established that prohibits unauthorized updates by controlling and authorizing updates only after thorough evaluation and test. Microcode update procedures are repeatable, consistent, and include methods for verifying the correctness of the installed update. Updates do not have a detrimental impact on SCF processing.

15. The MCP hardware interrupt configuration and design is identified. Multicore processor hardware interrupts are serviced in a safe and deterministic manner that does not impact other non-interrupted processing cores (and VMs as applicable) that are operating in the MCP environment. Interrupt routing and servicing schemes utilized provide for deterministic, stable SCF operations. Shared interrupts are not utilized for SCF operations (exception: a full MCP

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reset interrupt). Interrupts supporting a non-SCF process or processing task do not impact SCF processing.

16. The design, routing, and arbitration scheme of the processor interconnect and other resources used to interface processing cores and resources within the MCP ensure the integrity of SCF operation. The design and routing utilized do not introduce unacceptable latency or cause unacceptable jitter. Arbitration of messages support deterministic SCF operation compliant with latency and performance requirements. Failures in non-safety-critical processes or processing tasks do not adversely impact the ability of the processor interconnect or resources used for interfacing, their routing, or arbitration to support the integrity of SCF operations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination. Examination verifies that the HESSs of the SPA are identified and defined. Hazard analysis ensures hazards are traced to applicable HESSs. Verify through analysis and test that SPA HESSs and applicable components have the following performed as appropriate:

a. Acceptable level of burn-in time completed without failure.
b. Supplier acceptance testing (e.g., parts screening, sample lot acceptance testing).
c. Failure modes, effects, and criticality analysis performed.
d. Electronic components supporting safety-critical applications are tested to safety-critical standards.
e. Failure modes and effects testing on hardware supporting SCFs.
f. Qualification testing required for expected flight envelopes.

Examination and analysis ensure that all testing conducted at vendor and prime developer facilities adequately verifies hardware performance.

In addition, for designs utilizing MCP environments:

2. Analysis verifies that all MCP errata, deficiencies, and issues have been analyzed for safety impact, and have no adverse impacts to the integrity of SCF operation or have been safely mitigated.

3. Examination of MCP configuration documentation and settings verifies that processing cores that are hardware enabled and hardware disabled have been identified and confirms how processing cores will be hardware enabled or hardware disabled. Analysis and test verify that the mechanism to enable and disable processing cores maintains the configuration as set. Analysis and test verify that inadvertent changes to state are detected and safely accommodated to maintain the integrity of SCF operation.

4. Examination of MCP design and configuration documentation verifies that lowest level cache is dedicated in its entirety to its respective processing core. Examination verifies that higher levels of cache are dedicated to individual processing cores either in their physical entirety or through a partitioning approach. Analysis and test verify that the cache architecture and any partitioning scheme(s) utilized support the integrity of SCF operation.

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5. Analysis and test verifies that the data coherency mechanization maintains the integrity (value and timing) of data used by SCFs and ensures the integrity of SCF operation.

6. Examination of MCP settings confirm how processing cores will be functionally activated or deactivated, and that unused processing cores are hardware disabled or functionally deactivated. Analysis and test verify that the mechanism to activate or deactivate the processing cores maintains the configuration as set. Testing verifies that the MCP environment periodically monitors the state of processing core activation or deactivation and detects any changes in the activation state. Testing verifies that fault-handling techniques maintain the integrity of SCF operation for failures that change the activation state.

7. Failure modes and effects analysis and FMECA document failure modes of the MCP and identify their potential impact to SCFs. Analysis, testing, and simulation of chip-level failures verifies that failure mitigations are in place to prevent adverse impacts to the integrity of SCF operation. Analysis, testing, and simulation of chip-level failures verifies that failures of processing cores, chip resources, and components not supporting SCFs do not adversely impact the integrity of SCF operations, or that their effects have been mitigated to prevent adverse impacts to the integrity of SCF operation.

8. Analysis and test verifies that the hardware provides deterministic operation and meets latency and performance requirements for SCF operation.

9. Analysis verifies that the interface management approaches utilized by the MCP (between processing cores and external to MCP) have been identified and provide adequate rates, priorities, routing, error management, interference resolution, and methods for ensuring consistency across shared memory supporting the integrity of SCF operation. Testing, including FMET, verifies that the functional operation and performance of the interface management approaches meet requirements and are adequate for SCF processing.

10. Analysis of the MCP hardware identifies all interference channels and areas of resource contention, processing conditions that can contribute to interference or resource contention, and associated mitigations. Testing verifies that the MCP responds as expected under conditions that can produce interference or resource contention and that mitigations employed are adequate to prevent adverse impacts to the integrity of SCF operation. The testing is performed over a comprehensive set of expected operating conditions including the known worst-case operating and timing conditions that can result in interference or resource contention. The testing covers all known interference channels, areas of possible resource contention, and associated mitigations.

11. Failure analysis (e.g., FMEA, FMECA) verifies the MCP failures have been identified, the detectability of each failure, and whether each failure is a critical failure. Analysis and testing, including FMET, verify the detection of failures and the system response supports fault tolerance requirements.

12. Examination of MCP design and configuration documentation verify that the following are identified: configurable features, dynamically modifiable performance features, and the defined configuration of the settings for all features. Analysis and test verify that the deactivation of processing cores and resources within an MCP through features and settings do not adversely impact the integrity of SCF operation. Analysis and test verify that the MCP settings mechanization scheme maintains the settings per the standard and that the settings utilized support deterministic, stable SCF processing and disable features that can influence deterministic

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operation (e.g., power saving features), cause interference channels, or resource contention. Analysis and testing show that unintended changes in settings are detected and safely accommodated.

13. Analysis and testing verify that the MCP supports an AMP or BMP scheme for processing cores processing a SESS and that other schemes utilized for processing cores not processing a SESS have no safety impact on SESS processing. The analysis also shows that the MCP will not dynamically allocate a process or VM to another processing core processing a SESS.

14. Examination of CM documentation and microcode update procedures verify that control plans and update procedures are in place for safely handling microcode updates, including fully evaluating the impact to SCF processing. Testing verifies that microcode updates can be performed repeatedly, consistently, and verify the installation is correct. Testing verifies that the microcode update does not adversely impact the integrity of SCF operation.

15. Examination of MCP design and interrupt configuration documentation verifies that the configuration approach for hardware interrupts has been identified. Analysis and test verify that the routing and servicing scheme facilitates deterministic, stable SCF operations; does not impact non-interrupted processing cores (and VMs as applicable); and that non-safety-critical interrupt handling does not impact safety-critical processing.

16. Examination of the MCP design documentation verifies that the design, routing, and arbitration schemes of the processor interconnect and other resources used for interfacing have been defined. Analysis and test verify that the design and routing employed do not introduce unacceptable latency or unacceptable jitter into messages supporting SCF operations. Analysis and test also verify that the arbitration scheme utilized prioritizes those messages supporting safety-critical processing to ensure compliance with latency and performance requirements and that failures with non-safety-critical processes or processing tasks do not adversely impact the ability of the processor interconnect or resources used for interfacing to support the integrity of SCF operations.

15.3.2 Separation of safety-critical electronic hardware components.

Standard:

1. Verify that all safety-critical electronic hardware components are physically and functionally separated from non-safety-critical electronic hardware components. If not separated, verify that the non-safety-critical electronic hardware components are developed and qualified to the same CSIL as the HESS with which they are integrated. All safety-critical electronic hardware components are designed, developed, integrated, and tested to appropriate CSILs. Electronic hardware components of HESSs are physically and functionally separate from non-safety-critical electronic hardware components. Non-safety-critical electronic hardware components that are integrated with the electronic hardware of a HESS are designed, developed, integrated, and tested to the CSIL of the HESS.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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1. Verification methods include analysis, test, and examination. Verify through examination of hardware design documentation that safety-critical electronic hardware components of HESSs and non-safety-critical electronic hardware components are identified and documented. Analysis, test, and examination of design documentation verify that the safety-critical electronic hardware components are physically and functionally separated from non-safety-critical electronic hardware components. Verify through examination of documentation identifying CSIL designations for hardware that non-safety-critical electronic hardware components, which are integrated with the electronic hardware of a HESS, are designed, developed, integrated, and tested to the CSIL of the HESS. Analyses (e.g., FMECA) and tests (e.g., FMET) verify that non-safety-critical electronic hardware components do not adversely impact the integrity of SCF operation.

15.3.3 Data bus redundancy.

Standard:

1. Verify that HESSs have redundant buses that are physically separated. Buses which carry safety-critical data are identified as safety-critical. These buses are redundant, implemented with proper redundancy management, bus control, and are physically separated (e.g., run through separate connectors and separate wiring harnesses).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination of design documentation. Verification of safety-critical buses and physical separation is through analysis and examination of SPA drawings and other documentation. Analysis and examination identify the safety-critical bus controller(s). Testing verifies the redundancy management and bus control functionality.

15.3.4 Environmental qualification.

Standard:

1. Verify that all hardware elements are capable of safely operating within planned operational environments. All hardware elements have been identified. The environmental requirements (e.g., environmental criteria document) have been defined for natural and induced environmental conditions expected for air system locations, envelopes, and operational environments. All hardware elements meet defined environmental requirements and component integrity standards (e.g., MIL-STD-1796 and MIL-STD-810).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and test. Verify that environmental life requirements are met by hardware elements through test and analysis. All hardware elements are qualified to

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defined environmental requirements and those requirements reflect current environments expected.

15.4 Software development processes.

15.4.1 Software processes.

Standard:

1. Verify that the software development process for SESSs is fully documented (e.g., SDP, Software Safety Plan), followed, and is sufficiently suitable to produce software supporting SCFs. Identify all SESSs via a documented SCFTA. SESSs adhere to a rigorous development process which is suitable, comprehensive, well defined, documented, and consistently applied. The processes and activities are applied to all software intended for use (e.g., new development, COTS, GOTS, reuse, legacy). The processes and associated life cycle activities are based on a development standard that represents a high integrity process for safety-critical software. The development process is documented in an SDP (or set of SDPs) that addresses all activities required for the developing organizations. The activities are identified to support the SESS requirements, design, code, integration, verification, validation, and release of products. The SDP may reference other documents and data as part of identifying all processes and related activities (e.g., coding standards, system safety plan, risk management plan, CM plan, and quality assurance plan). Development processes for SESSs and SENSSs may be addressed in the same SDP. The software development processes account for developing SESS software to operate with the technologies utilized (e.g., MCPs).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes examination of process and product documentation. Verify that the SESSs are identified and documented. Verify that the SESS development processes address the following areas and the defined processes are followed:

a. Functional analysis (e.g., SCFTA).
b. Safety-critical function testing methodology.
c. Quality assurance.
d. Configuration management.
e. Coding standards.
f. System safety.
g. Peer reviews.
h. Software test methodology (e.g., unit-level, software integration, software requirements).
i. Regression test methodology.
j. Verification environment (e.g., SIL).
k. Failure modes and effects testing.
l. Risk management.
m. Software Review Board composition.
n. Traceability.

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o. Metrics relevant to data used to support AW criteria (e.g., throughput utilization, structural coverage, requirement verification coverage).
p. Software discrepancy reports.
q. Change impact analysis.
r. SESS architecture support of SPA.
s. Requirement definition and allocation.
t. Software engineering environment.
u. Software reuse.
v. Areas of auto code.
w. Processes adequately account for the technologies utilized (e.g., MCPs).

15.4.2 Traceability.

Standard:

1. Verify that each ESS has sufficient bidirectional traceability established for requirements (performance and interface), design, implementation, source code, and test data. Bidirectional requirement traceability exists from the air system level to the lowest requirement level. ESS bidirectional traceability also exists from the ESS requirements to the detailed design, and to the implementation and source code and all levels of test. Safety-critical function bidirectional traceability exists both vertically (system level to the lowest level) and horizontally (functionally across the system). Requirements supporting SCFs are tagged with an identifier of the specific SCF(s) supported. Bidirectional traceability of all software ensures that unwanted or unapproved code is not present (e.g., unused code, malicious code).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes analysis. Verify that requirements have full bidirectional traceability from the system level to the lowest level requirements. Verify that bidirectional traceability also exists for the ESS design, implementation, and source code. Full SCF bidirectional traceability (both vertically and horizontally) from the system level to the ESS design level is verified through analysis of traceability data. Verify through bidirectional traceability that requirements and functions are coupled to all levels of testing. Verify that requirements supporting SCFs are tagged with an identifier of the specific SCF(s) supported. Verify that no unwanted code exists.

15.4.3 Configuration management.

Standard:

1. Verify that the configuration and change control management process is fully documented, followed, and is sufficiently suitable to control all computer systems and software items across the system. Configuration management and change control processes are defined and documented for all hardware and software. Hardware configuration items and CSCIs are defined and the associated CSIL is identified. The configuration and change control management process ensures that the physical and functional characteristics of a HWCI and CSCI are identified,

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documented, and verified. The CM process identifies means to record the specific configurations of HWCIs and CSCIs and establishes baselines for each CI. Processes are defined and documented to control changes to the baselines and associated documentation of all HWCIs and CSCIs under CM control throughout the life cycle. The CM process accounts for managing the configuration of the technologies utilized (e.g., MCPs) including modifiable parameters and microcode.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes examination of process documentation, product configuration data, and quality assurance reports. Verify that the configuration and change control management process maintains the integrity of hardware and software configurations. Verify that all configurations are properly documented.

15.5 Software architecture and design.

15.5.1 Software architecture.

Standard:

1. Verify that the software architecture and design are defined, properly implement the system and software requirements, and are safe. Computer system software architecture is defined. System-level requirements are allocated to the subsystem and software requirements. All software and associated safety requirements are defined and allocated to software components. Software architecture is consistent with the software functional requirements; especially functions that ensure system integrity (e.g., partition schemes). The software architecture and design mechanizations implement proven techniques and are safe. The software architecture is compatible with the target hardware architecture.

In addition, for designs utilizing MCP environments:

2. The software architecture implemented across the processing cores of an MCP is documented.

3. The software architecture accounts for the unique attributes associated with operation in the MCP environment. The software architectural design identifies: all cross-process and cross-processing core dependencies, within and external to, the MCP; processing scheme (e.g., AMP, symmetric multi-processing planned for each processing core; SESS allocation to processing cores (and VMs as applicable); MCP resource allocation to software architectural components; resource control and management (e.g., bus control software); hypervisor implementations; VMs utilized; partitioning schemes utilized; and the executive and control structure capability to support multicore operation to comprehensively show the software architecture's interaction and operation with MCP hardware.

4. Any given instance of a process or processing task that is executing in a processing core (and VM as applicable) processing a SESS is assigned to only one processing core (and VM as applicable) and the assignment remains static during operation. The dynamic allocation of a

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SENSS process or processing task to a processing core or VM not processing a SESS does not adversely impact the integrity of SCF operation.

5. MCP resources needed for SESS operation are allocated and identified as shared or dedicated resources.

6. Single processing core SESSs that have been ported over for reuse on the MCP are identified and properly modified for use in the MCP environment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination of documents. Examination of software architecture design documentation ensures that the software architecture is defined. Analysis and test of the software architecture and design mechanization ensures software and system requirements are properly allocated, are met, and provide a safe implementation for the processes and processing tasks supported. Analysis ensures software design supports overall software architecture and does not conflict with system integrity requirements.

In addition, for designs utilizing MCP environments:

2. Examination of software architecture design documentation ensures that the software architecture is defined.

3. Examination of software and MCP design documentation verifies that the software architecture accounts for operation within the MCP environment. Analysis verifies that the software architectural design accounts for the items identified in the standard and that the software architecture and hardware interaction and operation is fully understood.

4. Analysis of the software architecture within the MCP environment verifies that the assignments of process and processing task instances to a processing core (and VM as applicable) processing a SESS are appropriate and remain static during operation. Analysis and test verify that the dynamic allocation of a SENSS process or processing task to a processing core (and VM as applicable) not processing a SESS does not adversely impact the integrity of SCF operation.

5. Analysis verifies that MCP resources needed by each SESS are identified and designated as shared or dedicated.

6. Examination of software process documentation verifies that a sound approach was used to port single processing core SESSs to the MCP environment. Analysis of the SESS identifies the attributes that are accounted for to port to the MCP environment, or that no modifications are needed. Testing verifies that the function and performance of the ported software meets functional and performance requirements.

15.5.2 Software control structure and execution rates.

Standard:

1. Verify for each SESS that the execution rates provided by the executive and control structure (considering priority assignments and interrupt design) are consistently obtainable and sufficient

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to safely provide the required performance for all SCFs supported. Software element supporting safety-critical function(s) flow and execution is deterministic. All processes and processing tasks meet their deadlines under all conditions. Processing frame rates are compatible with real-time system performance requirements and support execution rate requirements. System processing architecture mechanization including process and processing task priority assignments, interrupt structure, and overall processing control structure supports the integrity of SCF operation. The executive and control structure is designed, developed, integrated, and tested to the same CSIL (or higher) commensurate with the highest functional criticality of the system supported by the executive and control structure.

In addition, for designs utilizing MCP environments:

2. This standard is applicable per each processing core (and VM as applicable) within an MCP supporting SCF processing. All instances of executive control coupling between processing cores (and VMs as applicable) are identified and do not adversely impact the integrity of SCF operation.

3. For each processing core (and VM as applicable) utilized within the MCP environment, the software control and execution structure of the hosted software elements are identified, to include:

a. Processing rates for each process and processing task.
b. Foreground vs background process and processing task structure and prioritizations.
c. Major and minor frame structures and scheduling.
d. Memory management approach for each process and processing task.
e. Synchronization techniques used to synchronize processing cores (and VMs as applicable) to other processing cores (and VMs as applicable).
f. Synchronization techniques used to synchronize processing cores (and VMs as applicable) to redundant processes and processing tasks external to the MCP.

The software control and execution structure are designed to meet system performance requirements and support the integrity of SCF operation.

4. The executive and control structure enforces an AMP or BMP scheme for the operation of all processing cores processing a SESS.

5. The software control and execution structure accounts for interference channels and resource contention, and provides adequate mitigations to maintain the integrity of SCF operation.

6. All software interfaces with MCP hardware are identified and support the integrity of SCF operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination. Analysis verifies execution rates directly correlate to required processing frame rates and allowable data latencies. Examination of documentation verifies that the executive and control structure is developed to the same CSIL (or higher) commensurate with the highest functional criticality of the system supported by the

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executive and control structure. Software development lab tests verify that the executive and control structure meets required execution rates under fully loaded, worst-case timing conditions.

In addition, for designs utilizing MCP environments:

2. Examination of software design documentation verifies the software executive structure for each processing core (and VM as applicable) is identified, and that all instances of executive control coupling between processing cores (and VMs as applicable) have been identified. Testing verifies that the executive and control structure is adequately verified and validated for use in the MCP environment. Analysis and testing verify that hypervisors are adequately verified and validated and shown to be safe and compatible with the chosen operating system(s). Analysis and testing under nominal and off-nominal conditions verify that all instances of executive control coupling between processing cores (and VMs as applicable) support the integrity of SCF operation.

3. Examination of software design documentation verifies that the software control and execution structure attributes identified in the standard have been identified. Analysis and testing verifies that the software control and execution structure meets requirements and is safe.

4. Examination of software architecture requirements and design verify that an AMP or BMP scheme is utilized for cores processing a SESS. Analysis, demonstration, and test verify that the executive and control structure enforces an AMP or BMP scheme for processing cores processing a SESS.

5. Analysis verifies that interference channels and resource contention are identified for the MCP environment and their potential impact on the integrity of SCF operation. Analysis and testing verify that the software control and execution structure provide adequate mitigations for every possible interference channel and resource contention identified that can impact the integrity of SCF operation.

6. Examination of MCP hardware-software interface design documentation verifies that the software interfaces with the MCP hardware have been identified. Testing verifies that the interface design supports SCF processing requirements and there are no issues with the compatibility of the interface design.

15.5.3 Software architecture attributes and performance.

Standard:

1. Verify that the software architecture and design, including the initialization, synchronization, timing, data flow, control flow, interrupt structure, and data structures for all SESSs are safe and sufficient to support the required processing performance for all SCFs supported. The software architecture and design, software functional control flow mechanization, and data structures are suitable and compatible with the system and subsystem SCF processing without unacceptable latencies. Source code matches the data and control flow defined in software architecture. Real-time architectures are designed to meet processing requirements. All processes and processing tasks and interrupts are defined and prioritized. Processes and processing tasks and interrupt mechanizations are appropriately implemented. Software, data initialization, and synchronization requirements are defined and meet specified performance.

In addition, for designs utilizing MCP environments:

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2. This standard is applicable per processing core (and VM as applicable) within the MCP.

3. The MCP software contribution to the processing of SCFs is shown to be safe, deterministic, and in compliance with functional, timing, and performance requirements.

4. Data and control flow are identified for all internal (processing core (and VM as applicable)-to-processing core (and VM as applicable)) interfaces, external interfaces, and cross-process and cross-processing task dependencies. The methods used for data sharing and messaging are identified. All instances of data and resource sharing have been identified between all:

a. Multicore processor processes.
b. Processing tasks.
c. Processing cores (and VMs as applicable).

Processing cores not processing a SESS do not have write capability to shared safety-critical data. All functional coupling between processes and processing tasks running in the MCP environment are identified and support the integrity of SCF operation.

5. Data integrity and coherency mechanisms are incorporated for data and messaging management within the MCP environment. Data integrity and coherency mechanisms meet design requirements and are adequate to support the integrity of SCF operation.

6. Software within the MCP environment that can contribute to the occurrence of, or mitigate the impacts of, interference channels and resource contention events (e.g., shared resources, deadlocks, race conditions) is identified. Mitigations are employed to address identified interference channels and resource contention to maintain the integrity of SCF operation in the event of occurrence.

7. Within the MCP environment, the methods for synchronizing safety-critical processes and processing tasks across processing cores (and VMs as applicable), internal and external to the MCP, are documented and support integrity of SCF operation.

8. Software elements supporting safety-critical functions do not implement redundant functionality across processing cores (and VMs as applicable) within the same physical MCP as a replacement for, or means to eliminate, physical redundancy.

9. Timing requirements for processes and processing tasks have been established for each active processing core within the target MCP environment. The WCET has been identified for each process and processing task and meets latency and performance requirements. All interference channels; interfaces with other processes, processing tasks, and processing cores; resource contentions; and signal, data, and timing dependencies have been accounted for. Worst-case execution times are acceptable for safe operation of supported SCFs, and account for any cumulative WCET conditions supporting an SCF.

10. Software allocation and partitioning within each processing core (and VM as applicable) is shown to be safe. Partitioning implementations for each processing core (and VM as applicable) are identified and ensure the integrity of SCF operation. The partitioning scheme ensures that all throughput and data resources for SESS processing are fully protected. The CSIL of the software performing the partitioning scheme is equal to or greater in criticality to the highest level CSIL running within any partition.

11. The MCP environment initialization approach is identified. The approach correctly initializes the MCP environment upon startup or restart.

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12. All buses supporting SCF operation (i.e., safety-critical buses) have bus messaging traffic software (e.g., bus controllers) running on processing cores (and VMs as applicable) that are designed to run a SESS. Redundant bus control functionality for a safety-critical bus is implemented across separate MCPs; a physical MCP does not contain more than one processing core (and VM as applicable) for controlling the same safety-critical bus. MCP bus resources utilize a deterministic bus control mechanization for SESS operations, while providing adequate performance to support the integrity of SCF operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination. Examination ensures source code matches the data and control flow defined for the software architecture and design. Analysis and test verify processing performance requirements for SCFs are supported by the software architecture and design. Areas analyzed include processor and data initialization, synchronization of data and channels, overall flow of the software architecture and design implementation, data dependencies, process and processing task prioritization, and interrupt structure.

In addition, for designs utilizing MCP environments:

2. This MoC is applicable per processing core (and VM as applicable) within the MCP.

3. Analysis verifies that the MCP software design does not have unacceptable hazards; is designed to provide deterministic operation; and addresses allocated functional, timing, and performance requirements needed to ensure the integrity of SCF operation. Testing verifies deterministic operation, adequacy of hazard mitigations, and that requirements are met by the software design.

4. Examination of software design documentation verifies that data sharing and messaging methods have been identified. Analysis verifies that data and control flow have been identified per the standard, and all instances of data and resource sharing between all:

a. Multicore processor processes.
b. Processing tasks.
c. Processing cores (and VMs as applicable).

Analysis and test verify that processing cores not processing a SESS do not have write capability to shared safety-critical data. Functional coupling analysis verifies that all inter-process coupling (i.e., direct and indirect via data or control coupling) within the MCP environment has been identified and that safety-critical processing does not have any coupling dependencies on non-safety-critical data or control flow.

5. Examination of software design documentation verifies the data integrity and coherency mechanisms are identified for data and messaging. Analysis and testing verify that the mechanisms used provide the integrity and coherency needed to meet design requirements and support the integrity of SCF operation.

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6. Analysis verifies that all software within the MCP environment that contributes to or mitigates occurrences of interference channels or resource contention have been identified. The analysis addresses all software within the MCP environment (e.g., application software and software related to memory management schemes, processing frame structures and timing, data and control flow, and other internal and external dependencies). Analysis and test verify that the mitigations employed to address interference channels and resource contention maintain the integrity of SCF operation.

7. Examination of software design documentation verifies that the synchronization approach for the MCP environment is documented. Analysis and test verify that the safety-critical processes and processing tasks are properly synchronized and meet SCF processing and fault tolerance requirements.

8. Examination of design documentation verifies that SESSs do not implement redundant functionality across processing cores (and VMs as applicable) within an MCP as a replacement for, or means to eliminate, physical redundancy.

9. Examination of requirement documents verify that timing requirements for processes and processing tasks running in the MCP environment have been established. Analysis verifies that WCETs have been identified for each process and processing task in the target MCP environment, accounts for all applicable contributors, and identifies applicable cumulative WCET conditions that could impact the integrity of SCF operation. The analysis assesses a process and processing task WCET with respect to all other processes and processing tasks running in the MCP environment, taking into account any impacts that the other processes and processing tasks may have on the WCET. Analysis and testing verify that the WCETs (including cumulative conditions) support latency and performance requirements.

10. Examination of design documentation verifies the partitioning approach that is utilized for each processing core (and VM as applicable) is identified and that CSIL levels are appropriately assigned to the software performing the partitioning scheme. Analysis and test verify that software operating (nominally or in a failed state) in other partitions do not have the ability to influence partitions running a SESS. The analysis and test show that time and data processing of the SESSs are protected. Testing includes nominal operation testing and FMET to ensure failure conditions in other partitions do not propagate or effect functionality or performance of partitions running a SESS.

11. Examination of design documentation verifies that the initialization approach is identified. Testing verifies that the MCP environment initializes correctly upon startup or restart.

12. Examination of design documentation verifies safety-critical buses are identified, are controlled by processing cores (and VMs as applicable) that are designed to run a SESS, and that a physical MCP does not contain more than one processing core (and VM as applicable) for controlling the same safety-critical bus. Testing verifies that MCP bus resources utilized by a SESS are deterministic and support the integrity of SCF operation.

15.5.4 Dynamic operation.

Standard:

1. Verify that the following are designed to safely operate under all dynamic conditions anticipated: mode inputs, operational flight modes, failure monitoring and detection techniques,

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failure management functions, redundancy management, voting schemes, self-checks, BITs, safety interlock mechanizations, SCF interfaces, health status interfaces, reconfiguration capabilities, and switchover of C2 data links. The SESS designs account for all dynamic conditions anticipated for the system. Ensure the transient effects of mode switching and condition changes are acceptable and do not introduce unacceptable hazards. Switchover of C2 data links does not result in loss of control. Design techniques employed in the software, hardware, and computer system architecture mechanization of the system do not introduce unacceptable hazards. Typical areas to address are the techniques for detecting, monitoring, isolating, and accommodating failures; the entire redundancy management and fault tolerance scheme (from the lowest level through the system level); the techniques for assessing self-health; the techniques employed for determining other channels and external dependent subsystem and system health status; the voting scheme mechanization; the flow of all mode unique inputs through the system; and the safe implementation of internal, functional, subsystem, and system interfaces throughout all dynamic conditions, modes, and envelopes expected.

Note: “SCF interfaces” refers to interfaces supporting SCFs.

In addition, for designs utilizing MCP environments:

2. This standard is applicable per processing core (and VM as applicable) within an MCP supporting SCF processing.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination of test plans. Verification testing includes extensive system and subsystem integration tests, and FMET at various levels. Analyses and tests ensure the system is safe under all expected dynamic conditions and modes throughout the design envelope. Verify through examination that test plans cover the testing of all dynamic conditions expected.

In addition, for designs utilizing MCP environments:

2. This MoC is applicable per processing core (and VM as applicable) within an MCP supporting SCF processing.

15.5.5 Failure management and redundancy management.

Standard:

1. Verify the safe operation of BIT, redundancy management, and failure management algorithms. For each subsystem’s first failure instance, the fault detection and accommodation implementation is designed to detect, isolate, and accommodate 100% of critical failures to maintain the integrity of SCF operation. For subsystems supporting SCFs that have been determined to require three or more channels of redundancy, the fault detection and accommodation implementation is designed to detect and accommodate 95% of second like critical failures to maintain the integrity of SCF operation. The fault detection and accommodation implementation meets specified safety requirements (e.g., PLOC). The fault detection and accommodation implementation is compatible with the levels of redundancy,

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redundancy management mechanizations, and failure management mechanizations employed. The system design addresses acceptable transient levels, prevention of propagation of failures, and re-admittance of failed elements.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, demonstration, and examination. Testing includes system integration testing with actual flight hardware, FMET, and ground test. A combination of analysis, demonstration, test, and examination verifies meeting detection, isolation, and accommodation requirements such that, given a failure, the system will maintain the integrity of SCF operation. System and subsystem integration testing, ground testing, and FMET ensure that the design accommodates transient levels, prevention of propagation of failures, and re-admittance of failed elements.

15.5.6 Digital system failures.

Standard:

1. Verify that the SESSs are adequately designed to detect and accommodate digital system failures. The computer system architectural design, through techniques employed in the software and associated hardware, is fault tolerant. Typical digital system areas to address are initialization, reconfiguration, processor halt detection, transient suppression, underflow and overflow protection, deadlock prevention, race conditions prevention, divide-by-zero protection, not-a-number protection, anti-aliasing, saturation protection, memory protection and utilization, self-check, and protection techniques against failure propagation to preclude safety hazards. Protection techniques to consider include but are not limited to processor error detection and recovery, watchdog monitor implementation, and processing frame over-run protection.

In addition, for designs utilizing MCP environments:

2. The MCP environment is capable of detecting and accommodating digital system failures (including all unique MCP failure modes) to meet safety and fault tolerance requirements.

3. The watch dog monitor designs and protection mechanisms for each process and processing task and processing core (and VM as applicable) are fully documented, prevent failure propagation within SCF operation, and do not induce safety hazards. The triggering of watch dog monitors or protection mechanisms by a processing core not processing a SESS or by a non-safety-critical process or processing task does not adversely impact the integrity of SCF operation.

4. The MCP is designed to detect and accommodate MCP hardware failures with a graceful shutdown of the entire MCP environment. If it can be assured that the failure detected is a hardware failure limited to a single processing core and the accommodation will not have any impact to SCF processing in other processing cores, then this particular failure mode may allow the MCP environment to continue processing while only gracefully shutting down the processing core that failed.

Method of Compliance:

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Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination of design documentation. Complete software testing is accomplished from the unit level to the system level. Digital hardware FMEA and FMET verify that the system's design is able to detect and accommodate digital system failures. Verify that the design mitigates hazards due to digital system failures. Verify, through analysis and test, that data and real time processing integrity are maintained.

In addition, for designs utilizing MCP environments:

2. Analysis verifies that the MCP failure modes are identified. Analysis and testing (e.g., FMET) verify that the identified MCP failure modes are detected and accommodated to meet fault tolerance requirements.

3. Examination of design documentation verifies that the monitoring and protection mechanisms have been documented. Analysis and test verify that the watch dog monitors and protection mechanisms prevent failure propagation; do not induce hazards; and ensure that triggering by a processing core not processing a SESS, or due to a non-safety-critical process or processing task, does not adversely impact the integrity of SCF operation.

4. Analysis verifies failure modes that are clearly hardware failures of the MCP are identified and whether the failure can be isolated to drive an accommodation that shuts down one processing core or the whole MCP. Simulation and testing verifies that hardware failure modes result in graceful shutdowns of MCP (or processing core when applicable).

15.5.7 Restart and reset capabilities.

Standard:

1. Verify that SESS designs have the necessary provisions to restart and/or reset the system safely while in flight operations. The system SESSs are designed in conjunction with the digital hardware to reset and/or restart the computer system safely without inducing unacceptable effects (e.g., transients). Aspects of the design include channel and data resynchronization, the systems interrupt structure, the system reinitialization, re-check of system health, and reconfiguration to safe states. The design of flight-critical functions accommodates transient time limits dependent on the air system's inherent stability and safety margins, envelope, and flight maneuvers before unrecoverable departure.

In addition, for designs utilizing MCP environments:

2. This standard is applicable per processing core for the implementation of restart and/or reset capability within the MCP environment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and test. Testing is accomplished at all levels (from software to integrated system) and includes FMET. Analysis coupled with test verifies that the

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reinitialization and reconfiguration capability, channel and data resynchronization, and system's interrupt structure meet the system reset and/or restart safety requirements. System integration tests ensure all safety related state transitions meet required timelines, are stable and do not result in loss of safety-critical data.

In addition, for designs utilizing MCP environments:

2. This MoC is applicable per processing core for the implementation of restart and/or reset capability within the MCP environment.

15.5.8 Unsafe techniques.

Standard:

1. Verify that SESSs do not utilize or include unsafe techniques or attributes (e.g., patches, deactivated code, lab test functionality). Program process documents identify the prohibition of unsafe techniques. Code does not implement unsafe techniques. Unsafe techniques and attributes include:

a. Patching of object code.
b. Existence of deactivated code, flight test unique software, training system unique software, dead code, lab unique functionality.
c. Reuse of software with unknown or questionable pedigree.
d. Undocumented functions.
e. Dynamic memory allocation after initialization.
f. Unverifiable source code statements and structures.
g. Alteration of source code to achieve verification.
h. Any other unsafe techniques identified through established policy by the respective armed services.

Techniques that can adversely impact the integrity of SCF operation are prohibited, examples include the following:

a. Dynamic SCF process or processing task assignment across MCP cores.
b. Process or processing task parallelization within a processing core (e.g., Simultaneous Multi-Threading) for SCF processing.
c. Virtual machines containing SCF processing that rely on hypervisors that require software emulation of individual processing core operations.
d. Run-time configuration changes once initialization is complete.
e. Dynamic clock rate modifications.
f. Power saving modes.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, demonstration, and examination of process and design documentation. Verify, by analysis and examination of processes and code, that unsafe techniques are not employed. Perform testing and demonstration on the software product to

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ensure that employed techniques and attributes do not adversely impact the integrity of SCF operation.

15.5.9 Resource capacity.

Standard:

1. Verify that there is sufficient capacity and design margin for all processors, data channels (e.g., I/O, buses), and data storage devices. The system is designed to accommodate the software implementation with enough capacity (throughput, memory, bus, and I/O capacity) to complete all critical processing tasks. Each processing component is designed with enough capacity that any limits (e.g., in major and minor frames) are not reached (for Air Force, exceeded) under peak, worst-case, full loading. Systems analysis develops throughput, memory, bus, and I/O utilization allocations. The processing throughput capacity (as measured at the minor frame boundary) of each processing core (e.g., a single core of an MCP) does not exceed 90% utilization under peak, worst-case, fully loaded conditions. For internal processor buses, utilization does not result in lost data, erroneous data, unacceptable delays, or other effects that can adversely impact the integrity of SCF operation. For all other buses (e.g., internal LRU, external LRU), bus loading capacity does not exceed 90% utilization under peak, worst-case, fully loaded conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and test. Throughput is measured as a series of benchmarks of the implemented system. Throughput allocations are verified in test under peak, worst-case, fully loaded conditions (e.g., MCP environment with all software running simultaneously) that the air system could realistically encounter. Analysis and test verify that all processing and data channel rates are acceptable. Analysis verifies that load image size does not exceed available memory. Analysis and test verify that the memory usage and structure ensure the integrity of SCF operation under worst-case scenarios.

15.5.10 Software element supporting safety-critical function(s) performance.

Standard:

1. Verify that all SESSs provide acceptable performance and safety. All performance and safety requirements are identified, documented, and traceable to software specifications. Software requirements and interfaces are accurate, consistent, unambiguous, stated in quantifiable terms with tolerances, sufficiently detailed, and verifiable. Ensure that the implementation satisfies the performance and safety requirements. Performance deficiencies have been identified and fully documented. The design implementation has not introduced unacceptable hazards.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

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1. Verification methods include analysis, test, and examination of requirements. Analysis and examination are performed to determine proper flow down of system and safety requirements to the software specification. Hazard analysis is performed to ensure no new or unacceptable risks have been introduced by the design. Analyze discrepancy reports to determine failure trending and disposition. All performance and safety requirements are verified.

15.6 Software qualification and installation.

15.6.1 Software test methodology.

Standard:

1. Verify that each SESS is tested and integrated in a multi-level approach from the software component level to the integrated system level and that there is adequate test coverage at each level of testing. The system is integrated and verified using an established, documented, and proven process which includes thorough testing of SCFs with complete test and structural coverage. Testing activities include:

a. Unit testing.
b. Component testing.
c. Computer software and configuration item testing on target hardware.
d. Computer software and configuration item integration testing.
e. Subsystems testing.
f. Systems integration testing, including human-in-the-loop testing.
g. Safety-critical function thread testing.
h. Iron bird testing.
i. Flying test bed testing.
j. Air vehicle-level testing.
k. Ground testing.
l. Flight testing.

Types of testing include:

a. Requirements verification.
b. Failure condition testing (out-of-bounds, off-nominal and robustness).
c. Regression testing (required at all levels for software modification and revisions).
d. Functional validation.
e. Design validation.

Each build released for flight operations has all the functionality necessary to ensure integrity of SCF operation.

In addition, for designs utilizing MCP environments:

2. The integration methodology adequately accounts for the safety aspects of utilizing the MCP environment. Examples of areas addressed include:

a. Resource, process, and processing task allocation to processing cores.
b. Processing core, process, and processing task dependencies, integration, and test.
c. Interference channels and resource contention.
d. Performance of shared resources.

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e. Configuration and design validation of MCP operation with all loaded software components.
f. Processing core-level failure mode coverage.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis and examination of documentation. Examination of planning and test documentation ensures adequacy of the verification approach for all levels of testing. Verify design dependencies of the SPA are identified in planning documentation and are coupled to the overall test coverage strategy. Verify that build plans show a reasonable build-up approach. Examination ensures test documents reflect all levels and types of testing executed for each level of the architecture. Verify all functionality is in place to support SCFs and safe flight operations. Test coverage analysis verifies that no gaps exist in the test coverage. Test coverage analysis is performed on all levels of testing of the system and its SCFs.

In addition, for designs utilizing MCP environments:

2. Examination of software development and software integration plans verify that proper planning for utilizing and verifying software running on MCP environments has been accomplished.

15.6.2 Full qualification of software.

Standard:

1. Verify that all SESSs released for flight operations have been fully qualified. For each SESS, identify the associated CSCI. Full FQT is performed on target hardware for configurations of CSCIs being released for flight operations.

In addition, for designs utilizing MCP environments:

2. Software elements supporting safety-critical function(s) running on MCPs are fully qualified with the MCP configured and running with the full flight configuration of software loaded on the MCP and running in a flight representative environment that is realistically stressing the configuration under test. Test environment simulations and stubs can be utilized to create the full flight configuration of loaded software if those simulations and stubs are qualified to requirements and validated to show that their performance is equitable to the actual flight configuration.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination of process documentation. Examination verifies that 100% requirements testing (or other approved verification method, if required) is planned for SESSs. Verify that SESSs have 100% of their requirements verified for every CSCI version release for flight operations. Analysis and examination verify that all failed

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requirements have been dispositioned to assure safe flight operations. Ensure FQT reports are submitted for each CSCI.

In addition, for designs utilizing MCP environments:

2. Examination of SESS qualification plans, procedures, and reports verify that the qualification was performed with a full flight configuration of software planned for the MCP environment. Analysis and testing verify that any simulations and stubs meet acceptable performance within the qualification environment. Analysis of test results verify that the simulations and stubs functioned as expected during the qualification process.

15.6.3 Software build process.

Standard:

1. Verify that the software build process for SESSs is safe. A sound, repeatable process is defined and implemented for generating the air system's software load images. The build process addresses the build environment (including what is necessary to verify the build), identification of optional build processes (e.g., nominal, emergency release), safety, and CM. The process ensures that the build product is not corrupted, is compatible (hardware and software), and is complete. The scope of the build process includes executables and data files. All air system loadable and non-loadable software elements are identified. The software build process accounts for unique build aspects of the hardware utilized (e.g., MCPs) and verifies that the resulting build meets all safety objectives required to operate safely in the hardware environment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include analysis, test, and examination of build process documentation. The software load image build process is verified to ensure that the generated products are complete, compatible, and are not corrupt. Configuration and process documents are reviewed for completeness to ensure that all CSCI developers' products are included, and version descriptions and configuration data are identified (e.g., unique CSCI constraints, sizing information, load image link data, data files). Verify the correctness and completeness of the compatibility matrix through analysis and test. Ensure that the process for generating build products is reliably repeatable. Verify that the build process addresses all air system, manufacturer, and depot loadable software and data files. Examination of build process documentation verifies that the build process has properly accounted for unique hardware (e.g., MCPs) build requirements. Analysis and testing of the software build verifies that the build is correct and has incorporated all necessary safety attributes to meet required safety objectives.

15.6.4 Software load compatibility.

Standard:

1. Verify that adequate CM controls are in place to ensure proper and functionally compatible software loading for the intended use on the air system. Multiple versions and configurations of hardware and software in support of varying missions (e.g., service related) are carefully

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controlled. A matrix is established to identify compatibility for software and hardware configurations. A means for checking the correct software configuration (executables and data files) for loading is provided. Configuration management controls preclude loading an incorrect software version onto the air system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification method includes analysis, test, and examination. Verify that the hardware, software, and data file configuration for the flight release configuration is correctly defined and the installed configuration is compatible. Verify that CM controls are in place to prevent incorrect version selection and loading of incompatible or incorrect configurations.

15.6.5 Software load process.

Standard:

1. Verify that the software loading and load verification processes for all software are safe and correct. A sound, repeatable process is used to load software and associated data files onto the air system, ensuring all software load images have been properly loaded onto the corresponding hardware. The process ensures the loaded software and data files are correct, have not been corrupted, have not compromised compatibility (hardware and software), and are completely loaded. The software load process accounts for unique loading aspects of the hardware (e.g., MCPs) environment and verifies that the load result meets all safety objectives that are required for the software to operate safely in the hardware environment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

The number next to an MoC line item corresponds to the numbered line item in the standard that is being verified:

1. Verification methods include test and examination. Verify through examination that the software loading and load verification process is defined and implemented. Verify through test that the correct software load images and data files are consistently loaded by the software load process onto the air system. The correct loading of all manufacturer and depot loadable software (box loadable) is verified through examination and test. Examination of software load process documentation and loading instructions verifies that the load process has properly accounted for unique hardware (e.g., MCPs) loading requirements. Analysis and testing of the software load procedures validates that the load process is correct and has incorporated all necessary safety attributes to meet required safety objectives.

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16. MAINTENANCE

Although MIL-STD-1629 has been officially canceled, it is used as a foundational guide and includes Tasks 101 FMEA, 102 Criticality Analysis, and 103 FMECA-Maintainability Information Analysis. A FMECA-Maintainability Information Analysis includes maintenance and maintainers in the design process. The FMEA, Criticality Analysis, and FMECA-Maintainability Information Analysis are all necessary to make the FMECA useful for Section 16.

Section 16 includes criteria that use FMECA as an artifact to prove compliance of:

a. 16.1.2: Failure modes that cause hazards to maintainers or failure causes from maintenance that creates hazards will have cautions and warnings in the maintenance publications.
b. 16.1.6: Failure modes, effects, and criticality assessment end effects that cause hazards to personnel or air vehicle and are testable items will have operational, functional or system tests in the maintenance publications.
c. 16.2.4: Detection methods that require inspections will have corresponding inspections in the maintenance publications.

Reliability-Centered Maintenance is a data-driven process used to develop an air vehicle's failure management strategies resulting in a scheduled maintenance program. Reliability-centered maintenance employs a decision logic that considers the risk of failure with respect to safety, environmental, and cost consequences in determining the appropriate maintenance strategy for a given failure mechanism, mode, and effect. The RCM process utilizes SAE JA1011 including A4A MSG-3 Volume 1 and Volume 2 (previously ATA MSG-3).

Section 16 includes criteria that use RCM analysis as an artifact to prove compliance of:

a. 16.1.1: Servicing instructions will be identified from the RCM analysis scheduled restoration tasks.
b. 16.2.1: Preflight and post-flight daily inspections will be identified in the RCM analysis scheduled failure-finding tasks.
c. 16.2.2: Special inspections will be identified from the RCM analysis Event Driven tasks.
d. 16.2.3: Life-limited parts will be identified in the RCM analysis discard tasks.
e. 16.2.4: Maintenance inspections and intervals will be identified in the RCM Analysis Failure-Finding tasks and derived from On-Condition tasks.

16.1 Maintenance manuals and checklists.

16.1.1 Servicing instructions.

Standard: Maintenance publications include servicing instructions for replenishing consumables depleted during normal operations (e.g., fuel, engine oil, hydraulic fluid).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the maintenance publications to ensure they include servicing instructions derived from the RCM analysis servicing task requirements.

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(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct RCM analysis that meets SAE JA1011 to identify servicing task requirements. A4A MSG-3 Volume 1 and Volume 2 (previously ATA MSG-3) meet the intent of this MoC.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the servicing instructions are complete, accurate, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle servicing procedures on production-configured hardware using government maintenance personnel to ensure servicing procedures do not introduce AW hazards.

16.1.2 Cautions and warnings.

Standard: Maintenance publications and crewmember checklists document cautions and warnings for potential AW hazards introduced to the crewmembers or air system, IAW MIL-STD-38784, Warnings, cautions and notes paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the maintenance publications and crewmember checklists to ensure they include cautions and warnings related to the prevention of AW hazards derived from the FMECA, FHA, and SSHA and ensure the cautions and warnings are formatted IAW MIL-STD-38784, Warnings, cautions and notes paragraph.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct a FMECA to identify potential AW hazards.

(2) Conduct an FHA IAW SAE ARP4761, Functional Hazard Assessment paragraphs to identify potential AW hazards.

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(3) Conduct a SSHA IAW MIL-STD-882, Task 204 or a similar hazard analysis method to identify potential AW hazards.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(4) Perform a desktop analysis of cautions and warnings using the government verification team to ensure AW hazards identified through the FMECA, FHA, and SSHA have cautions and warnings in the maintenance publications and crewmember checklists.

16.1.3 Maintenance checklists.

Standard: Maintenance checklists do not introduce AW hazards to the air vehicle and are derived from the detailed procedures contained in the maintenance publication. Maintenance checklists are IAW MIL-DTL-5096, Checklists paragraphs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine maintenance checklists to ensure no deviations exist between checklists and procedures within the maintenance publications.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the maintenance checklists are complete, accurate, do not deviate from procedures within the maintenance publications, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle maintenance checklist procedures on production-configured hardware using government maintenance personnel to ensure maintenance checklists do not introduce AW hazards.

16.1.4 Support equipment.

Standard: Maintenance publications identify support equipment used to perform maintenance functions on the air vehicle. The support equipment used during maintenance will not introduce AW hazards.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examine the maintenance publications to verify that air vehicle specific operating parameters and procedures used with support equipment are included.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm maintenance publication procedures associated with support equipment are complete, accurate, can be performed as written, and the use of support equipment does not introduce AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle maintenance procedures that require support equipment on production-configured hardware using government maintenance personnel to ensure maintenance procedures using support equipment do not introduce AW hazards.

16.1.5 Removal and installation procedures.

Standard: Maintenance publications include procedures for removals and installations for each level of maintenance. Field level and depot level maintenance publications contain information to remove and install components, LRUs, subsystems, and systems.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Ensure maintenance publications have removal and installation procedures for each component, LRU, subsystem, and system.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the removal and installation procedures are complete, accurate, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

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Demonstration.

(2) Perform all air vehicle removal and installation procedures on production-configured hardware using government maintenance personnel to ensure removal and installation procedures do not introduce AW hazards.

16.1.6 Operational, functional, and system testing.

Standard: Maintenance publications include procedures for operational, functional, and system tests and identify support equipment, supplies, and tools to perform the test.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the maintenance publications to ensure they include operational, functional, or system test procedures and performance parameters for testable items that have an impact to AW.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct a FMECA to determine if failure of testable items have an impact to AW.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the operational, functional, and system tests are complete, accurate, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle operational, functional, and system test procedures on production-configured hardware using government maintenance personnel to ensure procedures do not introduce AW hazards.

16.1.7 Troubleshooting procedures.

Standard: Maintenance publications include procedures for troubleshooting at each level of maintenance. Troubleshooting procedures describe the probable malfunctions and remedial actions for those malfunctions or isolate faulty item for repair or replacement. Troubleshooting procedures identify items (e.g., support equipment, supplies, tools) needed to perform the troubleshooting activities.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the maintenance publications to ensure troubleshooting procedures are included for each level of maintenance.

(2) Examine the troubleshooting procedures to ensure information to identify the fault and remedial action is included.

(3) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the troubleshooting procedures are complete, accurate, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle troubleshooting procedures on production-configured hardware using government maintenance personnel to ensure troubleshooting procedures do not introduce AW hazards.

16.1.8 Non-destructive inspection procedures.

Standard: Maintenance publications identify NDI procedures and associated intervals, damage limits, and repair procedures.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine maintenance publications to ensure NDI procedures and associated intervals, damage limits, and repair procedures are identified.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct a FMECA to identify failure modes that require NDI detection.

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(2) Conduct SSHA IAW MIL-STD-882, Task 204 or a similar hazard analysis method to identify AW hazards that require mitigation by NDI.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the NDI procedures are complete, accurate, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle NDI procedures on production-configured hardware using government maintenance personnel to ensure NDI procedures do not introduce AW hazards.

16.1.9 Maintenance augmented reality.

Standard: Augmented reality devices used in the performance of maintenance procedures are consistent with approved configuration and maintenance publications.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review augmented reality air vehicle maintenance procedural materials for consistency to those in the maintenance publications.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the use of augmented reality does not introduce AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle maintenance procedures using augmented reality on production-configured hardware using government maintenance personnel to ensure use of augmented reality does not introduce AW hazards.

16.1.10 Critical safety item installation, handling, repair, and overhaul.

Standard: The maintenance publications document the CSIs and their CCs which are susceptible to damage from maintenance or handling. The maintenance publications include the statement "CRITICAL SAFETY ITEM" preceding procedural step(s) that could affect a CSI's CCs IAW

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MIL-STD-40051, Critical Safety Items paragraph and include instructions to preserve CSIs and their CCs.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the complete CSI CCs list with rationale for determination of susceptibility of CCs to damage from maintenance or handling to ensure there is engineering assessment for each determination.

(2) Examine maintenance publications to ensure that CSIs and their CCs susceptible to damage from maintenance or handling are identified and include instructions to preserve CSIs and their CCs.

(3) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct Operating and Support Hazard Analysis IAW MIL-STD- 882, Task 206 or a similar hazard analysis of all CSIs and their CCs to determine their susceptibility to damage from maintenance or handling.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm CSI installation, handling, repair and overhaul instructions are complete, accurate, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle CSI installation, handling, repair, and overhaul procedures on production-configured hardware using government maintenance personnel to ensure maintenance procedures do not damage CSI CC’s.

16.2 Inspection requirements.

16.2.1 Preflight and post-flight checklists.

Standard: Preflight checklists used by the ground crew contain steps to prepare and inspect an air vehicle for the first operation of the day, and post-flight checklists used by the ground crew contain steps to service and inspect an air vehicle after the final operation of the day. Preflight and post-flight checklists do not cause AW hazards.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examine ground crew and flight crew preflight checklists to ensure hazardous areas are clear prior to execution of preflight procedures.

(2) Examine ground crew post-flight checklist to ensure installation of protective items and locking devices that should be installed prior to performing maintenance are documented (e.g., covers, plugs, and safety pins).

(3) Examine ground crew preflight checklist to ensure removal of protective items and locking devices that should be removed prior to operation are documented (e.g., covers, plugs, and safety pins) unless contained in the flight crew or end-of-runway checklists.

(4) Examine ground crew preflight checklist to ensure security of items that should be secured prior to operation are documented, unless contained in the flight crew checklist.

(5) Examine ground crew preflight and post-flight checklists to ensure air vehicle configuration equipment requiring additional inspection or servicing tasks are addressed.

(6) Examine maintenance publications to ensure items that are subject to damage by environmental forces (e.g., wind, rain, or corrosive environments) have inspections documented in the checklist.

(7) Examine maintenance publications to ensure items that normally require service or verification of service prior to the first operation of the day have inspection or servicing tasks documented in the checklist.

(8) Examine post-flight checklists to ensure items that are subject to loss of lubricants, fluids, or compressed gases during operation have inspections.

(9) Examine maintenance publications to ensure items that are subject to damage from impact or foreign objects have inspections documented in the post-flight checklist.

(10) Examine post-flight checklists to ensure items that require daily inspections have documented inspections.

(11) Examine the maintenance publications to ensure they include preflight and post-flight checklists derived from the RCM analysis.

(12) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct analysis to determine if AW hazards exist due to actions being performed simultaneously by flight crew and ground crew while executing their respective preflight procedures.

(2) Conduct RCM analysis that meets SAE JA1011 to identify inspection requirements for items that are subject to loss of lubricants, fluids, or compressed gases during operation and for items that require daily inspections. A4A MSG-3 Volume 1 and 2 (previously ATA MSG-3) meet the intent of this MoC.

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Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(1) Perform all air vehicle preflight and post-flight checklist procedures on production-configured hardware using government maintenance personnel to ensure preflight and post-flight checklist procedures do not introduce AW hazards.

16.2.2 Special inspections.

Standard: The maintenance publications have special inspections for air vehicle exceedances as reported by the flight crew or diagnostic system, and abnormal events as reported by the flight crew or ground crew. Special inspection procedures define parameters and abnormal events which establish the need for the special inspection procedure to be performed. Damage criteria delineate serviceability, repair, or discard requirements. Special inspection procedures identify items (e.g., support equipment, supplies, and tools) needed to perform the inspections.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine maintenance publications to ensure instructions for special inspections establish allowable service limits, and standards for determining maintenance action.

(2) Examine maintenance publications to ensure exceedances in speed, temperature, g loads, landing sink rate, and vibrations which are documented in the flight manual have corresponding inspection or maintenance actions.

(3) Examine the maintenance publications to determine if use of emergency or alternate fuels requiring inspection after such usage per the operational materials have special inspection procedures documented.

(4) Examine maintenance publications to determine if sudden stoppage events (e.g., rotor, propeller, and compressor) have special inspection procedures documented.

(5) Examine maintenance publications to determine if FOD events (e.g., bird strikes, hail, and lost tool) have special inspection procedures documented.

(6) Examine maintenance publications to ensure lightning strikes to the air vehicle have special inspection procedures documented.

(7) Examine maintenance publications to ensure fires and use of firefighting equipment on the air vehicle have special inspection procedures documented.

(8) Examine maintenance publications to ensure all RCM analysis event driven tasks have special inspection procedures documented.

(9) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

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Analysis.

(1) Conduct RCM analysis that meets SAE JA1011 to identify event driven task special inspections derived from abnormal events. A4A MSG-3 Volume 1 and Volume 2 (previously ATA MSG-3) meet the intent of this MoC.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the special inspection procedures are complete, accurate, and can be performed as written without introducing AW hazards. Maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Demonstration.

(2) Perform all air vehicle special inspection procedures on production-configured hardware using government maintenance personnel to ensure special inspection procedures do not introduce AW hazards.

16.2.3 Life-limited parts.

Standard: Maintenance publications identify LLPs and their replacement intervals.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the maintenance publications to ensure they include LLPs derived from the RCM analysis.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct RCM analysis that meets SAE JA1011 to identify LLPs. A4A MSG-3 Volume 1 and 2 (previously ATA MSG-3) meet the intent of this MoC.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(2) Perform a desktop analysis of LLPs using the government verification team to ensure LLPs identified from the RCM analysis are in the maintenance publications.

16.2.4 Maintenance inspections and intervals.

Standard: Maintenance publications identify air vehicle and ground control station items requiring periodic inspections and the corresponding inspection intervals.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine the maintenance publications to ensure maintenance inspections and intervals are derived from the FMECA and RCM analysis.

(2) Examine the artifacts that demonstrate contractor certification or contractor validation to ensure the maintenance publications have undergone a quality check for currency and accuracy IAW TO 00-5-3, Contractor Certification Process and Certification Requirements paragraphs for the Air Force; NAVAIR 00-25-604, Contractor Validation Plan and Validation paragraphs for the Navy; and DA PAM 25-40, Validation paragraphs for the Army.

Analysis.

(1) Conduct RCM analysis that meets SAE JA1011 to identify maintenance inspections and intervals. A4A MSG-3 Volume 1 and Volume 2 (previously ATA MSG-3) meet the intent of this MoC.

(2) Conduct a FMECA to identify failure modes that require inspections to detect the failure.

Demonstration.

(1) Perform air vehicle maintainability demonstrations to confirm the inspection tasks are complete, accurate, and can be performed as written without introducing AW hazards. Prior to entering flight test, maintenance publications will receive updates to address maintenance issues discovered during maintainability demonstrations.

Prior to entering operational test or operations, verify the standard by the following method(s):

Analysis.

(3) Perform a desktop-analysis of inspection intervals using the government verification team to ensure intervals from the RCM analysis are in the maintenance publications.

Demonstration.

(2) Perform all air vehicle inspection procedures on production-configured hardware using government maintenance personnel to ensure inspection procedures do not introduce AW hazards.

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17. ARMAMENTS AND STORES INTEGRATION

Section 17 covers the integration, installation, interface, carriage, and employment of armament and stores on air systems. The armament system includes weapons and equipment that enable stores and weapons to operate (e.g., to deliver aerial munitions to a target). Stores, as defined in this section, are expendable and non-expendable items carried internally or externally to the air vehicle, including nuclear weapons, mines, fuel and spray tanks, torpedoes, sonobuoys, pods (e.g., refueling, gun, and electronic countermeasures), targets, decoys, drones, and air vehicle survivability equipment (e.g., chaff and flare). Weapons are actual munitions, including guns, guided and unguided rockets, missiles, dispensed munitions, bombs, DEWs, and lasers and laser designators. Armament equipment is associated with suspension and release equipment (pylons, racks, adapters, launchers, and countermeasure dispensers). The armament system also includes the SMS, which manages the interface between the air vehicle, the armament equipment, and the weapons themselves. It controls the selection, arming, and release of stores. The armament system also includes the fire control system, which is responsible for aiming and firing the weapons. It takes data from various sensors to calculate a firing solution to accurately hit the target. Not all stores are directly controlled by the armament system. In cases where the standard identifies “store/weapon”, the associated armament system enabling the store operation is also included.

The SMS manages communication between air vehicle stores and other air system subsystems and provides the user interface to the crewmembers. The SMS can be standalone or highly integrated with air system avionics and fire control systems. Evaluating AW for store carriage and employment requires a multidisciplined approach to ensure safe store operation and air vehicle-to-store compatibility. Store compatibility requires assessing the entire armament system from initial store loading through mission employment or recovery. Airworthiness assesses a subset of store compatibility that addresses the ability of an air vehicle and the stores to coexist and operate safely without degradation to either the air vehicle or store. In this context, compatibility is the ability of an air vehicle, stores, SMS, and suspension and release equipment to coexist without unacceptable effects of one on the aerodynamic, structural, electrical, or functional characteristics of the others under all flight and ground conditions experienced by the air vehicle-to-store combination.

MIL-HDBK-244, MIL-HDBK-1763, and ADS-44 provide detailed guidance on stores compatibility and armament system and subsystem AW qualification. Familiarity with the handbooks and their additional references is needed to evaluate Section 17 criteria. Section 17 criteria are applicable if there are changes in the configuration usage or environment of the air system or stores that impact compatibility. Section 17 covers armament system and store/weapon integration for any application from newly developed air systems and stores/weapons to minor modifications.

Section 17 is divided into five sections:

a. Aeromechanical: addresses physical fit and function, store structural integrity, store separation, stability and control, exhaust effects, and safe escape.
b. Stores environments: addresses electrical compatibility, induced environments, and E3.
c. Stores integration: addresses stores management functions, store safe operating conditions, system health, air vehicle-to-store interface (logical), software compatibility, and positive control of weapons.

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d. Laser integration: addresses crewmember exposure, laser boresight, laser fire control, latching, display indications, airframe contact, and ground lasing.
e. Stores technical documentation: addresses stores configuration, handling and loading, loadouts and limitations, and stores operating and delivery procedures.

17.1 Aeromechanical.

17.1 focuses on aeromechanical engineering disciplines that affect the ability to safely carry and employ stores.

17.1.1 Physical fit and function.

17.1.1 criteria assess air vehicle and stores physical fit and function. This includes ground fit and compatibility, mechanical function (e.g., carriage systems, weapon bay doors, store wing deployments), routing of cables, lanyards and umbilicals, loading, unloading, handling, and alignment. Alignment is an AW concern as poor alignment can result in degraded separation characteristics, increased drag effects, failure to meet minimum clearance for gunfire operations, and impingement of store antenna emissions.

17.1.1.1 Stores clearances.

Standard: Stores meet minimum clearance requirements IAW MIL-STD-1289, Clearances section during all phases of ground and flight operations.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analysis shows the store OML is the same, or has minimal differences, from a previously certified store.

(2) Virtual or analytical fit check shows clearance requirements are met IAW MIL-STD-1289, Clearances section.

(3) Dynamic deflection analysis ensures stores do not contact the air vehicle, pylon, ground or other stores during all phases of flight.

(4) For rotary-wing air vehicles, analysis verifies that projectiles do not contact rotor blades during dynamic maneuvers IAW MIL-STD-1289, Clearances section.

Test.

(1) Ground test: Verify that physical fit and clearance requirements are met IAW MIL-STD-1289, Clearances section and MIL-HDBK-1763, Test 101 for all store configurations.

(2) Ground test: Verify that fired munitions and projectiles have clearance from the air vehicle or other stores IAW MIL-STD-1289, Clearances section and Installation Test Methods section and MIL HDBK-1763, Test 160.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(3) Flight test: Verify stores do not contact the air vehicle or adjacent stores throughout the operational envelope.

(4) Flight test: Verify that fired munitions and projectiles do not contact the air vehicle or other stores IAW MIL HDBK-1763, Test 273 and Test 274.

17.1.1.2 Mechanical function.

Standard: Air vehicle and store mechanical interfaces allow for full range of motion and function of air vehicle and store movable equipment (e.g., guns, shutters, fin deployment, rotary launchers, flaps, slats, speed brakes).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct digital function checks of air vehicle and store movable equipment to verify operation throughout range of motion.

Test.

(1) Ground test: Verify store mechanical functions IAW MIL-HDBK-1763, Test 101.

(2) Ground test: Verify ammunition loading, misfire handling, store malfunctions, and gun jams will not cause damage to the air system or injury to crewmembers, IAW MIL-HDBK-1763, Test 160.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Verify store mechanical functions operate as intended in the flight environment.

(4) Flight test: Verify ammunition loading, misfire handling, store malfunctions, and gun jams will not cause damage to the air system or injury to crewmembers, IAW MIL-HDBK-1763, Test 273 and Test 274.

17.1.1.3 Loading, handling, and accessibility.

Standard: All stores are capable of being uploaded, downloaded, and handled under all operating conditions without damaging the air system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that store loading and installation procedures are documented and support loading and handling of stores without degrading accessibility IAW MIL-STD-1289, Accessibility section to other air systems or inhibiting crewmember egress in the event of an emergency.

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(2) Verify that store technical documentation covers all upload and download procedures, store maintenance and troubleshooting procedures, weapon station checks, installation of arming lanyards and clips, and removal of safing pins.

(3) Verify that ground support equipment required for loading and weapon station checks have been evaluated for compatibility IAW MIL-HDBK-1763, Test 101 and Test 102, without compromising the AW of the air system.

Demonstration.

(1) Verify through demonstration that stores can be uploaded and downloaded under all operating conditions while maintaining clearances with the air vehicle and any adjacent stores IAW MIL-HDBK-1763, Test 101 and Test 102, without compromising the AW of the air system.

17.1.1.4 Lanyard and umbilical routing.

Standard: Lanyards, wires, hoses, and store umbilicals are of the correct length for routing and can be secured and retained for all carriage and release operating conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze models or drawings for proper routing (e.g., correct length, bend radius, clamp and retention points) and security of all lanyards, wires, hoses, and umbilicals for all applicable weapon stations and mixed load configurations. If lanyard routing is significantly different than previous cleared, then endurance testing IAW MIL-HDBK-1763, Test 253, is required.

Demonstration.

(1) Verify routing and security of all lanyards, wires, hoses, and umbilicals for all weapon stations and mixed load configurations.

Test.

(1) Lab test: Verify correct function and release of all arming lanyards, retention lanyards, pull-out switches, and store umbilicals IAW MIL-HDBK-1763, Test 110.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Lanyard and umbilical continues to function and is not damaged when exposed to carriage environments. For expendable stores, verify that lanyards and umbilicals safely separate as intended during store release sequences and the air vehicle retains all hardware designated for retention. Flight test footage enables observation of lanyards and umbilicals during flight to identify snags, chafing, stowage, or retention issues.

17.1.1.5 Store alignment.

Standard: Store(s) and store components are capable of being accurately aligned or boresighted to a sight axis or datum line and retain sufficient alignment with any aiming or fire control

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system so as not to degrade store pointing accuracy or flight operations within the scope of AW (e.g., internal or external guns, external rocket pods, laser systems, seekers, fire control radars, pylons).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Test.

(1) Ground test: Perform alignment checks via ground fit and ground fire test(s) IAW MIL-HDBK-1763, Test 101, Test 102, Test 161, and Test 272 and MIL-STD-1289, Alignment of Stores and Installation Test Method sections.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Alignment or boresight procedures and realignment intervals are included in the pertinent documentation required for fielding.

Analysis.

(1) Verify through analysis of ground test reports and post-flight test inspection that stores can retain alignment, boresight, and the inspection intervals are established to prevent stores from falling out of alignment IAW MIL-HDBK-1763, Test 273.

17.1.2 Stores structural integrity.

17.1.2 criteria assess store structural integrity. This includes static and dynamic loads, store strength, durability, damage tolerance, aeroelasticity, and aeroservoelasticity. Store induced environments and vibration, shock (mechanical and pyrotechnic), and acoustics are covered in 17.2. Airframe structural integrity is addressed in Section 5.

Prior to carriage and employment of a store, an analysis of the strength of the store itself, the airframe, and all suspension and release equipment are required. The analysis covers the structural integrity of the store alone, the structural loading effects of the store on the airframe (Section 5) and suspension and release equipment, and the effects of the air vehicle and suspension equipment on the store. The analysis is performed for all conditions throughout the desired flight envelope in all operating environments, including takeoff (catapult), carriage, separation and landing and recovery (including arrested landing), and hangfire (e.g., hung, partially released stores, or missile thrust without release from air vehicle) loads. This also includes any potential sloshing effect of liquid within external tanks.

17.1.2.1 Store loads.

Standard: Loads used to verify the structural integrity of store safety-critical structure reflect all possible worst-case load combinations of loading conditions that result from the intended operation, usage, and environment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Analysis.

(1) Verify initial loads analyses report is complete and representative of all expected loads IAW MIL-STD-8591, Carriage Design Limit Loads appendix and MIL-M-8856, Analyses, Data, and Reports appendix. Models are correlated with data collected through wind tunnel testing for verification of structural integrity.

(2) Conduct analysis (e.g., FEA, CFD, other modeling and simulation) to predict all aerodynamic and mechanical loads used to verify the structural integrity of the store to flight environment for all intended operations and all mixed load configurations.

Test.

(1) Ground or Lab test: Test data (e.g., from wind tunnel or static ejection testing) captures the stores loads profile to validate actual loads are correlated with the results predicted by analytical methods.

(2) Ground test: Loads calibration ground tests of the instrumented test vehicles validate instrumentation is installed to fully characterize the store exposure to the flight environment and verify instrumentation is functioning correctly.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(3) Analyze flight test reports to verify the loads analysis is complete, all models are updated and correlated to the flight environment, and results have been incorporated into all store structural integrity assessments.

Test.

(3) Flight test: Instrumented measurement vehicle that is representative of all fielded store configurations captures loads experienced in the flight environment IAW MIL-STD-8591, Carriage Design Limit Loads appendix and MIL-M-8856, Analyses, Data, and Reports appendix.

17.1.2.2 Store strength.

Standard: The store and armament equipment withstand all design yield loads without yielding or detrimental deformation and can withstand design ultimate loads without rupture or collapsing failures for all operating environments in the flight envelope throughout the specified service life of the store.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct structural analysis to verify that the store structure can withstand all DLLs without yielding or detrimental deformation and can withstand design ultimate limit loads without failure. Loads and minimum margins of safety are compliant IAW MIL-STD-8591, Design Strength section and MIL-M-8856, Design Strength section. The analysis should describe the critical load cases and should account for relevant A, B, or S-basis and environmental

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knockdowns in material properties and allowables IAW MMPDS or CMH-17, as needed, for composite materials.

Test.

(1) Lab test: Determine store strength capability, to include store subassembly, component, and joining methods. Measure store deformation (strain) and deflection under all loads and substantiate pre-test analysis with lab test results.

(2) Lab test: Conduct full-scale static test with test articles that are structurally representative of the store to ultimate load IAW MIL-HDBK-1763, Test 130 and MIL-M-8856, Structural Demonstration Tests appendix and update strength analyses or supplement with additional lab tests as necessary.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct loads test with structurally representative IMV to measure and validate design loads against flight conditions representative of the full flight envelope IAW MIL-HDBK-1763, Test 200. Compare measured flight test loads to strength analyses, lab test, and ground test and update analyses or re-accomplish lab or ground testing as necessary.

17.1.2.3 Store durability.

Standard: The store and armament equipment are durable such that they resist cracking, corrosion, thermal degradation, delamination, and wear to maintain structural integrity and mechanical operation on all critical structure(s) throughout the specified service life and envelope of the store IAW MIL-M-8856, Durability section and MIL-STD-8591, Service Life section.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Initial durability analysis is conducted to verify store structural integrity and mechanical operation for all critical structures commensurate with the specified service life IAW MIL-M-8856, Analyses, Data, and Reports appendix and MIL-STD-8591, Carriage Design Limit Load appendix.

(2) As an alternative to Analysis MoC (1), if the store has been previously cleared for use in similar flight environments, the existing data can be leveraged for a by-similarity comparison to support first flight.

Test.

(1) Lab test: Test to expected repeated flight loads (determined via analysis or loads survey IAW 17.1.2.1) has been accomplished up to or exceeding 2X the intended number of captive carriage hours necessary for flight test. Sub-assembly or component-level testing of critical structures is permitted in lieu of AUR testing to support initial flight test when there is sufficient design margin.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Verify the updated durability analysis and FSDT demonstrate durability within the intended service life IAW MIL-M-8856, Structural Demonstration Tests appendix and MIL-STD-8591, Carriage Design Limit Load appendix.

(2) Verify that tracking mechanisms are in place to prevent stores exceeding the service life and identify when stores require inspection or repair to maintain durability within the service life.

(3) If there is no tracking mechanism (design feature or procedure) to determine the cumulative flight history of the store, then verify that inspection intervals are capable of identifying signs of cracking or structural degradation. Verify that inspection criteria specify non-serviceability limits or end of life.

Analysis.

(3) Verify the durability analysis is updated with the repeated loads and dynamic environments measured during flight test, correlated to the FSDT, and demonstrates structural integrity and mechanical operation of critical structures commensurate with the specified service life.

Test.

(2) Ground test: Conduct full-scale durability ground test IAW MIL-M-8856, Laboratory and Ground Tests appendix and MIL-STD-8591, Carriage Design Limit Load appendix using representative loading spectra to 2X the requested service life. Full-scale durability test shows no crack nucleation, delamination, or wear such that store structural integrity is commensurate with the specified service life.

(3) Flight test: In tandem with 17.1.2.1 and 17.2.2.1, conduct flight test with a structurally representative instrumented test vehicle that is capable of measuring repeated loads and dynamic environments experienced in flight.

17.1.2.4 Store damage tolerance.

Standard: The armament and store critical structure are damage tolerant (assuming defects) and maintain residual strength throughout the specified service life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Initial damage tolerance analysis verifies the design demonstrates residual strength commensurate with the specified service life.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(2) Validate the damage tolerance analysis with loads collected from flight test and update damage tolerance analysis if necessary.

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(3) Verify that the damage tolerance analysis is correlated to all damage tolerance tests, and that the system demonstrates residual strength commensurate with the specified service life.

Test.

(1) Lab test: For primary composite structures, verify damage tolerance of the store using representative, damaged structure at or below the visible damage threshold IAW MIL-M-8856, Laboratory and Ground Tests appendix. Store exhibits capability to withstand full static ultimate load with no damage growth under sustained and repeated loads commensurate with the environmental conditions expected during the specified service life.

(2) Flight test: In tandem with 17.1.2.1 and 17.2.2.1, conduct flight test with structurally representative IMV to measure repeated loads and dynamic environments experienced in the flight environment.

17.1.2.5 Aeroelasticity.

Standard: The store and armament equipment can operate in the intended usage environment without structural failure due to aeroelastic instability.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the initial flutter and divergence analysis report is complete; the models are updated and correlated with all lab tests and GVTs; and analytical results show no instability with a 15% margin on airspeeds in the approved envelope.

Analysis.

(1) Conduct an initial flutter and divergence analysis IAW MIL-M-8856, Aeroelasticity section and verify that there are no signs of instability within required margins of safety for the full envelope in all operating environments.

Test.

(1) Lab test: Perform test to substantiate aerodynamic, mechanical, and mass models used for the flutter and divergence predictions.

(2) Ground test: Verify the freeplay and rigidity of control surfaces.

(3) Ground test: Conduct aeroelastic GVT IAW MIL-HDBK-1763, Test 120 to verify modal response of the air vehicle and store. Correlate models used for flutter and divergence analyses with stiffness response and modal parameters measured during GVT.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Verify the final flutter and divergence analysis report is complete and shows no instability with a 15% margin on airspeeds in the approved envelope.

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Analysis.

(2) Analyze manufacturing tolerances measured for the store and store control surface actuators and interfaces to define the largest sources for freeplay and ensure limits or engineering rationale are incorporated in final analysis.

17.1.2.6 Aeroservoelasticity.

Standard: The weapon control system has gain and phase margins to operate without instabilities induced through coupling of aerodynamic, structural, thermal, and inertial forces throughout the intended employment environment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the ASE analysis shows the weapon control system has gain and phase margins to operate without instabilities induced through coupling of aerodynamic, structural, thermal, and inertial forces throughout the intended employment environment.

Analysis.

(1) Conduce ASE analysis using models correlated with lab developmental and ground structural coupling tests. Analytical results show no instabilities.

Test.

(1) Lab test: Conduct developmental tests on control systems to substantiate the aerodynamic, mechanical, flight control, and mass models used for the control system phase and gain predictions.

(2) Ground test: Conduct full-scale structural coupling tests to update and correlate the analysis for modal coupling with the FCS.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Verify the ASE analysis report is complete with the models updated for all flight control software updates from flight test and analytical results show no instabilities.

Analysis.

(2) Update the ASE analysis for all flight control software updates derived from flight test.

17.1.3 Store separation and post-separation flyout.

17.1.3 criteria assess stores for safe separation from the air vehicle, and store autopilot and post-release behavior for any potential re-contact with the host air vehicle. Guided and powered stores may introduce post-separation hazards and are referred to as within or out of the store separation regime. Store safe separation boundaries and safe post-separation boundaries are not standardized to a predetermined volume or distance in any standard or handbook. The distinction between store separation envelopes and post-store separation envelopes and the methodology

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used to determine these volumes and distances are determined based on the controllability and potential trajectory of the store.

Showing compliance with criteria in this section starts with the understanding of the fundamental requirements, guidance, and best practices contained in MIL-STD-1289, MIL-HDBK-1763, and MIL-HDBK-244. These documents inform and provide guidance to SMEs to evaluate safety and assess if systems can be evaluated by similarity. The separation phase is defined as both the region of store first motion until the store is outside the aerodynamic influence and immediate flight path of the air vehicle while positively accelerating away from the airframe, continuing to increase miss distance between the store and air vehicle and controlled with successful handoff to guidance and control algorithms responsible for targeting (if applicable).

Store separation characteristics are also different for external environments verses internal environments, as internal environments may be more benign for carriage but present thermal shock and shear layers outside of the weapons bay that may inhibit safe separation if not considered.

17.1.3.1 Safe separation.

Standard: Separation of stores does not result in store-to-store or store-to-air vehicle contact, damage, or exceedance of store or air vehicle design loads during jettison and employment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Perform analysis to assure positive miss distance is expected and maintained during the store separation phase. Assure analysis methods utilize best practices (wind tunnel testing or CFD) for determination of aerodynamic loads.

Test.

(1) Lab test: Perform separation wind tunnel testing of the store and the air vehicle to characterize store separation behavior for all expected modes of separation (e.g., jettison, release) for all mixed load configurations IAW MIL-HDBK-1763, Test 140 and Test 144. Determine employment envelope that prohibits release conditions where there exists potential to have store-to-store or store-to-air vehicle recontact during store separation.

(2) Ground test: Characterize bomb rack ejection dynamics and effects from lanyards, arming wires, or other external mechanical forces IAW MIL-HDBK-1763, Test 110. Update separation analyses with data collected from ground test.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(2) Perform post-flight separation analysis validating model results to test data. Analysis identifies configurations and flight envelopes IAW MIL-HDBK-1763, Separation Analysis paragraph.

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Demonstration.

(1) Show store-to-store and store-to-air vehicle safe separation compliance through execution in a relevant environment IAW MIL-HDBK-1763, Test 271 and Test 272. The number of separation events is informed by the safe separation analysis to determine regions of the employment envelope that show potential for store-to-store or store-to-air vehicle recontact.

Test.

(3) Flight test: Conduct store separation test to measure store trajectory and observe store-to-store and store-to-air vehicle miss distance IAW MIL-STD-1289, Clearances Section. Separation testing is accomplished IAW MIL-HDBK-1763, Test 144, Test 270 and Test 280 with separation test vehicle(s) that match system attributes of store variants being certified for fielding or operational use. System attributes include OML, general mass properties (e.g., weight, CG, moments of inertia), aerodynamic performance, and deployment of control surfaces.

17.1.3.2 Autopilot.

Standard: For guided stores with control mechanisms for post-launch actions (e.g., rocket motor ignition, control surface deployment, guidance and control) that are active during the store separation phase, assure control parameters are sufficient to avoid store-to-store and store-to-air vehicle contact (e.g., inhibits, functional limitations, delay timers, pre-planned maneuvers) for authorized store loadout and release conditions.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Perform analysis to identify store properties (e.g., store attitudes, body rates) that can result in store-to-store and store-to-air vehicle contact.

(2) Perform analysis of the autopilot logic to identify failure modes that can result in store-to-store and store-to-air vehicle contact. Analysis should include transition from safe separation controller to guidance and navigation controller to sensor control, as applicable.

Test.

(1) Lab test: Verify store inhibits actions that result in store-to-store and store-to-air vehicle contact post-launch and release.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(3) Conduct analysis showing that the store autopilot maintains controlled flight through separation and transition to guidance ensuring minimum store-to-store and store-to-air vehicle miss distance through store separation in a relevant operating environment. The number of separation events is informed by the safe separation analysis to determine regions of the employment envelope that show potential for store-to-store or store-to-air vehicle recontact.

(4) Conduct analysis of post-flight data documenting autopilot behavior and correlate to pre-flight test analysis to validate the autopilot model for use in evaluation of the full release

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envelope for all mixed load configurations and release conditions. This analysis will include store functionality such as control surface behavior and store body rates as functions of time and distance away from air vehicle.

Test.

(2) Flight test: Measure store-to-store and store-to-air vehicle miss distance to verify compliance to minimum separation distance and validate separation predictions provided by analytical methods. Flight test data measures timing of post-release events such as rocket motor and engine ignition, control surface deployment, fin unlock, and initial guidance and control maneuvers.

17.1.3.3 Post-separation collision.

Standard: For stores with the potential to re-contact the air vehicle post-safe separation (e.g., deployed systems such as first stage rocket boosters or inlet covers, loitering munitions, air launched effects and powered stores with loft trajectories), the store includes design features, system lockouts and limitations, or processes to avoid store-to-store and store-to-air vehicle contact. For systems without collision avoidance mechanisms, procedures exist to prevent contact post-separation (e.g., air vehicle deconfliction maneuver).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine store post-launch control systems to identify timing of post-launch events and potential sources of post-separation collision during intended stores operations.

Analysis.

(1) Perform analysis of the store fly-out model to identify limitations or procedures necessary to avoid store-to-store or store-to-air vehicle collisions, including errant engagements.

(2) Perform FMECA of store systems to determine if failure modes increase likelihood of post-separation collision with other stores or the host air vehicle.

Demonstration.

(1) Evaluate the store OFP and flight control features using lab demonstration to verify that post-launch store features preventing store-to-store or store-to-air vehicle collision function as intended.

Test.

(1) Lab test: Perform and measure system properties (e.g., ejection forces, aero loads) to characterize mechanics related to store separation and post-separation events (e.g., staging mechanisms, deployed systems).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(2) Demonstrate the air vehicle can perform maneuvers to avoid potential debris, errant loitering munitions, or other post-separation recontact.

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Test.

(2) Flight test: Perform with instrumented vehicles to characterize mechanics related to store separation and characterize fly out model (e.g. Time Space and Position Information data) of post-separation events (e.g., staging mechanisms, deployed systems) to ensure store-to-store and store-to-air vehicle collision does not occur IAW MIL-HDBK-1763, Test 290.

(3) Flight test: Perform to measure the actual separation distances and timing of events to show systems meet minimum store-to-store and store-to-air vehicle miss distances for all post-separation phases of flight. Validate predicted miss distances with measured flight test data to substantiate the full flight envelope for all release conditions and mixed load configurations.

17.1.4 Store characteristics.

This section focuses on the key store characteristics that affect the ability of the air vehicle to maintain sufficient stability and control as a direct result of the carriage and employment of stores. Carriage and release of stores can result in substantial and sudden changes to air vehicle weight and CG. Parasitic drag effects and aerodynamic effects induced by stores can alter the aerodynamic characteristics of the air vehicle (affecting range, endurance, and handling qualities), and asymmetric carriage of stores (especially as a direct result of release a store) can further exacerbate these effects. Changes to the air vehicle weight and CG or center of pressure may also reduce the control authority of air vehicle control surfaces and FCS. Section 6 is responsible for the overall aerodynamic characteristics of the air vehicle while carrying and employing stores. Section 17 is responsible for the critical store characteristics that contribute to these effects including mass properties, physical properties (e.g., OML), store drag, and flutter.

Internal carriage of stores may predominantly impact air vehicle total weight and CG by eliminating parasitic drag effects and reducing effects due to asymmetry by placing weapon bays closer to the centerline of the air vehicle, where externally carried stores may be likely to contribute to higher levels of parasitic drag and lateral asymmetry.

17.1.4.1 Store mass and physical properties.

Standard: Store mass properties (including but not limited to weight, length, height, width, suspension spacing, CG, and moments of inertia) are defined for all variants intended for flight test and operational use.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct analysis of store, store components, and store materials to produce accurate estimates of store mass properties (e.g., weight, length, height, width, CG, moment of inertia) for all store configurations and allowable tolerances.

Test.

(1) Ground test: Perform ground measurements to measure store mass properties and validate estimates produced via analytical methods.

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17.1.4.2 Store drag index and aerodynamics.

Standard: Store aerodynamic properties (e.g., drag index) are defined for all stores and carriage locations (e.g., weapon stations), for each unique store configuration, loadout and flight condition (e.g., air vehicle AOA, store incident angle, dynamic pressure) that changes the store drag, lift and aerodynamic responses.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze store aerodynamic properties through CFD to determine drag and lift indices for all mixed load configurations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(2) Analyze production representative variant(s) of store to verify that all pre-test methods of determining drag and lift indices are representative of fielded configurations.

Test.

(1) Lab test: Conduct wind tunnel testing to define freestream drag and lift indices and to measure air vehicle and store aerodynamic interactions.

17.1.5 Store exhaust effects.

Store exhaust produced by thrust-augmented systems (e.g., rockets, air-to-air and air-to-ground missiles, chaff and flare dispensers, ejection systems, and gunfire operation) may be comprised of gases, fluids, and particulates at various temperatures and concentrations. This can lead to damage of the air vehicle skin and structure through direct contact and impingement of surfaces. It also has the potential to damage engine systems or significantly degrade engine performance if ingested by the air vehicle inlet(s). Stores should be located in areas that minimize engine ingestion to the greatest extent possible, and any potential engine ingestion and impact to engine systems is addressed in Section 7. Ingestion of store exhaust may also impact ECS and the air quality for systems that use bleed air (e.g., crewmember air intake systems) due to toxic exposure, which may impact criteria in Sections 8 and Section 9.

Impingement effects on the air vehicle and other stores includes residue build-up and potential for corrosion, blast overpressure, shock waves, thermal effects, and surface erosion due to the velocity of exhaust by-products.

17.1.5.1 Store exhaust impingement.

Standard: Store exhaust (e.g., gases, fluids, and particulates) and recoil forces do not obscure transparencies, damage structure or erode surfaces, ablate faces of flight-critical sensors, or degrade any other flight-critical systems (e.g., ECS, antenna, air data system). The air vehicle is capable of withstanding the impingement effects of store exhaust without weakening the skin of the air vehicle, degrading critical structure of the airframe, or degrading critical crew systems necessary for sustainment of flight.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine store design documentation to determine the potential for store exhaust to impinge air vehicle surfaces and flight-critical systems.

(2) Verify the chemical composition, temperature, pressure and toxicity of any exhaust gases, fluids, or particulates to determine the impingement characteristics of the exhaust and plume.

Analysis.

(1) Conduct analysis to verify that store exhaust does not obscure transparencies, erode surfaces, ablate faces of sensors, or degrade other flight-critical systems.

(2) Conduct analysis to verify the store exhaust gas and plume does not impinge on flight-critical sensors, target acquisition and designators (e.g., radar, laser) to ensure employment of stores does not result in degraded performance, unsafe conditions, or hazards.

(3) Conduct analysis to verify the store exhaust, muzzle blast, overpressure, temperature, and recoil does not impinge or damage air vehicle structure, surfaces, or injure crewmembers.

Test.

(1) Ground test: Verify through static ground fire testing that gases and ejecta from gunfire and rocket motor operations do not obscure, ablate, erode or damage air vehicle surfaces or flight-critical sensors.

(2) Flight test: Conduct live fire testing to measure store exhaust, muzzle and blast overpressure, temperature, and recoil forces to determine effects on transparencies, air vehicle surfaces and structure, flight-critical systems, and crew members IAW MIL-HDBK-1763, Test 160.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Verify through ground or flight demonstration that stores can operate without obscuring transparencies, eroding surfaces, ablating faces of flight-critical sensors, or degrading other flight-critical systems or crew members.

17.1.5.2 Engine effects.

Standard: Store exhaust (gases, particulates, and fluids) from firing, employing, or dispensing stores (e.g., gun operations, rocket motors, countermeasure dispensers) is not ingested by air vehicle engine.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine placement of stores with proximity to engine inlets and weapon event timeline through store employment to determine if potential exists for store exhaust to be ingested.

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(2) Examine store delivery procedures and identify post-launch maneuvers that may prevent ingesting store exhaust.

Analysis.

(1) Conduct analysis of exhaust plume composition to identify potential to damage engine systems or degrade engine combustive performance.

(2) Conduct analysis of store exhaust plume dispersion and proximity to engine inlet systems to verify that engines do not ingest exhaust.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Demonstration.

(1) Conduct flight demonstration of store employment to verify that exhaust is not ingested by the engines.

17.1.6 Safe escape.

17.1.6 focuses on evaluating the air vehicle ability to safely escape both intended weapon effects (e.g., warhead fragmentation and blast pressure effects) and unintended effects due to possible failure modes and degraded conditions (e.g., early burst, rocket motor overpressure).

Safe escape uses probabilistic analysis to determine the level of risk that the air vehicle will be exposed to during an energetic event and subsequent delivery conditions for air vehicle/weapon/fuze combinations based on the proximity of air vehicle and weapon at detonation and resultant fragment motion.

Safe escape efforts aim to protect air vehicle from fragments from own weapons. Safe escape analysis includes time-spatial calculations of bursts at corresponding release conditions for air launched weapons systems. These bursts consider warhead ground burst, warhead air burst, as well as early burst (detonation at minimum fuze arm delay time). Fuze arm delay time is defined as the time elapsed, or distance traveled from launch to arming, which differs from a fuze function delay. Additional efforts may include gun round ricochet and premature deflagration or overpressure of the rocket motor on ignition. One key result is safe arm values (minimum arm time or arm distance).

Safe escape analyses are required on air-launched munition systems which could potentially be hazardous to the air vehicle due to munition fragmentation. Safe escape calculates the probability that a weapon fragment could impact the delivery air vehicle after detonation. The Ph is evaluated and compared to the platform or service safe escape criteria to determine if the delivery scenario is safe or unauthorized. The SEA should consider munitions functioning within design specifications as well as potential munition failure modes. Any potential hazard predicted by the analyses due to munition fragmentation or proposed air vehicle/munition configuration should be either prohibited or limitations will be provided. Limitations consist of air vehicle delivery conditions (altitude, dive angle, and maneuver) necessary to attain safe escape criteria.

The SEA is informed by fragmentation characteristics data gained from static arena testing. Data on existing munitions can be found in the related Joint Munitions Effectiveness Manual created by the Joint Technical Coordinating Group for Munitions Effectiveness. Environment effects (usually wind or altitude air density), blast pressure and air vehicle survivability may also be

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taken into account. Live fire weapons flight testing is typically conducted in conjunction with other weapon release flight testing to relieve the requirement for stand-alone instrumented test assets.

For newly developed, unqualified rocket motors, a risk analysis using safe escape tools may be performed to evaluate the potential risk to air vehicle if the rocket motor fails due to overpressure or deflagration. This analysis is typically done in support of developmental test only when flight test occurs in parallel with rocket motor development and qualification for the purpose of providing programs and flight test personnel with associated probability of hazard to or loss of air vehicle and the appropriate mitigations including safe escape maneuvers to increase separation distance between the rocket and air vehicle. Rocket motors should be fully qualified with demonstrated reliability prior to operational use and fielding.

17.1.6.1 Ground burst, air burst, and gun ricochet.

Standard: The safe release envelope (e.g., safe release altitude, safe release airspeed) is defined such that releasing or employing a store will ensure the air vehicle is not critically damaged by weapon fragmentation resulting from intended downrange detonation. Ricochet from gunfire operations do not have the potential to strike the air vehicle within the safe firing envelope.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Determine the threshold for Ph to launch air vehicle from fragmentation effects (ground fragmentation, terrain, and air burst fragmentation separately) that would result in critical damage to the air vehicle.

(2) Establish the minimum fuze arm time required to achieve safe escape.

(3) Conduct SEA by modeling the air vehicle, weapon, and fragment trajectories that include the point of detonation. Burst analyses determine if an air vehicle can safely release a weapon with a fuze that functions properly, (e.g., that detonates on the ground or target (impact fuze) or at a HAT or distance to target (proximity fuze). The Ph by fragmentation to the air vehicle is modeled considering factors such as the air vehicle release conditions and flight path, the weapon's trajectory and orientation after launch, the warhead's fragmentation data informed by static arena testing, and if applicable, fuze reliability data, specifically fuze failure rate.

Types of burst analyses to be conducted:

a. Fragment minimum altitude: The lowest altitude or target separation distance where the risk to the air vehicle due to fragments from ground weapon detonations does not exceed the risk criteria.
b. Terrain minimum altitude: The lowest altitude that an air vehicle can release a weapon and avoid impact with the terrain (ground and terrain clearance factor).
c. Ricochet minimum altitude: The lowest open fire/cease fire altitude where air vehicle is safe from gun round ricochet or fragmentation.

(4) Evaluate the probability that a warhead or gun fragment (where applicable) can impact the delivery air vehicle upon detonation on the ground or air target (impact fuze) or at a HAT or distance to target (proximity fuze).

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(5) Evaluate the probability that the delivery air vehicle will impact the terrain during weapon delivery or recovery.

(6) Identify the set of flight conditions (altitude, speed and engagement range) that will provide the air vehicle protection from the munition debris detonation at the target area given normal munition functions. This set of flight conditions, at a minimum, should include the planned flight test and release points as part of the live fire testing. Ideally, the considered envelope should characterize the entire employment envelope and weapon flight modes and fuze arm times.

(7) In cases where the Ph does not meet minimum safe escape criteria, determining Pk of the launch air vehicle may be conducted and correlated to the damage tolerance of the air vehicle to determine an overall risk. Probability of kill is determined through modeling the vulnerable areas of the launch air vehicle obtained from vulnerability analyses.

Test.

(1) Ground test: Conduct static arena test with production representative warhead to characterize the warhead fragmentation pattern, fragment size, fragment travel velocity, and fragment travel distance. Static arena test data will inform the SEA.

(2) Flight test: Conduct instrumented live fire flight testing to verify weapon flyout behavior, validate weapon flyout models, and measure fuze arm time and reliability prior to testing with a live AUR. Live fire flight testing may inform safe and arm device reliability in order to update reliability and failure probability assessment.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Verify that the safe release envelope necessary to maintain safe escape for store employment is defined and is accessible to flight crew (e.g., safe escape tables in weapon delivery procedures, safe escape tools in mission planning).

Analysis.

(8) Determine if there is protection to prevent critical damage to the air vehicle from the munition ricochet/fragmentation and detonation effects given a downrange detonated munition.

(9) Complete SEA for the entire flight envelope (flight maneuvers, weapon flight modes, and fuze arm times) or conduct a tailored SEA for a subset of conditions that were not previously analyzed prior to flight test. Use the SEA to identify safe and unsafe employment conditions and subsequent escape maneuvers.

17.1.6.2 Safe arm and early burst.

Standard: Store payloads maintain the minimum fuze arm delay required to meet all safe escape criteria. Store minimum arm time (or distance) after release does not fall within the minimum fuze arm time required to maintain safe escape criteria (or is otherwise considered an early burst). Any potential for early burst does not result in fragmentation to the launch air vehicle that exceeds a 1 in 10,000 threshold for overall Ph (the product of the Ph given a premature detonation and the probability of a premature detonation).

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine design documentation for safe and arming devices (e.g., fuze) to determine minimum arm time, to include any failure modes and demonstrated reliability of safe and arming devices and fuzes. This may be determined by service specific munitions safety review boards.

Analysis.

(1) Conduct SEA by modeling the air vehicle, weapon, and fragment trajectories that include the point of detonation. Early burst analysis assumes that the weapon detonates at the minimum fuze arm time and subsequent distance from the launch platform along a weapon flight trajectory. Probability of hit by fragmentation to the air vehicle is modeled considering factors such as the air vehicle release conditions and flight path, the weapon's trajectory and orientation after launch, and the warhead's fragmentation data informed by static arena testing, and if applicable, fuze reliability data, specifically fuze failure rate.

(2) Evaluate the minimum safe arming distance or fuze arm time setting that is possible once the munition has left the air vehicle. If applicable for the analysis, including fuze reliability (failure rate), as well as if potential munition failure modes should be considered. Determine the probability that a warhead fragment could impact the delivery air vehicle in the event of an early burst.

(3) Determine if evaluated probability meets the minimum threshold for escape criteria, 1 in 10,000. The probability of a single fragment hit to the delivery air vehicle may not exceed the established Ph and Ph/EB risk criteria for early burst. Determine if there is protection to prevent critical damage to the air vehicle from the munition fragmentation if early detonation should occur.

(4) Identify the set of flight conditions (altitude, speed, and engagement range) that will provide the air vehicle protection from the munition detonation given fuze failure. This set of flight conditions, at a minimum, should include the planned flight test and release points as part of the live fire testing. Ideally, the available data should characterize the entire employment envelope, weapon flight modes, and fuze arm times.

(5) In cases where the Ph does not meet minimum safe escape criteria, determining Pk of the launch air vehicle may be conducted and correlated to the damage tolerance of the air vehicle to show compliance (or inform an overall risk). Probability of kill is determined through modeling the vulnerable areas of the launch air vehicle obtained from vulnerability analyses.

Test.

(1) Ground test: Conduct static arena test with production representative warhead to characterize the warhead fragmentation pattern, fragment size, fragment travel velocity, and fragment travel distance. Static arena test data will inform the SEA.

(2) Flight test: Instrumented live fire flight testing may be conducted to verify weapon flyout behavior, validate weapon flyout models, and measure fuze arm time prior to testing with a live AUR. Live fire flight testing may inform distance at early burst or assess safe and arm device reliability in order to update early burst probability assessment.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Verify that the safe release envelope necessary to maintain safe escape for store employment is defined and is accessible to flight crew (e.g., safe escape tables in weapon delivery procedures, safe escape tools in mission planning).

17.1.6.3 Rocket motor overpressure.

Standard: Solid rocket motors are qualified for carriage and operation in the air vehicle flight environment during all operating conditions and do not fail prematurely (deflagrate, rupture, or explode) due to overpressure or design deficiencies while in proximity to the air vehicle. For unqualified rocket motors, if the potential for overpressure failure or deflagration of solid rocket motors exists, mitigations are established to ensure safe separation distance between the air vehicle and the rocket motor prior to ignition.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine design documentation for the rocket motor ignition system and firing sequence to determine the minimum time and distance between store release and rocket motor ignition.

(2) Examine design documentation to identify potential failure modes in the ignition system.

Analysis.

(1) Conduct rocket motor overpressure SEA by modeling the air vehicle, rocket motor, and fragment trajectories that include the time of predicted failure. rocket motor overpressure SEA assumes that the rocket motor deflagrates or ruptures at rocket motor ignition unless there is credible data verifying time of rupture as demonstrated through modeling or during static fire testing. Ph to the air vehicle or Pk by fragmentation modeling considers factors such as the air vehicle release conditions and flight path, the weapon's trajectory and orientation after launch, and the rocket motor fragmentation characteristics data due to rocket motor overpressure or deflagration. Rocket motor fragmentation may be informed by design data, modeling, or static arena testing. If available, rocket motor failure/reliability can be taken into consideration as part of the overarching Ph and Pk calculation.

(2) Evaluate the Ph from a rocket motor fragment that could impact the air vehicle.

(3) Evaluate the Pk from fragmentation. Probability of kill calculation should include modeling the vulnerable areas of the launch air vehicle obtained from vulnerability analyses.

(4) Evaluate mitigations that ensure safe separation distance between the air vehicle and rocket motor ignition point (e.g., delayed rocket motor ignition, safe escape maneuver).

(5) Identify the set of flight conditions (altitude, speed and engagement range) that will provide the air vehicle protection from the munition deflagration in the event of rocket motor failure. This set of flight conditions, at a minimum, should include the planned flight test and release points as part of the live fire testing. Ideally, the data set should characterize the entire employment envelope and weapon flight modes and rocket motor ignition times.

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Test.

(1) Lab test: Complete rocket motor qualification testing IAW NAVSEAINST 8020.5 to include static firing of rocket motors with no anomalies that could allude to premature rocket motor failure prior to developmental test use. Provide demonstration of rocket motor reliability, specifically failure rate, through the rocket motor ignition and burn phase. Static fire test data will inform the SEA.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Complete rocket motor qualification testing to include static firing of rocket motors with no anomalies that could allude to premature rocket motor failure prior to operational use and fielding. Provide demonstration of rocket motor reliability, specifically failure rate, through the full phase of flight to termination.

17.2 Stores environments.

The Store Environments section contains three additional sections.

The first section addresses the electrical compatibility of stores. The intent of 17.2.1 is to treat the store as an extension of the air vehicle if there are any shared electrical interfaces and dependency of the store on air vehicle power. Electrical compatibility includes preventing electrical hazards to either the air vehicle or store(s). This includes armament subsystems, suspension and release systems, and stores management and fire control systems. By treating the store as an extension of the air vehicle, electrical compatibility requires assessing the modification IAW any applicable standards contained in Section 12. This includes but is not limited to fault isolation, power quality, circuit protection, component selection and sizing, grounding and bonding, prevention of ignition, avoidance of single points of failure, and qualification of batteries. Stores and store modifications that impact electrical compatibility should refer to Section 12 for all matters of air vehicle flight safety and refer to 17.2.1 for material that is supplemental to Section 12 when Section 12 does not sufficiently address the AW of stores with respect to electrical compatibility, specifically, electrical interfaces and power quality. It is important to note that the scope of this section is limited to the AW of the air system, and there is typically additional rigor required to assess the safety of stores and ground equipment to comply with service specific mandatory system safety processes and policy.

The second section addresses induced environments. The intent of this section is to assess the environments induced by the store(s) on the air vehicle, and by the air vehicle on the store during all phases of flight. This includes consideration of the qualifications of stores in operating environments, to include transportation and storage. It is important to note that modifications to the air vehicle may affect induced environments to unmodified, previously certified stores. The scope of this section includes: dynamic environments (shock, vibration, fatigue, acoustic), temperature, humidity, altitude environments, cooling, explosive atmosphere, and fire protection.

The third section addresses E3 with special consideration to armament and stores. This section addresses HERO, and the source-victim relationship between air vehicle and store for EMI and EMC. This includes external RF EME.

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17.2.1 Electrical compatibility.

For all required stores configurations, the air vehicle and store are electrically compatible such that carriage, operation, and employment of stores do not adversely affect the AW of the air vehicle.

Electrical compatibility refers to the ability to safely carry and operate stores on the air vehicle, whether these systems rely on air vehicle power or alternative power sources internal to the store. Compatibility covers power quality and any gaps in air vehicle or store qualification to various versions of MIL-STD-704, all electrical interfaces, and sufficiency of design. It may be required to assess the sufficiency of the design for any electrical systems and components in armament systems and stores given the potential AW hazards these systems may present to the air vehicle.

Note: Use the ELA generated from Section 12 to verify that the air vehicle has sufficient power capacity available to meet the power requirements during all modes of operation for steady-state and peak transient loads.

17.2.1.1 Electrical Interfaces.

Standard: Physical interfaces for air vehicle-to-store electrical connections are defined and prevent unintended operation, inadvertent release (launch or jettison), and inadvertent detonation of stores or store devices (e.g., squibbing batteries, control surfaces, or other electrically initiated devices).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that electrical interfaces are defined for the air vehicle and store systems and meet applicable design standards. Electrical interfaces are defined in standards (e.g., MIL-STD-1760, SAE AS5726 for micro-munitions). For nonstandard interfaces, verify that the design cannot cause unintended or inadvertent operation and release of stores.

Test.

(1) Ground or Lab test: Conduct testing to verify electrical interfaces comply to design standards and prevent inadvertent operation of stores, to include inadvertent release (launch or jettison) and inadvertent activation of squibs or other electrically initiated devices.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Conduct testing to verify that electrical interfaces prevent inadvertent operation of stores, to include inadvertent release (launch or jettison) and inadvertent activation of squibs or other electrically initiated devices.

17.2.1.2 Power quality and electrical compatibility.

Standard: Operation of stores does not degrade power quality below levels that might present SOF issues for the air vehicle for all permissible weapon stations and load combinations under all operating conditions.

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Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that mitigations exist through either procedures or software implementation that inhibit or stagger power application to prevent degrading air vehicle power via voltage drops and peak power loads that exceed operating limits.

Analysis.

(1) Analyze MIL-STD-704 qualification reports for the air vehicle and store(s) to identify gaps in permissible voltage transients and power quality levels that may result in electrical incompatibility and risk of damage to safety-critical systems.

(2) Analyze the ELA to verify that the air vehicle has sufficient power capacity available to meet the power requirements during all modes of store operation for steady-state and peak transient loads.

(3) Analysis of ground test data shows sufficient electrical compatibility and that the air vehicle is capable of providing power quality to stores for all mixed load configurations under all operating conditions IAW MIL-STD-704, Detailed Requirements paragraphs.

Test.

(1) Lab test: Conduct testing to measure the timing of power application and worst-case peak loads to each and all weapon stations to identify potential to degrade safety-critical systems (e.g., voltage drops, overcurrent, power surges, brown-outs).

(2) Ground test: Conduct testing to characterize air vehicle power quality under worst-case store power draw for all mixed load configurations to determine if voltage drops, bus switching, and power transients are within store compatibility limits.

17.2.2 Induced environments.

The intent of this section is to assess the environments induced by the store(s) on the air vehicle, and by the air vehicle on the store during all phases of flight. This includes consideration for the qualification of stores to operating environments, to include transportation and storage as documented in the Life Cycle Environmental Profile referenced in MIL-STD-810. It is important to note that modifications to the air vehicle or store or installation of store may affect induced environments to unmodified, previously certified stores. The scope of this section includes: dynamic environments (shock, vibration, fatigue, acoustic), which should consider special environments such as gunfire and store ejection (e.g., pyrotechnic shock), temperature, humidity, altitude environments, cooling, explosive atmosphere, fire protection, corrosion, erosion, ablation, gas and plume hazards to primary sensors, and harmful gas accumulation. This section may not address all operating environments that stores need to consider for test and operations. Please reference Section 4 for coverage of operating environments. See also Section 19 for corrosion protection.

The dynamic environment for all flight conditions for stores is often undefined, especially for externally carried systems, and requires an incremental approach of qualifying systems to dynamic environments that progress in fidelity to the intended operational environment.

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This approach starts with assessing the store to currently certified stores to identify if the appropriate environment has already been defined and otherwise working to characterize the flight environment if it has not already been captured in a platform environmental specification.

If the flight environment has not been characterized for shock, vibration, and acoustics in the localized region where the subject armament systems and stores are to be carried, the following steps establish a process to safely proceed to flight test and then fielding or operational use of these systems.

Stores may be designed and qualified to a particular air vehicle and flight environment that may differ from additional air vehicles that integrate these systems. Air vehicles may also be certified to a known service life, and as mission usage changes with respect to armament and stores the stores induced environment on the air vehicle may change and may require re-evaluation.

The purpose of the IMV is to capture data during flight test that can bound the dynamic environment the store and subsequent variants will be qualified to. Data collection should follow acceptable measurement engineering practice (e.g., IEST-RD-DTE012.2) as well as the dynamic data instrumentation measurement guidelines established by MIL-STD-810, Vibration and Shock methods. This includes use of a minimum 2,000 Hz bandwidth for vibration and shock and a minimum 10,000 Hz bandwidth for acoustics. Special-case shock environments (e.g., pyrotechnic shock) may require adjustments to the bandwidth as appropriate.

While the data collected from IMV testing characterizes the store airframe dynamic environments, dynamic measurements are made throughout the store in order to characterize the environment for critical sub-assemblies (e.g., guidance systems, warhead and payload, propulsion systems, control actuator systems) and will be used to develop sub-assembly and component test specification flow-down criteria.

High cycle fatigue qualification for stores is accomplished by testing tactical hardware configurations to at least 2X the lifetime (twice the number of captive carriage necessary for the design life of the store).

In general, dynamic environment test criteria should consider a fatigue scatter factor of two and should envelope the appropriate MIL-STD-810 or tailored dynamic environment pertinent to the air system. Dynamic environment testing should at a minimum stress the first three structural elastic modes of the store airframe and any local mode frequencies for subsystem structure of significant mass. The input excitation should represent the expected dynamic environment at all frequencies up to the first flexible store bending mode frequency for ensuring the platform suspension equipment to store interface is stressed.

17.2.2.1 Dynamic environments.

Standard: Stores and store components are able to function within the operational service life without inducing flight or safety-critical failure modes under all operating conditions in the natural and induced dynamic environments (mechanical and pyrotechnic shock, thermal, vibration, acoustics).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Verify that the store mass properties, OML, and other pertinent design attributes captured in the store configuration data set are qualified to the platform specific dynamic environment. Qualification can be by similarity if the store is considered similar to store(s) that have already been qualified to the platform specific dynamic flight environment. For similar stores, verify through examination that stores have been qualified to expected shock, vibration, thermal, and acoustic loads for the intended operational life of the system, with fatigue limits otherwise identified for carriage on the intended air vehicle.

Analysis.

(1) Use analytical methods to develop estimates of dynamic environment criteria (shock, vibration, thermal, and acoustics) as applicable for the store(s) over the desired range of loadouts for all flight environments over the intended service life of the system.

(2) Conduct analysis to verify that stores are designed to operate in all dynamic environments present in the design envelope for flight.

Test.

(1) Lab test: Conduct test to verify that an IMV or equivalent test asset has sufficient design to survive the flight test program necessary to define the dynamic flight environment. This includes vibration testing IAW MIL-STD-810, Vibration method or the tailored dynamic environment for the air system based on similarity to other stores. In order to proceed to IMV flight testing, laboratory vibration testing should be accomplished up to or exceeding 2X the intended number of captive carriage hours necessary for IMV flights. This testing may not be necessary if the store's airframe fatigue capabilities have already been successfully demonstrated for a similar platform.

(2) Ground test: Conduct static fire ground test to measure dynamic loads (e.g., shock, vibration, acoustics, thermal) induced onto the store from store operations (e.g., gunfire operations, ejection/launch).

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Analysis.

(3) Analyze data collected through IMV testing to verify that store qualification testing encompasses the platform specific dynamic environment (shock, vibration, and acoustics).

Test.

(3) Flight test: Conduct test with an IMV or equivalent test asset that contains data collection systems and instrumentation while maintaining sufficient similarity to the store variants being assessed, including matching geometric dimensions, mass properties, stiffness, and structural modes.

Dynamic similarity of the IMV and store variants are verified through modal test or analysis IAW MIL-STD-810, Similarity section. Key items of consideration to determine dynamic similarity are:

a. Difference in total mass is within ±10%.

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b. Difference in CG is within ±10% in all directions.
c. The first three structural elastic modes are within ±5%.

(4) Lab test: If measured IMV data exceeds qualification levels for tests previously accomplished against the store, then verification is accomplished by re-testing the store to levels matching or exceeding levels measured during IMV flight testing.

17.2.2.2 Store life cycle environmental profile.

Standard: Stores are qualified to all operating conditions for the specific air vehicle application and store environments for the intended operational service life.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify through analysis that the life cycle environmental profile of the store or equivalent stockpile-to-target sequence contains coverage of the air vehicle usage environment to include temperature, humidity, and altitude as well as environmental operating conditions that impact the serviceability of the store while installed on the air system (e.g., maritime, corrosion, erosion, and ablation).

(2) Verify through comparison on basis of similarity to previously qualified stores that the store is capable of safe carriage and operation in the usage environment.

Test.

(1) Lab test: Verify through environmental qualification test that the store is capable of safe carriage and operation in all usage environments. Test is accomplished IAW, but not limited to, applicable MIL-STD-810, Temperature, Humidity, Vibration and Altitude, Climatic Category, Salt Fog, Sand and Dust, and Acidic Atmosphere sections tailored to the air vehicle environmental profile.

(2) Ground or Lab test: Conduct test to measure the thermal environment created through store operations (e.g., gunfire operations, dispensing of countermeasures, rail launched rocket motors) as applicable to the specific air vehicle usage environment. Update analysis to include thermal environment measured through lab and ground testing. The tailored environments may also be measured through flight test with an IMV.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Verify through examination that all environmental qualification data supports the intended operating life in all operating environments.

17.2.2.3 Explosive atmosphere and fire protection.

Standard: Harmful gases, particulates, and fluids from gun, rocket, or other store-related systems operation do not accumulate beyond prescribed allowable concentrations (e.g., explosive atmosphere, overpressure).

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Stores are capable of operating without producing ignition sources (e.g., sparks, arcing, auto-ignition, surface temperatures exceeding flash points), as fuel-air explosive atmospheres may contain harmful, flammable, and explosive vapors that may originate either from stores or from an external source.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine store-related systems (e.g., weapon bays/compartments that contain internal gun systems or store ejection systems/suspension and release equipment; store internal systems or subsystems) to determine if the potential exists for gases, particulates, or fluids to accumulate and identify potential sources of ignition.

(2) Evaluate any purge and ventilation systems to verify they prevent gases from gun, rocket, or other store-related systems operation from accumulating beyond prescribed levels for harmful or explosive concentrations within expected flight test environments.

(3) Examine the explosive atmosphere volume for fire detection and fire suppression systems.

Analysis.

(1) Conduct analysis to verify that gases, particulates, and fluids from gun, rocket, or other store-related systems operation do not accumulate beyond prescribed allowable concentrations.

(2) Analyze stores and compartments/weapon bays to verify that there are no sources of ignition within proximity to any accumulated gases, particulates, or fluid.

(3) Verify that fire detection and suppression systems are located where needed and are capable of detecting expected concentration levels and informing flight crew.

Test.

(1) Ground test: Conduct testing that verifies purge and ventilation systems sufficiently prevent gases, particulates, and fluids from guns, rockets, or other store-related systems operation from accumulating beyond allowable concentrations and pressures. Fire detection and suppression systems function as intended.

(2) Lab test: Conduct explosive atmosphere lab testing that verify stores demonstrate the ability to operate in fuel-air explosive atmospheres without causing ignition, and that stores demonstrate that an explosive or burning reaction occurring within the store will be contained and will not propagate outside the test item. Testing is accomplished IAW MIL-STD-810, Explosive Atmosphere method.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct testing that verifies purge and ventilation systems prevent gases from guns and rocket operations from accumulating beyond prescribed levels for explosive concentrations for operations that cannot be tested on the ground. Verify that sensors can function in the flight environment and inform the flight crew of hazard.

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(4) Flight test: Conduct testing to verify that gases, particulates, and fluids from gun and rocket operation do not accumulate beyond prescribed allowable concentrations and pressures.

17.2.3 Armament and stores electromagnetic environmental effects.

This section assesses E3 analysis for the store based on the lifecycle environmental profile of the store, store subsystems, and application. 17.2.3 assesses the differences between the store E3 analysis and the air vehicle E3 analysis addressed in Section 13. 17.2.3 assesses the operation of the store and hazards induced to the air vehicle during employment and continued AW of the store.

The following store E3 standards and MoC represent the minimum criteria needed to establish, verify, and maintain the store's AW design while integrated into the air system. These standards ensure that the store operates within defined E3 safety-critical limits, focusing on its electromagnetic environment and compatibility with the host air system. Compliance with these standards is intended to prevent potential E3 hazards to the store, air system, and crewmembers during flight operations in the intended environment.

To avoid confusion and ensure the cognizant stores engineer understands that compliance involves more than just examination and analysis, the Test MoCs outlined in criteria 17.2.3.1, 17.2.3.2, and 17.2.3.4 should be closely coordinated with Section 13 SMEs, which pertains to EMI, EMC, and lightning testing. This coordination ensures clarity regarding safety-critical limits, test setups, and modes of operation. The results from these tests will be integral in determining E3 mutual compatibility and will be used to inform a comprehensive risk assessment, ensuring the stores engineer fully understands the compliance requirements.

17.2.3.1 Armament and stores subsystem and component electromagnetic interference.

Standard: Store subsystems and components comply with their defined EMI control requirements, ensuring emissions and susceptibility remain within the safety-critical limits as specified in MIL-STD-461, Detailed Requirements section. All store modes and functions (e.g., inventory, carriage, employment, flight termination, and jettison) will be assessed to ensure no EMI anomalies are introduced that could disrupt or cause potential hazards to store and air system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design documentation (e.g., installation drawings, ICDs, operational modes and characteristics) to verify the applicable store subsystem configurations of components, software, modes, and functions undergo the required EMI testing IAW MIL-STD-461, Detailed Requirements paragraphs.

Analysis.

(1) Verify the store subsystems and components are developed under the same set of EMI control requirements as the host air system. If not, conduct a gap analysis to compare the store's equipment against the host air system's MIL-STD-461 requirements and applicable test limits.

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This analysis will identify EMI qualification differences and potential EMI hazards to the stores and host air system.

Test.

(1) Lab test: For store equipment that is flight-critical or associated with SCFs, conduct the following EMI test IAW MIL-STD-461 test requirements: CE102, RE102, CS101, CS114, CS115, CS116, CS117, RS103, and if applicable for RF transceivers CE106, CS103, CS104, CS105. Verify the applicable store configuration(s) (e.g., hardware, software, modes, and functions) complete the required EMI testing and meet the safety-critical limits as specified by their EMI control requirements.

(2) Lab test: For store equipment that is not flight-critical or associated with SCFs, conduct the following EMI tests IAW MIL-STD-461 test requirements: CE102 and RE102. Verify the applicable store configuration(s) (e.g., hardware, software, modes, and functions) complete the required EMI testing and meet the safety-critical limits as specified by their EMI control requirements.

17.2.3.2 Armament and stores intra-system electromagnetic compatibility.

Standard: The store is electromagnetically compatible with the host air system and adjacent store configurations, ensuring that the air system and store flight-critical systems, and associated SCFs are not degraded and have no potential EMI hazards, such as unintentional detonation, release, launch, flight termination, or jettison of stores during concurrent intra-system operations (e.g., inventory, carriage, and employment) IAW MIL-STD-464, Intra-system Electromagnetic Compatibility paragraph. This includes stores configured or modified with RF transmitters and receivers (e.g., EMC, flight test, training, or telemetry kits).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review store subsystem and component-level EMI test data to assess test anomalies to be evaluated during system-level EMC ground test for potential EMI or hazards to the store and host air system.

(2) Review AUR, modified, or instrumented store design documentation (e.g., installation drawings, ICDs, operational modes and characteristics) to verify intended capabilities and modes of operation during pre-launch and post-launch are evaluated during the system-level EMC ground test.

(3) Review store EMC test plan to verify store and host air system configuration (e.g., hardware and software), operating modes, source/victim procedures, and pass/fail criteria to be executed during EMC testing.

Analysis.

(1) For stores configured with RF transmitters and receivers, conduct an RF antenna co-site analysis to verify store and air system RF compatibility. Identify worst-case store location(s) and load outs from air system and stores safety-critical antenna-connected receivers.

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Test.

(1) Ground test: Conduct a store and air system EMC ground test to verify store compatibility. For instrumented or flight test configured stores, a ground station or mobile test vehicle may be required to support EMC testing and data collection.

(2) Ground test: Verify store emission EMI test results for RE102 meet the safety-critical limits as specified by their EMI control requirements. If not or no test data is available, conduct a spectrum analyzer test on the air system safety-critical receivers to verify store compatibility.

17.2.3.3 Armament and stores external radio frequency electromagnetic environments.

Standard: Stores can safely operate and are compatible with their defined external RF EME, as specified in MIL-STD-464, External RF EME paragraphs and MIL-HDBK-240-4, External RF EME Evaluation, External RF EME Test Planning, Conducting the Test, and Data Analysis sections, such that the store and host air system's SOF systems and associated SCFs are not degraded, and no potential EMI hazards such as unintentional detonation, release, launch, flight termination, or jettison of stores during inter system EME operations (e.g., inventory, carriage, employment, and jettison).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review store subsystem box-level EMI test results to establish a 200 v/m hardness immunity level for shielding effectiveness analysis. Verify SOF store subsystems are tested to CS114 (curve 5) and RS103 (200 v/m) safety-critical limits.

Analysis.

(1) Verify the store is developed under the same external RF EME requirements as specified for the host air system. If not, conduct a gap analysis of stores and host air system external RF EME requirements to identify differences to be assessed.

(2) Conduct a shielding effectiveness analysis of the store. Identify and assess system performance where specified external RF EME levels are higher than box-level EMI test and shielding effectiveness analysis results.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Ground test: As required, conduct a high-level RF EME test of the store system integrated on the air system (with EMC instrumented store installed if applicable). Analyze test results and anomalies. Alternatively, if the store has previously operated and been employed on a similar mission type air system that was previously tested to the specified external RF EME requirements, a similarity analysis may be conducted, provided the configurations are comparable.

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17.2.3.4 Armament and stores lightning.

Standard: Store can safely operate (e.g., inventory, carriage, employment, and jettison) and does not cause a hazard (e.g., unintentional detonation, release, or FOD) to the air system when exposed to direct and indirect effects of lightning IAW MIL-STD-464, Lightning paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review grounding and bonding specification documentation, drawings, and manufacturing bonding process for verification of mechanical interfaces to the store and air system.

Analysis.

(1) Conduct air system lightning zoning analysis to include external stores. Identify applicable lightning threat levels to be assessed for store testing.

Test.

(1) Lab test: Conduct direct effects lightning testing on store radome or non-conductive surfaces (if applicable), skin or panel, and variable or fixed control surface representative coupons.

(2) Ground test: Take store bonding measurements as installed on the air system to verify bonding performance requirements for applicable mechanical interfaces.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test: For external fuel tanks, ensure ignition sources are eliminated by conducting high current testing on fuel system representative components IAW SAE ARP5416B Ignitable Mixture (Flammable Gas) Test Method paragraph or an equivalent, such as the RTCA DO-160 Ignitable Mixture (Flammable Gas) Test Method.

(4) Ground test: Conduct a passive store, installed on air system or wing mockup, high-level lightning test or low-level CW test. Conduct post-test analysis to extrapolate current and voltage measurements to the applicable waveforms and threat levels to be assessed for additional testing. Alternatively, if the store has previously operated and been employed on a similar mission type air system that was previously tested to the specified lightning requirements, a similarity analysis may be conducted, provided the configurations are comparable.

(5) Ground test: Verify high-level lightning or low-level CW store test results show safety margin to prior subsystem MIL-STD-461 CS117 test results. If not, conduct active systems direct drive testing to store interfaces as installed on the air system or wing mockup to verify safety-critical systems can safely operate during the applied extrapolated lightning waveforms and test levels.

17.2.3.5 Armament and stores electrostatic charge control.

Standard: Store systems control and dissipate electrostatic charge buildup resulting from various sources, including p-static, fluid flow, air flow, vertical lift ESD, and other triboelectric charge-generating mechanisms IAW MIL-STD-464, Electrostatic Environments paragraphs; MIL-

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HDBK-240-3, Personnel-Borne Electrostatic Discharge (PESD) and Helicopter-Borne Electrostatic Discharge (HESD) paragraphs; and JOTP-062, PESD Requirements paragraphs and HESD Requirements paragraphs. This is critical to prevent ordnance hazards, personnel shock risks, degradation of SOF systems, including RF safety-critical receivers, during store inventory, carriage, employment, and jettison when applicable. Additionally, this applies to in flight reloading of munitions (e.g., guns, cannons, common launch tubes) for specific air systems and crewmembers.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review grounding and bonding specification documentation, drawings, and manufacturing bonding process for verification of mechanical interfaces to the store and air system.

Test.

(1) Ground test: Take store bonding measurements as installed on the air system to verify bonding performance requirements for applicable mechanical interfaces.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Lab test: For air systems enabling crewmembers to reload munitions during flight operations, conduct ESD testing IAW MIL-HDBK-240-3, Personnel-Borne Electrostatic Discharge (PESD) section, to assess potential hazards to exposure of ordnance items, electrical interfaces, and related systems such as firing circuits and control electronics (e.g., weapon release, intent to launch), to crewmember handling during flight operations. If store is ESD sensitive, verify hazards are documented in the applicable maintenance and flight manuals.

(3) Lab test: For vertical lift air systems enabling crewmembers to reload munitions during flight operations, conduct ESD testing IAW MIL-HDBK-240-3, Helicopter-Borne Electrostatic Discharge (HESD) section, to assess potential hazards to exposure of ordnance items, electrical interfaces, and related systems such as firing circuits and control electronics (e.g., weapon release, intent to launch) to crewmember handling during vertical flight operations. If store is ESD sensitive, verify hazards are documented in the applicable maintenance and flight manuals.

(4) Ground test: Conduct air system level p-static test with store system installed on the air system. Verify that the store dissipates charges in a timely manner and does not generate discharge (e.g., arching or sparking) that could potentially create EMI hazards to safety-critical antenna-connected receivers.

17.2.3.6 Armament and stores hazards of electromagnetic radiation to ordnance.

Standard: Stores with EIDs, which includes stores with FTSs, as well as adjacent store configurations, are not inadvertently activated, nor experience degraded performance, due to exposure to external radiated EME. This includes both direct RF-induced actuation and coupling to associated firing circuits during air system and store operations (e.g., inventory, carriage, employment, and jettison). Compliance with this standard is supported by HERO assessments, as

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specified in MIL-STD-464, Hazards of Electromagnetic Radiation to Ordnance (HERO) paragraph, and MIL-HDBK-240-2, HERO Test Process, Test Plan, Conducting the Test, and Data Analysis sections.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Perform HERO Safe, Unsafe, and Susceptible analysis for armament and stores configured with EIDs and its corresponding circuitry IAW MIL-HDBK-235-1, Calculations section, and NAVSEA OP 3565 Volume 1, Calculation of Electromagnetic Fields appendix. This includes the worst-case load out location to air system RF transmitters and adjacent stores configured with RF transmitters. Verify calculated HERO Safe, Unsafe, and Susceptible standoff distances are documented in the applicable maintenance and flight manuals.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Ground test: Conduct high-level HERO testing on an instrumented store installed on the applicable air systems to assess the worst-case store loadout locations, ensuring a 16.5 dB safety margin for EIDs and a 6 dB margin for the associated firing circuitry IAW MIL-STD-464, Hazards of Electromagnetic Radiation to Ordnance (HERO) paragraph.

17.3 Armament (stores and weapons) integration.

This section establishes the AW criteria for the integration, interface, and operation of armament systems, including the software, firmware, and hardware components of the air vehicle and its stores.

The scope of this section encompasses all armament components, features, and system attributes essential for maintaining safe operation and managing the state of stores through all phases of flight, based upon the definition of AW as the property of an air system to attain, sustain, and complete flight IAW approved usage limits. The focus is on the air system-to-store integration at both the system and subsystem levels. Aeromechanical and flight sciences are addressed in 17.1 and environmental integration in 17.2, respectively.

For the purpose of this section, the following definitions apply:

a. Safe Operation: An operating condition, selectable by a crewmember or inherent to the store, where the store operates predictably and poses no hazard to the crewmembers or air vehicle.
b. Armed State: An operating condition where the store is ready to operate as intended, with the potential to release energy, be employed, or be released from the air vehicle or detonate.
c. Inadvertent: A system-controlled failure occurring without crewmember consent or intent (e.g., the system fails to prevent an uncommanded store/weapon release).
d. Unintentional: A failure resulting from crewmember action (e.g., a crewmember accidentally engages the release command).

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This section addresses the following critical areas:

a. 17.3.1 Safe Operating Condition: Evaluates the SMS, safety interlocks, and inhibits that prevent hazardous store operation.
b. 17.3.2 System Health: Details the system's ability to monitor the health and status of the armament system and stores, and to alert crewmembers to malfunctions or degraded modes.
c. 17.3.3 Air Vehicle Stores Interface and Safe Operations: Addresses software compatibility, management of store-induced hazards, and the integrity of electrical and logical interfaces.
d. 17.3.4 Positive Control: Ensures positive C2 of stores during all employment conditions, including the crewmember interface, command paths, and lost-link responses.

Compliance requires coordination with Systems Engineering (Section 4) and Computer Systems and Software (Section 15) to identify SCFs and address the safety impact of failure modes. Verification may be achieved through system safety processes and coordination with relevant safety boards. Regression testing for configuration changes may be tailored based on the criticality of the change and an analysis of the impacted functions.

17.3.1 Safe operating condition.

The safe operating condition is maintained by the SMS, stores management function, and a combination of software, hardware, and firmware controls. These controls manage store inventory, arming and safing, employment (launching, firing, dispensing, jettisoning), and status communication. Although SMS traditionally refers to weapons, in the context of 17.3, it is intended to include the subsystems that C2 any store such as pods or lasers. Key elements include safe operation of the SMS functionality; robust safety interlocks to prevent inadvertent operation; inhibits, limits, and interrupts to enforce operating envelopes; and reliable targeting data to ensure safe employment. Each element will be verified to ensure the armament system properly protects the air system.

Safety interlocks are discrete signals that are designed to interface with the armament system to prevent inadvertent and unintentional state changes such as arming fuzes, firing weapons, or employing electromagnetic radiation during ground or flight operations. Safety interlocks can be any combination of hardware, software, or firmware within air vehicle or control station (e.g., weight-on-wheels, weapon on station, master arm, bay door switches).

Weapon inhibit, limit, and interrupts are conditional design features that prevent weapon launch or firing and store operation when limits or constraints are exceeded, ensuring air system safety. Weapon inhibit, limit, and interrupts are categorized as "performance" or "safety". Performance WILIs prevent firing outside optimal launch conditions, minimizing target miss probability; these are typically not evaluated for AW unless directly impacting safety. Safety WILIs prevent firing under hazardous conditions. Crewmembers can typically override performance WILIs but not safety WILIs. Weapon inhibit, limit, and interrupts define fire and no-fire conditions within areas such as launch acceptability regions, weapons engagement zones, keep-out zones, and no-fire zones.

The fire control system, encompassing targeting and mission systems, critically impacts armament operations and flight path. Data acquisition, processing, and communication for stores employment will be accurate; errors, latency, or fidelity reductions can endanger the air system

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during store release. The accuracy of sensor and fused targeting data provided to the SMS for store engagement is also crucial to prevent hazards to the air vehicle and ensure its safe recovery. Flight control and fire control systems are tightly coupled, often via software, using tailored control laws and symbology to expedite fire control processes across the air vehicle's operational range. Accumulated errors within the fire control system, including tolerance variations and latency in cueing systems (e.g., helmet cueing, sensors, barrel erosion, launcher alignment, turret backlash, ballistic calculations), can compromise the firing solution and create a hazard.

17.3.1.1 Stores management functionality.

Standard: The SMS(s) manages the SCFs for controlling and operating stores (e.g., managing store inventory with associated mass properties, issuing squib fire voltages to release equipment, employment of armament, and release of countermeasures and electromagnetic radiation).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Examine SMS development documentation to verify that all SCFs and safety-critical components within the SMS architecture are identified, allocated with traceability, and that their required integrity levels are defined.

(2) Evaluate the interface between the SMS and other air vehicle subsystems (e.g., flight controls) and each individual store to identify all safety-critical data paths and communication threads. Verify compliance to the ICD for each store and intermediate interface between armament systems.

Analysis.

(1) Conduct an analysis to evaluate safety-critical functionality within the SMS and store(s). Analyze failure modes and hazards attributed to individual contributions of the software, firmware, and hardware components within the SMS and store(s) to determine where the SMS requires redundancy mechanisms. Analyze the FMECA SCFs for redundancy (i.e., single points of failure) and verify that mitigating procedures exist to address in flight failures.

Test.

(1) Lab test: Conduct HITL and software-in-the-loop end-to-end testing to verify stores management and store functionality during carriage, conditioning (e.g., power-up, initialization, transfer alignment, pre-arming, targeting), and employment (e.g., release and jettison, pod operation). Conduct fault insertion testing (e.g., flipped bits, corrupted data sets) to verify functions are fault tolerant and demonstrate correct failure response.

(2) Ground test: Perform on air vehicle functional test of station and suspension and release equipment (e.g., release and control) using associated armament support equipment. Verify that any deficiencies identified in lab testing have been mitigated. Verify that data pertinent to the air vehicle-to-store interface (e.g., air vehicle identification and configuration, lever arms, store mass properties) is valid and complies to the ICD for each store and intermediate interface between armament systems IAW MIL-HDBK-244, Stores Management Equipment Test paragraph.

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Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Flight test: Conduct flight testing to verify stores management functionality (e.g., store conditioning, managing of store inventory, pre-arming, firing, jettison, release, launch).

17.3.1.2 Safety interlocks.

Standard: The air system safety interlocks prevent inadvertent store operations (weapon or expendable release or launch, firing of guns, laser or other directed energy system firing, or pod operation resulting in RF emissions).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Ensure that safety interlocks within the air system architecture are designed to prevent inadvertent store operation that would result in a hazard to the air system (e.g., inadvertent release of store(s) due to stray voltage on the squib fire line for the ejection rack, inadvertent deployment of store inlet cover or wings, inadvertent radiation of EW stores).

(2) Evaluate system documentation for all signal paths that enable store operations and determine if the air system has redundancy mechanisms in place to prevent inadvertent store operations. Evaluate system documentation to ensure no single point failure can result in an inadvertent store operation.

Analysis.

(1) Evaluate maintenance panel and control design to ensure crewmembers can “SAFE” the store system.

(2) Conduct analysis of signals that enable or control store operation to identify redundancy mechanisms and determine where safety interlocks are required to eliminate single points of failure and prevent inadvertent store operations.

Test.

(1) Lab test: Verify that at least two separate crewmember actions are required to employ or operate stores in high energy states.

(2) Lab test: Conduct HITL and software-in-the-loop testing with interlock hardware, software, and firmware for the purpose of end-to-end testing to verify the safety interlocks function as designed.

(3) Lab test: Conduct FMET to verify interlocks and redundancies protect against single mechanical or logical failure resulting in inadvertent store operation.

(4) Ground test: Conduct testing on air system to verify safety interlocks functionality, including that safety interlock overrides can be returned to nominal state.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(5) Lab test, ground test, or flight test: Verify interlocks prevent inadvertent arming of the fuze, launching, firing, jettisoning, and employment of the store and that two separate actions are required to employ munitions or operate stores in high energy states.

17.3.1.3 Inhibits, limits, and interrupts.

Standard: For store limits implemented via software (e.g., WILIs), the air system cannot release or launch stores, fire guns, or operate pods outside of those limits. Examples include maintaining minimum release intervals for salvos or ripple release programs or withholding release outside of a safe operating condition (e.g., exceeding maximum airspeed, below minimum release altitude).

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Identify conditions where stores engagement criteria and air vehicle limit exceedances exist within the entire flight envelope. For limitations implemented via software control as an inhibit or interrupt, examine the design criteria for SMSs to verify that the limit has been successfully implemented. This includes examining the design to ensure that the system can detect the conditions that would trigger the inhibit or interrupt.

Analysis.

(1) Perform a WILI analysis IAW ADS-44-HDBK, Weapon Inhibits, Limits and Interrupts (WILI) Analysis paragraph, to define the parameters, safety constraints, and software/firmware necessary to limit or prevent firing of guns, release or launch of weapons, and operation of pods under conditions that present an AW hazard to the air vehicle. If software/firmware WILIs are not implemented (i.e. no software inhibits or interrupts), ensure procedural indications (e.g., warnings, cautions, notes) and limits are identified during the WILI analysis.

Test.

(1) Lab test: Verify the stores management defined store operating limits and inhibit zones (e.g., radar masking) prevent store operations outside of SMS controlled limits.

(2) Ground test: Verify that software controlled inhibits and interrupts prevent the operation, employment, and separation (e.g., release, jettison) of stores when air vehicle or store limits and constraints are exceeded. Verify that SMS controlled keep-out zones accommodate air vehicle OML and all potential peripheral configurations.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(3) Lab test, ground test, or flight test: Verify software controlled inhibits and interrupts prevent store operations outside of defined limits. Testing includes captive carriage (e.g., simulated stores, captive flight trainers, and inert ordnance) and may include employment of stores.

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17.3.1.4 Target and fire control data.

Standard: The acquisition and designation of targets and the fire control systems that provide information to the store cannot induce errors that would result in a flight or safety-critical hazard during store operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis

(1) Evaluate targeting system and target data set (e.g., target handover) fidelity and error sources including sensor, sensor boresight, and latency for all operating conditions to verify it does not exceed the design tolerances for store employment.

(2) Evaluate the fire control system that provides gun, rocket, and missile cueing for timing, sequencing, accuracy, and errors that may result in loss of control or errant firing that might lead to damage to or recontact with the host air vehicle.

Test.

(1) Lab test: Evaluate the fire control integration and interface data integrity, timing, and error for impact to store IAW JSSSEH, Critical Timing and Interrupt Functions section, and AOP-52, System Integration paragraph. For integrated or remote fire control systems, evaluate the source, controller, data links, and operation for failure modes and verify that the system control can be overridden by crewmembers.

Verify the positional reference (e.g., boresight, transfer alignment) and target data set of the fire control sensor and the store to characterize the error, aimpoint, and fly-out behavior to prevent recontact to the air vehicle. Evaluate armament/sensors cueing and slaving as it applies to store aiming. Evaluate sensor fidelity as compared to measured truth data and miss distances.

(2) Ground test: Verify that the air system provides cueing and control data to enable store operation, targeting and firing solutions (e.g., aiming, control and employ guns, rockets or missiles including the operation, duty cycle, turret and articulating system control), timing and accuracy to ensure that the employment of the store does not present a hazard to the launch air vehicle.

17.3.2 System health.

The purpose of this section is to verify that if the store or interface to air system fails to operate as intended, the air system can continue to operate safely in a degraded state and will indicate to the crewmember when a degraded state or store malfunction occurs. This section is in support of 9.2.8 and 11.3.1 which cover alerting systems for crewmembers and their avionics architecture.

The system is able to detect and respond to faults or failures that could lead to hazardous conditions. A "degraded mode " refers to a failure in the air system, SMS, or store interface that results in abnormal store behavior (e.g., degraded GPS data causing target drift, inadvertent clearing of store mass properties in flight). A "malfunctioning store" refers to a critical failure within the store itself that poses a hazard to the air vehicle (e.g., store inertial measurement unit failure, seeker gimble failure, hung store, thermal battery runaway).

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Hazard analyses (e.g., preliminary hazard analysis, SHA, FMECA) are critical for identifying these conditions. The system provides clear WCAs to enable the crewmembers to take appropriate corrective or emergency action.

17.3.2.1 Health status monitoring for degraded modes.

Standard: The air system monitors the health and status of the store(s) and interfacing air vehicle equipment and alerts the crewmembers of any detected or reported degraded modes (e.g., WCAs, limitations) to enable corrective action.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Ensure degraded modes in the air system are identified through examination of technical data (e.g., ICDs, preliminary hazard analysis, FTA, FMECA).

(2) Ensure degraded modes of the store are identified through examination of store technical data.

(3) Verify through examination that stores report health status and degraded modes, and that the air system monitors stores for reported health status and provides crewmembers with store health status.

Analysis.

(1) Determine the severity of each degraded mode and criticality impact of continued operation of the air system or store. Determine the conditions of continued operation during degraded mode to include definition of degraded modes, contributors, and cause and effects of system failures. Determination of required health status and degraded modes supporting this analysis may include output from System Safety Working Group and Crew System Working Group.

(2) Verify that store health status consists of current state of armament system and store IAW MIL-HDBK-244, General Design Criteria paragraph, including IBIT, CBIT, periodic BIT, and degraded mode.

Test.

(1) Lab test: Verify that the air system is capable of determining the condition of the air vehicle and stores. Verify the air system transitions to correct degraded mode operations when store or air vehicle failures occur. System integration laboratory should be capable of simulating degraded modes of the air system.

(2) Lab test: Verify the air system capability to provide health status and degraded modes to crewmembers. Ensure crewmembers can execute troubleshooting and emergency procedures in a degraded mode scenario to maintain safe operation during flight and recovery of the air vehicle

17.3.2.2 Malfunctioning stores.

Standard: A malfunctioning store does not result in inadvertent store operations or inadvertent transition of store state (e.g., powered off to on). In the case of a malfunctioning store, the air

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system or crewmembers have the ability to power down the store, transition the store to a safe state, or jettison the store.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Ensure safety-critical store malfunctions are identified and the severity of impact to the air vehicle has been determined.

Analysis.

(1) Verify that the PVI design allows the crewmember to have positive indication, display and control of the store inventory, selection, status, C2 of critical state and mode transitions.

(2) Conduct failure mode analysis to identify the store failure modes, contributors, effect on the air system, fault detection and handling within the store, and reporting within the store OFP and to the air system, and to verify the failure accommodation within the air system. Determine methods for continued safe operation during malfunctioning stores, including but not limited to failure indication, removing and cycling power, and emergency jettison.

Test.

(1) Lab test: Verify the air system capability to determine condition of the store and detect the malfunctioning store to include hung store, hangfire, misfire, or other critical store malfunctions. Verify that malfunctioning stores can be transitioned to a safe state (e.g., powered off or released if required to protect the air vehicle) IAW JSSSEH, Software Systems Safety Tests section.

(2) Lab test: Verify the store ability to sense a critical malfunction and provide status to the air system. Verify that the air system capability to command, control, and power down malfunctioning stores or transition the air system to degraded mode.

(3) Lab test: Verify the air system can indicate the condition of store to crewmembers. Ensure crew members can execute troubleshooting and emergency procedures in malfunctioning store scenario (e.g., hangfire, hung store, thermal battery runaway) to maintain safe operation during flight and recover the air vehicle.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(4) Flight or Ground test: Conduct testing to ensure the SMS controls and conditions the malfunctioning stores properly to a safe state and facilitates jettison of malfunctioning stores when commanded by the crewmember. Crewmembers are able to execute emergency procedures to recover the air vehicle with malfunctioning stores (e.g., jettison, droop gun recovery, uncontrolled ‘runaway’ gun, off-set landing, thermal battery rundown).

17.3.3 Stores interface and safe operations.

The air system functional integration of the store software ensures safe operation during all phases of operation. Store software and firmware are compatible with all air system safety-critical subsystems, including armament, to ensure safe operation during all phases,

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encompassing state changes, arming, release, launch, jettison, employment, and other critical functions. The software compatibility testing includes flight representative software for the air system and store. If the store fails to operate as intended, it does not create or induce a hazard to the air vehicle or crewmembers during all phases of flight.

The software documentation describes the architectural design, detailed design, and interface design necessary to integrate the software and firmware systems in the manner that they function together for safe operation. The software and firmware changes may influence SCF as well as the data communications bus traffic and loading, timing, throughput, and latency. Key software documentation provided by the store vendor are the software version description, VDD, FQT, SOF letter. System-level laboratory integration test is conducted to determine if the air system and store function, perform and interface as intended to prevent any adverse safety conditions and to verify that the store software functionality will not degrade the air vehicle under all stages of flight.

Software integration tests are conducted in an avionics integration lab, software integration lab, or on a "hot bench" configured with air system representative components including sensors, store emulators, “weapon in a box” or other representative engineering tools. It is important to ensure that the software integration lab environment is representative of the air vehicle and flight test environment and that the store lab test unit is representative of the store at the functional interface level (with the exception of energetic or radiation components) to certify results of testing are representative of the on air vehicle test (e.g., simulators, emulators, fault insertion devices are representative of flight test vehicles).

In cases where lab environments do not have the fidelity to represent the air vehicle and flight test environments, air vehicle ground testing can be used to supplement lab testing. Evaluation of store critical function considers the specific air system application and operational environment against the store's hazard analysis and baseline qualification. This includes identifying hardware, firmware, and software hazards arising from handling, storage, operations, and operational lifecycle exposure that could lead to AW hazards (e.g., exceedances, environmental exposure, MP changes, or altered interfaces). This evaluation also leverages findings from other government agencies specializing in energetics, munitions, and commodities (e.g., Weapon System Explosive Safety Review Board, Non-Nuclear Munitions Safety Board, Air Force Nuclear Weapon Center) to determine their applicability to AW. Identified hazards include those affecting critical store control functions such as fuze operation, arming logic, motor and engine start, wing and fin deployment, nose cover deployment, pre and post-launch events, carriage band release, parachute operation, pod operation, and store operation in proximity to the air vehicle.

17.3.3.1 Software compatibility.

Standard: The air system is functionally and logically compatible with the store to ensure SCFs between the air vehicle and the store are not negatively impacted.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Identify air system to store interface requirements that have implications to the AW of the air vehicle.

(2) Review store software documentation (e.g., OFP software version description or VDD) to verify identified store unique hazards are not inadvertently realized due to functional integration.

(3) Examine store software documentation to verify that Software Trouble Report and software anomaly reports have been corrected.

Analysis.

(1) Conduct an integration analysis, including air system databus communications and electrical interfaces, to ensure proper store operation.

Test.

(1) Lab test: Conduct software verification testing on the store OFP for each CSCI.

(2) Lab test: Conduct compatibility testing at the air system level.

(3) Lab test or ground test: Perform safety regression testing IAW JSSSEH and AOP-52, Regression Testing sections as required due to hardware or software changes that affect functional integration of the store.

(4) Lab test: Conduct fault insertion stress testing to ensure continuity, timing, sequence, fault tolerance, and ICD compliance.

(5) Ground test: Perform a functional test of the air system and the store for software compatibility and SCF IAW MIL-HDBK-1763, Test 102 and Flight Test Procedures appendix. Evaluate the interface between air system and store to verify proper communication and electrical interfaces between air system and store.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Review all store software documentation to ensure it is updated to reflect the configuration for operational use including Software Trouble Report, FQT, VDD, software version description and software deficiency reports (e.g., problem reports, deficiency reports, software anomaly reports) and review for residual AW hazards.

Test.

(7) Lab test: Evaluate FQT conducted by the original equipment manufacturer on the store for operational store OFP IAW JSSSEH, Formal Safety Qualification Testing paragraph and AOP-52, Functional Qualification Testing and Independent Verification and Validation section to ensure SCFs and data integrity between the air system and store interface do not result in AW hazard.

(8) Flight test: Perform captive carriage flight test prior to live fire flight test to evaluate the air system software and the store OFP to verify the data integrity, communication, and interface. Verify that the store software and air system software are compatible and there are no software anomalies that affect the AW prior to store operation.

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17.3.3.2 Stores critical function.

Standard: Stores execute critical control functions (e.g., fuze function, arming, motor and engine start, wing and fin deployment) when commanded without inducing flight or safety-critical hazards to the air vehicle or crewmembers during all phases of flight.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) The store hazards associated with critical functions identified have been individually assessed for each store and the contributors to AW impact have been evaluated to ensure there are no store induced hazards that could result in a flight or safety-critical hazard during the all phases of flight.

(2) Ensure any hazard related to software, hardware, or energetic exceedances, generated as part of store or laser safety review boards (e.g., Non-Nuclear Munitions Safety Board, WSESRB, Army Weapon System Safety Review Board, Joint Explosive Safety Review Board, Laser Safety Review Board, Air Force Nuclear Weapon Center, Nuclear Surety) that impact the AW of the air system application are mitigated by the design.

(3) Review safety review board concurrence memorandums to identify store induced hazards and to ensure recommended procedures, limitations, or mitigations have been documented in flight manuals.

(4) Analyze approved store government agency certification and qualification artifacts for residual hazards including but not limited to:

a. Final type qualification of energetics.
b. Insensitive munition assessment of AUR specific to air vehicle environment.
c. Interim hazard classification and final hazard classification.
d. Handling, loading, and downloading.
e. Ship suitability testing and ship weapon integration testing.
f. Basic ship safety.

Test.

(1) Ground or Lab test: Verify air vehicle design features mitigate any induced flight or safety-critical hazard from the store.

17.3.3.3 Interfaces.

Standard: The air system complies to all logical (digital and analog) interfaces (wired and wireless) that control armament and store SCFs. These interfaces prevent adverse effects from erroneous or spurious data, detect interface failures, maintain a defined state during power-up, operational transitions, and system errors or failures, and are verified for compliance with the ICD.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Examine interface documentation (e.g., ICD, interface change notices, store supplements, platform annexes, schematics) to identify all safety-critical interface requirements IAW JSSSEH and AOP-52, Interface Design Requirements sections that cover paths of communication and control (analog and digital) within the armament system as well as between the air vehicle and the store.

Analysis.

(1) Analyze the functional interfaces to ensure SCFs are traced through the entire air vehicle to store interface.

Test.

(1) Lab test: Conduct compatibility testing in a HITL environment with air system and store representative hardware, firmware, and software for all critical signals within the electrical signal set and logical message set for all relevant air system and store configurations. Verify that the desired loadouts and store configuration can operate safely with the air system's logical and electrical interface through inventory testing (e.g., power on multiple stores at once using air system representative interface, firing sequences). Verify critical interfaces function as designed to support transition from "safe" to operational ("armed") state (e.g., intended release of energy and stores).

(2) Lab test: Conduct testing of electrical interface and components to ensure continuity, timing, sequence, fault tolerance, and ICD compliance.

(3) Ground test: Conduct functional store station checks to verify wiring continuity and correct pinout of armament system cabling from SMS through the store umbilical.

(4) Ground test: Conduct ground communication testing on air system with the store-in-the-loop to verify air system to store communication, BIT checks, and power management for all electrical signal sets and applicable logical message sets IAW MIL-HDBK-1763, Test 102.

(5) Ground test: Conduct ground communication testing with support equipment designed to test the air system or store interface (e.g., perform release and control checks) to ensure accurate stimulation of SCFs, interlocks, and compliance with the platform to store interface.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(6) Flight test: Conduct captive carriage flight test for final verification of compliant interfaces.

17.3.4 Positive control.

The air system maintains communication with the store to ensure positive C2 of the store. If communication between the host platform and store or off-board controller and store are degraded or broken, the air vehicle and store will transition to a predetermined state to prevent incurrence of hazard to the air vehicle or crewmembers. In situations where the store is controlled autonomously, software and hardware controls will be evaluated to ensure safe operation throughout employment of the store and proper "human-in-the-loop" oversight. Failure modes and vulnerabilities will be assessed for fail-safe or default to known state/condition.

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This section also addresses the safety of C2 of stores controlled in non-traditional ways such as with off-board fire control, tablets, autonomous dispensing systems, FTSs, or using BLOS/LOS data links or control stations.

17.3.4.1 Crewmember interface controls and displays.

Standard: The controls and displays associated with the air system armament system enable the crewmembers to maintain oversight of the control and monitor all state and mode transitions during the employment of stores to prevent uncommanded operation.

Note: Evaluation of this standard is supported by Section 9.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Verify that the PVI design allows the crewmember to have positive indication, display, and control of the store inventory, selection, status, C2 of critical state and mode transitions IAW JSSSEH, Operator/Computing System Interface section and AOP-52, Human Interface section. Identify the system safe states and recovery modes and evaluate the conditions for safe state return IAW AOP-52, System Design Requirements section.

(2) Verify the design of the PVI is integrated with the crew alerting system and displays state and mode transition, degraded modes, and failure modes IAW JSSSEH, Safety-Critical Displays paragraph and Safety-Critical Alerts paragraph.

Test.

(1) Lab test: Evaluate the PVI (to include control station as applicable) in the SIL to demonstrate that the crewmember can control all modes of the armament and store operation. Evaluate hazards and mitigation features IAW JSSSEH, User Interface Tests paragraph and AOP-52, Human Interface section by simulating hazardous conditions, fault injection, and verifying the air system's applicable display or alert response. Verify that all armament symbology and associated states and modes are correctly displayed on the air vehicle and control station displays, including but not limited to state and mode transition from "safe operating" condition to "armed", launch conditions and employment zones, and that all controls operate and function as designed to prevent unintended operation of the store. Conduct robustness testing to evaluate state and mode transition.

(2) Ground test: Evaluate the PVI (to include control station as applicable) to demonstrate that the crewmember can control all modes of the armament operation IAW the air system specification and MIL-HDBK-244, Cockpit Controls, Displays and Management section, JSSSEH, User Interface Tests paragraph, and AOP-52, Human Interface section. Conduct testing to verify that all armament symbology is correctly shown on the air system displays (e.g., threat symbology, BIT status, expendable inventories) and that all safety-critical controls and indicators operate and function as designed to prevent unintended operation of the store.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

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Test.

(3) Flight test: Evaluate the PVI (to include control station as applicable) to demonstrate that the crewmember can control all modes of the armament operation IAW the air system specification and MIL-HDBK-244, Cockpit Controls, Displays and Management section. Conduct testing to verify that all armament symbology is correctly shown on the air system displays (e.g., threat symbology, BIT status, expendable inventories) and that all safety-critical controls and indicators operate and function as designed to prevent unintended operation of the store.

Verify that in auto modes of operation or auto track, the store transitions to a safe state of operation if manual shutoff override command is engaged.

17.3.4.2 Command and control path.

Standard: All C2 paths (internal and external) for autonomous and semi-autonomous control systems of suspension and release equipment, laser systems, weapons, or other stores that control safety-critical armament functions operate as designed without instability, corruption, or failure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze C2 path architecture and design between the host air vehicle and store including any federated components or off-board controllers for hazards associated with loss of communication, instability, corruption, vulnerability, and failure modes.

(2) Identify failure modes within C2 path architectures and mitigation design features to prevent inadvertent store operation.

(3) Evaluate software and hardware for autonomous and semi-autonomous functionality that could result in hazards and ensure controls or design mitigations are in place to prevent AW hazards. Verify that all autonomous control systems have crewmember oversight and overrides to control store operation and state transitions.

(4) Evaluate the C2 design to ensure safety precepts (e.g., Unmanned System Safety Engineering Precepts Guide for DoD Acquisition), mitigations and human-in-the-loop controls have been implemented to mitigate AW hazards associated with autonomous and semi-autonomous operation.

Test.

(1) Lab test: Conduct SIL testing to evaluate all hardware and software interfaces of the armament system C2 for intended function during store operation, including fault insertion and failure modes.

(2) Lab test: Evaluate all autonomous modes of store operation for failure modes that could result in AW hazards induced by store operation to the host air vehicle.

(3) Lab test: Verify all C2 paths of the armament system impacting store operation do not create an AW hazard, including but not limited to:

a. Host air vehicle.
b. Autonomous state change.

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c. Electronic kneeboard and laptops.
d. External entities that controls (disrupt) fire control system.
e. Beyond line of sight input into fire control system.
f. Automatic EW (dispense).

17.3.4.3 Lost link.

Standard: In the event of loss of C2 links between a control station and armed UAS or between launch platform and store or laser, the system transitions to predetermined states that prevent AW hazards due to inadvertent store or laser operation.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Analyze the lost link condition during all store or laser operational states to verify that the design prevents inadvertent store or laser operation.

(2) Evaluate software and hardware design documentation to confirm necessary design mitigation features are implemented.

Test.

(1) Lab test: Conduct SIL testing of the system and communication interface to verify design mitigation features function as designed for all store and laser operational states.

(2) Ground test: Evaluate system and communication interface to verify that control and design mitigation features function as intended under lost link condition.

17.4 Laser integration.

This section addresses optical radiation (electromagnetic energy within the ultraviolet, visible or regions of the spectrum) as specified in MIL-STD-1425, Definitions and Units section. This section is primarily focused on emission sources based on laser physics. However, this section also applies to non-laser high power optical radiation sources that pose a hazard to personnel or equipment. Examples include bright incoherent emitters such as spotlights and illuminators. This section also addresses advances in optical DEW such as high-power weapons-grade optical radiators (laser or non-laser). The term laser will be used in all criteria, but this section should be understood to apply to all hazardous non-laser optical radiation sources that are to be integrated onto the air vehicle. Key aspects of laser system safety include maximum permissible exposure, nominal ocular hazard distance, and nominal skin hazard distance. This section does not apply to RF DEWs such as electronic attack or high power microwave.

Definitions for laser firing actions in this section are as follows:

a. Manual Mode: Laser fires only with continuous crewmember input (e.g., pressing a dead-man switch). Stops immediately when input is released.
b. Auto Mode: Laser fires automatically after activation. Stops at a preset condition (e.g., time, pulse count). Manual shutoff available.

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c. Auto-Track Mode: Auto Mode with automated target tracking. Laser system autonomously controls laser pointing and firing while tracking a target. Stops at a defined condition or state (e.g., new target, track lost, mask). Manual shutoff available.
d. Latching: Unattended, continuous laser firing after activation. Not permitted.

17.4.1 Crewmember exposure.

Standard: The integrated laser hardware and software design assures, during all modes of operation, crewmembers are not exposed to optical radiation emissions above the maximum permissible exposure limits, IAW ANSI Z136.1, Criteria for Exposures of Eye and Skin sections. Crewmembers will not operate laser(s) within the nominal ocular hazard distance or nominal skin hazard distance limits as defined by the service LSRA without the use of required personal protective equipment.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the flight manual documents the laser system's hazard warning signs and labels, operational safety procedures, and personal protective equipment (if applicable) per the systems laser hazard classification.

(2) If applicable, verify the LEP issued to crewmembers complies with the specifications defined in the LSRA's laser hazard assessment. The LEP will be rated for the laser system's operating wavelength(s) and will provide an optical density equal to or greater than the minimum optical density values identified to ensure eye protection.

Test.

(1) Ground or Lab test: The LSRA will conduct a ground or laboratory test to characterize integrated laser system IAW ANSI Z136.4, Laser Measurements section. Test parameters at a minimum will include measurements of maximum output power or pulse energy, peak wavelength, pulse duration and repetition characteristics, minimum beam divergence, and laser beam diameter at the output aperture. Following data collection, LSRA will perform a laser hazard analysis to determine maximum permissible exposure limits, nominal ocular hazard distance for both unaided and aided (e.g., binocular) viewing scenarios, minimum required optical density for LEP, nominal skin hazard distance limits, and assign an appropriate laser hazard classification IAW ANSI Z136.1, Hazard Evaluation and Classification section.

17.4.2 Laser boresight.

Standard: The laser system retains boresight alignment with the airframe LOS to ensure laser pointing angular accuracy prevents laser energy from entering masked or cutout zones IAW MIL-STD-1425, Boresight section. The laser system alignment remains within tolerance for the full flight profile or provides a means to detect when it does not.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review design documentation (e.g., ICD, specifications, drawings, descriptions, flight manual) to verify boresight alignment process with airframe LOS, alignment accuracy errors, boresight drift, misalignment indications, and procedure for inflight alignment.

Test.

(1) Ground test: Verify laser boresight alignment performance maintains clear separation between alignment and masking error bounds under static controlled conditions. Verify crewmembers can perform a laser boresight alignment procedure.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify laser boresight alignment and retention performance continues to maintain clear separation between alignment and masking error bounds during dynamic flight maneuvers. Verify misalignment indications and that crewmembers can perform an inflight laser boresight alignment procedure.

17.4.3 Laser fire control.

Standard: The laser fire control system requires two distinct and sequential actions to prevent inadvertent laser firing: an arming action, followed by a firing action, defined as a continuous crewmember input (e.g., pressing a dead-man switch) IAW MIL-STD-1425, Unintentional Output and Laser Fire Switch sections.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify laser system design (e.g., ICD, specifications, drawings, descriptions, flight manual) has two distinct and sequential actions required to fire the laser.

Demonstration.

(1) Verify that two distinct and sequential actions are required to fire the laser to prevent inadvertent firing of the laser in either ground or lab environments.

17.4.4 Latching.

Standard: The laser system incorporates hardware and software safeguards to prevent long-term or unattended continuous lasing (latching). In Manual Mode, laser emissions immediately cease fire upon release of the dead-man switch. In both Auto and Auto-Track Modes, the system initiates automatic shutoff upon reaching a predefined condition or state. A manual shutoff, single push switch, is provided for both Auto and Auto-Track Modes to allow immediate deactivation of the laser to prevent inadvertent lasing or emissions IAW MIL-STD-1425, Laser Fire Switch, Operation Switch, and Beam Pointing sections.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Verify laser system design (e.g., ICD, specifications, drawings, descriptions, flight manual) prevents the laser from latching, incorporates a dead-man switch for Manual Mode, and manual shutoff switch for Auto and Auto-track Modes.

Demonstration.

(1) Verify that the laser does not latch, incorporates a dead-man switch for Manual Mode, and manual shutoff switch for Auto and Auto-Track Modes.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(1) Flight test: Verify that in Auto and Auto-Track modes, the laser system automatically shuts off upon reaching predefined conditions or states to prevent unintended emissions and the manual single-push shutoff switch instantly deactivates the laser.

17.4.5 Display indications.

Standard: The laser control display provides continuous indication of all safety-critical system states. At a minimum, the following are displayed: system armed status, active firing status, current pointing position, and any firing inhibition conditions (e.g., masking, safety interlocks) IAW MIL-STD-1425, Emission Indicator section. These indications are clearly visible to crewmembers to prevent inadvertent laser emissions or exposure.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review design (e.g., ICD, specifications, drawings, descriptions, flight manual) to verify laser system display indications, armed status, active firing status, current pointing position, and any firing inhibition conditions.

Demonstration.

(1) Verify that the laser control display indications are clearly visible and show when the laser is armed, actively firing, pointing position, and when firing is inhibited due masking or safety interlocks.

17.4.6 Airframe contact.

Standard: The laser system inhibitor, comprising of both software and hardware, controls the laser's FOR to prevent laser emission from striking any part of the air vehicle or externally mounted equipment (e.g., stores, sensors, pods) IAW MIL-STD-1425, Beam Pointing section.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

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Examination.

(1) Review design (e.g., ICD, specifications, drawings, descriptions, TO) to verify the laser inhibitor software and hardware controls are implemented to prevent laser emission from striking the air vehicle.

Analysis.

(1) Conduct masking analysis of the air vehicle's OML for worst-case mission store configurations to identify laser sensor line of sight obstructions. Define masking cutout and warning zones with sensor FOR for azimuth and elevation viewing angles. Create a parametric masking matrix for all mission store configurations. Include laser system pointing errors by expanding cutout zones with worst-case angular margins to ensure full coverage of potential beam deviations.

Test.

(1) Ground test: Verify masking matrix accuracy, laser cutout zones, and cockpit visual or aural indications. Document discrepancies and update masking profiles as required.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Flight test: Verify the real-time laser inhibitor triggers and visual or aural indications are provided to the flight crew within designated laser masking sectors during dynamic air vehicle maneuvers. Document discrepancies and update masking profiles as required.

17.4.7 Ground lasing.

Standard: The laser system is inhibited from arming and firing while the air vehicle is on the ground. An override for ground lasing is only available for maintenance operations IAW MIL-STD-1425, Airborne Lasers section.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify the laser system design (e.g., ICD, specifications, drawings, descriptions, flight manual) includes a ground fire interlock to prevent arming and firing when the air vehicle is on the ground.

Demonstration.

(1) Verify that the ground fire interlock (e.g., weight on wheels, software airspeed and altitude inhibitor) prevents the laser from arming and firing when the air vehicle is on the ground.

17.5 Armament and stores technical documentation.

This section addresses armament and stores technical documentation including store configuration, store handling and loading procedures, store loadouts and limitations, and armament and store operation, and delivery procedures. This data set specifically includes all

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stores configurations (hardware and software), safe release envelopes, data for operational mission planning, and flight limitations for carriage, employment, and jettison of stores.

17.5.1 Armament and store configuration.

Standard: The full configuration (hardware and software) of armament system components and stores is defined in applicable technical data to prevent unauthorized flight configurations. Configuration data set defines the physical characteristics, store SCFs, authorized inspection criteria, store assembly, buildup of armament and stores systems, and operational service life limit definition. This includes any individual store configuration buildup manual and electronic tools that dictate the authorized configuration of the store or air vehicle armament system.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that the configuration data set, including mass properties, drawings, schematics, and VDDs define the system.

(2) Verify that the system complies to the configuration data set.

(3) Verify that only authorized store configurations are defined in applicable stores limitation sheets.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Verify support equipment and test sets used to configure and load the store and store assembly publications are authorized to load the store and accurately depict the store configuration authorized for loading.

(5) Verify approved configuration data set(s) are documented in USG verified flight manual(s) or equivalent supplement(s).

17.5.2 Stores handling and loading procedures.

Standard: Procedures for the handling, release and control, loading, and maintenance of armament systems and stores are defined and prevent damage to the air vehicle during loading, operation, or delivery of stores. This includes any individual store or air vehicle installation manuals and electronic tools that dictate how stores are installed and removed from the air vehicle under any operational condition.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that procedures for release and control and loading and unloading the store are available prior to loading non-inert stores or powering on the air vehicle.

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(2) Verify that all safety and emergency procedures associated with handling, loading and downloading, and emergency disposal are technically accurate, meet all operating environment restrictions, and have been provided to the test team.

(3) Verify all loadout configurations to ensure there are no unique and undocumented loading sequences that need to be adhered to prevent unsafe loading, specifically associated to ordnance or energetics.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Verify that applicable release and control and store loading and unloading procedures have been verified by the appropriate technical authority and are documented in USG verified technical documents or equivalent supplement(s).

(5) Verify that all electronic tools supporting store loading procedures are verified and comply with the USG verified technical documents.

17.5.3 Store loadouts and limitations.

Standard: Air vehicle and store allowable mixed loadout combinations and their associated carriage and release limitations are defined for the safe carriage, operation, and release of stores. This includes information published in (or supplemental to) store loadout and air vehicle configuration manuals and electronic tools (e.g., mission planning) that dictate which stores are authorized on the air vehicle for a given loadout or mission.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that all approved air vehicle store loadout combinations and their associated carriage and release limitations are derived from store compatibility engineering source data and documented in technical data.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(2) Verify approved store loadout combinations and applicable limitations and procedures have been validated and verified by the appropriate technical authority and are documented in USG verified flight manual(s) or equivalent supplement(s).

(3) Verify that all electronic tools supporting store loadout and limitations are verified and comply with the USG verified technical documents.

17.5.4 Armament and store operation and delivery procedures.

Standard: Procedures for armament and stores operation and weapon delivery are defined, have been verified, and prevent unsafe operations, employment, or release/delivery. This includes functional and emergency procedures for the fire control systems, necessary pre-launch and post-

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launch support, all available safe ripple release intervals, salvo tables, safe escape tables and safe release envelopes, surface danger zones, masking profiles and exclusion zones, and emergency and combat jettison procedures. This also includes any flight manuals, operating procedures, and electronic tools that dictate the safe operation, employment, and release of stores that enable in-flight weaponeering.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify that technical data contains store operating and delivery procedures.

(2) Verify technical data contains all operating procedures, warnings, cautions, notes, and limitations.

(3) Technical data contains employment limitations including safe escape, salvo, surface danger, and exclusion zones.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(4) Verify that applicable store operating and delivery procedures have been validated and verified by the appropriate technical authority and are documented in USG verified technical documents or equivalent supplement(s).

(5) Verify that all electronic tools supporting store operating and delivery procedures are verified and comply with the USG verified technical documents.

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19. MATERIALS AND MATERIAL INTEGRITY

Materials comprise the entire air vehicle including air vehicle structure, air vehicle subsystems, propulsion systems, EPSs, mission systems, crew systems, and armament/stores systems.

Revision D has streamlined Section 19 by leveraging up-to-date military and industry specifications where possible. For corrosion prevention, NDIPs (described as an NDI Program Plan in AS6870), and materials selection, this section relies primarily on MIL-STD-1568, SAE AS6870, and MIL-STD-1587 respectively. If a program complies with these documents, it will comply with those Section 19 criteria.

Many of the MoCs in this section rely on laboratory data for substantiation. Based on the MIL-STD-961 definition, the reliance on laboratory data falls under the “Analysis” category. For legacy materials and corrosion prevention schemes, much of the laboratory data already exist. For new materials, or in cases where legacy material databases have a gap, new laboratory data are required to comply with the “Analysis” MoCs.

Tailoring of the MoC in this section by the government cognizant engineering authority is likely necessary. The tailoring will likely be driven by the phase of the program and the type of system being evaluated. For example, 19.1 requires verification of corrosion design per MIL-STD-1568 testing and through service history. At least one of these should be tailored out for every evaluation. For example, there is no reason to test if the corrosion design has been demonstrated through service life in a sufficiently comparable application. Another example is the use of MIL-STD-1587 for all air vehicles, and munition attachments using MIL-STD-186. In rare cases, weapons may be subject to AW evaluation and may use MIL-STD-186. Other examples of necessary tailoring of methods can be found by close examination of the MoCs.

This revision attempted to encompass both developmental and pre-existing systems. However, there are requirements specific to systems in development that had to be included for completeness and would be difficult to meet with a pre-existing system. Heavy tailoring is recommended for pre-existing systems. The intent of this tailoring should be to show that the pre-existing system meets the intent of the criteria through successful service history or other alternate means.

The authors of this revision intend that 19.1, 19.2, and 19.3 apply in some form to all AW evaluations. Use of 19.4 should be limited to the special processes under consideration in the AW evaluation that could pose an inherently new risk to the system. For example, if a modification is being evaluated for AW and the modification involves drilling new holes in an existing component, the special processes used to fabricate the component should not be considered, but the process controls on drilling the new holes (e.g., speeds, feeds) should.

Users of this handbook should be aware that additional materials requirements specific to air vehicle propulsion systems are in Section 7, electrical wiring insulation are in Section 12, and additional NDIP manual requirements are in Section 16.

19.1 Materials selection, processing, and properties.

Standard: Material used in the air vehicles are fully characterized and meet the technical requirements for materials, materials processing, and materials properties established in MIL-STD-1587.

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Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) No prohibited or commonly restricted materials listed in MIL-STD-1587 or MIL-STD-1568 are used in SOF components in the air vehicle design.

(2) Material properties are based on the specific process controls used for component production for process intensive materials such as welding and additive manufacturing, IAW MIL-STD-1587.

Analysis.

(1) The materials selected for use in the air vehicle meet the technical requirements established in MIL-STD-1587.

(2) The materials properties used in the air vehicle are design allowables developed IAW MIL-STD-1587 and meet the requirements of ASIP (MIL-STD-1530), PSIP (MIL-STD-3024), RSIP (MIL-STD-3063), or MECSIP (MIL-STD-1798), as applicable.

(3) The materials properties used in the air vehicle account for processing variation IAW MIL-STD-1587.

(4) For electrical wiring, insulation is IAW MIL-STD-1587.

19.2 Corrosion prevention and control.

Standard: Corrosion prevention and control processes IAW MIL-STD-1568 are established in a CPCP to maintain structural integrity for the operational and maintenance capabilities throughout the service life of the air vehicle.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Review documentation to confirm the CPCP for the system meets the requirements IAW MIL-STD-1568.

(2) Review the system design to ensure corrosion control and prevention measures are defined IAW MIL-STD-1568.

(3) System corrosion design is verified by service history on systems with more than 5 years of service.

(4) Ensure design finish specifications in the CPCP are documented IAW MIL-STD-1568.

(5) Corrosion-prone areas requiring NDI are identified and access for inspection throughout the system lifecycle (both usage and calendar time) has been assessed during system development IAW MIL-STD-1568.

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(6) Non-inspectable areas have increased corrosion control and prevention measures IAW MIL-STD-1568.

Analysis.

(1) Review the system design to ensure selected materials and corrosion preventative treatments meet the technical requirements of MIL-STD-1568, are an integral part of design, and they are included in all manufacturing, test, and usage support activities.

(2) Ensure finishes and finish processes specified in the engineering drawings and associated technical specifications conform to the definition of finishes and their associated selection criteria IAW MIL-STD-1568.

(3) Review the system corrosion design verification data to verify functionality per MIL-STD-1568 is achieved.

(4) The requirements for NDI are established during system development and test or via operational corrosion history, and the CPCP includes procedures to modify the requirement IAW MIL-STD-1568.

(5) Elastomeric sealing compounds for electrical systems are IAW with SAE AS50881.

19.3 Non-destructive inspection plan.

Standard: An NDIP exists, ensures that any damage on the air vehicle is smaller than the acceptable limits necessary to support the air vehicle design intent and maintain system integrity, and is IAW SAE AS6870.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) All initial and recurring inspection intervals are included in the NDIP.

(2) Non-destructive inspection process and procedures described in the NDIP are included in the maintenance publications.

(3) Any special NDI demonstration programs conducted to satisfy the durability and/or damage tolerance requirements are included in the NDIP.

Analysis.

(1) All inspections required for continuing AW over the expected system life on mission-critical, durability-critical, or SOF parts have a quantified capability that meets the requirements of ASIP (MIL-STD-1530), PSIP (MIL-STD-3024), RSIP (MIL-STD-3063), or MECSIP (MIL-STD-1798) as applicable.

(2) Required inspection intervals are based on component failure modes, the selected life management approach, the governing integrity program requirements, and, if necessary, a validated inspection capability for the specific component.

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(3) When flaw sizes are necessary for the life management approach of a component, and those flaw sizes are based on inspections, the inspections have a validated Probability of Detection IAW MIL-HDBK-1823.

19.4 Special processes and process integrity.

Special processes are those that directly affect the properties of the air system's materials, and the impact of the process on those properties cannot be or is difficult to validate after completion of the process. Because these processes cannot be validated after completion, additional controls and oversight are needed to ensure that the materials achieve their expected design performance and therefore provide the expected design performance and AW of the air system. Examples of special processes include heat treatments, welding and bonding, composite fabrication, additive manufacturing, shot peening, nondestructive inspection, sealant application, and organic and inorganic finish application.

The dependency of materials properties on the manufacturing processes and the inability to validate those properties upon completion of the processing requires critical process variable controls achieved via a special process control plan. The purpose of the special process control plan is to ensure that the material used in the air system design produced by a special process achieve their design performance for the duration of their design life. The special process plan also provides an integrated management structure to ensure ongoing and effective coordination between engineering, production and quality. A critical aspect of special processes and the special process control plan is that it requires special process manufacturing specifications that define for the special process the critical process variables, the methods to monitor and control those variables, the manner in which to qualify the special process for use, the manner in which to verify process capability, and guidance on what changes to a qualified special process will require requalification of the process and the required extent of the requalification.

19.4.1 Special process control plan.

Standard: A special process control plan that ensures that material used in the air system design produced by special processes achieve their design performance for the duration of their design life exists for the air vehicle.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Review documentation to confirm a special process control plan that meets the standard exists for the air vehicle.

(2) Review documentation to confirm special processes are defined and controlled IAW SAE AS9100.

19.4.2 Special process integrity.

Standard: Each special process is controlled by a process specification that defines the critical process variables, the methods to monitor and control those variables, the manner in which to qualify the special process for use, the manner in which to verify process capability, and

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guidance on what changes to a qualified special process will require requalification of the process and the required extent of the requalification, so that the materials used in the air system design produced by the special process achieve their design performance for the duration of their design life.

Method of Compliance:

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Examination.

(1) Review documentation to confirm that each special process in the special process control plan has a corresponding process specification that meets the standard.

Analysis.

(1) Review process specifications for special processes to confirm they will result in continual process integrity, thus meeting the standard.

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20. AERIAL DELIVERY: AIR TRANSPORT AND AIRDROP

The introduction of new air vehicle designs, modifications to existing air vehicles, and changes to air vehicle mission sets all require careful consideration of the unique challenges associated with aerial delivery operations. For new air vehicle designs, it is essential to ensure that the air vehicle is optimized for safe and efficient airworthy cargo transport, with design features and systems that meet AW standards and MoC governing aerial delivery. It is critical to recognize that new, outsized, and non-standard cargo may change the capability and performance of the air vehicle cargo system; as such, new, outsized, and non-standard cargo is subject to the same standards and MoC as the air vehicle itself. Similarly, when an air vehicle's mission set is changed to include aerial delivery capabilities, the air vehicle and its systems should be evaluated and certified to ensure compliance with relevant standards and MoC.

The sections below provide a detailed overview of the standards and MoC that apply to aerial delivery capabilities, covering topics such as air vehicle design, cargo restraint systems, weight and balance, and personnel airdrop systems. In addition, the following standards and MoC assess the AW of the air vehicle system’s aerial delivery capability.

20.1 Clearances.

Standard: Air vehicle dimensional envelope is defined and provides physical clearance with transported aerial delivery items approved for use in its technical data IAW MIL-STD-1791, Loading Unloading paragraph and Dimensions for Flight paragraph, and MIL-STD-1791-1, Tip-off Curve paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct dimensional analysis between air vehicle clearance limits and transported cargo capability to show clearance is maintained.

Demonstration.

(1) Perform loading of representative cargo capability to verify air vehicle clearances are maintained.

20.2 Air vehicle restraint system structure.

Standard: The structural integrity of the restraint systems and their supporting components can withstand the required static loads and stresses imposed by aerial delivery, IAW MIL-STD-1791, Restraint Conditions paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify restraint limits are included in applicable cargo loading manual and maintenance manuals.

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Analysis.

(1) Perform loads and strength analysis to verify restraint system and respective structure will support the identified cargo loads.

(2) Perform loads and strength analysis to verify the restraint system and structure will restrain air vehicle maximum payload capacity under flight loads for each principal direction.

Demonstration.

(1) Conduct a ground demonstration to verify that restraint devices can be anchored and applied to the maximum payload capability of the air vehicle cargo system without interference.

Test.

(1) Ground test: Perform testing to verify restraint system and respective structure will withstand forces at the rated capacities of the restraint system IAW JSSG-2006, Static Strength paragraph.

(2) Ground or Lab test: Perform component testing to verify restraint system and respective back up structure will withstand forces at the rated capacities of the restraint system IAW JSSG-2006, Static Strength paragraph.

(3) Ground or Lab test: Perform interface testing to verify restraint system and respective back up structure will withstand forces at the rated capacities of the restraint system IAW JSSG-2006, Static Strength paragraph.

20.3 Cargo restraint system structure.

Standard: The air vehicle cargo restraint system is designed to resist dynamic loads and prevent movement of the cargo and payload in each principal direction during aerial delivery (air transport and airdrop), ensuring the cargo remains securely positioned and restrained throughout the operation, IAW MIL-STD-1791, Loads and Restraint paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Conduct loads and strength analysis to verify the air vehicle restraint system can restrain entire cargo and payload capacity of the air vehicle to meet aerial delivery requirements.

Demonstration.

(1) Conduct a ground demonstration to verify air vehicle restraint system can restrain the entire load capacity in all required directions and configurations of the cargo in the air vehicle.

(2) For items intended for airdrop operations, demonstration includes airdrop systems and release mechanisms.

Test.

(1) Ground test: Perform restraint system testing to verify the air vehicle restraint systems and restraint subsystem components (mechanisms) required for operation of those systems will withstand forces required to secure the cargo and payload IAW MIL-STD-1791, Loads and Restraint paragraph.

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20.4 Cargo and payload restraint provisions.

Standard: The air vehicle's cargo restraint system meets the air transportability requirements in the principal directions IAW MIL-STD-1366, Cargo Restraint Requirements paragraph for the cargo and payload capacity of the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review air vehicle capabilities documents, cargo system design data packages, and air transportability requirements to ensure air vehicle, cargo, and payload have defined tiedown provisions.

Analysis.

(1) Conduct loads and strength analysis to verify the defined provisions can restrain entire cargo and payload to meet aerial delivery requirements

Demonstration.

(1) Conduct a ground demonstration to verify cargo and payload restraint provisions can be anchored to the air vehicle cargo system without interference.

Test.

(1) Ground or Lab test: Perform crash hazard acceleration testing to verify tiedown provision restraint system will withstand forces required to secure the cargo and payload IAW MIL-STD-1791, Verification Methods paragraph.

Prior to entering operational test/operations, verify the standard by the above and following method(s):

Test.

(2) Ground or Lab test: Simulate load on tiedown provision restraints to meet aerial delivery requirements.

20.5 Weight and balance, center of gravity movement and jettison capability.

20.5.1 Loaded cargo weight and balance.

Standard: Air vehicle cargo and payload systems account for cargo CG static placement and does not result in an out of air vehicle CG envelope during aerial delivery. Air vehicle design ensures overall cargo weight, size, and CG location are compatible with air vehicle structural, weight, and balance limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Verify loading locations are specified to satisfy air vehicle CG and structural limits.

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(2) Review air vehicle capabilities documents, cargo system design data packages, and air transportability requirements to determine cargo location that will meet air vehicle structural, weight, and balance limits.

Analysis.

(1) Perform computational analysis to verify cargo is within the air vehicle weight and balance limits.

20.5.2 Cargo and payload CG movement during aerial delivery and dynamic operations.

Standard: The air vehicle accounts for cargo and payload weight shifts during airdrops and external load operations, ensuring that the air vehicle's CG always remains within published limits.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Ensure stability and control analyses are conducted to show air vehicle CG can be maintained during all flight phases.

Analysis.

(1) Conduct high fidelity 6-DOF flying quality simulations of taxi, takeoff, climb, cruise, descent, approach, landing, and dynamic aerial delivery, including transients.

(2) Simulate the dynamic forces of airdrop on air vehicle weight and balance during airdrop operations.

20.5.3 Cargo jettison capability.

Standard: Jettison emergency airdrop operations do not cause damage to the air vehicle.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review air vehicle capabilities documents, cargo system design data packages, and air transportability requirements to ensure air vehicle weight and balance limits are not exceeded during jettison operations.

Analysis.

(1) Conduct high fidelity 6-DOF flying quality simulations of dynamic cargo jettison, including transients.

Demonstration.

(1) Conduct a ground demonstration of emergency release procedures to show that crewmembers can safely jettison cargo in case of emergencies.

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20.6 Aerial delivery manuals.

Standard: Air vehicle cargo and payload loading and airdrop manuals provide all procedures necessary for aerial delivery.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Demonstration.

(1) Perform ground demonstration to verify cargo and payload loading and airdrop manuals provide the level of instruction that permits crewmembers to transport and air drop cargo without damage to the air vehicle.

20.7 Personnel airdrop system structure.

Standard: Air vehicle personnel airdrop system is designed to withstand the loads imposed by personnel during airdrop and possible malfunctions of personnel airdrop equipment IAW MIL-STD 1791-1, Personnel Airdrop paragraph.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Analysis.

(1) Perform loads and strength analysis to verify airdrop systems, including towed jumper systems, will support the specified numbers and weights of paratroopers and personnel recovery systems defined for the air vehicle.

(2) Simulate calculated forces on personnel airdrop systems including towed jumper retrieval systems.

Demonstration.

(1) Conduct a ground demonstration to verify personnel airdrop system including towed jumper retrieval of the specified numbers and weights of paratroopers and personnel recovery systems IAW MIL-STD-1791-1, Personnel Airdrop paragraph.

Test.

(1) Ground or Lab test: Perform system testing to verify personnel airdrop system, including towed jumper retrieval system, will withstand forces greater than the rated capacities of the personnel airdrop system.

(2) Lab test: Perform component testing to verify personnel airdrop system, including towed jumper retrieval system, and respective structure will withstand forces greater than the rated capacities of the personnel airdrop system.

(3) Ground or Lab test: Perform interface testing to verify personnel airdrop system, including towed jumper retrieval system, and respective back up structure will withstand forces greater than the rated capacities of the personnel airdrop system.

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20.8 Cargo ignition sources.

Standard: Cargo is loaded and handled in a manner to prevent it from either becoming an ignition source itself or from contacting any ignition sources.

Method of Compliance:

Prior to entering flight test, verify the standard by the following method(s):

Examination.

(1) Review air vehicle engineering drawings to verify sources, locations, and configurations of possible ignition sources.

(2) Integrate cargo engineering drawings with air vehicle engineering drawings to verify existence of cargo clearances and means of preventing cargo contact with ignition sources.

Demonstration.

(1) Conduct ground demonstration to verify cargo clearances and preventive means of contacting ignition sources exist.

Test.

(1) Lab test: Perform testing, including but not limited to explosive atmosphere testing, to show cargo items do not become ignition sources IAW MIL-STD-810, Explosive Atmosphere paragraph.

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21. NOTES

21.1 Intended use.

This handbook outlines the AW criteria for assessing the design of all fixed-wing and rotary-wing air systems, both manned and unmanned. It is intended for use at any stage of an air system's life cycle when a design-based AW assessment is necessary. The criteria represent a minimal subset of design standards and are not intended to establish strict requirements. Rather, users are expected to review the criteria, determine their applicability, and tailor them as needed to ensure an equivalent level of safety for their specific system once approved. A program may use these tailored MIL-HDBK-516 criteria to establish system requirements. The AW criteria should be integrated into the broader systems engineering and requirements development process, rather than treated in isolation, and should be used in conjunction with other specifications, standards, and engineering practices to fully implement OSS&E.

21.2 Subject term (key word) listing.

Aerial delivery

Aerial refueling

Air system

Air vehicle

Airworthiness

Armament and stores

Avionics

Computer systems and software

Crew systems

Diagnostic systems

Electrical systems

Electromagnetic environmental effects

Failure modes

Flight technology

Hazard analysis

Maintenance

Materials

Propulsion systems

Risk management

Safety

Structures

Subsystems

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Systems engineering

System safety

Verification and validation

21.3 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A
AIRWORTHINESS CERTIFICATION CRITERIA POINTS OF CONTACT

A.1 Scope.

This appendix provides points of contact for the AW certification criteria contained within this document.

A.2 Points of contact.

A.2.1 U.S. Air Force.

ALFCMC/EN-EZ Engineering Directorate

2145 Monahan Way, Bldg 28

Wright Patterson Air Force Base, OH 45433-7017

Email: USAF.Airworthiness.Office@us.af.mil

A.2.2 U.S. Army.

U.S. Army Combat Capabilities Development Command Aviation and Missile Center (DEVCOM AvMC)

Army Airworthiness

RDMR-AE

Bldg 4488 Martin Road, C-100

Redstone Arsenal, AL 35898-0001

Phone: 256-313-8400

Email: usarmy.redstone.devcom-avmc.mbx.airworthiness@army.mil

Website: https://avmc.devcom.army.mil/

A.2.3 U.S. Navy.

Airworthiness & CYBERSAFE Office

Naval Air Systems Command

22244 Cedar Point Road, B460

Patuxent River, MD 20670-1163

Airworthiness Global Customer Support Team

Phone: 301-757-0187

Email: usn.patuxent.nawcad.mbx.AIRWORTHINESS@us.navy.mil

Website: https://airworthiness.navair.navy.mil

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Air Force -18, 20, 24, 40, 84, 171, 184
Navy - CG

Civil agencies:
FAA - ACO
NASA - NA

Preparing activity:
Air Force - 11
(Project No. SESS-2026-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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